

MKM Research Labs

Weather Patterns to Physical Risk Swaps

MKM RESEARCH LABS

David K Kelly

Edited and Foreword by Johnny Mattimore

Second Edition: TBA

A Practitioner's Guide to Flood Risk Management and Other Hazards

Second Edition

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About the Author



David K Kelly is a seasoned investment banking professional with 30 years of experience, having held senior leadership positions in the front office and risk departments at global Tier 1 banks.

His entrepreneurial ventures include his most recent role as co-founder and Chief Science Officer of MKM Research Labs, a research company focused on developing and donating open-source business solutions for finance, particularly in the physical risk space.

Throughout his career, David has demonstrated expertise in implementing advanced modelling solutions for capital requirements under new regulatory frameworks. He is a leading advocate for Model Risk Governance for vendors that provide data and analytics to Finance.

David has held influential roles in the Model Risk industry, writing the definitive book entitled “Model Risk Management for Vendors”, published in August 2025.

David’s educational background includes a first-class degree in Mathematics from the University of Bristol and Honours in Advanced Studies (Part III) in Pure Mathematics from the University of Cambridge.

About the Editor

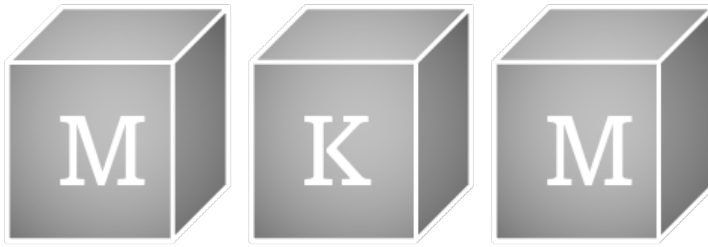
Johnny Mattimore has over 35 years in financial services. He originally trained as a mathematician and markets economist working on major trading floors of capital markets firms. He worked for the first half of his career in investment banking in analysis, trading and structuring.

He then spent the second half of his career in hedge funds and asset management across a range of roles running money, risk management, and trading systems design and integration. More recently, he has focussed his efforts on the integration of cutting-edge risk solutions into the financial ecosystem for banking, insurance and asset management.

He is a leading proponent of open-source standards and has a high level of expertise in the integration of data, methods and models into finance for lines of business and group functions. As well as having held mainstream roles in large organisations, he has a long history in developing new businesses, particularly in hedge funds and fintech.

Johnny graduated from the University of Bristol, UK, in 1988 with a BSc in Mathematics. He has written extensive research throughout his career, including major works for risk, emerging markets, structured debt and investment methodologies, ranging from discretionary to systematic.

About MKM Research Labs



MKM Research Labs

MKM stands at the forefront of open-source innovation in banking technology, mainly focused on addressing the banking sector's critical challenges of physical risk management. Founded in January 2025 by Johnny Mattimore, David Kelly, and Fearghal McGoveran, MKM consolidated years of prior open-source work into a cohesive entity dedicated to developing and donating

open-source business solutions for finance.

“We are not in the weather prediction business. Rather, we quantify banking risk by modelling the probability distributions of extreme but plausible weather events that trigger floods. We focus on assigning accurate probabilities and banking impact severities to these tail events, enabling banks to assess their exposure to hazard-related risks properly.” - MKM Exco.

The company's flagship initiative focuses on transforming flood risk management through open-source banking technology. Their approach applies capital markets technology to physical risk management, enabling flood risk to be packaged, priced, and traded in real-time, similar to interest rate and credit default swaps. Central to their work is developing a Common Domain Model (CDM) implementation that standardises flood and hazard risk data and models, including integrating fluid dynamics for flood modelling.

MKM leads crucial projects in Physical Risk & Resilience while spearheading the extension of the Common Domain Model to property and physical hazards. Their work extends beyond traditional risk management to include the OS-SFT (Open-Source Sustainable Finance Taxonomy) project, which aims to harmonise regulatory interpretation across jurisdictions. Through their open-source approach under the FINOS umbrella, MKM is building a new market infrastructure that promises to ensure the resilience of property finance for decades to come.

MKM actively participates in the Banking Services Open-Source movement under the auspices of ISDA and FINOS.

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For the Common Domain Model, the Author would like to thank Ian Sloyan, who has provided invaluable guidance on our initial designs for the Physical Risk CDM, which can be found at <https://github.com/MKM-Research-Labs/Physrisk-cdm>.

Finally, there are many organisations who have contributed to the thinking behind this book over recent years, including, but not limited to: ISDA (International Swap Dealers Association), ICMA (International Capital Markets Association), ISLA (International Securities Lending Association), BNP Paribas, Citi, Goldman Sachs, JP Morgan, Morgan Stanley, UBS, Fannie Mae, Red Hat, NVIDIA, JBA Risk Management, OASIS Loss Modelling Framework, and Imperial College London.

To my lovely wife, Victoria, and my wonderful children, David Henry, Georgina and Angus

To Johnny Mattimore, whose infectious enthusiasm for this book series and thoughtful critique has kept me (nearly) sane.

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Foreword - Editor, Johnny Mattimore

David Kelly and I have come a long way together on the journey of integrating physical risk into the banking system. While this journey is quite advanced, it is far from over. The completion of the first edition of this book marks a meaningful milestone in our journey. It represents the first major publication from our new firm, MKM Research Labs, which was founded by the three of us - Johnny Mattimore, David Kelly, and Fearghal McGoveran - in January 2025 to consolidate our decades of experience in banking and finance.

“This book provides a visionary guide to integrating Physical Risk into the banking system, at high speed, large scale and reduced cost for all.”- Johnny Mattimore, MKM.

Introduction

The motivation for writing the book is identical to that for creating the firm: to combine cutting-edge artificial intelligence, data, models and deep domain expertise to solve some of the most challenging problems in finance. One of those problems is how to integrate physical risk into the banking ecosystem at high speed, at large scale and at reduced cost for all.

The book is crafted from many years of real-world experience and scholarly excellence, combined with the essential reflection and vision necessary to have a meaningful impact on the reader.

Context for the Book

Put simply, this book makes the subject of physical risk easy and accessible for bankers to understand. It provides bankers with what they need in order to understand physical risk without the need to become experts in every underlying skill. Uniquely, in my view, it does this as a single source, as Kelly has taken the best of current research and models from hundreds of sources and combined them with his individual skill and knowledge, into an accessible guide for bankers.

To achieve this simplicity, it has three common threads running throughout it, taken from great minds of history: Newton, Aristotle and Michelangelo. All of these threads are familiar to us and are identifiable in all works of great success for humankind.

Giants

"If I have seen further, it is by standing on the shoulders of giants."- Sir Isaac Newton, (1675).

In this phrase, Newton acknowledges that his discoveries were made possible by building on the work of great thinkers before him - such as Galileo Galilei, Johannes Kepler, and René Descartes.

Following this tradition, this book illustrates how a solution for the bankers' problem is only possible by building on the work of others. So, the book draws together many other areas of skill developed over many decades in the modern fields of physical risk, including: weather forecasting, hydrology, hydraulics, specialist common data models, artificial intelligence and the imperative of timeseries models for pricing risk in banks. Through these guiding steps, Kelly illuminates the pathway for solving the bankers' problem of physical risk integration into its existing core system with minimal friction.

Value

"The whole is greater than the sum of its parts"- Aristotle, circa (340 BC).

While the exact wording may not appear verbatim in Aristotle's surviving works, the idea reflects a key concept from his *Metaphysics*, where he discusses how complex systems and organisms have properties that cannot be explained solely by analysing their individual components.

Thus, the brilliance of this book lies in Kelly's ability to connect the many areas of expertise in an accessible way. Each component is demystified so that by the end, the reader feels confident that the formerly opaque and confusing becomes clear and lucid.

Moreover, it then becomes evident how these individual components alone do not solve the bankers' integration problem of estimating the financial risk in terms of real money. Instead, it shows that only by combining these components, with defined sequencing and coupling can one extract the required value. Thus, it delivers on the promise to the banker of proving that "the whole is greater than the sum of the parts".

Ambition

"The greater danger for most of us lies not in setting our aim too high and falling short, but in setting our aim too low and achieving our mark."- Michelangelo Buonarroti (1475-1564).

While possibly apocryphal, this phrase nevertheless captures the ambitious spirit of Michelangelo's life and work which gave rise to his monumentally daring projects. Today, it is still a powerful motivational idea associated with his legacy.

Rephrasing this for our era and for bankers, an appropriate rewrite might be: "In banking, as in life, the greater danger lies not in bold targets unmet, but in modest goals too easily achieved." So, one could play it safe and get modest results, but as any financier knows, it is bold goals that drive meaningful change to deliver outsized growth while maintaining stability and repeatability.

Kelly outlines the opportunity that I coined, and which he and I have jointly developed for a new tradeable asset to manage, hedge and transfer physical risk: the Physical Risk Swap (PRS).

We developed this idea for a capital markets PRS over the last five years. Our first publication was in November, 2023, as a LinkedIn post. This was where we first coined the term Physical

Risk Swap (PRS), comparing it to a Credit Default Swap (CDS), first used in Collateralised Debt Obligations (CDO), and with a securitised form of a Parametric Insurance Contract.

Our invention of PRS is truly ambitious: it not only transforms the world of physical risk into a tradeable asset, but it is predicated on the ambition of creating an open source architectural framework for the entire banking ecosystem.

In this book, Kelly's explanation takes a huge step forward compared with other related literature detailing how to achieve the ambition of PRS across a stack of models. In my view, the creation of a market in PRS is the pinnacle of our joint ambition and the ambition of our firm, MKM Research Labs.

Kelly explains this transformational leap forward to PRS by leveraging the achievements of others in different domains to deliver value faster; by combining the sum of the parts to deliver even greater total value; and, finally, by setting the ambition high. In our vision, PRS creates a brand new growth opportunity while enhancing systemic stability.

This combination of Giants, Value and Ambition is what makes this book a wonderful piece of craftsmanship and a truly outstanding contribution to banking and society.

Final Comments

Throughout reading this book, perhaps remember this one enduring fact: without a functioning, solvent banking system, almost all that we depend upon, and all that we take for granted is at risk of being taken away from us unless we protect ourselves from extreme physical risks. This book contributes substantially to the body of accessible knowledge to help us preserve all those things that we hold dear.

Enjoy the read and then the ride as you witness how managing physical risk in banking will change the world for the better, for the benefit of all and forever.

Johnny Mattimore

CEO and Co-founder of MKM Research Labs

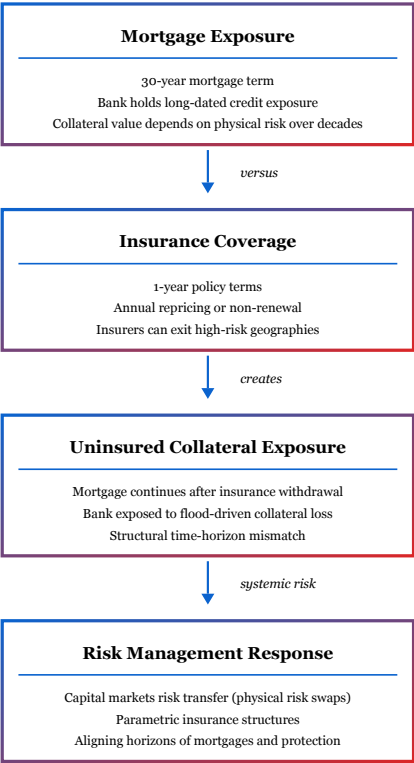
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Prologue - Author, David K Kelly

The banking world stands at a critical inflexion point. A threat more insidious than market volatility, more persistent than credit cycles, and more fundamental than technological disruption is reshaping the risk landscape. Once considered a future concern, physical risk has evolved into an immediate crisis that threatens the very foundations of our banking system.

Consider this stark reality: The US property market, valued at \$52.5 trillion and burdened by \$13.6 trillion in debt, faces an unprecedented challenge. In 2024, we observed a mass exodus of insurers from coastal markets, with 42% of US coastal insurers withdrawing from Florida and the Carolinas. This situation left \$4.2 trillion in mortgage collateral exposed and uninsured. The reduction in available insurance is not a gradual shift; flood insurance premiums skyrocketed 58% yearly, pricing out homeowners and destabilising property valuations across entire regions.

Figure 1 – Mortgage–Insurance Time Horizon Mismatch



The scale of exposure to the UK property market is staggering. Along the Gulf and Atlantic Coasts alone, 7.3 million homes face the threat of storm surge flooding. Nationwide, 17% of all properties risk flood damage, with the highest concentrations in Florida, Louisiana, California,

New York, and New Jersey. In California, 4.6 million properties stand in wildfire zones, while Texas faces approximately 137 tornadoes annually, affecting over 13 million people. These are not distant threats – they are present realities affecting properties that underpin mortgages, secure investments, and represent lifelong savings for millions of Americans.

We face a fundamental mismatch in time horizons that threatens the stability of our banking system. Banks write mortgages with 30-year terms, while insurers underwrite policies annually. As physical risks intensify, insurers can and do retreat from markets they deem too risky, leaving banks exposed to decades-long commitments on potentially worthless assets. This temporal disconnect echoes the opacity that preceded the 2008 banking crisis, but with a crucial difference – physical risk is not a complex banking instrument that can be traded away. Flood risk is tied to the immutable realities of geography, weather patterns and infrastructure, and the risk of banking loss is tied to individual property value, flood resilience, and outstanding mortgage debt.

More recently, banking regulators in various jurisdictions have been working on incorporating climate risk more explicitly into regulatory frameworks, but these are generally still developing and vary by region. The Basel Committee on Banking Supervision has published principles for the effective management and supervision of climate-related banking risks, but these haven't yet translated into standardised capital requirements. However, there are concerns that individual regulators could use their discretionary powers under Pillar 2 to impose capital charges of up to 15% for unhedged physical climate risk exposures.

Some countries are taking more direct approaches to managing climate-related physical risks. For instance, the Italian government has implemented measures requiring all corporations to obtain specific flood-related insurance coverage, shifting some of this risk management responsibility directly to businesses rather than leaving it entirely within the banking regulatory framework.

Many lenders are considering such regulatory risk and exiting mortgage provisions backed by any property in a heightened flood-risk area.

The traditional bank risk management playbook offers no solutions. Our banking systems, in handling market and credit risk, remain primitive in their approach to physical risk. The tools, models, and frameworks that served us well in managing traditional banking risks prove inadequate in the face of this new challenge.

Yet, within this crisis lies opportunity. The same forces driving insurance retreat could catalyse innovation in risk transfer, valuation, and resilience investment. When properly directed, the banking sector's capacity for innovation could transform physical risk from a threat to a frontier for new banking products, improved risk management, and enduring property finance.

This book presents the MKM Framework for Physical Risk, a transformational approach that bridges the gap between academic-driven scientific research and banking mathematics, property-level analysis and systemic risk management, and current challenges and future solutions. The stakes could not be higher. The integrity of our mortgage markets, the stability of our banking system, and the resilience of our communities all hang in the balance.

To provide some context, the Pew Research Centre outlines that home equity is a significant component of household wealth, accounting for 45% of net worth among UK homeowners in 2021. While the UK and EU have slightly lower concentrations, Chinese households, according to East-West Property, hold 65% of real estate.

The time for incremental solutions has passed. We need a fundamental reimagining of measuring, managing, and transferring physical risk. The future of finance depends on it. This book aims to forge a path from forecasting weather patterns to live revaluation of a portfolio of banking assets.

The ultimate goal is to apply Capital Markets technology as a bridge into the insurance industry, triggering a recovery in insurance capacity and leading to lower and more available underwriting capacity.

Before we can design and scale these capital markets solutions, we need a shared language for the underlying physical threats. The same way credit risk models rest on clear definitions of borrower characteristics and default events, any framework for physical risk swaps must start with a disciplined view of the hazards themselves—what they are, how they behave, and how they enter bank balance sheets. The next section sets out this hazard landscape, creating the taxonomy on which the rest of the MKM Framework for Physical Risk is built.

The Hazard Landscape

Physical risk in banking is not a single phenomenon but a family of hazards that share common drivers, behaviours, and transmission channels into balance sheets. For the purposes of this framework, it is useful to group them into four broad classes that align with how risk data are generated, modelled, and ultimately securitised.

Hydrological risks – water as balance-sheet risk.

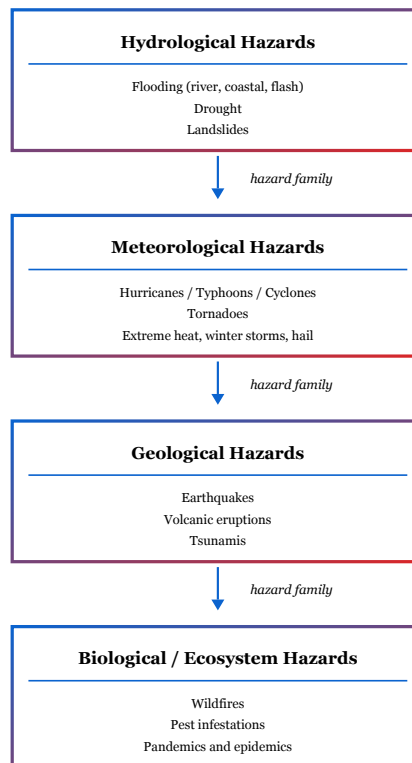
Hydrological hazards arise when water is in the wrong place, at the wrong time, and in the wrong quantity. River, coastal, and flash flooding directly damages residential and commercial property, disrupts business operations, and impairs infrastructure that underpins local economies. Drought sits at the other end of the spectrum, eroding agricultural productivity, stressing water utilities, and weakening the cash flows that support project finance and sovereign debt. Landslides, often triggered by prolonged rainfall or soil saturation, can destroy transport links and housing clusters, creating concentrated collateral losses within a small geographic area. For banks, hydrological risks translate into simultaneous hits to collateral value, borrower income, and local economic activity, making them central to any discussion of mortgage and property-backed exposures.

Meteorological risks – atmosphere-driven shocks.

Meteorological hazards are driven by atmospheric dynamics and include hurricanes, typhoons and cyclones, tornadoes, extreme heat, severe winter storms and hail. These events are typically

acute, high-intensity shocks with clear timestamps, which makes them amenable to parametric triggers and event-linked structures. A single landfalling cyclone can combine multiple perils—wind damage to roofs and transmission lines, storm surge flooding of coastal property, and heavy rainfall overwhelming urban drainage systems—creating complex loss patterns across general insurance, mortgage books and infrastructure lending. Heatwaves and cold snaps are slower-burn hazards, stressing power systems, increasing mortality, and affecting labour productivity, with second-order implications for consumer credit, SME lending and municipal finance. In all cases, meteorological hazards generate rapid changes in default probabilities and recovery rates that traditional, backward-looking credit models struggle to capture.

Figure 2 – Physical Hazard Taxonomy



Geological risks – low-frequency, high-severity events.

Geological hazards such as earthquakes, volcanic eruptions and tsunamis are not directly driven by climate dynamics but sit within the broader physical risk set because of their catastrophic loss potential. They tend to exhibit low frequency but very high severity, with damage concentrated in specific regions that share geological features. When a major earthquake strikes a metropolitan area, entire portfolios of mortgages, commercial real estate loans and infrastructure financings can experience simultaneous collateral impairment, business interruption and, in extreme cases, permanent out-migration. Tsunamis and volcanic ash clouds can further amplify losses by

disrupting ports, airports and logistics hubs, affecting trade finance and supply-chain-dependent borrowers far from the epicentre. For capital markets, geological risks resemble tail events in credit portfolios, demanding tools that can absorb and redistribute extreme but infrequent losses.

Biological and ecosystem risks – living systems as risk vectors.

Biological and ecosystem-related hazards, including wildfires, pest outbreaks and pandemics, represent channels through which changes in climate and land use feed into financial outcomes. Wildfires combine meteorological precursors (heat, drought, wind) with ecosystem conditions (fuel load, land management) to generate fast-moving events that can devastate peri-urban housing, disrupt power grids and leave long-lasting scars on local property markets. Pest infestations and plant diseases affect crop yields and forestry assets, weakening borrowers in agricultural and rural economies and introducing volatility into commodity-linked revenues. Pandemics and epidemics operate primarily through health and mobility, but their impacts on employment, consumption, and government balance sheets feed directly into default rates across retail and corporate lending.

“These hazards remind us that physical risk does not only arrive as “storm damage on a single property” but can propagate through biological systems into broader credit and market risk factors.” - David Kelly, MKM

Taken together, these four hazard families provide a structured map of the physical risk universe that banks must navigate. In the remainder of this book, we narrow our focus to hydrological risk—specifically flood—because it offers the clearest pathway from scientific hazard data to property-level valuation, mortgage portfolio re-pricing, and ultimately to the design of physical risk swaps in capital marketse book, it is worth outlining a classification.

Although this book reflects physical risk as a present danger, for completeness, the Intergovernmental Panel on Climate Change (IPCC) of the United Nations, in its sixth assessment report, classifies physical risks related to climate change into three main categories:

About this book - A focus on flood risk

The world of finance, meteorology, fluid dynamics, and instrument pricing is awash with academic articles and dense mathematical formulae. The purpose of this book is not to dive into the maths but to help capital markets practitioners make sense of the problem of flood risk in finance and forge a pathway to implementation that will need to straddle multiple academic expertise.

At some point, we will introduce mathematical equations. The author believes they are helpful for explanation but not essential for an overall appreciation of the concept. Therefore, non-mathematicians can skim through with a knowing nod, leaving it to the quants to enjoy. The sections, including mathematical equations are highlighted in blue.

There are many hazards that groups like OS-Climate continue to lead with excellent research. Hazards such as wildfire, soil erosion, drought and pestilence. Maybe not the last. We are

focused on floods arising from weather formations to set up the correct framework for risk transfer within a capital markets context. To that end, there is some overlap with wind and coastal damage as hazards arising from cyclones.

The author and the MKM Exco leave the modelling of potential climate changes for horizons out to 2100 to academic bodies under the IPCC's umbrella. MKM is dedicated to implementing a framework that complies with how banking market risk management is adopted by all banks that warehouse risk. The management of hazard risk follows the same four steps currently applied to instrument pricing: Identifying hazard risk, Measuring hazard risk, Monitoring exposure to hazards, and building Control tools to affect risk transfer.

Impact of Flood Risk on UK Properties

While the focus here is on the UK property sector, the underlying issues apply straightforwardly to the EU, US, China, and Japan.

Flood risk has emerged as a critical factor reshaping property markets, regulatory frameworks, and banking stability across the UK, with profound implications for the broader banking sector. Recent comprehensive assessments reveal an escalating threat that demands urgent attention from banking institutions, policymakers, and property owners alike.

The latest National Assessment of Flood Risk (NaFRA) identifies 6.3 million properties in England currently classified as flood-prone, with 4.6 million at risk from surface water flooding and 2.4 million from rivers/sea flooding. This represents a substantial portion of the nation's housing stock, with climate change projections suggesting this figure could rise dramatically to 8 million properties (1 in 4) by mid-century, driven by increasingly extreme weather patterns, heavier rainfall, and accelerating sea-level rise.

Regional disparities in flood risk exposure are significant and economically consequential. East Anglia, the North West, and Yorkshire face the highest exposure rates, with 13–18% of properties in high-risk zones. Urban areas with ageing or inadequate drainage infrastructure, such as parts of London and Birmingham, are particularly vulnerable to surface water flooding, which now accounts for 57% of all flood insurance claims.

The Bayes Business School's "Residential Property Flood Risk UK 2023" research report reveals that flood risk creates measurable valuation penalties in the property market. Properties affected by flood risk sell at an 8.14% average discount compared to equivalent non-affected properties, escalating dramatically to 31.3% for very high-risk areas. This price differential correlates with flood probability, with each 1% increase in risk-reducing property values by 0.13–0.19%.

The market dynamics reflect rational economic behaviour: high-risk homes experience a 50% slower price growth than properties in safer areas. Critically, the average flood risk of sold properties between 2006 and 2021 was 8.01%, compared to an unsold property's 8.63%, indicating that flood exposure significantly impacts marketability. Analysis shows that zero exposure to flooding can increase a property's saleability from 63.3% to 65.6%.

Insurance Challenges and Banking Stability Risks

The UK's government-backed insurance scheme, Flood Re, in their Transition Plan Report 2023, currently protects approximately 200,000 households in high-risk areas. Still, its scheduled expiration in 2039 raises serious concerns about future insurance affordability and availability. With the Association of British Insurers pointing out that the average flood claims cost £32,000 per incident, insurers are increasingly factoring climate projections into their risk models and pricing.

This creates a potential systemic risk to banking stability through collateral devaluation. Properties in high-risk areas face growing loan defaults as declining property values outpace mortgage balances, potentially creating clusters of negative equity in vulnerable regions. JBA Risk Management estimates the annual cost of flood damage to UK properties at £527 million, a figure projected to rise substantially without significant adaptation measures.

Regulatory Response and Adaptation Measures

Regulatory frameworks have evolved to incorporate stricter building codes in response to these growing risks. New floodplain developments must now implement elevated foundations, flood barriers, and sustainable drainage systems (SuDS). The Sequential Test mandates avoiding high-risk zones for new construction unless no viable alternatives exist.

In March 2025, the UK Government and Environment Agency announced a record £2.65bn investment in flood defences, aimed at building or repairing over 1,000 flood schemes and protecting 66,500 properties by 2026, focusing on vulnerable regions like the Thames Estuary and Humber Basin. Despite these efforts, climate projections by the National Assessment of Flood and Coastal Erosion Risk in England 2024 indicate that 637,600 properties could face high river/sea flood risk without accelerated adaptation by 2070 (a 73% increase), and 1.8 million may be exposed to severe surface water flooding (a 66% increase).

Future Outlook and Banking Sector Implications

The banking sector's response to escalating flood risk represents a fundamental restructuring of how financial institutions assess, price, and manage climate-related exposures across their mortgage portfolios and lending operations. This transformation extends far beyond regulatory compliance, encompassing innovative financial products, technological integration, and a complete reimagining of risk transfer mechanisms that will reshape property markets and banking practices for decades.

Parametric Insurance as a Banking Innovation

Parametric flood insurance has emerged as a transformative mechanism that fundamentally alters the relationship between flood events and financial recovery. Unlike traditional indemnity-based insurance that requires lengthy damage assessments and claims processing, parametric solutions provide automatic payouts when predetermined thresholds are exceeded, such as rainfall

intensity, river water levels, or satellite-detected flood extent. This approach delivers prompt compensation through a swift, transparent process that eliminates the need for lengthy claims procedures.

The market for parametric flood insurance is experiencing rapid growth, with projections indicating the global market will reach \$3,000 million in 2025 and expand at a compound annual growth rate of 13.4% through 2032. This swift expansion reflects fundamental shifts in how financial institutions and property owners perceive and manage flood risk. The technology utilises machine learning and real-time monitoring from satellite imagery and IoT sensors to enable businesses to recover more quickly from climate-related disruptions.

Parametric products can be structured as Flood-at-Location coverage using on-site sensors, River Gauge monitoring of water level indices, or Excess Precipitation detection using gridded rainfall data. These different trigger mechanisms enable banks to tailor protection based on specific geographic and property-level vulnerabilities within their mortgage portfolios.

Property Flood Resilience and Valuation Transformation

Property flood resilience measures have transitioned from optional upgrades to essential valuation components affecting property prices and mortgage terms. Integrating flood resilience features—including flood barriers, water-resistant materials, elevated electrical systems, and improved drainage—now commands measurable premiums in property transactions. This shift reflects a market repricing of flood risk that acknowledges both the frequency of flood events and the demonstrated effectiveness of mitigation measures in reducing damage.

European banking stress tests have revealed that high and medium flood risk exposures bear half of the projected losses despite representing only 31% of overall exposures. In flood scenarios, real estate collateral experiences severe damage that increases loan-to-value ratios by 5-20% depending on flood risk classification, ultimately driving up loss-given-default rates and expected losses. Mortgages appear particularly vulnerable to flood risk materialisation compared to corporate exposures secured by real estate, with LTV ratio deterioration showing marked increases within one-year projection horizons.

The valuation impact extends to mortgage availability and pricing. Properties in flood-prone areas without adequate resilience measures face higher mortgage rates, and in extreme cases, complete loss of mortgage availability in certain localities. This creates a bifurcated property market where flood-resilient homes maintain value and accessibility to financing, while vulnerable properties experience systematic devaluation and reduced liquidity.

Banking Sector Integration and Risk Management Frameworks

Financial institutions are developing frameworks that integrate flood risk assessment throughout the lending lifecycle, from origination through portfolio management. This integration requires massive technological infrastructure capable of processing geospatial data, climate projections, and property-level characteristics at scale. Industry initiatives demonstrate this ambition—current projects aim to process flood risk assessments for over one million

mortgages in Irish markets, 10+ million mortgages across UK lenders, and ultimately scale to 300+ million mortgages in the U.S. market.

The operational framework combines regulatory reporting data on physical risk (flooding, forest fire) with AI-based clustering on geo-location and hazard rates to identify portfolio-level concentrations. Banks then model individual mortgage and aggregate revaluations across different time horizons for at-risk clusters, establishing strategic concentration-reduction targets. This enables scenario analysis comparing different mitigation approaches—from flood defence installation costs to emission reduction through EPC improvements—against pathway projections without mitigation under various climate scenarios.

Fewer than 25% of banks currently include private insurance in their flood risk assessments, with only half of those considering substantial coverage (more than 50% of collateral loss). Public natural disaster relief schemes are even less frequently factored into most banking models. This underestimation of protective mechanisms may result in overly cautious risk evaluations but also underscores the important opportunity for banks to collaborate with insurers on comprehensive risk transfer solutions.

Regulatory Evolution and Market Structure

Mandatory climate risk disclosures for mortgages represent regulatory recognition that flood risk constitutes material financial information requiring transparent communication to investors and borrowers. These disclosure requirements compel banks to develop granular property-level risk assessments and communicate potential vulnerabilities explicitly. The regulatory framework acknowledges that if banks suddenly price in climate-related risk over short periods, it could create shocks to the financial system and broader economy, though gradual implementation timelines reduce the likelihood of systemic disruptions.

The pricing adjustment brings forward societal costs of physical risk, creating near-term costs especially for holders of vulnerable assets and banks holding such assets as collateral. Mortgage holders in flood-prone areas experience higher rates or reduced availability, while properties in especially vulnerable locations may face complete loss of insurance coverage. However, mechanisms like the UK's Flood Re scheme counterbalances to upward pressure on homeowner insurance premiums, demonstrating how public-private partnerships can maintain market stability during the transition.

Systemic Implications and Investment Decision-Making

Parametric structures offer vital advantages as flood frequency and severity rise, providing risk and price transparency that benefits both policyholders and insurers. The different types of floods—surface water, coastal, and river flooding—require tailored product approaches, and the availability of data limits the ability to develop solutions for highly localised risks. Nonetheless, parametric insurance is broadening into previously neglected areas that traditional insurance deemed too risky or lacked sufficient infrastructure.

For banking institutions, this evolution creates opportunities to develop climate-resilient lending practices and innovative products that incentivize adaptation. Banks can offer preferential rates for properties with documented flood resilience measures, structure loans that include financing for mitigation improvements, and partner with parametric insurers to provide bundled protection-financing packages. The capacity for parametric flood coverage extends up to \$70 million per contract, enabling comprehensive protection for commercial properties, infrastructure projects, and large residential developments.

The comprehensive integration of flood risk into banking frameworks highlights that physical climate risks are significant financial threats requiring the same careful analysis as credit, market, and operational risks. Estimated combined credit and market risk losses from short-term physical risk scenarios across major European banks are around €70 billion, showing the potential scale of flood risk effects on banking stability. As urbanisation advances and climate change affects precipitation patterns, the property sector's ability to accurately price and effectively manage flood risks will decide whether banks keep their portfolios resilient or face increasing losses from poorly assessed exposures.

The transition toward climate-resilient banking practices demands substantial investment in data infrastructure, modelling capabilities, and risk management frameworks, but also positions forward-thinking institutions to gain competitive advantages through superior risk selection, innovative product offerings, and stronger relationships with borrowers seeking comprehensive solutions to flood vulnerability. This represents not merely a defensive response to emerging risks, but a strategic opportunity to reshape banking-property market dynamics around sustainability and resilience principles.

History and evolution of the MKM solution framework

The approach proposed by MKM Research Labs tackles the double jeopardy of increased insurance payouts caused by hazard risks and their effect on assets, particularly properties and mortgages. The insurance industry's response is to reduce capacity and concentrate only on low-risk areas. The challenge for banks and mortgage providers is made worse by vendor solutions that are not truly effective for them.

Therefore, the answer is to expand capacity for underwriting such risk by utilising capital markets under the guidance of ISDA and FINOS, which offer a shared legal and data framework from which the industry can develop models that accurately price (as shown in this book) flood risk and transfer risk through new derivative instruments like physical risk swaps.

The efficiency of risk transfer will enable a higher degree of assurance around underwriting hazard (flood) risk and its impact on the current portfolio of assets. With capital markets technology that understands how to manage derivative-type exposure, the increase in capacity will lead to lower insurance premiums.

A critical societal side effect of improved hazard risk discipline, thanks to ISDA and the standardisation of data through FINOS CDM, is the drive for a more effective link between resilience investment and lower mortgage and insurance costs.

“Wouldn’t it be nice if the mortgage provider, insurance underwriter, and local government spoke the same language based on common data? Crazy, I know, but we have to start somewhere, and all great journeys start with a single step.” - David Kelly, MKM.

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Chapter 1: Physical Risk in Finance

We are about to start Weather Patterns to Physical Risk Swaps with some research on the current state of physical risks, including climate risks as of 2025. Since this is more research-focused, you can find a broad cross-section of references from the major players in this space at the end of the chapter.

The banking sector's exposure to physical climate risks has intensified dramatically, with 2024 marking a watershed year for both the scale of natural catastrophe losses and the sophistication of regulatory responses. Global economic losses from natural disasters reached USD 320 billion in 2024, a 19% increase from 2023's USD 268 billion, while insured losses climbed to USD 140 billion, representing only 44% of total economic damage.

This persistent protection gap of approximately USD 180 billion underscores the growing disconnect between escalating physical risks and the financial system's capacity to absorb them. Global natural catastrophe losses are showing the persistent insurance protection gap, with 2024 reaching record highs in both economic and insured losses.

Recent surveys reveals an alarming acceleration across multiple physical risk dimensions. Medium-sized natural catastrophe events have grown by an average of 7.5% annually since 1994, while severe convective storms show the most pronounced escalation with insured losses rising 9.7% annually in inflation-adjusted terms. The Network for Greening the Financial System (NGFS) has substantially revised upward its chronic physical risk projections, with the latest Phase V scenarios forecasting 15% GDP losses globally by 2050 under current policies—triple the previous estimate of 5%.

Perhaps most striking is the historical context: hydrometeorological disasters have increased nearly fivefold between 1971-1980 and 2001-2010, rising from approximately 750 incidents to around 3,500 incidents per decade, with cumulative economic losses escalating more than fivefold from USD 156 billion to USD 864 billion per decade. This exponential growth pattern fundamentally challenges traditional risk modelling assumptions based on historical stationarity.

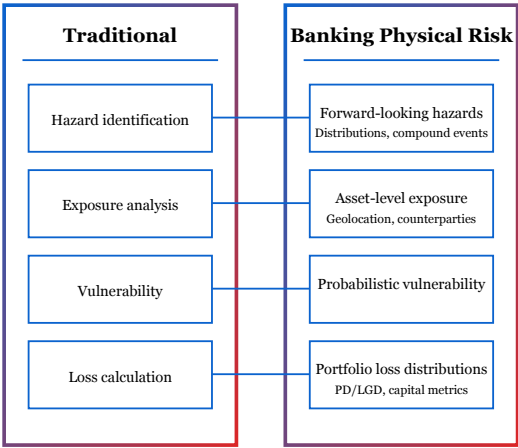
Banking Sector Exposure and Vulnerabilities

European banks face particularly acute exposure, with over 70% of corporate credit exposures directed towards high-emitting firms concentrated in manufacturing, real estate, and retail sectors. Recent analysis suggests that 80% of European bank loan portfolios are exposed to present and projected physical risk drivers, representing approximately USD 2.9 trillion in exposures. The ECB's latest assessments reveal that while transition risks alone are manageable, the combination with physical risks and macroeconomic stress could generate losses exceeding EUR 630 billion for the banking sector—nearly 90% higher than baseline conditions.

The mortgage sector represents a critical vulnerability vector. New research demonstrates that banks are beginning to incorporate climate risk premiums into residential mortgage rates, with

institutions identified as "adequately" considering climate risk by ECB Banking Supervision charging higher risk premiums that have increased particularly following supervisory expectations. However, significant heterogeneity exists in bank practices, with some institutions showing no evidence of climate risk pricing despite material exposures.

Figure 3 – From Cat Modelling to Banking Physical Risk



EU climate stress test results show varying vulnerability across financial sectors, with insurance and investment funds facing higher potential losses than banks in severe scenarios.

The regulatory landscape has evolved substantially, with the ECB's inaugural economy-wide climate stress test revealing sector-specific vulnerabilities. Under the most severe scenario combining "Run-on-Brown" investor behaviour with macroeconomic stress, the banking sector faces losses of 10.9%, while insurance companies could experience losses of 18.8% and investment funds up to 25% when including cross-sectoral amplification effects.

The Bank of England's 2025 climate disclosure indicates that sovereign bond holdings could decrease by over 9% in the most adverse climate scenario, comparable to previous years but using enhanced NGFS Phase V scenarios. Meanwhile, over 90% of European banks now consider themselves materially exposed to climate-related risks, a significant increase from 50% in 2021.

Asset-level physical risk assessment remains constrained by fundamental data limitations. Banks require precise geolocation data, with floor-level information becoming critical for accurate flood risk assessment. The European Banking Authority's Fit-for-55 data collection from 110 EU banks has highlighted the need for counterparty-level data to assess concentration risks and capture amplification mechanisms.

Advanced climate risk modelling now incorporates compound extreme climate events—simultaneous occurrences of multiple bundled hazards such as floods and storms, or heatwaves, droughts, and wildfires. New methodological frameworks assess over 330,000 physical assets

linked to companies and securities, providing unprecedented granularity for portfolio risk assessment.

Emerging Risk Transfer Mechanisms

The catastrophe bond market has demonstrated remarkable resilience, with USD 9.4 billion in new issuance in the first half of 2018 despite record losses from 2017's hurricane season. Weather derivatives trading has expanded significantly, with CME Group's temperature contracts showing monthly averages increasing from 11,500 contracts in 2021-2022 to 42,052 in 2023.

Credit Default Swap markets are increasingly reflecting carbon risk premiums. Explicit carbon pricing regimes result in firms incurring three times the additional credit protection costs compared to other regulatory approaches. This market-based pricing mechanism provides early indicators of transition risk materialisation across different regulatory jurisdictions.

Insurance Protection Gaps

The insurance protection gap continues expanding across regions, with only 5-7% of overall natural hazard losses insured in emerging and developing Asia-Pacific countries between 2000-2023. In 2024, the global protection gap reached approximately 57% of total economic losses, despite slight improvements in some regions.

This protection gap creates systemic vulnerabilities for the banking sector. Uninsured losses can trigger widespread defaults on outstanding loans and mortgages, particularly in high-risk coastal areas where insurance availability is declining. The IAIS estimates that 92% of properties remain uninsured against natural catastrophes in some jurisdictions, leaving financial institutions exposed to concentrated default risks.

Mortgage Default and Migration Patterns

Emerging research reveals that high-temperature events trigger elevated mortgage default and prepayment rates, particularly in high-amenity locations vulnerable to sea-level rise. Households experiencing temperature shocks demonstrate increased relocation likelihood, suggesting that climate events catalyze adaptive migration decisions that can impact regional mortgage portfolios.

Climate stress testing frameworks for mortgage portfolios now incorporate return periods up to 1-in-1,000 years for tropical cyclones, with damage estimates ranging 10-15 percentage points of collateral value under severe scenarios. This represents a 7-10 percentage point reduction in profitability for newly originated mortgages over 10-year periods.

Asset-level risk assessment capabilities have advanced significantly, with new platforms providing forward-looking climate and geospatial indicators for over 330,000 physical assets. Ordnance Survey location data enables property-level risk identification, recognizing that single postcode areas can cover over 6,000 square miles, making granular assessment essential.

Integrated Assessment Models now incorporate four acute physical hazards (floods, cyclones, heatwaves, droughts) alongside chronic climate impacts through enhanced damage functions. This multi-hazard approach better captures the compounding effects of simultaneous climate events on financial portfolios.

Policy and Market Implications

The research reveals fundamental inadequacies in current approaches. Traditional insurance models designed for annual policy terms prove ill-suited for decades-long mortgage exposures, creating temporal mismatches in risk assessment. The inability to adequately capture cascading effects through banking networks when insurers retreat from high-risk areas represents a critical modelling gap.

Market developments suggest increasing sophistication in risk transfer mechanisms. Physical Risk Swaps (PRS) and specialised mortgage-backed securities are emerging as analogues to traditional Credit Default Swaps and Collateralised Debt Obligations. The challenge lies in balancing collaborative frameworks through initiatives like FINOS against proprietary intelligence that provides competitive advantages.

The transformation of physical risk from an "unquantified threat to a manageable, transferable element of banking risk management" requires fundamental advances across multiple dimension. David Kelly, MKM

Banks must develop science-based approaches for quantifying physical risk with sufficient granularity to support client-specific assessments rather than relying on sector-level statistical approaches.

The evolution toward short-term climate scenarios through 2030 reflects the need for decision-useful outputs aligned with business and electoral cycles. These scenarios must incorporate cross-regional transmission of shocks through trade and financial linkages, recognising the interconnected nature of global financial systems.

The banking sector's response to these challenges necessitates unprecedented integration of meteorological science, hydrological modelling, and banking mathematics. Success requires not only analytical capabilities but also robust model risk governance structures that can translate complex climate projections into actionable risk management decisions across portfolios spanning millions of assets.

As the industry moves forward, the critical question shifts from whether physical risks will materialize to how quickly and effectively the financial system can adapt its risk management frameworks to address them. The statistics presented here suggest that this adaptation must accelerate substantially to match the pace of physical risk evolution.

The banking sector's challenge is to enhance risk measurement and develop new frameworks for understanding how physical risks propagate through banking networks. We require innovative ways to model the relationship between weather events and asset values, new data standards to

capture physical risk exposure, and novel banking instruments to effectively transfer and manage these risks. This transformation necessitates bridging multiple disciplines, from weather prediction through hydrology and loss modelling to banking mathematics and machine learning.

This chapter investigates the key dimensions of physical risk in finance and examines why traditional approaches are inadequate. We analyse how emerging technologies and methodologies facilitate a more understanding of physical risk and why this evolution is crucial for ensuring banking stability in an era of evolving physical risks. Most importantly, we delineate a pathway for transforming physical risk from an unquantified threat into a manageable, transferable element of banking risk management.

“The banking world's approach to physical risk is undergoing a fundamental transformation - from a future consideration to an immediate driver of value, from an unquantified threat to a tradable asset, from a peripheral concern to a core determinant of banking stability.” - David Kelly, MKM.

Overview of Physical Risk Challenge

Physical risk in the banking sector has evolved from a peripheral concern to a central challenge in modern risk management. The landscape of physical risk has transformed dramatically, characterised by the increasing frequency and severity of chronic physical events on regional and global scales. What was once considered idiosyncratic risks are now manifesting as systemic challenges, fundamentally altering how we approach risk assessment and management.

The Swiss Re Institute sigma No. 1/2024 report highlights that the number of medium-sized natural catastrophe events has grown by an average of 7.5% annually since 1994, with severe convective storms (SCS) showing the most significant increase in insured losses, rising by 9.7% annually in inflation-adjusted terms.

The Swiss Re Institute sigma No. 1/2023 report also notes a long-term upward trend in insured losses from natural catastrophes, driven by economic growth, urbanisation, and increased storm activity. Insured losses have grown 5–7% annually since 1992, with severe weather events being a key driver.

The complexity of physical risk assessment demands increasingly granular analysis at the asset level. Banking institutions must now evaluate thousands or millions of assets in their portfolios, considering precise physical locations and projecting expected losses across multiple physical hazards. This granular approach represents a significant departure from traditional portfolio-level risk assessment methods.

The data requirements for such detailed analysis present substantial challenges. Banking institutions require robust data on asset locations, hazard projections, and potential damage functions. Significant uncertainties and knowledge gaps in weather projections and impact modelling complicate these data needs. Integrating diverse data sources, from high-resolution terrain models to real-time weather measurements, demands new data management and standardisation approaches.

The banking sector is exploring innovative risk transfer mechanisms in response to these challenges. New securities, derivatives, and risk transfer products are being developed, including specialised mortgage-backed securities (MBS) and physical risk swaps (PRS). These instruments aim to provide more effective means of managing and transferring physical risks analogous to specialised Collateralised Debt Obligations (CDO) and Credit Default Swaps (CDS).

The industry's response to these challenges requires a fundamental evolution in risk management approaches. Traditional frameworks, built on historical data and statistical relationships, struggle to capture the changing nature of physical risks. The non-linear characteristics of weather patterns, feedback loops, and fractal-esq interactions necessitate new modelling paradigms that can handle these complexities while remaining viable for practical implementation.

Modern risk management requires banks to connect the science of physical risk with banking risk assessment to be practical in daily situations. This involves integrating advanced statistical methods, including domain-specific AI and Bayesian approaches, to better capture the complexity of weather patterns and their implications for banking.

Developing standardised data frameworks and robust governance structures is essential to supporting consistent risk assessment across institutions.

Implementing these new approaches presents significant operational challenges. Banks are required to develop and validate complex models while managing vast amounts of data. The computational demands of physical risk assessment require scalable technological solutions that can handle increasingly analysis requirements.

As we move forward, the industry needs a comprehensive approach to addressing these interrelated challenges. Given that physical risk presents a present issue for banks, a key aspect of any solution is to decide on the mix between the level of collaboration through groups such as FINOS, where the output is open source, and what is considered proprietary. The need for two counterparties to operate under the same legal, data and model framework encourages market liquidity, whereas proprietary intelligence provides a profitability edge.

This book offers an approach that progresses from fundamental weather prediction to banking impact assessment while prioritising practical implementation considerations. We start by exploring the elements of physical risk assessment and then create an integrated framework that merges these elements into a cohesive whole suitable for application in modern banking institutions.

Current Approaches and Limitations

The banking sector's traditional approach to physical risk assessment has evolved from catastrophe modelling developed for the insurance industry. This heritage reveals both the strengths and limitations of current methodologies. Insurance companies have invested decades in developing models to price annual policies' flood, fire, and storm risks. However, they did not

design these models to address the broader needs of banking institutions managing long-term exposures through mortgages, infrastructure investments, and structured products.

Current approaches typically follow a four-step process:

- Hazard identification.
- Exposure analysis.
- Vulnerability assessment.
- Loss calculation.

Hazard identification relies heavily on historical data. Exposure analysis focuses on asset location and characteristics, while vulnerability assessment examines the relationship between hazard intensity and potential damage. Finally, loss calculation combines these elements to estimate possible banking impacts. However, this seemingly straightforward process masks significant methodological weaknesses that limit its effectiveness for modern risk management.

A fundamental limitation lies in how these models handle risk quantification. Many existing models do not explicitly separate frequency and severity modelling, departing from established best practices in property insurance. Furthermore, they often produce deterministic expected loss values rather than complete probability distributions, severely limiting their utility for risk management applications.

Specifically, the vulnerability models that map hazard characteristics to potential damage typically employ deterministic rather than probabilistic approaches, failing to capture the full spectrum of uncertainty inherent in physical risk assessment.

This approach, while logical, faces significant limitations when applied to modern banking risk management. Relying on historical data becomes increasingly problematic as weather patterns' frequency and severity of extreme events follow random paths. The assumption of stationarity—that past patterns reliably indicate future risks—no longer holds.

The current models struggle with the temporal mismatch between insurance and banking time horizons. Insurance models are calibrated for annual policy terms, making them ill-suited for assessing the decades-long exposure of a mortgage portfolio. The models cannot adequately capture how physical risks might evolve over a 30-year mortgage term, nor how these risks interact with property values, insurance availability, and mortgage affordability.

“Existing approaches fail to capture how physical events trigger cascading effects through banking networks. When insurers retreat from high-risk areas, they leave properties uninsured and impact property values, mortgage default probabilities, and, ultimately, the stability of mortgage-backed securities markets. This limitation is exacerbated by the lack of clear attribution between risk factors and their economic consequences, making it difficult to conduct meaningful stress testing or design optimal insurance solutions.” - David Kelly, MKM.

Even worse, Climate models have a time horizon out to 2100. So how can asset managers and banks make sensible decisions on lending criteria based on such a maturity mismatch?

Many current models in physical risk are academic in their nature and outlook and thus function as "black boxes" without clear documentation of their methodologies and assumptions. This lack of transparency makes thorough model validation nearly impossible. Such behaviour contradicts the banking industry, where the model engines are similar, and the intellectual property is embedded in their implementation. In all areas of physical risk, the underlying models originate from academia and are publicly accessible.

For instance, fluid dynamics is covered in the first semester of a physics degree. The lack of transparency regarding applications or benchmarks for flood risk modelling further complicates the situation, hindering coherent comparison and systematic improvement of various modelling approaches.

Different vendors use varying assumptions, are very guarded about their methodologies, and use public and private data sources differently, making it difficult for banks to compare and aggregate risk assessments. This fragmentation hampers the development of market-based solutions for transferring and managing physical risk.

The limitations become particularly chronic when examining flood risk, which exemplifies the complexity of modern physical risk assessment. Flood models are required to integrate high-resolution terrain data, precipitation forecasts, infrastructure conditions, and property characteristics. To be remotely accurate, they must account for gradual changes in flood patterns and sudden shifts in insurance market behaviour. The recent exodus of insurers from Florida and the Carolinas demonstrates how quickly risk assessments can become outdated when market conditions change.

Basel III/IV regulations now explicitly require banks to account for physical risk in their capital calculations, but the tools for doing so remain primitive. Current approaches often resort to simple overlays or adjustment factors rather than an integrated assessment of how physical risks affect the probability of default, loss-given default, and other key risk metrics.

A particularly problematic aspect of current approaches is their origin outside the insurance domain. Many existing models were initially developed for land-use planning or other purposes, making adapting to insurance pricing and risk management applications problematic. This misalignment between the original design and the current application creates fundamental model reliability and appropriateness challenges.

The following chapters present such a framework, building from weather prediction to banking impact assessment while addressing the limitations of current approaches. The goal is not to replace existing catastrophe models but to extend and adapt their capabilities to the broader needs of modern banking risk management.

The curious case of Flood Hazards and EBITDA

Probably unfairly, I am going to use the desire to measure the impact of EBITDA based on potential hazard events. EBITDA (Earnings Before Interest Taxes Depreciation and Amortisation) is particularly useful metric as it feeds into corporate credit ratings/spreads and also enterprise value, extensively used by Private Debt and Private Equity respectively.

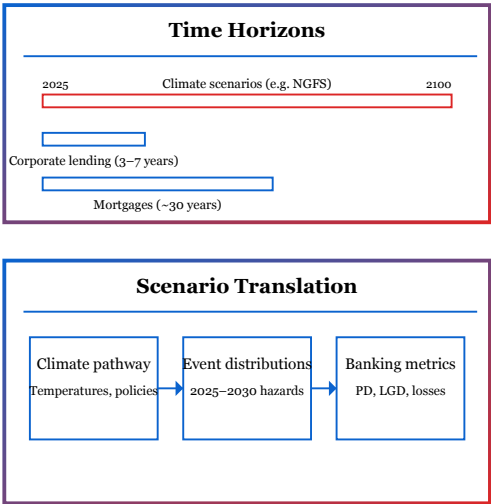
The challenge of translating physical hazards outlined by NGFS hazard pathways into quantifiable EBITDA impacts represents one of the most vexing problems in modern climate risk management. While the theoretical pathway from weather event to financial consequence appears straightforward, the practical implementation reveals layers of complexity that challenge traditional risk assessment frameworks.

The Maturity Mismatch Problem

A fundamental disconnect exists between climate model horizons and banking decision timeframes. Climate scientists project risks through 2100, yet banks make lending decisions on 3-7 year horizons and mortgage portfolios mature over 30 years. This temporal misalignment renders long-term climate projections practically useless for immediate banking decisions.

The solution lies in reframing the question entirely. Rather than asking "what will flood risk look like in 2100," we should ask "what distribution of precipitation events might we experience through 2030, and which tail events could trigger floods that materially impact our portfolio?"

Figure 4 – Maturity Mismatch and Scenario Translation



This shift from long-term projection to near-term probability distribution fundamentally changes the analytical approach. We move from deterministic scenarios to probabilistic assessments, from absolute predictions to relative likelihoods, from precision to pattern recognition.

From Scenarios to Distributions

The NGFS pathways provide structured scenarios for climate evolution, but translating these into actionable banking intelligence requires an intermediate step: converting pathway assumptions into distributions of physical events.

Consider a coastal manufacturing facility. Rather than stating "under the NGFS Current Policies scenario, sea level will rise X centimetres by 2050," we construct a probability distribution of precipitation events through 2030. This distribution captures not just the central tendency but critically, the tails—those extreme events that trigger the physical damages we care about.

The distribution might show that historical 1-in-100 year precipitation events could occur with 1-in-50 year frequency by 2030 under current policies. More importantly, it reveals how the tail thickens—extreme events become more probable, and the truly catastrophic events become less unthinkable.

This probabilistic approach immediately clarifies the nature of the risk. We're not predicting specific flood events; we're characterising how the likelihood of different rainfall intensities evolves. When rainfall exceeds local drainage capacity or overwhelms flood defences, floods occur. The distribution indicates how often we might encounter those threshold-crossing events.

The Six-Step Framework Reconsidered

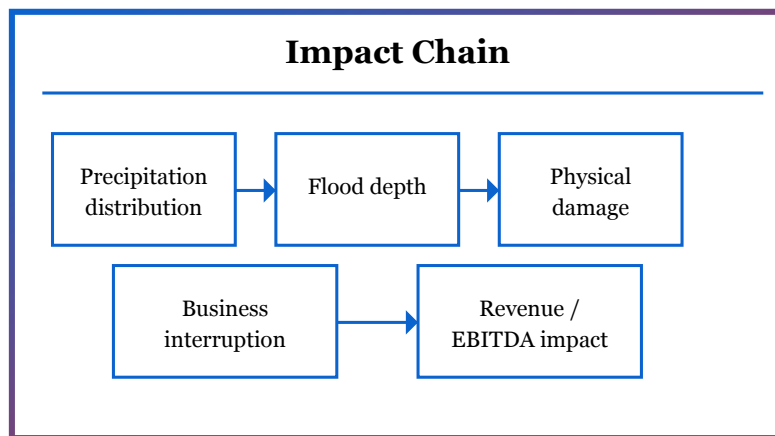
The traditional approach to calculating EBITDA impact follows a deceptively simple sequence:

- Step 1: Flood depth at facility = X meters
- Step 2: Damage cost = Property value × Damage curve(X)
- Step 3: Business interruption = Assume facility down Y days
- Step 4: Revenue loss = Daily revenue × Y × "availability factor"
- Step 5: Apply industry-average multipliers
- Step 6: Output: "EBITDA impact = \$Z million"

This framework appears rigorous because it is straightforward but hides uncertainties that increase at each stage. The uncertainty cascade starts with converting precipitation distributions into specific flood depths—a process requiring high-resolution hydrological modelling that considers local infrastructure, drainage capacity, and compound events where multiple hazards intersect.

Even if we determine flood depth with confidence, the damage response varies enormously across seemingly similar assets. Construction quality, building age, flood preparation measures, and even the specific contents of a facility create wide variation in damage for identical flood depths. Damage curves are typically deterministic, providing single point estimates when reality demands probability distributions.

Figure 5 – From Hazard to EBITDA Impact



Business interruption duration—Step 3's "Y days down"—represents perhaps the greatest source of uncertainty. Company responses to identical physical events vary by orders of magnitude. Organizations with robust business continuity plans, diversified operations, and strong balance sheets may resume operations within days. Others face months of disruption or never fully recover.

The factors driving this heterogeneity include supply chain resilience, availability of contractors and materials, workforce accessibility, and management capabilities. Geographic and temporal factors matter immensely—the same flood event has vastly different recovery profiles depending on whether it affects one facility in isolation or dozens simultaneously across a region.

The Nature of Uncertainty

The compounding uncertainty through this framework reveals a fundamental truth: we cannot predict precise EBITDA impacts from physical hazards with any meaningful confidence at the individual facility level. The error bounds on the final estimate inevitably dwarf the estimate itself.

But this realisation, rather than being cause for despair, liberates us to focus on what we can actually know: cause and effect relationships, relative magnitudes, and threshold behaviours.

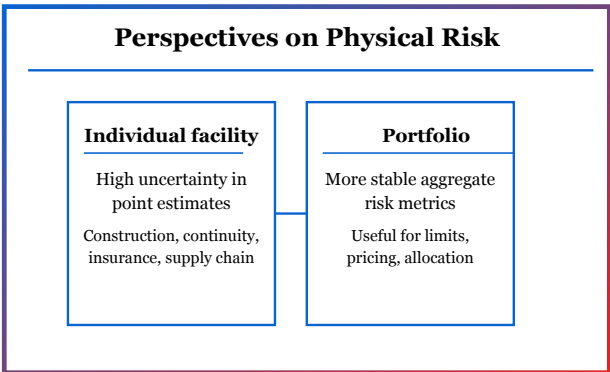
We cannot predict that a specific flood will cause \$5.47 million in EBITDA impact. But we can establish that facilities in coastal flood zones face materially higher revenue disruption risk than those on elevated terrain. We can identify that companies without business continuity plans suffer longer interruptions than those with robust preparations. We can recognise that industries with just-in-time inventory systems experience more severe cascading effects than those with buffer stocks.

The shift from absolute prediction to relative assessment fundamentally changes how we use these models. Rather than producing precise financial forecasts, we're identifying risk concentrations, threshold exposures, and relative vulnerabilities.

Portfolio-Level Insights from Facility-Level Uncertainty

Here lies the paradox: the framework that fails to provide reliable facility-level predictions can generate valuable portfolio-level insights. This seeming contradiction resolves when we understand that portfolio analysis asks fundamentally different questions than facility-level assessment.

Figure 6 – Facility vs Portfolio Perspective



“At a portfolio level, we're not trying to predict specific losses. We're asking questions like: "Do we have excessive concentration in flood-prone assets?" or "Which geographic regions present the highest relative risk?" or "What exposure threshold should trigger enhanced due diligence?"”- David Kelly, MKM

These questions don't require precise EBITDA calculations for each facility. They require relative risk rankings, identification of concentrations, and recognition of threshold conditions where multiple facilities could face correlated disruptions.

Statistical aggregation across multiple assets reduces idiosyncratic uncertainties. If we have one thousand facilities, the average response becomes more predictable even as individual facility

impacts remain highly uncertain. Diversification provides its usual benefits—not by eliminating risk but by making aggregate outcomes more stable than individual outcomes.

This portfolio perspective aligns naturally with strategic asset allocation decisions. Should we reduce exposure to manufacturing in flood-prone regions? Should we avoid purchasing mortgage portfolios in coastal South Florida? Should we establish exposure limits for facilities in areas facing insurance market retreat?

These decisions benefit from directional insights about aggregate exposure, not from facility-specific EBITDA forecasts with false precision. The goal is pattern recognition at scale, identifying systematic vulnerabilities rather than predicting individual outcomes.

The Direct and Indirect EBITDA Pathway

Physical hazards affect EBITDA through multiple channels that traditional models often conflate or ignore. Understanding these distinct pathways clarifies both the strengths and limitations of EBITDA as a risk metric.

Direct operating impacts flow immediately to the income statement. Revenue disruption from facility downtime reduces the earnings component directly. Increased operating costs—temporary facilities, overtime labor, expedited shipping—reduce operating margins. Supply chain disruptions create cost increases that flow through to EBITDA even when the physical event doesn't directly damage company assets.

These direct effects appear relatively straightforward to model, though the business interruption duration remains highly uncertain. More problematic are the indirect valuation impacts that affect enterprise value through channels EBITDA doesn't capture.

Assets now recognized as prone to physical hazards face multiple valuation pressures. Direct valuation adjustments occur when increased climate risk exposure forces asset writedowns. Enterprise value contracts through reduced EBITDA multiples as investors demand higher returns for perceived risk. Insurance premiums increase while coverage becomes less comprehensive, creating ongoing operating expense increases that reduce future EBITDA.

Perhaps most significantly, capital expenditures for reconstruction—often the largest cost following a physical event—don't directly appear in EBITDA calculations. When a flood destroys a facility, the \$50 million reconstruction cost represents a balance sheet event, not an income statement charge. EBITDA only captures the business interruption revenue loss and any increased depreciation on replacement assets.

This limitation means EBITDA substantially understates the financial impact of physical hazards. The metric works reasonably well for quantifying operational disruption but fails to capture the full economic cost of physical risk.

Company Response Heterogeneity

Perhaps the greatest source of uncertainty in translating physical hazards to EBITDA impact lies in company-specific response capabilities. Two seemingly identical facilities facing identical flood events can experience order-of-magnitude differences in financial impact based solely on organisational capabilities.

Prepared companies demonstrate several characteristics: robust business continuity plans with documented procedures and regular testing, diversified operations that allow production shifting between facilities, strong balance sheets providing financial resilience to absorb reconstruction costs, and established relationships with contractors and suppliers enabling rapid response.

These organisations may resume partial operations within days and full operations within weeks. Revenue disruption remains limited, customer relationships survive intact, and the EBITDA impact, while significant, stays manageable relative to overall financial capacity.

Vulnerable companies present the opposite profile: concentrated operations with single points of failure, weak business continuity planning, constrained financial positions leaving little margin for unexpected costs, and limited supplier relationships forcing dependence on spot markets during reconstruction.

For these organisations, identical physical events create existential threats. Months-long disruptions erode customer bases, financial constraints delay reconstruction, and the competitive position may never fully recover. The EBITDA impact can represent a substantial fraction of annual earnings or even trigger insolvency.

Industry characteristics create additional variation. Utilities and real estate companies have very different response profiles than manufacturing or retail operations. Capital-intensive industries with long-lived assets face different challenges than service businesses with minimal physical infrastructure.

This heterogeneity means that generic multipliers or industry-average assumptions introduce enormous error. The "availability factor" in Step 4 of the traditional framework tries to capture these differences with a single parameter, but in reality, assessing organisational capabilities and resilience requires a company-specific approach. Converting physical risk from an unquantified threat into a managed element of banking risk management means accepting that perfect prediction is impossible, but useful insights can be gained when the methodology aligns with the application.

We cannot predict exact EBITDA impacts from physical hazards, but we can establish cause-and-effect relationships, identify concentration risks, and recognise threshold exposures. The aim is not to eliminate uncertainty — that's impossible given the complex interactions between physical systems, organisational responses, and financial outcomes. Instead, it involves using probability distributions to explicitly characterise uncertainty and directing analytical efforts where they can produce actionable insights.

A bank that openly acknowledges uncertainty and identifies systematic vulnerabilities makes better decisions than one that claims unachievable precision. The subsequent chapters develop the technical frameworks necessary to progress from physical hazard projections and hydrological modelling to portfolio-level risk assessment, always maintaining an honest understanding of uncertainty and the limits of predictability.

Vendor Disclosure Trap and Gaps in Model Governance (TBR)

Recent benchmarking reveals considerable divergence among physical risk vendors. This article suggests the leading cause is insufficient model governance, which limits climate risk to disclosures rather than affecting capital adequacy.

In October 2025, “A Risk Professional’s Guide to Physical Risk Assessments”, a benchmarking study of 13 climate physical risk vendors conducted by the GARP Risk Institute on behalf of the Climate Financial Risk Forum. First of its kind, the review highlighted significant dispersion in hazard and damage estimates across vendors, with some properties assessed as highly exposed by one vendor but not by another.

For example, flood risk vendors can trace their model capabilities back to a few academic models, such as the Saint-Venant Equation for hydraulics, and to public datasets created by government organisations such as the National Oceanic and Atmospheric Administration (NOAA). The above dispersion is therefore determined by how these models are employed in a production-like setting. Flood vendors have so far been successful in providing insights into such risks for insurance companies without formal model governance standards, which are increasingly demanded as they expand into banking, where the current requirement is limited to disclosure.

The Basel Pillar Progression Paradox

The Basel framework’s three-pillar architecture was designed to create a progression: Pillar 3 disclosure creates transparency, which informs Pillar 2 supervisory review, which eventually feeds into Pillar 1 minimum capital requirements. For climate physical risk, this progression has stalled.

Pillar 3 climate disclosures are now mandatory in Europe under the EBA’s Implementing Technical Standards, requiring banks to report on physical risk exposures, energy efficiency of collateral, and climate-aligned portfolios. Yet the Basel Committee’s June 2025 framework for international Pillar 3 climate requirements became entirely voluntary, primarily due to US regulatory resistance.

Notably, progress on Pillar 2 capital requirements has halted. Less than half of the institutions currently include climate scenario analyses in their Pillar 2 procedures. The anticipated shift from transparency to capital adequacy remains impeded by what regulators describe as “fundamental methodological shortcomings”. Current disclosures are criticised for being ‘very granular and complex’ and for providing limited reliable data—particularly regarding physical risk assessments.

Why has this happened? Traditional stress testing frameworks, designed for short- to medium-term horizons, are fundamentally incompatible with physical climate risks that manifest as

sudden geographic shocks or unfold over decades. This mismatch has frustrated regulators, but the fault does not lie with the underlying science—the core academic frameworks have been validated over decades and work effectively for insurance applications.

What Banking Model Governance Requires

Since SR 11-7, the key US regulatory guidance on model risk management issued in 2011, banks have operated under a clear principle—they remain responsible for model risk even when they do not own the models themselves. Outsourcing model development or operation does not transfer responsibility for model risk. The regulators' 'caveat emptor' stance means that when a vendor model fails, the bank bears the consequences.

This imbalance in ownership and responsibility for failure creates an uncomfortable reality for physical climate risks and their interaction with banks' model validation teams. Banks must validate vendor models, but vendors often cannot provide the transparency needed for validation. External models may not allow scrutiny of underlying methodologies, data sources, or computational processes.

When model-specific documentation is not forthcoming, banks must spend considerable time extracting the necessary information to make an assessment. This places vendors on the receiving end of what feels like a blizzard of questions and data requests. This cycle recurs with every new client and every model update, consuming resources that could otherwise be directed toward product development.

The core issue is that most climate vendors confuse testing with validation. Testing checks that the code executes correctly according to specifications. Validation challenges the choices made by model owners and assesses whether those choices are appropriate for the intended use.

A model can pass every test and still fail validation if the vendor cannot articulate why specific methodological choices were made, what assumptions underpin the outputs, and—critically—where the model in a production environment might fail or produce misleading results.

Bank validators ask nine fundamental questions that emerge directly from SR 11-7. Does the model do what it claims? Is it reliable, timely, and stable? Does it consume reasonable computational resources? Are there controls over manual overrides? How is output consumed for risk decisions? Is the output used as input to other models? What are the model's weaknesses and limitations of use? What monitoring is required?

Climate vendors often prioritise protecting what they perceive as proprietary model IP, hoping to withstand the banks' scrutiny without full disclosure. But this approach does not build trust. In banking, where derivative model literature from Black-Scholes onwards sits in the public domain, competitive advantage has never resided in the models themselves—it comes from how they are implemented at scale. Climate vendors who struggle with transparency, resorting instead to providing test results and user guides, fail to recognise that validators are not trying to steal their methods. They are trying to assess whether the vendor and their model offering can be trusted

The latest from the Bank of England

The Bank of England's Prudential Regulation Authority has published Supervisory Statement 4/25 (SS4/25), a carefully constructed and extensively consulted document that replaces SS3/19 and reflects genuine supervisory effort to raise the bar on how UK banks and insurers should manage climate-related financial risks.

The statement consolidates years of thematic feedback, aligns with international standards from BCBS and ISSB, and applies a thoughtful proportionality framework that acknowledges firm heterogeneity. On governance, risk appetite setting, and disclosure expectations, SS4/25 is substantive and an improvement on its predecessor.

Yet, beneath this well-intentioned supervisory architecture lies a critical blind spot: the complete absence of model governance frameworks to ensure coherence across the hydrology, damage function, and asset impairment methodologies that underpin physical climate risk quantification. This absence makes SS4/25 a process-heavy, judgment-intensive regime that institutionalises methodological incoherence instead of making the leap a rigorous risk standard consistent with existing, transparent quantitative capital computation methodologies.

Hence, in the absence of such risk standards, this creates a supervisory arbitrage opportunity for firms that may mask such risks until material events unravel to show asset impairment. In order of severity this starts with portfolio insolvency, grows to individual bank insolvency and ultimately may lead to wider national or international banking systemic solvency through abrupt capital depletion - specifically in Pillar 1.

The PRA's supervisory statement makes several genuine advances over SS3/19. It explicitly demands that firms integrate climate-related risks into risk registers, linked to existing financial and operational risk types (credit, market, liquidity, operational) with clearly articulated transmission channels. This requirement, while conceptually straightforward, is more operationally rigorous than vague exhortations to "monitor" climate risk. A mortgage lender cannot hide flood exposure in governance language; it must enter flood-driven credit losses into the risk register, assign materiality, categorise response and present this to the board.

SS4/25 acknowledges the three distinctive characteristics of climate-related risks—systemic, uncertain yet foreseeable, and path-dependent—and uses this framing to justify scenario analysis and forward-looking assessment beyond historical data. This is conceptually sound and provides intellectual cover for the non-standard, judgment-based methodologies that firms will inevitably employ.

The proportionality framework allows smaller or less-exposed firms to use narrative-based scenarios with expert judgment and simpler data proxies, while holding larger or more-exposed firms to higher standards. This is pragmatic and acknowledges genuine capacity constraints across the industry. Finally, the governance architecture—board oversight, senior management accountability, risk appetite setting informed by stress scenarios, and periodic review—is textbook good practice. If implemented rigorously, it creates accountability loops and prevents climate risk from remaining siloed in sustainability teams.

SS4/25 assumes firms can coherently translate physical climate hazards (floods, heatwaves, storms) into financial risk (credit losses, collateral impairment, operational disruption) using existing methodologies. This is a material gap in assumption given the extent of the model transmission chains. Hydrological modelling involves flood depth and frequency projections at property level, incorporating climate scenarios, adaptation measures, and uncertainty bands.

Damage functions map flood depth to structural damage percentage, content loss, and repair costs, differentiated by building type, age, construction, and ground floor exposure. Collateral valuation translates damage costs and structural impairment into collateral value haircuts, incorporating market stigma, liquidity discount, and insurance coverage gaps.

Credit risk integration embeds collateral haircuts into loss-given-default (LGD) models, probability-of-default (PD) adjustments, and ICAAP capital calculations. Natural catastrophe (NatCat) vendors such as AIR, RMS, and PERILS operate across the first two steps but rarely into the latter ones. They use proprietary, opaque hydrological and damage modules that diverge significantly—often producing 30–50% variance in peak industry loss estimates for the same peril in the same region.

Insurance firms and reinsurers know this and manage it through vendor triangulation, internal validation, and explicit margining to reflect such uncertainty. For banks, uncertainty in the first two steps becomes amplified in the latter ones. SS4/25's guidance states that "Firms should use toolkits that incorporate models appropriate for assessing physical impacts, such as natural catastrophe (NatCat) models," and should "further consider their exposures to cross-border spillovers and incorporate additional analytical tools that draw on external modelling and scientific expertise."

Such declaration sidesteps the model-related gaps of NatCat models, which deal with catastrophes rather than the tail-event floods that banks need to assess. Damage functions are averages across actual insurance claims; therefore, the impact of such damage post-insurance claim on asset impairment can be modelled in multiple ways to capture "strategic default," where the borrower hands back the keys and walks away—something that occurred in California when insurers refused to cover houses built close to forests with excessive fuel.

SS4/25 asks banks to "balance appropriate use of data from external suppliers with the appropriate development of in-house capabilities" and to maintain "an effective system of governance to oversee and integrate any data from external suppliers, including understanding any limitations." Without specifying what "effective" means, supervisors have no lever to challenge implausible assumptions or results in banks' internal assessments. A firm using an idiosyncratic set of assumptions (such as a 20-year-old damage curve, outdated hydrological data, and a 2% LGD haircut) can document these choices, gain board sign-off, and meet the PRA's expectations simply because the PRA has not provided any quantitative benchmarks or model validation standards in its guidance.

The consequence of not being specific is that supervisors have allowed firms to use different methods for measuring the same risk (heterogeneity), but without a common yardstick

(comparability), firms can pick the method that shows the least risk exposure—called supervisory arbitrage and heterogeneity without comparability that undermine fair regulation.

Consider the following reasonable outcomes of this guidance:

- Bank A may use Vendor X hydrology plus Vendor Y damage functions plus a 5% collateral haircut and reverse stress testing.
- Bank B may use local authority flood maps plus insurer loss triangulation plus a 1% haircut and narrative scenario analysis.
- Bank C may use ESG climate scores and expert judgment.

Each bank then documents its approach, gains board sign-off, and claims proportionate application as per regulatory guidance. However, because the PRA has provided no model governance standard:

None of the banks' approaches or methods are comparable - no consistency. None of the banks can be challenged as they align with regulatory guidance - no consequence. To address model governance that should be central in this space, SS4/25 tells firms to "understand limitations" and "interpret prudently", which effectively become euphemisms for "do what you think is reasonable, document it, and hope supervision is lenient."

In short, this guidance will never be a substitute for robust, consistent and proportionate quantitative risk modelling that is used for all other material risks in the global banking system.

The PRA's SS4/25 represents a genuine and coherent effort to elevate climate risk governance, compelling firms to build capabilities, integrate physical and transition risks into management frameworks, and strengthen board accountability. These steps merit praise. Yet they fall short without governing the underlying models—hydrology chains, damage functions, collateral valuation, and credit linkages—which remain heterogeneous, fragmented, and often locked in vendor black boxes or highly subjective and unreliable ESG proxies.

A more robust statement would mandate two additions:

- **Model Governance Requirements:** setting standards for vendor vetting, damage function validation, and collateral impairment protocols.
- **Supervisory Benchmarking:** with PRA aggregation of peer loss distributions and materiality thresholds to facilitate cross-firm comparability and outlier challenge.
- **The core flaws persist:** zero model risk oversight across disjointed model strings, plus PRA's framing of flood risk as transitional (leading to credit impacts) rather than a direct risk factor—curiously recognized in Solvency II for insurance but absent from Basel III for banking

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Chapter 2 - Weather Data

Weather data represents the systematic collection and measurement of atmospheric conditions at specific locations and times. At its core, it is information that tracks and predicts weather conditions and patterns by measuring parameters such as temperature, air quality, wind speed, precipitation level, and atmospheric pressure. More formally, weather data encompasses any information or statistics about the state of the atmosphere, including temperature, wind direction, rain or snow, moisture, and pressure.

It's important to distinguish weather data from climate data. Weather represents the state of the atmosphere at a given moment in time at a particular location—for example, the air temperature on a specific day. Climate, by contrast, reflects the expected state of weather variables defined as averages over extended time periods, typically 30 years. Weather data form the building blocks that eventually become incorporated into climate values through long-term averaging.

Figure 7 – Weather and climate

Weather and climate
Weather State of the atmosphere at a specific time and place, for example the air temperature at 3 pm today.
Climate Statistics of weather variables over long periods, typically 30-year averages for a given location.

Weather data can be categorised along several dimensions. Local weather data provides information about conditions in specific localised areas such as counties or cities, whilst global weather data covers weather and climate patterns for the entire planet. Additionally, weather data can be divided temporally: real-time weather data delivers the most current information on conditions as they occur, allowing hour-by-hour tracking, whereas historical weather data provides intelligence on patterns and conditions from previous days, months, years, or even decades.

This data is collected through an extensive international network using standardised instruments and methods. Meteorologists employ cutting-edge equipment including satellite signals, airport observation stations, drone technology, and sophisticated mapping devices to gather remarkably accurate and current information. The measurement toolkit includes thermometers for temperature, barometers for air pressure, radar systems for tracking precipitation, wind vanes for wind direction, anemometers for wind speed, transmissometers for atmospheric visibility, and hygrometers for humidity. Weather satellites track everything from snow cover to ocean tidal

patterns, whilst radiosondes—balloons equipped with instruments—measure atmospheric characteristics as they ascend through the atmosphere, communicating data via radio.

A critical aspect of weather data is its standardisation across international boundaries. The World Meteorological Organisation (WMO), encompassing more than 175 nations, ensures that observation procedures remain consistent globally and facilitates the international exchange of weather data. This standardisation is essential because observations must be comparable across different locations and institutions. Weather information is transmitted electronically using internationally agreed codes—specifically, a series of five-figure numbers in a standard sequence—allowing data to be shared rapidly via cable or radio amongst weather bureaux worldwide.

Figure 7 – Weather and climate

Weather and climate
Weather State of the atmosphere at a specific time and place, for example the air temperature at 3 pm today.
Climate Statistics of weather variables over long periods, typically 30-year averages for a given location.

Modern weather data extends beyond direct observations to include synthesised data produced by numerical weather prediction (NWP) models. These physics-based models can estimate atmospheric conditions at locations and times where direct observations are unavailable. NWP models produce data that adheres to physical laws governing atmospheric motions and processes, generating information on convenient regular geographic grids with even temporal spacing. This capability enables the production of datasets representing the distribution of past weather conditions and capturing the chronological evolution of atmospheric scenarios in realistic ways. Numerical weather prediction uses mathematical models of the atmosphere and oceans to predict weather based on current conditions, creating comprehensive datasets that serve as the foundation for sophisticated weather analysis and risk assessment.

Operational NWP Systems: The Data Source Landscape

Numerical Weather Prediction (NWP) systems represent the primary source of weather data for modern risk assessment applications. Unlike simple observational data that records current conditions, NWP systems generate comprehensive datasets by solving the fundamental equations of atmospheric physics across three-dimensional grids. These models produce data that adheres to physical laws governing atmospheric motions and processes, creating information on regular geographic grids with consistent temporal spacing—precisely the format required for sophisticated risk analysis.

However, no single NWP system can simultaneously optimise for all desirable characteristics. The atmospheric phenomena relevant to flood risk span an enormous range of spatial scales, from continental weather patterns extending thousands of kilometres to individual thunderstorms measuring just a few kilometres across. Temporal scales prove equally diverse, ranging from the sudden intensification of a convective cell over minutes to the multi-day evolution of a major storm system. The computational resources required to simulate atmospheric processes scale dramatically with spatial and temporal resolution, creating fundamental tradeoffs that shape the operational NWP landscape.

Figure 8 – Weather data categories

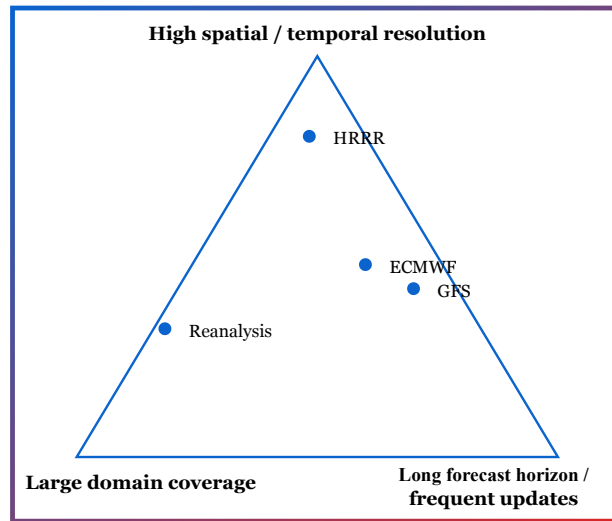
	Real-time	Historical
Local	Local real-time observations and nowcasts (e.g.airports, local radar products).	Local historical records and climatology (for example decades of daily station data).
Global	Global operational model output and satellite products updated frequently.	Global reanalysis and long-term gridded datasets reconstructing past atmospheric conditions.

The Fundamental Tradeoffs

Three primary constraints govern NWP system design, creating an unavoidable tension between competing objectives:

- **Resolution vs. Computational Cost:** Halving the grid spacing in all three dimensions requires more than a 100-fold increase in computing power due to both the increased number of grid cells and the need for smaller time steps to maintain numerical stability. A model with 3-kilometre horizontal spacing demands vastly more computational resources than one running at 25-kilometre spacing, limiting how frequently forecasts can be updated and how far into the future they can extend.
- **Domain Coverage vs. Spatial Detail:** Computational budgets force a choice between global coverage at coarser resolution or regional detail at higher resolution. A model covering the entire planet at 10-kilometre resolution requires similar computational resources to a model covering North America at 3-kilometre resolution. This constraint drives the development of regional models nested within global systems, where the global model provides boundary conditions for more detailed regional predictions.

Figure 9 – Fundamental NWP trade-offs



Each system occupies a different region of the trade-off space defined by resolution, domain coverage and forecast horizon.

- **Forecast Horizon vs. Update Frequency:** Longer forecast periods require more computation, limiting how frequently models can be updated with new observations. A system producing 15-day global forecasts might update four times daily, whilst a regional system focusing on 18-hour forecasts can update every hour. The value of frequent updates for short-range applications—incorporating the latest radar observations to predict severe weather—differs fundamentally from the value of extending forecasts further into the future for planning purposes.

These constraints create distinct niches in the operational forecasting landscape, with different systems optimised for different applications. Rather than competing, modern operational systems complement each other, with each system excelling in its designed domain whilst relying on others to fill gaps in its capabilities.

Categories of Operational Systems

The spectrum of operational NWP systems can be organised into three broad categories, each serving distinct purposes in the weather data ecosystem:

High-Resolution Regional Systems focus on spatial detail rather than broad coverage, using grid spacing fine enough to explicitly resolve convective processes that coarser models must parameterise. Systems like the High-Resolution Rapid Refresh (HRRR) operate at a 3-kilometre resolution over limited areas, updating forecasts hourly with short forecast horizons of 18-48 hours. This resolution allows for "convection-allowing" prediction, where the model directly

simulates individual thunderstorms, mesoscale convective systems, and other features essential for severe weather and flash flood prediction.

The computational cost of maintaining such high resolution restricts these systems to regional domains. HRRR covers the contiguous United States but requires boundary conditions from coarser global models that provide information about weather systems entering its domain. These systems excel at short-range forecasting where recent observations—particularly radar data assimilated every 15 minutes—provide crucial initialisation for rapidly evolving weather situations. However, their limited forecast horizons restrict their utility for medium-range planning applications.

Global Medium-Range Systems sacrifice spatial detail for planetary coverage and extended forecast horizons. Systems like the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Global Forecast System (GFS) operate at 9-18 kilometre horizontal resolution globally, producing forecasts extending 10-16 days into the future with updates four times daily. These systems capture synoptic-scale weather patterns—frontal systems, jet stream configurations, atmospheric blocking patterns, and large-scale storm systems—that govern weather evolution at timescales of days to weeks.

The coarser resolution prevents explicit representation of convective processes, requiring parameterisation schemes to approximate their statistical effects. This limitation manifests as systematic underestimation of extreme precipitation intensity and excessive spatial smoothing of rainfall patterns. However, for applications requiring advance notice of approaching storm systems, understanding of large-scale atmospheric configurations, or boundary conditions for regional models, global systems prove indispensable.

Reanalysis Systems reconstruct historical atmospheric conditions by combining all available observations with NWP models in a consistent framework. Systems like ERA5 employ modern data assimilation techniques and model physics retrospectively, creating homogeneous datasets spanning decades. These products prove valuable for model training, validation against historical extreme events, and understanding climatological patterns. However, reanalysis data cannot provide real-time forecasts and may under-represent extreme events where observational data was sparse.

Complementary Roles in Risk Assessment

Modern weather data applications for risk assessment typically integrate multiple NWP systems, leveraging the strengths of each whilst mitigating individual limitations. A comprehensive flood risk assessment framework might employ:

Global models (ECMWF, GFS) to identify approaching weather systems days in advance, providing advance warning of potentially hazardous conditions and supplying boundary conditions for regional models

High-resolution regional models (HRRR) for detailed precipitation forecasts at scales relevant to individual catchments, capturing the intense, localised events that coarser models smooth away

Reanalysis data (ERA5) for training statistical models, calibrating downscaling relationships, and validating that synthetic event scenarios exhibit realistic spatial and temporal characteristics

This hierarchical approach mirrors how operational meteorologists work: global models provide the "big picture" of evolving weather patterns, whilst regional models add crucial local detail. The challenge for risk assessment applications lies in properly combining information across different resolutions, forecast horizons, and uncertainty characterisations—a challenge addressed through the frameworks described in subsequent chapters.

System Selection Considerations

Selecting appropriate NWP data sources for specific applications requires careful consideration of several factors:

- **Spatial Scale of Interest:** Applications focused on individual properties or small catchments require the fine resolution that only regional high-resolution systems provide. Portfolio-level assessments spanning multiple regions might rely primarily on global model output, using regional models selectively for locations with highest exposure or where local topographic effects prove crucial.
- **Temporal Scale Requirements:** Short-term operational decisions (hours to days) benefit from the frequent updates and latest observational data assimilation that regional systems provide. Medium-range planning (days to weeks) requires the extended forecast horizons that only global systems offer. Long-term risk assessment might rely primarily on reanalysis data for characterising historical patterns and training statistical models.
- **Phenomenon of Interest:** Convective precipitation—thunderstorms, mesoscale convective systems, intense localised rainfall—requires convection-allowing resolution that only high-resolution regional systems provide. Frontal precipitation, widespread stratiform rain, and large-scale storm systems are well-captured by coarser global models. Orographic precipitation enhancement proves challenging for all systems but benefits from the finer resolution regional models offer in mountainous terrain.
- **Computational and Data Management Constraints:** High-resolution model output generates vast data volumes that require substantial storage and processing infrastructure. A single HRRR forecast hour produces approximately 100 MB of data in GRIB2 format, accumulating to terabytes for comprehensive historical archives. Applications with limited computational resources might rely more heavily on global model output or employ cloud-optimised formats like Zarr that enable selective data extraction.
- **Access and Latency Requirements:** Some applications require the absolute latest forecast information, whilst others can tolerate several hours of latency. Real-time operational systems like HRRR provide data within minutes of model completion, whilst some reprocessed formats may lag by hours. Public availability differs across systems—GFS data is freely available with minimal restrictions, whilst some international models have access limitations or commercial licensing requirements.

Data Standardisation and Interoperability

Despite diversity in resolution, domain, and forecast horizon, modern NWP systems share important commonalities that facilitate their integration. The World Meteorological Organisation (WMO) has established standardised definitions for atmospheric variables, ensuring that "temperature," "precipitation rate," "wind components," and hundreds of other parameters maintain consistent meanings across different modelling systems. This standardisation extends to data formats, with GRIB2 serving as the international standard for gridded meteorological data, though cloud-optimised formats like Zarr increasingly supplement it for specific applications.

This standardisation enables seamless comparison between different NWP systems, validation against observations, and integration into downstream applications. A flood risk model can ingest precipitation forecasts from ECMWF, GFS, or HRRR with minimal modification, treating them as alternative data sources that can be combined, compared, or selected based on their demonstrated skill for particular applications and forecast horizons.

Preview of Detailed System Descriptions

- **High-Resolution Rapid Refresh (HRRR)**, the primary data source for this framework's approach to flood risk assessment. HRRR's 3-kilometre resolution and hourly updates make it particularly valuable for capturing the localised extreme precipitation events that drive flooding, whilst its convection-allowing capability provides more realistic representation of storm structure than coarser models can achieve.
- **European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System**, widely regarded as producing the world's most accurate medium-range forecasts. ECMWF's global coverage and 10-15 day forecast horizon provide crucial advance warning of approaching weather systems and supply boundary conditions for regional models.
- **Global Forecast System (GFS)**, America's primary global forecast model. GFS's free public availability and extensive output make it widely used across government, commercial, and research applications, providing a complementary data source to ECMWF with particular strengths in tropical weather prediction.

Together, these operational NWP systems provide the weather data foundation upon which modern physical risk assessment builds. Their complementary capabilities—high-resolution regional detail, global medium-range forecasts, and historical reanalysis—enable comprehensive characterisation of weather patterns across the spatial and temporal scales relevant to banking risk assessment. Understanding each system's strengths, limitations, and appropriate applications proves essential for developing robust frameworks that properly leverage available weather data whilst acknowledging unavoidable uncertainties and gaps in coverage.

Before we consider each NWP, we have to provide an outline of level of data standardisation:

Industry Standard Parameters and Variables

The HRRR model implements the World Meteorological Organization's (WMO) standardised parameter definitions, representing the foundational variables used throughout the weather industry as the global standard:

Atmospheric State Variables

- Temperature (K): Measures the thermal state of the air at different altitudes or near the ground.
- Pressure (Pa): Indicates the weight of the overlying atmosphere at various levels.
- Relative Humidity (%): Expresses how close the air is to saturation with water vapour.
- Specific Humidity (kg/kg): Quantifies the actual mass of water vapour in a unit mass of air.
- Wind Components (U, V, m/s): Represent the horizontal wind speeds in the east-west (U) and north-south (V) directions.
- Vertical Velocity (Pa/s): Describes upward or downward air motion in the atmosphere.

Precipitation and Hydrology

- Total Precipitation Rate ($\text{kg/m}^2/\text{s}$): The instantaneous rate of rainfall and snowfall reaching the surface.
- Accumulated Precipitation (kg/m^2): The total mass of precipitation accumulated over a time period.
- Snow Water Equivalent (kg/m^2): The liquid water content contained within the snowpack.
- Frozen Precipitation Types: Categorises different snow and ice forms such as sleet or hail.
- Precipitation Type Probabilities (%): Indicates the likelihood of each precipitation form occurring.
- Surface Runoff (kg/m^2): The amount of water flowing across the surface after precipitation.

Radiative and Energy Parameters

- Downward Short-wave Radiation (W/m^2): The incoming solar energy reaching the surface.
- Upward Long-wave Radiation (W/m^2): The infrared energy emitted back to the atmosphere from the Earth's surface.
- Surface Heat Fluxes (W/m^2): Measure the energy exchange between the land surface and the atmosphere.
- Ground Heat Flux (W/m^2): Represents heat transfer between the surface and subsurface soil layers.
- Cloud Radiative Properties: Describe the influence of clouds on absorption, reflection, and emission of radiation.

Surface and Boundary Layer

- Surface Temperature (K): The temperature measured directly at the Earth's surface.
- Soil Moisture Content (kg/m²): Amount of water contained within the soil column.
- Surface Roughness (m): A measure of terrain texture affecting wind flow near the surface.
- Planetary Boundary Layer Height (m): The altitude to which turbulence from the surface mixes the atmosphere.
- Surface Wind Components (m/s): Horizontal wind speeds measured near the ground.
- Friction Velocity (m/s): Represents the shear stress between air and surface influencing turbulence.

Derived Meteorological Products

- CAPE (J/kg): Quantifies the potential energy available for convective updrafts in thunderstorms.
- CIN (J/kg): Measures the energy barrier that suppresses convection.
- Storm Relative Helicity (m²/s²): Assesses the potential for rotating updrafts in storms.
- Stability Indices: Summarize atmospheric stratification and storm potential.
- Echo Top Height (m): Highest altitude where radar detects precipitation-sized particles.
- Radar Reflectivity (dBZ): Indicates intensity of precipitation based on radar echo strength.

The Weather Industry's Universal Language

All NWP models speak the universal language of operational meteorology by implementing standardised parameters from the World Meteorological Organisation (WMO). These carefully defined variables represent decades of international collaboration and consensus in the weather industry, forming a comprehensive atmospheric measurement and prediction framework. This long-standing collaboration leading to standard metrics is where banks can stand on the shoulders of meteorological giants to advance their banking risk management.

At the foundation of NWP's capabilities lie the essential atmospheric state variables. The model tracks temperature variations across multiple pressure levels, from the surface to the upper atmosphere, while simultaneously monitoring pressure systems and humidity patterns. Wind patterns are captured through detailed U and V vector components, providing a complete picture of atmospheric motion at every level. These fundamental measurements form the backbone of all weather prediction and analysis.

The model's precipitation and hydrological parameters offer crucial insights for operational forecasting. Beyond simple rainfall measurements, NWP provides a detailed analysis of precipitation rates, accumulation patterns, and precipitation type classifications. The system distinguishes between rain, snow, and frozen precipitation while tracking crucial hydrological factors like surface runoff and snow water equivalent. These parameters prove especially valuable for flood prediction and winter weather operations.

Energy transfer within the Earth's atmosphere is a dynamic process involving multiple pathways of radiation and surface-level exchanges. These mechanisms collectively govern Earth's energy balance and influence temperature regulation, particularly in urban and agricultural settings.

Shortwave radiation from the sun enters Earth's atmosphere, with about 30% reflected due to the albedo effect from clouds and surface features, while 70% penetrates further. Of this penetrating energy, approximately 20% is absorbed by atmospheric gases, and the remainder reaches Earth's surface as direct and diffuse radiation. This solar energy varies regionally, with humid areas experiencing reduced solar heating due to cloud cover, while dry regions receive higher solar flux.

At Earth's surface, energy interacts through multiple processes: absorption converts solar radiation into heat energy; sensible heat flux transfers energy between the surface and atmosphere through conduction and convection; latent heat flux moves energy through evaporation and transpiration (especially in vegetated areas); and ground heat flux transfers energy into or out of the subsurface depending on temperature differentials between the surface and ground below.

The surface and boundary layer parameters provide critical information about the interface between Earth's surface and the atmosphere. NWP measures surface temperature, wind patterns, soil moisture content, and surface roughness characteristics. Calculating the planetary boundary layer height offers crucial insights into pollution dispersion and convective potential, making these parameters especially valuable for air quality forecasting and aviation operations.

This parameter set embodies the weather industry's standardised approach to atmospheric science. The parameters maintain uniform definitions across different modelling systems worldwide, ensuring seamless integration with existing weather visualisation and analysis tools.

This standardisation extends beyond mere measurement to encompass computation methods and quality control procedures, making NWP data immediately applicable within existing weather industry workflows. NWP maintains backward compatibility with historical datasets through this standardisation while pushing the boundaries of weather prediction capabilities.

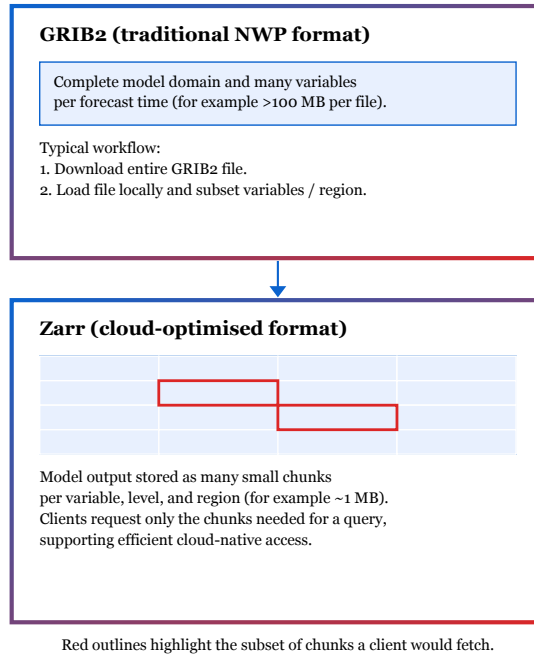
Industry Standard Formats

GRIB2 Format: The Traditional Standard

GRIB2 (Gridded Binary Second Edition) has long been the standard format used by numerical weather prediction centres to compress and disseminate model output as two-dimensional grids efficiently. Every hour, the HRRR model produces 18- or 48-hour surface forecast files in GRIB2 format, with each file approximately 100 MB in size. The GRIB2 format stores data in 64-bit float format, providing high precision for meteorological variables.

The primary advantage of GRIB2 lies in its efficient compression and widespread adoption across meteorological organisations. However, the format presents significant challenges for certain applications. When users need to access many GRIB2 files routinely, processing time and storage requirements become substantial. If a user wants only a small subset of data—for example, surface air temperature forecasts for Alabama—they must download the entire GRIB2

Figure 10 – Industry standard formats: GRIB2 and Zarr



file containing forecasts for all of CONUS and all variables (approximately 140 MB per timestep), then load the complete file before extracting the desired portion.

Zarr Format: The Cloud-Optimised Solution

Zarr was developed specifically to address GRIB2's bottlenecks by reformatting model output into a cloud-optimised storage format. The format was originally created by graduate student Taylor Gowan at the University of Utah's MesoWest group and is now supported through the Amazon Sustainability Data Initiative.

The key innovation of Zarr lies in its chunked data storage architecture. The NWP model output is reorganised so that each variable and vertical level is stored in Zarr stores of approximately 1 MB, with chunks around 10 kB containing all forecast lead times for 150×150 gridpoint subdomains. This structure enables users to query the full dataset in the cloud but only download their specific selection. Using the Alabama temperature example, a Zarr user would download less than 1 MB per timestep compared to 140 MB with GRIB2.

Additional advantages of Zarr include its support for parallel read/write operations, compatibility with cloud object stores like Amazon S3, and straightforward integration with open-source libraries such as xarray and Dask. The format uses JSON-based metadata that is human-readable and easily inspected.

Why Two Formats Coexist

The parallel availability of both formats reflects fundamentally different use cases and computational approaches. GRIB2 remains essential for applications requiring complete datasets, such as initialising high-resolution simulations that need all variables at multiple levels. Many traditional meteorological workflows are built around GRIB2, and the format continues to serve users who need full spatial coverage or prefer established tools.

Zarr, conversely, excels in scenarios involving data subsetting, machine learning applications, and cloud-native computing. When users need frequent, high-throughput access to specific variables or geographic regions, Zarr's architecture reduces data transfer times and costs by approximately 40 times compared to GRIB2. The format is particularly valuable for real-time forecasting applications and analytical work that benefits from rapid access to targeted data subsets.

However, Zarr does have limitations. The format is updated approximately three hours after each NWP run due to the processing time required to convert GRIB2 files. Some variables, such as GHI (Global Horizontal Irradiance), DNI (Direct Normal Irradiance), and DHI (Diffuse Horizontal Irradiance), are missing from the Zarr archive due to processing errors. Additionally, Zarr is relatively new and not as universally supported as GRIB2.

Both formats are publicly accessible through NOAA's Big Data Programme on Amazon Web Services, allowing users to select the format that best matches their computational needs and workflow requirements.

Now that we have all the standardisation out of the way we can have a look at the three NWPs.

Introduction to HRRR

The HRRR weather model began as an experimental system aimed at enhancing forecasts for aviation and has since evolved into one of the world's most accurate high-resolution continental weather models. Its development represents a significant advancement in numerical weather prediction, leveraging innovative technologies in computing, data assimilation, and atmospheric modelling.

The HRRR's origins lie in the Weather Research and Forecasting (WRF) model, a multiagency initiative that led to the development of the Advanced Research WRF (ARW) during over a decade of collaborative efforts. The ARW's open access made it popular among model developers applying it to various forecast problems across the globe. This foundation provided the technical infrastructure on which the HRRR was subsequently built.

The technical architecture of HRRR builds upon the foundation established by the Rapid Refresh (RAP) model, another NWP system. The RAP model provides the boundary conditions that HRRR requires to operate, creating a nested relationship where HRRR focuses on higher-resolution predictions within the broader domain covered by RAP. The HRRR system is initialised from RAP data one hour prior to each forecast start time, ensuring that each forecast

cycle benefits from the most current atmospheric analysis available. This relationship between RAP and HRRR exemplifies how modern weather data systems employ hierarchical approaches, with broader-scale models providing context for more detailed regional predictions.

The HRRR officially transitioned to NOAA National Weather Service operations on 30 September 2014. This change marked the culmination of experimental work by the Global Systems Laboratory (GSL) to develop a 3-km, convection-resolving model that updates forecasts hourly across the entire lower 48 United States. The model's resolution results in nearly two million surface grid points and predicts weather at 50 different elevations above each grid point.

At its core, HRRR is a convection-allowing numerical weather prediction model operated by the National Oceanic and Atmospheric Administration (NOAA). Its remarkable spatial resolution of 3 kilometres across the contiguous United States sets it apart, providing a level of detail that captures local weather phenomena crucial for risk assessment. This granularity allows risk managers to identify and evaluate location-specific weather patterns that might affect physical assets, from individual buildings to entire infrastructure networks.

The model's "rapid refresh" capability delivers frequently updated forecasts, enabling near-real-time risk assessment and decision-making. This combination of high-resolution and frequent updates makes HRRR particularly valuable for analysing chronic physical risks, such as severe weather events that could impact property portfolios or infrastructure investments.

The HRRR model is a 3-kilometre resolution weather model covering the continental United States (CONUS) that updates hourly and assimilates radar data every 15 minutes.

Introduction to ECMWF

The European Centre for Medium-Range Weather Forecasts (ECMWF) represents the pinnacle of global numerical weather prediction, consistently producing the most accurate medium-range forecasts available worldwide. Established in 1975 as an independent intergovernmental organisation, ECMWF has evolved from producing its first operational forecasts in 1979 to becoming the global benchmark against which all other forecasting systems are measured.

Unlike regional models such as HRRR that prioritise spatial detail over limited domains, ECMWF's mission focuses on providing global coverage with extended forecast horizons. Where HRRR excels at capturing convective-scale weather features over North America for 18-48 hours ahead, ECMWF provides the broader atmospheric context—the large-scale weather patterns, frontal systems, jet stream configurations, and storm tracks that govern weather evolution at timescales of days to weeks. This complementary relationship proves essential for comprehensive flood risk assessment: ECMWF identifies approaching weather systems days in advance, whilst HRRR provides the fine-scale precipitation details when those systems arrive.

ECMWF gained particular recognition in 2012 when it was the only major forecasting centre that correctly predicted Hurricane Sandy's unprecedented westward turn into the U.S. East Coast several days in advance—a forecast that proved crucial for emergency preparations. This

performance exemplifies ECMWF's consistent superiority in medium-range prediction, particularly for forecast lead times of 5-10 days where its advanced data assimilation, sophisticated physics, and comprehensive ensemble system provide clear advantages over competing global models.

Model Structure and Resolution

All ECMWF operational forecasts are generated using the Integrated Forecasting System (IFS), which is among the most advanced numerical weather prediction systems worldwide. The IFS uses a spectral atmospheric model with a hybrid sigma-pressure vertical coordinate system, combined with sophisticated models of other Earth system components. This design enables seamless integration of observations with physical models to produce highly accurate forecasts ranging from hours to weeks ahead.

The IFS atmospheric model uses 137 vertical levels extending from the surface to about 80 kilometres altitude, offering exceptional resolution throughout the troposphere and stratosphere. This vertical resolution surpasses HRRR's 51 levels, reflecting ECMWF's necessity to accurately depict stratospheric processes that influence medium-range forecast performance. The finest vertical resolution occurs in the planetary boundary layer, where weather directly affects human activities, becoming gradually coarser in the upper atmosphere. The hybrid coordinate system ensures model levels never intersect the ground, even over steep mountainous terrain—a vital consideration for global modelling that must manage the Himalayas, Andes, and other major mountain ranges.

The horizontal resolution of ECMWF's operational systems has improved over decades of development. The current high-resolution deterministic forecast (HRES) operates at approximately 9 kilometres grid spacing using a cubic octahedral grid (designated TC01279). This grid configuration, implemented in 2016, provides more accurate forecasts at equivalent computational cost compared to previous spectral truncation approaches, representing a significant technical breakthrough in global modelling.

The ensemble prediction system (ENS) also operates at 9-kilometre resolution for its 51 members—an increase from the previous 18-kilometre spacing implemented in 2024. This resolution enhancement required substantial computational resources but significantly improved ensemble forecast skill, especially for precipitation and other variables sensitive to spatial resolution. At 9 kilometres, ECMWF's ensemble approaches the effective resolution of many operational regional models, blurring traditional distinctions between "global" and "regional" forecasting whilst maintaining planetary coverage.

However, even at 9-kilometre resolution, ECMWF cannot explicitly resolve convective processes like HRRR does at 3 kilometres. Individual thunderstorms, mesoscale convective complexes, and other sub-10-kilometre features must be parameterised rather than directly simulated. This fundamental limitation means ECMWF systematically underestimates extreme precipitation intensity and produces spatially smoother rainfall patterns than observed—this same systematic bias affects all global models, although ECMWF's advanced parameterisation schemes help reduce these errors compared to competing systems.

Physics Parameterisation and Model Cycles

The IFS uses parameterisation schemes to represent processes that occur at scales smaller than the 9-kilometre grid can resolve. These schemes cover convection, cloud microphysics, boundary layer turbulence, radiation transfer, land surface processes, and other physics vital for realistic atmospheric development. ECMWF invests significantly in developing and refining these parameterisations through dedicated research programmes and regular model upgrades called "cycles."

Each IFS cycle represents a substantial package of improvements incorporating advances in physical parameterisations, numerical methods, resolution, and data assimilation. Recent cycles include Cycle 49r1, implemented in November 2024, which improved wind and temperature predictions, and Cycle 48r1 in June 2023, which increased skill and unified medium-range resolutions. These regular upgrades, typically occurring annually or biannually, ensure ECMWF maintains its position at the forefront of numerical weather prediction as scientific understanding advances and computational capabilities expand.

The convection parameterisation poses specific challenges for global models. ECMWF uses a scheme that models the statistical effects of unresolved convective clouds based on larger-scale atmospheric conditions. When atmospheric profiles show instability and adequate moisture, the scheme predicts vertical transport of heat, moisture, and momentum by updrafts and downdrafts. However, no parameterisation can fully capture the complexity and variability of real convective systems, especially during extreme events. This is why ECMWF forecasts, while excellent for synoptic-scale weather patterns, tend to underestimate peak precipitation intensities during intense convective episodes—precisely where HRRR's convection-allowing resolution offers better guidance.

Four-Dimensional Variational (4D-Var) System

One of ECMWF's most significant contributions to numerical weather prediction has been the operational implementation of four-dimensional variational (4D-Var) data assimilation, which became operational on November 25, 1997. This technique combines observations made over a 12-hour time window with a short-range forecast to produce the optimal estimate of the atmospheric state serving as initial conditions for new forecasts. The 4D-Var system represents a quantum leap beyond simpler assimilation methods, extracting maximum information from observations by accounting for how atmospheric features evolve during the assimilation window.

The system operates by minimising a cost function that assesses how much the analysis deviates from both observations and a short-range forecast (the background). The minimisation process employs linearised versions of the forecast model and physical parameterisation schemes to effectively establish the best three-dimensional representation of the atmospheric state at the analysis time. This method allows observations taken at different times within the 12-hour window to influence the analysis appropriately, considering atmospheric motion and change instead of treating all observations as if they occurred simultaneously — a vital distinction that enhances the quality of initial conditions and improves subsequent forecast accuracy.

The inclusion of comprehensive linearised physics parameterisations in 4D-Var represents processes such as radiation, convection, clouds, and turbulence within the linearised model used during assimilation, ensuring observations can influence the analysis in physically consistent ways. For example, a satellite observing outgoing longwave radiation provides information not just about upper-level temperature but also about cloud properties and their vertical structure, with the linearised physics determining how to optimally distribute information among model variables.

Observational Data Streams

ECMWF processes an extraordinary volume and variety of observational data to constrain its forecasts. The Centre routinely ingests data from approximately 90 satellite instruments as part of daily operations, receiving 800 million observations daily. Of these, approximately 60 million quality-controlled observations are available for assimilation into the IFS each day, with the vast majority coming from satellite measurements. This massive data processing infrastructure, combined with sophisticated quality control procedures, ensures ECMWF leverages the full global observing system whilst rejecting erroneous or inconsistent measurements.

Satellite observations provide global coverage that has transformed weather prediction, particularly improving forecast skill in data-sparse regions such as the Southern Hemisphere and over oceans where conventional observations remain sparse. ECMWF assimilates various types of satellite data including radiances from microwave and infrared sounders, atmospheric motion vectors derived from tracking cloud and water vapor features, scatterometer wind observations over oceans, and radio occultation measurements providing precise temperature and humidity profiles. The Centre pioneered direct assimilation of satellite radiances—a technique extracting maximum information from these complex observations by modeling radiation transfer through the atmosphere rather than relying on simplified retrievals.

In addition to satellite data, ECMWF benefits from all available in-situ observations including surface weather stations, radiosondes (weather balloons), aircraft reports, ships, buoys, and specialized networks. Whilst satellites now contribute most to global forecast skill measured by overall metrics, in-situ observations remain critically important for calibrating remote sensing instruments and providing direct measurements of atmospheric conditions. The combination of satellite and in-situ data, integrated through the sophisticated 4D-Var system, enables ECMWF to create analyses representing the best possible estimate of Earth system state—analyses that subsequently drive both deterministic and ensemble forecasts.

The quality of initial conditions fundamentally limits forecast skill, particularly at longer lead times where errors in initial conditions amplify through atmospheric chaos. ECMWF's superior data assimilation capability explains much of its consistent forecast advantage over competing global models. Even when using identical atmospheric model physics, forecasts initialized from ECMWF's 4D-Var analyses typically outperform those starting from simpler assimilation systems, demonstrating that "getting the initial conditions right" matters as much as sophisticated model physics for medium-range prediction.

Ensemble Methodology and Construction

A defining characteristic of ECMWF's forecasting approach is its pioneering use of ensemble prediction, operational since November 24, 1992—making ECMWF one of the first centers to provide operational probabilistic forecasts. The ensemble prediction system (ENS) generates 51 forecasts: one control forecast plus 50 forecasts with slightly perturbed initial conditions and slightly altered model physics. This approach recognizes that the atmosphere is a chaotic system where small uncertainties in initial conditions and imperfections in model physics can grow over time, creating irreducible forecast uncertainty that deterministic forecasts cannot represent.

The ENS is designed to sample this uncertainty space, providing probabilistic forecasts indicating the range of possible weather conditions and likelihoods of specific events such as heavy precipitation, strong winds, or extreme temperatures. When ensemble members cluster tightly, the forecast is considered more confident; when they diverge substantially, forecast uncertainty is high and multiple scenarios remain plausible. This probabilistic information proves invaluable for decision-makers who must assess risks of adverse weather impacts—precisely the requirement for banking applications where tail risk characterization matters more than most-likely outcomes.

Initial condition perturbations for the ENS are generated using singular vector techniques identifying the fastest-growing analysis errors. These mathematically sophisticated perturbations are designed to sample uncertainty in the analysis consistent with observation and background errors, ensuring ensemble spread reflects actual forecast uncertainty rather than arbitrary variations. The perturbations target atmospheric features and regions where small initial differences can amplify rapidly—storm tracks, regions of strong baroclinicity, and areas with limited observational coverage.

Model uncertainty is represented through stochastic perturbations to physical parameterisation schemes, acknowledging that the model cannot perfectly represent all atmospheric processes. These perturbations introduce controlled randomness into convection, clouds, radiation, and other parameterised physics, ensuring ensemble spread encompasses not just initial condition uncertainty but also structural model uncertainties. The ensemble system also employs an Ensemble of Data Assimilations (EDA) providing flow-dependent estimates of analysis uncertainty, ensuring perturbations remain physically consistent with estimated initial condition errors.

At 51 members, ECMWF's ensemble is smaller than some research ensembles comprising hundreds of members, reflecting computational constraints of running global 9-kilometre resolution forecasts. However, ECMWF's ensemble demonstrates superior skill to larger ensembles from other centres, showing that quality—sophisticated perturbation methods, high resolution, coupled Earth system components—matters more than quantity alone. For banking risk assessment requiring tail event characterization, even 51 members proves insufficient for robustly estimating low-probability events, necessitating the distribution modeling approaches described in subsequent chapters. Nonetheless, ECMWF's ENS provides crucial information about forecast uncertainty at medium-range timescales unavailable from deterministic forecasts alone.

Medium-Range Deterministic Forecasts (HRES)

ECMWF's core mission focuses on medium-range forecasts spanning up to 10 days ahead for the high-resolution deterministic HRES forecast and 15 days for probabilistic ENS forecasts. These timescales fill a critical gap in the forecasting spectrum: too long for convection-allowing regional models like HRRR that typically extend only 18–48 hours, yet short enough that deterministic weather prediction retains meaningful skill. At 5–7 day lead times, ECMWF forecasts achieve skill levels that other centres typically reach only at 3–5 days—a two-day advantage that can prove crucial for planning and emergency preparation.

The HRES provides a single detailed description of one possible weather evolution, serving users requiring highest spatial and temporal resolution. HRES forecasts extend to 10 days with temporal resolution varying by lead time: hourly output for the first 90 hours, then 3-hourly to 144 hours (6 days), and 6-hourly thereafter. This temporal resolution reflects typical use patterns—users need more frequent updates for short-range planning, whilst daily or twice-daily guidance suffices for longer ranges where forecast uncertainty grows and smaller temporal variations become less reliable.

HRES products include comprehensive atmospheric fields (temperature, wind, humidity, pressure at multiple levels), surface conditions (2-meter temperature, 10-meter wind, precipitation, cloud cover), derived quantities (precipitation totals, wind gusts, severe weather indices), and specialized forecasts for applications like aviation (turbulence, icing), marine (wave heights, sea state), and renewable energy (wind power, solar radiation). The diversity of products reflects ECMWF's role serving national meteorological services that generate public forecasts, specialized users in transportation and energy sectors, and research communities requiring detailed atmospheric state information.

For flood risk assessment, HRES provides crucial advance warning of approaching storm systems, moisture transport patterns, and large-scale atmospheric configurations conducive to extreme precipitation. A forecast showing a developing extratropical cyclone approaching a region 7 days ahead, for example, enables advance positioning of resources, pre-releases from reservoirs if appropriate, and heightened monitoring as the event approaches. However, HRES cannot provide the precise location and intensity of convective precipitation that drives many flash flood events—precisely where HRRR's convection-allowing capability becomes essential as forecast lead time contracts and the threatening weather system enters HRRR's domain.

Operational AI Forecasting System (AIFS)

In a landmark development, ECMWF made its Artificial Intelligence Forecasting System (AIFS) operational on February 25, 2025, becoming the first major forecasting center to deploy a machine learning weather prediction model with 24/7 operational support alongside traditional physics-based forecasting. The AIFS represents a fundamentally different approach to weather prediction: rather than explicitly integrating atmospheric physics equations, it uses machine learning trained on decades of historical forecasts to predict how initial conditions will evolve, learning statistical patterns of atmospheric evolution from vast training datasets.

The AIFS demonstrated remarkable skill during extended testing, with the deterministic version (AIFS Single) exceeding physics-based IFS skill across the vast majority of lead times and variables when evaluated on independent data from 2024. This performance validated years of research and development, showing that machine learning could match or exceed traditional numerical weather prediction for many applications. Following deterministic success, ECMWF deployed the ensemble version (AIFS ENS) on July 1, 2025, providing 51-member probabilistic forecasts that outperform state-of-the-art physics-based models for numerous measures, with gains up to 20% for surface temperature.

Critically, AIFS uses the same initial conditions as IFS, generated through ECMWF's sophisticated 4D-Var data assimilation system. This hybrid approach leverages strengths of both methodologies: physics-based data assimilation provides optimal initial conditions incorporating all available observations whilst maintaining physical consistency, whilst machine learning provides rapid, energy-efficient forecast generation. AIFS generates forecasts over 10 times faster than physics-based models whilst reducing energy consumption by approximately 1,000 times—dramatic efficiency gains that could transform operational meteorology as the technology matures.

However, AIFS faces inherent limitations common to machine learning approaches. The system is constrained by training data coverage, potentially struggling with weather patterns or extremes not well-represented in historical datasets. Rare events like "bomb cyclones" (rapidly intensifying extratropical cyclones) require careful evaluation, as pure machine learning systems might not capture these phenomena as reliably as physics-based models explicitly representing the relevant dynamical processes. ECMWF views AIFS and IFS as complementary tools rather than competitors, providing a range of products allowing users to select what best suits their needs.

Data Access and Distribution

ECMWF provides forecast data to Member and Co-operating States in real time, typically within hours of initialization times. Member states receive comprehensive model output including full atmospheric fields at all levels, surface forecasts, ensemble members, and specialized products. This data flows to national meteorological services that generate public forecasts, severe weather warnings, and specialized guidance for aviation, marine, agricultural, and other sectors. Co-operating States receive similar access though with some limitations compared to full Member States.

For non-member users, ECMWF provides open data access to selected forecast products, reflecting a transition toward more open data policies aligned with European open science initiatives. High-resolution deterministic forecasts (HRES), ensemble forecasts (ENS), and numerous derived products are now freely available through web services and APIs. However, some specialized products and higher-frequency output remain restricted to members, and commercial applications may face licensing requirements depending on use case and distribution.

Data formats follow international standards, with GRIB2 serving as the primary format for gridded meteorological fields. ECMWF also provides NetCDF format for some products, and web services offering JSON and other formats suitable for modern applications. The

Meteorological Archival and Retrieval System (MARS) provides access to historical forecasts and analyses, supporting research, model validation, and statistical applications requiring long records. Cloud-based access is being developed to facilitate efficient data retrieval for users not requiring complete global datasets.

For banking risk assessment applications, ECMWF's data accessibility enables systematic integration of global medium-range forecasts into risk frameworks. The combination of real-time operational forecasts and extensive historical archives through MARS supports both operational risk monitoring (using latest forecasts) and model development activities (using historical data for training and validation). However, potential licensing complexities for commercial applications warrant careful attention to ensure compliance with ECMWF's terms of use, particularly for applications involving redistribution or financial products.

Integration with Regional Models

A crucial aspect of ECMWF's role in global forecasting is providing boundary conditions for regional models like HRRR. Regional models must specify atmospheric and oceanic conditions at their domain boundaries—information that can only come from coarser global models with broader coverage. ECMWF's superior medium-range skill makes it the preferred choice for boundary condition provision when regional models seek best possible guidance for weather systems entering their domains.

The process involves extracting ECMWF forecasts along the regional model's domain boundaries at appropriate temporal frequency (typically 3-hourly or 6-hourly) and vertical resolution. The regional model then uses this information to constrain atmospheric evolution at domain edges whilst allowing the high-resolution model physics to develop fine-scale features in the domain interior. However, errors in global model boundary conditions inevitably affect regional forecasts, particularly for weather systems entering the domain. A regional model cannot correct systematic biases or completely compensate for inaccurate synoptic-scale patterns in boundary conditions.

This hierarchical relationship between global and regional models exemplifies their complementary nature. ECMWF excels at representing large-scale atmospheric patterns and providing advance warning of approaching systems. Regional models like HRRR excel at resolving local-scale processes and extreme precipitation intensities once those systems arrive. For comprehensive flood risk assessment, both capabilities prove essential: ECMWF for medium-range context and regional model boundary conditions, HRRR for short-range precipitation details driving hydrological responses.

The integration also matters for ensemble forecasting. Regional ensemble systems can be driven by global ensemble boundary conditions, enabling probabilistic forecasting at convection-allowing resolution. However, computational costs typically limit such approaches to limited domains and shorter forecast periods compared to global ensembles. The most common pattern involves using ECMWF ensemble forecasts for medium-range probabilistic guidance, then transitioning to regional deterministic or limited-area ensemble systems as forecast lead time contracts and fine-scale details become resolvable.

Known Biases and Limitations

Despite ECMWF's superior performance, the model shows systematic biases and limitations that users need to understand for proper forecast use. Precipitation intensity is consistently underestimated for extreme events, with observed precipitation surpassing forecast values by factors of 1.5-2.0 or more for the most intense convective systems. This bias primarily arises from coarse resolution and parameterised convection, affecting all global models, although ECMWF's advanced physics schemes minimise the issue compared to competitors.

The model also shows regional and seasonal biases in temperature, precipitation, and cloudiness that vary geographically and temporally. Some biases reflect limited observational coverage in remote regions where initial conditions remain uncertain. Others stem from systematic physics errors—perhaps convection schemes triggering precipitation too readily in certain atmospheric profiles, or cloud schemes producing excessive cloudiness in specific regions. ECMWF continuously works to reduce these biases through physics improvements, but eliminating all systematic errors remains an ongoing challenge.

Forecast skill deteriorates with lead time due to atmospheric chaos amplifying initial condition errors and model imperfections. Beyond approximately 10 days, deterministic forecasts lose most useful skill for specific weather events, though some skill remains for weekly-averaged conditions and large-scale patterns. The ensemble helps quantify this uncertainty growth, showing forecast spread increasing with lead time as initial small differences amplify into fundamentally different weather scenarios. Users must understand this predictability horizon, recognising that 10-day forecasts provide general guidance rather than precise predictions.

ECMWF's Role in Physical Risk Assessment

ECMWF provides the global medium-range context essential for comprehensive flood risk assessment. Its forecasts identify approaching weather systems days in advance, provide boundary conditions for regional models, and quantify forecast uncertainty through ensemble prediction. The Centre's consistent superiority in forecast accuracy makes it the preferred global model for applications where skill matters, whilst operational reliability enables systematic integration into banking workflows.

However, ECMWF complements rather than replaces high-resolution regional systems. For banking applications requiring the characterisation of extreme precipitation that drives flood risk, the hierarchical approach is essential: ECMWF for early warning and large-scale context, regional models like HRRR for convective-scale details, and distribution modelling frameworks for tail risk quantification beyond what any operational ensemble can offer. Understanding each system's strengths and limitations allows for the best integration of weather forecasting capabilities into physical risk assessment frameworks that serve banking needs.

Introduction to GFS

The Global Forecast System (GFS), operated by the National Centers for Environmental Prediction (NCEP) within the National Oceanic and Atmospheric Administration (NOAA),

serves as America's primary global numerical weather prediction model. Running operationally since the early 1980s, GFS has evolved through multiple generations of development, establishing itself as one of the world's major operational forecasting systems alongside ECMWF and serving as the backbone of weather prediction across the United States and many international partners.

GFS operates as a freely available global forecasting system, producing comprehensive atmospheric predictions four times daily at 00, 06, 12, and 18 UTC. This unrestricted access distinguishes GFS from some international models that restrict data availability or impose licensing requirements for commercial use. The model's open data policy has fostered widespread adoption across government agencies, commercial weather services, research institutions, and numerous downstream applications requiring global atmospheric forecasts without access barriers or licensing costs.

Operational Context and Evolution

NCEP operates GFS as part of its environmental prediction suite, which includes regional models, ensemble systems, and specialised forecast products. The operational GFS underwent significant modernisation in recent years, transitioning to the Finite-Volume Cubed-Sphere (FV₃) dynamical core in June 2019—a complete replacement of the model's computational engine representing the most substantial GFS upgrade in decades. This FV₃ dynamical core, developed through collaboration between NOAA and the Geophysical Fluid Dynamics Laboratory (GFDL), provides improved numerical accuracy and computational efficiency compared to the previous spectral dynamical core.

The GFS development process balances competing demands: maintaining operational reliability for users depending on consistent forecast delivery, incorporating scientific advances to improve forecast skill, and managing computational costs within available supercomputing resources. Unlike ECMWF, which benefits from multi-national funding supporting world-class infrastructure, NCEP operates within NOAA's federal budget constraints, creating persistent tension between desired model capabilities and available resources. Despite these constraints, GFS maintains its position as a capable global forecasting system serving critical national and international forecasting needs.

In recent years, GFS development has been accelerated through the Unified Forecast System (UFS) initiative, which aims to create a community-based, coupled, comprehensive Earth system modelling framework. This approach leverages contributions from multiple agencies and research institutions, potentially accelerating GFS improvements through broader participation than NCEP could achieve alone. The UFS framework provides a common infrastructure for weather, subseasonal-to-seasonal, and seasonal-to-decadal prediction, enabling more consistent modelling across timescales.

Resolution and Grid Structure

The operational GFS runs at approximately 13-kilometre horizontal resolution (technically 0.25-degree latitude-longitude grid spacing, varying from about 13km at the equator to approximately

7km near the poles due to meridional convergence). This resolution positions GFS between ECMWF's 9-kilometre spacing and older global model standards of 25-50 kilometres, providing reasonable representation of synoptic-scale weather features whilst remaining coarse enough that convective processes require parameterisation rather than explicit simulation.

Vertically, GFS employs 127 layers extending from the surface to approximately 80 kilometres altitude, comparable to ECMWF's 137 levels. This vertical resolution adequately captures tropospheric and stratospheric processes important for medium-range forecast skill, with finer spacing near the surface where atmospheric conditions most directly impact weather. The FV₃ cubed-sphere grid structure represents a significant technical advance over previous spectral models, reducing computational costs whilst improving representation of small-scale features and eliminating pole problems that plagued earlier grid configurations.

The model's computational domain encompasses the entire globe with no nesting or limited-area configurations, though NCEP operates separate regional models including NAM (North American Mesoscale) and HRRR for higher-resolution coverage over North America. GFS forecasts extend to 16 days (384 hours), with the deterministic forecast considered most reliable through approximately 10 days. Temporal resolution varies by forecast hour: 1-hourly output through 120 hours, 3-hourly from 120-240 hours, and 12-hourly thereafter, reflecting reduced forecast precision at longer ranges.

Physics Parameterisations

GFS employs comprehensive physics parameterisation schemes that represent processes occurring at scales smaller than the 13-kilometre grid can resolve. These include convection schemes (covering cumulus clouds and convective precipitation), microphysics (explicit cloud and precipitation processes), planetary boundary layer schemes (turbulent mixing near the surface), radiation transfer (both shortwave and longwave), land surface processes, and gravity wave drag (orographic and non-orographic effects on atmospheric momentum). The current operational parameterisation suite reflects ongoing development aimed at reducing systematic biases and enhancing forecast skill.

However, GFS physics schemes generally lag behind ECMWF's sophistication, contributing to systematic differences in forecast quality between the two systems. Convection parameterisation proves particularly challenging, with GFS tending to trigger convection too readily in certain atmospheric profiles, producing excessive light precipitation while underestimating extreme intensities — a common global model issue exacerbated by coarser resolution and less refined physics compared to ECMWF.

Recent physics upgrades have focused on improving tropical cyclone prediction, precipitation forecasts, and representation of cloud-radiation interactions. The transition to FV₃ enabled physics improvements that were impractical with the previous spectral core, though full realization of FV₃'s potential requires continued physics development over multiple operational upgrade cycles. The UFS framework facilitates physics innovation by enabling research groups to test improvements in a common framework, potentially accelerating the rate of operational physics advances.

Data Assimilation System

GFS employs a hybrid 4D Ensemble-Variational (4D-EnVar) data assimilation system, which combines ensemble-based covariance estimation with variational optimisation. This system represents NCEP's effort to approach ECMWF's 4D-Var sophistication whilst leveraging ensemble methods for computationally efficient covariance estimation. The hybrid approach updates the atmospheric analysis by combining a 6-hour forecast with incoming observations, producing optimal initial conditions for each forecast cycle.

The data assimilation system processes observations from multiple sources including satellites (radiances, atmospheric motion vectors, scatterometer winds, GPS radio occultation), conventional observations (radiosondes, surface stations, aircraft reports, ships, buoys), and radar data for regional applications. NCEP benefits from close coordination with NOAA's satellite programs (GOES, JPSS) and international data exchange through WMO, accessing comparable observational datasets to ECMWF though with some differences in satellite instrument coverage and data processing.

However, GFS data assimilation generally trails ECMWF's 4D-Var system in sophistication and demonstrated impact on forecast skill. The hybrid 4D-EnVar approach, whilst computationally efficient, may not extract quite as much information from observations as ECMWF's full 4D-Var with its more comprehensive linearized physics. Initial condition quality differences between GFS and ECMWF analyses explain a substantial portion of the forecast skill gap between the two systems, with ECMWF's superior data assimilation providing persistent forecast advantages particularly evident at 5-10 day lead times.

Ongoing development efforts focus on improving the hybrid system, incorporating additional observations (particularly from new satellite instruments), refining observation error characterization, and enhancing the ensemble-based covariance estimation. These incremental improvements gradually narrow the data assimilation quality gap with ECMWF, though achieving full parity would require sustained investment in data assimilation research and development infrastructure.

Ensemble Prediction System (GEFS)

The Global Ensemble Forecast System (GEFS) provides probabilistic forecasting capability complementing the deterministic GFS. GEFS comprises 31 ensemble members (including one control) run at slightly coarser resolution than the deterministic GFS—currently 0.5-degree spacing (approximately 25-28 kilometres) through 10 days, degrading to 1.0-degree spacing (approximately 50-56 kilometres) for days 11-16. This resolution reduction compared to the deterministic forecast reflects computational constraints, with running 31 members at full GFS resolution proving prohibitively expensive given available computing resources.

GEFS initialization employs Ensemble Transform with Rescaling (ETR) to generate initial condition perturbations, combined with stochastic physics perturbations during model integration to represent model uncertainty. The ensemble extends to 16 days like the

deterministic forecast but with broader temporal spacing at longer ranges. Updates occur four times daily (00, 06, 12, 18 UTC) for the full 16-day forecast, providing regular uncertainty updates as new observations become available.

Verification studies consistently show GEFS trailing ECMWF's ENS in ensemble forecast quality across most variables and lead times. The coarser resolution partially explains this skill gap—25–28 kilometres proves less capable than ECMWF's 9-kilometre ensemble for representing mesoscale features and precipitation patterns. Additionally, the smaller ensemble size (31 members versus ECMWF's 51) provides less thorough sampling of forecast uncertainty, particularly problematic for rare events where limited ensemble size may miss plausible but low-probability scenarios.

Despite these limitations, GEFS provides valuable probabilistic guidance for medium-range forecasting, particularly for users requiring free, unrestricted ensemble data. The ensemble spread indicates forecast confidence, with tight clustering suggesting reliable predictions whilst large spread signals high uncertainty requiring caution in decision-making. Products derived from GEFS include ensemble mean forecasts, probability forecasts for threshold exceedance, percentile forecasts, and anomaly forecasts comparing predictions to climatology.

Forecast Performance and Verification

Independent verification consistently demonstrates GFS trailing ECMWF in forecast accuracy across most variables, regions, and lead times. For the standard 500-millibar geopotential height metric, ECMWF typically achieves skillful forecasts approximately one to two days longer than GFS, with this gap widening for precipitation and other variables more sensitive to model physics and resolution. Northern Hemisphere forecasts show smaller skill gaps than Southern Hemisphere, reflecting denser observational coverage over land areas where both systems benefit from more observations constraining initial conditions.

However, GFS has shown gradual improvement over recent upgrade cycles, particularly following the FV3 implementation in 2019. Physics upgrades in 2021 and subsequent cycles further enhanced performance, narrowing though not eliminating the skill gap with ECMWF. Some specific applications show competitive GFS performance, particularly tropical cyclone track prediction where GFS demonstrates strong skill comparable to leading international models including ECMWF.

Precipitation verification reveals typical global model challenges: systematic underestimation of extreme intensities, excessive spatial smoothing, and difficulties with convective precipitation timing and location. GFS shares these biases with ECMWF and other global models operating at similar resolutions, though GFS's coarser 13-kilometre spacing and less refined physics produce somewhat larger systematic errors than ECMWF's 9-kilometre system. Both models require downscaling or bias correction for quantitative precipitation forecasts at scales relevant to flood prediction.

Regional and Seasonal Variations

GFS performance varies by geographic region and season, reflecting differences in forecast challenges and observational coverage. Winter season Northern Hemisphere forecasts generally show higher skill than summer forecasts, as wintertime synoptic-scale features prove more predictable than summer convective activity. Tropical forecasts present particular challenges for all global models, with smaller spatial scales, weaker steering flows, and active convection creating difficulties for parameterised physics and coarse resolution.

The model demonstrates particular strength in tropical cyclone track prediction, often matching or exceeding ECMWF's performance for Atlantic and East Pacific storms. This tropical cyclone skill reflects sustained NCEP investment in improving tropical prediction capabilities, including physics schemes specifically targeting warm-core vortex representation. However, intensity forecasts remain challenging for GFS as for all global models, with rapid intensification events particularly difficult to predict accurately.

Systematic biases vary regionally and seasonally, with known issues including excessive precipitation in certain tropical regions, cold biases in polar areas, and biases in cloud cover affecting radiation budget and surface temperatures. NCEP continuously works to reduce these biases through physics improvements and data assimilation refinements, though some systematic errors persist across multiple model versions reflecting fundamental limitations of parameterised physics at current resolutions.

Data Access and Format

GFS distinguishes itself through completely free, unrestricted public access to all forecast products. Full model output is available through NOAA's operational servers typically within 2-4 hours of initialization time, with no registration requirements, access fees, or usage restrictions beyond reasonable limits preventing server abuse. This open data policy enables widespread GFS adoption across commercial weather services, research institutions, educational applications, and individual users worldwide requiring global forecast data without licensing complications.

Data formats follow standard meteorological conventions, with GRIB2 serving as the primary distribution format for gridded fields. NOAA provides multiple access mechanisms including direct download via HTTPS/FTP, NOMADS (NCEP's operational data server), and cloud-based access through partnerships with commercial providers. Subsets, regridding, and format conversions are available through various tools and services, facilitating efficient data retrieval for users not requiring complete global datasets.

The extensive GFS archives maintained by NCEP and NOAA provide historical forecast data supporting research, model validation, and machine learning applications. These archives extend back decades for key fields, enabling long-term verification studies and development of statistical post-processing methods. The combination of real-time operational access and comprehensive historical archives makes GFS particularly attractive for applications requiring long data records without licensing constraints or access barriers.

Integration with Downstream Applications

GFS serves as the foundation for numerous downstream forecast products and applications across government and commercial sectors. The National Weather Service relies on GFS for medium-range forecast guidance, tropical cyclone prediction, and severe weather outlooks. Commercial weather providers incorporate GFS into their forecast processes, often blending it with ECMWF and other models to optimize predictions. Aviation, marine, renewable energy, and agricultural sectors utilize GFS forecasts extensively for operational planning and risk management.

Regional models including HRRR use GFS boundary conditions when ECMWF data proves unavailable or when operational workflows prioritize domestic data sources. This boundary condition role demonstrates GFS's continued importance even when not achieving best-in-class global forecast skill—the model provides reliable, freely available global forecasts serving essential infrastructure functions across numerous applications where ECMWF access restrictions or licensing costs prove prohibitive.

GFS data feeds numerous value-added services including specialized forecast products for specific industries, statistical post-processing systems correcting systematic biases, downscaling applications generating local-scale forecasts, and ensemble-based decision support tools. This extensive downstream ecosystem reflects decades of applications development building upon GFS's consistent operational delivery and unrestricted data access, creating substantial inertia favoring continued GFS use even when alternative models demonstrate superior raw forecast skill.

Summary

The Global Forecast System provides freely available global forecasts serving critical national and international needs. While generally trailing ECMWF in forecast accuracy, GFS maintains its position as a capable operational system demonstrating particular strengths in tropical cyclone prediction and providing essential services where unrestricted data access matters more than absolute forecast skill. The model continues improving through sustained development efforts, particularly the FV3 implementation and ongoing UFS initiative promising accelerated advancement through community-based development. For users requiring global forecast data without access restrictions or licensing complexity, GFS remains an essential resource complementing proprietary systems like ECMWF in a diverse operational forecasting ecosystem.

Common Applications

- Mapping & Visualisation: Extracting only the relevant region for interactive maps.
- Machine Learning on Geospatial Data: Reducing data size before training models.
- Satellite & Remote Sensing: Processing only the necessary geographic area to save computational resources.

- **Logistics & Supply Chain:** Optimising delivery routes by selecting relevant city or regional subsets.
- **Application - Regional Climatology:** Regional climatology represents the application of weather data and climate principles to specific geographic areas

Application - Mapping and Visualisation

NWP provides access to its data through a combination of official NOAA-developed tools, web-based interfaces, and third-party software applications, rather than offering a single comprehensive visualisation platform. This distributed approach reflects the model's primary function as a data-producing system, with visualisation capabilities developed to serve diverse user requirements across different technical expertise levels.

NOAA provides the Weather and Climate Toolkit (WCT), free mapping and data conversion software designed specifically for visualising weather and climate data, including NWP outputs. This platform includes source code and allows users to visualise various meteorological datasets with considerable flexibility. Additionally, the NWP HDW Map Interface provides direct visualisation of NOAA NWP forecast and analysis data, allowing users to select specific locations, variables, and time periods for examination.

A web-based NOAA NWP Browser application offers interactive features including real-time weather data display, an interactive map for panning and zooming, time navigation with date pickers and hour sliders, and automatic detection of the most recent available data. The application displays composite reflectivity data at 3-kilometre resolution and functions on both desktop and mobile devices, making NWP data accessible without requiring specialised software installation.

For users requiring more advanced visualisation capabilities, third-party solutions have emerged to fill specific needs. QGIS has become a particularly powerful option for visualising NWP meteorological data, allowing users to load NWP files as either raster or mesh layers and overlay them on base maps like OpenStreetMap to examine specific atmospheric variables at particular locations. The Herbie Python package provides programmatic access to NWP data through xarray with basic plotting capabilities using Cartopy, though these visualisation features remain in early development stages compared to the package's primary data access functionality.

This ecosystem approach enables different user communities—from operational forecasters requiring rapid web-based access to researchers needing sophisticated spatial analysis capabilities—to work with NWP data using tools appropriate to their specific requirements and technical workflows.

Application - Machine Learning on Geospatial Data

Several artificial intelligence systems have emerged using NWP as their primary training dataset. NWPCast represents a data-driven emulator for regional weather forecasting, trained on three years of NWP data and considered a data-driven sibling to the physics-based NWP model. This

approach demonstrates some accuracy whilst providing ensemble forecasting capabilities that would be computationally impractical with traditional physics-based methods.

Similarly, YingLong, an AI-based limited area model, was constructed using NWP outputs spanning 2015 to 2021, encompassing data from four different NWP versions with significant differences in physics schemes and data assimilation methods. When evaluated against NWP forecasts, YingLong-NWP showed higher skill scores for predicting wind components and particularly excelled in forecasting surface-specific humidity.

The applications of machine learning on NWP data extend beyond weather forecasting into air quality prediction, wind power forecasting, and climate risk assessment. Researchers have developed frameworks for PM_{2.5} concentration estimation, applying timeseries machine learning methods to NWP meteorological parameters for hourly air quality predictions. Wind power forecasting leverages NWP's high temporal resolution and 15-minute forecast intervals for operational planning. The model's real-time availability makes it particularly suitable for nowcasting applications where rapid prediction turnaround proves essential.

NVIDIA Earth-2 and CorrDiff

NVIDIA's Earth-2 platform represents a particularly sophisticated application of HRRR data in artificial intelligence workflows. The platform leverages HRRR data both as a reference dataset for training and as a target resolution for downscaling operations, combining artificial intelligence with physical simulations and GPU acceleration to create high-resolution weather and climate prediction tools that operate at unprecedented speed and scale compared to traditional physics-based approaches.

The primary application using HRRR within Earth-2 is CorrDiff (Corrector Diffusion), a generative AI model designed explicitly for weather downscaling. CorrDiff takes coarse-resolution weather forecasts and downscales them to high resolution, making HRRR both a training target and an output specification. The model accepts 38 surface and atmospheric variables from the Global Ensemble Forecast System (GEFS) at 25-kilometre resolution, including wind components at multiple atmospheric levels, temperature at various altitudes, relative humidity, surface pressure, and total column water vapour.

The model then produces downscaled predictions at 3-kilometre resolution matching the HRRR grid used by NOAA, generating eight critical surface and atmospheric variables: 10-metre wind components, 2-metre temperature, total precipitation, and precipitation type probabilities for snow, ice pellets, freezing rain, and rain. This represents a substantial increase in spatial resolution—from GEFS's 129×301 grid points to CorrDiff's $1,056 \times 1,792$ grid points over the contiguous United States.

CorrDiff was trained on GEFS and HRRR data spanning from December 2020 through December 2023, and evaluated on data from early 2024 through mid-2024. By learning the statistical relationship between lower-resolution global forecasts and high-resolution HRRR observations, the model enables cost-effective high-resolution predictions without the computational expense of running full physics-based regional models. The operational

advantages prove substantial: CorrDiff generates forecasts approximately 1,000 times faster and uses 3,000 times less energy than conventional weather models, producing 15-day global forecasts in approximately 90 seconds compared to traditional models that require hours of computation.

Beyond CorrDiff, Earth-2 includes StormCast, which uses ERA5 reanalysis data and HRRR observations to predict extreme weather phenomena at mesoscale resolution. This enables the prediction of severe storms, hurricanes, and other high-impact weather events. The model has demonstrated approximately 10% greater accuracy than operational HRRR in predicting storm-related variables such as temperature, wind, and radar reflectivity. This superior performance for extreme weather prediction stems from the generative AI approach's ability to capture spatial detail and non-linear dynamics that traditional models sometimes fail to represent accurately.

The Earth-2 platform's approach addresses fundamental limitations of traditional physics-based forecasting. Running high-resolution physics-based simulations continuously proves prohibitively expensive for many applications, whilst AI downscaling provides an economical alternative that reduces infrastructure burden whilst maintaining forecast quality. This enables smaller organisations and developing regions to access high-resolution weather information previously available only to institutions with substantial computational resources. Additionally, the speed advantage enables applications requiring rapid decision-making, such as emergency response to extreme weather, dynamic renewable energy forecasting, and near-real-time hazard detection.

Operational adoption demonstrates that AI-augmented weather prediction has matured from research to operationally viable technology. The Weather Company, Taiwan's Central Weather Administration, and various energy companies are amongst the early adopters leveraging Earth-2's APIs to generate high-resolution simulations and visualisations at the 2-kilometre scale and beyond.

The integration of HRRR data with machine learning through platforms like Earth-2 represents a convergence of high-resolution numerical weather prediction and data-driven modelling, enabling more efficient regional weather forecasting whilst maintaining competitive accuracy against physics-based approaches. As HRRR data continues to accumulate and cloud-native access improves, machine learning applications on this dataset will likely expand into new domains requiring high-resolution atmospheric information for climate risk assessment and operational decision-making.

Application - Mapping and Satellite Imagery

NOAA provides the Weather and Climate Toolkit (WCT), free mapping and data conversion software designed specifically for visualising weather and climate data, including HRRR outputs. This platform includes source code and allows users to visualise various meteorological datasets with considerable flexibility. Additionally, the HRRR HDW Map Interface provides direct visualisation of NOAA HRRR forecast and analysis data, allowing users to select specific locations, variables, and time periods for examination.

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Application - Logistics and Supply Chain

Modern logistics route optimisation systems collect diverse data including GPS coordinates, traffic reports, historical transportation patterns, and real-time demand forecasts. The optimisation process applies advanced algorithms that evaluate various possible routes whilst integrating real-time data such as live traffic updates, weather conditions, and road closures. When NWP weather data is incorporated into these systems, the algorithms can dynamically adjust routes to ensure optimal efficiency even when facing unforeseen weather challenges. For example, if NWP forecasts indicate adverse weather conditions along a planned route, the system can proactively reroute deliveries to avoid delays and potential safety issues.

For practical logistics applications, companies typically require weather data for specific operational areas rather than complete continental coverage. The process of selecting relevant city or regional subsets involves extracting the necessary geographic boundaries from the full NWP dataset. The Zarr format organises data into 96 small chunks for every variable in each model run, facilitating efficient extraction of specific geographic regions. This subset selection proves particularly valuable for companies with concentrated delivery networks in specific metropolitan areas or regions. Rather than processing weather data for the entire United States, logistics operations can focus on their specific service territories, reducing computational overhead whilst maintaining forecast accuracy.

The application of NWP data in route optimisation delivers measurable operational improvements. Studies of route optimisation systems using comprehensive data inputs, including weather patterns, have demonstrated a 15% reduction in average distance travelled by vehicles and a 20% decrease in fuel consumption compared to pre-optimisation routes. These systems can achieve near-perfect order fulfilment rates through data-driven approaches. The hourly update frequency of NWP proves particularly advantageous for logistics operations, as delivery schedules typically operate on similar timeframes. Companies can access updated weather forecasts aligned with their delivery windows, enabling more precise planning and real-time adjustments.

Downstream forecast products that rely on NWP data include various operational systems used in transportation planning, demonstrating the model's proven reliability for time-sensitive

decision-making applications. The continuous feedback loop in route optimisation systems allows data from executed routes to refine future planning, improving the accuracy and efficiency of subsequent route optimisations. For logistics companies operating across multiple facilities or distribution centres, NWP data can inform location-allocation decisions and help determine optimal service areas based on weather patterns and their impact on delivery reliability. This capability proves especially valuable for operations requiring strict adherence to delivery schedules or when transporting weather-sensitive goods.

Application - Regional Climatology

Regional climatology involves applying climate principles to specific areas, studying how atmospheric conditions develop and interact within defined spaces over long periods. While global climatology examines worldwide patterns and processes, regional climatology centres on understanding the unique features that arise from the interaction of latitude, topography, land-water distribution, and prevailing circulation patterns within particular regions. Climate itself is the combination of weather in a specific area, defined quantitatively by expected values of meteorological elements during certain months or seasons. This includes variables such as temperature, precipitation, wind, pressure, cloudiness, and humidity, measured by national meteorological agencies like the UK Met Office, NOAA, and coordinated through the World Meteorological Organisation (WMO).

The foundation of regional climatology rests on comprehensive, long-term datasets that span multiple decades, provided by institutions such as NOAA's National Centers for Environmental Information (NCEI), the European Centre for Medium-Range Weather Forecasts (ECMWF), and national meteorological services worldwide. These datasets provide a statistical framework essential for distinguishing normal variability from anomalous conditions. The WMO establishes standards for 30-year climatological normals (currently 1991-2020), which serve as baseline reference periods for climate information.

Climate represents far more than simple statistical averages; it encompasses the aggregate of atmospheric conditions involving heat, moisture, and air movement, including the extremes, frequencies, trends, and sequences of weather elements that occur from year to year. This temporal depth allows climatologists to move beyond the snapshot perspective of operational meteorology and instead capture the full spectrum of atmospheric behaviour characteristic of a region. Such extended records become particularly valuable when assessing the significance of recent observations, planning infrastructure designed to withstand expected extremes, or detecting emerging shifts in regional climate patterns.

Regional climatology distinguishes itself through attention to the mechanisms that create local differentiation within broader climatic zones. The climate of any region depends fundamentally on latitude, altitude, and orientation relative to water bodies, mountains, and prevailing wind directions. These factors interact to produce the distinctive atmospheric signatures that define regional climates, from the moisture-laden conditions of coastal areas to the continental extremes of interior regions, and from the modified temperatures of valley floors to the harsh conditions of mountain peaks. Understanding these regional characteristics requires examining how large-

scale atmospheric circulation patterns such as ENSO (El Niño-Southern Oscillation) and the North Atlantic Oscillation (NAO) translate into local weather phenomena, how surface characteristics modulate atmospheric responses including urban heat island effects, and how geographic features channel or block air masses to create persistent patterns.

Global climate datasets that support regional analysis are produced by several key institutions: NASA's Goddard Institute for Space Studies provides GISTEMP (1880-present), the UK Met Office Hadley Centre and Climatic Research Unit at the University of East Anglia jointly produce HadCRUT5 combining CRUTEM5 land data and HadSST4 sea surface temperatures (1850-present), NOAA NCEI produces multiple datasets including the Global Historical Climatology Network Daily (GHCNd), and the Copernicus Climate Change Service operated by ECMWF provides the Climate Data Store with ERA5 reanalysis data covering 1940 to present at 31km resolution with 137 vertical levels. Regional climate services are delivered through networks such as NOAA's six Regional Climate Centers covering the United States, the GCOS Surface Network of approximately 1,000 baseline stations providing uniform spatial coverage globally, and individual national meteorological services maintaining weather station networks for synoptic and climate observations.

Weather Data and the Banking Risk Architecture Challenge

The sophistication of modern weather data systems, exemplified by NWP's high-resolution forecasting capabilities and cloud-native data architectures, represents a remarkable achievement in atmospheric science and computational meteorology. Yet for banking institutions seeking to integrate physical climate risks into their credit and market risk frameworks, a fundamental question emerges: should weather data enter banking risk architecture at all?

Banks operate within established risk architectures built around probability of default, loss given default, exposure at default, and value-at-risk calculations. These frameworks rely on structured financial data, historical loss experience, and standardised risk ratings that translate directly into capital requirements, provisioning decisions, and portfolio management actions. Weather data, by contrast, arrives as atmospheric state variables—temperature in Kelvin, pressure in Pascals, precipitation rates in kilograms per square metre per second. The disconnect extends beyond mere units of measurement to encompass temporal scales, spatial granularity, and probabilistic frameworks that bear little resemblance to conventional banking risk metrics.

The temporal dimension reveals the deeper structural problem. NWP provides forecasts extending 18 to 48 hours ahead—a timeframe potentially valuable for tactical hedging decisions but largely irrelevant to the fundamental credit assessments that drive banking risk management. When a hurricane approaches landfall, banks can respond to the event through established channels: monitoring news reports, activating business continuity protocols, and potentially hedging exposures through catastrophe bonds or weather derivatives.

They need not integrate hourly atmospheric updates into their credit valuation models to take protective action. The information arrives through external news feeds, just as banks respond to

corporate earnings announcements, regulatory changes, or geopolitical developments without embedding real-time news data streams into their core risk systems.

This raises a critical distinction between responding to weather events and absorbing weather data into risk architecture. Banks have always managed acute weather risks through operational responses: extending payment deadlines for borrowers in disaster zones, reassessing collateral values following flood damage, or adjusting underwriting standards in hurricane-prone regions after major storms. These responses flow from observed events rather than atmospheric forecasts.

A borrower's creditworthiness changes when their factory floods, not when precipitation rates exceed certain thresholds in GRIB2 files. The connection between weather data and credit spreads remains sparse because the causal chain passes through physical damage, business interruption, and financial distress—intermediate steps that weather models do not directly quantify.

Even for longer-term climate risk assessment, the temporal mismatch persists. Banking credit exposures typically span years or decades, whilst operational weather forecasts provide hourly updates for days ahead.

This transformation challenge suggests that banks should not position themselves as consumers of raw weather data. The expertise required to convert atmospheric forecasts into credit risk metrics—geographic information systems, hydrological modelling, damage assessment, actuarial analysis—falls well outside core banking competencies. More fundamentally, building infrastructure to ingest, process, and maintain weather data streams would burden risk architectures with information that rarely translates directly into actionable risk management decisions.

The sheer volume and quality of weather data is both an achievement and a challenge for banking risk management. However, the link between atmospheric science and credit risk remains fragile because weather patterns affect financial outcomes only through complex chains of physical, operational, and economic impacts. Banks need information several steps downstream in this causal chain — not because they lack access to weather data, but because weather data in its raw form offers little actionable intelligence for risk managers who are used to thinking in terms of default probabilities, loss severities, and capital requirements.

“The challenge for the financial industry, and the purpose of this book, is to develop through bodies such as ISDA and FINOS/CDM standardised analytical infrastructure to integrate flood risk into banking risk architecture. Weather data can speak eloquently about atmospheric conditions, and banks speak of credit spreads. The journey we shall embark upon in the subsequent chapters is to bridge these two worlds, ultimately arriving at Physical Risk Swaps. - David Kelly, MKM

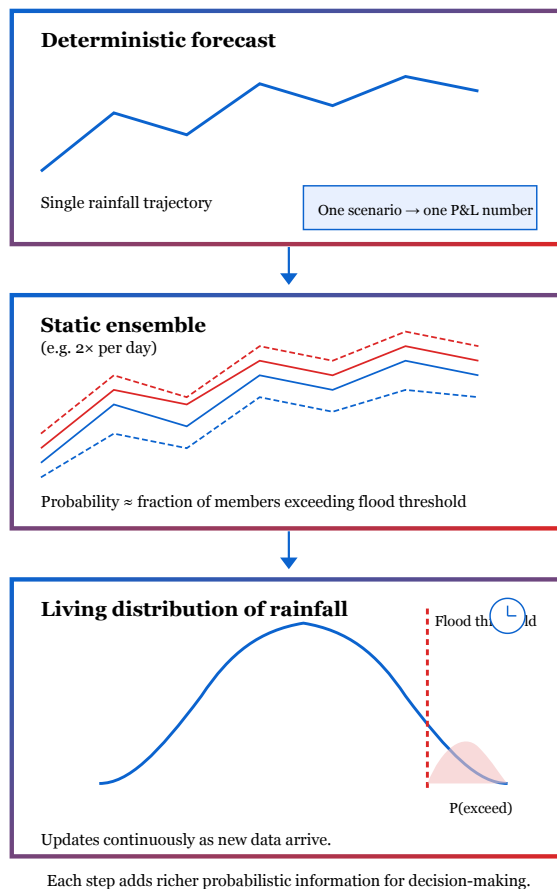
Chapter 3: Probabilistic Weather

Why banks need probabilistic weather

Banks do not need to know exactly what the weather will be at 3 p.m. next Thursday. They need to understand the range of plausible conditions and how those conditions could affect collateral values, cash-flows and solvency. A single deterministic forecast—“50 mm of rain on Thursday”—is almost never the right object for credit and capital decisions. What matters is the probability that rainfall will exceed the thresholds that trigger flooding, business interruption or infrastructure failure.

Traditional climate and weather use in finance is largely scenario-based. Institutions choose a handful of historical events or climate pathways, shock their portfolios to those cases, and infer losses. This approach is simple but brittle. It cannot answer questions such as: “What is the

Figure 11 – From single forecasts to living distributions



chance that a 1-in-100 year rainfall event will occur over this catchment in the next five days?” or “How does that probability change hour by hour as new radar and satellite data arrive?” It also struggles with extremes, because the most consequential events are precisely those that almost never occur in the historical record.

Modern numerical weather prediction systems already recognise that the atmosphere is chaotic and that small uncertainties in initial conditions grow rapidly over time. Operational centres therefore run ensembles: many slightly different forecasts, each starting from a perturbed version of the analysed state. For meteorologists, ensembles are a way to express forecast confidence. For banks, they can become the raw material for a probabilistic view of physical risk—if they are transformed into the right units and updated in the right way.

The key shift for banking is to move from static, one-off scenarios to what this book calls a living distribution of weather. Instead of freezing a handful of plausible futures once per quarter, we maintain an evolving probability distribution over thousands of physically realistic weather trajectories. As new information arrives—from high-resolution models such as HRRR, from surface stations, from radar and satellites—the distribution adjusts. Scenarios that become inconsistent with observations lose weight, and scenarios that match emerging patterns gain weight.

For a bank, this living distribution can be wired directly into risk systems. Collateral values for flood-exposed properties no longer depend on a fixed “1-in-100” stress, but on the current probability that rainfall over the relevant catchment will exceed the threshold that drives damaging inundation. Expected credit losses on a portfolio of mortgages or corporate loans can be updated intraday as the tails of the distribution fatten or thin. Capital allocation and hedging decisions—for example, buying protection via physical-risk swaps or parametric insurance—can respond to changes in the probability of severe weather with the same agility that trading desks already show in response to market news.

The remainder of this chapter explains how to construct such a living distribution in a way that remains anchored in atmospheric physics and usable in banking practice. We first describe how ensembles and Monte Carlo methods turn the butterfly effect from a liability into an asset: by exploring not one future but many. We then show how Bayesian updating allows the distribution itself to learn continuously from new data. Finally, we explain how these probabilistic weather patterns are converted into rainfall time series that plug into flood models and, through them, into credit and capital metrics. Readers interested mainly in the concepts can skim the mathematical expressions; they are included for completeness rather than as a prerequisite for understanding the narrative.

From single forecasts to living distributions

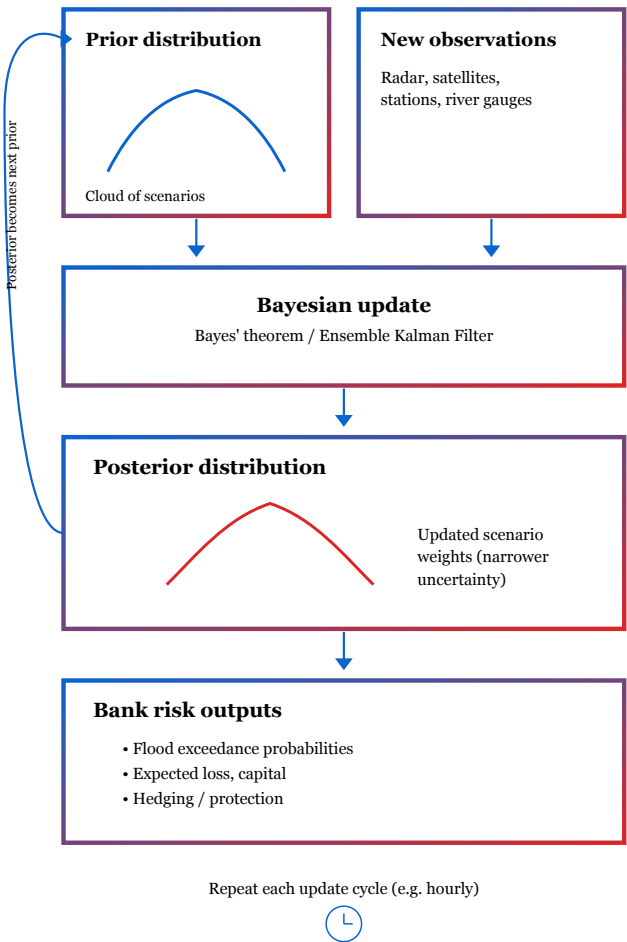
Weather forecasting has traditionally aimed to answer a deceptively simple question: what will happen? A numerical model is initialised from the best estimate of the current atmosphere and integrated forwards to produce a single trajectory. For many operational purposes—planning an outdoor event, scheduling an airline rotation—that is good enough. For banking risk, it is not. A

single trajectory hides the fact that many other outcomes are almost equally plausible, particularly several days ahead, and that some of those plausible outcomes involve damaging extremes.

Monte Carlo and ensemble methods replace the single trajectory with a cloud of possibilities. Instead of one forecast, we run many simulations, each starting from a slightly different initial state or with slightly perturbed model parameters. These perturbations are small, but in a chaotic system, they grow quickly, leading to a spread of outcomes. If we run 1,000 such simulations and 700 produce catchment-wide rainfall above a flood-relevant threshold, we can treat that as an estimate that the probability of exceedance is about 70%. The exact equations used to generate the perturbations matter to model developers, but for a risk manager the key idea is simple: probability is estimated by counting how often an outcome occurs across the ensemble.

However, even a large ensemble generated once or twice per day is still a static object. It reflects the information available at the time the model was initialised and nothing that has happened

Figure 12 – Bayesian updating loop for weather and risk



since. Banking applications need something more dynamic. As new HRRR runs complete, as radar detects developing convective cells, as river gauges and surface stations report updated observations, the probability of different rainfall scenarios can change materially within hours.

A framework that simply regenerates a fresh ensemble from scratch at each update wastes information and offers no coherent link between “what we believed this morning” and “what we believe now.” - David Kelly, MKM

The framework developed in this book instead treats the probability distribution over weather scenarios as an object that learns over time. At any point we have a prior distribution that summarises everything known so far about the atmospheric state and its likely evolution. As new data arrive, we compare them to what the current distribution expected to see. Outcomes that were previously deemed unlikely but are now observed cause the distribution to shift—scenarios consistent with the data gain weight, while those that implied very different observations lose weight. This is the essence of Bayesian updating: prior beliefs are adjusted in proportion to how well they explain incoming evidence, producing a posterior distribution that immediately becomes the new prior for the next round of observations.

In practice, this Bayesian mechanism sits inside a continuous loop. Each update cycle ingests the latest observations, refines the distribution, and regenerates or reweights a set of ensemble trajectories. The bank never works with a stale snapshot; it always sees the current best estimate of the full range of possible weather outcomes and their probabilities. A risk desk monitoring flood exposure can therefore watch the distribution evolve through the day—seeing, for example, how the probability mass in the upper tail for a given catchment thickens as a slow-moving convective system stalls, or thins as a frontal system accelerates away. The mathematics that implement this loop are straightforward for a statistician, but a non-technical reader can simply think of it as a disciplined way of saying: “as the evidence changes, we change our minds, and we quantify exactly how much.”

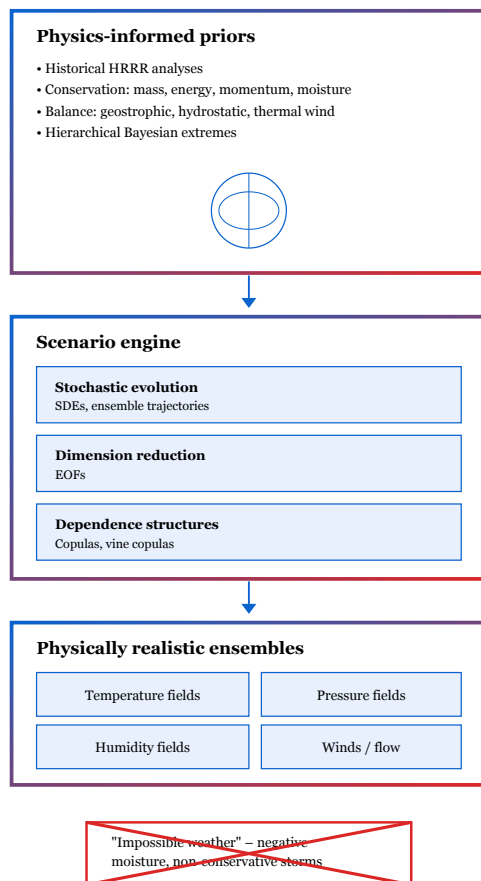
Building physically realistic weather scenarios

A probabilistic framework is only as useful as the scenarios it contains. If the ensemble includes trajectories that violate basic physics—air appearing from nowhere, storms that never obey energy constraints, rainfall intensities that cannot be supported by the available moisture—then any probabilities derived from it are misleading. For banking applications, this is not an abstract concern. Capital models, collateral valuations and hedging decisions will ultimately rest on these scenarios. They must therefore be both diverse enough to explore genuine uncertainty and constrained enough to remain meteorologically realistic.

The framework addresses this by starting from physics-informed priors rather than from purely statistical fits to past data. Historical analyses from high-resolution models such as HRRR provide kilometre-scale information on temperature, pressure, humidity and winds across many years and seasons. From these archives we infer typical patterns and dependencies, but we also encode non-negotiable constraints: mass and energy must be conserved, moisture cannot go negative, winds must be broadly consistent with pressure gradients and Earth’s rotation. The details of the governing equations are provided explicitly later in the chapter for readers who

want to see them. A non-mathematical reader can think of them as rules that stop the simulation from proposing “impossible weather.”

Figure 13 – Physics-informed scenario generator



A second challenge is that the events of greatest interest to banks—truly extreme rainfall and flooding—are by definition rare. No single catchment has centuries of high-quality observations. To overcome this, the framework uses a hierarchical Bayesian structure that pools information across space and scale. Global-scale climate constraints inform regional behaviour, and regional statistics inform local behaviour, with each level allowed to deviate where local data justify it. Extreme value theory provides the mathematical machinery for extrapolating beyond what has actually been seen, using specialised distributions for block maxima and threshold exceedances. Again, the precise forms of these distributions and their parameters are discussed later; at this stage, the key point is that the tails of the scenario distribution are shaped by both physics and carefully pooled data, not by naïve curve-fitting to a handful of storms.

Finally, the framework must reflect the fact that atmospheric variables do not move independently. Heavy rainfall is usually accompanied by particular patterns of humidity,

instability, pressure and wind; changes in one field alter the others in systematic ways. To capture this, the scenario generator uses dependency structures that go beyond simple linear correlations. In practical terms, the model learns the joint behaviour of multiple variables under different regimes—calm anticyclones, frontal passages, convective outbreaks—and samples new scenarios that preserve those relationships.

The machinery behind this relies on copulas and related multivariate techniques, but a reader can safely regard it as a way of ensuring that “when one thing looks like storm conditions, everything else does too.” The result is an ensemble of weather trajectories that are varied, physically plausible and statistically consistent with both ordinary and extreme conditions—the raw material for the banking applications developed in the remainder of the book.

From atmospheric patterns to flood-relevant rainfall

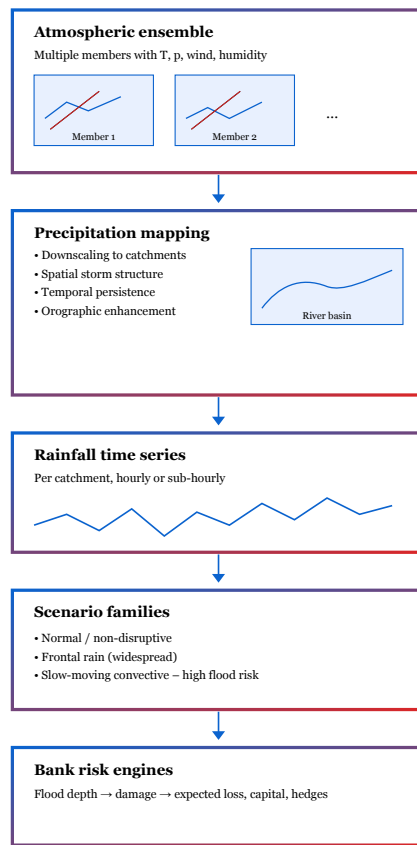
Even a well-constructed ensemble of atmospheric trajectories is not yet in the right currency for flood risk. Banks, and the hydrological models they rely on, work with rainfall over specific catchments and time windows, not with abstract fields of temperature, pressure or humidity. The next step in the framework therefore translates each physically realistic weather scenario into a precipitation pattern that has the right spatial and temporal structure to drive flood models. That means preserving how rain systems organise themselves over river basins, how long they persist, and how they evolve as fronts, convective cells or atmospheric rivers.

This translation is not a simple point-by-point conversion. A catchment floods when rainfall is intense and coherent across the upstream basin, not when a single grid cell spikes briefly. The framework therefore pays particular attention to spatial dependence: neighbouring locations in a scenario are strongly correlated when the underlying dynamics demand it, and less so when storms are more scattered. Temporal persistence is treated with the same care. A scenario that delivers 100 mm in 24 hours might do so as a single intense burst or as a long, moderate event; these have very different hydrological consequences. The scenario generator produces rainfall time series that reflect these distinctions, so that models of river flow, surface runoff and inundation see inputs that behave like real storms rather than artificial noise.

A raw ensemble can contain thousands of such rainfall scenarios, which is more than most risk teams can work with directly. To make this probabilistic information usable, the framework groups related trajectories into a small number of scenario families. One family might represent typical, non-disruptive weather; another might represent frontal systems that bring widespread but manageable rain; a third might capture slow-moving convective systems that pose high flood risk to specific catchments. Within each family, member scenarios differ in detail but share a common structure and impact profile. Probability weights from the underlying distribution carry through to these families, allowing a risk manager to say, for example, “there is currently a 15% chance that our portfolio will face a slow-moving convective pattern over the next 72 hours.” This is the form in which probabilistic weather information begins to look like a familiar risk input.

The final step is to express the rainfall associated with each scenario or scenario family as time series that can be ingested by existing banking models. For each catchment, and for each scenario, the framework outputs sequences of rainfall at the temporal resolution needed by

Figure 14 – From atmospheric ensembles to flood-relevant rainfall and risk



Each step translates raw ensemble output into actionable risk metrics.

downstream hydrological and credit-risk components—typically hourly or sub-hourly. These time series can be fed into flood depth models, damage functions, and ultimately into the expected loss and capital engines used elsewhere in the bank. Because the time series are tied back to an underlying probability distribution, the bank can move seamlessly from statements about “1-in-100 year” rainfall to statements about the current probability that a given threshold will be exceeded over the life of a loan or a trading position.

What risk managers should take away

The framework described in this chapter does not try to eliminate uncertainty about weather; it organises that uncertainty in a way that is usable for banking decisions. Instead of relying on a handful of fixed scenarios or a single deterministic forecast, the bank works with a living distribution of physically realistic weather trajectories, each carrying an explicit probability. That distribution evolves as new observations arrive, so the institution’s view of flood-related risk is never more than one update cycle out of date.

For day-to-day practice, this means that key questions can be reframed in probabilistic terms. Rather than asking “will it flood?”, portfolio and risk managers can ask “how likely is it that rainfall

over this catchment will exceed the level that causes flooding in the next week?” and “how does that likelihood change over the life of these loans or hedges?” The answers arrive not as opaque model scores but as collections of rainfall time series, grouped into meaningful scenario families and linked directly to expected loss, capital and hedging metrics. Readers who are comfortable with the mathematics can inspect the underlying equations and statistical machinery in the following chapter; those who are not can still rely on the narrative described here, which is designed so that the core ideas—living distributions, physics-informed scenarios and flood-relevant outputs—remain clear without following every algebraic detail.

Roadmap to the Technical Chapters

The conceptual framework described in this chapter is implemented through a five-phase mathematical structure detailed in Chapters 3.1 through 3.5. Each chapter addresses a specific component of the Weather Monte Carlo system, progressing from initialization through to precipitation outputs. Readers comfortable with mathematical formalism will find complete specifications of the methods, equations, and algorithms; those focused on applications can return to these chapters as references when specific implementation details are needed.

- Chapter 3.1: Initialization establishes the foundation by constructing physics-informed priors over atmospheric states. Beginning with a compact mathematical description of the atmosphere as a high-dimensional state vector, it imposes physical constraints through conservation laws (mass, energy, momentum, moisture) and balance relationships (geostrophic, hydrostatic, thermal wind). The hierarchical Bayesian treatment of extreme events ensures the framework can estimate flood probabilities even for unprecedented events beyond the historical record, addressing the fundamental challenge that 100-year events rarely appear in 50-year datasets.
- Chapter 3.2: Stochastic Evolution shows how atmospheric states evolve through time using stochastic differential equations. The drift term captures deterministic physics—advection, pressure gradients, radiative heating, and moisture transport—while the diffusion term quantifies irreducible uncertainties from incomplete observations, model approximations, and chaotic amplification. The Euler-Maruyama numerical scheme discretizes these continuous equations, and repeated integration with different initial states and random perturbations generates the ensemble of weather trajectories used throughout the framework.
- Chapter 3.3: High-Dimensional Sampling and Dependence addresses the practical challenge that atmospheric states contain millions of degrees of freedom. Naïve Monte Carlo sampling would be both computationally prohibitive and meteorologically unrealistic. The chapter explains how quasi-Monte Carlo methods (Sobol sequences) explore uncertainty spaces more efficiently than pseudo-random sampling, how dimension reduction via empirical orthogonal functions concentrates effort on the atmospheric patterns that matter most, and how copula-based dependence structures ensure that when one variable signals storm conditions, related variables do too. Vine copulas decompose the full multivariate dependence into tractable bivariate building blocks, enabling realistic covariation across temperature, pressure, humidity, and winds.

- Chapter 3.4: Bayesian Updating formalizes the continuous learning mechanism that transforms static probabilistic forecasts into a living distribution. As new observations arrive from weather stations, radar, satellites, and numerical weather prediction analyses, Bayes' theorem adjusts the probability distribution: scenarios consistent with the data gain weight, while inconsistent scenarios lose weight. The Ensemble Kalman Filter provides a computationally efficient approximation to the full Bayesian update, making the approach tractable for high-dimensional atmospheric states. The chapter specifies observation operators that map between model variables and sensor readings, likelihood functions that account for measurement errors, and the temporal propagation between updates that maintains physical consistency as uncertainty grows between observation times.
- Chapter 3.5: Precipitation Mapping translates atmospheric ensemble members into the rainfall time series needed by hydrological and credit-risk models. Catchment-scale precipitation depends not just on local atmospheric conditions but on spatial correlation structures across river basins, temporal persistence of storm systems, and orographic enhancement where topography amplifies rainfall. The chapter details how atmospheric fields are downscaled to catchment grids, how spatial and temporal covariance structures preserve realistic storm characteristics, and how intensity-duration-frequency relationships connect continuous ensemble trajectories to the discrete flood thresholds used in banking risk models. Pattern family identification groups thousands of ensemble members into manageable scenario families with common impact profiles, enabling risk teams to work with statements like "15% probability of slow-moving convective pattern" rather than thousands of individual trajectories.

Together, these five chapters provide a complete specification of the Weather Monte Carlo framework. Each builds on the previous, progressing from the physical foundation through numerical methods to practical banking outputs. Readers can engage with them at different levels: as detailed implementation guides for model developers, as validation references for model risk and supervisory reviewers, or as conceptual anchors for understanding how the non-technical descriptions in this chapter translate into executable code.

Mathematical foundations of Weather Monte-Carlo

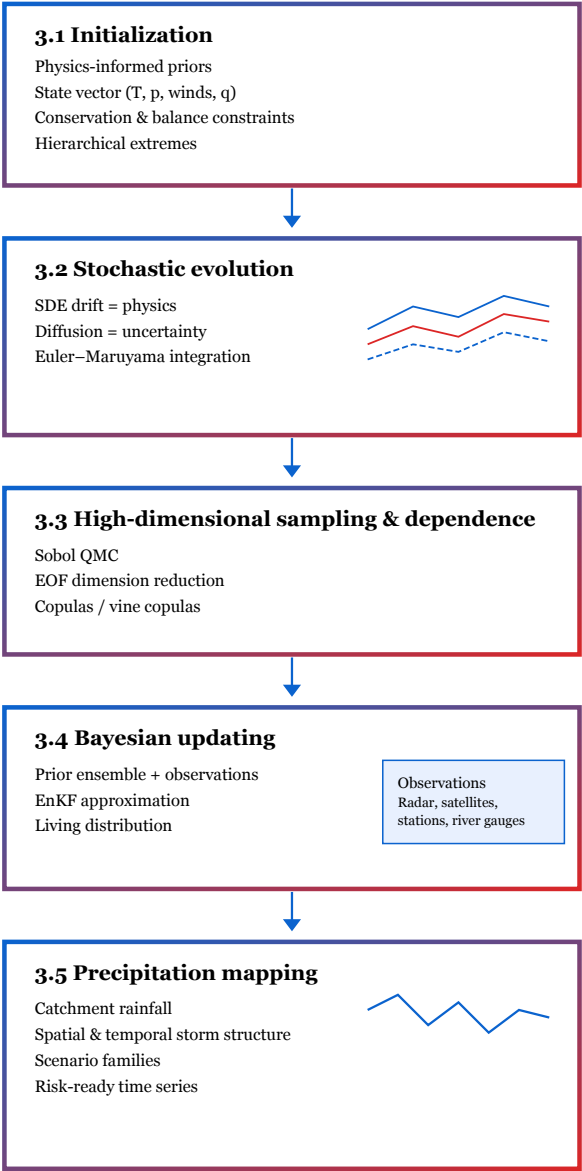
This chapter has described the framework in the language of risk management: living probability distributions, physically realistic scenarios, and rainfall time series that can drive flood and credit-risk models. The next chapter develops the same ideas in mathematical form. It is written for readers who need to see the equations behind the narrative—quantitative developers, model-risk teams and technically minded supervisors. A non-mathematician can safely skim or skip that chapter; all of the core concepts already appear in words and diagrams here.

- The first part of The following chapter defines the atmospheric state vector and the physics-informed priors that underpin every scenario. It introduces the notation for temperature, pressure, winds and humidity on a grid, and specifies a prior probability distribution for this high-dimensional state. Conservation and balance laws—mass, energy, momentum and moisture, together with geostrophic and hydrostatic balance—appear

explicitly in equation form. The intent is not to add new concepts, but to show exactly how the physical realism described in this chapter is enforced mathematically.

- The second part sets out the stochastic evolution equation that generates ensembles. It writes the atmospheric dynamics as a stochastic differential equation with a drift term (the deterministic physics) and a diffusion term (the uncertainty around that physics). Numerical schemes such as Euler–Maruyama are introduced to show how discrete time steps produce individual trajectories, and how repeating the procedure with different random inputs gives the “cloud of possibilities” we have already discussed. Readers who are comfortable with calculus will recognise familiar structures; others can treat this section as a detailed justification for why the ensemble really does reflect both our knowledge and our ignorance.
- The third part explains how the framework handles dependence and high dimensionality. It formalises the use of quasi-Monte Carlo sampling, dimension reduction and copulas to generate scenarios in which atmospheric variables move together in realistic ways. The chapter shows how low-discrepancy sequences, empirical orthogonal functions, and vine copulas are combined to efficiently cover the relevant state space while preserving observed relationships among temperature, pressure, humidity, winds, and precipitation. This is the mathematical counterpart of our earlier discussion about ensuring that when one feature of the system looks storm-like, the rest of the system does too.
- The final part of the following chapter presents Bayesian updating and its translation to rainfall in full detail. It writes down the update rules that adjust probability distributions as new observations arrive, clarifying how prior beliefs and likelihoods combine to form posteriors that become the next prior. It also formalises the mapping from atmospheric fields to catchment-scale precipitation, the construction of rainfall time series for each scenario, and the empirical estimators used to derive exceedance probabilities and other risk metrics from the ensemble. These sections do not introduce new ideas beyond those in this chapter; they simply provide the algebra that sits behind the concepts. Readers interested mainly in practical implications can rely on the explanations already given here and treat the following chapter as a reference, dipping into particular subsections only when they want to see exactly how a piece of the machinery is implemented.

Figure 15 – Weather Monte Carlo framework:
Chapters 3.1–3.5



Chapter blocks 3.1–3.5 in sequence

Chapter 3.1: Weather - Initialization

This chapter establishes the mathematical foundation for the Weather Monte Carlo framework by constructing physics-informed priors over atmospheric states. Every scenario generated in the framework begins life as a sample from this prior distribution, making its specification critical to all subsequent results. The challenge is to encode both our knowledge and our uncertainty: we know that the atmosphere must obey conservation laws and balance relationships, yet we remain uncertain about the exact state at any given time and location. We know that extreme precipitation events can occur, yet we have limited historical observations of the most consequential extremes.

The chapter addresses these challenges through three interconnected components. First, we define the atmospheric state vector—temperature, pressure, winds, humidity, and related fields on a numerical grid—and specify a multivariate prior distribution that captures both typical conditions and their spatial and temporal covariation. Second, we impose physical constraints through conservation laws for mass, energy, momentum, and moisture, along with balance approximations that prevent the prior from generating meteorologically impossible states. Third, we employ a hierarchical Bayesian structure to handle extreme events, pooling information across global, regional, and local scales to estimate tail probabilities even when local observations are sparse.

The resulting physics-informed prior $p(X_0)$ provides a coherent starting point for ensemble generation, dependence modeling, and Bayesian updating developed in subsequent chapters. Readers will find explicit mathematical specifications of all distributions, constraints, and parameters; those focused on conceptual understanding can rely on the narrative descriptions while treating the equations as formal documentation of the methodology.

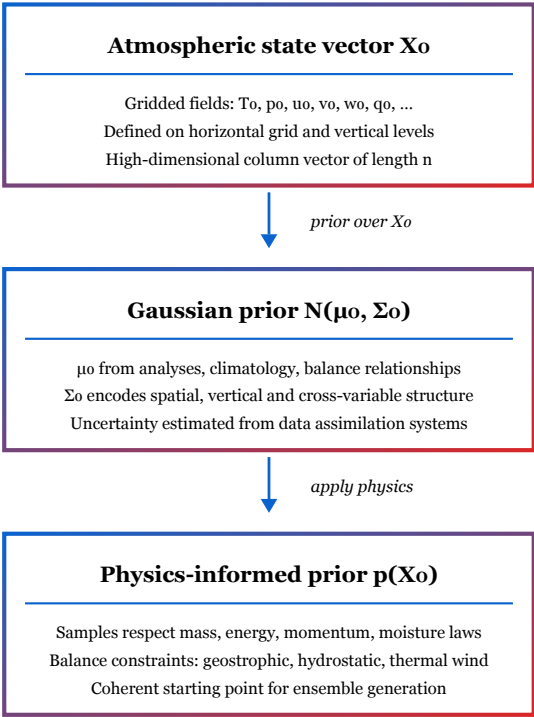
State vector and physics-informed priors

The starting point for the mathematical framework is a compact description of the atmosphere as a state vector. At any given time t , we collect the key prognostic variables—temperature, pressure, wind components, vertical velocity, humidity and any additional fields—into a single high-dimensional object. Each element is itself a field defined on the numerical grid (horizontal locations and vertical levels). For convenience, we often treat $\mathbf{x}$ as a column vector of length N , where N can easily reach millions once all grid points and variables are included.

We then place a probability distribution over this initial state. The simplest and most tractable choice is a multivariate normal prior, with mean vector $\bar{\mathbf{x}}$ and covariance matrix Σ . The mean encodes our best estimate of the current atmosphere, typically derived from recent numerical weather prediction analyses and climatological information where observations are sparse. The covariance encodes both uncertainty and structure: it captures how errors co-vary across space, between variables, and between levels. Nearby grid points are strongly correlated, vertical levels are linked, and cross-correlations between variables reflect known relationships—for example, between temperature and pressure, or between humidity and temperature.

Crucially, this prior is not an arbitrary statistical construct. It is constrained by the governing physical laws. At a minimum, the atmospheric states drawn from it must be consistent with conservation of mass, energy, momentum and moisture, and with standard balance approximations. In differential form these constraints appear as the continuity equation for mass, the first law of thermodynamics for energy, the momentum equations (including Coriolis, pressure gradient, gravity and friction terms), and a moisture conservation equation that links humidity to sources, sinks and transport. Balance conditions such as geostrophic and hydrostatic balance impose additional relationships between pressure, winds and vertical structure. In practice, these laws enter the construction of and through dynamical balance operators and through filters that rule out states violating basic constraints.

Figure 16 – Atmospheric state vector and Gaussian prior



The prior must also represent extreme events realistically, despite the limited length of observational records at any single location. To achieve this, we use a hierarchical Bayesian structure in which parameters controlling tail behaviour are linked across spatial scales. At the top level, global-scale parameters governing, for example, maximum precipitable water and large-scale circulation impose broad bounds on how heavy the tails can be. Regional parameters then vary around these global constraints to reflect climate differences between, say, maritime and continental regimes. Local parameters for individual catchments vary around their regional parents, allowing for orographic enhancement, urban effects and other site-specific features. Extreme value theory provides the parametric forms—typically Generalised Extreme Value

distributions for block maxima and Generalised Pareto distributions for threshold exceedances—while the hierarchy determines how their parameters are shared and updated.

The combination of the high-dimensional Gaussian prior for the state vector, the physical constraints that shape its mean and covariance, and the hierarchical treatment of extremes gives us a physics-informed prior. Every scenario generated later in the framework begins life as a sample from this distribution. Because it encodes both ordinary weather and plausible extremes, and because it enforces fundamental conservation and balance relationships, it provides a coherent foundation for the stochastic evolution, dependence modelling and Bayesian updating developed in the following sections.

Atmospheric State Variables

The state vector X_0 at initial time comprises spatially distributed fields:

$$X_0 = [T_0, p_0, u_0, v_0, w_0, q_0, \dots]$$

Where:

- T_0 represents temperature
- p_0 pressure
- u_0 and v_0 horizontal wind components
- w_0 vertical velocity
- q_0 specific humidity

Each variable receives a prior distribution, typically multivariate normal for computational tractability:

$$X_0 \sim N(\mu_0, \Sigma_0)$$

The mean vector μ_0 derives from:

- Recent numerical weather prediction analyses (providing current atmospheric state estimates)
- Seasonal climatological normals (for components without recent observations)
- Dynamical balance relationships (ensuring geostrophic balance between wind and pressure)
- The covariance matrix Σ_0 encodes:
 - Spatial correlation structures (nearby locations have correlated atmospheric states)
 - Variable cross-correlations (temperature and pressure exhibit negative correlation in many regimes)
 - Vertical correlation structures (adjacent atmospheric levels strongly correlated)
- Uncertainty estimates from data assimilation systems

Mass Conservation

The continuity equation ensures that atmospheric mass neither appears nor disappears arbitrarily. For an atmospheric column, this manifests as:

$$\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{v}) = 0$$

Where:

- ρ represents air density
- $\mathbf{v}$ the wind velocity vector.

Prior distributions must produce atmospheric states that satisfy this constraint, preventing physically impossible mass accumulations or depletions.

Energy Balance

Thermodynamic consistency requires that energy transformations follow the first law of thermodynamics. For atmospheric parcels:

$$c_p(dT/dt) = Q - (RT/p)(dp/dt)$$

Where:

- c_p denotes specific heat T temperature
- Q heating rate
- R gas constant
- p pressure.

Priors ensure that temperature changes relate appropriately to diabatic heating and adiabatic compression/expansion.

Momentum Conservation

Wind evolution follows Newton's second law applied to atmospheric motion:

$$d\mathbf{v}/dt = -2\boldsymbol{\Omega} \times \mathbf{v} - (1/\rho)\nabla p + \mathbf{g} + \mathbf{F}$$

Where:

- Coriolis force ($2\boldsymbol{\Omega} \times \mathbf{v}$)
- pressure gradient force ($-(1/\rho)\nabla p$)
- gravity ($\mathbf{g}$)
- friction ($\mathbf{F}$).

Prior distributions for wind fields must produce states consistent with observed pressure distributions and Earth's rotation.

Moisture Conservation

Water substance (vapour, liquid, ice) obeys conservation principles with phase transformations:

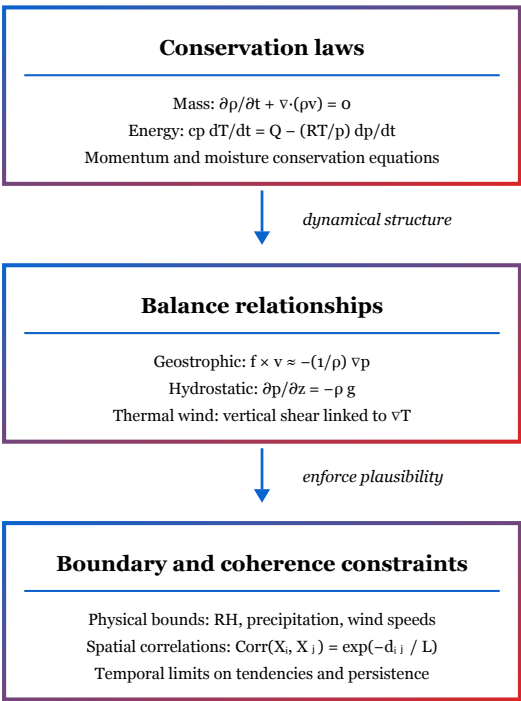
$$dq/dt = E - P + \nabla \cdot (\text{moisture flux})$$

Where:

- q represents specific humidity
- E evaporation
- P precipitation.

Priors link humidity distributions to temperature (via the Clausius-Clapeyron relationship) and ensure realistic moisture transport patterns.

Figure 17 – Conservation, balance and boundary constraints



Hierarchical treatment of extremes

At the global level, parameters governing tail behaviour—for example, those controlling maximum precipitable water and large-scale circulation—set broad physical limits. Regional parameters are modelled as variations around these global values, capturing differences between climates such as maritime west coasts and continental interiors. Local parameters for individual catchments are then modelled as variations around their regional parents, allowing for orographic enhancement, urbanisation and other site-specific effects.

The resulting physics-informed prior $p(X_0)$ combines all these elements: a multivariate distribution over the atmospheric state vector with mean and covariance shaped by conservation and balance laws, and tail behaviour structured hierarchically across space. Every scenario generated later in the framework begins life as a sample from this distribution.

Boundary Conditions

Beyond incorporating physics through parameter choices, the initialisation phase structures priors to enforce physical plausibility:

Certain atmospheric quantities possess physical bounds:

- Relative humidity: $0\% \leq RH \leq 100\%$ (with supersaturation possible in cloud-free conditions)
- Precipitation rates: $R \geq 0$ mm/hr (cannot have negative rainfall)
- Wind speeds: typically bounded by observed climatological maxima for the region

Priors employ truncated distributions or transformations ensuring samples respect these bounds. For example, relative humidity might use a beta distribution naturally confined to the zero-to-one interval, whilst precipitation employs gamma or mixed distributions with support on positive reals.

Balance Relationships

The atmosphere exhibits approximate balance between forces in many conditions:

Geostrophic Balance

Away from the equator and outside strong convection, horizontal wind approximates geostrophic flow:

$$f \times v \approx -(1/\rho) \nabla p$$

Where:

- f denotes Coriolis parameter

Priors constrain wind and pressure field relationships to satisfy this balance approximately, preventing initialisation with meteorologically unrealistic states where strong winds exist without corresponding pressure gradients.

Hydrostatic Balance

Vertical pressure variation follows:

$$\partial p / \partial z = -\rho g$$

Priors ensure vertical pressure profiles consistent with temperature distributions (via the ideal gas law), preventing states with impossible pressure-temperature-height relationships.

Thermal Wind Balance

Vertical wind shear relates to horizontal temperature gradients:

$$\partial v / \partial z = (g / f T) k \times \nabla T$$

Where:

- k represents the vertical unit vector. Initialisation maintains consistency between three-dimensional temperature and wind structures.

Spatial and Temporal Coherence

Atmospheric states exhibit coherence across space and time that priors must capture:

Spatial Correlation Functions

The prior covariance matrix Σ_0 employs correlation functions that decay with distance:

$$\text{Corr}(X_i, X_j) = \exp(-d_{ij} / L)$$

where d_{ij} represents distance between locations i and j , and L the correlation length scale (typically 50-200 kilometres for synoptic features, smaller for convective processes).

Temporal Evolution Constraints

Priors restrict how rapidly atmospheric states can change:

Tendency Limits

Rates of temperature, pressure, and wind change are bounded by physical processes:

Temperature: typically $|dT/dt| < 10^\circ\text{C}/\text{hour}$ except in extreme frontal passages

Pressure: synoptic changes $|dp/dt| < 5 \text{ hPa}/\text{hour}$ except in explosive cyclogenesis

Wind: acceleration constrained by pressure gradient force magnitude

Persistence

Current conditions inform expectations for near-future states through autocorrelation:

$$E[X(t+\Delta t) | X(t)] \approx X(t) + a(X,t)\Delta t$$

For small time increments, atmospheric state changes slowly, providing strong constraints on initial forecast distributions.

Predicting Extreme Events

Predicting extreme precipitation events confronts a profound data scarcity challenge. By definition, extreme events occur rarely—a 100-year rainfall event has, on average, only a 1% chance of occurring in any given year. For a location with 50 years of observational records, the probability of having observed even one such event is less than 40%. For a 500-year event, the chances drop to approximately 10% with 50 years of data.

This creates an impossible situation: the events we most need to predict (the rare, consequential extremes that drive flood risk) are precisely those for which we have the least empirical evidence. Traditional statistical approaches founder on this reality—how can you estimate the tail of a distribution when you've never observed values in that tail?

The hierarchical Bayesian approach solves this through borrowing strength: leveraging information from related contexts where data may be more abundant, whilst respecting the unique characteristics of the specific location of interest.

Extreme Event Detection

Pattern families with elevated flood risk receive special attention through multiple detection mechanisms:

- **Historical Analogue Matching:** Compare current patterns against historical flood-producing events. High similarity indicates current patterns resemble past flood configurations.
- **Compound Extreme Detection:** Identify patterns combining multiple risk factors—high precipitation, saturated soil, unfavourable timing, spatial concentration.
- **Precursor Recognition:** A library of atmospheric states that preceded significant floods by 12-48 hours enables early warning. Current ensemble members matching precursor patterns trigger alerts even before extreme conditions develop.
- **Trajectory Divergence Analysis:** Ensemble members starting similar but diverging toward extreme outcomes mark "butterfly effect" regions requiring careful monitoring.

The Architecture of Knowledge Pooling

The hierarchical structure recognizes that atmospheric processes operate simultaneously at multiple spatial scales, with information flowing between them. A precipitation event at a specific location results from the interaction of:

- Global-scale dynamics: Planetary circulation patterns, jet stream configurations, atmospheric thermodynamics
- Regional-scale patterns: Synoptic weather systems, orographic effects, maritime influences
- Local-scale features: Topographic details, land surface characteristics, urban modifications

Rather than treating these scales as independent, the hierarchical framework explicitly models their dependencies, creating a structure where information cascades from broader to narrower scales whilst allowing local observations to refine understanding at all levels.

The joint probability distribution across all hierarchical levels is expressed as:

$$p(\theta_{\text{global}}, \theta_{\text{regional}}, \theta_{\text{local}}, y) = p(y | \theta_{\text{local}}) \times p(\theta_{\text{local}} | \theta_{\text{regional}}) \times p(\theta_{\text{regional}} | \theta_{\text{global}}) \times p(\theta_{\text{global}})$$

Where:

- y represents observed precipitation data
- θ_{global} represents global-scale physical parameters
- θ_{regional} represents regional-scale meteorological parameters
- θ_{local} represents location-specific parameters

This factorisation explicitly shows how information flows through the hierarchy: global parameters constrain regional parameters, which in turn constrain local parameters, while observations at the local level influence all upstream levels through Bayesian updating.

Extreme Value Theory Integration

The hierarchical structure proves particularly powerful when combined with extreme value theory, which provides the mathematical machinery for extrapolating beyond observed data to estimate probabilities of unprecedented events. At each hierarchical level, the framework employs extreme value distributions to characterise tail behaviour, with parameters at each level informed by the level above.

Generalised Extreme Value (GEV) Distribution

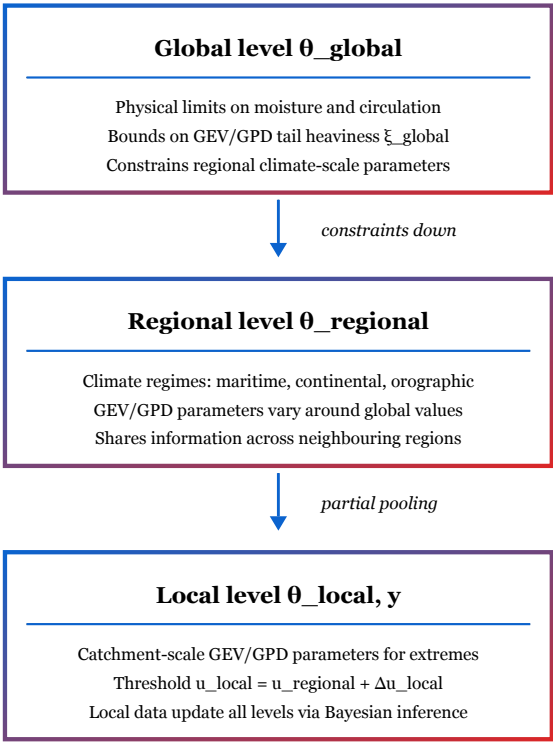
For block maxima (e.g., annual maximum precipitation), the GEV distribution characterizes the upper tail:

$$p(x) = (1/\sigma) \exp[-(1 + \xi((x-\mu)/\sigma))^{(-1/\xi)}] \times (1 + \xi((x-\mu)/\sigma))^{(-1/\xi - 1)}$$

Where:

- μ (location parameter): determines the central position of the distribution
- σ (scale parameter): controls the spread of extreme values
- ξ (shape parameter): governs tail heaviness ($\xi > 0$ indicates heavy tail, $\xi < 0$ bounded tail, $\xi = 0$

Figure 18 – Hierarchical Bayesian structure for extremes



corresponds to Gumbel distribution)

Within the hierarchical framework, these parameters cascade across scales:

- **Global Level:** Physical constraints establish bounds on ξ_{global} (atmospheric moisture limits prevent infinitely heavy tails)

- Regional Level: $\xi_{\text{regional}} \sim N(\xi_{\text{global}}, \tau^2_{\text{regional}})$, allowing regional climate characteristics to create variations around global expectations
- Local Level: $\xi_{\text{local}} \sim N(\xi_{\text{regional}}, \tau^2_{\text{local}})$, with local observations refining the shape parameter for specific locations

Generalised Pareto Distribution (GPD)

For threshold exceedances, the GPD provides tail estimates beyond observed maxima:

$$p(x) = (1/\sigma)(1 + \xi(x-u)/\sigma)^{-(1/\xi+1)}$$

for exceedances above threshold u , where σ and ξ have analogous interpretations to the GEV parameters.

The hierarchical structure determines both the threshold u and the GPD parameters:

$$u_{\text{local}} = u_{\text{regional}} + \Delta u_{\text{local}}$$

Where:

- u_{regional} derives from regional precipitation climatology
- Δu_{local} accounts for site-specific factors (orographic enhancement, urban effects).

Similarly:

$$\sigma_{\text{local}} \sim \text{Gamma}(\alpha_{\text{regional}}, \beta_{\text{regional}}) \quad \xi_{\text{local}} \sim N(\xi_{\text{regional}}, \tau^2_{\text{local}})$$

These distributions provide principled extrapolation beyond historical observations, enabling probability estimates for events more extreme than anything in operational archives. The hierarchical structure ensures that even locations with no observed extremes receive meaningful tail estimates by borrowing strength from regional and global understanding, whilst locations with abundant extreme event data can deviate from regional patterns when evidence supports it.

Validation Against Historical Events

Before proceeding to Monte Carlo generation, initialised priors undergo validation to ensure the hierarchical structure produces realistic atmospheric states:

Sample atmospheric states from priors at all hierarchical levels:

$$X_{\text{sample}} \sim p(X | \theta_{\text{global}}, \theta_{\text{regional}}, \theta_{\text{local}})$$

Verify samples satisfy conservation laws and balance relationships established in Phase 1, ensuring hierarchical pooling hasn't introduced physical inconsistencies.

Confirm extreme value tail behaviour matches historical event climatology by comparing:

Empirical quantiles:

$Q_{\text{emp}}(p)$ from historical data Prior quantiles:

$Q_{\text{prior}}(p) = F^{(-1)}(p | \theta_{\text{GEV}})$ from hierarchical GEV

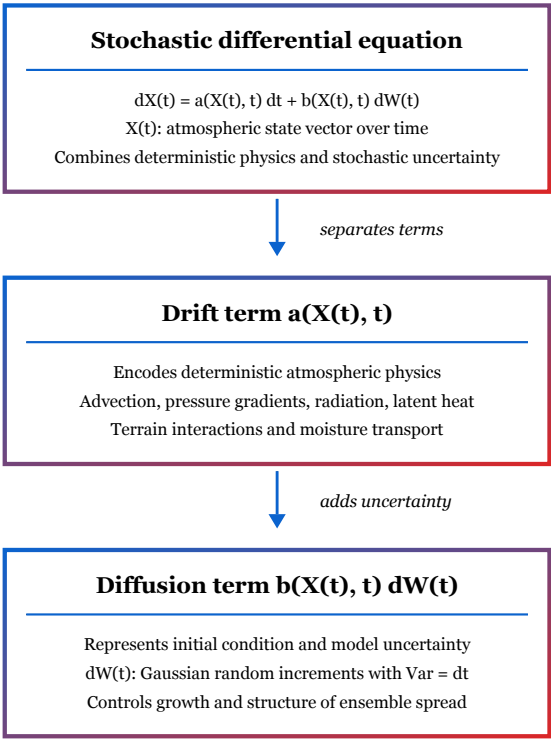
This validation prevents pathological initial conditions that might compromise subsequent phases, while confirming that the hierarchical structure successfully pools information without violating physical constraints or meteorological realism.

Chapter 3.2: Weather - Stochastic Evolution

Once the initial atmospheric state has been described by the physics-informed prior developed in Chapter 3.1, we require a mechanism to evolve that state forward in time while accounting for both deterministic physical laws and irreducible uncertainties. This chapter specifies the stochastic differential equation (SDE) that governs atmospheric evolution and shows how numerical integration of this equation generates the ensemble of weather trajectories used throughout the framework.

The SDE combines two essential components: a drift term capturing deterministic atmospheric physics (advection, pressure gradients, radiative transfer, latent heat release, terrain interactions), and a diffusion term capturing everything we cannot resolve—initial condition errors, model approximations, and chaotic amplification of tiny perturbations. The drift ensures that, in the absence of uncertainty, the system behaves like a realistic numerical weather prediction model. The diffusion ensures that uncertainty grows in a meteorologically plausible manner, reflecting known limits on atmospheric predictability.

Figure 19 – Atmospheric SDE: drift and diffusion



The Euler-Maruyama numerical scheme discretises the continuous SDE into time steps that can be implemented computationally. Each integration from the prior distribution produces one possible atmospheric trajectory. Repeating this process many times, with different initial draws or different random perturbations, builds the ensemble that represents our “cloud of possibilities.” This chapter provides the complete mathematical specification of drift and diffusion terms, their parameterisation from historical data and physical principles, and the numerical algorithms that transform these abstract equations into executable forecasts.

Stochastic Differential Equations: The Mathematical Foundation

The ensemble generation rests upon the stochastic differential equation specified during initialisation:

$$dX(t) = a(X(t),t)dt + b(X(t),t)dW(t)$$

This equation governs how each atmospheric state vector $X(t)$ evolves through time, combining deterministic physics with stochastic uncertainty.

The Drift Term: Deterministic Physics

The drift term $a(X(t),t)$ encodes the deterministic physical laws governing atmospheric behaviour:

- **Thermal Dynamics:** Temperature gradients drive heat transfer through advection and diffusion. Warm air rises, cold air sinks, creating vertical circulations. Solar heating warms the surface, terrestrial radiation cools the atmosphere, and latent heat release from condensation provides energy to storm systems.
- **Pressure Gradient Forces:** Spatial variations in atmospheric pressure generate forces that accelerate air parcels. High pressure regions export air mass whilst low pressure regions import it, creating the wind patterns that transport heat, moisture, and momentum across the globe.
- **Moisture Transport:** Water vapour moves with the wind field, evaporates from surfaces, condenses in rising air, and precipitates when saturation occurs. The moisture distribution fundamentally determines where and when precipitation develops.
- **Terrain Interactions:** Topography forces air to rise over mountains (producing orographic precipitation), channels flow through valleys, and creates local circulations through differential heating of slopes. These terrain effects prove crucial for flood-prone catchments where orographic enhancement amplifies precipitation.

Advection and Diffusion.

Advection is the transport of a substance or property (like heat, pollutants, or momentum) by the bulk motion of a fluid. It's essentially "transport by flow" — the substance is carried along by the moving fluid itself, like leaves floating down a river. Quick comparison:

Advection:

- Driven by bulk fluid motion (organized flow)
- Directional transport following the flow field
- Can move substances against concentration gradients
- Typically much faster than diffusion
- Described by velocity field: $\partial C / \partial t = -\mathbf{v} \cdot \nabla C$

Diffusion:

- Driven by random molecular motion (Brownian motion)
- Non-directional, spreading from high to low concentration
- Always moves down concentration gradients
- Typically slower, especially over large distances
- Described by Fick's laws: $\partial C / \partial t = D \nabla^2 C$

Combined Effects

In real systems, both often occur simultaneously, described by the advection-diffusion equation:

$$\partial C / \partial t = -\mathbf{v} \cdot \nabla C + D \nabla^2 C$$

The relative importance is characterised by the Péclet number ($Pe = vL/D$), where high Pe means advection dominates.

The drift term incorporates these processes through the dynamical core equations validated during Phase 1 initialisation.

For temperature evolution, this includes:

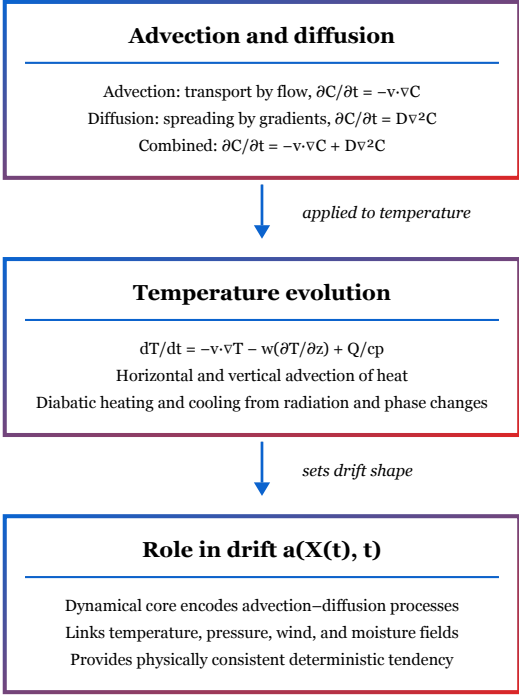
$$dT/dt = -\mathbf{v} \cdot \nabla T - w(\partial T / \partial z) + Q/c_p$$

Where:

- $\mathbf{v} \cdot \nabla T$ represents horizontal advection
- $w(\partial T / \partial z)$ vertical advection
- Q/c_p diabatic heating.

Similar equations govern pressure, wind, and moisture evolution.

Figure 20 – Advection–diffusion and temperature evolution



The Diffusion Term: Quantifying Uncertainty

The diffusion term $b(\mathbf{X}(t), t)dW(t)$ acknowledges that atmospheric prediction contains irreducible uncertainties:

- **Initial Condition Uncertainty:** Despite sophisticated data assimilation, the true atmospheric state remains incompletely known. Observations contain measurement errors, spatial coverage proves incomplete (particularly over oceans and remote regions), and retrieval algorithms for satellite data introduce additional uncertainties.

- **Model Error:** Numerical weather prediction models approximate reality through discretisation, parameterisation, and simplification. Sub-grid-scale processes cannot be explicitly resolved, physical parameterisations contain empirical relationships rather than exact physics, and numerical schemes introduce truncation errors.
- **Chaotic Amplification:** Even if initial conditions and model physics were perfect, atmospheric chaos means that infinitesimal perturbations grow exponentially. This fundamental limit on predictability manifests through the diffusion term's structure.

The Wiener process

$dW(t)$ represents random fluctuations with specific statistical properties:

Zero mean: $E[dW(t)] = 0$

Variance proportional to time increment: $\text{Var}[dW(t)] = dt$

Independent increments: $dW(t_1)$ independent of $dW(t_2)$ for $t_1 \neq t_2$

Gaussian distribution: $dW(t) \sim N(0, dt)$

The matrix $b(X(t), t)$ scales and correlates these random fluctuations according to the uncertainty structure established during initialisation.

The diffusion grows with forecast lead time, reflecting the expanding envelope of possible atmospheric states as predictability decreases.

Numerical integration and ensemble generation: Euler-Maruyama Scheme

Stochastic differential equations (SDEs) rarely admit closed-form analytical solutions, necessitating numerical approximation methods for practical applications. The Euler-Maruyama scheme represents the stochastic analogue of the classical Euler method for ordinary differential equations, extending the deterministic framework to accommodate random fluctuations. This method provides a straightforward yet powerful approach to simulate sample paths of SDEs by approximating both the deterministic drift and stochastic diffusion components. The continuous-time SDE requires discretisation for computational implementation.

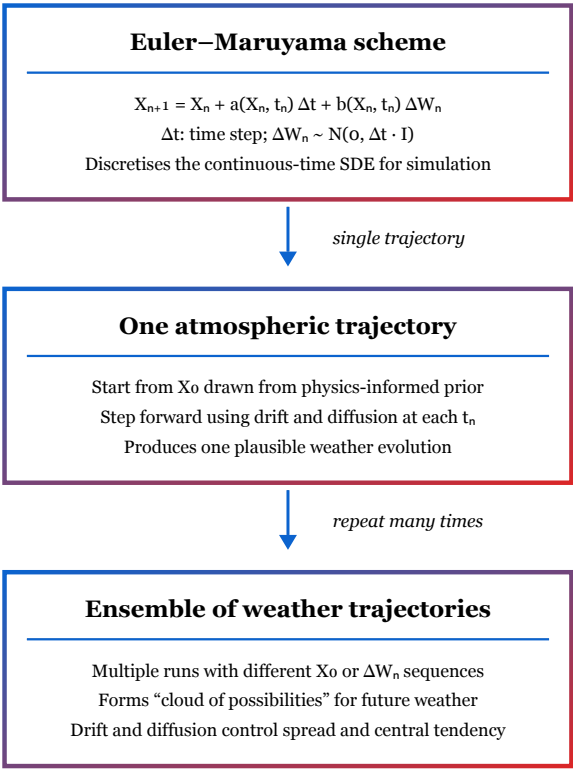
The Euler-Maruyama scheme provides a straightforward approach:

$$X_{n+1} = X_n + a(X_n, t_n)\Delta t + b(X_n, t_n)\Delta W_n$$

Where:

- X_n represents the atmospheric state at time t_n
- Δt denotes time step
- $\Delta W_n \sim N(0, \Delta t \cdot I)$ represents random increments

Figure 21 – Euler–Maruyama ensemble generation



where ΔW_n is a vector of random increments with the appropriate variance and correlation structure. Higher-order schemes, such as Milstein’s method, can be used in strongly nonlinear regimes, but the basic logic is the same: at each time step, we apply the deterministic drift and add a stochastic perturbation drawn from the diffusion structure.

A single integration of this discrete equation, starting from an initial draw X_0 from the prior, produces one possible atmospheric trajectory. To build an ensemble, we repeat the integration many times, each time using either a different initial state, a different sequence of random increments, or both. The collection of resulting trajectories forms the set of scenarios that, at the conceptual level, were described earlier as the “cloud of possibilities”. The values attached to the drift and diffusion parameters, therefore, control not only the central tendency of these trajectories but also the breadth and shape of the uncertainty around them.

For a reader mainly interested in the banking application, the key message is that the SDE and its parameters give a disciplined way to generate many plausible future weather evolutions from the physics and uncertainties encoded in the prior. The following sections add the machinery needed to sample efficiently in such a high-dimensional space and to keep the dependencies between variables realistic.

Chapter 3.3 : Weather - Sampling and Dependence

The atmospheric state vector contains an enormous number of degrees of freedom: temperatures, winds, humidities and pressures at every grid point and level. Sampling directly in this full space with naïve Monte Carlo methods would be both inefficient and misleading. It would waste many samples on redundant variations, and it would struggle to reproduce the tight dependencies between variables and locations that characterise real weather. This section describes how the framework addresses these issues by combining quasi-Monte Carlo sampling, dimension reduction and copula-based dependence modelling.

The first step is to design the sampling scheme itself. Rather than drawing independent pseudo-random points, the framework uses low-discrepancy sequences such as Sobol sequences to explore the space of uncertainties more evenly. These sequences fill the unit hypercube in a way that avoids clumping, which improves convergence when mapping them through the prior distributions for the atmospheric state. Each Sobol point is transformed, via the relevant cumulative distribution functions and covariance structure, into a perturbation of the initial state or of key parameters. This gives a set of candidate states that reflect the marginal distributions specified in the prior, but it does not yet ensure that all variables co-move in a meteorologically realistic way.

To make the sampling practical in such a large space, we then work in a reduced representation. Historical analyses are decomposed into dominant modes of variability—often via empirical orthogonal functions or related techniques—so that most of the variance in the atmospheric fields is captured by a relatively small number of coefficients. The Sobol-based perturbations are applied in this reduced space, and the resulting states are projected back to the full grid while enforcing dynamical balance constraints. The effect is to concentrate sampling effort on the patterns that matter most for weather evolution, rather than on arbitrary grid-point combinations that the atmosphere rarely occupies.

Even in this reduced space, however, the variables are not independent. Heavy rainfall tends to occur with certain patterns of humidity, instability, winds and pressure; large-scale anomalies in one field are mirrored in others. To capture these relationships, the framework uses copula-based dependence structures. Copulas separate the marginal distributions of individual variables from the way they are linked, allowing us to choose appropriate marginals (for example, normal for some variables, skewed or heavy-tailed for others) while modelling their joint behaviour explicitly. A dedicated subsection on copula parameters follows here, specifying how copula families are chosen, how their parameters are estimated from historical data, and how they are conditioned on weather regimes so that dependence strengthens or weakens as the atmosphere shifts between patterns.

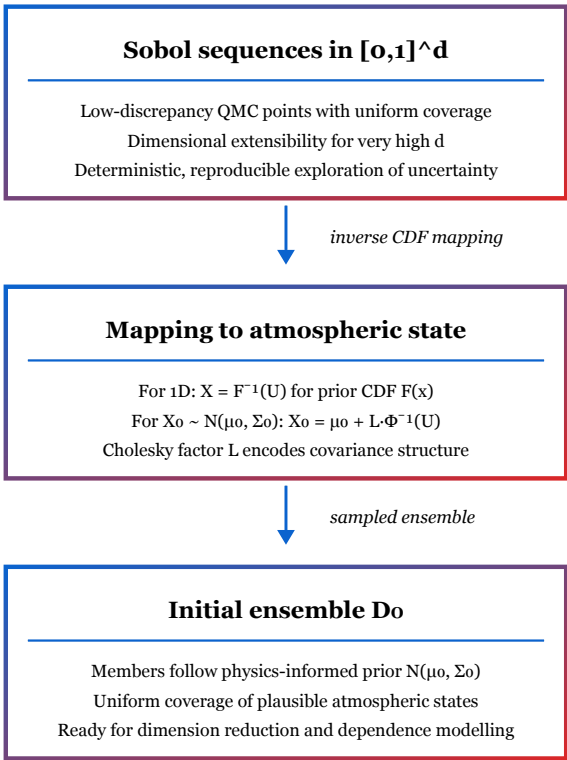
For high-dimensional settings, simple multivariate copulas are not flexible enough, so the framework employs vine constructions that assemble the full dependence structure from bivariate building blocks. This allows different pairs of variables, or different levels and regions, to use

different copula families and parameter values while still forming a coherent whole. Sampling proceeds by drawing uniform variables from the Sobol sequence, transforming them through the vine structure into correlated uniforms and then mapping these into physical units through the chosen marginal distributions. The result is an ensemble of atmospheric states in which the main modes of variability are well covered, the key dependencies between variables are respected, and the marginal behaviour of each quantity matches both physical expectations and empirical evidence.

With the initial Monte Carlo ensemble D_0 generated, the framework now enters its most distinctive feature: the continuous Bayesian update cycle that transforms static probabilistic forecasts into a living, evolving distribution. As new atmospheric observations arrive hourly from ground sensors, satellites, radar systems, and numerical weather prediction analyses, the framework systematically refines its understanding of current conditions and future evolution pathways.

This phase implements the heart of the MKM framework's adaptive capability—the mechanism through which yesterday's posterior distribution becomes today's prior, enabling continuous learning without information loss while maintaining rigorous probabilistic foundations.

Figure 22 – Quasi-Monte Carlo sampling of atmospheric priors



Quasi-Monte Carlo Sampling

Traditional Monte Carlo methods using pseudo-random numbers can leave significant coverage gaps in high-dimensional spaces. Two trajectories might explore very similar regions whilst large portions of state space remain unsampled. Such inefficient sampling risks missing critical pathways of weather pattern evolution in atmospheric systems with millions of degrees of freedom.

The framework addresses this through quasi-Monte Carlo (QMC) methods that employ low-discrepancy sequences, achieving more uniform coverage than pseudo-random sampling.

Sobol Sequences

Sobol sequences provide deterministic points in high-dimensional unit hypercubes with superior uniformity properties. Unlike pseudo-random sequences, where coverage improves slowly (proportional to $N^{(-1/2)}$ for N samples), Sobol sequences achieve convergence rates approaching $N^{(-1)}$ or better.

The sequence construction uses direction numbers based on primitive polynomials over finite fields, generating points that systematically fill the unit hypercube. For atmospheric applications, the key advantages include:

- **Dimensional Extensibility:** Sobol sequences maintain low-discrepancy properties in arbitrarily high dimensions—crucial for atmospheric systems where state vectors contain hundreds to millions of components.
- **Stratification:** Points distribute uniformly across all marginal projections, ensuring that each atmospheric variable receives adequate sampling coverage regardless of dimension.
- **Determinism:** The sequence is reproducible, enabling validation and debugging whilst maintaining uniform coverage properties.

Mapping to Atmospheric State Space

Sobol sequences generate points in the unit hypercube $[0,1]^d$. Mapping these to physically meaningful atmospheric states requires transformation through the cumulative distribution functions specified during initialisation:

For a single atmospheric variable with prior distribution $F(x)$, the transformation:

$$X = F^{-1}(U)$$

Where:

$U \sim \text{Uniform}[0,1]$ produces samples X following distribution $F(x)$.

This inverse transform method extends to multivariate distributions through copula structures (detailed below).

For the multivariate normal prior $X_0 \sim N(\mu_0, \Sigma_0)$:

$$X_0 = \mu_0 + L \cdot \Phi^{-1}(U)$$

Where:

- L represents the Cholesky decomposition of Σ_0 ($LL^T = \Sigma_0$)
- Φ^{-1} denotes the inverse standard normal CDF
- U is the Sobol vector.

In Monte Carlo simulations requiring correlated random variables, the Cholesky decomposition accelerates the generation process by efficiently transforming independent random samples into correlated ones. Given a covariance matrix Σ , the Cholesky decomposition factors it as $\Sigma = LL^T$, where L is a lower triangular matrix.

To generate a sample with the desired correlation structure, you simply multiply a vector of independent standard normal variables by L —a computationally efficient operation due to L 's triangular form. The decomposition itself needs to be computed only once and can be reused across millions of simulation runs, making it far more efficient than repeatedly inverting or factoring the covariance matrix.

This transformation ensures that:

- Each ensemble member has atmospheric state distributed according to the prior $N(\mu_0, \Sigma_0)$
- The ensemble collectively provides uniform coverage of probable atmospheric states
- Physical constraints encoded in μ_0 and Σ_0 (balance relationships, conservation laws) are satisfied

Dimension Reduction for Practical Implementation

Atmospheric state vectors contain millions of components (temperature, pressure, winds, humidity at each grid point and vertical level). Although theoretically possible, generating Sobol sequences in such extreme dimensions presents practical challenges.

The framework employs dimension reduction to focus QMC sampling on the most influential atmospheric variations:

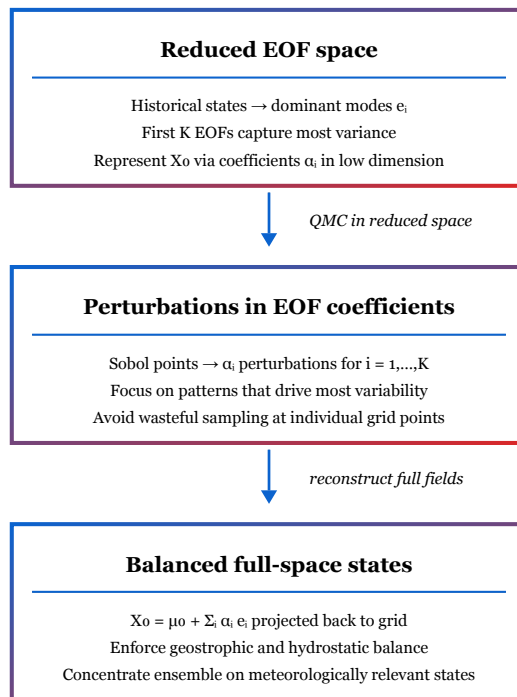
- Empirical Orthogonal Functions (EOFs): Principal component analysis of historical atmospheric states identifies the dominant modes of variability. Typically, the first 50-200 EOFs capture 90%+ of atmospheric variance. The QMC sampling generates perturbations in this reduced EOF space:

$$X_0 = \mu_0 + \sum_{i=1}^K \alpha_i e_i$$

Where:

- e_i represents the i -th EOF
 - K the retained modes (e.g., 100)
 - α_i the coefficients generated via Sobol sequences.
- **Balanced Sampling:** The reduced-space perturbations are then expanded back to full atmospheric state space whilst maintaining dynamical balance relationships (geostrophic balance, hydrostatic balance) through projection onto balanced modes.

Figure 23 – Dimension reduction and balanced reconstructions



This approach concentrates computational resources on atmospheric variations that:

- Explain the majority of historical weather pattern diversity
- Satisfy physical balance constraints
- Cover the meteorologically relevant state space efficiently

Copula framework for multivariate dependence

A copula is a multivariate distribution with uniform marginal distributions on. Sklar's theorem states that any joint distribution can be written as a copula applied to its marginal cumulative distribution functions. In this framework, that separation is useful because it allows us to choose margins that match the empirical behaviour of each variable—normal for some, skewed or heavy-tailed for others—while capturing their dependence structure through a copula. Temperature, pressure, humidity, wind components and precipitation can therefore each follow the marginal distributions implied by the physics-informed prior, but their joint behaviour is governed by a separate object that can be tuned to reproduce observed co-movements, asymmetries and tail dependence.

The atmospheric dependence structure is far from purely linear. Relationships between variables strengthen and weaken under different regimes; extremes in one field may be associated with extremes in another, even when their correlation under typical conditions is modest. Copulas handle this naturally. Families such as Gaussian and Student-t capture symmetric dependencies with differing tail behaviour, while asymmetric families such as Gumbel or Clayton can represent situations in which one tail is more strongly coupled than the other. By selecting appropriate copula families for different pairs of variables and different regimes, the framework can reproduce both everyday and extreme-event dependence patterns seen in historical data. The following copula-parameter subsection specifies how those choices are made and how the parameters are linked to the broader atmospheric state.

Vine Copulas and Sampling Procedure

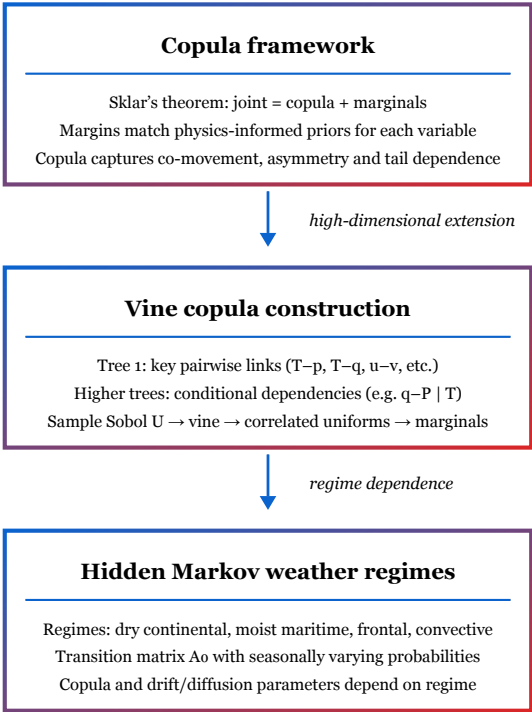
In high dimensions it is not practical to specify a single copula over all components of the state vector. Vine copulas address this by decomposing the full dependence structure into a sequence of bivariate copulas arranged on a tree-like structure. The first tree captures direct pairwise dependencies between selected variables, such as temperature and pressure or humidity and vertical velocity. Subsequent trees capture conditional dependencies, for example between humidity and precipitation conditional on temperature. Each edge in this vine is associated with a chosen copula family and parameter set, as described in the previous subsection.

Sampling from this structure begins with a vector of independent uniform variables, typically generated from the quasi-Monte Carlo sequence used elsewhere in the framework. These uniforms are transformed sequentially through the vine, using the conditional distributions implied by each bivariate copula, to produce a set of correlated uniforms that embody the desired

dependence structure. Finally, each component is mapped through the inverse of its marginal distribution to obtain physical variables such as temperature, pressure or humidity.

When combined with the dimension-reduction and balance-enforcement steps described earlier, this procedure yields atmospheric states that populate the ensemble with realistic variability and co-movement, providing a sound basis for the probabilistic weather scenarios used in later chapters.

Figure 24 – Copulas, vines and regime-dependent dependence



Vine Copula Construction

High-dimensional copula modelling faces the curse of dimensionality—direct specification of multivariate copulas with $d > 3$ becomes intractable. Vine copulas address this through hierarchical decomposition into bivariate building blocks.

The decomposition proceeds sequentially:

Tree 1: Model direct pairwise dependencies. For atmospheric states, this might include:

- Temperature and pressure (negative correlation in many regimes)
- Temperature and humidity (constrained by Clausius-Clapeyron)
- Wind components u and v (reflecting rotational and divergent flow patterns)

Each pair receives a bivariate copula $C(F(x_1), F(x_2))$ chosen from families suitable for the observed dependency (Gaussian, Student-t, Gumbel, Clayton).

Tree 2: Model conditional dependencies. Given the variables linked in Tree 1, what additional dependencies exist? For example:

- Humidity and precipitation, conditional on temperature
- Vertical wind and precipitation, conditional on humidity

Tree 3: Continue building conditional dependencies until the vine structure adequately captures observed multivariate behaviour.

The resulting vine specification allows efficient sampling:

- Generate independent uniform random variables $U_1, U_2, \dots, U_n$ from Sobol sequences
- Transform through the vine structure following the specified conditional distributions
- Apply inverse marginal CDFs to obtain atmospheric variable values

This produces multivariate samples exhibiting the complex dependencies observed in atmospheric data whilst maintaining computational tractability.

Hidden Markov weather regimes

The dependence structures described so far capture how atmospheric variables relate to one another at a given time. They do not yet describe how the large-scale weather pattern itself tends to persist or transition over time. In practice, many variables co-evolve because the atmosphere sits for hours or days in a particular regime—such as a dry anticyclone, a moist tropical inflow, or a frontal passage—and then transitions to another regime. To represent this, the framework uses a Hidden Markov Model (HMM) defined over a finite set of weather regimes.

Historical atmospheric states are first clustered into a small number of characteristic regimes based on patterns in pressure, winds, moisture and other fields. Each regime corresponds to a recurring configuration, for example “dry continental”, “moist maritime”, “frontal transition” or “convective instability”. Once these regimes have been identified, we estimate a transition probability matrix A_0 whose entries a_{ij} represent the probability of moving from regime i at one time step to regime j at the next. These probabilities are obtained by counting transitions in the historical record and then smoothing them to avoid zero probabilities for rare but possible moves.

Seasonal variation can be captured either by estimating separate matrices for different seasons or by allowing the transition probabilities to depend on time of year.

Within the HMM, the regime sequence itself is not directly observed; what we observe are the atmospheric fields, whose distribution depends on the underlying regime. The copula parameters and, if desired, some aspects of the drift and diffusion terms are allowed to depend on the current regime, so that dependence structures strengthen, weaken or become more asymmetric when the atmosphere enters particular patterns. During simulation, the regime evolves according to the transition matrix, and at each time step the atmospheric state is sampled conditional on that regime. This combination of regime-dependent copulas and Markovian regime dynamics enables the ensemble to reproduce not only realistic instantaneous relationships between variables, but also the observed persistence and sequencing of large-scale weather patterns that drive flood risk

For capturing weather regime persistence and transitions, Hidden Markov Model transition matrices receive initial specifications:

$$A_0 = [a_{ij}]$$

Where:

$$a_{ij} = P(\text{regime } j \text{ at } t+1 \mid \text{regime } i \text{ at } t)$$

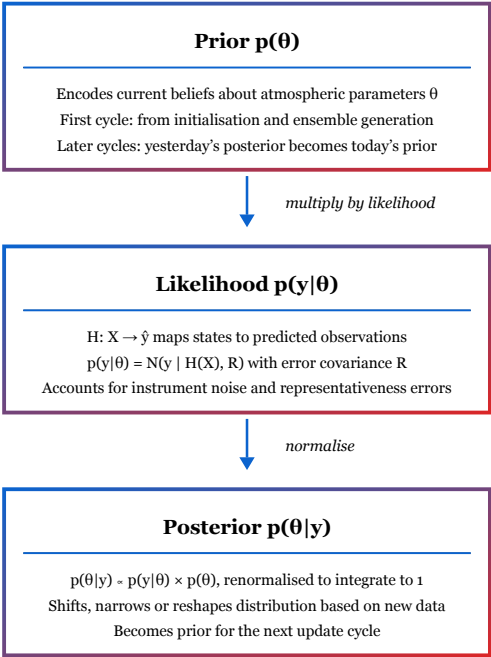
Initialisation proceeds through:

- Regime identification: Cluster historical atmospheric states into distinct weather patterns (e.g., "dry continental", "moist tropical", "frontal passage", "convective instability")
- Transition counting: Calculate empirical transition frequencies from historical sequences
- Smoothing: Apply Bayesian smoothing to prevent zero probabilities for rare but possible transitions
- Seasonal variation: Allow transition probabilities to vary by season (summer convection differs from winter dynamics)

Chapter 3.4 : Weather - Bayesian Update Cycle

Static probabilistic forecasts, no matter how carefully constructed, lose relevance as new atmospheric observations arrive. A distribution generated at 6 AM may be materially outdated by noon if radar detects developing convection, if surface stations report unexpected warming, or if a high-resolution model run reveals a stalling frontal boundary. Banking applications require a framework in which probability distributions evolve continuously, adjusting to new evidence without discarding accumulated knowledge. This chapter formalises the Bayesian updating mechanism that transforms the Weather Monte Carlo ensemble from a static snapshot into a living, learning distribution.

Figure 25 – Bayesian update from prior to posterior



Bayes' theorem provides the mathematical foundation: prior beliefs about atmospheric parameters are multiplied by a likelihood function quantifying how probable the new observations would be under those parameters, producing a posterior distribution that becomes the prior for the next update cycle. The challenge lies in implementation—the atmospheric state is high-dimensional, observations arrive from diverse sensors with different error characteristics, and computational efficiency is critical for operational use. The Ensemble Kalman Filter (EnKF)

addresses these challenges by approximating the full Bayesian update through ensemble statistics, scaling linearly with ensemble size rather than cubically with state dimension.

This chapter specifies the complete update cycle: observation operators that map atmospheric states to sensor readings, likelihood functions that account for measurement precision and representativeness errors, the EnKF equations that adjust ensemble members in response to observations, and the temporal propagation between updates that allows the distribution to evolve according to the stochastic differential equation developed in Chapter 3.2. Adaptive ensemble management techniques—inflation, localization, regeneration—ensure that the ensemble maintains appropriate spread and avoids collapse despite repeated updates. The result is a continuous filtering process in which yesterday's posterior becomes today's prior, enabling banking risk systems to respond to emerging weather patterns with the same agility that trading desks show in response to market news.

Fundamental Structure

The core mathematical relationship governing the update process follows Bayes' theorem:

$$p(\theta|y) \propto p(y|\theta) \times p(\theta)$$

This deceptively simple equation encapsulates a profound shift in how weather pattern distributions evolve:

- **Prior $p(\theta)$:** Represents current beliefs about atmospheric parameters θ before observing new data. At the first update cycle, this derives from Phase 1 initialisation and Phase 2 ensemble generation. At subsequent cycles, it equals the posterior from the previous update—creating a continuous chain of learning.
- **Likelihood $p(y|\theta)$:** Quantifies how probable the newly observed data y would be if the parameters were θ . This function encodes the relationship between theoretical atmospheric states and actual measurements, accounting for instrument characteristics, measurement errors, and observation operator properties.
- **Posterior $p(\theta|y)$:** Represents updated beliefs after incorporating the new observations. This distribution reflects the synthesis of prior knowledge and new evidence, weighted according to their relative certainties.

The proportionality constant (normalising factor) ensures the posterior integrates to unity:

$$p(\theta|y) = [p(y|\theta) \times p(\theta)] / \int p(y|\theta) \times p(\theta) d\theta$$

In practice, this normalisation often proves computationally challenging, leading to approximate inference methods detailed below.

Observational Data Streams

The framework assimilates diverse atmospheric observations, each with distinct characteristics and information content:

- **Surface Weather Stations:** Direct measurements of temperature, pressure, humidity, wind, and precipitation at fixed locations. High temporal frequency (hourly) and excellent accuracy (temperature $\pm 0.5^\circ\text{C}$) but sparse spatial coverage.
- **Weather Radar:** Doppler radar measures precipitation intensity with high spatial (1-4 km) and temporal (5-10 minute) resolution. Exceptional coverage of precipitation systems but requires calibration and suffers from terrain blocking.
- **Radiosondes:** Weather balloons launched twice daily provide vertical profiles of temperature, humidity, and wind through the troposphere—invaluable for characterising atmospheric stability.
- **Satellites:** Geostationary satellites provide continuous imagery (5-15 minute intervals) showing cloud patterns and derived atmospheric motion vectors. Polar-orbiting satellites carry advanced sensors that penetrate clouds to observe temperature and humidity profiles.

Observation Operators and Likelihood Functions

The likelihood function $p(y|\theta)$ depends critically on observation operators that relate model state to observations:

$$H: X \rightarrow y_{\text{predicted}}$$

Where:

H maps the atmospheric state vector X to predicted observation values.

For direct observations like surface temperature, H extracts the appropriate model variable with spatial interpolation. For indirect observations like satellite radiances, H involves complex radiative transfer calculations.

The likelihood function accounts for observation uncertainty:

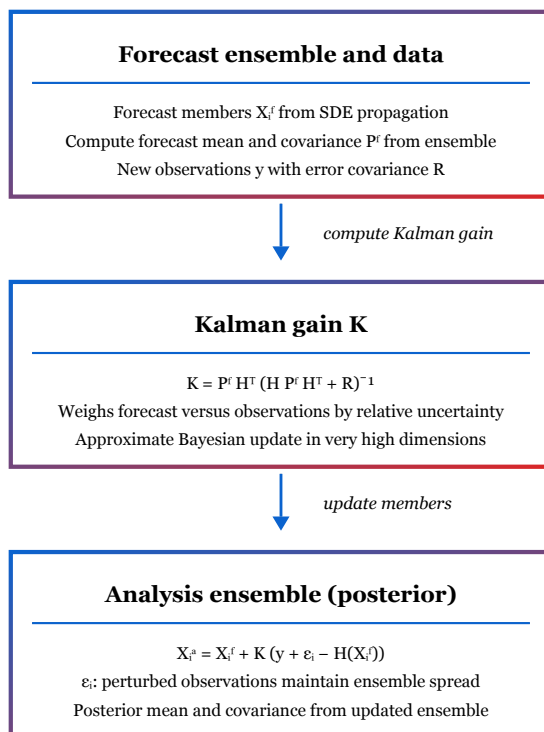
$$p(y|\theta) = N(y | H(X), R)$$

Where:

- R represents the observation error covariance matrix, encoding:
 - Instrument measurement precision
 - Representativeness errors (point observation versus gridded model)

- Forward model errors in observation operators
- Ensemble Kalman Filter Implementation

Figure 26 – Ensemble Kalman Filter update step



The Ensemble Kalman Filter: High-Dimensional Bayesian Updating

For operational efficiency with high-dimensional atmospheric states, the framework employs the Ensemble Kalman Filter (EnKF), which approximates the Bayesian update through ensemble statistics. The EnKF represents one of the most significant methodological innovations in modern data assimilation, enabling tractable Bayesian inference for systems with millions of state variables—a feat that would be computationally prohibitive using classical Kalman filter formulations.

The classical Kalman filter, developed by Rudolf Kálmán in 1960, provides the optimal solution for linear systems with Gaussian uncertainties. However, it requires explicit computation and

storage of the full state covariance matrix P , which for atmospheric applications would be a matrix of size $n \times n$ where n represents the state dimension (typically 10^6 to 10^7). This leads to a computational impossibility:

- Storage: A covariance matrix for 1 million state variables requires storing 10^{12} elements (1 trillion numbers), consuming approximately 4 terabytes of memory
- Computation: Matrix operations scale as $O(n^3)$, making a single update cycle computationally intractable—requiring approximately 10^8 floating-point operations

The EnKF circumvents these limitations through a Monte Carlo approximation: rather than tracking the full covariance matrix, it maintains an ensemble of N possible atmospheric states (typically $N = 50-500$), from which covariances are estimated empirically. This reduces computational complexity from $O(n^3)$ to $O(nN)$, making real-time operational data assimilation feasible.

For operational efficiency with high-dimensional atmospheric states, the framework employs the Ensemble Kalman Filter (EnKF), which approximates the Bayesian update through ensemble statistics:

Kalman Gain Calculation:

$$K = P^f H^T (H P^f H^T + R)^{-1}$$

Where:

- P^f represents the forecast error covariance (estimated from ensemble spread)
- H the observation operator (linearised)
- R the observation error covariance

Ensemble Update:

$$X_i^a = X_i^f + K(y + \varepsilon_i - H(X_i^f))$$

Where:

ε_i represents perturbed observations ensuring proper ensemble spread maintenance.

The EnKF's computational efficiency scales linearly with ensemble size rather than cubically with state dimension, making it tractable for atmospheric applications with millions of state variables.

The Kalman gain K represents the optimal weighting between prior knowledge (forecast) and new information (observations), balancing their relative uncertainties.

Distribution Evolution Mechanics

The Bayesian update transforms the prior distribution into the posterior through three mechanisms:

- **Location Shift:** When observations indicate conditions warmer, wetter, or windier than the prior ensemble predicts, the posterior distribution shifts accordingly. The magnitude depends on the Kalman gain, which balances prior and observational confidence.
- **Variance Reduction:** Observations always reduce uncertainty in the observation space:

$$P^a = (I - KH)P^f$$

The posterior covariance is smaller than the prior covariance. Uncertainty decreases most along well-observed dimensions; poorly observed dimensions retain prior uncertainty levels.

- **Shape Transformation:** Non-Gaussian updates can alter distribution shape substantially. Radar detecting light precipitation eliminates ensemble members predicting no precipitation, skewing the posterior. Observations of extreme conditions increase tail probabilities relative to the prior.

Temporal Propagation Between Updates

Between observation times, the posterior distribution propagates forward using the SDE from Phase 2:

$$dX(t) = a(X,t)dt + b(X,t)dW(t)$$

This propagation increases uncertainty as forecast lead time grows (small initial differences amplify through chaotic dynamics) whilst maintaining physical consistency through conservation laws. The propagation-update cycle continues indefinitely:

Update → Propagate → Update → Propagate → ...

creating a sequential filtering process where the distribution continuously evolves.

Adaptive Ensemble Management

As the update cycle progresses, ensemble characteristics may require adjustment:

- **Ensemble Inflation:** Systematic underestimation of uncertainty causes ensemble spread to collapse. The framework monitors spread-skill relationships and applies adaptive inflation when needed:

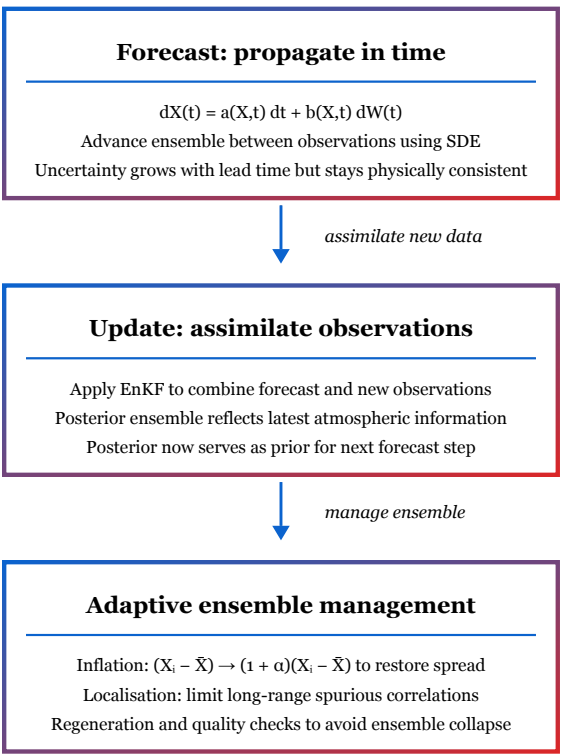
$$(X_i - \bar{X}) \rightarrow (1 + \alpha)(X_i - \bar{X})$$

Where:

$\alpha \in [0.05, 0.20]$ increases spread whilst preserving the ensemble mean.

- Localisation: Spurious long-range correlations in finite ensembles can cause observations to inappropriately influence distant regions. Localisation limits observation impact to meteorologically relevant distances (typically 200-500 km), ensuring surface pressure observations in one region do not incorrectly adjust temperature fields thousands of kilometres away.

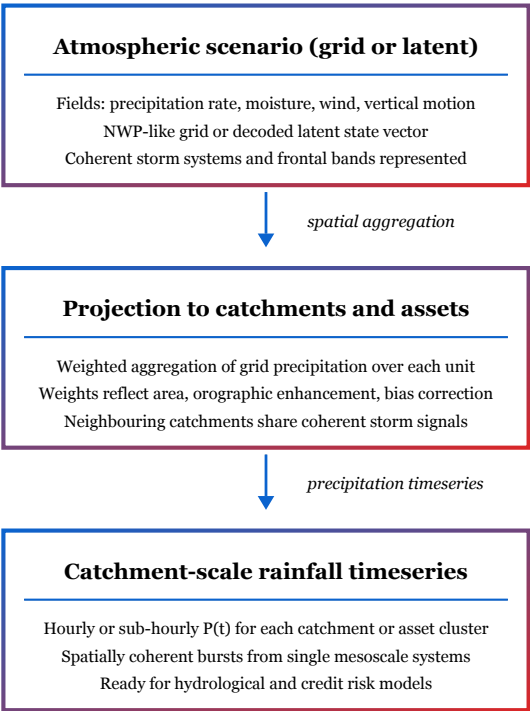
Figure 27 – Propagate–update cycle and ensemble management



Chapter 3.5 : Weather - Precipitation Mapping

To be useful for hydrology and credit risk, each atmospheric scenario must be translated into rainfall at the level of individual catchments, river reaches, or asset clusters rather than remaining as abstract pressure, humidity, and wind fields. This section describes how the framework converts physically plausible, dynamically updated atmospheric states into ensembles of precipitation time series that preserve spatial coherence, temporal clustering, and extreme event behaviour consistent with modern numerical weather prediction systems.

Figure 28 – From atmospheric fields to catchment rainfall



From atmospheric fields to catchment-scale rainfall

The starting point for this mapping is an atmospheric scenario expressed either directly on an NWP-like grid or as a latent state vector that can be decoded into fields of precipitation rate, moisture, wind, and vertical motion. For each time step and scenario, these fields are first

projected onto the spatial units relevant for downstream risk analysis, typically river catchments, administrative regions, or buffers around critical assets.

This projection is implemented as a weighted aggregation of grid-scale precipitation rates over the target unit, with weights reflecting fractional area coverage, orographic enhancement, and, where necessary, simple bias-correction factors derived from reanalysis or gauge records. The use of physically based fields rather than independent point draws ensures that neighbouring catchments share coherent rainfall signals when they lie under the same mesoscale convective system or frontal band, mirroring the behaviour seen in high-resolution regional models such as HRRR. In practice, a single synthetic storm cell in the atmospheric state will therefore generate correlated rainfall bursts across an entire river reach rather than unrealistically isolated spikes at individual grid points.

Temporal structure and storm lifecycles

Once spatial aggregation is defined, the framework must represent the temporal evolution of rainfall within each scenario so that hydrological models see realistic storm lifecycles and wet-dry alternation. The atmospheric evolution described by the stochastic differential equation in Section 4.2 naturally generates sequences of synoptic patterns and convective triggers; these, in turn, determine when and where rainfall is initiated, intensifies, and decays over time.

At the temporal resolution required by downstream models (for example hourly or sub-hourly), each scenario is thus expressed as a continuous precipitation time series for every spatial unit, capturing both the duration of events and the structure of intra-storm intensity. Temporal dependence is handled explicitly so that rainfall does not appear as a sequence of independent pulses, but rather as clustered episodes whose persistence and inter-event dry spells are calibrated against historical records and high-resolution NWP output. This prevents underestimation of flood risk arising from multi-day accumulation or back-to-back systems, a known limitation of simple daily resampling approaches.

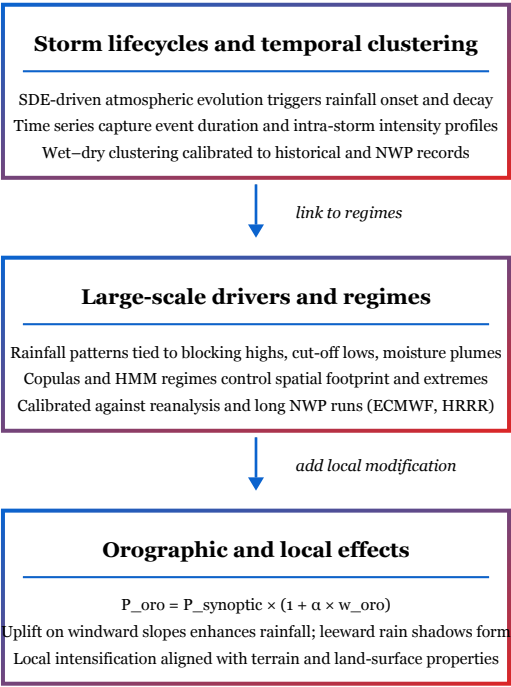
Consistency with large-scale drivers and regimes

The mapping from atmospheric state to rainfall is constrained to remain consistent with the large-scale circulation patterns and weather regimes that drive real extreme events. Synoptic-scale structures such as blocking highs, cut-off lows, and moisture plumes are captured by the state vector and by the dependence structures introduced via copulas and hidden Markov regimes in Section 4.3, ensuring that each synthetic rainfall footprint can be traced back to a plausible atmospheric configuration.

To enforce this consistency, the framework is calibrated against a combination of reanalysis products and long historical runs of global and regional NWP systems, which provide joint samples of circulation patterns and resulting precipitation fields. This calibration step aligns the spatial patterns, storm tracks, and typical catchment-level rainfall responses of the synthetic scenarios with those observed in systems like ECMWF and HRRR, even though the framework operates in a reduced, risk-oriented state space. As a result, when the Bayesian engine favours a

particular regime or teleconnection pattern, the associated rainfall scenarios display the expected geographic focus and footprint magnitude rather than arbitrary, regime-independent noise.

Figure 29 – Temporal structure, regimes and orographic effects



Scenario rainfall ensembles and empirical probabilities

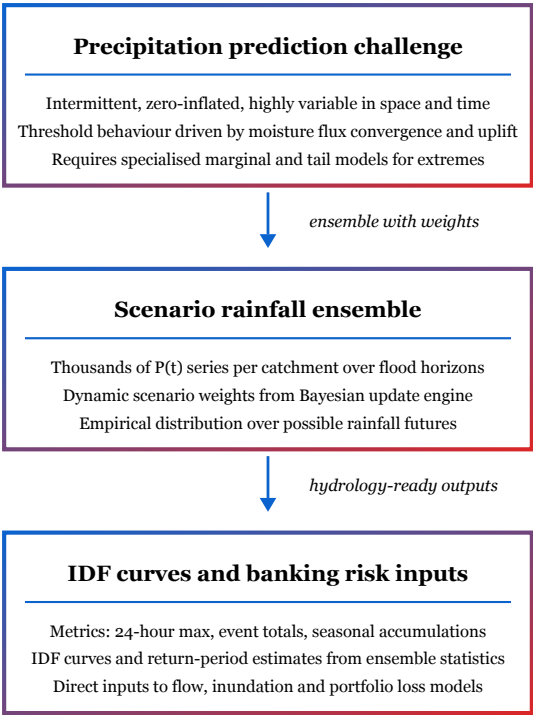
For each atmospheric scenario in the ensemble, the procedures above deliver a complete precipitation time series for every catchment or asset over the horizon of interest, typically spanning both the lead-up to and the evolution of potential flood events. Section 4.4 then provides a set of dynamically updated scenario weights, derived from Bayesian assimilation of new observational information, which convert this ensemble of rainfall histories into an empirical probability distribution over possible futures.

These weights allow the construction of exceedance probability curves and return-period estimates for a wide range of rainfall metrics, such as maximum 24-hour depth, event-total precipitation over a storm window, or cumulative rainfall over a season. Because each rainfall time series is already spatially aggregated to the units used in exposure and loss models, the same

scenario weights can be applied downstream to generate distributions of river flows, inundation depths, or portfolio-level losses. In this way, the chapter closes the loop from physical atmospheric dynamics through stochastic modelling and Bayesian updating to a probabilistic rainfall input that can directly support pricing and risk management of physical risk derivatives.

This phase bridges the gap between meteorological prediction and banking application that we will cover later, producing thousands of probabilistic precipitation scenarios that maintain spatial coherence, temporal persistence, and physical plausibility. The precipitation timeseries constitute the critical deliverable for banking risk systems—the format that existing infrastructure can ingest, the inputs that hydrological models require, and the foundation upon which all subsequent flood risk calculations depend.

Figure 30 – Scenario rainfall ensembles and IDF products



The Precipitation Prediction Challenge

Precipitation represents perhaps the most difficult atmospheric variable to predict accurately. Unlike temperature or pressure, which vary continuously and smoothly, precipitation exhibits:

- Intermittency: Most locations experience no precipitation most of the time, creating distributions with large point masses at zero
- High Spatial Variability: Intensity can vary by orders of magnitude over just a few kilometres
- Temporal Variability: Rates fluctuate rapidly—from zero to intense within minutes for thunderstorms
- Threshold Behaviour: Many processes activate abruptly when atmospheric conditions exceed critical values

These characteristics demand specialised approaches beyond simple interpolation of atmospheric state variables.

From Atmospheric State to Precipitation

The atmospheric state vectors from Phase 4 provide the foundation for precipitation inference, but the relationship requires careful specification through physical constraints:

- Moisture Budget: Precipitation forms when moisture flux converges:

$$P = -\nabla \cdot \mathbf{Q} + E - \partial W / \partial t$$

Where:

Convergence of moisture flux combined with limited evaporation produces precipitation.

- Vertical Motion: Precipitation intensity relates to uplift carrying moist air:

$$P \propto w \times q$$

Where:

- w denotes upward velocity
- q specific humidity.

Marginal Distributions

The precipitation distribution at each location features:

- Zero-Inflation: A point mass at zero representing dry conditions, with occurrence probability depending on atmospheric state (precipitable water, vertical velocity, instability).

- **Conditional Intensity:** Given precipitation occurs, intensity typically follows gamma distributions, with shape and scale parameters varying by atmospheric regime—convective situations produce higher variability than stratiform precipitation.
- **Extreme Tails:** For flood-relevant heavy precipitation, Generalised Pareto Distributions characterise tail behaviour, with parameters derived from historical extremes, atmospheric state, and pattern family characteristics.

Spatial Correlation Structure

Rain does not fall independently at neighbouring locations. Spatial correlation reflects the physical coherence of atmospheric systems—frontal passages produce precipitation bands extending hundreds of kilometres, whilst convective systems create more localised patterns.

Spatial correlation decays with separation:

$$\theta_{ij} = \theta_0 \exp(-d_{ij} / L_{\text{corr}})$$

Where:

L_{corr} represents the correlation length scale—typically 10–50 km for convective precipitation, 50–200 km for stratiform systems.

Regime-Dependent Structure

Different weather patterns exhibit distinct correlation structures:

- Atmospheric rivers show strong correlation along moisture transport direction
- Convective systems exhibit shorter-range correlations with intense local maxima
- Stable anticyclones produce weak correlations (little precipitation anywhere)

Orographic Enhancement

Topography profoundly influences precipitation through mechanical uplift:

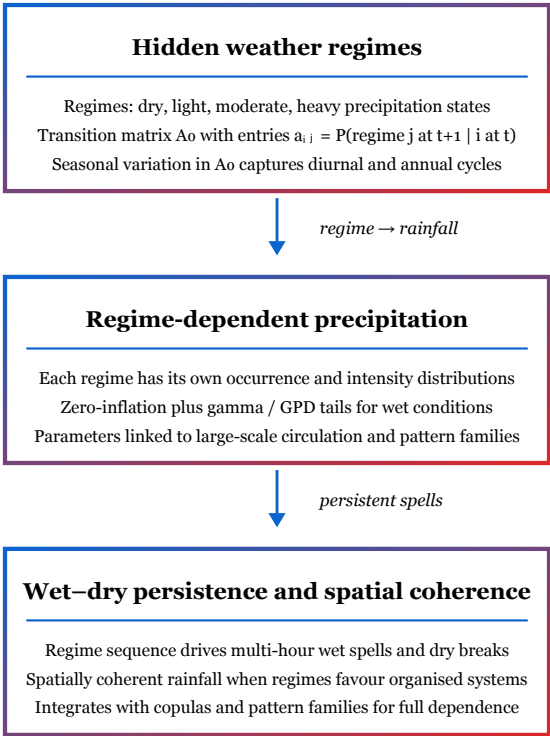
$$P_{\text{oro}} = P_{\text{synoptic}} \times (1 + \alpha \times w_{\text{oro}})$$

Where:

- P_{oro} — the orographically-enhanced precipitation rate at a given location. This is the actual precipitation after accounting for terrain effects.

- P_{synoptic} — the large-scale "background" precipitation rate determined by the synoptic weather pattern (frontal systems, moisture transport, atmospheric instability). This is what precipitation would be over flat terrain under the same atmospheric conditions.
- α — the orographic efficiency parameter, which controls how effectively uplift converts to additional precipitation. This depends on atmospheric stability and moisture content: a moist, unstable atmosphere (high α) converts uplift to precipitation efficiently, whilst a dry or stable atmosphere (low α) shows weaker orographic response.
- w_{oro} — the orographic vertical velocity, representing the upward motion of air forced by terrain. Calculated as the dot product of horizontal wind and terrain gradient ($w_{\text{oro}} = \mathbf{v} \cdot \nabla h$), stronger winds hitting steeper slopes produce larger values. Positive values indicate uplift on windward slopes; negative values indicate descent on leeward slopes.

Figure 31 – Hidden Markov weather and precipitation regimes



When moist air encounters rising terrain, it is forced upward. This uplift cools the air, causing water vapour to condense and fall as enhanced precipitation on the windward side. Once the air crosses the ridge and descends the leeward slope, it warms and dries, producing the characteristic "rain shadow" where precipitation is suppressed.

The multiplicative structure $(1 + \alpha \times w_{oro})$ means that with no orographic effect ($w_{oro} = 0$), precipitation equals the synoptic value. Positive uplift enhances precipitation above the background rate; descent reduces it below.

Hidden Markov Models

Precipitation occurrence exhibits persistence—rainy periods tend to continue, as do dry periods. Hidden states represent weather regimes with distinct characteristics (dry, light, moderate, heavy precipitation), with transition probabilities governing regime evolution. High atmospheric moisture increases transitions toward wet states; weak synoptic forcing promotes transitions toward dry conditions.

Diurnal (daily) and Seasonal Cycles

Generated timeseries incorporate systematic temporal variations—afternoon convection peaks over land in summer, whilst winter frontal precipitation shows weaker diurnal dependence.

Pattern Family Integration

Each weather pattern family from Phase 4 exhibits distinct precipitation characteristics. Timeseries generation employs pattern-specific parameters:

- **Occurrence Probability:** Atmospheric river patterns have high precipitation probability (~ 0.7); anticyclone patterns have low probability (~ 0.1)
- **Intensity Distribution:** Slow-moving convective systems produce heavier-tailed distributions than broad frontal systems
- **Spatial Correlation:** Organised systems show longer correlation lengths than disorganised convection
- **Temporal Persistence:** Quasi-stationary patterns exhibit longer persistence than rapidly translating systems

Timeseries generation samples across pattern families proportionally to their Phase 4 probabilities, ensuring common patterns contribute many timeseries whilst extreme event patterns receive representation proportional to their elevated probabilities.

Catchment-Scale Products

Banking applications will focus on flooding impacts at specific catchment scales:

Raw precipitation fields at 3 km resolution are aggregated to catchment-relevant scales:

$$P_{\text{catchment}}(t) = (1/A_C) \iint_C P(x,y,t) \, dx \, dy$$

with area-weighting reflecting sub-catchment runoff coefficients—urban areas and steep slopes receive higher weights than vegetated areas with infiltration capacity.

Intensity-Duration-Frequency Relationships

The ensemble provides IDF statistics essential for flood risk assessment:

- Maximum intensities for standard durations (1, 3, 6, 12, 24 hours)
- Return period estimates: the 100-year 24-hour precipitation corresponds to the 99th percentile across the ensemble
- Parametric IDF curves fitting ensemble statistics

Chapter 4: Climate Risk Modelling

The preceding chapter examined weather prediction—the art and science of forecasting atmospheric conditions over hours to days. This chapter addresses a fundamentally different domain: climate modelling and the regulatory frameworks that have emerged to address long-term physical risks. While weather and climate both involve atmospheric science, they serve distinct purposes, operate on different timescales, and require entirely different modelling approaches.

"Weather prediction asks 'what will happen next week?' Climate modelling asks, 'What patterns will emerge over decades?' These are not merely different timeframes—they represent fundamentally different scientific questions requiring different methodologies, different validation approaches, and different applications in banking risk management."
- David Kelly, MKM

Understanding this distinction proves crucial for banking institutions navigating physical risk assessment. Weather models inform operational decisions about imminent risks to specific assets, and climate models inform strategic decisions about portfolio composition, long-term capital allocation, and regulatory compliance. Confusing these purposes leads to misapplication of tools and flawed risk assessments.

This chapter proceeds in three parts. First, we establish the fundamental differences between weather and climate modelling, clarifying why tools designed for one purpose prove inadequate for the other. Second, we examine the Intergovernmental Panel on Climate Change (IPCC) and its assessment reports—the scientific foundation underpinning global climate policy. Third, we explore the Task Force on Climate-related Financial Disclosures (TCFD) and how it translates climate science into banking requirements. Finally, we position academic frameworks like CLIMADA within this regulatory ecosystem, explaining their complementary role.

Weather vs. Climate: Fundamental Distinctions

The atmosphere exhibits behaviour at multiple timescales, each governed by different physical processes and predictable through different methods. Understanding these distinctions prevents the common error of treating climate models as "long-range weather forecasts"—a conceptual confusion that leads to inappropriate tool selection and flawed risk assessments.

Temporal Scope and Predictability

Weather prediction focuses on atmospheric states over hours to days, with practical limits as anyone who has relied on a forecast to organise a trip around 10-14 days. Beyond two weeks, specific weather patterns become unpredictable due to chaotic dynamics—small measurement errors or modelling approximations compound exponentially, making precise forecasts impossible. This fundamental limitation stems not from inadequate models or insufficient data or

computation capacity but from the inherent mathematics of chaotic systems covered by Fractal Geometry.

Climate modelling addresses entirely different questions. Rather than predicting whether it will rain on a specific Tuesday in 2040, climate models characterise statistical properties of weather patterns over decades. - David Kelly, MKM

Will average annual precipitation increase or decrease? How will the frequency of extreme rainfall events evolve? What seasonal patterns will emerge? These statistical properties prove predictable even when individual weather events remain chaotic.

This distinction mirrors the difference between predicting individual coin flips and the distribution of outcomes across thousands of flips. Individual events remain unpredictable; aggregate statistics become highly predictable. Banking applications require both perspectives: weather forecasting for near-term asset management and climate modelling for long-term strategic planning.

Figure 32 – Resolution & Scale Distinction



Note: Weather captures tactical hazards; Climate captures strategic trends.

Physical Processes and Resolution Requirements

Weather models must capture processes at scales of kilometres to meters—individual thunderstorms, local terrain effects, urban heat islands. High temporal resolution (hourly or sub-hourly) proves essential for representing rapidly evolving weather systems. The HRRR data described earlier exemplifies this approach: 3-kilometre horizontal resolution with hourly updates, capturing detailed atmospheric dynamics relevant to flood forecasting.

Climate models operate at coarser resolutions—generally 100–300 kilometres horizontally, with monthly or seasonal temporal aggregation. This coarseness reflects different priorities: climate models must simulate decades of Earth system evolution, requiring computational efficiency and comprehensive representation of slow processes such as ocean circulation, ice sheet dynamics, and carbon cycle feedbacks. Processes occurring at sub-grid scales are parameterised—represented through simplified relationships rather than explicit simulation.

These different resolution requirements reflect varying validation approaches. Weather models confirm their accuracy by comparing predictions with observations made hours to days later—for example, whether the forecast rainfall occurred. On the other hand, climate models are validated through comparisons with paleoclimate records, historical trends, and physical consistency—such as whether simulated ice ages correspond with geological evidence.

Initialisation vs. Boundary Conditions

Weather forecasting depends on precise atmospheric initialisation—accurately representing current temperature, pressure, humidity, and wind fields across the atmosphere. Small errors in initial conditions quickly spread, reducing forecast accuracy within days. To this end, advanced data assimilation techniques integrate observations from satellites, weather balloons, aircraft, and ground stations to create optimal initial states.

Climate modelling depends primarily on boundary conditions and external forcings rather than initial conditions. Starting from an arbitrary atmospheric state (say, January 1, 1850), climate models spin up for decades until reaching equilibrium with specified greenhouse gas concentrations, solar radiation, volcanic aerosols, and land use patterns. After equilibrium, the specific weather patterns prove unpredictable, but climate statistics (average temperatures, precipitation patterns, seasonal cycles) emerge reproducibly.

This distinction has profound implications for banking applications. Weather forecasting enables tactical responses to near-term hazards—alerting mortgage borrowers about approaching storms, adjusting collateral valuations ahead of extreme weather, and activating emergency procedures for at-risk properties. Climate modelling informs strategic decisions, such as geographic portfolio rebalancing, long-term capital allocation, and product development for changing risk landscapes.

Uncertainty Characterisation

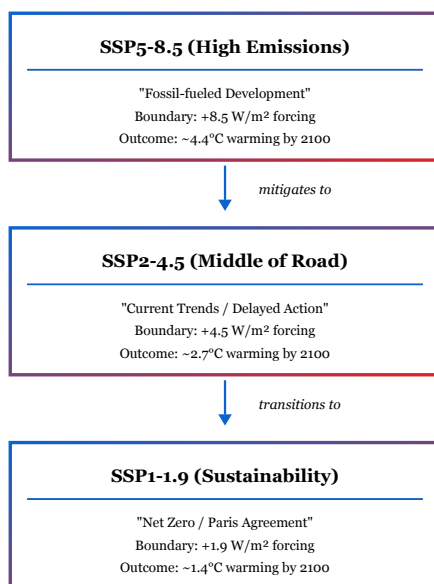
Weather forecast uncertainty stems primarily from initial condition uncertainty and model imperfection. Ensemble forecasting addresses this by running multiple forecasts with slightly perturbed initial conditions or different model configurations. The spread among ensemble members quantifies forecast uncertainty—a wide spread indicates high uncertainty, a narrow spread suggests confident predictions.

Climate model uncertainty arises from different sources: parameter uncertainty within models, structural uncertainty across different modelling approaches, scenario uncertainty about future human choices (emissions trajectories, land use, technology deployment), and internal climate

variability (natural fluctuations unrelated to external forcings). Climate assessments typically employ multi-model ensembles to characterise these uncertainties, producing probabilistic projections across plausible futures.

Banking risk management requires understanding both uncertainty types. Weather forecast uncertainty affects near-term operational decisions—should we delay construction? Shall we implement emergency procedures? Climate scenario uncertainty affects long-term strategic decisions— which geographic markets should we enter or exit? How should capital allocation shift across portfolios? The methodologies for addressing these distinct uncertainty types differ fundamentally.

Figure 34 – IPCC Representative Pathways



The IPCC: Scientific Foundation for Climate Policy

The Intergovernmental Panel on Climate Change (IPCC) is the authoritative source for global climate science assessments. Understanding the IPCC's structure, methodology, and conclusions provides essential context for banking institutions navigating climate-related financial risks and regulatory requirements.

Origins and Structure

The IPCC was established in 1988 by the United Nations Environment Programme (UNEP) and the World Meteorological Organisation (WMO), endorsed by the UN General Assembly in

December 1988. This timing proved significant—emerging from the Advisory Group on Greenhouse Gases (AGGG) established in 1986, the IPCC reflected growing recognition that climate change required international coordination beyond what small scientific advisory groups could provide.

The organisational structure balances scientific rigour with governmental oversight. The IPCC comprises 195 member states that govern the organisation and elect a bureau of scientists to serve through six to seven-year assessment cycles. Three working groups and a task force conduct scientific work: Working Group I addresses the physical science basis, Working Group II examines impacts and adaptation, and Working Group III focuses on climate change mitigation.

Critically, the IPCC conducts no original research. Instead, it examines all relevant scientific literature on climate change, compiling findings into Assessment Reports through what experts describe as the biggest peer review process in the scientific community. Reports must be backed by leading relevant scientists while member governments endorse findings through consensus agreement—creating a unique hybrid of scientific assessment and intergovernmental approval.

Assessment Reports and Historical Impact

Since 1988, the IPCC has completed six assessment cycles, each producing comprehensive Assessment Reports that have directly shaped international climate policy:

- The First Assessment Report (1990) underlined climate change as a global challenge requiring international cooperation, playing a decisive role in creating the United Nations Framework Convention on Climate Change (UNFCCC). This report established that human activities could change the climate with potentially severe economic and social consequences.
- The Second Assessment Report (1995) provided crucial material for governments negotiating the Kyoto Protocol, adopted in 1997. This assessment strengthened evidence for human influence on climate and began quantifying potential impacts across different warming scenarios.
- The Third Assessment Report (2001) focused on climate change impacts and adaptation needs, expanding beyond pure physical science to examine vulnerabilities across sectors and regions. This report catalysed early thinking about climate adaptation financing and institutional responses.
- The Fourth Assessment Report (2007) laid the groundwork for post-Kyoto agreements, emphasising the importance of limiting warming to 2°C above pre-industrial levels. The IPCC received the Nobel Peace Prize in 2007, recognising its role in building and disseminating climate knowledge. This report marked a turning point in mainstream acceptance of climate science and the urgency of action.
- The Fifth Assessment Report (2013-2014) provided scientific input for the Paris Agreement, strengthening evidence for the feasibility and importance of limiting warming to 1.5°C. This

assessment introduced Representative Concentration Pathways (RCPs) for scenario analysis and enhanced treatment of climate risk management frameworks.

- The Sixth Assessment Report (2021-2023) represents the most comprehensive assessment to date, establishing with high confidence that climate change is unequivocally caused by human activities and is already causing widespread, irreversible damage to natural and human systems globally.

Key Conclusions from the Sixth Assessment Report

The IPCC Sixth Assessment Report (AR6) is not a climate model but a comprehensive scientific assessment synthesising and evaluating thousands of peer-reviewed studies and climate projections. Completed between 2021 and 2023, it represents an eight-year collaborative effort involving 782 scientists organised into three specialised working groups: 234 scientists examined the physical science basis (Working Group I), 270 analysed impacts and adaptation (Working Group II), and 278 evaluated mitigation strategies (Working Group III). The entire assessment spans approximately 8,000 pages and was formally approved at the IPCC's 58th Session in Interlaken, Switzerland in March 2023.

Rather than generating original climate projections, AR6 draws from an enormous database of 3,131 scenarios produced by over 95 different model families—ranging from global circulation models to regional and sectoral pathways. This represents a massive expansion from the previous Fifth Assessment Report (AR5), with the AR6 data template tracking approximately 2,000 variables compared to just 135 in AR5.

The assessment introduced significant methodological advances, including a fully integrated temperature-based classification system for mitigation scenarios and using observational constraints to select appropriate model subsets for warming projections. This framework allows scientists to evaluate which emissions pathways are consistent with specific temperature targets (like limiting warming to 1.5°C or 2°C) and to assess the feasibility and implications of different policy responses across the entire climate system.

The output from the academic working groups feeds into the stakeholders that represent the governments involved. Their conclusion that evolves into a narrative that all participants agree upon is mouth-fed to all governments and their media. The conclusions are as follows and thus much of the subtlety of the academic output has been lost to enable widespread comprehension. The author sees this as a material failure of model risk governance that society is currently living through.

- **Definitive Human Causation:** Climate change results directly from over a century of human activities, particularly greenhouse gas emissions from energy use, land-use changes, and consumption patterns. Global fossil fuel consumption more than doubled in the past 50 years, rising from approximately 4 billion metric tons of oil equivalent in 1971 to over 8 billion metric tons in 2018. This unprecedented increase, combined with deforestation rates of approximately 10 million hectares of primary forest annually since 2015, creates the current climate crisis.

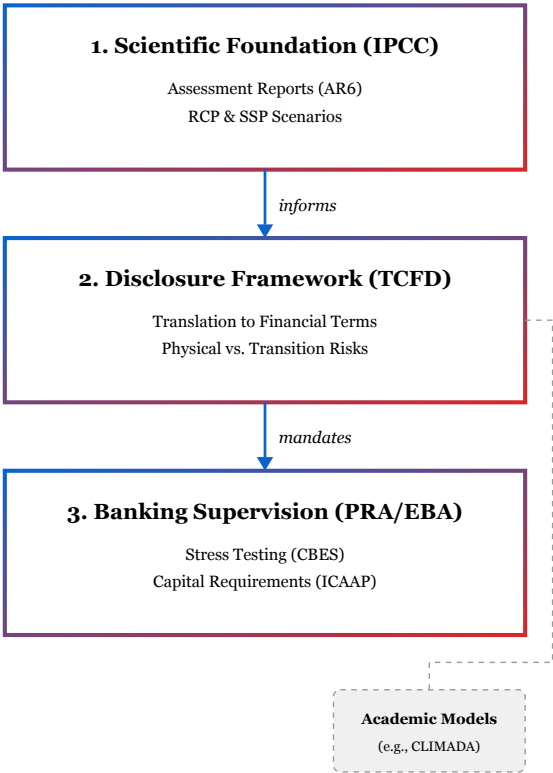
- **Substantial and Accelerating Damages:** Climate change has already caused substantial damages and increasingly irreversible losses in terrestrial, freshwater, coastal, and marine ecosystems. Impacts are already more far-reaching and extreme than previously anticipated. Even limiting warming to 1.5°C would expose 950 million people in drylands to water stress, heat stress, and desertification, while increasing flood exposure for 24% of the global population.
- **Biodiversity Crisis:** Terrestrial and freshwater species face 3-14% extinction risk at 1.5°C warming, with approximately 50% of tropical marine species facing extinction at 4°C warming. Mass extinctions represent not distant threats but likely consequences of current trajectories.
- **Human Vulnerability:** Between 3.3 and 3.6 billion people currently live in areas highly vulnerable to climate change impacts. Climate change has already exposed millions to acute food and water insecurities, particularly affecting populations in Africa, Asia, Central and South America, small islands, and the Arctic. Approximately 50% of the world's population experiences severe water insecurity for at least one month per year.
- **Critical Timing:** This decade proves critical for climate action, with choices made before 2025 being essential for limiting warming to 1.5°C. A resilient and livable future remains available only through deep, rapid, and sustained greenhouse gas emission cuts during this critical decade. The window for limiting warming to 1.5°C remains open but rapidly closes.
- **Emission Budget Constraints:** Limiting global warming to 1.5°C requires restricting total emissions to 510 gigatonnes of carbon dioxide by the 2050s. However, existing and planned fossil fuel infrastructure already exceeds this budget by 340 GtCO₂, necessitating retirement of existing facilities, cancellation of planned projects, and implementation of carbon capture and storage systems.
- **Adaptation Financing Gaps:** Developing countries require \$127 billion annually by 2030 and \$295 billion annually by 2050 for adequate adaptation. Actual adaptation funding reached only \$23.46 billion during 2017-2018, demonstrating a critical financing gap that must be addressed to meet Paris Agreement adaptation goals.

For banking institutions, these findings translate into material physical risks across portfolios, regulatory requirements for climate risk assessment and disclosure, and strategic imperatives for portfolio repositioning and product development aligned with transition pathways.

TCFD: Translating Climate Science into Banking Requirements

While the IPCC provides scientific assessment, the Task Force on Climate-related Financial Disclosures (TCFD) translates climate science into practical frameworks for financial institutions. Understanding TCFD's origins, structure, and requirements proves essential for compliance and strategic climate risk management.

Figure 33 – The Climate Risk Regulatory Chain



Origins and Formation

The TCFD was established in December 2015 by the Financial Stability Board (FSB). This international body monitors and makes recommendations about the global financial system, collaborating with the Group of 20 (G20). The formation was primarily driven by the 2015 Paris Agreement, which established nationally determined contributions but revealed significant gaps in transparency and international standards for demonstrating climate commitments.

Michael Bloomberg, founder of Bloomberg LP and former Mayor of New York City, was appointed to chair the Task Force. The initiative comprised 32 members from diverse organisations, including large banks, insurance companies, asset management companies, pension funds, credit rating agencies, and accounting and consulting firms. This diversity ensured recommendations would prove practical across financial services sectors.

The FSB recognised the urgent need for better climate-related financial information to support informed investment, lending, and insurance underwriting decisions. The Task Force was explicitly designed to "promote more informed investment, credit, and insurance underwriting

decisions" and enable stakeholders to "understand better the concentrations of carbon-related assets in the financial sector and the financial system's exposures to climate-related risks."

In June 2017, the TCFD released its final recommendations report, establishing what became known as the TCFD framework. The Task Force issued 11 voluntary recommendations providing a structured approach for companies and financial institutions to disclose climate-related financial information transparently.

Four Core Pillars

The TCFD framework organises around four thematic areas representing core elements of organisational operations:

- **Governance** addresses how the organisation's board and management oversee climate-related risks and opportunities. This includes board-level responsibility for climate risk, management's role in assessing and managing climate risks, and organisational processes for informing the board about climate risks.
- **Strategy** examines climate-related risks and opportunities' actual and potential impacts on the organisation's businesses, strategy, and financial planning. This pillar requires describing climate risks and opportunities over the short, medium, and long term, detailing impacts on business, strategy, and financial planning, and articulating strategy's resilience under different climate scenarios.
- **Risk Management** involves the methods used by the organisation to identify, evaluate, and handle climate-related risks. This includes procedures for recognising and assessing climate risks, strategies for managing these risks, and integrating climate risk procedures into the overall risk management framework.
- **Metrics and Targets** reviews the metrics and targets for evaluating and managing relevant climate-related risks and opportunities. Organisations should disclose metrics used for assessing climate risks in accordance with strategy and risk management processes, greenhouse gas emissions (Scope 1, 2, and 3), and targets used to manage climate risks and measure performance against those targets.

Key Principles and Universal Application

The TCFD recommendations are built on fundamental principles guiding adequate climate-related financial disclosures:

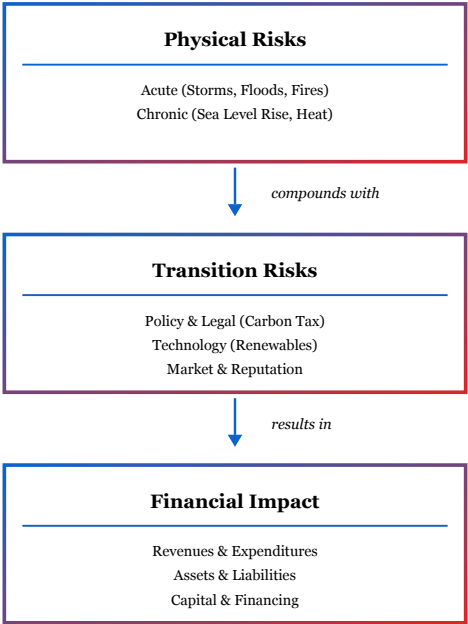
- **Decision Usefulness:** Climate-related financial disclosures should provide relevant information to help investors and stakeholders make informed financial decisions. The information should be material, clear, and actionable rather than comprehensive or voluminous.
- **Consistency and Comparability:** Organisations should provide disclosures that are consistent over time and comparable among companies within sectors. This enables stakeholders to track

performance and benchmark against industry standards, supporting capital allocation decisions.

Reliability and Verifiability: Disclosed information should be supported by appropriate internal governance processes, robust methodologies, and reliable data, with the aim of ensure it can be independently verified.

The TCFD recommendations apply universally across all sectors and industries, making them broadly applicable while allowing for sector-specific implementation through complementary frameworks. This comprehensive approach has led to widespread global adoption, with over 120 regulators and governments becoming TCFD supporters and many jurisdictions incorporating TCFD-aligned disclosure requirements into legislation.to

Figure 35 – TCFD Risk Classification



Scenario Analysis: The Critical Challenge

The most challenging TCFD requirement involves scenario analysis under the Strategy pillar. Organisations must describe the resilience of their strategy under different climate scenarios, including a 2°C or lower scenario. This requirement forces institutions to grapple with uncertainties about future climate pathways and their business implications.

Scenario analysis typically employs IPCC scenarios (RCPs or SSPs) combined with Integrated Assessment Models, translating climate pathways into economic implications. Banks must assess how different warming scenarios affect credit risk (borrower default probabilities), asset values (collateral valuations), and strategic positioning (market opportunities and threats).

One aspect of TCFD that is underappreciated is its readability and thoughtfulness. Many supervisory bodies have adopted their core recommendations relatively unchanged and certainly unchallenged into legal frameworks.

CLIMADA: Architecture and Methodology

This requirement drives demand for tools like CLIMADA (CLIMate ADaptation) that enable quantitative scenario analysis across portfolios. However, as we will explore, significant methodological challenges remain in translating global climate scenarios into property-level risk assessments suitable for banking applications. The weather prediction approaches described throughout this chapter represent one path toward physical risk assessment in banking. However, understanding the broader landscape of climate risk modelling frameworks—particularly those emerging from academic research—provides essential context for evaluating different methodologies and their appropriate applications.

The past decade has witnessed remarkable growth in open-source, inter-governmental-backed physical climate risk assessment frameworks. These tools, developed primarily by research institutions and made freely available to the global community, serve complementary purposes to proprietary solutions. Understanding where academic frameworks excel and where they fall short for banking applications enables institutions to make informed decisions about tool selection and integration strategies.

The Rise of Academic Climate Risk Frameworks

Academic frameworks for physical climate risk assessment emerged from a fundamental need: the scientific community required transparent, reproducible methodologies for quantifying the implications of climate change. Traditional catastrophe models developed by the insurance industry, while operated as proprietary black boxes—their methodologies hidden behind commercial confidentiality. This opacity and specific use cases - notably catastrophe insurance - prevented limited academic advancement and created barriers for developing nations.

The response came through several major academic initiatives. ETH Zurich's Weather and Climate Risks Group developed CLIMADA, quickly becoming the leading open-source framework for probabilistic climate risk assessment. FEMA created HAZUS for United States-focused multi-hazard analysis. Regional initiatives like CAPRA (Central America Probabilistic Risk Assessment) and New Zealand's RiskScape addressed specific geographic needs with locally calibrated methodologies.

These frameworks share common characteristics that distinguish them from commercial alternatives. Complete transparency enables any researcher to examine methodology and

replicate results. Peer-reviewed foundations ensure scientific rigour through publication in academic journals. Global consistency allows cross-regional comparisons using standardised approaches. Community development leverages collective expertise for continuous improvement. Most importantly for banking applications, regulatory acceptance of peer-reviewed methodologies provides credibility for compliance purposes.

CLIMADA represents the most comprehensive and widely adopted academic physical climate risk assessment framework. Developed at ETH Zurich and supported by over 170 scientific citations, CLIMADA provides a globally consistent platform integrating climate science with economic impact modelling. Understanding its architecture illuminates the capabilities and limitations of academic approaches to risk assessment.

The framework implements a fully probabilistic risk assessment model based on the IPCC's foundational definition: Risk equals Hazard multiplied by Exposure multiplied by Vulnerability. This deceptively simple relationship masks considerable sophistication in how each component is modelled and how they interact.

CLIMADA: An Event-Based Probabilistic Framework

CLIMADA employs an event-based probabilistic approach that is fundamentally different from the pattern-based methodology described earlier in this chapter. Rather than characterising weather pattern distributions as the MKM Framework does, CLIMADA generates stochastic events by analysing the frequency and intensity characteristics of historical events. This represents a fundamental shift in how climate risk is conceptualised and quantified.

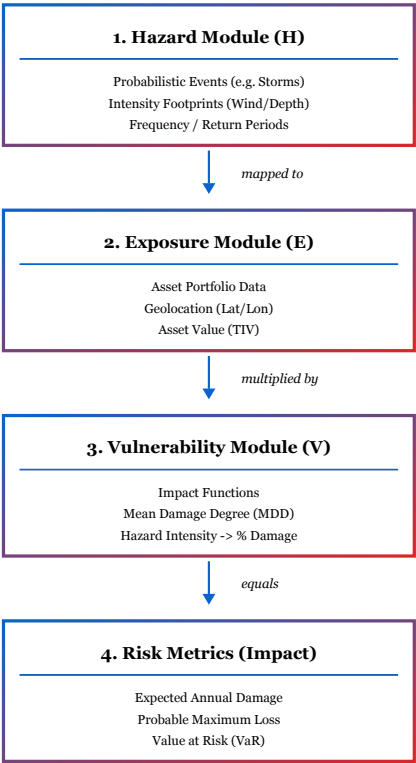
The Event-Based Probabilistic Paradigm

The event-based probabilistic framework represents a modelling paradigm where each potential hazard occurrence is treated as a discrete event with specific physical characteristics and an associated probability of occurrence. In this approach, the system creates a comprehensive catalogue of possible hazard events, assigning each event a specific rate or probability of occurrence rather than relying on aggregate statistical distributions.

This methodology differs markedly from deterministic models, which only quantify the impact of specific scenarios without describing the full range of possible future events, and from purely statistical approaches that might use predefined probability distributions without grounding in physical event characteristics. The probabilistic nature means that instead of producing a single predicted outcome, the model generates a probability distribution as its solution, explicitly acknowledging the inherent randomness and uncertainty in predicting future hazardous events.

The power of this methodology lies in its ability to construct stochastic event sets—large ensembles of synthetic events, often exceeding 10,000 individual events, that capture not only what has been historically observed but also what is physically plausible based on the underlying atmospheric and geophysical processes. These event sets extend beyond the limitations of the

Figure 36 – CLIMADA Modelling Framework



historical record, which may span only decades or centuries, to encompass the full spectrum of events that could occur given our understanding of the physical systems involved.

Synthetic Event Generation and Historical Data

The methodology begins with historical hazard data as its foundation. CLIMADA draws upon track data from IBTrACS (International Best Track Archive for Climate Stewardship) for tropical cyclones, which provides comprehensive global records of storm positions, intensities, and characteristics. For flood hazards, the framework leverages simulations from ISIMIP (Inter-Sectoral Impact Model Intercomparison Project). At the same time, wildfire modelling incorporates records from global fire datasets that document historical burn patterns, intensities, and frequencies.

For each historical event in these datasets, CLIMADA creates ensembles of probabilistic events that maintain the statistical properties of the historical record while exploring variations in track, intensity, timing, and spatial extent. This synthetic event generation process is not merely a random perturbation of historical events. Instead, it employs statistical and physical models that understand the constraints and relationships governing hazard behaviour. For instance, a synthetic tropical cyclone must respect the relationship between sea surface temperatures and

maximum potential intensity, the influence of wind shear on storm development, and the typical tracks associated with large-scale atmospheric circulation patterns.

The result is a catalogue of thousands of possible events, each with an assigned probability of occurrence. For each synthetic event in the set, CLIMADA calculates the spatial distribution of hazard intensity and combines this with exposure and vulnerability information to determine potential impacts. The events are generated by analysing patterns in historical data—such as the tracks, intensities, and frequencies of past events—and then using this understanding to create additional synthetic events that respect the same physical constraints and statistical characteristics. This allows the model to include rare but consequential events that may not have occurred in the limited historical record but remain physically possible.

River Flooding Methodology

CLIMADA has a methodology for each hazard. Here we will focus solely on river flood.

River flood modelling leverages ISIMIP simulations driven by multiple Global Climate Models under different emissions scenarios, providing global coverage at approximately 4-kilometre resolution. These simulations incorporate similar representations of hydrological processes, including precipitation, snowmelt, infiltration, and runoff routing through river networks. By using ensembles of climate models, CLIMADA can capture uncertainty in future climate projections and assess how flood risk might evolve under different greenhouse gas concentration pathways.

The ISIMIP framework provides pre-computed hazard intensity maps for various return periods, which CLIMADA integrates into its probabilistic event sets. This allows the framework to assess flood risk under both current and future climate conditions, accounting for how changing precipitation patterns and hydrological regimes might alter flood frequencies and intensities.

Risk Metrics and Decision Support

By assigning occurrence rates to each event in the stochastic set, the framework can compute critical risk metrics such as annual exceedance probabilities and return periods for different loss levels. The rate of exceedance for any given impact threshold is calculated by summing the occurrence rates of all events that exceed that threshold, then converting these rates into probabilities that inform risk management decisions.

For example, to calculate the 100-year return period loss (the loss level that has a 1% probability of being exceeded in any given year), CLIMADA identifies all events in the stochastic set that produce losses above various thresholds, sums their occurrence rates, and finds the loss level where this cumulative rate equals 0.01 per year. This event-by-event accounting provides granular insights into how different hazard scenarios contribute to overall risk, enabling more nuanced analysis than pattern-based approaches that aggregate conditions into broader categories.

This granularity supports better risk management applications. Decision-makers can explore questions such as: Which specific event types drive tail risk? How does risk change if certain protective measures are implemented? What is the probable maximum loss at different confidence levels? The probabilistic framework also facilitates cost-benefit analysis of adaptation measures by quantifying risk reduction in monetary terms across the full probability distribution.

Framework Advantages and Considerations

The event-based probabilistic approach offers several distinctive advantages. It provides a natural framework for incorporating physical understanding of hazards while maintaining computational tractability for global applications. It explicitly quantifies uncertainty and rare events that purely statistical approaches might miss. And it generates outputs—such as exceedance probability curves—that directly support decision-making in insurance, infrastructure planning, and climate adaptation.

However, the framework's global scope necessitates compromises. Local-scale processes may be inadequately represented, parametric models may oversimplify complex physical phenomena, and the assumption that historical relationships will persist into the future becomes increasingly uncertain as the climate system moves into unprecedented states. These considerations do not invalidate the approach but highlight the importance of understanding its assumptions and appropriate applications.

The Exposure Component

Exposure representation in CLIMADA centres on the LitPop methodology—an approach combining nightlight satellite data with GDP and population statistics to estimate asset values at high resolution globally. This technique proves remarkably effective for portfolio-level analysis where precise individual property values matter less than understanding aggregate exposure across regions.

The framework structures exposure using standard categories: residential buildings, commercial properties, infrastructure assets, and agricultural outputs. Each category is assigned suitable valuation methods and damage functions. For banking purposes, the capacity to quickly evaluate exposure across entire mortgage portfolios or infrastructure loan books offers immediate benefit.

However, the LitPop methodology's reliance on proxy data introduces systematic biases. Nightlight intensity imperfectly correlates with property values—historic districts with strict lighting regulations appear undervalued, while industrial zones may appear overvalued. GDP downscaling assumes uniform wealth distribution within administrative regions, missing the stark variations evident in actual property markets. These limitations matter most when analysing individual properties or small portfolios where proxy errors cannot average out across large samples.

The Vulnerability Component

Vulnerability functions—often called damage curves or impact functions—translate hazard intensity into expected losses. CLIMADA maintains libraries of impact functions derived from empirical post-disaster data, engineering models, and expert judgment. A typical flood impact function might indicate that one meter of water depth causes 40% damage to residential structures, 60% damage to contents, and requires four months for recovery.

These functions vary by building type, construction era, occupancy class, and geographic region. However, CLIMADA generally employs relatively generic impact functions due to limited availability of detailed empirical damage data globally. This genericity represents a conscious trade-off: broader applicability with less local specificity versus higher accuracy with limited geographic coverage.

The framework allows users to customise impact functions based on local knowledge or proprietary data—an essential capability for banking applications where institution-specific vulnerability assessments may improve accuracy. Many bank implementations combine CLIMADA's hazard modelling with proprietary vulnerability functions calibrated to their specific portfolio characteristics.

Risk Calculation and Metrics

CLIMADA computes risk through matrix multiplication across events, locations, and impact functions. For each of thousands of synthetic events, the model calculates impacts at each exposure location, producing detailed loss distributions. These distributions enable the calculation of critical risk metrics that banking institutions require.

- Expected Annual Impact (EAI) represents the probability-weighted average loss at each location—essentially the actuarial fair premium for perfect insurance.
- Average Annual Impact (AAI) aggregates EAI across entire portfolios, providing top-level loss expectations for strategic planning.
- Probable Maximum Impact (PMI) quantifies tail risk by identifying losses exceeded with specified low probabilities, typically 1-in-250-year or 1-in-1000-year return periods corresponding to regulatory stress scenarios.

Impact exceedance curves visualise complete loss distributions, showing how frequently different loss levels might be exceeded. These curves are essential for pricing risk transfer instruments, setting risk appetite thresholds, and communicating risk profiles to senior management and boards of directors.

CLIMADA in Banking Applications: Strengths and Limitations

The past five years have witnessed rapid adoption of CLIMADA-based methodologies within banking, driven primarily by regulatory requirements for climate risk assessment and disclosure—

understanding where CLIMADA excels and where it falls short illuminates appropriate use cases and necessary complementary approaches.

Where CLIMADA Falls Short

- **Resolution Limitations:** CLIMADA's typical 4-kilometre resolution proves insufficient for property-level underwriting decisions. Urban neighbourhoods separated by a few hundred meters may face dramatically different flood risks due to local drainage, elevation differences, or protective infrastructure—variations invisible at 4km resolution. While the framework technically supports finer resolutions, computational requirements increase exponentially, and data availability constrains practical implementation.
- **Static Vulnerability Functions:** Impact functions in CLIMADA generally remain static over time, failing to reflect evolving building codes, adaptation measures, or changing construction practices. A property improved with flood-resistant materials receives the same vulnerability assessment as an unimproved property of similar type—missing crucial risk reductions that matter for pricing and capital allocation.
- **Generic Impact Functions:** Limited empirical damage data globally forces CLIMADA to employ relatively generic vulnerability functions. These may substantially misestimate losses in specific contexts—Mediterranean construction differs from Northern European, affecting flood vulnerability; Japanese earthquake engineering produces different damage curves than Turkish construction.
- **Compound and Cascading Risks:** While CLIMADA handles multiple hazards independently, modelling simultaneous or cascading events remains challenging. A hurricane bringing both wind and flood damage, followed by prolonged power outages affecting recovery, involves complex interactions not fully captured by independent hazard modelling.
- **Calibration and Validation Gaps:** CLIMADA validation studies demonstrate damage estimates "correct within an order of magnitude for 63% of past events"—impressive for a global academic model but insufficient precision for pricing individual mortgages or setting property-specific insurance premiums. The framework excels at identifying high-risk regions but struggles with fine-grained distinctions within those regions.
- **Computational Requirements:** Uncertainty analysis using CLIMADA's `unsequa` module requires substantial computational resources. Comprehensive uncertainty quantification for a large portfolio might require thousands of model runs—feasible for research projects or annual strategic assessments but impractical for near-real-time risk monitoring or transaction-level analysis.

The Broader Open-Source Landscape

While CLIMADA represents the most comprehensive academic framework, several other open-source tools serve specific niches within physical risk assessment. Understanding this broader landscape enables institutions to select appropriate tools for different applications.

- **HAZUS (FEMA):** This agency focuses exclusively on the United States and provides exceptional detail for domestic applications. It covers earthquakes, floods, hurricanes, and tsunamis, calibrating region-specific vulnerability functions to U.S. building codes and construction practices. This agency provides standard methodology for U.S. disaster planning and insurance applications but offers no international coverage.
- **CAPRA (ERN-AL):** Developed for Latin America and the Caribbean, with exceptional calibration to regional building practices, hazard characteristics, and socioeconomic conditions. The probabilistic approach is similar to CLIMADA but optimised for particular regional contexts. Limited adoption outside Latin America due to regional specialisation.
- **RiskScape (New Zealand):** Multi-hazard framework with particular strength in earthquake and volcanic risk. Modular architecture similar to CLIMADA but with deeper integration to New Zealand-specific data sources and validation against local events. Demonstrates the importance of regional specialisation.

The common pattern across successful open-source frameworks: deep validation within specific geographic or hazard domains, transparency enabling independent verification, and active research communities continuously improving methodologies. The challenge for global banks: integrating multiple regional frameworks or accepting reduced accuracy from globally consistent but regionally generic approaches.

Integration with Weather Prediction: A Unified Perspective

Weather patterns inform operational decisions about near-term risks to specific assets. The MKM Framework's probabilistic approach to weather pattern distributions enables precise quantification of flood risk over days to weeks—critical for underwriting decisions, collateral valuations, and portfolio risk management. High temporal and spatial resolution proves essential for these applications.

Climate scenarios inform strategic decisions about portfolio composition, long-term capital allocation, and regulatory compliance. Academic frameworks like CLIMADA enable scenario analysis under different warming pathways, supporting TCFD disclosures and supervisory stress tests. Consistency across portfolios and time periods proves more important than property-level precision for these applications.

Neither perspective is sufficient on its own. Weather prediction without considering climate context overlooks long-term trends and structural changes in risk environments. Climate modelling without short-term weather forecasts offers inadequate detail for operational risk

management. Banking institutions need both operational accuracy for immediate decisions and strategic consistency for long-term planning.

However, climate scenarios inform how flood risk evolves over mortgage terms and investment horizons. Under certain climate pathways, a property with acceptable flood risk today might face materially higher risk in 20 years. Combining near-term weather-based flood forecasting with long-term climate-informed risk trajectories provides a comprehensive assessment supporting both operational decisions and strategic planning.

The Practical Reality: Data, Models, and Decisions

Understanding the theoretical distinctions between weather and climate proves necessary but insufficient. Banking institutions must navigate the practical realities of data availability, model limitations, regulatory requirements, and commercial constraints.

The Resolution Challenge

This resolution gap cannot be eliminated through simple downscaling. Physical processes operating at meter scales (local drainage, building characteristics, protective infrastructure) involve fundamentally different mechanisms than processes operating at kilometre scales. Empirical downscaling relationships calibrated to historical periods may break down under climate change. Dynamic downscaling through nested regional models remains computationally prohibitive for routine banking applications.

The practical solution involves hybrid approaches: global climate models for long-term scenarios, regional weather models for near-term forecasting, and local-scale augmentation using terrain data, building information, and historical damage observations. Each scale contributes different information; integration requires careful attention to consistency and uncertainty propagation.

The Uncertainty Communication Challenge

Climate scientists carefully communicate uncertainty through ensemble projections, confidence intervals, and likelihood assessments ("likely," "very likely," "virtually certain"). This approach suits scientific assessment but proves awkward for banking decisions requiring binary outcomes—approve or deny the mortgage? Accept or reject the collateral?

Weather forecasters face similar challenges but have developed pragmatic solutions: probabilistic forecasts (30% chance of rain) that quantify uncertainty while supporting decisions. Banking risk management requires similar approaches: probability distributions over possible losses rather than single-point estimates.

Weather prediction and climate modelling are distinct scientific enterprises that address different questions through different methodologies. Weather models forecast near-term atmospheric states with high spatial and temporal resolution. Climate models project long-term statistical

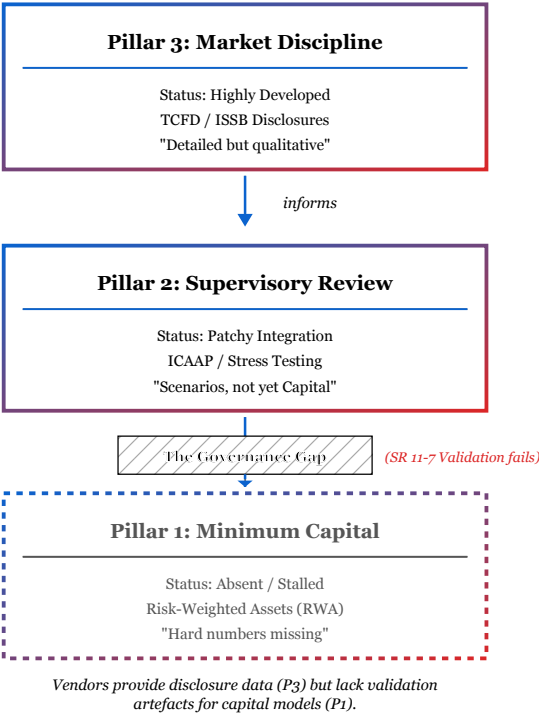
properties of climate systems under different forcing scenarios. Both prove essential for comprehensive physical risk assessment in banking.

The IPCC provides a scientific foundation for understanding climate change—its causes, consequences, and potential responses. These assessments inform international policy and establish the context within which banking institutions operate. The TCFD translates climate science into practical frameworks for financial institutions, requiring disclosure of climate governance, strategic resilience, risk management processes, and performance metrics.

Academic frameworks like CLIMADA occupy a specific niche within this ecosystem—enabling scenario analysis and regulatory compliance through peer-reviewed, transparent methodologies. These tools excel at portfolio-level assessment and strategic planning but prove inadequate for property-level operational risk management.

Specialised frameworks like MKM address complementary needs—operational precision for underwriting, pricing, and portfolio management. These tools optimise for different objectives: near-term forecast accuracy, property-level granularity, and integration with banking systems rather than scientific credibility for regulatory compliance.

Figure 37 – The Basel Pillar Progression Paradox



As we move on to hydrological modelling in the next chapter, understanding different frameworks and their suitable applications remains valuable. Flood risk assessment involves combining weather forecasting (short-term rainfall), terrain analysis (local hydrology), and climate context (long-term risk changes)—the very multi-scale integration that defines physical risk assessment in banking.

The Basel Pillar progression paradox

The Basel three-pillar design assumes a progression from transparency to capital: Pillar 3 disclosures inform Pillar 2 supervisory review, which in turn shapes Pillar 1 minimum capital requirements. For physical climate risk, this sequence has stalled: disclosures have become detailed, but Pillar 2 integration is patchy and no Pillar 1 standard yet uses vendor hazard chains in a capital-relevant way.

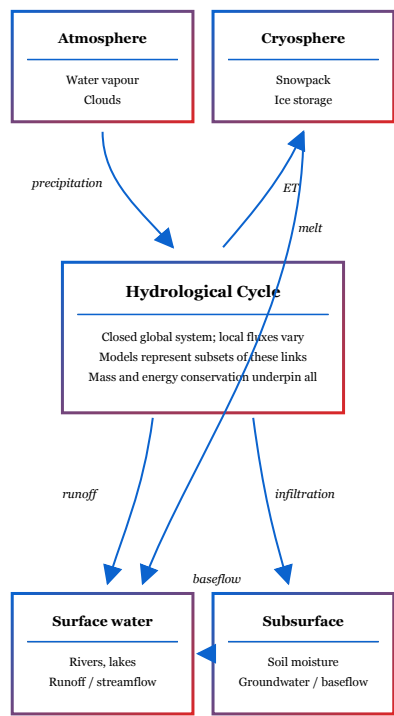
A key reason, explored in depth in the chapter on Model Risk Governance, is the governance gap: most physical risk vendors can support climate reporting, but cannot yet provide the transparency, limitations-of-use, and validation artefacts that SR 11-7 style frameworks require for capital and pricing decisions. Until that gap is closed, physical risk will remain highly visible in Pillar 3 templates but largely absent from the hard numbers that drive risk appetite and capital.

Chapter 5 - Hydrological Modelling

Assessing physical risk in water-related hazards fundamentally depends on our ability to understand and model how water moves through the environment. Hydrological modelling is the critical bridge between weather events and their impacts on the flood risk assessment. It translates meteorological data into actionable insights about flood risks, water scarcity, and other hazards. This chapter explores the complex world of hydrological modelling, examining how different approaches can simulate water movement through natural and built environments.

"All hydrological models, regardless of their complexity or simplicity, are ultimately expressions of the same fundamental physical laws established by Newton. We may dress them differently, adapt their form to various scales and purposes, but beneath these surface variations lie the immutable principles of mass and energy conservation." - David Kelly, MKM.

Figure 38 - The Hydrological Cycle (stores and transfers)



Hydrological modelling is grounded in fundamental physical principles - primarily Newton's laws of motion and thermodynamics. These universal physical laws govern how water moves and changes state throughout its journey, from the moment it falls as precipitation until it reaches its ultimate destination, whether that is the ocean, groundwater, or the atmosphere through evaporation.

While models may vary significantly in complexity and approach, they all ultimately derive from these same physical principles. This process involves multiple interconnected components: precipitation patterns, surface runoff, infiltration into the soil, groundwater movement, and channel flow dynamics. The accuracy of physical risk assessment depends critically on how well we can model these processes.

Hydrological modelling has evolved significantly with advances in computational power and data availability, though all approaches - from the simplest to the most complex - remain fundamentally rooted in classical physics. Whether a model is lumped, distributed, empirical, or physically based, we can easily trace their underlying equations to the same core physical principles of mass, energy, and momentum conservation.

Tracing back to Newton is a crucial aspect of such modelling. The intellectual property of those who model such outcomes for their insurance clients does not stem from their original creation of the models, as these models originate in academia and are therefore in the public domain. The intellectual property should align with that of banks and their instrument pricing models, which are based on implementation rather than solely on accuracy, granularity, and scalability.

Early models were necessarily simple, treating entire watersheds as single units with uniform characteristics. Modern approaches can incorporate detailed spatial variations in terrain, land use, and soil properties, thereby enabling more accurate predictions of water movement and accumulation.

Hydrological Model Overview

Hydrological models can be classified along several dimensions, each offering different advantages for specific applications:

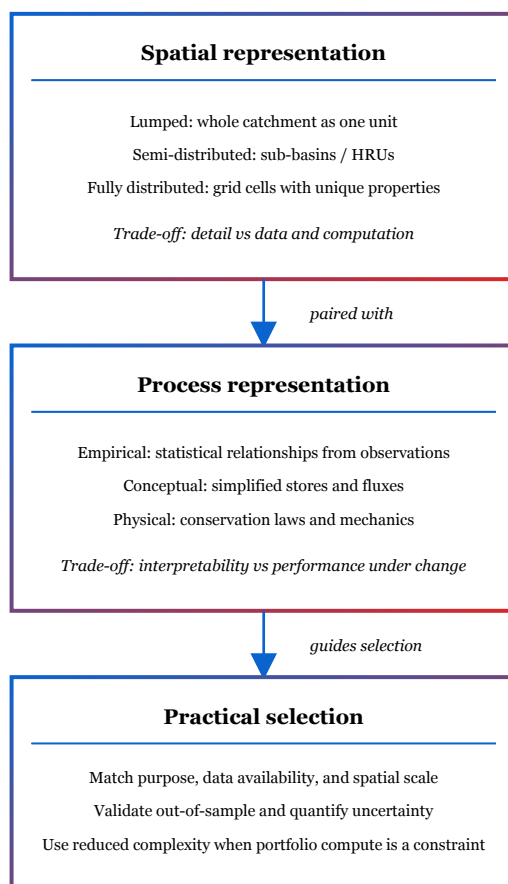
Spatial Representation

- **Lumped Models:** These models treat the watershed as a uniform unit with uniform characteristics. While computationally efficient and requiring minimal data, they sacrifice spatial detail in favour of simplicity. They remain valuable for rapid assessments and preliminary analyses.
- **Semi-Distributed Models:** Representing a middle ground, these models divide watersheds into sub-basins, each with its own characteristics. This approach balances computational efficiency with the need to represent spatial variability. It makes them particularly useful for medium—to

large watersheds where some spatial detail is essential but complete distribution isn't necessary.

- **Fully Distributed Models:** These models represent the ultimate level of spatial detail, dividing the study area into a fine grid in which each cell can have unique properties. While computationally intensive, they provide the most detailed representation of spatial processes and are essential for applications requiring high spatial resolution.

Figure 39 - Hydrological model classification



Process Representation

The way models represent physical processes varies significantly, reflecting different philosophical approaches to hydrological modelling.

- **Empirical Models:** Built on statistical relationships derived from observed data, these models excel in stability and simplicity but may struggle when conditions differ from their training data. They serve well in operational settings where quick, reliable results are needed for familiar conditions.
- **Conceptual Models:** These models balance empirical and physical approaches, representing key processes through simplified equations. They maintain physical meaning while remaining computationally manageable, making them popular for many practical applications.
- **Physical Models:** These models offer the most robust theoretical foundation. They demand data and computation based on fundamental water movement and conservation equations. Still, they provide the best framework for understanding process changes under novel conditions.

When integrating backtesting into hydrological model applications for physical risk assessment, it becomes critical to address the following enhanced considerations:

- **Robustness to non-stationarity:** Backtesting against historical extreme floods helps validate model performance under shifting climatic regimes. Traditional backtesting assumes stationarity, so newer frameworks combine historical validation with stress testing against synthetic extremes beyond the bounds of observational records.
- **Uncertainty quantification:** Backtesting enables empirical estimation of prediction intervals by analysing model predictive capability across diverse historical scenarios. This directly informs risk pricing through metrics like reliability, sharpness, and skill scores.
- **Computational efficiency:** While backtesting requires intensive scenario runs, reduced-complexity models can maintain usability if they demonstrate consistent skill across validation periods, stability in parameter transferability, and scalable performance when applied to large portfolios of risk locations.
- **Implementation challenges:** Long-term observational datasets are a prerequisite for meaningful backtesting, non-stationarity may render older events less representative of future hazards, and trade-offs exist between model complexity and computational demands for ensemble backtesting.

The Hydrological Cycle

The hydrological cycle represents the continuous circulation of water through Earth's atmosphere, land surface, and subsurface systems. This perpetual, solar-driven process operates as a closed system with no net inputs or outputs at the global scale, though water constantly moves between different reservoirs and undergoes phase changes. Understanding this cycle is fundamental to hydrological modelling, as every model—regardless of complexity—attempts to represent some portion of these interconnected processes.

The cycle comprises two principal phases. The atmospheric phase describes water movement as gas (water vapour) and liquid or solid forms (rain, snow, hail, sleet) in the atmosphere. The terrestrial phase describes water movement in, over, and through the Earth, further subdivided into the surface water phase (runoff and streamflow) and the groundwater phase (subsurface storage and flow).

Storage Components

Water resides in several major reservoirs within the hydrological cycle. Oceans contain approximately 86% of global water and serve as the source for 86% of global evaporation. Terrestrial storage includes rivers, lakes, and reservoirs on the surface; soil moisture and vadose zone water in the unsaturated subsurface; groundwater in aquifers; snow and ice in cryospheric regions; and atmospheric water vapour and clouds. Each reservoir has characteristic residence times ranging from days for atmospheric moisture to thousands of years for deep groundwater. For banking applications, understanding these residence times proves crucial—flood risk responds to short-residence-time components (surface water, atmospheric moisture), while drought risk relates more to longer-residence-time components (groundwater, snowpack).

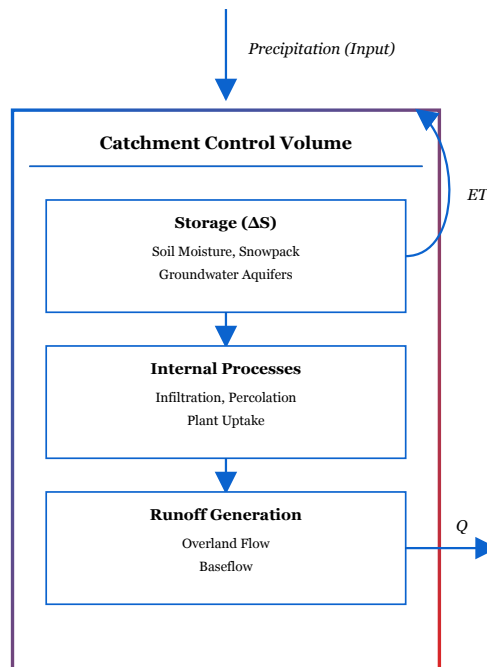
Transfer Processes

Water moves between reservoirs through several interconnected processes:

- Evaporation transforms liquid water into water vapour using energy from the sun, occurring primarily from ocean and land surfaces. This phase change absorbs latent heat of vaporisation, playing a critical role in energy transfer from the tropics toward the poles.
- Transpiration releases water vapour from plants and soil into the atmosphere. Plants absorb water through their roots, use it for photosynthesis, and release excess water through stomata as water vapour. The combined process of evaporation and transpiration—termed evapotranspiration—represents a major output from terrestrial systems.
- Condensation occurs when water vapour cools and changes back into liquid form, forming clouds in the atmosphere. This process releases latent heat, contributing to atmospheric dynamics and energy distribution.
- Precipitation represents water falling to Earth's surface in various forms including rain, snow, sleet, and hail. Precipitation can occur over land or bodies of water and represents the primary input to terrestrial hydrological systems.
- Infiltration describes the process by which water enters the soil surface. The infiltration capacity—the maximum rate at which soil can absorb water—depends on soil properties, including texture, structure, and antecedent moisture conditions.
- Percolation involves water moving vertically downward through the soil profile and underlying rocks under the influence of gravity. This process replenishes groundwater reservoirs.

- Surface runoff (also called overland flow) occurs when excess water that does not infiltrate the ground flows over the surface, forming streams and rivers that eventually reach the ocean. Runoff generation depends on rainfall intensity, duration, soil infiltration capacity, and landscape characteristics.
- Throughflow represents the lateral movement of water through the soil matrix, typically occurring in the upper soil layers above less permeable horizons. This subsurface flow can contribute to stream discharge without appearing as surface runoff.
- Groundwater flow describes water movement through the saturated zone under the influence of gravity and pressure gradients. Groundwater moves slowly and can remain in aquifers for thousands of years before discharging to springs, streams, or directly to the ocean. Groundwater discharge sustains baseflow in perennial streams.

Figure 40 - Water Balance: $P = Q + ET + \Delta S$



Mass Balance: $\text{Inputs} = \text{Outputs} + \text{Storage Change}$

Water Balance Principles

The water balance equation represents a fundamental application of the law of conservation of mass to hydrological systems. For any defined area or water system, the inflows must equal the outflows plus any change in storage during a specified time interval. This principle provides the

foundation for quantitative hydrology and water resources assessment—and, crucially for banking applications, for flood risk quantification.

The general water balance equation for a watershed or basin is expressed as:

$$P = Q + ET + \Delta S$$

Where:

- P is precipitation
- Q is streamflow or runoff
- ET is evapotranspiration
- ΔS represents the change in storage (in soil, groundwater, surface water bodies, or snow).

This equation can be expanded to include additional terms for specific applications, such as irrigation inputs, groundwater exchanges between basins, or water withdrawals for human use.

Water balance analysis serves multiple critical functions in hydrology and water resources management. It enables assessment of water availability for various uses, evaluation of the impacts of climate variability and land use change on water resources, comparison of hydrological characteristics across different watersheds, and quantification of individual hydrological processes that may be difficult to measure directly.

Evapotranspiration Estimation

Evapotranspiration (ET) represents a major component of the water balance, often accounting for 60-90% of precipitation in many watersheds. Accurate ET estimation is essential for water resources management, irrigation scheduling, and hydrological modelling. Numerous methods have been developed to estimate ET, ranging from empirical temperature-based equations to physically based energy balance approaches.

The Penman-Monteith Equation

The Penman-Monteith equation represents the most widely accepted physically based method for estimating reference evapotranspiration. This combination equation considers both the energy available for evaporation and the aerodynamic processes that remove water vapour from the evaporating surface. The Food and Agriculture Organisation (FAO) recommends the FAO Penman-Monteith method as the standard for computing reference evapotranspiration. The method requires meteorological data including solar radiation, air temperature, humidity, and wind speed.

The FAO Penman-Monteith equation for reference evapotranspiration is:

$$ET_0 = [0.408\Delta(R_n - G) + \gamma(900/(T + 273))u_2(e_s - e_a)] / [\Delta + \gamma(1 + 0.34u_2)]$$

Where:

- ET_0 = reference evapotranspiration (mm/day)
- R_n = net radiation at the crop surface (MJ/m²/day)
- G = soil heat flux density (MJ/m²/day)
- T = mean daily air temperature at 2m height (°C)
- u_2 = wind speed at 2m height (m/s)
- e_s = saturation vapour pressure (kPa)
- e_a = actual vapour pressure (kPa)
- $(e_s - e_a)$ = saturation vapour pressure deficit (kPa)
- Δ = slope of the vapour pressure curve (kPa/°C)
- γ = psychrometric constant (kPa/°C)

The 0.408 coefficient converts MJ/m² to mm of water equivalent. The 900 is an empirical coefficient for the aerodynamic term when using daily time steps with a grass reference surface (0.12m height).

Alternative Methods

The Hargreaves equation provides a temperature-based method suitable when climatological data are limited, computing reference crop evapotranspiration on time intervals of one month or longer.

Hargreaves Equation:

$$ET_0 = 0.0023 \times (T_{\text{mean}} + 17.8) \times (T_{\text{max}} - T_{\text{min}})^{0.5} \times Ra$$

Where:

- ET_0 = reference evapotranspiration (mm/day)
- T_{mean} = mean daily air temperature (°C)
- T_{max} = maximum daily air temperature (°C)
- T_{min} = minimum daily air temperature (°C)
- $(T_{\text{max}} - T_{\text{min}})$ = daily temperature range, a proxy for solar radiation
- Ra = extraterrestrial radiation (mm/day equivalent)

The 17.8 constant is an empirical adjustment that improves estimates across different climates. The elegance of this equation lies in requiring only temperature data—making it invaluable for data-sparse regions or historical analyses where only temperature records exist.

The Priestley-Taylor method simplifies the Penman approach by eliminating the aerodynamic term, making it applicable in humid regions where advective effects are minimal.

Priestley-Taylor Equation:

$$ET = \alpha \times (\Delta / (\Delta + \gamma)) \times (R_n - G) / \lambda$$

Where:

- ET = evapotranspiration (mm/day)
- α = Priestley-Taylor coefficient (dimensionless, typically 1.26 for well-watered surfaces)
- Δ = slope of the saturation vapour pressure curve (kPa/°C)
- γ = psychrometric constant (kPa/°C)
- R_n = net radiation (MJ/m²/day)
- G = soil heat flux (MJ/m²/day)
- λ = latent heat of vaporisation (approximately 2.45 MJ/kg)

The α coefficient of 1.26 represents an empirical enhancement factor accounting for the contribution of advection under equilibrium conditions. In arid environments, α may exceed 1.26; in very humid conditions, it approaches 1.0.

Direct Measurement Techniques

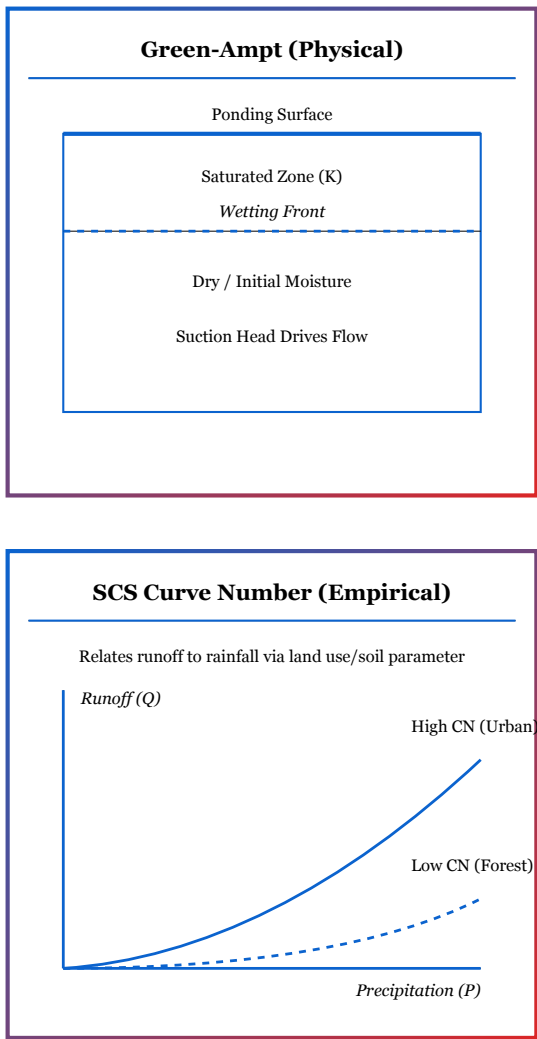
- Lysimeters—containers filled with soil and vegetation that allow precise measurement of water inputs and outputs—provide direct ET measurements and serve as validation standards for other methods. Eddy covariance systems measure turbulent fluxes of water vapour and energy above the canopy, providing high-frequency ET estimates at the field scale. These systems use water vapour analysers and sonic anemometers operating at frequencies faster than 10 times per second.
- Remote sensing approaches for ET estimation have advanced significantly. Methods like the Two-Source Energy Balance (TSEB) model estimate instantaneous ET from thermal and optical remote sensing data, which must then be extrapolated to daily values using approaches such as the evaporative fraction method or the solar radiation approach.

Infiltration Processes

Infiltration—the process by which water enters the soil surface—controls the partitioning between surface runoff and subsurface flow. Infiltration rates depend on soil texture, structure, porosity, hydraulic conductivity, antecedent moisture conditions, vegetation cover, and rainfall

intensity. Understanding and modelling infiltration is fundamental to predicting runoff, recharge, and flood generation—making it a critical process for physical risk assessment.

Figure 41 - Infiltration Approaches



The Green-Ampt Infiltration Model

The Green-Ampt infiltration model provides a physically based approach developed in the early 1900s. The model conceptualises infiltration as a sharp wetting front advancing into the soil, with saturated soil above the front and initial moisture content below. The model assumes Darcy's law

applies to flow through the wetted zone, and the driving force for infiltration is the hydraulic gradient created by the saturated hydraulic conductivity, capillary suction at the wetting front, and depth of ponded water.

The Green-Ampt equation for cumulative infiltration at time t is:

$$F = Kt + \psi \Delta\theta \ln(1 + F/(\psi \Delta\theta))$$

Where:

- K is saturated hydraulic conductivity
- ψ is the matric potential at the wetting front
- $\Delta\theta$ is the difference between saturated and initial moisture content.

The model works best in coarse-textured soils with distinct wetting fronts and when air entrapment, surface crusting, and soil swelling do not substantially influence infiltration.

The SCS Curve Number Method

The SCS Curve Number (CN) method, developed by the USDA Soil Conservation Service (now Natural Resources Conservation Service), provides an empirical approach widely used for infiltration and runoff estimation due to its simplicity and extensive validation. The method is based on water balance principles and relates direct runoff to total rainfall through a dimensionless curve number parameter that accounts for soil type, land use, hydrologic condition, and antecedent moisture conditions.

The fundamental equations of the SCS-CN method are:

$$Q = (P - I_a)^2 / (P - I_a + S)$$

Where:

- Q is direct runoff depth, P is total rainfall
- I_a is initial abstraction (interception, surface storage, and infiltration before runoff begins)
- S is potential maximum retention. The parameter S is related to the curve number by:
$$S = (25400 - 254CN) / CN \quad (\text{in mm})$$

Curve numbers range from approximately 30 for highly permeable soils with high infiltration rates to 100 for water bodies and impervious surfaces. The method includes adjustments for antecedent moisture conditions, with separate CN values for dry (CN I), normal (CN II), and wet (CN III) conditions. Research demonstrates that the Green-Ampt and SCS-CN methods can be analytically linked, allowing transformation from one model to the other.

Groundwater Recharge Modelling

Groundwater recharge—the process by which water infiltrates through the vadose zone to replenish aquifers—represents a critical component of the hydrological cycle and water resources sustainability. Estimating recharge is essential for groundwater management, particularly in regions where groundwater provides primary water supplies. However, direct measurement of recharge is extremely difficult, necessitating indirect estimation methods.

The water table fluctuation (WTF) method infers recharge from groundwater-level rises, requiring only specific yield estimates and groundwater-level data. This method benefits from the abundance of available groundwater-level data but assumes no preferential flow paths and requires accurate specific yield estimates.

Physical-based unsaturated zone models simulate water movement through the vadose zone, calculating recharge as water flux at the bottom of the root zone. These models require detailed soil hydraulic properties, boundary conditions, and initial moisture states.

Transfer function-noise (TFN) models provide an alternative approach that uses groundwater-level data to estimate recharge by modelling the aquifer response to precipitation and evaporation inputs. Research has shown that nonlinear recharge models significantly outperform linear models, particularly under soil moisture-limited conditions.

Snowmelt Processes

In many regions, particularly mountainous areas and high latitudes, snowpack accumulation and melt dominate the hydrological cycle. Snow represents an important hydrological reservoir, with snowmelt providing the primary water input to streams during spring and early summer. Accurate simulation of snow processes is essential for water resources forecasting, flood prediction, and hydropower operations.

Two main approaches exist for snowmelt modelling. Temperature-index (degree-day) methods use air temperature as a proxy for the energy available for melt, providing computational simplicity and modest data requirements. These empirical approaches perform adequately when energy fluxes other than incoming solar radiation have minor effects. However, they cannot explicitly account for rain-on-snow events or variations in shortwave radiation, limiting their applicability in complex terrain.

Energy balance snowmelt models explicitly compute energy fluxes to and from the snowpack, including shortwave and longwave radiation, sensible and latent heat exchange with the atmosphere, heat from precipitation, ground heat flux, and energy released or absorbed during phase changes. The energy balance determines snowpack temperature, liquid water content, and melt rate.

Rainfall-Runoff Modelling

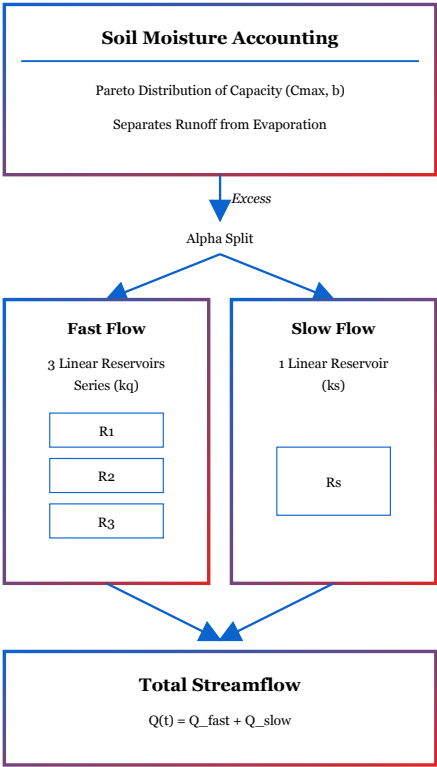
Rainfall-runoff models describe how precipitation is converted into streamflow in a drainage basin, producing a surface runoff hydrograph in response to a rainfall event represented as a hyetograph. These models form the core of most hydrological modelling systems and vary widely in complexity from simple empirical relationships to detailed physically based simulations.

Linear and Non-Linear Reservoir Models

One of the most fundamental rainfall-runoff models represents the watershed as a linear reservoir. The linear reservoir assumption leads to exponential recession of the hydrograph after rainfall ceases. While simple, the linear reservoir has limited applicability and is primarily suited to small catchments with relatively uniform characteristics.

Building on the linear approach, non-linear reservoir models utilise power-law relationships between storage and discharge, offering broader applicability. The model exhibits a delayed response characteristic of many natural watersheds. Calibration establishes the parameters for specific catchments.

Figure 42 - Hymod Model Structure



HYdrological MODel

Here's the expanded Hymod section with the key formulas:

The Hymod Model

The Hymod model provides a flexible conceptual structure that is increasingly adopted for its ability to fit diverse practical applications. Originally proposed by Boyle (2000) and formalised by Wagener et al. (2001), Hymod is a five-parameter lumped rainfall-runoff model with two main components: a nonlinear soil moisture accounting store, and a routing module with one slow and three fast linear reservoirs.

Parameters:

- C_{max} : maximum soil moisture storage capacity
- b : shape parameter controlling spatial variability of storage capacity
- α : fraction of effective rainfall sent to fast reservoirs (remainder to slow reservoir)
- k_q : residence parameter for fast reservoirs (0–1)
- k_s : residence parameter for slow reservoir (0–1)

Soil Moisture Accounting:

Hymod assumes each point in the basin has a local soil water storage capacity that varies spatially according to a probability distribution:

$$F(c) = 1 - (1 - c/C_{max})^{(1+b)}, 0 \leq c \leq C_{max}$$

Where:

$F(c)$ is the fraction of area with capacity $\leq c$.

This distribution determines how rainfall is partitioned between storage replenishment and runoff generation—areas with lower capacity saturate first and contribute to runoff earlier.

Routing Structure:

The model divides effective rainfall into fast and slow runoff components:

$$\text{Fast flow: } Q_{fast} = \alpha \times P_{effective}$$

$$\text{Slow flow: } Q_{slow} = (1 - \alpha) \times P_{effective}$$

Fast runoff is routed through a cascade of three linear reservoirs in series, each with residence parameter k_q , producing the quick-response surface flow. Slow runoff passes through a single linear reservoir with residence parameter k_s , representing delayed baseflow contributions. The total discharge is the sum of outputs from both pathways:

$$Q_t = Q_{\text{fast},t} + Q_{\text{slow},t}$$

This parsimonious structure—only five parameters—captures both quick-response surface flow and delayed baseflow contributions whilst remaining computationally efficient for uncertainty analysis and multi-objective calibration studies.

SCS Unit Hydrograph

The Soil Conservation Service dimensionless unit hydrograph provides a standardised approach for estimating direct runoff timing. Combined with the SCS Curve Number method for computing runoff volume, this integrated approach requires minimal calibration and is widely used for design applications in ungauged watersheds. Parameters include lag time (often estimated from watershed characteristics) and the curve number reflecting soil type, land use, and antecedent conditions.

Major Hydrological Modelling Systems

SWAT (Soil and Water Assessment Tool)

SWAT is a river basin scale model developed by the USDA Agricultural Research Service to quantify the impact of land management practices in large, complex watersheds. As a continuous time, semi-distributed model operating on a daily time step, SWAT simulates weather, surface runoff, return flow, percolation, evapotranspiration, transmission losses, pond and reservoir storage, crop growth and irrigation, groundwater flow, reach routing, nutrient and pesticide loading, and water transfer.

SWAT uses a two-level disaggregation scheme. Watersheds are divided into subbasins based on topographic criteria and stream network configuration. Each subbasin is further discretised into Hydrologic Response Units (HRUs)—areas with homogeneous combinations of soil type, land use, and slope class. This HRU-based approach assumes that areas with identical soil, land use, and slope characteristics have identical hydrologic response, allowing computational efficiency while maintaining some spatial detail.

The model operates using a water balance equation with four storage volumes: snow, soil profile (0-2m depth), shallow aquifer, and deep aquifer. Surface runoff can be estimated using either the SCS Curve Number method for daily simulations or the Green-Ampt infiltration method for subdaily simulations. As of 2012, more than 1,000 peer-reviewed articles had documented SWAT applications worldwide, demonstrating its widespread adoption.

HEC-HMS (Hydrologic Engineering Center - Hydrologic Modeling System)

Developed by the US Army Corps of Engineers, HEC-HMS provides a comprehensive framework for simulating precipitation-runoff processes in both dendritic and non-dendritic watersheds. The model is designed for both event-based and continuous simulation applications, with particular strengths in flood forecasting and design flood estimation.

HEC-HMS represents watersheds as interconnected components including subbasins, reaches, junctions, reservoirs, and diversions. For each subbasin, the user selects methods for computing losses (infiltration), transforming excess precipitation to direct runoff (unit hydrograph or kinematic wave), and simulating baseflow. Common method combinations include the SCS Curve Number loss method with the SCS Unit Hydrograph transform for design applications, or the deficit-constant loss method with Clark Unit Hydrograph for continuous simulation.

A distinctive feature of HEC-HMS is integration with GIS software through HEC-GeoHMS, enabling automated watershed delineation, stream network extraction, and parameter estimation from digital elevation models and spatial datasets.

TOPMODEL (Topography-Based Hydrological Model)

TOPMODEL is a semi-distributed conceptual model using topographic information to predict the extent of contributing areas in runoff production and calculate water storage deficit at any location. Originally developed by Keith Beven and Mike Kirkby in the late 1970s, TOPMODEL aims to reproduce hydrological behaviour in a computationally efficient manner by exploiting the relationship between topography and hydrological response.

The model is based on three main assumptions: (1) the hydraulic gradient of the water table equals the surface slope, (2) the distribution of transmissivity with depth follows an exponential decline, and (3) steady-state conditions prevail. TOPMODEL's key innovation is the topographic index $\ln(a/\tan\beta)$, where a is the upslope contributing area per unit contour length and $\tan\beta$ is the local slope. Points with identical topographic index values are assumed to have similar hydrological behaviour in terms of soil moisture and surface saturation, allowing spatial aggregation.

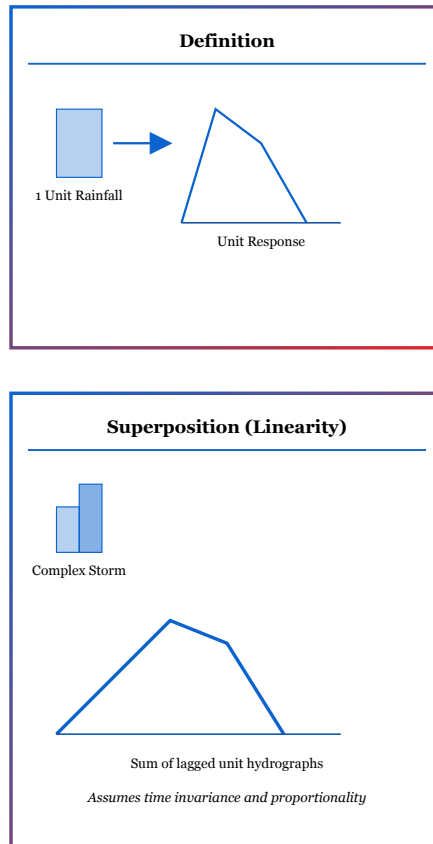
TOPMODEL has proven particularly effective for catchments where topography strongly controls runoff generation, especially in humid climates where saturation-excess overland flow dominates. The model requires relatively few parameters compared to fully distributed models, making it attractive for applications in data-limited settings.

Unit Hydrograph Theory

The unit hydrograph (UH) represents a fundamental concept in hydrology, defined as the direct runoff hydrograph resulting from one unit depth (typically 1 cm or 1 inch) of effective rainfall uniformly distributed over a watershed at a uniform rate during a specified duration. First

proposed by L.K. Sherman in 1932, the unit hydrograph provides a simplified representation of watershed response that remains widely used in hydrological modelling, flood forecasting, and water resources engineering.

Figure 43 - Unit Hydrograph Theory



Fundamental Assumptions

Unit hydrograph theory rests on two principal assumptions:

- **Time Invariance:** The direct runoff hydrograph from a given pattern of effective rainfall in a drainage basin is always the same regardless of when the rainfall occurs. This implies that basin characteristics controlling runoff generation remain constant over time, even though this assumption may be violated seasonally due to vegetation changes, frozen ground, or variations in antecedent moisture.

- **Linearity and Proportionality:** The ordinates of direct runoff hydrographs for the same basin and storm duration are directly proportional to the total amount of direct runoff represented by each hydrograph. If 2 cm of effective rainfall falls instead of 1 cm, the resulting hydrograph ordinates will be exactly twice those of the unit hydrograph. This linear response assumption also enables superposition—hydrographs from successive rainfall bursts can be lagged and added to construct the response to complex rainfall patterns.

Construction and Application

Deriving a unit hydrograph from observed rainfall-runoff data involves several steps. First, an isolated storm with relatively uniform distribution over the watershed is selected from historical records. The corresponding streamflow hydrograph is analysed to separate baseflow from direct runoff using graphical or analytical techniques. The direct runoff hydrograph is then adjusted to represent response to exactly one unit depth of effective rainfall by dividing all ordinates by the depth of effective rainfall that produced the observed hydrograph.

Geospatial unit hydrograph (GUH) methods use digital elevation models to compute travel time distributions from each grid cell to the watershed outlet, creating synthetic unit hydrographs without requiring observed data. Such approaches enable unit hydrograph estimation for ungauged watersheds purely from topographic and landscape characteristics.

Model Calibration, Validation, and Uncertainty

Model calibration refers to the process of adjusting model parameters to achieve optimal agreement between simulated and observed outputs, typically streamflow. Calibration represents an inverse problem—inferring parameter values from system observations—and faces challenges from parameter identifiability, equifinality (multiple parameter sets producing similar results), and limited information content in observations.

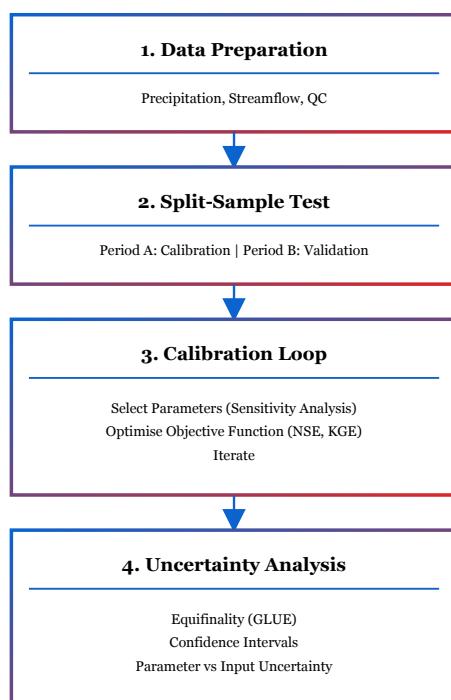
Effective calibration requires several components:

- **Objective Functions:** Quantitative metrics measure the agreement between simulated and observed values. Common choices include the Nash-Sutcliffe Efficiency (NSE), coefficient of determination (R^2), root mean square error (RMSE), percent bias (PBIAS), and Kling-Gupta Efficiency (KGE). Each metric emphasises different aspects of model performance.
- **Parameter Sensitivity Analysis:** Before calibration, sensitivity analysis identifies which parameters most strongly influence model outputs. This reduces the dimensionality of the calibration problem by focusing on parameters that matter most and helps understand model behaviour. Global sensitivity analysis methods like Sobol indices or Morris screening provide robust identification of influential parameters across the entire parameter space.
- **Optimisation Algorithms:** Various algorithms search the parameter space for optimal or near-optimal parameter sets. Methods include manual calibration (expert-guided parameter adjustment), automatic algorithms (gradient-based methods, evolutionary algorithms like

genetic algorithms), and Bayesian approaches that estimate probability distributions for parameters.

- **Split-Sample Testing:** Calibration and validation should use different time periods to test model performance on independent data. The differential split-sample test suggests calibrating on one climatic period and validating on another period with different hydroclimatic characteristics to ensure model robustness.

Figure 44 - Calibration and Validation Workflow



Performance Evaluation Metrics

- **Nash-Sutcliffe Efficiency (NSE):** The Nash-Sutcliffe efficiency coefficient, introduced in 1970, has become the most widely used metric for assessing predictive skill of hydrological models. An NSE of 1 indicates perfect agreement, 0 indicates the model performs no better than using the observed mean as a predictor, and negative values indicate the observed mean is a better predictor than the model.
- Subjective thresholds suggested in the literature include $NSE > 0.50$ for satisfactory performance, $NSE > 0.65$ for good performance, and $NSE > 0.75$ for very good performance, though these should be interpreted in context. Limitations of NSE include sensitivity to

extreme values (giving high weight to peak flows), potential insensitivity to systematic bias, and studies have shown NSE is sensitive to maximum daily runoff simulation but less sensitive to minimum flows.

- **Kling-Gupta Efficiency (KGE):** Developed to address some NSE limitations, the KGE decomposes model performance into correlation, bias, and variability components. A KGE of 1 indicates perfect performance. The KGE's decomposition allows for diagnosing whether poor performance stems from timing errors, over- or under-prediction of variability, or systematic bias.

Uncertainty Analysis

Hydrological models face multiple sources of uncertainty that affect predictive reliability. Understanding, quantifying, and communicating these uncertainties is essential for informed water management decisions—and critically important for banking applications where physical risk assessment informs capital allocation and pricing.

Sources of Uncertainty:

- **Input Data Uncertainty:** Precipitation measurements contain errors from gauge wetting, wind effects, spatial interpolation, and radar calibration. These errors propagate through the nonlinear model structure, often amplifying in their effect on streamflow predictions.
- **Validation Data Uncertainty:** Streamflow observations, used for model calibration and validation, contain errors from rating curve uncertainty, stage measurement errors, and flow measurement errors. At extreme high or low flows, rating curve extrapolation beyond measured conditions introduces additional uncertainty.
- **Model Structure Uncertainty:** All hydrological models simplify reality and cannot perfectly represent the true physical processes. Structural uncertainty arises from incomplete process understanding, necessary simplifications, and scale mismatches between model concepts and observable reality.
- **Parameter Uncertainty:** Most hydrological model parameters cannot be measured directly at the watershed scale and must be inferred through calibration. Parameter uncertainty stems from limited information content in observations, measurement errors, equifinality, and spatial/temporal variability in true parameter values.

Methods for Uncertainty Quantification:

- **Monte Carlo Analysis:** The Monte Carlo approach generates an ensemble of model realisations by sampling parameter values from specified probability distributions. Each parameter set is used to run the model, creating an ensemble of outputs whose spread characterises predictive uncertainty.

- **Generalised Likelihood Uncertainty Estimation (GLUE):** GLUE acknowledges that multiple model structures and parameter sets may be 'behavioural' (capable of adequately simulating the observed system). The approach defines likelihood measures that weight different model realisations according to their agreement with observations.
- **Bayesian Approaches:** Formal Bayesian methods combine prior information about parameters with observational data to derive posterior probability distributions for parameters and predictions. Bayesian methods provide a rigorous statistical framework but require specification of prior distributions and likelihood functions.

Advances in Data-Driven Modelling

The proliferation of hydrological data and advances in machine learning have enabled purely data-driven approaches to hydrological prediction. Deep learning architectures such as LSTM (Long Short-Term Memory), convolutional neural networks (CNNs), and attention mechanisms excel at learning patterns in sequential hydrological data.

Advantages of data-driven approaches include the ability to capture nonlinear relationships without explicit process representation, potential for real-time learning and adaptation, and applicability when process understanding is incomplete. However, these methods require substantial training data, may not extrapolate well beyond training conditions, and lack physical constraints that can prevent unrealistic predictions. Hybrid approaches combining data-driven learning with physical constraints represent an active research frontier.

For banking applications, data-driven approaches offer particular promise for rapid assessment across large portfolios, where computational efficiency may outweigh the interpretability advantages of process-based models. However, the 'black box' nature of machine learning models creates challenges for model governance and regulatory acceptance—issues that the later chapter on Model Risk Governance addresses directly.

Precipitation Modelling

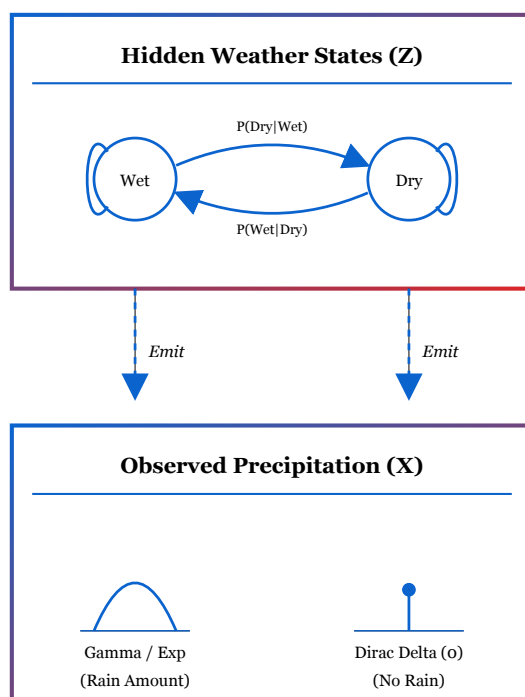
Precipitation modelling is one of the most challenging aspects of weather prediction, owing to the complex interplay of multiple physical processes across various spatial and temporal scales. The fundamental challenge is that many critical precipitation processes occur at scales significantly smaller than typical model grid cells, necessitating larger parameter schemes to represent their collective effects.

Modern precipitation modelling approaches encompass a spectrum of methodologies, ranging from purely statistical frameworks that leverage historical data patterns to entirely mechanistic models that simulate fundamental physical processes. At one end of this spectrum, statistical approaches treat precipitation as a stochastic process, using probability theory and historical observations to capture temporal and spatial patterns without explicitly modelling the underlying atmospheric physics.

These methods include traditional timeseries analysis, which can identify cyclical patterns and trends, and more involved machine learning techniques, which detect complex, nonlinear relationships in precipitation data.

As we move along the spectrum, semi-empirical models combine statistical relationships with basic physical constraints, balancing computational efficiency and physical realism. At the entirely mechanistic end, models directly simulate atmospheric dynamics, thermodynamics, and microphysics, requiring substantial computational resources but providing explicit representations of the physical processes driving precipitation formation.

Figure 45 - Hidden Markov Model for Precipitation



Separates occurrence (state) from amount (emission)

Hidden Markov Models (HMM)

Hidden Markov Models mark a paradigm shift in how precipitation processes are understood and modelled statistically. Instead of viewing rainfall as a simple stochastic process or a direct result of observable variables, HMMs recognise that precipitation arises from an underlying system of atmospheric states that influence the statistical properties of observed rainfall events. This framework offers both diagnostic insight into rainfall variability and predictive power for

simulation and forecasting purposes, making it one of the most advanced tools for capturing the complex temporal dynamics of rainfall patterns.

The fundamental innovation of HMMs lies in their recognition that observable precipitation patterns arise from latent atmospheric conditions that persist over time and evolve according to probabilistic rules. The model introduces a small number of discrete rainfall states—typically ranging from three to six—that capture the essence of observed rainfall variability in terms of distinct rainfall patterns. These hidden states might represent large-scale atmospheric regimes such as "dry continental conditions," "moist tropical flow," "frontal system passage," or "convective instability." Each state remains hidden from direct observation, representing latent atmospheric conditions that produce characteristic precipitation signatures.

While we cannot directly observe these atmospheric states through precipitation measurements alone, their manifestation in precipitation patterns allows us to infer their presence and dynamics through statistical methods. - David Kelly, MKM

This hidden nature reflects physical reality: we observe rain at the ground, but the atmospheric circulation patterns driving that rain operate at larger spatial and temporal scales. The temporal evolution of these hidden states follows a Markov chain, meaning the state active on any given day depends only on the state from the previous day according to fixed transition probabilities. This first-order Markov property substantially simplifies the model structure while maintaining sufficient complexity to capture realistic precipitation dynamics.

Dual-Layer Structure

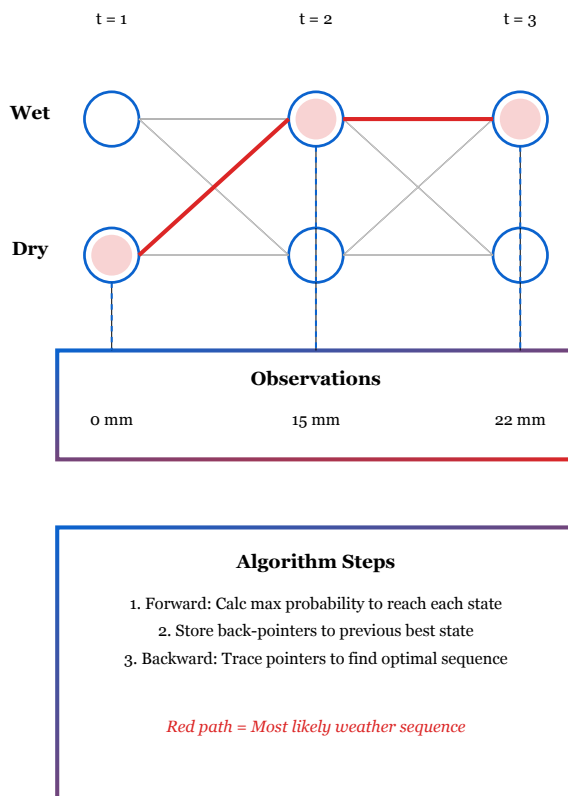
The HMM framework consists of two interconnected probabilistic structures that work in concert to model precipitation processes.

- The first component is the transition probability matrix, which governs how the system moves between hidden weather states. Elements of this matrix represent the probability of transitioning from state i to state j , encoding the persistence and frequency of different weather regimes. These transition probabilities allow the model to naturally represent phenomena such as dry spells and wet periods, capturing the tendency of weather patterns to persist before transitioning to different conditions.
- The second component comprises the emission probability distributions, which specify how precipitation is generated given the current hidden state. Each hidden state possesses its own emission distribution that governs the statistical properties of observed precipitation when that state is active. These distributions typically employ a mixed approach: a discrete probability mass at zero to represent dry conditions, combined with a continuous distribution (often gamma or mixed exponential) to model non-zero precipitation amounts. This mixture naturally accommodates the highly skewed and zero-inflated nature of precipitation data, where many days record no rainfall while precipitation events exhibit substantial variability in intensity.

Different states can be associated with distinct characteristics such as rainfall occurrence probabilities, intensity distributions, and spatial correlation patterns across multiple gauge

locations. The hidden Markov model assumptions simplify the temporal and spatial structures requiring parameterisation, since the common weather state accounts for the temporal correlation that would otherwise need explicit modelling.

Figure 46 - Viterbi Algorithm: Decoding Hidden States



The Viterbi Algorithm and Model Learning

Extracting meaningful information from HMMs requires computational algorithms. The Viterbi algorithm serves as one of the cornerstone techniques for HMM analysis, designed to identify the most probable sequence of hidden states that could have generated an observed precipitation record. Named after Andrew Viterbi who developed it in 1967, this dynamic programming algorithm efficiently solves what would otherwise be a computationally intractable problem.

The Viterbi algorithm works by recursively computing the probability of the most likely path to each state at each time step. Rather than enumerating all possible state sequences (which would

grow exponentially with sequence length), it exploits the Markov property to build up the solution incrementally. At each time step, for each possible state, the algorithm calculates the maximum probability of any path that ends in that state, considering both the emission probability for the observed precipitation and the transition probabilities from all previous states. It also maintains a record of which previous state yielded this maximum probability.

After processing the entire observation sequence forward in time, the algorithm then traces backwards through these recorded optimal predecessors to reconstruct the most probable state sequence. This backwards pass reveals which weather regimes were most likely active on each day based on the entire precipitation record.

For precipitation modelling, the Viterbi algorithm allows researchers to understand how various atmospheric conditions have influenced regional precipitation variability and to monitor changes in the frequency and persistence of these regimes over time. By estimating the most likely sequence of states, researchers can identify periods dominated by different weather regimes and analyse how transitions between regimes relate to synoptic-scale atmospheric conditions.

Complementing the Viterbi algorithm, the Baum-Welch algorithm offers a maximum likelihood method for estimating model parameters from observed precipitation sequences. This expectation-maximisation technique iteratively improves estimates of transition probabilities and emission distributions until it reaches convergence.

More recent developments incorporate Bayesian approaches, which allow the incorporation of prior physical knowledge and provide full posterior distributions rather than point estimates. Flexible spline-based Bayesian HMM approaches allow the model to capture non-stationary behaviour and temporal trends in precipitation characteristics, assessing trends in arbitrary measures such as dry spell length and precipitation intensity while maintaining computational tractability.

Spatial Dependencies and Multi-Site Modelling

A distinguishing feature of HMMs in precipitation applications is their natural capacity for representing spatial dependencies across rainfall networks. When multiple rain gauge stations are considered simultaneously, the hidden states provide an elegant mechanism for inducing spatial dependence. If two stations both experience precipitation because they share the same hidden weather state, their rainfall amounts will be correlated in a physically meaningful way. This state-driven correlation captures the reality that precipitation patterns often extend across regions, driven by synoptic-scale atmospheric features.

The model can simultaneously characterise precipitation at multiple gauge locations, treating the multi-site observations as joint realisations conditional on the active hidden state. This approach preserves the spatial correlation structure of rainfall events without requiring explicit spatial covariance modelling, as the hidden states implicitly account for the common atmospheric forcing that produces correlated rainfall across the region.

The HMM represents raw station observations directly without requiring spatial averaging or gridding procedures, and handles missing data in a probabilistic manner rather than requiring arbitrary cutoffs or interpolation—a significant practical advantage given the ubiquity of gaps in observational records.

Non-Homogeneous Extensions and Climate Applications

While classical HMMs assume constant transition probabilities over time, non-homogeneous hidden Markov models (NHMMs) represent an advancement that allows transition probabilities to vary temporally based on external covariates. This modification incorporates large-scale atmospheric predictors into precipitation models, enabling the model to respond dynamically to changing climatic conditions. The transition matrix becomes time-dependent, with probabilities modulated by covariates such as temperature, humidity, sea surface temperature anomalies, or atmospheric circulation indices.

NHMMs offer a relatively simple framework for simulating precipitation at multiple rain gauge stations conditional on synoptic atmospheric patterns, providing a mechanistic link between large-scale climate drivers and local precipitation patterns.

This capability makes NHMMs particularly valuable for downscaling general circulation model (GCM) simulations to the station scale, translating coarse-resolution climate model output into locally relevant precipitation scenarios. The ability to incorporate covariates representing large-scale climate states makes NHMMs essential tools for translating climate change projections into actionable information for water resource management and infrastructure planning.

Simulation, Stochastic Generation, and Practical Applications

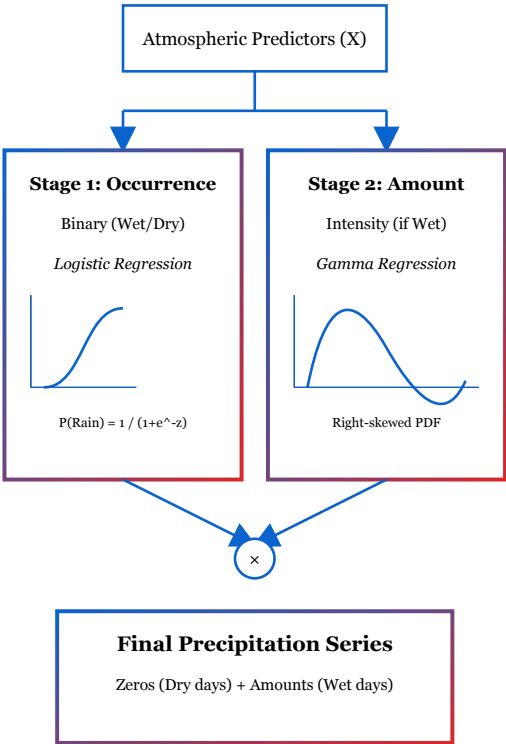
Beyond their analytical capabilities, HMMs serve as stochastic weather generators capable of producing synthetic precipitation sequences that retain key statistical properties of observed records. The stochastic process begins by simulating the sequence of hidden states based on the estimated transition probabilities, then drawing precipitation amounts from the emission distributions linked to each active state. This two-stage process naturally replicates the temporal clustering of wet and dry periods seen in real precipitation records, as persistence in the hidden state sequence translates into persistence in rainfall patterns.

The simulated rainfall matches observed characteristics, including occurrence probabilities, intensity distributions, dry and wet spell length distributions, and inter-site correlation patterns. This simulation capability proves invaluable for generating large ensembles of precipitation scenarios for applications such as water resource planning, agricultural modelling, and hydrological risk assessment. The generated sequences can feed directly into hydrological models, crop simulation models, or flood risk assessments, providing probabilistic inputs that account for weather persistence and variability.

Introduction to Generalised Linear Models

While HMMs provide statistical frameworks for precipitation modelling, other approaches offer complementary capabilities. Generalised Linear Models (GLMs) give more direct relationships between predictors and precipitation, often with greater interpretability and simpler parameter estimation. GLMs excel when clear relationships exist between atmospheric covariates and precipitation, and when the focus is on explaining variance rather than generating realistic temporal sequences.

Figure 47 - GLM Two-Stage Framework



GLMs have established themselves as a cornerstone statistical framework for precipitation modelling, offering a mathematically rigorous yet practically implementable approach to one of the most challenging aspects of weather prediction. The fundamental appeal of GLMs in this domain stems from their ability to handle the unique distributional characteristics of precipitation data while maintaining interpretability and computational efficiency—qualities essential for both research and operational forecasting applications.

The standard GLM approach to precipitation modelling employs a two-stage framework that elegantly separates the occurrence and intensity components. This decomposition reflects physical understanding: the atmospheric mechanisms governing whether rain occurs differ fundamentally from those controlling rainfall intensity once precipitation has initiated.

The first stage models precipitation occurrence using logistic regression with a Bernoulli distribution. The probability of precipitation on a given day is expressed through the logit transformation as a linear combination of atmospheric predictors.

The second stage addresses precipitation amounts on wet days using gamma regression with a logarithmic link function. The gamma distribution proves particularly suitable for rainfall amounts because it accommodates positive values and exhibits the right-skewed distribution characteristic of precipitation intensity. The separation of occurrence and amount allows each component to be modelled using the most appropriate distribution and link function, enhancing overall model performance.

Formula 3: Generalised Linear Models

For Generalised Linear Models (GLMs), the precipitation process is often separated into occurrence and intensity components:

$$\text{logit}(P(R_t > 0)) = X\beta + \varepsilon$$

Where X represents predictor variables and β their coefficients. For non-zero precipitation:

$$\log(R_t | R_t > 0) = Z\gamma + \eta$$

Where Z may include both atmospheric predictors and temporal dependencies.

In physically-based models, the fundamental equations include the conservation of water vapour:

$$\partial q / \partial t + v \cdot \nabla q = S - C + D$$

Where q is specific humidity, v is wind velocity, S represents sources (evaporation), C represents sinks (condensation), and D represents diffusion.

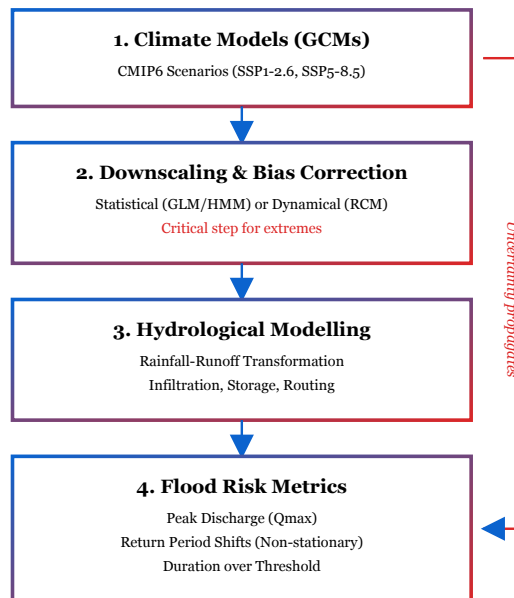
Before examining the statistical foundations of precipitation downscaling in detail, we must step back to consider how these precipitation inputs—whether from climate models or stochastic generators—flow through the complete hydrological system to produce actionable climate risk assessments.

Climate Model Outputs to Flood Forcing

Flood risk assessment begins by translating projected climate conditions into the rainfall (and, where relevant, temperature) inputs that drive rainfall–runoff and routing models. Because climate-model output is produced at scales and with error characteristics that do not match catchment-scale hydrology, a practical workflow uses before hydrological simulation:

- Scenario selection
- Downscaling
- Bias correction

Figure 48 - Climate-to-Flood Risk Workflow



Scenario frameworks such as CMIP6 ScenarioMIP provide ensembles under SSP pathways (e.g., SSP1-2.6, SSP2-4.5, SSP5-8.5), allowing flood assessments to span a range of plausible future forcings rather than a single “best guess”. Ensemble use matters because model structural differences are a dominant source of uncertainty in projected regional precipitation change, particularly for extremes that control flood peaks.

Downscaling converts large-scale climate information into local precipitation fields suitable for hydrological modelling, using either dynamical approaches (RCMs) or statistical approaches (including GLM-based methods and regime models such as hidden Markov formulations). Bias correction then adjusts systematic errors in precipitation frequency, intensity distribution, and temporal sequencing so that hydrological models are not forced with inputs that would create unrealistic runoff responses. For flood work, it is worth stating explicitly that bias correction must preserve the upper tail (extremes) as well as the wet-day structure, because peak discharge is disproportionately sensitive to the short-duration, high-intensity part of the precipitation distribution.

A flood-focused bridge benefits from making the “input → runoff → flood metric” chain explicit: precipitation occurrence and intensity determine effective rainfall, effective rainfall determines fast/slow runoff partitioning and hydrograph shape (via infiltration, soil storage, and routing), and the hydrograph determines metrics used in flood hazard analysis (peak flow, event volume, duration above threshold). This framing sets up why the statistical choices that follow (distributions and link functions) are not merely technical details: they shape the probability and structure of the precipitation events that ultimately drive flood peaks.

Non-Stationary Flood Extremes

Classical flood frequency analysis assumes that the probability distribution of annual maxima (or peaks-over-threshold) is time-invariant, enabling return periods to be estimated from historical records. Under climate change, the distribution of extreme precipitation—and therefore flood-driving inputs—can evolve over time, which motivates non-stationary frequency analysis where distribution parameters vary with time or climate covariates. Non-stationary generalized extreme value (GEV) models are a common way to implement this idea by allowing location and/or scale (and sometimes shape) parameters to change as functions of time or predictors such as climate indices.

A concise flood-risk narrative should also connect the thermodynamic basis for intensifying rainfall extremes to hydrological implications. A widely used first-order expectation is that moisture availability increases with warming at roughly 7% per °C (Clausius–Clapeyron scaling), which provides a physical rationale for intensification of extreme precipitation under higher temperatures. For flood assessment, the key point is not that every storm scales at the same rate, but that changes in the upper tail of precipitation can produce nonlinear changes in runoff peaks once infiltration capacity and catchment storage are exceeded.

Return period language can then be reframed for a changing climate without overcomplicating the text. One practical way is to describe how an event defined as a “T-year” flood under a historical baseline may correspond to a shorter effective return period under future conditions, because exceedance probabilities increase as the extremes distribution shifts. This keeps the narrative aligned with the chapter’s emphasis on backtesting, non-stationarity, and uncertainty—while staying focused on floods rather than broad hydroclimatic impacts.

Flood Process Sensitivities

Flood outcomes depend on how precipitation is partitioned into infiltration, storage, and runoff, and on how that runoff is routed to the outlet. Climate change affects flood hazard primarily through (i) changes in rainfall intensity and sequencing, and (ii) temperature-driven shifts that alter snow and melt dynamics in cold regions, both of which modify the timing and magnitude of effective rainfall.

The infiltration and loss representations described earlier in the chapter (e.g., Green–Ampt, SCS Curve Number) are the immediate “gatekeepers” that convert rainfall to excess rainfall. For flood peaks, this gatekeeping role means that small changes in storm intensity (or in antecedent moisture distributions) can move a catchment across the threshold where infiltration-excess or saturation-excess runoff dominates, producing disproportionate increases in peak discharge. It is therefore helpful to connect precipitation downscaling choices to runoff-generation regimes: downscaling that misrepresents short-duration intensity or wet-day persistence will misrepresent the very conditions that trigger rapid runoff and flashy hydrographs.

In snow-influenced basins, flood hazard can be strongly shaped by the rain–snow partition and melt processes. Temperature-index melt methods offer computational simplicity, while energy-balance approaches better represent melt under complex conditions, including situations where warm storms coincide with snowpack and generate rapid runoff. Keeping the flood focus, the narrative can highlight that warming tends to increase the frequency of “marginal” conditions (near-freezing), which can raise the chance that precipitation arrives as rain on an existing snowpack and yields unusually large event runoff volumes.

Routing choices then translate excess rainfall into the flood hydrograph, and the chapter’s discussion of fast/slow reservoirs (e.g., Hymod’s fast and slow pathways) provides a useful conceptual hook. A flood-oriented bridge can state plainly that climate-driven shifts toward more intense precipitation place more mass into the fast-response component, steepening rising limbs and increasing peaks, even if total seasonal rainfall changes modestly.

Uncertainty and Validation for Flood Projections

Flood projections inherit uncertainty from each link in the modelling chain: scenario choice, climate model structure, downscaling and bias correction, hydrological model structure, and parameter uncertainty. CMIP6 analyses highlight that model structural differences are a dominant source of uncertainty in projected temperature and precipitation patterns, reinforcing the need to treat flood assessment as an ensemble problem rather than a single deterministic run. Within hydrology, the chapter’s coverage of equifinality and parameter uncertainty is directly relevant to floods because different parameter sets can match historical hydrographs yet diverge substantially in simulated extremes.

Validation for future flood risk cannot be done directly against observations that do not yet exist, so credibility is built through targeted tests that focus on the flood-relevant behaviours. The chapter’s emphasis on split-sample testing and robustness to non-stationarity can be adapted by calibrating on one hydroclimatic regime and validating on another (e.g., wetter/drier periods,

warmer/cooler decades) to assess whether a model preserves peak-flow skill under regime shifts. Backtesting against historical extreme floods remains essential, but the narrative should be explicit that backtesting must be paired with stress-testing beyond the observed range when the goal is forward-looking flood risk under changing extremes.

A concise way to close this bridge—while teeing up the GLM section—is to restate the dependency: flood hazard estimates are only as credible as the precipitation extremes fed into the hydrological model, and statistical precipitation models are a principal tool for generating those extremes in a controlled, interpretable way. With that in place, the reader is prepared to move from “why precipitation statistics matter for floods” to “how distributions and link functions encode precipitation occurrence and intensity”.

Statistical Distributions and Link Functions

The choice of distributions and link functions in GLM-based precipitation modelling reflects consideration of data properties and computational requirements. The logistic link function maps the unbounded linear predictor and the constrained probability space $(0,1)$ for occurrence modelling. This link function also offers practical advantages in interpretation, as coefficients can be readily converted to odds ratios, facilitating communication of results to non-technical stakeholders.

For amount modelling, the gamma distribution's flexibility in capturing various shapes of skewed distributions makes it ideal for precipitation intensity. The log link function ensures that predicted amounts remain positive and allows for multiplicative effects of predictors, which often align better with physical intuition about how atmospheric variables influence rainfall intensity. Alternative distributions, such as the lognormal or Weibull, can also be employed within the GLM framework when data characteristics suggest their appropriateness.

Applications in Climate Downscaling

GLMs have proven especially useful for translating coarse-resolution climate model outputs to local scales needed for impact assessments and decision-making. Global circulation models usually operate at spatial resolutions of 50-100 kilometres, which is far too coarse for many practical purposes. GLM-based downscaling creates empirical relationships between large-scale atmospheric variables from reanalysis datasets or climate models and local precipitation observations, allowing projection of future precipitation patterns under different climate scenarios.

More variants include Generalized Hierarchical Linear Models (GHLM), which extend standard GLMs to manage multi-site precipitation by incorporating both within-site and between-site variability through a hierarchical structure. GHLMs can pool information from neighbouring stations to build more robust relationships between large-scale climate variables and local precipitation, demonstrating superior performance compared to simple approaches. These models maintain the interpretability of GLMs while capturing spatial dependencies essential for realistic multi-site precipitation simulation.

Model selection typically employs information criteria such as the Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) to identify optimal predictor combinations while balancing model complexity against goodness-of-fit. Cross-validation techniques, including leave-one-station-out and k-fold approaches, enable robust assessment of model performance and transferability to ungauged locations—a critical consideration for practical applications.

However, GLMs also face limitations. The linear relationship assumed between transformed response and predictors may prove restrictive when authentic relationships are highly nonlinear.

Modern precipitation modelling increasingly integrates GLM approaches with other methodologies within comprehensive forecasting frameworks. GLMs serve as essential components in ensemble prediction systems, providing probabilistic forecasts through their natural expression of uncertainty. They also interface effectively with mechanistic models, offering bias correction and statistical post-processing of physically-based model outputs.

Figure 49 - Comparison of Statistical Downscaling Approaches

Approach	Key Features	Pros / Cons
Standard GLM	Empirical fit Linear logic Single-site	+ Fast & Simple + Interpretable - Linearity limit
Hierarchical (GHLM)	Multi-site pooling Spatial structure Robust for ungauged	+ Spatial consistency + Handles missing data - Complexity
Hybrid / ML Approaches	Nonlinear mapping Remote sensing data Ensemble mix	+ Highly flexible + High skill - Black box / Data

Selection Criteria

*Choose based on data availability, need for interpretability vs. skill,
and computational resources (AIC/BIC validation).*

Current research focuses on enhancing GLM flexibility through machine learning techniques, incorporating additional covariates from remote sensing platforms, and developing more spatial-temporal dependency structures.

Comparative Context and Methodological Considerations

Mechanistic models based on radar data and physical principles offer high temporal and spatial resolution by explicitly representing physical processes such as cloud formation, microphysics, and atmospheric dynamics. These models provide process-level understanding but require substantial computational resources and detailed initialisation data. They typically outperform statistical models for short-term forecasting but may struggle with longer-term climate applications where structural uncertainties dominate.

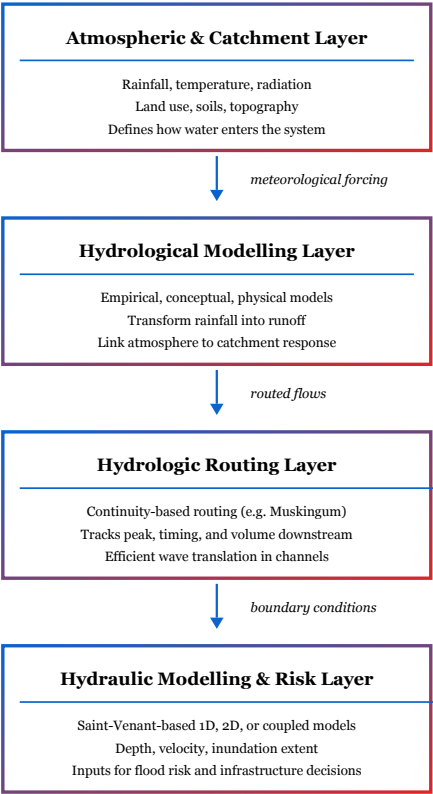
The choice among these approaches depends on specific application needs, available data, and computational resources. HMMs occupy a valuable middle ground, providing physically interpretable structures (the hidden states representing weather regimes) while maintaining computational tractability and the ability to generate realistic stochastic ensembles. Their learning capability makes HMMs particularly valuable for regions with complex precipitation patterns where mechanistic understanding remains incomplete.

The HMM framework essentially performs cluster analysis on multi-site precipitation observations with explicit temporal dependence through the Markov chain structure. This combination of clustering (identifying distinct weather regimes) with temporal dynamics (modeling transitions between regimes) captures fundamental aspects of how atmospheric systems actually behave, making HMMs powerful tools for precipitation modeling across diverse climatic regions and applications, from operational forecasting to climate change impact assessment.

Chapter 6 - Hydraulic Modelling

"Hydrologic modelling shows us the journey of every raindrop across watersheds—predicting how much water arrives and when. Hydraulic modelling reveals how water surges through channels, spills across floodplains, and exerts a force on everything in its path. One tells us what's coming, the other shows us what happens when it gets here."-
David Kelly, MKM

Figure 50 – Water’s Journey from Rainfall to Flood Risk



This is a good point to take stock of the central themes of this chapter, which has an end product of a flood.

The hydrologic and hydraulic routing field embodies the fundamental tension in environmental modelling between theoretical rigour and practical utility. From the most straightforward Muskingum calculations to the most two-dimensional Saint-Venant implementations, each approach balances physical realism, data requirements, computational feasibility, and decision-making needs.

This balance is not merely a technical consideration but a philosophical one—it recognises that models serve as tools for understanding and decision-making rather than perfect representations of reality. The ongoing challenge of finding appropriate applications continues to drive innovation in the field, from methodological advances to computational techniques and uncertainty characterisation.

“Three distinct yet interconnected approaches form the foundation of our understanding of water's movement through landscapes: hydrological modelling, hydrologic routing, and hydraulic modelling. Each represents a different scale of analysis, a different set of simplifications, and a different perspective on water's journey.” - David Kelly, MKM

Hydrological Modelling

Hydrological modelling embraces the entire water cycle within a watershed. It begins with rain falling from the sky. As it is intercepted by vegetation, it follows that water infiltrates into soils, percolates to groundwater, evaporates back to the atmosphere, or becomes surface runoff. This holistic approach asks: Of all the water that falls as precipitation, how much becomes runoff, and when does it reach the stream network?

The watershed resembles a canvas showcasing details of land use, soil types, topographic features, and vegetation patterns. Each element affects the fate of water—urban areas shed runoff quickly, forests capture and slowly release moisture, and agricultural fields change with the seasons. Hydrological models integrate these complex interactions, ranging from simple empirical equations to physics-based formulations.

Hydrology modelling connects the atmospheric drivers (precipitation, temperature, solar radiation) to the terrestrial response, creating the crucial link between meteorology and hydrologic routing.

Hydrologic Routing

As water converges into channels and reservoirs, hydrologic routing takes centre stage. This approach focuses on how the shape and timing of flood waves change as they move downstream—a process fundamental to flood forecasting and reservoir operations.

Hydrologic routing embraces the principle of continuity: water cannot be created or destroyed, only stored or released. The elegant simplicity of the continuity equation—where the difference between inflow and outflow equals the rate of change in storage—belies the complex interplay between channel characteristics and flow dynamics.

Methods like the Muskingum approach represent channels as a series of natural reservoirs, using empirical relationships to describe how flood waves attenuate and lag as they travel downstream. Reservoir routing similarly tracks the balance between inflows, outflows, and changes in storage, enabling operators to manage releases for flood control, water supply, hydropower generation, and environmental flows.

These approaches sacrifice the detailed mechanics of fluid flow for computational efficiency and practical applicability. They recognise that for many applications, we need not resolve the intricacies of hydraulics to predict the essential characteristics of downstream flow—the peak, timing, and volume that drive decision-making.

Hydraulic Modelling

Hydraulic modelling is a crucial next step when we need to understand not only the volume of water passing a point and the timing but also its depth, velocity, energy, and spatial distribution. Based on the principles of fluid mechanics—particularly the conservation of mass, momentum, and occasionally energy—these models clarify the detailed behaviour of water in channels, floodplains, and hydraulic structures.

Hydraulic models address whether water will reach a gauge's alert level, resulting in flooding. How effective is this proposed flood control structure? Which areas will be inundated during a 100-year flood event? Which bridge designs can withstand the force of floodwaters? The answers influence billions in infrastructure investments and numerous decisions impacting public safety and property.

This tension between the mathematical precision of the models and the chaotic reality of natural systems demands a particular mindset—one that balances confidence in scientific principles with humility about their limitations.

We will come back to model governance in a later chapter, at this moment we need to absorb the following and move on.

The role of model governance is to ensure that the mindful modeller understands that models tools for approaching measurements that might be useful. Those that use such models to make decisions have to appreciate there are considerable tensions around what can be measured, processed and reported - David Kelly, MKM

As weather patterns alter hydrological regimes and urbanisation transforms watersheds, the importance of robust routing methods only grows. Integrating traditional process understanding with modern computational techniques and data science approaches promises a future where we can better understand, predict, and manage water's journey through our environment – an endeavour with profound implications for infrastructure resilience, ecological sustainability, and human well-being.

Like water itself, our exploration will flow from the broad watershed scale to the detailed mechanics of channel flow, always seeking to understand the essence of water's movement through our world.

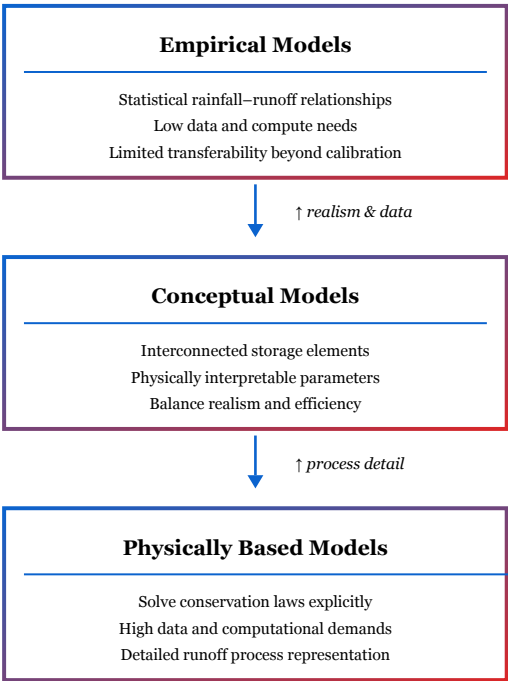
Runoff Modelling

The transformation of precipitation into runoff represents a critical link in the hydrological cycle, with profound implications for physical risk assessment. While conceptually straightforward, this

process embodies remarkable complexity in practice, encompassing multiple interacting pathways through which water moves through a catchment system. Understanding these pathways and their relative importance under different conditions forms the foundation of modern runoff modelling approaches.

The evolution of runoff modelling reflects a continuous tension between complexity and practicality, leading to diverse approaches ranging from purely empirical to conceptual to fully physically based frameworks.

Figure 51 – Runoff Modelling Spectrum



From empirical to physical: increasing realism, data, and complexity in runoff models

Empirical approaches draw purely from statistical relationships between rainfall inputs and runoff outputs. They offer computational efficiency but sacrifice explicit representation of physical processes. While limited in their transferability to conditions outside their calibration domain, these models continue to find practical application in risk assessment scenarios where rapid computation is essential.

Moving along the spectrum of complexity, conceptual models attempt to bridge the gap between empirical and physical approaches by representing key hydrological processes through simplified mathematical frameworks. These models typically conceptualise catchments as collections of interconnected storage elements, each representing different components of the hydrological system. The popularity of conceptual models in practical applications stems from their ability to maintain physical interpretability while remaining computationally tractable.

At the most end of the spectrum, physically-based models attempt to represent the mechanics of water movement through the catchment system. These models solve fundamental mass, momentum, and energy conservation equations, providing detailed representations of infiltration, subsurface flow, and channel routing processes. While theoretically more robust, their practical application often faces challenges related to data requirements and computational demands.

The spatial representation of catchment processes introduces another dimension of complexity in runoff modelling. Lumped models, treating entire catchments as single units, offer simplicity but may miss critical spatial variations in catchment response.

Semi-distributed models partition catchments into hydrologically similar units, providing some predictive capability and a pragmatic balance between spatial detail and computational efficiency.

While fully distributed models offer the most detailed spatial representation, they also demand extensive data and computational resources that improvements in predictive capability may not always justify.

The runoff generation process encompasses multiple mechanisms operating at different temporal and spatial scales. Surface runoff, generated when rainfall intensity exceeds soil infiltration capacity or when soil becomes fully saturated, represents the most rapid response component. Subsurface flow through soil layers provides a slower response mechanism, while groundwater contribution to streamflow represents the slowest component. The relative importance of these mechanisms varies with catchment characteristics and storm properties, necessitating careful consideration in model development.

Unit Hydrograph Concept

Within this framework, the unit hydrograph concept has emerged as a foundational approach to understanding and predicting catchment response to rainfall.

A hydrograph serves as the fundamental graphical representation of a watershed's response to rainfall, depicting how streamflow changes over time following a precipitation event. Developed by Sherman in 1932, the unit hydrograph models the direct runoff response to a standardised rainfall input (typically 1 inch or 1 cm) distributed uniformly across a watershed during a specified duration. This elegant concept transforms the complex rainfall-runoff relationship into a predictable, reproducible pattern based on several key principles: time invariance (consistent watershed response over time), linearity (proportional relationship between rainfall and runoff), and superposition (ability to combine responses from sequential rainfall events).

The beauty of the unit hydrograph lies in its practicality—it establishes a watershed's unique "signature" or response pattern that can be scaled and applied to predict runoff from storms of varying intensities and durations. This makes it an invaluable tool for flood forecasting, water resource management, and infrastructure design.

Despite its simplifying assumptions—uniform rainfall distribution and consistent watershed conditions—the unit hydrograph concept remains one of hydrology's most enduring and valuable

tools for translating the complex interactions between precipitation and watershed processes into practical, quantifiable predictions.

This method, while simplified, provides crucial insights into how catchments transform excess rainfall into runoff. The conventional approach derives unit hydrographs from observed rainfall and runoff data for single storm events, carefully separating baseflow from the total hydrograph to isolate the direct runoff component. This empirical foundation, however, faces practical limitations in ungauged catchments, leading to the development of synthetic unit hydrograph methods.

SCS Dimensionless, Snyder and GUH

Hydrologists derive unit hydrographs through various methods, including direct analysis of observed rainfall-runoff data, synthetic approaches like Snyder's method that relate hydrograph characteristics to watershed properties, or the dimensionless hydrograph that standardises the shape based on peak flow timing.

These synthetic approaches, including the widely adopted SCS Dimensionless Unit Hydrograph and Snyder's Method, enable hydrograph estimation without observed data. More techniques, such as the Geomorphological Unit Hydrograph (GUH), forge explicit connections between hydrograph shape and catchment physical characteristics, incorporating stream order statistics and network width functions. The time-area method offers another perspective, conceptualising catchment response through isochrones representing zones of equal travel time to the outlet.

Synthetic Unit Hydrographs: Practical Solutions for Ungauged Catchments

The challenge of predicting hydrological responses in watersheds without gauging stations has driven the development of ingenious synthetic approaches. These methods represent different philosophical approaches to the same fundamental problem—how to translate watershed characteristics into expected runoff patterns without direct observations.

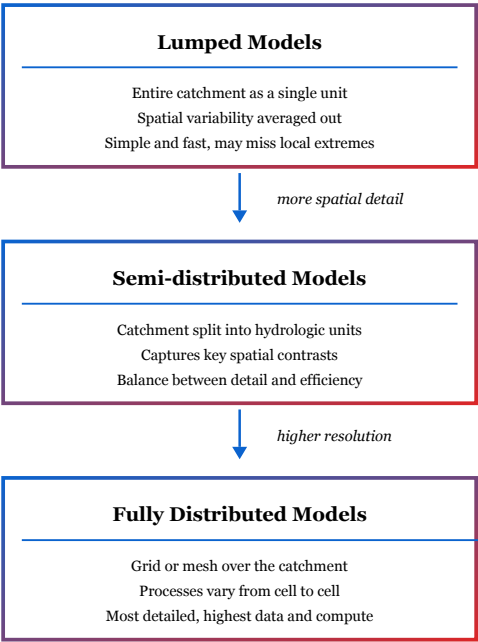
The Soil Conservation Service (now Natural Resources Conservation Service) approach embodies the power of standardisation in hydrological science. Rather than deriving unique solutions for each watershed, the SCS method offers a standardised template derived from analysing numerous catchments across diverse landscapes. This dimensionless curvilinear shape can be scaled to fit specific watersheds using two key parameters: time to peak and peak discharge.

The SCS method's enduring popularity stems from its pragmatic balance between simplicity and effectiveness. Assuming a consistent hydrograph shape across watersheds allows practitioners to focus on determining the appropriate scaling factors. The method relies on a Peak Rate Factor (PRF)—typically set at 484 in traditional units—which can be adjusted based on terrain characteristics, with higher values for steeper watersheds and lower values for flatter terrain.

While often approximated as a triangular shape for calculation simplicity, the complete SCS dimensionless hydrograph captures the characteristic asymmetry of natural runoff events—a rapid rise followed by a more gradual recession. This standardised approach makes the SCS method accessible to practitioners with limited data or hydrological expertise, explaining its widespread adoption in engineering practice and regulatory frameworks worldwide.

Despite its empirical nature and generalised assumptions, the SCS method continues to provide reasonable estimates for many practical applications. Its limitations primarily emerge in watersheds with unusual characteristics or situations requiring more nuanced representations of catchment behaviour. In these cases, modifications to the standard PRF value or more methods may be necessary.

Figure 52 – Spatial Representation of Runoff Models



Increasing spatial representation of catchment processes from lumped to fully distributed

Snyder's Method: Regional Empiricism in Hydrograph Synthesis

Snyder's approach, pioneered in the late 1930s, represents an early attempt to codify the relationship between watershed characteristics and hydrograph shape through regional empirical equations. Unlike the standardised shape of the SCS method, Snyder's approach directly estimates key hydrograph parameters—specifically basin lag time and peak discharge—based on physical watershed characteristics.

The cornerstone of Snyder's method is the relationship between basin lag time and watershed geometry, particularly length and centroid distance. By incorporating these physical features, Snyder recognised that watershed shape fundamentally influences the timing and magnitude of runoff. The regional coefficients in Snyder's equations acknowledge that watersheds in different geographic regions respond differently to rainfall, even when their physical dimensions are similar.

This regional sensitivity makes Snyder's method adaptable across diverse landscapes, from the original Appalachian watersheds where it was developed to mountainous regions and coastal plains. However, this adaptability comes with the challenge of determining appropriate regional coefficients, which typically requires calibration against observed data from similar watersheds in the region.

Snyder's approach bridges purely empirical and physically based methods. While not directly derived from physical principles, it acknowledges that watershed geometry meaningfully influences hydrological response, creating an intuitive connection between watershed characteristics and hydrograph parameters. The method has proven particularly valuable for medium to large watersheds and continues to influence modern hydrological engineering, including integration into contemporary software like HEC-HMS.

The practical implementation of Snyder's method involves not only calculating peak discharge and timing but also constructing the complete shape of the hydrograph. Parameters such as hydrograph width at 50% and 75% of peak flow help define this shape, enabling engineers to estimate not only when and how high a flood will peak but also how long elevated flows will persist—critical information for infrastructure design and emergency planning.

Geomorphological Unit Hydrograph: Catchment Form as Hydrological Function

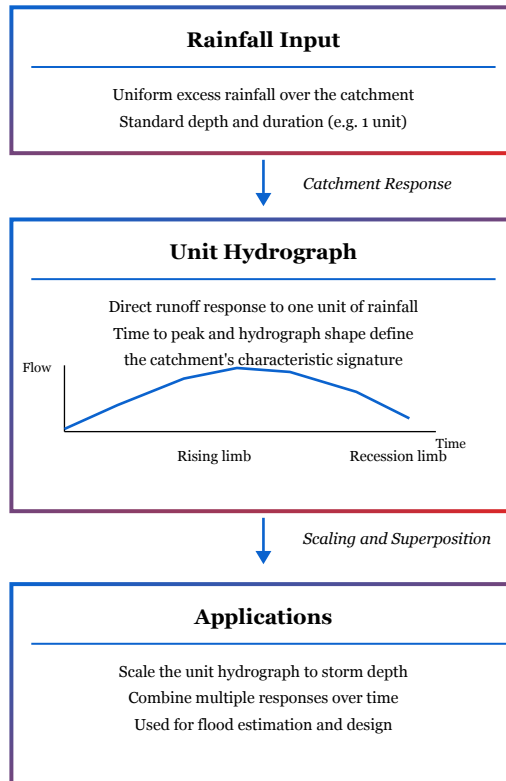
The Geomorphological Unit Hydrograph (GUH) represents a departure from earlier synthetic approaches. Rather than relying primarily on empirical relationships or standardised shapes, GUH explicitly connects hydrograph characteristics to the structure of drainage networks within watersheds. This approach, pioneered by Rodriguez-Iturbe and Valdes in 1979, recognises that river networks encode information about how watersheds process rainfall.

The GUH approach draws on Horton's stream ordering laws, which quantify river networks' hierarchical structure. When analysed through the lens of Horton-Strahler ordering, these networks reveal remarkable regularities across diverse landscapes—patterns in how streams branch, how their lengths increase with order, and how drainage areas expand. The GUH approach posits that these network characteristics fundamentally determine how rainfall is transformed into runoff.

Unlike other methods that treat watersheds as black boxes or simplified geometric shapes, GUH acknowledges the watershed as a complex system where water flows through a network of pathways with varying travel times. The approach models runoff as a probabilistic process where

Figure 53 – Unit Hydrograph Concept

Weather Pattern



The unit hydrograph links a standard rainfall pulse to a catchment's direct runoff response

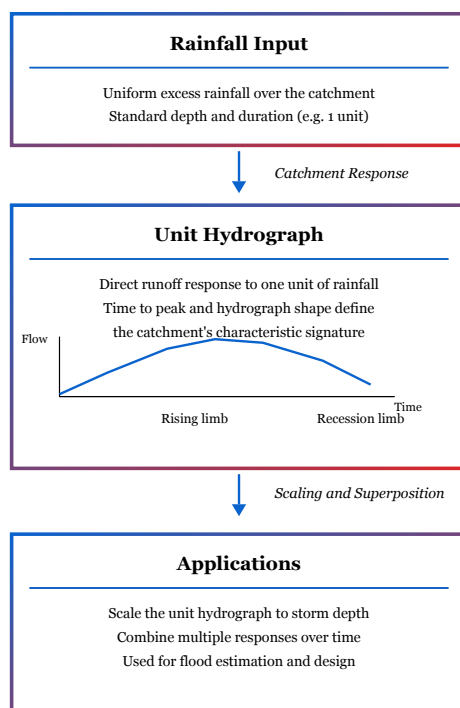
water particles move through different stream orders before reaching the watershed outlet, with each transition governed by geomorphological ratios derived from the stream network.

This statistical translation of watershed structure into runoff dynamics makes GUH particularly valuable for ungauged basins where traditional calibration data are unavailable. By extracting network characteristics from maps or digital elevation models, hydrologists can develop reasonably accurate hydrograph estimates without historical flow records—a significant advantage in data-sparse regions.

However, the GUH approach introduces complexity in both its conception and application. The original formulations involve probability theory and differential equations, which can be challenging to implement in practice. This complexity has led to various simplifications and approximations, such as using equivalent Nash cascade models that preserve key GUH properties while enhancing computational tractability.

The GUH approach also raises interesting theoretical questions about the linearity assumption inherent in unit hydrograph theory. Because GUH parameters like time to peak depend on flow velocity, which may vary during storm events, the method contains implicit non-linearities that challenge traditional unit hydrograph assumptions. Some hydrologists have suggested that "geomorphological response function" might be a more accurate term than "unit hydrograph" for this approach.

Figure 54 – Unit Hydrograph Concept



The unit hydrograph links a standard rainfall pulse to a catchment's direct runoff response

Despite these complexities, GUH represents an essential bridge between traditional hydrology and the emerging field of hydrogeomorphology, recognising that watershed form and hydrological function are intimately connected. Modern GIS capabilities have made GUH implementation more accessible, allowing the extraction of network parameters from digital elevation models rather than laborious manual map analysis.

In practice, these synthetic unit hydrograph approaches are not competing alternatives but complementary tools in the hydrologist's toolkit, each with distinct strengths and limitations. The SCS method offers accessibility and standardisation, Snyder's approach provides regional sensitivity and geometric consideration, while GUH establishes explicit connections to watershed structure and network properties.

Modern hydrological practice often involves hybrid approaches that combine elements of different methods. For example, regionalised SCS parameters might be developed through GUH analysis, or Snyder's coefficients might be adjusted based on stream network characteristics. Digital tools and geographic information systems have revolutionised all these methods, enabling rapid parameter extraction and reducing the historical barriers to implementation.

The evolution of synthetic unit hydrograph methods reflects a broader trend in hydrology toward physically based approaches that capture the underlying processes governing watershed response. While perfectly representing these complex processes remains challenging, each generation of methods has moved closer to linking hydrograph characteristics to fundamental watershed properties.

Despite computational models now available, these synthetic unit hydrograph approaches remain valuable for their balance of physical intuition and practical applicability. They offer conceptual frameworks that help practitioners understand watershed behaviour, not just predict it. This conceptual understanding is particularly valuable in ungauged basins where data limitations preclude more data-intensive approaches.

As land use changes and watershed characteristics modify, these synthetic methods evolve. Regional coefficients require periodic recalibration, and the fundamental relationships between watershed form and hydrological function may shift in response to changing conditions. Yet the core principles—connecting physical watershed characteristics to runoff patterns through standardised shapes, empirical relationships, or network properties—remain as relevant as ever in modern hydrological practice.

Mathematical representations have further enriched the hydrograph modelling toolkit, providing rigorous frameworks for understanding catchment response.

Nash Cascade: Visualising Catchment Response

Nash Cascade Catchment Response

The Nash cascade model conceptualises the catchment as a series of n linear reservoirs, each with storage coefficient K , yielding an impulse response function:

$$h(t) = \left(\frac{1}{K(n-1)!} \right) \left(\frac{t}{K} \right)^{(n-1)} e^{-(t/K)}$$

Where:

$h(t)$ represents the instantaneous unit hydrograph ordinate at time t , this elegant formulation captures the essential features of catchment response while maintaining mathematical tractability.

The convolution integral forms the mathematical backbone of unit hydrograph application:

$$Q(t) = \int_{[0 \text{ to } t]} I(\tau) h(t-\tau) d\tau$$

Where:

$Q(t)$ is the direct runoff hydrograph, $I(\tau)$ is the excess rainfall intensity, and $h(t-\tau)$ is the unit hydrograph ordinate. In discrete form, this becomes:

$$Q(t) = \sum_{j=1 \text{ to } m} P(j) U(t-j+1)$$

Where:

$P(j)$ represents excess rainfall in period j , and $U(t-j+1)$ is the unit hydrograph ordinate.

The Nash cascade model offers an elegant conceptual framework for understanding how watersheds transform rainfall into streamflow. Imagine a series of connected reservoirs—like a sequence of small ponds—each feeding into the next. This is essentially how the Nash model represents a catchment, with water flowing through multiple storage elements before reaching the outlet.

In this model, the number of reservoirs (n) and their storage behaviour (K) together determine how quickly or slowly the watershed responds to rainfall. More reservoirs create a smoother, more delayed response, much like how a complex watershed with many storage zones tends to delay and attenuate flood peaks. Similarly, larger storage values (K) represent a watershed that holds water longer before releasing it downstream.

When rain falls on the watershed, this model traces how that water pulse moves through the system. At first, little water reaches the outlet as the reservoirs begin filling. Gradually, flow increases as water cascades through the system. Eventually, the flow peaks and then recedes as the reservoirs progressively empty. This rising, peaking, and falling pattern creates the bell-shaped hydrograph observed in natural streams.

The beauty of the Nash model lies in its ability to capture complex watershed behaviour with just two parameters. A watershed with many storage zones (wetlands, lakes, soil layers) might have more reservoirs, while a flashy urban watershed might use fewer. Similarly, a watershed with high infiltration capacity and substantial groundwater storage would have a more significant storage coefficient than an impervious, steep catchment.

The Convolution Integral: Translating Rainfall Patterns to Streamflow

Using a unit hydrograph, the convolution integral represents the fundamental mathematical operation connecting rainfall to streamflow. Given its an integral, it accounts for how every bit of rainfall contributes to streamflow over time.

If we consider rainfall as a sequence of individual pulses, each pulse generates its own small streamflow response, resembling the unit hydrograph but scaled by the amount of rainfall. The convolution process effectively stacks these individual responses together, correctly offset in time, to yield the total streamflow.

When an intense rainstorm occurs over several hours, the initial rainfall begins generating streamflow immediately, while subsequent rainfall enhances this primary response. As time goes on, the impact of earlier rainfall decreases, while the effect of later rainfall increases. The convolution integral accurately tracks and adds these overlapping contributions together.

Hydrologists typically use a discretised version of this process, dividing rainfall into short time intervals (perhaps hourly or sub-hourly). For each rainfall interval, they calculate the resulting streamflow contribution across all future time steps. Then, at any specific time, they aggregate the contributions from all previous rainfall to determine the total streamflow.

Our mathematical framework enables hydrologists to convert complex, variable rainfall patterns into anticipated streamflow responses. It underpins nearly all rainfall-runoff modelling applications, from flood forecasting to water resource planning. It provides the crucial link between what falls from the sky and what flows in our rivers.

Modern deconvolution techniques enable the extraction of unit hydrographs from complex observed hydrographs through Fourier analysis or matrix operations, while data-driven approaches leverage machine learning to bypass traditional hydrograph separation requirements. Each method presents distinct advantages and limitations, with selection typically guided by data availability, catchment characteristics, and specific modelling objectives.

The outputs of runoff models provide essential information for risk assessment:

Hydrograph Characteristics:

- Peak flow rate (Q_p) in m^3/s .
- Time to peak (t_p) in hours.
- Hydrograph volume in m^3 .
- Flow duration at specific thresholds.

Derived Metrics:

- Annual maximum series.
- Flow duration curves.
- Flood frequency distributions.
- Base flow separation indices.

Risk Assessment Parameters:

- Inundation extent predictions.

- Alert exceedance probabilities.
- Above alert duration analysis.

These outputs form the foundation for various risk assessments, from flood plain mapping to infrastructure design and insurance pricing. Depending on the application and catchment characteristics, the temporal resolution of outputs typically ranges from sub-hourly to daily.

Model development follows a structured yet iterative process. It begins with formulating a perceptual model that captures the modeller's understanding of key catchment processes. This conceptual framework is then translated into mathematical expressions, leading to decisions about model structure and parameter relationships. The crucial calibration and validation steps follow, where model parameters are optimised against observed data, and performance is verified using independent datasets.

Runoff models are used in physical risk assessment in multiple domains, from insurance and reinsurance to infrastructure investment and property valuation. For infrastructure investment, they support design flood estimation and adaptation planning. In property valuation, runoff models contribute to flood risk zonation, influencing investment decisions.

The challenge of uncertainty pervades all aspects of runoff modelling, manifesting in parameter estimation, input data quality and model structure. Managing these uncertainties requires careful attention to model selection criteria, considering the purpose of analysis, data availability, required accuracy, and resource constraints. The communication of these uncertainties to stakeholders remains a critical aspect of model application in risk assessment contexts.

Recent Developments in Runoff Modelling

Integrating machine learning approaches has significantly enhanced recent advancements in runoff modelling. Long-short-term memory (LSTM) networks have demonstrated superior performance in capturing temporal dependencies compared to traditional methods, particularly during high-flow periods.

Hybrid approaches combining LSTM with signal decomposition techniques like Variational Mode Decomposition (VMD) have improved Nash-Sutcliffe Efficiency and reduced root-mean-square error across diverse conditions. Similarly, Random Forest models have outperformed traditional conceptual frameworks by leveraging ensemble techniques to produce more stable and precise runoff estimates, especially when incorporating antecedent runoff data to improve accuracy across seasonal variations.

Data assimilation techniques have transformed how hydrological models integrate real-time observations. The Ensemble Kalman Filter and variational techniques have been widely employed to incorporate near-real-time data such as streamflow and remotely sensed soil moisture, significantly enhancing prediction accuracy.

The development of frameworks like the Parallel Data Assimilation Framework (PDAF) has enabled better integration of high-resolution land surface and meteorological observations across large spatial domains. Despite these advances, challenges persist in addressing non-stationarity in hydro-climatic conditions and managing limited high-quality field data. Machine learning models have shown promise in effectively utilizing sparse datasets to partially mitigate these constraints.

Practical Implementation

The practical implementation of runoff models in risk assessment requires careful attention to best practices in model selection, data management, and result communication. Success depends not only on technical proficiency but also on a clear understanding of model limitations and uncertainties. Regular model reviews and updates are required to ensure continued relevance and accuracy in risk assessment applications.

The dynamic nature of weather and human development patterns ensures that runoff modelling remains an active area of research and development. As our understanding of hydrological processes deepens and computational capabilities expand, the challenge is to balance model sophistication with practical utility in risk assessment applications. This balance becomes particularly crucial as we face increasing uncertainty in future weather conditions and their implications for water-related risks.

Hydrologically Tracing Water's Journey

Hydrologic routing addresses the specific challenge of tracking how water moves through landscape features once it enters channels, rivers, and reservoirs. This approach emphasises pragmatic simplification while maintaining sufficient physical realism for practical applications.

Hydrologic routing rests upon fundamental principles of water movement. At its core lies the continuity equation – the mathematical expression of mass conservation that states the rate of change in storage equals the difference between inflow and outflow:

Formula 5: Hydrologic Routing Differential

$$dS/dt = I(t) - O(t)$$

Where:

S represents storage volume

I(t) captures the inflow rate at time t

O(t) describes the outflow rate at time t

This elegant relationship forms the mathematical backbone for tracking water's movement through the landscape. However, a solvable system requires complementary relationships between storage and discharge.

Model Parameters and Implementation

Effective implementation of hydrologic routing involves careful consideration of several parameter types:

Physical Parameters:

- Channel roughness coefficients.
- Cross-sectional geometry parameters.
- Channel bed slope.
- Expansion/contraction coefficients.

Numerical Parameters:

- Time step size (Δt).
- Space step size (Δx).
- Numerical scheme coefficients.
- Convergence tolerances.

Boundary Condition Parameters:

- Upstream flow conditions.
- Downstream stage relationships.
- Lateral inflow coefficients.
- Structure operation rules

The practical implementation involves careful attention to data requirements. Channel geometry data, typically obtained through field surveys or remote sensing, must capture relevant features while maintaining computational efficiency. Boundary conditions require careful consideration to ensure physical realism and numerical stability.

Hydrologic routing methods have evolved distinct approaches for different landscape features:

Water bodies with relatively horizontal surfaces – lakes, reservoirs, and wide floodplains – invite a simplification known as level pool routing. Here, the water surface remains horizontal throughout, allowing storage to be calculated directly from the water level.

The Modified Puls Method and Flow Routing

The Modified Puls Method represents one of the fundamental techniques in hydrologic engineering for predicting how flood waves move through storage facilities such as reservoirs, detention basins, and channel reaches. This hydrologic routing approach addresses a critical challenge in water resources management: understanding how the shape and magnitude of a flood hydrograph changes as water flows through a system where storage effects are significant.

At its core, flow routing involves tracking the movement of a flood wave through a river system or storage facility, accounting for both the delay in timing and the reduction in peak discharge that occurs as water temporarily accumulates before being released. When a flood enters a reservoir or detention basin, the incoming water spreads across the pool surface, causing the flood peak to be attenuated (reduced) and delayed compared to the inflow hydrograph. The Modified Puls method provides a computational framework for quantifying these transformations.

The Modified Puls method, also known as storage routing or level pool routing, rests on two fundamental principles: conservation of mass (continuity equation) and an empirical relationship between storage and discharge. The method makes a key simplifying assumption that the water surface in the storage facility remains essentially horizontal—a "level pool"—which allows storage to be calculated directly from the water surface elevation.

Modified Puls Flow Rate The Modified Puls method operationalises this approach through a finite difference approximation:

$$S(j+1) + (\Delta t/2)O(j+1) = S(j) + (\Delta t/2)O(j) + (\Delta t/2)(I(j) + I(j+1))$$

Where:

- $S(j)$ is the storage volume at time step j .
- $S(j+1)$ is the storage volume at the next time step $j+1$.
- $O(j)$ is the outflow rate at time step j .
- $O(j+1)$ is the outflow rate at the next time step $j+1$.
- $I(j)$ is the inflow rate at time step j .
- $I(j+1)$ is the inflow rate at the next time step $j+1$.
- Δt is the time step interval.

This method is particularly useful because it handles the relationship between storage and outflow. In natural systems, outflow typically depends on storage—more water stored means

higher pressure and faster outflow. The Modified Puls method accounts for this by using a weighted average of outflows (both current and future) rather than just a single value.

In practice, engineers first establish a relationship between storage and outflow for the specific system through direct measurement or hydraulic calculations. Then, for each time step in a storm event, they solve for the unknown future outflow using known values of current storage, current outflow, and inflows (both current and future).

The method is especially valuable because it preserves mass conservation while being computationally simple. It's widely used in flood routing applications, reservoir operations, and stormwater management because it balances accuracy with practicality. Modern software packages often implement Modified Puls routing because it provides reliable results without demanding excessive computational resources or complex data inputs.

Channel Routing: Muskingum

Rivers and streams present a different challenge for hydrologic analysis compared to reservoirs. Water moves longitudinally with varying speeds and depths, creating complex flow patterns that differ fundamentally from the level pool assumptions applicable to reservoirs.

The Muskingum method, originally developed by McCarthy in 1938 for flood control studies in the Muskingum River basin in Ohio, addresses this challenge through a semi-empirical approach that has become one of the most widely applied flood routing techniques in river engineering practice. The method's enduring popularity stems from its elegant conceptual foundation, which relates channel storage to weighted combinations of both inflow and outflow rather than to either value alone. This fundamental insight recognises that storage in a river reach depends not only on the water leaving the reach but also on the water entering it, capturing the dynamic nature of flood wave propagation.

The approach distinguishes between two types of storage within a river reach: prism storage, which represents the regular channel volume under uniform flow conditions, and wedge storage, which accounts for the additional volume created by the advancing flood wave when inflow exceeds outflow.

Since its introduction over 85 years ago, the Muskingum method has been extensively researched and refined, with much of this work focusing on establishing physical interpretations of its two key parameters. The storage time constant K approximates the travel time through the reach, while the weighting coefficient X determines the relative influence of inflow versus outflow on storage calculations.

The method's practical utility was significantly enhanced when Cunge in 1969 demonstrated that it could be derived as an approximation to the full Saint-Venant equations of open channel flow, effectively transforming it from a purely empirical technique into one with rigorous theoretical foundations. This extension, known as the Muskingum-Cunge method, introduced variable parameters that adapt to changing flow conditions, greatly expanding the method's applicability and accuracy across diverse hydrologic scenarios.

Formula 7: Muskingum River Basin

$$S = K[xI + (1-x)O]$$

Where:

- S is the storage volume in the river reach (m³).
- I is the inflow rate to the reach (m³/s).
- O is the outflow rate from the reach (m³/s).
- K represents a storage constant approximating travel time through the reach (seconds).

x serves as a weighting factor (between 0 and 0.5), balancing the influence of inflow and outflow

x = 0 represents pure translation (reservoir-type storage)

x = 0.5 represents pure translation (kinematic wave)

Typical values range from 0.1 to 0.3 for natural channels

This conceptualisation leads to an efficient routing equation:

$$O(j+1) = C_0 \cdot I(j+1) + C_1 \cdot I(j) + C_2 \cdot O(j)$$

Where:

- O(j+1) is the outflow at the next time step j+1.
- I(j+1) is the inflow at the next time step j+1.
- I(j) is the inflow at the current time step j.
- O(j) is the outflow at the current time step j.
- C₀, C₁, and C₂ are routing coefficients derived from K, x, and Δt (time step)

The routing coefficients are calculated as:

$$C_0 = (-Kx + 0.5\Delta t)/(K - Kx + 0.5\Delta t) \quad C_1 = (Kx + 0.5\Delta t)/(K - Kx + 0.5\Delta t) \quad C_2 = (K - Kx - 0.5\Delta t)/(K - Kx + 0.5\Delta t)$$

Routing coefficients (C₀, C₁, and C₂) are derived from the physical parameters K and x and the computational time step. This approach has become a workhorse for operational hydrology, offering accuracy when adequately calibrated.

Considering channels as having both "prism storage" (regular channel volume) and "wedge storage" (additional volume from flood waves), the Muskingum method captures how flood waves change shape as they move downstream. The method's conceptual foundation lies in relating the water stored in a river reach to both inflow and outflow rather than to either value alone.

The storage constant approximates how long water typically takes to travel through the river reach, representing the natural lag in the system. The weighting factor x indicates how much the storage in the reach depends on inflow versus outflow. When x equals zero, storage depends entirely on outflow (like a level pool), while higher values of x (though typically not exceeding 0.5) indicate a more significant influence of the inflow.

The Muskingum method transforms these physical concepts into a straightforward computational procedure. For each time step, the outflow is calculated using a weighted combination of current and previous inflows plus the previous outflow. This approach effectively tracks how flood waves attenuate (reduce in peak) and lag (shift in timing) as they move downstream.

River engineers and hydrologists value the Muskingum method for several reasons. It preserves the conservation of mass, runs efficiently even with limited computational resources, requires minimal data inputs, and conceptually aligns with physical river processes. Properly calibrating using observed inflow-outflow data provides reliable results for many natural river systems. These benefits have established the Muskingum method as a standard tool in flood forecasting, water resource planning, and reservoir operation studies.

Muskingum Calculations

To perform Muskingum calculations, you need the following data:

- Streamflow data is usually measured in cubic feet per second (cfs).
- A roughness coefficient, C , characterises the stream's resistance to flow.
- A time lag, T , represents the delay between changes in upstream conditions and downstream responses.
- Once you have this data, you can calculate the Muskingum parameters:
- Instantaneous discharge (Q) at each time step.
- Area-averaged velocity (V) at each time step.
- Storage (S) at each time step.
- Time of travel (T_t) between upstream and downstream locations.

Once you have these coefficients, you can use them to estimate the discharge at an ungauged location using the Muskingum-Cunge method. This method assumes that the relationship between discharge and upstream and downstream conditions is linear, which may not always be accurate but is often a good approximation.

You can calculate the Muskingum parameters using the following equations:

Instantaneous discharge: $Q = C \cdot A \cdot V^2 / (2g)$

Where:

- A is the cross-sectional area
- g is the acceleration due to gravity
- V is the average velocity.

$$\text{Storage } S = A \cdot (h_{\text{down}} - h_{\text{up}}) / 1000$$

Where:

- h_{down} and h_{up} are the water surface elevations downstream and upstream of a given cross-section.

$$\text{Time of travel (Tt): } Tt = L / V$$

Where:

- L is the reach length between the upstream and downstream locations.

$$\text{Area-averaged velocity (V): } V = Q / A.$$

After calculating these parameters, you can use them to compute the Muskingum coefficients:

$$\text{Coefficient a: } a = [(Q_{\text{down}} - Q_{\text{up}}) (Q_{\text{down}} + Q_{\text{up}})]^{-2}.$$

$$\text{Coefficient b: } b = [(S_{\text{down}} - S_{\text{up}}) (S_{\text{down}} + S_{\text{up}})]^{-2}.$$

$$\text{Coefficient c: } c = [(Tt_{\text{down}} - Tt_{\text{up}}) (Tt_{\text{down}} + Tt_{\text{up}})]^{-2}.$$

This method can help estimate discharge at locations where flow measurements are unavailable or difficult to obtain. However, the accuracy of the estimates depends on the accuracy and representativeness of the upstream and downstream data and the Muskingum coefficients.

Muskingum-Cunge

The evolution of the Muskingum method represents a significant advancement in hydrological flood routing techniques. The classical Muskingum method, originally developed by McCarthy in 1938, relied on calibration using measured flood hydrographs to determine its routing parameters. This fundamental limitation meant the method could only be effectively applied to gauged locations where sufficient streamflow data existed for parameter calibration, and even then, the calibrated parameters often performed poorly when applied to different flood events or reaches.

Cunge's transformative contribution in 1969 fundamentally changed the Muskingum method by establishing its connection to physical-numerical principles. Cunge demonstrated that the Muskingum method could be interpreted as a hydraulic method, wherein the routing parameters

could be directly calculated from physical channel characteristics rather than requiring calibration from historical streamflow data. This breakthrough meant that the routing coefficients could be determined based on measurable channel properties such as cross-sectional geometry and Manning's roughness coefficient (n).

The physical basis originates from Cunge's recognition that the method is a finite-difference solution to the diffusion wave equation. By relating the Muskingum parameters to the properties of the governing diffusion wave, the method gains both theoretical rigour and practical usefulness. The diffusion wave framework applies to a broader range of flow conditions than alternative approaches such as kinematic or dynamic wave models.

This physically-based parameterisation revolutionised the method's applicability to ungauged streams, which vastly outnumber gauged streams in practical engineering applications. Instead of requiring extensive historical streamflow records for calibration, engineers can now apply the Muskingum-Cunge method using readily available or measurable channel data including channel slope, cross-sectional area, top width, and roughness characteristics. This makes the approach particularly valuable for rivers with major tributaries that are not well gauged. The method demonstrates very good performance for all events within a given reach, limited only by the requirement that the geometric and hydraulic channel data must be representative of the actual reach conditions. This consistency across different flood events represents a major improvement over the classical Muskingum approach, which typically performs well only for the reach and event used during calibration.

The nonlinear or variable coefficient formulation of the Muskingum-Cunge method addresses a critical limitation of constant-parameter approaches by recognising that wave celerity and storage characteristics change with discharge magnitude. Rather than using fixed routing parameters throughout an event, the variable coefficient method recalculates the routing parameters at every time step based on the current flow conditions. This dynamic adjustment accounts for the fact that storage-discharge relationships, wave speeds, and channel characteristics vary significantly as flows transition from low to high stages and as flood plains become inundated.

The Muskingum-Cunge method requires several key inputs to characterise the routing reach. Essential data includes channel geometry (cross-sectional area, top width), reach length, channel slope, Manning's roughness coefficient, and the inflow hydrograph. The selection of spatial increment (Δx) and time step (Δt) also influences computational accuracy, though the method demonstrates notable consistency in producing similar outflow hydrographs across various selections of these intervals, provided they remain within reasonable ranges. Specifically, Δx should not be significantly smaller than the distance travelled by the flood wave during a single time step to maintain computational stability and physical representativeness.

The routing equation employs three coefficients that weight the influence of upstream and downstream flows at different time steps. These coefficients incorporate channel characteristics, wave celerity, and temporal-spatial discretisation parameters to simulate flood wave translation and attenuation through the reach. The method can be readily implemented using various computational tools including spreadsheets, programming languages, and commercial software,

offering significant advantages in computational efficiency and speed compared to full unsteady flow solutions while maintaining comparable accuracy across a wide range of flow situations.

Muskingum-Cunge Discharge

$$Q_{est} = Q_{up} (1 + a (1 - \exp(-b (h_{down} - h_{up}) T))) (1 - c \exp(-(h_{down} - h_{up}) T))$$

Where:

- Q_{est} : Estimated discharge at the ungauged location.
 - Q_{up} : Discharge at the upstream location.
 - h_{down} : Water surface elevation at the downstream location.
 - h_{up} : Water surface elevation at the upstream location.
 - T : Time constant, typically related to the travel time.
 - a, b, c : Muskingum coefficients calculated from observed data.
 - a : Coefficient related to the relative change in discharge.
 - b : Coefficient related to the relative change in storage.
 - c : Coefficient related to the relative change in travel time.
-

The outputs of hydrologic routing provide essential information for risk assessment applications. Water surface profiles enable inundation mapping, while calculated discharges support infrastructure design and floodplain management. The temporal evolution of these parameters offers crucial insights into flood wave propagation and its implications for infrastructure and property exposure.

Modern developments in hydrologic routing have embraced technological advances in computational capabilities and data availability:

- Enhanced computational methods allow for more detailed simulations.
- Improved remote sensing technologies provide better geometric data.
- Integration of real-time monitoring systems supports operational flood forecasting.
- Machine learning approaches complement traditional routing methods.

The intersection of physical process understanding and artificial intelligence presents the most exciting frontier in hydrological routing. Machine learning shows particular promise in addressing long-standing challenges:

Parameter estimation through machine learning can establish relationships between observable landscape features and difficult-to-measure parameters.

Model error correction using neural networks can identify and compensate for systematic biases.

Computational efficiency improvements through surrogate modelling can accelerate simulations dramatically.

Balancing Theory and Practice

The balance between theoretical rigour and practical utility remains a central consideration in hydrologic routing applications. Computational demands increase significantly with model complexity, necessitating pragmatic decisions about spatial resolution and temporal discretisation. Data requirements similarly scale with model sophistication, often necessitating simplifications where detailed information is unavailable.

This theoretical-practical tension manifests in selecting model simplifications appropriate to specific applications:

- More straightforward methods may offer sufficient accuracy for many flood forecasting applications.
- Detailed floodplain mapping may demand more approaches.
- Infrastructure design applications may require detailed local modelling and simplified routing for broader system representation.

Uncertainty Considerations

- Despite these advances, uncertainty remains an inherent aspect of all routing applications. This uncertainty stems from multiple sources:
- Roughness coefficients and geometry data uncertainty.
- Model structure uncertainty.
- Future land use changes.

Communicating these uncertainties to stakeholders remains crucial for informed decision-making. Modern routing applications increasingly embrace ensemble approaches, sensitivity analysis, and formal uncertainty quantification to provide decision-makers with a complete picture of what is known, what is assumed, and what remains uncertain.

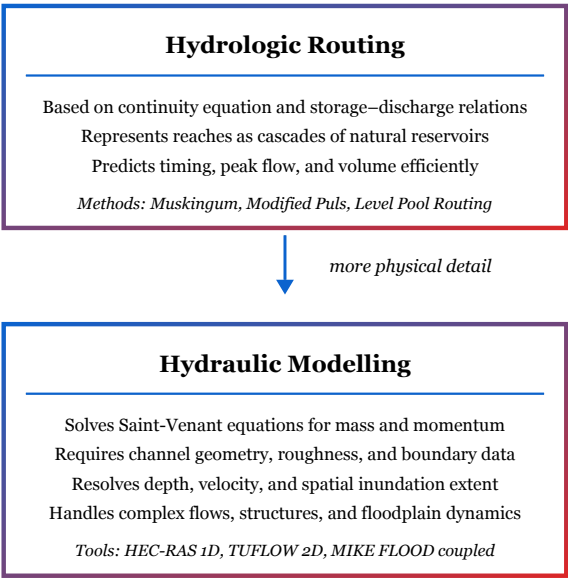
Despite being computationally intensive in some implementations, this advanced framework is an excellent candidate for AI development. It offers the detailed analysis needed for complex systems where more straightforward methods fall short. The ongoing challenge of balancing physical accuracy with computational efficiency continues to propel advancements in the field, highlighting its increasing importance in physical risk assessment applications.

Hydraulic Modelling

Hydraulic modelling is a fundamentally different approach to water flow analysis than hydrologic routing. The distinction is rooted in their mathematical foundations.

While hydrologic routing relies solely on the continuity equation and empirical storage-discharge relationships, hydraulic modelling is firmly grounded in the Saint-Venant equations, which incorporate mass conservation and momentum principles.

Figure 55 – Hydrologic versus Hydraulic Routing



*Hydrologic routing is efficient for wave translation,
while hydraulic models resolve detailed flood dynamics.*

“This critical difference means that hydraulic models solve the complete physics of water movement, accounting for acceleration terms, pressure forces, and complex flow interactions that hydrologic methods cannot capture.”- David Kelly, MKM

Consequently, hydraulic modelling demands significantly more detailed inputs—including precise cross-sectional geometry, roughness coefficients, and channel characteristics—but provides superior resolution of flow dynamics, water surface profiles, and velocity distributions.

Saint-Venant Equations

The Saint-Venant equations represent a landmark achievement in hydraulic engineering and fluid mechanics, providing the mathematical foundation for understanding how water flows through rivers, canals, and other open channels. When Adhémar Jean Claude Barré de Saint-Venant first presented these equations in his 1871 paper, he fundamentally transformed how engineers could analyse and predict unsteady flow behaviour in open channels. Prior to this development, the analysis of such complex flow phenomena lacked a rigorous mathematical framework that balanced physical accuracy with practical applicability.

Barré de Saint-Venant's work emerged during a period of rapid advancement in fluid mechanics and hydraulic engineering. His contribution extended beyond merely formulating equations; he established a systematic approach to describing flow that varies both spatially along a channel and temporally. The equations are alternatively known as the dynamic wave equations, reflecting their capacity to capture the complete dynamics of wave propagation through open channels. This mathematical framework has since become the cornerstone for numerous computational models used in modern hydraulic engineering, including widely-used software such as HEC-RAS.

The Saint-Venant equations derive their power from capturing two essential conservation laws that govern fluid motion. The continuity equation embodies the principle of mass conservation, ensuring that water is neither created nor destroyed as it flows through a channel reach. Mathematically, this accounts for how changes in cross-sectional area and flow velocity along the channel relate to changes in storage over time.

Formula 9: Saint-Venant Equations

The equations consist of:

Continuity equation (Conservation of Mass):

$$\partial h / \partial t + \partial(uh) / \partial x + \partial(vh) / \partial y = 0$$

Where:

- h : Water depth
- u, v : Velocity components in x and y directions

Momentum equations in the x -direction and y -direction:

$$\partial(uh) / \partial t + \partial(u^2h + \frac{1}{2}gh^2) / \partial x + \partial(uvh) / \partial y = gh(S_{0x} - S_{fx})$$

$$\partial(vh) / \partial t + \partial(uvh) / \partial x + \partial(v^2h + \frac{1}{2}gh^2) / \partial y = gh(S_{0y} - S_{fy})$$

Where:

- g : Gravitational acceleration.
 - S_{0x} , S_{0y} : Bed slopes in x and y directions.
 - S_{fx} , S_{fy} : Friction slopes in x and y directions.
-

These equations form the mathematical foundation for hydraulic modelling and are widely used in engineering applications for water resource management, flood prediction, and environmental studies.

The derivation of the Saint-Venant equations is built on a foundation of key assumptions that balance physical accuracy and practical applicability. The framework assumes a one-dimensional flow with a hydrostatic pressure distribution, relatively small channel slopes, and uniform velocity distribution within each cross-section. Regular channel geometry is presumed, while wind and turbulence effects are negligible.

While simplifying the mathematical treatment, these assumptions preserve the essential channel flow characteristics necessary for most practical applications, instilling confidence in the model's reliability.

One and Two-Dimensional Approaches

Hydraulic models are classified based on their spatial dimensionality, with important implications for applications in flood risk assessment. One-dimensional (1D) models represent flow along a single spatial dimension, typically the channel centerline. These models use cross-sections perpendicular to the flow direction to capture channel geometry and solve the 1D Saint-Venant equations for flow and water surface elevation.

They are most appropriate for channels with uniform floodplains and predominantly longitudinal flow. However, they are limited in representing complex lateral flows that often characterise extensive floodplains, which is an essential consideration in their application.

Two-dimensional (2D) models account for flow variations in longitudinal and lateral directions by solving the 2D shallow water equations, extending Saint-Venant principles to multiple dimensions. These models better represent complex terrain, meandering rivers, and irregular flows over floodplains, providing more detailed spatial information about flood characteristics. However, they require significantly more data and computational resources, which can constrain their application in resource-limited situations.

“Coupled 1D-2D models combine the strengths of both approaches by using 1D representation for channel flow and 2D for floodplain flow.”- Johnny Mattimore, MKM.

This hybrid approach allows computational efficiency in well-defined channels while capturing complex floodplain dynamics. Such models require careful interface treatment between 1D and 2D domains to ensure proper mass and momentum transfer.

As outlined in the door-stopper and seminal *Flood Handbook Analysis and Modelling*, edited by Saeid Eslamian and Faezeh Eslamian, recent advancements in hydraulic modelling have seen coupled 1D-2D models emerge as the preferred approach for flood risk assessment due to their ability to balance computational efficiency with spatial accuracy.

These hybrid systems integrate the strengths of both modelling approaches—employing 1D solutions for efficient channel routing while leveraging 2D frameworks for detailed floodplain dynamics. Research from the handbook demonstrates that coupled models can reduce simulation times by 30–50% compared to complete 2D models while improving flood extent accuracy by 15–25% over standalone 1D systems.

Technological progress has driven this evolution, including improved coupling algorithms that ensure momentum conservation and high-performance computing capabilities that enable basin-scale applications.

The increasing preference for coupled 1D-2D approaches is further evidenced by their successful implementation in complex environments, such as Vietnam's Mekong Delta, where the F28 model effectively captured intricate infrastructure-floodplain interactions while maintaining computational feasibility.

Similarly, regulatory bodies have recognised these advantages, with FEMA guidelines now specifically recommending coupled approaches for systems with disconnected floodplain flows. As computational resources expand, these integrated modelling systems have increasingly become the standard methodology for flood risk assessments requiring both hydraulic precision and operational practicality, particularly in lowland deltas and urbanised floodplains where infrastructure significantly alters natural flow patterns.

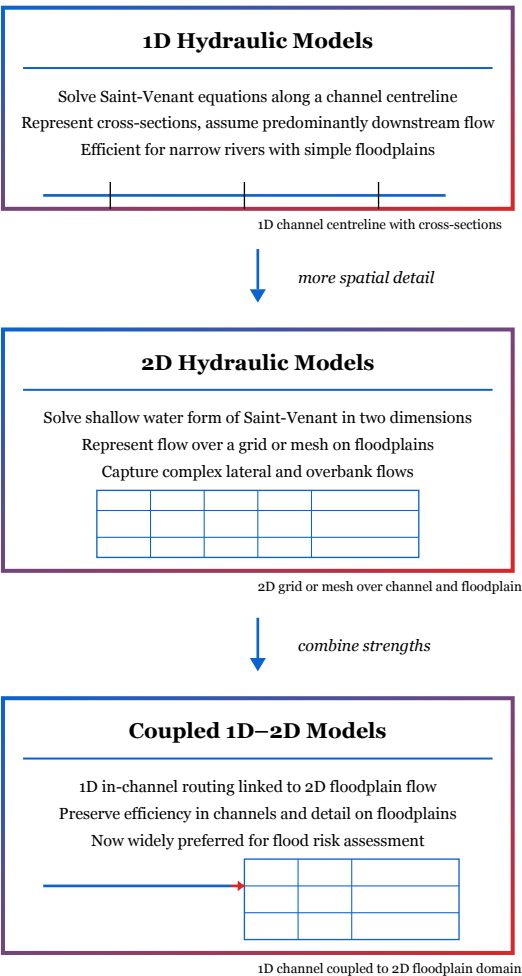
Model Requirements and Implementation

Successfully applying hydraulic modelling requires careful data inputs. Input quality directly impacts model reliability. Topographic data forms the foundation for channel and floodplain geometry. Digital Terrain Models (DTMs) represent terrain's bare earth surface. They remove all vegetation and human-made structures from the landscape. LiDAR technology generates these models efficiently. Light Detection and Ranging uses laser pulses to measure distances to Earth. It creates precise elevation measurements over large areas. Survey methods can also produce DTMs when LiDAR isn't available.

High-resolution DTMs significantly enhance flood predictions. They can improve accuracy by up to 30% compared to simpler models. Generating DTMs faces challenges in certain environments. Dense vegetation obstructs laser penetration, and water surfaces reflect signals unpredictably. Supplementary data often fills these gaps. Comprehensive data collection ensures reliable hydraulic modelling.

DTMs serve as bare-earth elevation models that capture the underlying terrain by filtering out vegetation and structures from raw data. This enables high-precision representation of riverbed morphology and floodplain features with 10–15 cm vertical accuracy.

Figure 56 – 1D, 2D and Coupled 1D–2D Hydraulic Models



LiDAR technology has revolutionised topographic data collection by using laser pulses to measure distances to the Earth's surface. This generates dense point clouds that can be processed to achieve spatial resolutions of 1m or finer. This level of detail allows hydraulic models to detect crucial microtopographic elements such as riverbanks, levees, and floodplain depressions that significantly influence water movement during flood events.

Channel and floodplain roughness parameters, expressed as Manning's n coefficients, characterise the resistance to flow and significantly affect predicted water levels and velocities.

Boundary conditions, including inflow hydrographs and stage-discharge relationships, define how water enters and leaves the model domain.

Detailed representation of infrastructure elements such as bridges, culverts, and levees ensures their hydraulic effects are appropriately captured. For unsteady flow simulations, initial conditions define the system's starting state.

The integration of high-resolution DTMs with these hydraulic parameters has transformed flood prediction capabilities. In 2019, Xia et al.'s National Water Model (NWM) improvement project documented significant gains when transitioning from simplified trapezoidal channels to realistic geometries derived from LiDAR data.

Xia demonstrated that replacing simplified channel assumptions with realistic geometries extracted from LiDAR-derived DTMs can improve flood stage and discharge predictions by up to 30%.

Despite their advantages, DTM generation faces challenges in areas where dense vegetation or water surfaces obscure ground returns. This often requires supplementary data from sonar surveys or ground measurements to fill gaps in riverbed bathymetry, highlighting the importance of comprehensive data collection approaches for robust hydraulic modelling.

Brief overview of Manning's Constant

Manning's constant, more formally known as Manning's roughness coefficient (denoted as "n"), is a dimensionless empirical value that indicates the roughness or friction affecting water flow caused by a channel's surface. This coefficient is essential to Manning's equation, which is used to estimate flow velocity and discharge in open channels such as rivers, canals, and drainage systems.

The Manning's roughness coefficient quantifies how surface characteristics impede water flow. Smoother surfaces have lower n values, while rougher surfaces have higher values. For example, finished concrete typically has an n value around 0.012, whereas natural streams with vegetation may have values around 0.035 or higher. The coefficient can range from approximately 0.01 for very smooth channels to 0.06 or more for channels with stones, debris, and significant vegetation.

The selection of an appropriate Manning's n value is crucial because it can affect computational results in hydraulic modelling. These values are typically selected from established tables based on the channel material and condition, though they can also be back-calculated from field measurements. The coefficient was introduced by Irish engineer Robert Manning in 1889 as part of his empirical equation for open channel flow.

Model Parameters:

- Physical parameters: roughness coefficients, expansion/contraction coefficients.
- Numerical parameters: time step size, space step size, convergence tolerances.

- Boundary condition parameters: upstream flow conditions, downstream stage relationships.

The computational implementation involves several important considerations that affect model performance and reliability. Spatial resolution must balance detail with computational efficiency. The finer resolution provides more accurate results but requires more computational resources.

Temporal discretisation significantly impacts unsteady simulation stability, accuracy, and computational efficiency. Time steps influence how flow changes are captured over time. Smaller time steps generally increase stability by better resolving rapid flow transitions. However, they also increase computational demands. Different numerical schemes offer various trade-offs. Explicit schemes solve equations directly for each time step but require smaller steps to maintain stability. Implicit schemes can use more significant time steps but involve more complex calculations at each step. The Courant-Friedrichs-Lewy (CFL) condition, which relates time step size to grid spacing and flow velocity, provides a key stability criterion.

Brief Overview of the Courant-Friedrichs-Lewy Condition

The Courant-Friedrichs-Lewy condition is a key stability rule for explicit numerical methods used in solving partial differential equations, especially the Saint-Venant equations that describe open channel flow. Named after Richard Courant, Kurt Friedrichs, and Hans Lewy, who first explained it in 1928, this condition defines an important link between time step size, spatial resolution, and flow velocity necessary to avoid numerical instability in explicit finite difference methods.

Temporal discretisation greatly influences the stability, accuracy, and efficiency of unsteady flow simulations. The CFL condition offers the mathematical basis for selecting suitable time step sizes based on grid spacing and flow features. Smaller time steps typically improve stability by more accurately capturing rapid flow changes, but they also significantly increase computational effort. The CFL condition precisely determines how small the time step must be to ensure numerical stability, preventing the simulation from generating non-physical oscillations or unbounded solutions.

The physical interpretation of the CFL condition is elegant: it requires that the computational time step not exceed the time needed for a disturbance to propagate across a single spatial grid interval. Essentially, information cannot travel faster in the numerical scheme than it does in the physical system being modelled. When this condition is violated, the numerical scheme attempts to advance the solution in time before accounting for disturbances arriving from neighbouring grid points, leading to instability and solution breakdown.

The CFL Condition Formula:

For the Saint-Venant equations governing open channel flow, the CFL condition is expressed as:

$$Cr = (|V| + c) \times \Delta t / \Delta x \leq 1$$

where:

- Cr is the Courant number (must be ≤ 1 for stability)
- V is the flow velocity
- c is the wave celerity ($\sqrt{gy}$), where g is gravitational acceleration and y is flow depth)
- Δt is the computational time step
- Δx is the spatial grid spacing

The term $(|V| + c)$ represents the maximum speed at which information propagates through the system—the sum of flow velocity and wave celerity captures both the advective transport of the flow itself and the propagation of surface gravity waves.

Practical implementation guidelines have emerged from extensive research. Studies suggest that time steps should be less than the hydrograph rise time divided by 20 for flood modeling applications. The theta weighting factor in finite difference schemes balances stability and accuracy in hydraulic software like HEC-RAS. Importantly, coarser computational grids enable stable simulations with longer time steps due to relaxed CFL conditions—for instance, using 500-meter cells instead of 5-meter cells dramatically increases the allowable time step, reducing computational demands while maintaining stability through techniques like subgrid bathymetry that preserve hydraulic accuracy.

Modern computational approaches have developed alternatives to address CFL limitations. Implicit and semi-implicit schemes offer unconditional stability, meaning they remain stable regardless of time step size, though they require solving simultaneous equations across all computational points at each time step. Adaptive time-stepping approaches dynamically adjust step size based on flow conditions and error estimates, maintaining stability while minimizing computational cost. GPU acceleration and parallel computing architectures have transformed simulation capabilities, enabling researchers to handle the smaller time steps required by explicit schemes while achieving dramatic speedups—studies demonstrate $100\times$ performance improvements using parallel CUDA architectures compared to traditional CPU implementations.

HEC-RAS: Industry Standard Implementation

Recent advancements in computing power have enabled more involved approaches. Research by Oleg Zikanov (2010) established practical guidelines stating that time steps should be less than hydrograph rise time divided by 20 for flood modelling. The theta weighting factor balances stability and accuracy in methods used by software like HEC-RAS (Hydrologic Engineering Center's River Analysis System)

GPU acceleration has transformed simulation capabilities. Studies by Eslamian and Eslamian (2017) demonstrated that parallel CUDA architectures enable million-particle Smoothed Particle Hydrodynamics simulations with $100\times$ speedups compared to CPUs. Adaptive time-stepping

approaches dynamically adjust step size based on error estimates, maintaining stability while minimising computational cost.

Modern workflows increasingly combine multiple techniques. Bengt Andersson's work (2011) on error transport equations co-solved with primary equations shows how localised discretisation errors can enable iterative correction for improved accuracy. Knight and Shamseldin's research (2009) on hybrid implicit-explicit schemes leverages GPUs for stiff terms while explicitly handling non-stiff components, optimising stability and speed. These developments allow hydraulic modellers to achieve high-fidelity simulations of complex phenomena like turbulence and multiphase flows within practical timeframes.

The HEC-RAS, developed by the U.S. Army Corps of Engineers, is the global industry standard for implementing hydraulic modelling principles. Its comprehensive capabilities, including one-dimensional steady and unsteady flow modelling, two-dimensional unsteady flow modelling, and combined 1D/2D approaches, have established it as a worldwide cornerstone tool for flood risk assessment. In addition to hydraulic calculations, it provides modules for sediment transport with movable boundary computations and water quality analysis, ensuring a holistic assessment of water resource systems.

The software's advanced 2D modelling features represent a significant evolution in hydraulic modelling capabilities. Subgrid bathymetry allows detailed terrain representation even with relatively coarse computational meshes, improving efficiency without sacrificing accuracy.

This innovative approach, detailed in the HEC-RAS 2D User's Manual version 3 (2024), enables computational models to represent fine-scale underwater terrain features while using coarse computational grids. The system works by preprocessing detailed bathymetric data into hydraulic property tables for each coarse grid cell. These tables store critical relationships between water elevation, wetted area, volume, and hydraulic roughness at subgrid scales.

A key advantage is the handling of partially submerged terrain. Subgrid bathymetry is a computational technique that incorporates high-resolution underwater terrain data (bathymetry) within larger computational grid cells, essentially storing detailed elevation information below the visible resolution of the model's mesh.

Rather than simplifying terrain by averaging elevations within each cell, this approach preserves critical small-scale features like narrow channels, depressions, and ridges by maintaining their hydraulic properties in lookup tables that inform calculations even when these features are smaller than the grid cell.

Cells can track narrow channels within larger grid elements using elevation-dependent volume curves, thus avoiding simplified binary "wet/dry" approximations. This preservation of flow dynamics in small features maintains model fidelity, even with larger cell sizes. The Global Modelling and Assimilation Office's 2025 study demonstrates that this approach enables flood modelling across entire basins while still resolving meter-scale hydraulic features.

Coarser grids dramatically lower computational demands—for example, using 500-meter cells instead of 5-meter cells reduces total cell count by a factor of 10,000. The HEC-RAS Model Library (2022) documents how these larger cells enable stable simulations with longer timesteps due to relaxed Courant–Friedrichs–Lewy conditions.

Despite this computational efficiency, accuracy is maintained through terrain fidelity. Small channels are less than 100 meters wide are preserved via subgrid elevation-volume curves, preventing artificial widening that plagued earlier modelling approaches. The software even tracks sediment movement at subcell resolution, avoiding homogenisation issues across large computational elements.

Implementation of rainfall-runoff modelling with infiltration enables direct simulation of pluvial flooding processes. Granular representation of hydraulic structures ensures that bridges, culverts, weirs, and similar features are appropriately integrated into the hydraulic calculations.

Levee and floodwall modelling capabilities permit the evaluation of flood defence systems under various loading conditions. Flexible hydraulic connections between 1D and 2D domains allow appropriate representation of complex flow paths. Wetting and drying algorithms accurately simulate the dynamic progression of inundation across initially dry areas, which is crucial for realistic flood mapping.

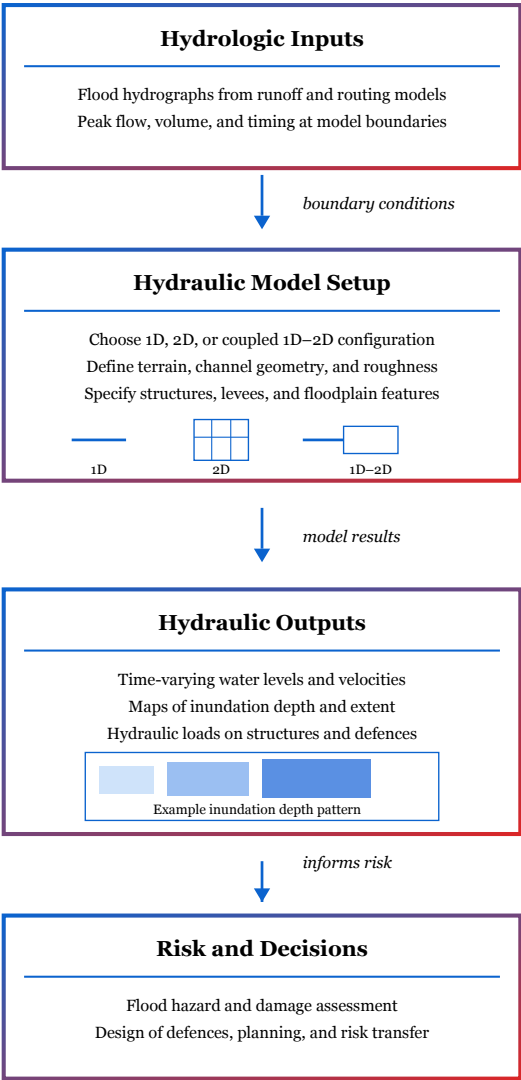
Through its RAS Mapper interface, HEC-RAS provides an integrated geospatial environment, supporting terrain model development, comprehensive spatial data management, and advanced visualisation of hydraulic results. This integration with GIS platforms facilitates seamless workflows from data preparation through result analysis and presentation. Computational enhancements, including parallel computing capabilities and GPU acceleration options, allow efficient execution of complex simulations. At the same time, adaptive time stepping improves computational efficiency by adjusting temporal resolution based on flow conditions.

The model setup process involves a systematic approach to represent the physical system accurately. Preparing terrain data incorporates LiDAR, survey data, and other sources to describe the model domain accurately. Developing the computational mesh establishes 1D cross-sections and/or 2D computation cells at appropriate locations and densities to capture relevant hydraulic features.

Specifying boundary conditions defines how water enters and exits the system while assigning hydraulic parameters characterising the flow resistance and structural properties. After executing the simulation, analysing results provides insights into flow characteristics relevant to risk assessment.

HEC-RAS provides a robust framework for implementing the Saint-Venant equations and their variations in practical applications, making it an essential tool for detailed flood risk assessment. Its combination of theoretical rigour, practical usability, and comprehensive features has established it as the preferred platform for hydraulic modelling in professional practice and research applications.

Figure 57 – Hydraulic Modelling in the Flood Risk Workflow



Hydraulic modelling, particularly with advanced tools like HEC-RAS, offers powerful capabilities for flood risk assessment but comes with important considerations. The detailed representation of flow physics enables numerous valuable applications in risk assessment and management. Detailed floodplain delineation provides spatial information on flood extent and characteristics, which are crucial for zoning, insurance, and emergency planning.

Infrastructure design and analysis ensures that bridges, culverts, levees, and other structures perform adequately under design flood conditions. Evaluation of flood mitigation alternatives

allows comparison of different intervention strategies to identify optimal solutions. When coupled with hydrologic models, real-time flood forecasting warns of flood conditions.

HEC-RAS Outputs for Flood Risk Assessment

HEC-RAS generates comprehensive hydraulic outputs that serve as fundamental inputs for flood risk assessment and decision-making. These outputs provide spatial and temporal information about flood characteristics across the modeled domain, enabling detailed evaluation of potential flood impacts on communities, infrastructure, and ecosystems.

Inundation Extent and Depth

Inundation extent defines the geographical area affected by flooding, while depth represents the vertical distance between the water surface and ground elevation at any given location. HEC-RAS computes these values for every computational cell in 2D models or at cross-sections in 1D models.

Inundation depths are typically classified into risk categories:

- Low hazard: 0-1 m
- Medium hazard: 1-2 m
- High hazard: greater than 2 m

These classifications directly inform emergency response planning and building design requirements. Studies have demonstrated that inundation depth can range from 0-3.2 m at different locations along river systems, with variations depending on return periods and topographic characteristics.

These outputs can be exported to GIS platforms like ArcGIS for spatial analysis, allowing risk managers to identify specific properties, critical infrastructure, and populations within flood-prone areas. The extent and depth information proves essential for determining flood insurance premiums, establishing building elevation requirements, and delineating floodway boundaries.

Flow Velocity Distributions

Velocity distributions describe the speed and direction of water movement throughout the inundated area. HEC-RAS provides velocity information everywhere within the computational domain, capturing both magnitude and directional components of flow. Maximum velocities at any location and velocities at specific time steps during the flood event can be extracted.

High-velocity zones present substantial hazards beyond simple inundation, as they can compromise structural stability, increase erosion potential, and create dangerous conditions for evacuation or rescue operations. Velocity data prove critical for embankment protection design, ensuring that erosion control measures can withstand anticipated flow forces.

The depth-velocity product, which combines these two parameters, indicates pedestrian and vehicle stability in floodwaters, informing evacuation planning and road closure decisions.

Water Surface Profiles

Water surface profiles depict the elevation of the water surface along the course of a river or channel. HEC-RAS calculates these profiles using the energy equation through an iterative method, where the total energy at any point equals the sum of potential energy (elevation head) and kinetic energy (velocity head). The calculation uses the direct step method for steady flow conditions or solves the full Saint-Venant equations for unsteady flow scenarios.

These profiles reveal how backwater effects from downstream obstructions, such as bridges or channel constrictions, influence upstream water levels. Profile lines can be created at any location within 2D modeling areas to extract water surface elevations at specific time steps.

Arrival Time of Flood Peaks

The temporal dimension of flooding—specifically when peak water levels or discharges arrive at different locations—provides crucial information for emergency management. HEC-RAS unsteady flow simulations track the propagation of flood waves through the river system, enabling determination of arrival times at critical locations. Timeseries data for depth and velocity at any point within the inundation area can be retrieved from the model, and flow hydrographs across specified profile lines show how discharge varies over time at specific locations.

Shear Stress for Erosion Potential

Shear stress indicates the tangential force per unit area that flowing water applies to the channel bed and banks. HEC-RAS calculates shear stress values across the modelled domain, with results available at peak values or at specific time steps. This parameter is directly linked to the erosion potential of various surfaces and the stability of channel boundaries during flood events.

High shear stress zones indicate locations where erosion is likely to occur, potentially leading to channel migration, undermining of structures, or failure of bank protection measures. These values inform the design of erosion control measures, including:

Sizing of riprap

Riprap is a layer of large, durable rocks or stones placed along riverbanks, shorelines, bridge foundations, and other hydraulic structures to prevent erosion from flowing water. The primary purpose of riprap is to protect soil and embankments from being washed away by water flow. The rocks absorb and dissipate the energy of moving water, reducing its erosive force on the underlying surface. This is particularly important in areas with high flow velocities where water can cause significant erosion damage.

Stream power, another related output available from HEC-RAS, combines shear stress with velocity to provide additional insight into erosive forces. This information proves essential for assessing infrastructure vulnerability, particularly for bridge foundations, pipeline crossings, and levee systems where scour and erosion pose significant risks.

Despite its capabilities, hydraulic modelling faces several limitations and challenges that influence its application. Extensive data requirements can constrain implementation in areas with limited topographic or hydrometric information. Significant computational demands may necessitate simplifications or reduced resolution in large or complex systems.

The need for skilled practitioner expertise means that model quality depends heavily on the modeller's experience and judgment. Uncertainty in boundary conditions and parameters propagates through the model, affecting the reliability of results. Model simplifications may not capture all relevant processes, particularly in highly complex or unconventional hydraulic situations.

Several promising developments are shaping the future of hydraulic modelling. Enhanced integration with real-time monitoring systems enables continuous model updating and improved forecast accuracy. Improved representation of complex hydraulic structures allows better simulation of engineered systems.

More efficient numerical schemes lower computational demands while preserving accuracy. Improved uncertainty quantification and communication aid decision-makers in grasping model results' limitations and confidence levels. Integration with artificial intelligence presents significant advancements in model calibration and prediction capabilities, potentially lowering the expertise barrier for effective modelling.

Hydraulic modelling provides a detailed analysis of complex systems where more straightforward hydrologic routing methods fall short. The ongoing challenge of balancing physical accuracy with computational efficiency continues to drive advancements in the field, highlighting its increasing importance in physical risk assessment applications. As computational capabilities advance and data availability improves, the scope and detail of hydraulic modelling applications will continue to expand, enhancing our ability to understand and manage flood risk in an increasingly complex and changing environment.

Chapter 7 - Flood Risk Assessment

While hydrological modelling provides the foundation for understanding water movement, assessing flood risk requires a further layer of analysis that bridges the gap between water dynamics and their impacts on the built environment. The preceding chapter established how we model the fundamental physics of water movement; this chapter examines how we translate that understanding into practical flood risk assessments that inform banking, insurance, and property decisions.

Flood risk assessment is where the abstract physics of water flow meets the concrete reality of human settlement. It translates hydrodynamic principles into the language of vulnerability, exposure, and banking impact.

Regarding property valuation, the eternal adage of location, location, location also applies to flood hazards.”- David Kelly, MKM

According to the World Health Organisation, floods are the most frequent natural disaster, affecting over 2 billion people between 1998 and 2017. They cause widespread devastation, loss of life, and significant economic damage. The International Association of Hydrological Sciences, in cooperation with UNESCO, emphasises that floods represent the most common natural disaster worldwide, with their frequency increasing dramatically over the past two decades. According to Eslamian and Eslamian (2017), flood events accounted for 43% of all recorded natural disasters between 1995 and 2015, affecting more communities than any other hazard type.

Research published in the "Flood Handbook" by Eslamian and Eslamian (2017) highlights that floods are among the most devastating natural hazards, with global costs significantly increasing due to factors such as climate change and expanded development in floodplains. Their analysis documents that floods regularly claim over 20,000 lives annually, impacting approximately 75 million people worldwide. Economic assessments from Pender and Faulkner's 2011 "Flood Risk Science and Management" indicate that annual global flood damages range from \$50 to \$60 billion, projected to increase substantially by 2050.

Knight and Shamseldin's comprehensive study (2009) demonstrates that flood-related losses have grown exponentially, with economic damages increasing sevenfold after adjusting for inflation between the 1960s and 2000s. Their research shows that these impacts disproportionately affect developing nations, where flood damages can represent up to 15% of annual GDP in severely impacted regions. Teegavarapu's analysis (2012) further quantifies that flood events account for approximately 40% of all global economic losses from natural disasters.

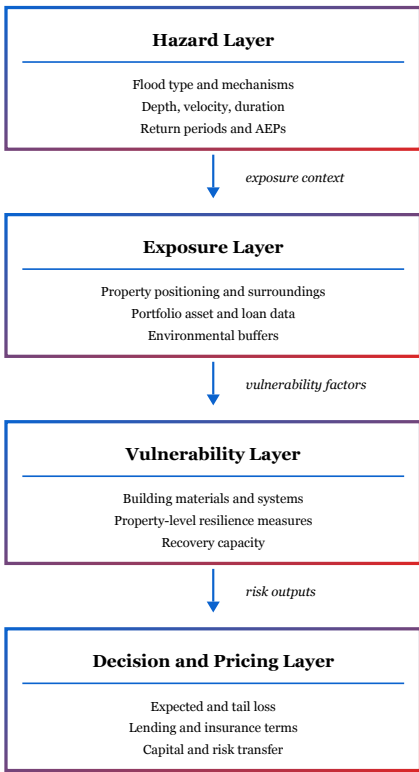
At its core, flood risk is the product of two essential components: the probability of a flood hazard occurring and the vulnerability or potential consequences stemming from that event. This dual nature of risk—combining likelihood with impact—forms the fundamental principle underlying all assessment methodologies.

Modern flood risk analyses leverage increasingly technology, including advanced software tools, high-performance computing, high-resolution datasets, LiDAR elevation models, and detailed building inventories.

These technologies enable unprecedented precision in evaluating flood impacts across large geographic areas and diverse property portfolios. The outputs of these analyses are often visualised through flood risk mapping, which represents hazard and risk information spatially and shows inundation extent, depth, and velocity for different probability scenarios.

The essential first step in this process is classifying flood types. While all floods involve excess water, their dynamics, predictability, and impacts vary dramatically depending on their source and characteristics. Riverine flooding follows different patterns than coastal storm surges, and flash floods present distinct challenges compared to gradual groundwater rise. Each classification entails different modelling approaches, warning timeframes, and mitigation strategies.

Figure 58 - Flood Risk Assessment Layers



Property-Specific Assessment

Property-specific assessment forms the crucial next layer of analysis. A structure's position within the landscape—its elevation, proximity to water bodies, and relationship to surrounding terrain—fundamentally determines its exposure to flood hazards. Practitioners must analyse this geospatial positioning and broader environmental factors, including watershed characteristics, natural and local flood defences, and weather pattern change projections that may alter historical patterns.

The resilience of individual properties completes this assessment picture. Structural characteristics, adaptation measures, and system redundancies all influence how a property will respond when flood waters arrive. This component of assessment bridges purely physical analysis with the practical concerns of property owners, insurers, and lending institutions.

As we explore these dimensions of flood risk assessment, we extend the hydrologic and hydraulic models for water flow into frameworks that support the initial assessment of property exposure to a hazard such as flood.

While significant progress has been made in flood risk assessment methodologies, standardised practices and agreed indicators for flood risk mapping, especially at local scales, continue to evolve across different regions and countries. This progression—from hydrological modelling to flood risk assessment—ultimately enables the quantification of impacts discussed in the following chapter.

Flood risk assessment becomes decision-useful when it is explicitly mapped to banking and insurance levers: loan origination, collateral valuation, portfolio limits, and the design of risk transfer structures such as Physical Risk Swaps. At property level, hazard, exposure, and vulnerability metrics feed directly into loan-to-value (LTV) adjustments, debt service coverage requirements, and covenants. At portfolio level, they inform geographic and peril concentration limits, capital allocation, and the cost of hedging residual risk through reinsurance or capital markets instruments.

For a lender, a “high” flood risk does not merely imply a larger expected loss; it also changes loss timing and correlation. Concentrated flood risk can create clustered defaults and abrupt collateral value write-downs, which is precisely the risk that PRS structures are designed to offload. Making this link explicit—hazard → property risk score → loan term ↔ PRS hedge—not only clarifies the role of Chapter 7 but also prepares the reader for later chapters on structuring. The accurate quantification of flood risk requires specialised analytical processes that go beyond simply modelling water movement. A comprehensive flood risk analysis quantifies the probabilities and consequences of potential flood events, serving as a critical subset of the broader flood risk management process. This management approach represents a cyclical, continuous effort to reduce flood impacts through structural and non-structural measures, with risk assessment providing the analytical foundation for decision-making.

Flood Classification Systems

The systematic categorisation of flood events is the foundation for practical risk assessment. Classification systems provide the necessary framework to differentiate between flood types, each with distinct characteristics influencing their modelling, prediction, and mitigation approaches. This section explores established and emerging flood classification methodologies, their applications, and limitations in the context of physical risk assessment.

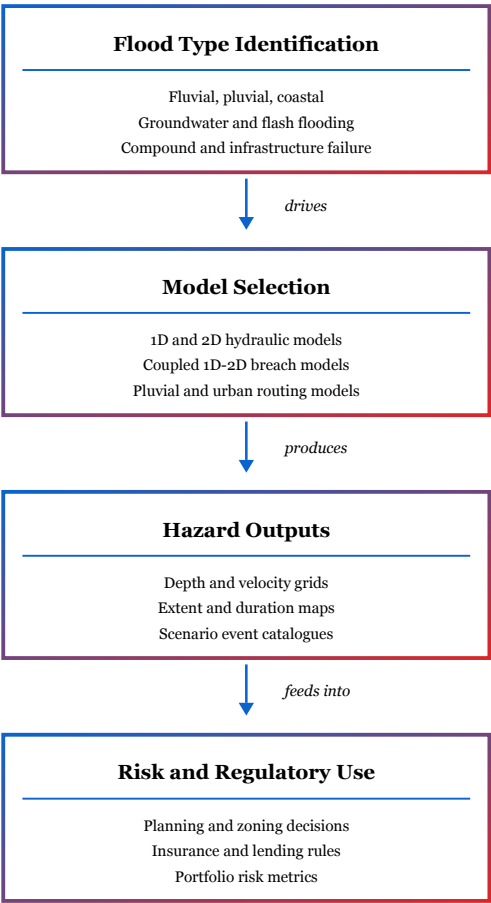
Traditional Classification Methods

Historically, floods have been classified primarily by their source mechanism, with each category representing fundamentally different hydrodynamic processes:

- **Fluvial (Riverine) Flooding:** Occurs when water exceeds the capacity of river channels, causing overbank flow. Riverine floods typically develop more gradually than other flood types, with warning times ranging from hours to days, depending on watershed characteristics. The dynamics of these floods are governed by river channel geometry, floodplain topography, and upstream hydrological conditions. Assessment methodologies typically incorporate river gauge data, precipitation records, and watershed models to establish probability distributions of flood magnitudes.
- **Pluvial (Surface Water) Flooding** occurs when rainfall overwhelms drainage systems' capacity or infiltrates the ground, causing water to pool or flow over the land surface. Unlike riverine flooding, pluvial events can happen anywhere—even far from water bodies—making them particularly challenging to predict and map. The risk is especially pronounced in urban environments with extensive impervious surfaces. Assessment requires high-resolution digital elevation models, detailed drainage system mapping, and precipitation statistics.
- **Coastal Flooding:** Primarily driven by storm surges, high tides, or tsunamis, coastal flooding presents distinct challenges due to its interaction with wave dynamics, tidal patterns, and coastal geomorphology. The combined effect of high water levels and wave action creates complex loading scenarios for coastal structures. Assessment methodologies must account for astronomical tides, barometric effects, wind setup, wave runup, and, increasingly, sea level rise projections.
- **Flash Flooding:** Characterised by rapid onset (typically within six hours of the causative event) and high water velocities, flash floods represent hazardous events with limited warning time. They commonly occur in steep watersheds (a drainage area or catchment where the land slopes sharply, leading to rapid water movement), urban areas with high imperviousness, or regions susceptible to intense rainfall. Assessment approaches must emphasise the temporal dimension of flooding, incorporating rainfall intensity-duration-frequency relationships and time-of-concentration calculations.
- **Groundwater flooding** occurs when the water table rises to reach the land surface, usually after extended periods of heavy rainfall. This type of flood typically lasts a long time but develops

gradually. Proper assessment involves understanding hydrogeological conditions, prior moisture levels, and subsurface water movement—factors often overlooked in standard flood models.

Figure 59 - Flood Classification to Modelling Choice



While these traditional classifications provide a helpful starting point, they often fail to capture the complexity of real-world flood events, which frequently involve multiple mechanisms operating simultaneously. This limitation has driven the development of more classification frameworks.

Regulatory Preamble

There is never a perfect time or place to mention regulations, but this is the best moment since how governments consider flood risk and their chosen responses is central to how insurance and mortgage lending operate.

The European Floods Directive

Before the establishment of comprehensive flood management frameworks in Europe, the continent experienced several devastating flood events that highlighted the need for coordinated action. The early 2000s saw particularly catastrophic flooding, with the August 2002 floods affecting Central Europe, causing over €15 billion in damages and claiming dozens of lives across Germany, Austria, the Czech Republic, and other nations.

These events, along with rising concerns about the effects of climate change on precipitation patterns and sea levels, prompted European policymakers to create a more integrated approach to flood risk management. Between 2000 and 2006, over 175 significant floods were recorded in Europe, emphasising the urgency for systematic assessment and management protocols that crossed national boundaries.

In response to these challenges, the European Parliament and Council adopted the European Floods Directive (2007/60/EC) on October 23, 2007, which came into force on November 26, 2007. The Directive established a framework requiring Member States to assess and manage flood risks coordinated across transboundary river basins.

Implementation followed a six-year cycle, starting with Preliminary Flood Risk Assessments completed by December 2011, then detailed flood hazard and risk maps developed by December 2013, and comprehensive Flood Risk Management Plans by December 2015. This revolutionary approach transcended traditional flood defence strategies, shifting towards more holistic risk assessment methodologies that acknowledged flood events' complex, multi-faceted nature and their impacts on human health, economic activities, cultural heritage, and the environment.

U.S. National Flood Insurance Program

Before the National Flood Insurance Program (NFIP) was established, the United States encountered a significant gap in disaster management policy, as many flood-prone properties were uninsurable in private markets, leaving residents primarily dependent on federal disaster assistance.

A series of catastrophic flood events in the 1950s and 1960s, including Hurricane Betsy in 1965, which caused over \$1 billion in damages (equivalent to approximately \$8.5 billion today), underscored the need for a systematic approach to flood risk. These events, coupled with rising federal disaster relief costs and increasing development in floodplains, prompted Congress to commission extensive studies on flood hazards and potential insurance solutions, culminating in the landmark 1966 "Insurance and Other Programs for Financial Assistance to Flood Victims" report that laid the groundwork for a national flood insurance mechanism.

In response to these challenges, the United States Congress established the National Flood Insurance Program through the National Flood Insurance Act of 1968, implementing a comprehensive approach that balanced insurance accessibility with floodplain management requirements.

The program became operational in 1969 under the Department of Housing and Urban Development before transferring to the Federal Emergency Management Agency (FEMA) in 1979 following its creation. The NFIP underwent significant expansions and reforms through subsequent legislation, including the Flood Disaster Protection Act of 1973, which introduced mandatory purchase requirements for properties with federally-backed mortgages in Special Flood Hazard Areas, and the National Flood Insurance Reform Act of 1994, which strengthened compliance mechanisms and established the Flood Mitigation Assistance Grant Program.

These developments transformed what began as a modest voluntary program into a cornerstone of American flood risk management, providing over 5 million policies with more than \$1.3 trillion in coverage nationwide by the early 2000s.

UK's Flood RE

In the decades preceding Flood Re, the United Kingdom experienced increasingly frequent and severe flooding events, highlighting significant market failures in flood insurance provision. The Easter floods of 1998, the widespread flooding of autumn 2000, and the catastrophic summer floods of 2007—which affected over 55,000 properties and caused approximately £3.2 billion in damages—demonstrated the urgent need for reform.

Before 2016, flood insurance in the UK was governed by a series of voluntary agreements between the government and insurers, starting with the informal "Gentleman's Agreement" in the 1960s and culminating in the Statement of Principles (2000-2013). However, this system became increasingly unsustainable as climate change heightened flood risk, and premium reached prohibitive levels for many homeowners in high-risk areas.

Following extensive consultation and negotiations between 2010 and 2014, the UK Parliament passed enabling legislation through the Water Act 2014, establishing Flood Re as a not-for-profit reinsurance scheme. The program officially launched on April 4, 2016, representing a novel approach to address the growing insurance protection gap.

In 2022, Flood Re enhanced its offerings by introducing the "Build Back Better" scheme. This innovative initiative provides up to £10,000 in additional funding for property-level flood resilience measures when claims are settled. This forward-thinking program enables homeowners to not repair flood damage and enhance their property's resilience against future flooding events through solid doors, raised electrical outlets, and water-resistant flooring.

Designed with a planned 25-year lifespan, Flood Re gradually created a transitional mechanism to move from subsidised to risk-reflective pricing by 2039. Its Build Back Better scheme exemplifies how insurance can be leveraged to drive adaptation and risk reduction rather than simply transferring financial risk, representing a significant evolution in approach compared to traditional flood insurance models.

UK's Hazard Rating System

The development of comprehensive hazard classification systems in the United Kingdom emerged from a growing recognition of the need to quantify flood risks more precisely for infrastructure planning and public safety purposes. In the early 2000s, following significant flood events across the UK, particularly the Easter floods of 1998 and the widespread flooding of autumn 2000, research institutions and government agencies began developing more approaches to assess flood hazards beyond simple inundation mapping.

This period coincided with the implementation of the Housing Act 2004, which introduced the Housing Health and Safety Rating System (HHSRS), signalling a broader shift toward risk-based assessment methodologies across multiple sectors, including flood management. The Flood Risk to People research project, commissioned by the Department for Environment, Food and Rural Affairs (DEFRA) in 2003, played a pivotal role in establishing the scientific basis for what would become the UK's standard hazard rating formula.

Building on this foundation, the UK Environment Agency and DEFRA jointly published "Flood and Coastal Defence R&D Programme: Flood Risks to People" in 2006, which formally introduced the depth-velocity hazard rating formula that would become widely adopted across the nation.

This methodology represented a significant advancement over previous approaches by explicitly incorporating multiple flood characteristics—water depth, velocity, and debris factor—into a single quantifiable metric directly linked to human stability thresholds in floodwaters.

The formula $HR = d \times (v + 0.5) + DF$ was subsequently incorporated into official flood risk assessment guidance documents, including the 2008 "Supplementary Note on Flood Hazard Ratings and Thresholds" and the Flood Risk Assessment Guidance for New Development (FD2320/TR2). By 2010, this approach had become embedded in standard practice for flood risk assessments throughout England and Wales. It provided emergency planners, local authorities, and developers with a consistent framework for evaluating and communicating flood hazards that directly corresponded to potential impacts on human safety and infrastructure vulnerability.

Australian Rainfall and Runoff Guidelines

The Australian Rainfall and Runoff (ARR) Guidelines represent one of the longest-standing and most comprehensive national frameworks for flood estimation and water management in the world. First published in 1958, these guidelines have undergone several major revisions to incorporate advancements in hydrological science, data collection techniques, and computational methods.

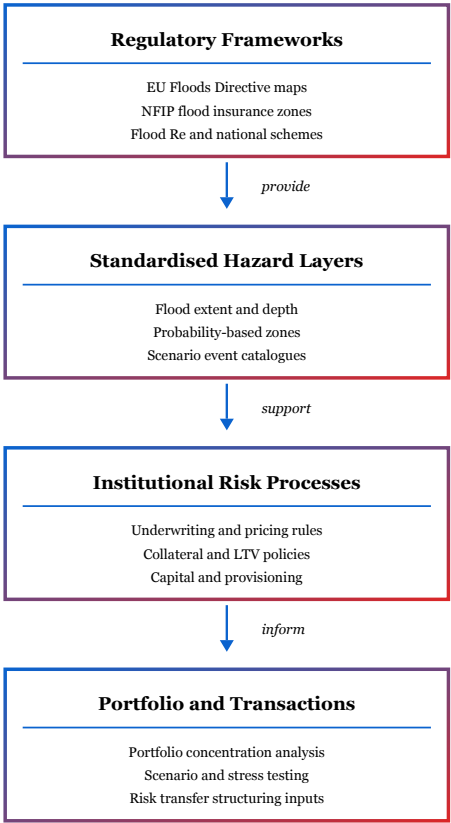
The most significant transformation came with the 4th edition, released in 2016. This edition represented a complete overhaul that integrated over 30 years of additional rainfall observations from more than 10,000 gauging stations across Australia's diverse climatic regions. This landmark update moved away from outdated methodologies like the Rational Method toward

more regional flood frequency estimation techniques that better reflect Australia's highly variable precipitation patterns.

Following the 2016 revision, Engineers Australia transferred ongoing management of ARR to Geoscience Australia, ensuring its continued development as a government-funded public resource.

ARR has maintained a foundational principle that effective flood risk management requires calibration to local conditions—acknowledging that Australia's diverse landscapes, from tropical northern regions to temperate southern zones, necessitate tailored approaches rather than one-size-fits-all solutions. This emphasis on local calibration has become increasingly important as climate change alters historical rainfall patterns, requiring practitioners to continuously adapt their flood classification methodologies to reflect evolving regional circumstances.

Figure 60 - Regulatory Mapping into Banking and Insurance



Advanced Classification Frameworks

Recognising the limitations of source-based classifications, more nuanced frameworks have emerged that incorporate multiple parameters to characterise flood events:

- **Multi-parameter Classification Systems:** These approaches classify floods based on combinations of factors, including source, temporal characteristics (onset speed, duration, seasonality), spatial extent, and driving meteorological conditions. The European Floods Directive, for example, employs a multi-parameter approach that enables more precise risk assessment and management planning. These systems recognise that a single flood event may encompass multiple source mechanisms, such as combined coastal and fluvial flooding during severe storms.
- **Duration-Depth-Velocity Matrices:** These three-dimensional classification schemes recognise that flood impacts depend on water depth, flow velocity, and event duration. High-velocity shallow flooding, for instance, can cause more damage than deeper, static flooding due to increased hydrodynamic forces on structures. Similarly, extended duration increases damage to building materials unsuitable for prolonged water exposure. These matrices support more accurate vulnerability assessments and mitigation planning.
- **Recurrence Interval Categorisation:** This approach, which prevails in catastrophe modelling, classifies floods by their statistical return period, such as 1-in-100-year events (1% annual exceedance probability) or 1-in-500-year events (0.2% yearly exceedance probability). While widely used in regulatory contexts and infrastructure design, these classifications face increasing challenges due to non-stationarity in weather patterns. While beneficial to note, this is not what we need to solve in a capital markets scenario.
- **Combined Hazard Classification Approaches:** These frameworks integrate multiple flood characteristics to produce comprehensive hazard ratings that directly connect to potential impacts. For instance, the UK's Hazard Rating system uses an innovative methodology that combines water depth, flow velocity, and a debris factor to calculate risk levels. This integrated approach produces a single numerical value directly correlating with floodwaters' human stability thresholds. By accounting for the combined effects of these variables rather than assessing them in isolation, the system provides a more nuanced and impact-focused classification that better reflects real-world flood dangers to both people and infrastructure.
- **International Standards and Regional Variations:** Classification systems vary significantly across jurisdictions, reflecting regional priorities, data availability, and historical flood experience. For example, the Australian Rainfall and Runoff Guidelines emphasise the importance of local calibration in flood classification. In contrast, the U.S. National Flood Insurance Program classifications focus primarily on regulatory floodplain delineation. These variations create challenges for consistent cross-border risk assessment but also reflect the importance of local context in flood risk management.
- **Urban-Specific Flood Typologies:** The unique characteristics of urban flooding—influenced by complex drainage networks, building configurations, and infrastructure dependencies—have

prompted specialised classification systems for urban environments. These typologies incorporate drainage capacity exceedance, surface water flow paths, and infrastructure failure modes. They often employ high-resolution modelling to capture micro-topographic features influencing urban flood routing.

- **Infrastructure Failure Flood Types:** In addition to natural flood mechanisms, increased attention has focused on classifying floods resulting from infrastructure failures, including dam breaches, levee failures, and urban drainage system malfunctions. These classifications incorporate cascade effects and system dependencies, recognising that infrastructure failures often produce flood characteristics significantly different from those of naturally occurring events, particularly onset speed and flow velocity.

Compound Flood Event Categorisation

Compound flood event categorisation is an emerging field focused on understanding the complexity of flooding mechanisms that occur simultaneously or in close succession. This research addresses the amplified impact of such occurrences due to their multidimensional nature, involving combinations of drivers like precipitation, storm surge, river discharge, high tides, and more.

The Intergovernmental Panel on Climate Change (IPCC) first formally addressed compound events in 2012, noting that they involve multiple drivers or hazards occurring simultaneously or sequentially, often amplifying the resulting damage (Seneviratne et al., 2012). A notable example is Hurricane Harvey (2017), where record-breaking rainfall, river discharge, and storm surge combined to cause catastrophic flooding.

Research into compound events has proposed several classification frameworks. For instance, one framework identifies four interaction types for compound events:

- **Preconditioned:** Saturated soil that amplifies flood impacts.
- **Multivariate:** Co-occurrence of two hazards like storm surge and river discharge.
- **Temporally Compounding:** Sequential hazards in a region.
- **Spatially Compounding:** Hazards in different locations interact to amplify impacts.

These categories help researchers analyse specific conditions and impacts of compound flood events.

In practical modelling, rainfall runoff, storm surge, and other mechanisms are often studied together to enhance flood hazard assessments tailored to specific geographic areas, like coastal and estuarine environments. Simplified models now aim to break down complex interactions to better understand their effects on flooding and provide actionable insights for flood resilience and management.

The categorisation of compound flood events has emerged as a critical area of study, with several classification dimensions:

- **Flood-Generating Process Combinations:** Compound events frequently involve interactions between different flood types. These combinations include coastal-fluvial events (where storm surge prevents river discharge, exacerbating upstream flooding), pluvial-fluvial events (where intense rainfall overwhelms drainage systems while simultaneously causing river flooding), and coastal-pluvial events (where storm surge coincides with heavy rain). Each combination produces distinct flood dynamics that standard single-process models may fail to capture.
- **Temporal Sequencing:** Compound events can be categorised based on their temporal characteristics—single events with multiple mechanisms, multiple events clustered closely in time, or events separated by insufficient recovery periods. This temporal dimension is particularly significant for infrastructure resilience and community recovery capacity. Systems still recovering from initial flooding often display amplified vulnerability to subsequent events, even of lesser magnitude.
- **Characteristic Combinations:** The specific combination of flood characteristics—magnitude, duration, timing, onset speed, and spatial extent—provides another classification dimension. For example, prolonged moderate flooding followed by a brief high-magnitude event creates different impact patterns than consecutive moderate events. Seasonal timing significantly influences impacts, particularly for agricultural systems with varying vulnerability throughout growing cycles.
- **Pathway and Mechanism Interactions:** Compound events often involve complex interactions between flood pathways—defence overtopping or breaching, surface water accumulation, groundwater emergence, and infrastructure failure. Each timeseries pathway may trigger or exacerbate others, creating cascade effects that traditional siloed assessments fail to capture. Classification frameworks increasingly incorporate these interaction pathways to better represent real-world flood complexity.
- **Vulnerability System Memory:** A critical dimension in compound event classification involves the "flood memory" of affected systems—how quickly natural and built environments recover their baseline resilience. Systems with long recovery periods (extended groundwater saturation, damaged flood defences, or depleted community resources) remain at elevated vulnerability, creating compound impacts even when subsequent events would not usually cause significant damage.

These multi-dimensional classification approaches represent a significant advancement beyond single-mechanism frameworks, better reflecting the complex reality of flood events that rarely occur in isolation. Comprehensive compound event classification requires considering hazard drivers, physical characteristics, failure mechanisms, and receptors' vulnerability and resilience characteristics.

Riverbank Breach Modelling

The modelling of riverbank breaches represents a specialised and critically important subset of flood risk assessment. These breaches—whether occurring in natural riverbanks or engineered levees—often result in rapid inundation of previously protected areas, creating distinct hazard characteristics compared to gradual overbank flooding. This section explores the methodologies to model such breaches and their implications for comprehensive flood risk assessment.

Riverbank breach modelling employs a range of approaches that balance physical accuracy with computational efficiency:

- **One-dimensional (1D) Hydraulic Models:** represent the river system as a series of cross-sections perpendicular to the flow direction, solving the Saint-Venant equations for conservation of mass and momentum. While 1D models significantly simplify the three-dimensional flow processes during breaching events, they remain widely used due to their computational efficiency and ability to model long river reaches. The effectiveness of these models depends critically on accurate stage-discharge relationships, particularly for overbank flows where the curve typically shows a distinct inflexion point as flow encounters the floodplain's higher roughness and storage capacity.
- **River schematisation for 1D models** necessitates a "broad brush" approach aligned with the overall river features. Cross-sections are positioned to capture significant changes in channel geometry, roughness, or potential breach locations. In these models, breaches are generally depicted as lateral structures with time-varying geometric properties. The breach initiation, widening rate, and final dimensions are either predefined based on geotechnical analysis or modelled dynamically from hydraulic parameters.
- **Two-Dimensional (2D) Hydraulic Models:** divide the floodplain into a grid or mesh of cells, solving the shallow water equations to determine water depth and velocity vectors throughout the domain. These models offer significant advantages for breach modelling by directly representing the spatial variation of flood propagation across the landscape. The explicit terrain representation allows for more accurate prediction of inundation patterns following a breach, particularly in urban areas where buildings and infrastructure create complex flow paths. Additionally, 2D models provide spatial distributions of flow velocity, which are crucial for assessing erosive forces during breach development and potential damages to structures in the breach flow path.
- **Coupled 1D-2D Approaches:** represent the main river channel in 1D while representing the floodplain in 2D, offering a balance between computational efficiency and physical realism. The treatment of the interface between domains is critical for accurate breach modelling, with various methods including lateral structures, vertical links, or horizontal links. Advanced coupled models dynamically activate the 2D domain when water levels exceed threshold values, efficiently focusing computational resources on areas experiencing inundation following a breach.

- **Computational Fluid Dynamics Applications:** These models provide a detailed representation of three-dimensional flow structures for site-specific analyses. They capture complex flow features around breach openings, including vortex formation, supercritical flow regions, and hydraulic jumps influencing breach development and downstream hazards. The extreme computational demands typically limit their application to relatively small spatial domains and short time intervals.
- **Dam Break Modelling:** a specialised subset of breach modelling, focuses on dam failures that can produce particularly catastrophic flooding. In many jurisdictions, dam owners must conduct dam break analyses to predict potential inundation areas and develop emergency plans. Specialised models have been developed for dam break wave propagation, accounting for the distinctive hydraulic characteristics of these extreme events, including supercritical flow transitions and bore formation. In river systems with multiple dams, models must account for potential cascade failures where the breach of an upstream structure leads to overtopping and failure of downstream structures.

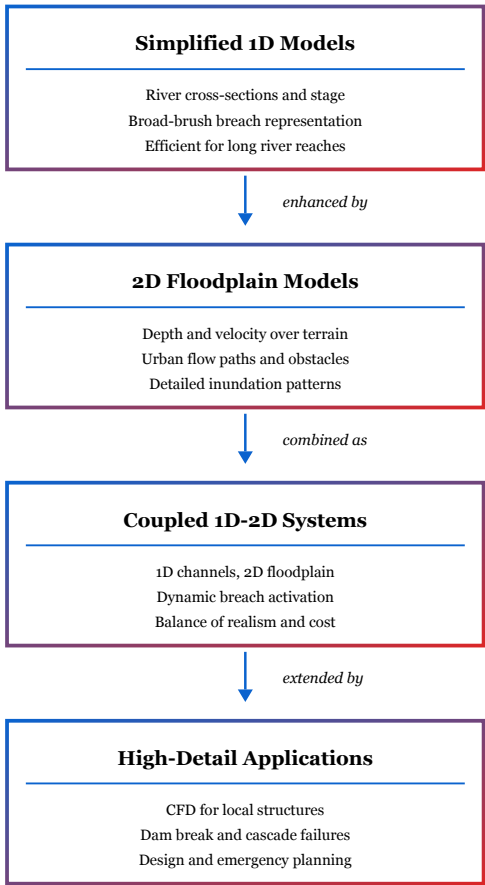
Geotechnical Factors in Breach Formation

The hydraulic modelling of breaches must be informed by a geotechnical understanding of the failure mechanisms that initiate and drive breach development:

- **Soil Composition and Erodibility:** fundamentally control breach susceptibility. Different soil types exhibit varying resistance to erosion, with cohesive clay soils generally more resistant than non-cohesive sandy soils. Direct measurement of soil erodibility through Jet Erosion Tests or Erosion Function Apparatus testing provides quantitative parameters for physically-based breach models. Natural riverbanks and even engineered levees often contain spatial variations in material properties, creating potential weak points that may initiate breach formation.
- **Vegetation Effects:** bank stability operates through both mechanical and hydrological mechanisms. Plant roots provide mechanical reinforcement to the soil, increasing erosion and mass failure resistance. The degree of reinforcement depends on root density, tensile strength, and architecture. Hydrologically, vegetation influences soil moisture regimes through rainfall interception, transpiration, and preferential flow paths along roots. These effects can either enhance stability (by reducing soil saturation) or reduce stability (by creating preferential seepage paths). The stabilising influence of vegetation often varies seasonally, creating temporal patterns in breach susceptibility.
- **Geomorphological Evolution:** recognises that rivers and their banks represent dynamic systems that evolve as a timeseries. Natural river meandering processes create zones of erosion on the outside of bends and deposition on the inside, continuously altering bank geometry. Long-term processes of channel incision (downcutting) or aggradation (sediment accumulation) alter the hydraulic loading on banks and levees, potentially creating conditions conducive to breaching even without changes in flow magnitude. Analysis of historical breach locations often reveals patterns related to these geomorphological features.

- **Advanced Soil Mechanics Applications:** Advanced soil mechanics applications increasingly incorporate principles into breach modelling. Models that combine hydraulic loading with transient seepage can forecast the development of internal erosion pathways and stability reduction caused by rising pore pressures. Combining slope stability analysis with hydraulic models allows for the evaluation of mass failure mechanisms that may trigger or hasten breach formation. Advanced models include unsaturated soil mechanics to represent the shift from unsaturated to saturated states during flood events, along with the related changes in strength and erodibility.

Figure 61 - Breach Modelling Approaches



Real-time Monitoring and Prediction

Beyond modelling for planning and design purposes, increasing emphasis has been placed on real-time monitoring and predicting potential breaches during flood events. Since the early 2010s, significant technological advances have transformed flood forecasting capabilities, with AI and machine learning integration revolutionising prediction accuracy and lead times.

Google's AI-driven Flood Hub exemplifies this evolution, providing forecasts up to seven days in advance by analysing global weather data, satellite imagery, and river gauge measurements to enhance situational awareness during flood emergencies.

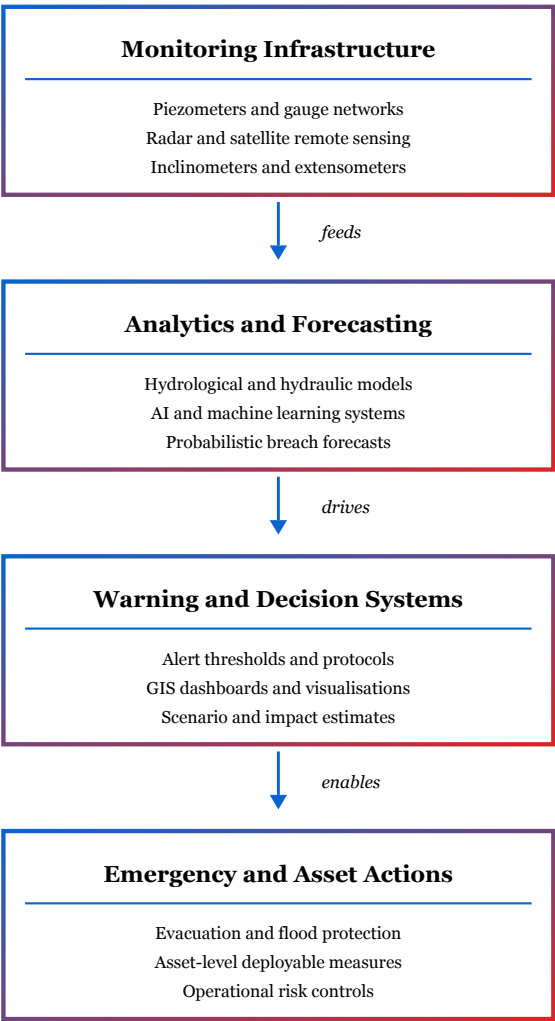
Modern integrated systems now combine GIS tools with hydrological data to produce detailed flood maps, hazard zones, and impact assessments that enable timely alerts to minimise property damage. Companies like Previsico have pioneered high-resolution modelling systems capable of predicting flood risks at the property level, utilising networks of sensors and "instacasting" to provide actionable warnings based on live radar data and hydrological measurements—dramatically improving the precision with which authorities can identify potential infrastructure failures and implement targeted emergency responses:

- **Sensor Networks:** deploy multiple monitoring technologies to detect precursors to breach formation. Piezometers (a device used to measure the pressure of a fluid, typically groundwater, in a specific location) embedded within levees measure pore water pressures, allowing the detection of anomalous seepage conditions that may precede internal erosion-driven breaches. Deformation monitoring through exotically named inclinometers (precision instruments used to measure angles of slope, elevation, or inclination relative to gravity's pull), extensometers (measure small changes in the distance between two points, specialising in detecting deformation, displacement, or strain in materials and structures), and remote sensing techniques measures bank or levee movement, providing early warning of potential mass failure mechanisms. These sensors are increasingly connected to integrated monitoring systems with automated alert thresholds and real-time data visualisation.
- **Remote Sensing Technologies:** provide broad spatial coverage to complement point-based sensor networks. Synthetic Aperture Radar systems can detect surface deformation and soil moisture changes associated with potential breach conditions, even during storm events when optical systems are limited by cloud cover. Thermal imaging can identify temperature anomalies related to seepage. In contrast, time-series analysis of remotely sensed data identifies progressive changes in bank or levee conditions that may indicate increasing breach susceptibility.
- **Machine Learning Approaches:** apply advanced analytical techniques to the complex, multi-parameter problem of breach prediction. Algorithms trained on historical breach data and precursor conditions identify patterns that may not be apparent in traditional analyses. These approaches can integrate disparate data streams from multiple sensor types, extracting meaningful signals from complex, noisy datasets. Bayesian methods provide probabilistic breach forecasts that update dynamically as new monitoring data becomes available, supporting risk-based decision-making during flood events.
- **Early Warning System Integration:** ensures monitoring and prediction capabilities translate into effective risk reduction. The definition of appropriate alert thresholds balances false positive risk against the need for sufficient warning time to implement emergency measures or evacuations. Clear communication protocols ensure monitoring data and breach predictions reach decision-makers and affected populations through multiple, redundant channels.

Integrating breach prediction into broader flood early warning systems provides comprehensive situational awareness for emergency management.

The modelling and monitoring of riverbank breaches continues to advance through integrating hydraulic modelling, geotechnical analysis, and real-time monitoring technologies. These developments support a more accurate assessment of breach-related risks and a more effective allocation of mitigation resources.

Figure 62 - Real-Time Monitoring into Early Warning



Property Positioning Analysis

The physical positioning of properties within the landscape fundamentally determines their exposure to flood hazards. While flood classification and breach modelling characterise the hazard, property positioning analysis translates these hazard characteristics into site-specific exposure assessments. This analytical approach draws from methodologies developed in real estate valuation—where the mantra "location, location, location" underscores positioning's critical importance—and adapts them to the specific requirements of flood risk assessment.

A practical property positioning workflow for lenders and insurers can be summarised in five steps:

- Screen with national / EU / FEMA-style hazard maps to identify whether the property lies within mapped floodplains for fluvial, coastal, or pluvial hazards.
- Extract high-resolution DEM or LiDAR-based elevation and compute elevation relative to local watercourses and mapped flood levels (e.g. 1-in-100 year depth).
- Analyse microtopography and street-network routing to identify whether the property sits on a local low point, flow path, or ponding area, using 2D surface flow models where available.
- Quantify distance and hydraulic connectivity to flood sources (rivers, coastlines, drainage outfalls), distinguishing direct inundation from backflow and infrastructure-mediated pathways.
- Translate these metrics into an exposure index (for example, 0–100), which can be plugged into loss models, pricing tools, and portfolio dashboards.

Topographical Analysis

Topography represents the most fundamental positioning factor for flood risk assessment, directly influencing flow paths, water accumulation, and inundation potential:

- **Elevation Relative to Flood Sources:** is the primary determinant of flood exposure. The vertical distance between a property and potential water sources—whether rivers, coastlines, or surface water accumulation zones—establishes the basic threshold for flooding. This relationship is complicated because absolute elevation alone is insufficient; the hydraulic connectivity between the property and flood sources must be considered. Properties at relatively high elevations may still experience flooding if hydraulic pathways (natural or artificial channels, drainage systems) connect them to flood sources.
- **Digital Elevation Models:** provide the foundation for topographic analysis in modern flood risk assessment. The resolution and accuracy of these models significantly influence assessment quality. While national or regional DEMs may be sufficient for broad-scale risk screening, property-specific assessment typically requires high-resolution data derived from LiDAR or

similar technologies capable of sub-meter vertical accuracy. Such precision is essential for identifying subtle topographic features that may significantly influence flood routing.

- **Microtopography:** is particularly important in urban environments, where insignificant elevation differences can significantly alter flood pathways. Features such as curbs, berms, garden walls, and small depressions may redirect or impound water in ways not captured by coarser elevation models. Advanced assessments incorporate these features through ultra-high-resolution elevation data or specific feature recognition and hydraulic representation.
- **Relative Positioning:** within the landscape context often matters more than absolute elevation. Even if their absolute elevation is relatively high, properties at local low points may be subject to ponding during intense rainfall events as surface water flows converge. Similarly, properties positioned along preferential flow paths—natural drainage lines, valleys, or urban street corridors that channel water—face elevated risk during high-intensity events regardless of their elevation relative to typical flood sources.

Mini-example (numbers illustrative)

Suppose a residential property is: 0.6 m above the 1-in-100 fluvial flood level, 80 m from a small river, located in a local depression identified by a 2D pluvial model as accumulating 0.25 m of water in a 1-in-30 rainfall event, and connected to the river via an undersized drainage system. A simple exposure index might assign 40 points for fluvial proximity, 30 for pluvial ponding, and 10 for infrastructure-mediated pathways, offset by a -10 point credit for moderate elevation, yielding an overall exposure score of 70/100 (high). This single number can then drive LTV caps, mandatory resilience upgrades, or the required protection via a PRS overlay at the portfolio level

Location Analysis

Comprehensive property positioning assessment requires analysis at multiple spatial scales, each revealing different aspects of flood risk exposure:

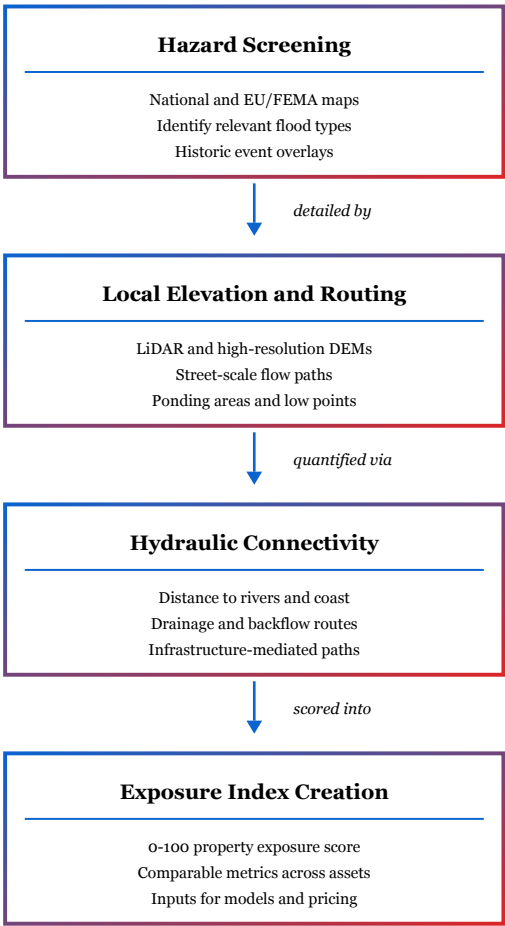
- **Regional Analysis:** examines the broader hydrological context within a property. This includes watershed characteristics, regional weather patterns, and large-scale flood control systems influencing the flood regime. Properties in regions with flat topography, poor regional drainage, or prone to large-scale weather systems capable of producing widespread precipitation (such as tropical cyclones or atmospheric rivers) face fundamentally different risk profiles than those in regions with efficient drainage networks or less extreme meteorological patterns.
- **Macro-location Analysis:** focuses on the property's position within its city or district and its relationship to significant flood sources. The traditional real estate categorisation of locations as central, semi-central, or peripheral takes on a specific meaning in flood risk assessment. Central urban areas often face different flood mechanisms (predominantly pluvial and drainage-system related) compared to peripheral regions (where riverine flooding may dominate). Historical development patterns frequently placed central districts near water bodies for

transportation and industrial purposes, creating legacies of flood exposure that modern developments must address.

- **Micro-location Analysis:** examines the immediate surroundings of a property, typically within a radius of several hundred meters, to identify localised flood risk factors. Critical considerations include:

The proximity to water bodies and drainage systems includes significant rivers, smaller streams, drainage channels, and stormwater infrastructure.

Figure 63 - Property Positioning Workflow



Local topographic features that might channel or impede water flow.

- Barriers and obstructions that may protect or, conversely, increase flood risk by redirecting flows.
- Neighbouring structures that may influence local flow dynamics.
- Impervious surfaces in the immediate vicinity that generate runoff.

Street-level Positioning: represents the finest scale of analysis, examining how a property's specific position on its street or block influences its flood exposure. Factors such as setback distance from the street, position relative to the street crown and drainage structures, driveway configuration, and even subtle grading around the building footprint can significantly influence how surface water interacts with the structure during flood events.

Built Environment Context

The built environment surrounding a property creates a complex system that significantly modifies natural flood dynamics:

Urban Density: alters hydrological processes by increasing impervious surface area, modifying drainage patterns, and creating the channelling effect of street networks. High-density urban environments typically result in more rapid runoff with higher peak flows than natural landscapes, generating complex flow paths through the built environment. Properties in dense urban areas may experience flooding mainly due to these altered urban hydrological processes rather than from natural water bodies.

- **Street Network Hydraulics:** play a critical role in urban flood routing. Streets often function as secondary drainage systems during intense rainfall events when conventional drainage infrastructure reaches capacity. The orientation, slope, width, and connectivity of streets create a network of flow paths that distribute floodwaters throughout urban areas. Properties at street intersections, low points in the street network, or downstream of large contributing street corridors face elevated exposure to these urban flow paths.
- **Infrastructure Systems:** significantly influence flood exposure through both intended and unintended effects. Stormwater drainage systems reduce flood risk when functioning properly by efficiently removing surface water. However, these same systems can become flood sources when they reach capacity or experience backwater effects. Subsurface infrastructure—including utility corridors, subway systems, and underground parking facilities—can create unexpected hydraulic connections that bypass surface topography, potentially bringing floodwaters to otherwise protected areas.
- **Building Arrangement Effects:** occur when nearby structures alter flow paths to increase or decrease risk to a specific property. Depending on their configuration relative to flow direction

and adjacent structures, buildings may block, channel, or redirect flows. In dense urban environments, the arrangement of buildings can create corridor effects, focusing flows between structures or shadow effects, where upstream buildings protect downstream properties. These complex interactions typically require detailed two-dimensional hydraulic modelling to capture accurately.

Exposure Assessment Integration

The comprehensive assessment of property positioning requires the integration of these multiple factors into a coherent exposure assessment:

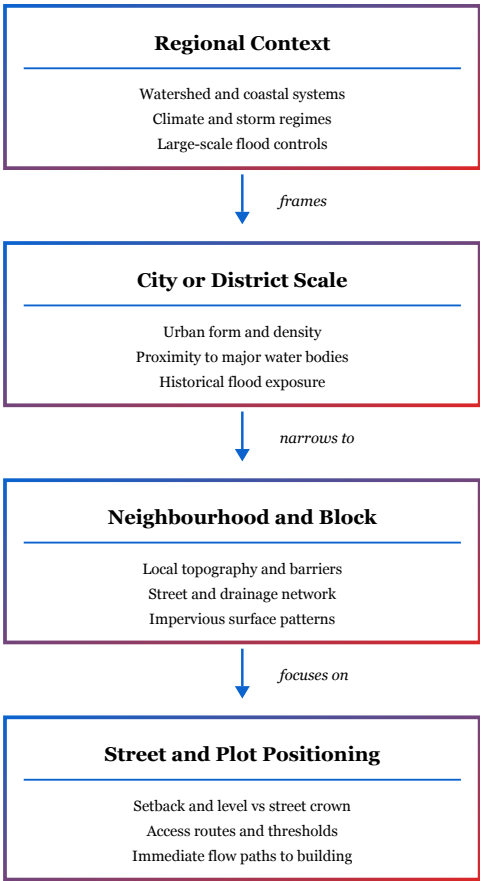
- **Exposure Pathway Identification:** systematically analyses all potential routes by which floodwaters might reach a property. This includes direct inundation from water bodies, surface flow across the landscape, backwater effects through drainage systems, groundwater rise, and infrastructure-mediated pathways. Each pathway requires specific positioning analysis techniques and may dominate under different flood scenarios.
- **Multi-hazard Positioning:** recognises that property location simultaneously influences exposure to multiple flood types. A comprehensive positioning analysis must consider exposure to riverine, coastal, pluvial, and groundwater flooding and infrastructure-related flood risks. Properties often face different positioning-based exposure levels for each flood type, creating complex multi-hazard risk profiles.
- **Temporal Dynamics:** in positioning assessment acknowledge that exposure changes over time through natural processes and human interventions. Coastal erosion, riverbed migration, urban development patterns, and infrastructure modifications all gradually alter a property's effective position relative to flood sources.
- **Quantitative Exposure Metrics:** translate qualitative positioning analysis into numerical inputs for risk assessment models. These metrics include elevation relative to various flood levels, distance from flood sources adjusted for hydraulic connectivity, flow accumulation indices from hydrological models, or compound indices that integrate multiple positioning factors. These quantitative metrics enable consistent comparison across properties and incorporation into broader risk assessment frameworks.
- **Property positioning analysis** represents the essential bridge between hazard characterisation and vulnerability assessment. By systematically analysing how a property's location influences its exposure to various flood mechanisms, this approach provides the foundation for targeted resilience strategies and accurate risk quantification. As urban development alters flood patterns, positioning analysis becomes increasingly critical for effective flood risk management across the built environment.

Environmental Resilience Factors

While property positioning establishes exposure to flood hazards, environmental resilience factors determine how natural systems buffer, absorb, or amplify these hazards before they reach

vulnerable assets. Resilience—the ability of a system to absorb disturbances while maintaining its essential functions and structure—represents a critical dimension of comprehensive flood risk assessment. Natural systems exhibit complex resilience characteristics that fundamentally influence the translation of meteorological events into flood impacts.

Figure 64 - Multi-Scale Location Analysis



Natural Flood Defence Systems.

The landscape surrounding developed areas provides the first line of defence against flooding through various natural mechanisms:

- **Wetland Buffer Capacity:** represents one of the most effective natural flood mitigation systems. Wetlands function as natural sponges, temporarily storing flood waters and releasing them gradually, thereby reducing downstream peak flows. The flood attenuation capacity of wetlands depends on their type, size, antecedent conditions, and position within the watershed.

Riparian wetlands directly connected to river systems provide immediate storage during high flows, while isolated wetlands may reduce the volume of water reaching rivers through groundwater recharge. Historical wetland loss—exceeding 50% globally and reaching 90% in some developed regions—has significantly reduced this natural resilience mechanism, amplifying flood risks downstream.

- **Forest and Vegetation Systems:** influence flood dynamics through multiple mechanisms operating at different time scales. The canopy intercepts precipitation, reducing the volume and velocity of water reaching the ground. Root systems enhance soil infiltration capacity, converting potential surface runoff into subsurface flow, moving more slowly toward water bodies. The forest floor, with its organic litter layer, provides additional water storage and slows surface flow. These mechanisms don't eliminate flooding during extreme events but modify the flood hydrograph by delaying and attenuating peak flows. The effectiveness of these systems varies by forest type, age, health, and management practice, with mature, diverse forests generally providing more significant flood mitigation benefits than young, monoculture plantations.
- **Soil Systems:** play a crucial but often overlooked role in flood resilience. Healthy soils with high organic content and well-developed structures can absorb and store significant water volumes, reducing runoff's volume and velocity. Infiltration capacity—the rate at which water can enter the soil profile—represents a critical parameter for flood mitigation. This capacity varies dramatically across soil types and conditions, with well-structured loamy soils potentially infiltrating ten times more water than compacted clay soils or those depleted of organic matter. Land management practices that maintain soil health contribute significantly to flood resilience, while soil degradation through compaction, organic matter depletion, or contamination reduces this natural defence.
- **Natural Channel Morphology:** influences how efficiently river systems convey flood waters. Naturally, meandering rivers with connected floodplains dissipate energy and store water during high flows, reducing downstream flood peaks. The complexity of natural channel forms—including pools, riffles, side channels, and woody debris—creates hydraulic roughness that slows flow velocities during floods. Channel straightening, artificial levee construction, and floodplain disconnection, while potentially protecting specific areas, often increase downstream flood risk by increasing flow velocity and peak discharge. River restoration efforts increasingly recognise the flood mitigation value of returning rivers to more natural configurations, providing ecological benefits and enhanced flood resilience.

Resilience Mechanisms and Processes

Beyond specific natural features, broader ecological and geomorphological processes contribute to environmental resilience against flooding:

- **Self-Organisation Capacity:** distinguishes resilient natural systems from engineered flood defences. While built infrastructure typically provides static protection until failure thresholds are exceeded, natural systems dynamically adjust to changing conditions. River systems redistribute sediment, vegetation communities adapt to hydrological changes, and coastal

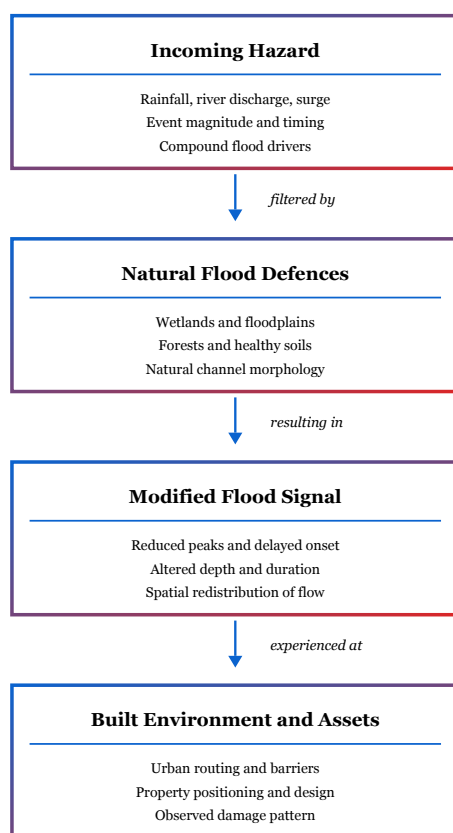
systems migrate in response to sea-level fluctuations. This self-organisation allows natural systems to maintain their flood mitigation functions across various conditions, though within certain thresholds. When these thresholds are exceeded—through extreme events or cumulative human alterations—natural systems may shift to alternative states with different flood response characteristics.

- **Disturbance Absorption Thresholds:** represent the magnitude of flood events that natural systems can accommodate while maintaining their essential functions. These thresholds vary across different ecosystem types and conditions. Healthy floodplain forests may withstand extended inundation during seasonal flooding but lose resilience when flooding becomes too frequent or prolonged. Similarly, coastal dune systems absorb the energy of moderate storm surges but may breach during extreme events, requiring time to rebuild naturally. Understanding these thresholds is essential for a realistic assessment of the protection provided by natural systems and for identifying points of failure where natural resilience may suddenly diminish.
- **Adaptive Learning Processes:** enable natural systems to incorporate previous disturbances into their structure and function, potentially increasing resilience to future events. Vegetation communities in frequently flooded areas develop specialised physiological, morphological, and reproductive adaptations that enhance survival during inundation. Geomorphic systems reorganise after significant floods, sometimes creating more stable configurations that better accommodate future events. These natural adaptive processes operate across multiple time scales, from seasonal adjustments to evolutionary adaptations spanning centuries. Unlike engineered systems designed for static conditions, natural systems continuously adjust to changing flood regimes, though human development time frames often fail to recognise these slower processes.
- **System Recovery Trajectories:** describe how natural flood defence systems return to functional states following disturbance. These trajectories rarely represent simple returns to previous conditions; instead, they often involve reorganisation around new equilibrium states that reflect both the disturbance and changing environmental conditions. For example, a coastal marsh may rebuild after a storm surge with altered channel networks that more efficiently dissipate future surge energy. These dynamic recovery processes contribute significantly to long-term resilience but complicate assessment efforts based on static environmental conditions.

Quantifying Natural Defences for Finance

From a financial perspective, natural flood defences can be treated as distributed infrastructure assets that modify depth–damage curves and expected loss at each property. Empirical studies show that wetland and floodplain restoration can significantly reduce downstream peak flows and damages, often with benefit–cost ratios above one when ecosystem services are included. For lenders and insurers, this effect can be approximated by applying location-specific adjustment factors to hazard intensities or by modelling alternative scenarios with and without planned nature-based interventions.

Figure 65 - Environmental Resilience along the Flood Path



Structurally, this opens the door to instruments where PRS coupons or payouts are conditional on the implementation and performance of nature-based solutions, similar to how green bonds link financial terms to environmental key performance indicators. For example, a PRS could be structured so that if a municipality delivers a specified amount of floodplain restoration verified via satellite and hydrological metrics, the protection buyer receives a premium rebate or enhanced coverage, reflecting the reduced modelled loss for the underlying mortgage portfolio.

Social-Ecological System Interactions

The resilience of natural systems cannot be fully understood in isolation from human systems, as the two interact in complex ways that influence overall flood risk:

- **Coupled System Dynamics:** recognise that human and natural systems interact through multiple feedback mechanisms relevant to flood resilience. Human modifications to natural systems—through development, resource extraction, pollution, or restoration—alter their flood mitigation capacity. Conversely, natural system responses to these modifications and flood events influence human vulnerability and subsequent adaptation decisions. These coupled dynamics create complex feedback loops operating across multiple spatial and temporal scales.

For instance, development in previously natural floodplains reduces flood storage capacity, potentially increasing downstream flooding, which may prompt further flood control measures that alter natural systems.

- **Resilience Trade-offs:** occur when enhancing one aspect of system resilience compromises another. Engineered flood control systems often increase short-term protection for specific assets while reducing the long-term resilience provided by natural systems. Flood walls may protect riverside developments but prevent the natural overbank flows that sustain floodplain ecosystems, providing flood mitigation. Similarly, drainage systems that effectively remove local surface water may increase downstream flood peaks. Comprehensive assessment requires recognising these trade-offs and their implications across different spatial scales and time horizons.
- **Knowledge Integration Challenges:** arise from how human and natural system resilience are conceptualised and measured. Engineering approaches typically evaluate performance against defined design standards, while ecological resilience encompasses more complex, sometimes qualitative characteristics like diversity, redundancy, and adaptive capacity. Practical assessment requires bridging these knowledge systems, incorporating quantitative hydraulic analysis and ecological understanding of system dynamics. This integration remains challenging but increasingly necessary as recognition grows of the substantial flood protection value provided by natural systems.
- **Governance System Alignment:** with natural resilience processes is critical in maintaining and enhancing environmental contributions to flood risk reduction. Governance frameworks—including regulations, incentives, planning processes, and management institutions—strongly influence whether natural flood mitigation systems are protected, restored, or degraded. Effective governance recognises the spatial misalignment often present between jurisdictional boundaries and watershed or coastal system boundaries. Systems that align governance with natural processes can significantly enhance overall flood resilience.

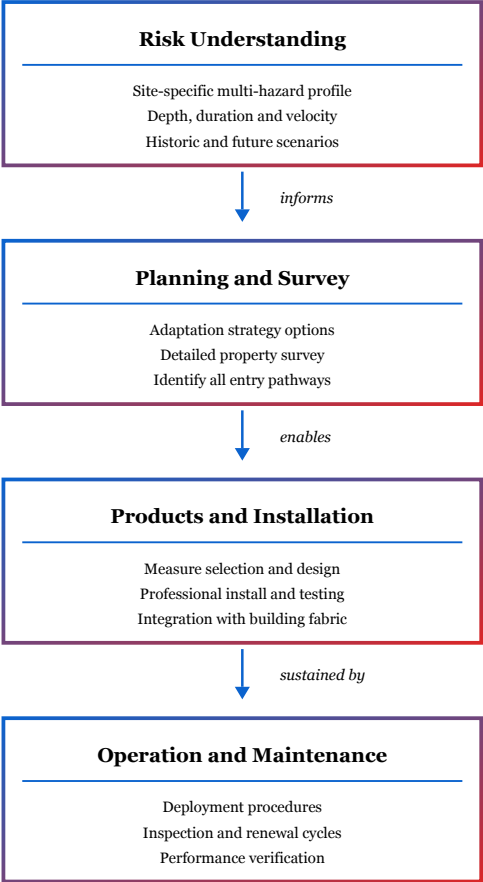
Weather Pattern Change Implications

The natural shift in weather patterns introduces new uncertainties to environmental resilience assessment, potentially altering both the hazards that natural systems must absorb and the system's capacity to provide protection:

- **Shifting Baseline Conditions:** that challenge the assumption that natural systems will continue to provide historical levels of flood protection. Changes in precipitation patterns, rising sea levels, and varying temperature regimes alter the environmental conditions to which local ecosystems are adapted. Depending on local circumstances and ecosystem types, these shifts may enhance or reduce natural flood mitigation capacity. For instance, increased precipitation may improve wetland function in some areas, while drought conditions in other regions may decrease soil infiltration capacity and vegetation cover, thus diminishing natural flood protection.

- **Ecosystem Transition Thresholds:** can be crossed, leading to rapid changes in natural flood defence systems. Many ecosystems display non-linear responses to changing environmental conditions, maintaining relative stability until critical thresholds are exceeded, then quickly shifting to alternative states. Coastal marshes may keep pace with moderate sea level rise but suddenly turn into open water when a critical rate is surpassed. Forested watersheds can sustain flood mitigation functions during moderate drought but lose them rapidly during extreme moisture stress events that cause widespread mortality. Identifying these potential transition points is a crucial area in resilience assessment.

Figure 66 - Property-Specific Resilience Assessment Steps



- **Adaptive Management Imperatives:** emerge from uncertainty when projecting future weather pattern intensity. This uncertainty necessitates approaches that monitor natural system conditions, detect early warning signals of resilience loss, and adjust management strategies accordingly. Adaptive management frameworks incorporate structured learning processes that treat interventions as experiments, generating knowledge to refine subsequent actions. For flood risk assessment, this approach recognises that environmental resilience factors are not

static features to be measured once but dynamic processes requiring ongoing evaluation and adjustment of protection estimates.

The assessment of environmental resilience factors completes the hazard component of flood risk analysis by addressing how natural systems mediate the translation of meteorological events into flood impacts on human settlements.

This assessment connects traditional hazard analysis with vulnerability assessment by acknowledging that flood characteristics in specific locations arise from the initiating event and the intricate environmental systems that influence the movement of floodwaters. Understanding and enhancing these natural resilience mechanisms becomes increasingly central to effective flood risk management as human development patterns evolve.

Property-Specific Resilience Assessment

The culmination of flood risk assessment occurs at the individual property level, where the analysis of hazard characteristics, positioning factors, and environmental resilience coalesce into property-specific vulnerability and adaptation assessments. This final analytical layer translates broader flood risk understanding into actionable property-level interventions that significantly reduce physical damage and banking impacts. A structured approach to property-specific resilience assessment provides the critical link between technical flood risk analysis and practical risk reduction outcomes.

Structural Vulnerability Analysis

The built characteristics of individual properties fundamentally determine their flood vulnerability through multiple physical mechanisms:

- **Material Vulnerability Differentiation:** reveals significant variation in how building materials Respond to flood exposure. Masonry structures generally withstand hydrostatic pressure better than timber-frame constructions, but may experience greater capillary action, drawing water upward beyond the visible flood line. Concrete structures typically offer superior structural resilience but may pose substantial drying challenges after flooding. These material-specific vulnerabilities also affect finishing elements — plasterboard deteriorates quickly when inundated, whereas water-resistant gypsum boards might retain their integrity during moderate flooding events. A comprehensive assessment requires evaluating the entire material assembly, from structural elements through insulation to interior finishes.
- **Critical System Placements:** significantly influence both damage potential and recovery timelines. Properties with elevated electrical systems—raised sockets, elevated consumer units, and ring mains routed through upper portions of walls rather than near floor level—typically experience less critical system damage and faster recovery. Similarly, positioning HVAC (Heating, Ventilation, and Air Conditioning) equipment, water heaters, and other mechanical systems relative to anticipated flood levels directly correlate with system survival rates. To

prioritise protection or relocation interventions, vulnerability assessment must identify these critical system elevations relative to property-specific flood risk characteristics.

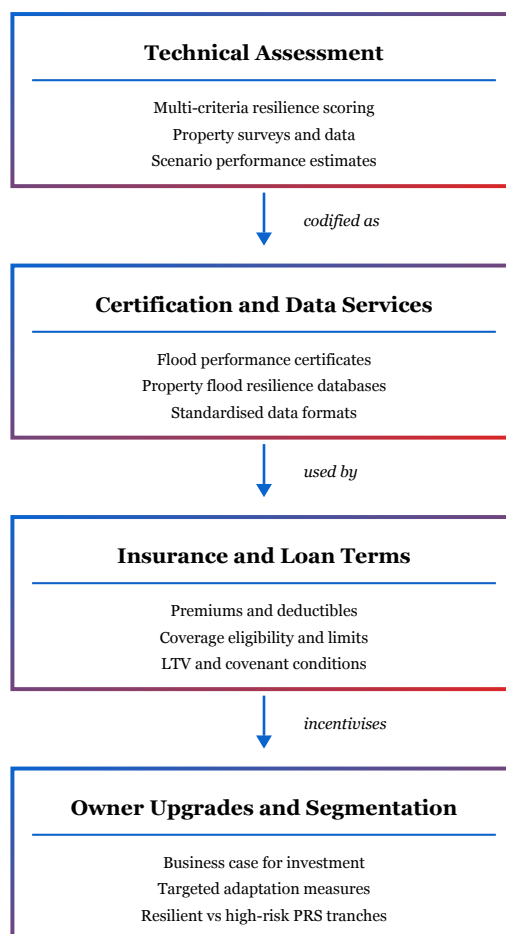
- **Foundation-type assessment** reveals different flood response traits across various foundation systems. Properties with solid floor construction encounter different vulnerabilities compared to those with suspended floors—while solid floors might resist structural damage more effectively, they also tend to pose greater drying challenges. Suspended floors facilitate underfloor drying but may be subjected to higher uplift forces and provide access points for floodwater. Crawlspace and basements add further complexities, potentially acting as beneficial buffer zones when well designed or as critical vulnerabilities if improperly protected. The assessment must examine the foundation type and specific details, such as the position of damp-proof courses and perimeter drainage systems.
- **Dynamic Pressure Considerations:** become particularly important in high-velocity flood scenarios. Properties in breach flow paths or flash flood zones face additional hydrodynamic forces beyond simple inundation. Structural elements require assessment for their capacity to withstand these lateral forces—particularly corner junctions, door and window openings, and non-load-bearing walls that may face disproportionate pressure. This assessment extends to external elements like garden walls and fences, which may provide beneficial shielding or create debris hazards depending on their construction and orientation relative to anticipated flow paths.

Property-Level Adaptation Measures

Practical resilience assessment extends beyond vulnerability identification to evaluation of adaptation options through a structured six-step process:

- **Risk Understanding:** provides the foundation for property-specific interventions. A comprehensive assessment requires evaluating the likelihood and potential severity of flooding for the specific property, incorporating all relevant flood mechanisms—fluvial, pluvial, coastal, groundwater, and infrastructure failure. This risk profile should consider historical events but must also include forward-looking projections. Risk understanding should go beyond simple inundation mapping to address anticipated flood characteristics, including depth, duration, velocity, water quality, and seasonal timing—each influencing the choice of appropriate resilience measures.
- **Adaptation Planning:** translates risk understanding into strategic decision-making. Property owners must evaluate the full spectrum of available approaches, from resistance strategies (preventing water entry) to resilience measures (minimising damage when water enters). This planning process requires a comprehensive cost-benefit analysis considering initial implementation costs, maintenance requirements, operational reliability, aesthetic implications, and potential insurance benefits. Effective planning aligns selected measures with specific property characteristics and owner capabilities, recognising that even the most interventions prove ineffective if improperly deployed or maintained.

Figure 67 - Resilience Scores into Insurance and Lending



- **Property Survey:** provides the technical foundation for adaptation implementation. Unlike generalised risk assessments, property-specific surveys identify all potential water entry pathways—doors, windows, air bricks, service penetrations, and construction joints or material interfaces. Effective surveys evaluate construction details that might not appear on standard plans, such as hidden service penetrations, legacy features from previous modifications, or subtle topographic features that influence local flow paths.
- **Product Selection:** matches identified vulnerabilities with appropriate technological solutions. Property-specific assessment must evaluate not only the theoretical performance of these products but also their appropriateness for the specific context, including user capabilities, deployment time requirements, aesthetic considerations, and compatibility with existing building features.

- **Professional Installation:** ensures theoretical protection translates to real-world performance. Assessment must consider the quality of installation and selected products, as even superior technologies fail when improperly integrated with existing structures. This evaluation extends to sealing methods, fixing approaches, and interface treatments between different protection system components. Post-installation testing provides critical verification, from simple hose testing of minor interventions to complete deployment exercises for more complex systems.
- **Maintenance and Operation:** completes the resilience assessment cycle. The most protection systems prove worthless without proper maintenance and timely deployment. Assessment must evaluate operational requirements against occupant capabilities, particularly considering warning time, physical abilities, and the potential need for assistance. Maintenance assessment extends beyond simple physical inspection to consider ageing effects, material degradation patterns, and operational testing protocols, particularly for less frequently used mechanical components that may seize or deteriorate while dormant.

Resilience Scoring Methodologies

Quantitative assessment of property-specific resilience enables more risk management and banking protection:

- **Multi-criteria Evaluation Frameworks:** generally include resistance measures (preventing water entry) and recovery-enhancing features (reducing damage when water enters). Effective scoring systems assign weights to different factors based on their importance for specific property types and flooding conditions. For instance, the resistance of building fabric might be assigned greater weight during short-duration, high-frequency events, while rapid-drying capability might be emphasised during prolonged flooding. These weighted scores offer a standardised way to compare properties, prioritise interventions, and track improvements over time.
- **Resilience Certification Programs:** translate technical assessments into accessible documentation for non-technical stakeholders. These programs, increasingly recognised by insurance markets, provide standardised verification of implemented measures and their expected performance across defined flood scenarios. Certification typically requires documentation review and physical inspection, often with periodic renewal requirements, to ensure ongoing maintenance. The resulting certifications provide objective evidence of reduced risk that can support insurance negotiations, property valuation, and regulatory compliance demonstrations.
- **Insurance Premium Integration:** highlights a key application of resilience scoring. Whereas past flood insurance methods often relied on simple risk categories, modern insurers increasingly factor in property-specific resilience features into underwriting. Properties with strong protection measures—especially those confirmed through recognised certification schemes—can secure more favourable coverage, lower premiums, reduced deductibles, or coverage in locations previously deemed uninsurable. This recognition encourages resilience investment, creating a positive cycle where savings on premiums support further protective efforts.

- **Valuation Methodology Alignment:** extends resilience benefits beyond insurance markets. Property valuation increasingly considers flood risk, with high-risk properties facing potential value penalties. Property-specific resilience assessment provides a mechanism to differentiate protected properties within high-risk areas, potentially preserving significant value through demonstrated risk reduction. Advanced valuation methodologies incorporate resilience scoring alongside traditional factors, recognising that properties with equivalent flood zone designations may face dramatically different actual risk profiles based on implemented protection measures.
- **Cost-benefit optimisation:** enables the targeting of limited resilience investments for maximum returns. A comprehensive assessment helps prioritise interventions based on their potential risk reduction relative to costs. This optimisation usually shows that specific low-cost measures—such as upgrading electrical systems or installing removable flood barriers—offer disproportionate benefits compared to more extensive structural changes. For properties frequently exposed to flooding, these cost-benefit analyses often reveal very short payback periods for resilience investments, especially when insurance savings and avoided disruptions are fully recognised.

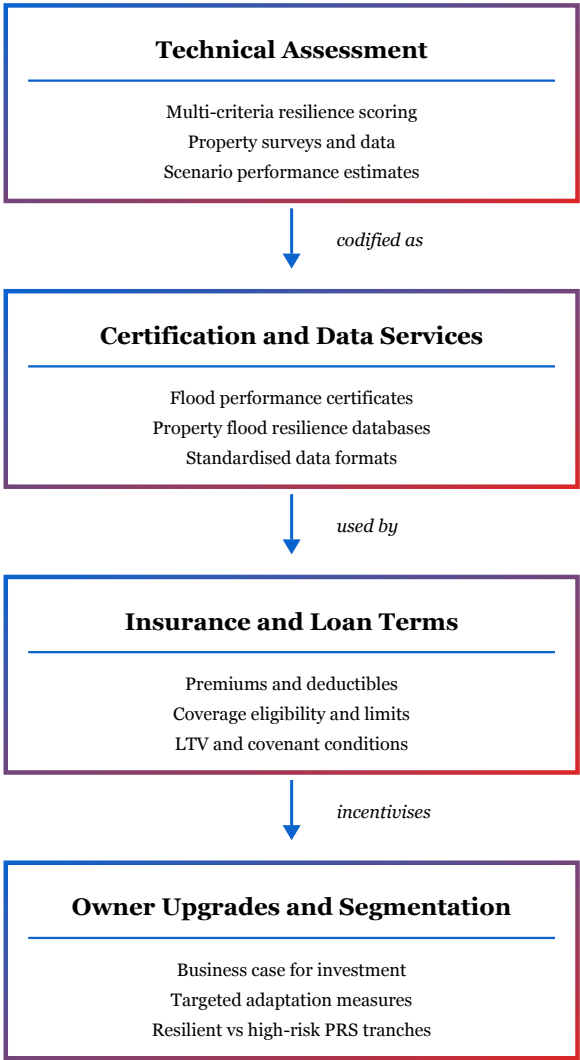
The assessment of property-specific resilience completes the risk assessment chain, translating broader flood hazard understanding into practical, location-specific protection strategies. As development pressures continue to alter flood risk profiles across the built environment, this property-level focus provides the most immediate and effective path to risk reduction.

While community-scale defences and watershed management remain essential components of comprehensive flood risk management, property-specific interventions offer immediate protection benefits regardless of broader defence implementation timelines or funding constraints. Integrating these property-level assessments with risk transfer mechanisms creates powerful banking incentives that accelerate adoption without regulatory mandates, creating a market-driven approach to enhanced resilience.

Recent work in the UK and internationally suggests that property-specific flood resilience can be standardised through scoring systems and certificates. For example, the proposed Flood Performance Certificate (FPC) concept aims to codify the level of protection a property has against a set of reference flood scenarios, providing consistent data to insurers, lenders, and buyers. Parallel initiatives such as Property Flood Resilience databases collect structured data on installed measures and their observed performance, closing the information gap that has historically limited risk-based pricing.

Incorporating such scores into insurance and mortgage pricing can create strong incentives, but it must be done carefully to prevent moral hazard. If resilience upgrades result in premium reductions or more favourable lending terms, homeowners have a clear business case to invest; if instead upgrades are overlooked or offset by higher deductibles, the incentive is lost. A practical approach is to link part of insurance and loan pricing directly to verifiable, certified resilience scores, with independent surveys and periodic re-certification to confirm that measures remain in place and operational. For PRS structuring, property-level resilience scores provide a way to divide the reference portfolio into more and less resilient groups, improving risk selection and

Figure 67 - Resilience Scores into Insurance and Lending



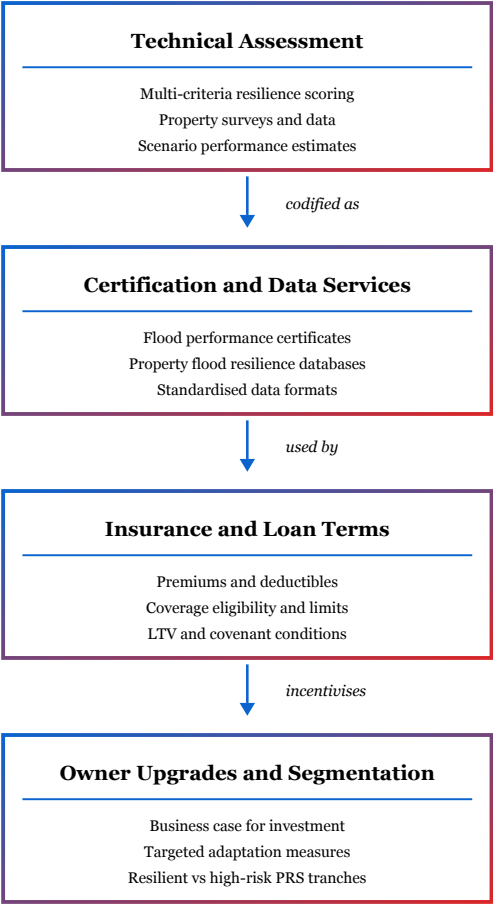
reducing basis risk. A bank might, for example, target PRS protection on the subset of exposures that combine high hazard, high exposure, and low resilience scores, while holding more resilient exposures on the balance sheet.

From Assessment to Transaction

All of the analytical layers developed in this chapter—hazard classification, property positioning, environmental resilience, and property-specific measures—ultimately feed into a single number for each loan or property: an adjusted, forward-looking flood loss distribution. In practice, PRS

structuring teams aggregate these distributions across thousands of exposures to build portfolio loss curves, then choose trigger metrics and attachment points that best align hedge payouts with portfolio losses. The quality of the resulting transaction depends directly on the depth and consistency of the assessments described here, which is why standardised mapping, positioning indices, and resilience scores are so valuable for scaling PRS beyond bespoke, one-off deals.

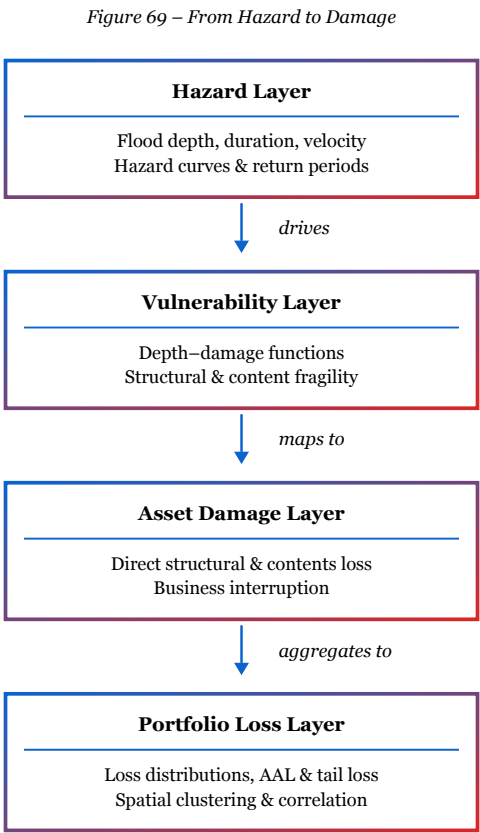
Figure 68 - Resilience Scores into Insurance and Lending



Chapter 8.1 - From Hazard to Damage

The quantification of physical risk impacts represents the culmination of our progression from fundamental principles to practical applications. Having established how water moves through the environment and how we assess the resulting flood risks, we now turn to converting these assessments into measurable impacts on property, infrastructure, and economic systems.

"The value of risk assessment ultimately lies in our ability to quantify potential impacts. Without this quantification, we cannot properly price risk, allocate resources for mitigation, or make informed decisions about development and adaptation."- David Kelly, MKM.



Physical hazard is translated into asset-level damage and portfolio loss tails, forming the bridge between scientific flood modelling and banking applications.

Impact quantification is where the physical world finally meets the banking book. Up to now, we have focused on understanding and simulating the hazard itself: how weather systems evolve,

how catchments respond, and how flood waters spread across the landscape. In this chapter, we translate those physical outcomes into financial impacts that can be used directly in risk, capital and pricing decisions.

Impact quantification represents the point where our physical understanding of floods becomes directly useful for decision-makers. Up to this point, we have focused on how weather systems evolve, how catchments respond, and how flood waters move through the landscape. In this section, we focus on what those physical processes actually do to assets and portfolios: how often damaging events occur, how severe they are, and how those severities accumulate across properties and neighbourhoods.

The value of this analysis is not in producing a single, definitive loss number for a given location, but in building a transparent chain from hazard to damage that others can interrogate and reuse. Engineers, insurers and bankers all need to understand how a “1-in-100-year” flood translates into a range of plausible damage outcomes, how those outcomes vary with building type and local conditions, and how uncertainty in the science propagates into uncertainty in loss estimates. By making each link in this chain explicit, we create a common language that later chapters can connect to credit metrics, capital calculations and market instruments without having to re-open the physical risk questions every time.

Before turning to the banking book and metrics such as probability of default, loss given default and exposure at default, we therefore spend time on the physical side of impact. We show how to construct hazard curves that describe the frequency of different flood intensities; how to use vulnerability or damage functions to translate those intensities into loss distributions for individual assets; and how spatial clustering and shared infrastructure turn individual asset risk into concentrated portfolio risk. Once this foundation is in place, we can then translate damage into credit and capital terms in the following sections with far greater confidence and clarity.

The central question is simple to state and hard to answer: given a distribution of future flood events, what happens to default rates, recoveries, exposure profiles and ultimately capital requirements for a real portfolio? The goal here is not to produce a single “right” number, but to build a transparent chain from rainfall and river flow to expected loss and tail loss that bankers, supervisors and investors can interrogate and trust

This chapter connects the hazard stack we have covered in previous chapters to the banking stack that we will cover, by:

- Defining impact metrics that matter for banks: shifts in PD, LGD, EAD and property valuations at loan and portfolio level.
- Showing how flood depth, duration and frequency feed into those metrics through damage ratios, business interruption and migration effects.
- Illustrating simple, implementable mappings from hazard outputs to credit risk parameters that can sit inside existing IRB, IFRS 9 and stress testing frameworks.

- Highlighting where current data and methods are robust enough for capital and pricing, and where impacts should remain in scenario analysis, disclosures or Pillar 2 overlays.

Brief Introduction to Credit Impact Metrics

The banking system manages risk through a small set of tightly defined credit metrics. They sit behind every lending decision, limit, price and capital calculation, and they are the language through which supervisors, investors and internal risk functions understand the health of a bank's balance sheet. When we talk about “quantifying impact” from physical risk, we are ultimately talking about how floods and other hazards move these metrics over time.

For a Basel III bank, four families of credit impact metrics are central:

- Probability of Default (PD)
- Loss Given Default (LGD)
- Exposure at Default (EAD)
- Risk-weighted Assets and Capital (RWA, CET₁)

Each of these metrics is already familiar to credit practitioners, but physical risk forces us to think about them in a new way: not just as functions of borrower behaviour and macroeconomics, but as conditional on the state of the physical world.

Probability of Default (PD)

PD measures the likelihood that a borrower will fail to meet their obligations over a given horizon, typically one year for regulatory capital and lifetime for IFRS 9 provisioning. It is the primary way a bank turns information about a counterparty, its sector and the economy into a single, comparable risk measure across a portfolio.

Physical risk enters PD through several channels. A severe flood can interrupt income (for households through job losses or illness, for businesses through lost revenues), increase unavoidable expenses (temporary accommodation, repairs) and trigger relocation or business closure. Even if insurance is in place, delays or disputes in claims settlement can create liquidity stress at exactly the wrong time. For a credit division, these mechanisms show up as higher arrears, more forbearance requests and, ultimately, higher default rates in affected regions and sectors. The role of impact quantification is to make those PD shifts explicit and forward-looking rather than waiting for them to appear in historical data.

Loss Given Default (LGD)

LGD measures the proportion of exposure that is lost when a borrower defaults, after accounting for recoveries from collateral, guarantees and other credit enhancements. In mortgage

and secured lending, LGD is heavily driven by the value and liquidity of property collateral at the time of default, as well as the cost and duration of the workout process.

Flood risk acts directly on LGD by damaging or destroying collateral. A property that suffers repeated inundation will typically experience both acute loss (the cost to restore habitability) and chronic devaluation (discounts applied by buyers and valuers once the risk is known). Insurance coverage mitigates part of the first channel, but only if it remains affordable and available over the life of the loan, and only to the extent that policy limits and exclusions do not bite. When insurers withdraw, cap coverage or raise deductibles, the unsecured share of the loss borne by the bank increases and LGD can rise sharply. From a Basel III perspective, even modest, systematic increases in LGD on a large mortgage or SME portfolio translate into material increases in capital requirements.

Exposure at Default (EAD)

EAD is the amount a bank is exposed to if a borrower defaults. For term loans and mortgages it is mainly driven by scheduled amortisation, prepayments and redraws; for revolving facilities and contingent lines it reflects both current usage and potential future drawdowns.

Physical risk influences EAD less visibly but still meaningfully. In the run-up to, and aftermath of, a flood event, businesses may draw down undrawn lines to cover working capital gaps, and households may use credit cards and overdrafts to bridge delays between loss and settlement. In stressed regions, refinancing and property sales may stall, extending the effective life of exposures and slowing down the natural run-off of risk-weighted assets. For a credit division managing limits and product design, understanding these dynamics is crucial: facilities that look low-risk in normal times can become highly utilised sources of loss in hazard-driven stress periods.

Risk-Weighted Assets and Capital (RWA, CET₁)

Under Basel III, PD, LGD and EAD feed into formulae that determine the risk-weighted assets (RWA) associated with a portfolio. Capital ratios such as CET₁/RWA are the core measures of solvency used by regulators and markets. Any systematic upward shift in PD or LGD, or any structural increase in EAD, therefore translates into higher RWA and higher capital demand for the same nominal book.

This linkage makes credit impact metrics central to the integration of physical risk into banking. Supervisors increasingly expect banks to demonstrate how climate- and hazard-related vulnerabilities will affect their capital position under plausible scenarios, not just in narrative terms but in terms of quantified PD, LGD and EAD paths and the resulting RWA profile. Credit divisions cannot treat physical risk as a separate “ESG topic”; it is a driver of the same Basel III metrics that already govern risk appetite, pricing and portfolio optimisation.

These metrics are the focus of this chapter because they are the precise points at which physical reality enters the machinery of banking. Once we have physically realistic flood scenarios, the central task is to understand how they alter probability of default, loss given default, exposure at default and, ultimately, risk-weighted assets and capital demand. In practice this means

translating observable hazard indicators such as flood depth, duration, recurrence and statistically derived damage ratios into systematic adjustments to PD and LGD at the level of individual loans and well-defined segments, rather than treating physical risk as an external narrative.

It also means describing how hazard stress changes the way facilities are used, so that changes in EAD under stress, particularly for revolving products and refinancing-dependent books, are captured in a way that is simple, transparent and compatible with existing credit frameworks. From there, the chapter follows the chain through to portfolio-level transformations, showing how these adjusted parameters propagate into RWA and capital within Basel III and ICAAP structures. By working explicitly in the currency of PD, LGD, EAD and RWA, physical risk ceases to be an add-on topic and becomes part of the same quantitative toolkit that credit committees, ALCOs and boards already use to steer the bank's risk appetite, pricing and balance-sheet strategy.

Vulnerability functions then connect these hazard characteristics to expected damages. These functions—often expressed as relationships between flood depth and percentage damage—vary significantly across building types, content categories, and occupancy classes. The development of these functional curves draws upon empirical data from past events, engineering principles, and, increasingly, advanced simulation techniques.

Applying these functions yields damage estimates across multiple dimensions: direct structural damage, content losses, business interruption, and broader economic impacts. Each category requires specialised methodologies that account for repair costs, replacement values, temporal disruption, and complex economic interdependencies. These estimates must further consider insurance coverage, including policy terms, exclusions, and risk transfer mechanisms that distribute impacts across stakeholders.

The ultimate expression of these impacts often appears in property valuations, where market participants price risk into their investment decisions. Understanding how physical risks translate into changes in market value requires consideration of capitalisation rates, market perception, regulatory influences, and adaptation potential.

As we explore these quantification methodologies, we complete the analytical pathway that began with fundamental physical principles. This progression—from Newtonian physics to banking economics—enables the effective management of physical risks in an increasingly uncertain world.

The central question is simple to state and hard to answer: given a distribution of future flood events, what happens to asset damage, recovery costs and portfolio-level loss tails, and how do these physical outcomes set the stage for subsequent changes in default rates, recoveries, exposure profiles and ultimately capital requirements for a real portfolio? The goal here is not to produce a single “right” number, but to build a transparent chain from rainfall and river flow to expected loss and tail loss that bankers, supervisors and investors can interrogate and trust.

In the remainder of this section, we focus on the physical side of that chain. We start with hazard curves that describe the frequency and intensity of flood events at a given location, then introduce damage and vulnerability functions that translate those intensities into loss distributions for individual assets and portfolios. Only once this foundation is in place do we turn, in the following chapter, to how these loss distributions are mapped into PD, LGD, EAD and regulatory capital metrics within a Basel III framework.

Hazard Curve Development

Hazard curves form the first hard connection between scientific hazard models and the quantitative machinery used in banking. They translate the language of “1-in-X-year floods” and climate scenarios into the probability distributions that both catastrophe models and credit-risk style engines can understand. In the FINOS physical risk projects under OS-Climate, these curves are the interface between upstream flood modelling and downstream damage and loss calculations; in the banking stack, the same curves become the natural analogue of hazard rates in a CDS framework, feeding into expected loss and tail loss calculations in a familiar mathematical form.

At their core, hazard curves describe how often different levels of physical intensity are expected to occur at a given location. For floods, the key intensity measure is typically water depth, sometimes supplemented by duration or velocity. The curve links each depth level to an annual probability or return period, allowing us to move from statements such as “this location is exposed to a 1-in-100-year flood” to a continuous function that can be integrated, stressed and combined with vulnerability functions. This step is essential if banks want to plug physical risk into existing risk engines rather than maintaining a separate, incompatible modelling stack.

Only once these hazard curves are in place can we systematically connect OS-Climate style flood maps and weather-conditioned scenarios to banking applications. They allow us to: extract location-specific exceedance probabilities from global and regional datasets; convert those probabilities into probability bins or continuous distributions; and, crucially, re-express them in the same hazard-rate language used for credit default swaps and other capital markets instruments. The following subsections set out how these curves are constructed in practice and how they provide the common foundation for both damage modelling and CDS-compatible pricing of physical risk.

“Return period (τ) refers to the average time between events with intensity higher than a specified threshold. Exceedance probability represents the probability that in a given period (typically one year), an event of intensity more significant than a specified threshold will occur”, David Kelly, MKM

Hazard Curve Development

The mathematical relationship between these concepts is expressed as the probability of at least one occurrence in time t :-

$$P(X^h_t \geq 1) = 1 - e^{-(t/\tau)}$$

Where:-

- $P(X^h_t \geq 1)$ is the probability of at least one hazard event occurring within time period t .
- X^h_t represents the number of hazard events of type h occurring in time period t .
- t is the time period of interest (typically 1 year).
- τ is the return period in years (average recurrence interval).

Convert to probability bins. Using the relationship

$$F'_S(h) = F'_S(s_q^{\text{(lower)}}) - F'_S(s_q^{\text{(upper)}})$$

Where:

- $F'_S(h)$ is the probability of the hazard intensity falling within a specific bin.
- $s_q^{\text{(lower)}}$ is the lower bound of the hazard intensity bin.
- $s_q^{\text{(upper)}}$ is the upper bound of the hazard intensity bin.
- F'_S represents the probability density function of the hazard intensity

In everyday parlance, a 100-year flood has approximately a 1% probability of occurrence in any given year.

Constructing Hazard Curves

The OS-Climate framework constructs hazard curves through the following process:

- Identify hazard indicators: The primary hazard indicator is flood depth (measured in meters) for floods.
- Obtain spatial hazard datasets: OS-Climate leverages multiple data sources, including:

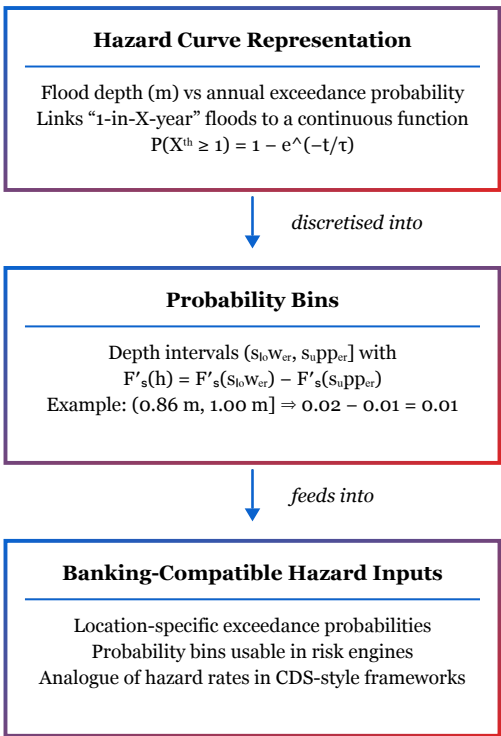
World Resources Institute (WRI) Aqueduct flood model for global coverage of coastal and riverine inundation. Higher resolution regional datasets (e.g., TU Delft pan-European dataset with 100m resolution)

Weather-conditioned future projections

Extract location-specific hazard data: Extract the relationship between hazard intensity and return period for a given location (latitude, longitude). For inundation, this might look like:

Construct probability bins for different hazard intensities. For example, the probability of occurrence of a flood with depth in the range $(0.86m, 1.00m]$ would be $0.02 - 0.01 = 0.01$.

Figure 70 – Flood Hazard Curves and Probability Bins



Hazard curves are converted into discrete probability bins so that flood information can be integrated directly into catastrophe and credit risk engines.

Interpolate between return periods. For comprehensive risk assessment, interpolation methods are applied to estimate hazard intensities between the defined return periods.

Account for weather pattern uncertainty. Compare hazard curves under baseline historical conditions with weather-conditioned curves for future periods to quantify hazard frequency and severity changes.

Connecting Hazard and Vulnerability

The hazard curve and vulnerability curve are connected in the following way:

- The hazard curve provides the probability of experiencing different hazard intensities at a location.
- The vulnerability curve takes those hazard intensities as inputs and translates them into expected damage/loss values for the exposed assets.

In risk analysis, the hazard curve is first used to determine the range of hazard intensities that must be considered based on their likelihood at that location. Then, the vulnerability curve maps each hazard intensity to a damage ratio or loss estimate for the asset being analysed.

By combining the hazard exceedance probabilities from the hazard curve with the expected damage/losses from the vulnerability curve, catastrophe models can calculate risk metrics like the probable maximum loss or average annual loss for the exposed portfolio of assets.

In summary, the hazard curve characterises the frequency of the peril. In contrast, the vulnerability curve characterises the fragility of the asset to that peril's intensity - both critical components in quantifying potential losses from any flood event.

Vulnerability Curve Modelling

Vulnerability curves represent the relationship between hazard intensity and an asset's resulting damage or disruption. While hazard curves capture the probability of physical phenomena, vulnerability curves translate these phenomena into actual impacts on assets.

The OS-Climate methodology implements vulnerability models through conditional probability distributions that capture the uncertainty in asset response to hazard events.

Vulnerability Curve

For a given hazard intensity s , the vulnerability curve provides the probability distribution of damage/impact d :

$f_{D|S}(d, s)$ = probability that given hazard intensity s , impact d occurs

Where:

- $f_{D|S}(d, s)$ is the conditional probability density function of damage/impact.
- d represents the level of damage or impact to the asset (often expressed as a percentage of total value).
- s represents the hazard intensity (e.g., flood depth in meters, wind speed in km/h)

The effective impact distribution is then derived through the convolution of the hazard and vulnerability distributions:

$$f_D(d) = \int_{-\infty}^{\infty} f_S(s) f_{D|S}(d, s) ds$$

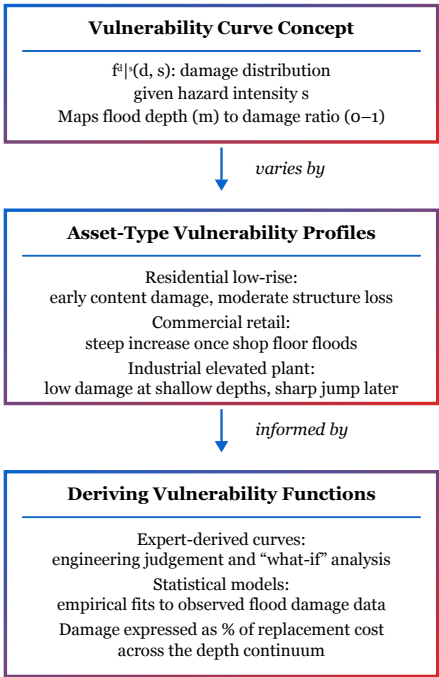
Where:

- $f_D(d)$ is the overall probability density function of damage level d .
- $f_S(s)$ is the probability density function of hazard intensity s .

The integration combines all possible hazard intensities to determine the total damage probability distribution.

The vulnerability curve shows the range of possible damage outcomes for a specific hazard level. Rather than predicting a single damage amount, it provides the probability of different damage levels occurring for a given flood level.

Figure 71 – Flood Vulnerability Curves by Asset Type



Different building types have distinct depth–damage relationships, captured through vulnerability curves derived by expert or statistical methods.

When discussing the effective impact distribution, we're combining two key pieces of information: how likely different hazard intensities are to occur in your area, and the potential damage each intensity might cause. This provides a complete picture of possible damages by considering every potential hazard scenario, weighing each by its likelihood, and summing all the resulting damage probabilities.

“Vulnerability curves account for how likely different flood depths are in your location and how the building’s resilience responds to each depth, producing your overall damage risk profile.”- David Kelly, MKM

Expert-Derived Flood Vulnerability Curves

Expert-derived flood vulnerability curves represent a critical methodological component in physical risk assessment for regions without a sufficiently rich dataset. These curves are developed through a structured analytical process integrating engineering principles with quantitative expert judgment.

The development methodology typically proceeds as follows:

- Structural and hydrological specialists systematically analyse how varying inundation depths affect building typologies and construction materials.
- Quantitative "what-if" analyses determine damage thresholds at specified flood depth intervals, identifying critical points where damage functions exhibit significant gradient changes.
- Mathematical relationships between inundation depth and structural response are established and calibrated to engineering principles and material behaviour characteristics.

The resulting functions express damage as normalised ratios (0 to 1) or percentage values of total replacement cost across the continuum of possible flood depths.

These expert-derived functions offer several methodological advantages within the risk assessment framework. They can be constructed without extensive empirical flood damage datasets, enabling vulnerability quantification in regions lacking recent flood event data. They incorporate building characteristics and flood parameters that may be absent from historical records while providing a technical foundation for forward-looking assessments under projected climate conditions.

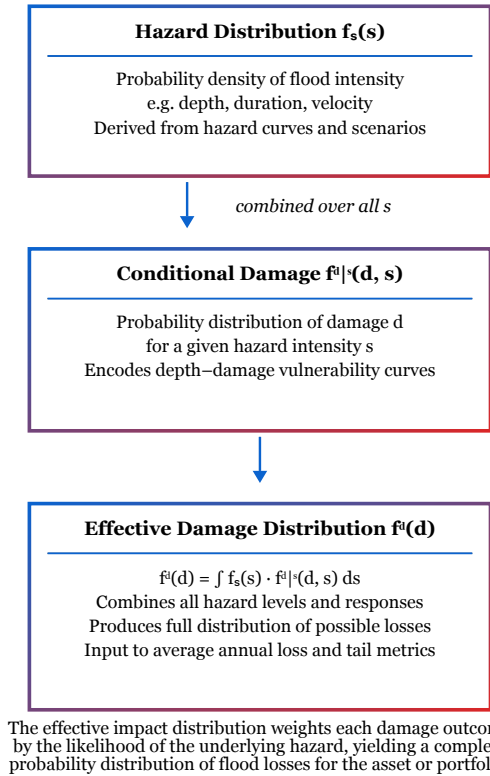
However, significant methodological limitations must be acknowledged when applying expert-derived vulnerability functions. They rely on expert judgment rather than observed damage data and may not capture all multivariate factors influencing flood vulnerability. These functions introduce subjectivity that may vary between expert groups, necessitating careful documentation of assumptions and methodological choices. When integrated with the hazard curves, such functions form the technical basis for quantitative risk assessment, informing infrastructure planning, insurance pricing models, and resilience investment decisions.

Statistical Vulnerability Models

Statistical vulnerability models provide a data-driven alternative to expert-derived curves. These models leverage historical damage data from past flood events to establish empirical relationships between hazard intensity variables and observed damages. Through regression analysis or machine learning techniques, these models can identify key vulnerability factors and provide more objective measures based on empirical evidence.

The OS-Climate framework implements several statistical approaches for flood vulnerability modelling. Random Forest models utilise environmental factors, building characteristics, and

Figure 72 – From Hazard and Vulnerability to Damage Distribution



historical damage data to predict potential losses at different flood depths. Regression-based approaches establish mathematical relationships between flood parameters (depth, duration, velocity) and observed damage ratios. Empirical damage functions are derived directly from historical loss data across different asset classes and regions, offering direct translation from hazard intensity to expected damage.

Statistical vulnerability models offer significant methodological advantages in risk assessment applications. They are based on actual observed damage rather than theoretical estimates, enabling them to capture complex, non-linear relationships between hazard intensity and damage that might not be apparent through expert assessment alone. They frequently reveal unexpected vulnerability factors not initially considered in expert-driven approaches, providing insights that can refine overall risk understanding.

However, these models face essential limitations in their application. They require sufficient historical damage data, which may not be available in all regions or asset classes. They may not adequately represent vulnerability under novel weather conditions outside historical experience. Additionally, they can be biased by the specific characteristics of the historical events depicted in the dataset, potentially limiting their generalizability to future conditions.

Parametric Vulnerability Functions

Parametric vulnerability functions represent a third approach in the OS-Climate framework, using standard mathematical distributions to model the relationship between hazard intensity and damage. These functions employ well-defined probability distributions such as Beta, Gamma, or Lognormal to express the damage response of assets to hazard intensities. The parameters of these distributions undergo calibration processes based on available empirical data or structured expert judgment. These functions systematically express damage as a percentage of asset value or absolute monetary loss, providing a standardised format for integration into risk assessment frameworks.

Beta distribution has emerged as a preferred mathematical form for modelling vulnerability in the OS-Climate methodology due to its specific properties and analytical advantages. The distribution is constrained to the interval $[0,1]$, which naturally corresponds to the range of possible damage ratios from no damage to total loss. Its shape parameters can be adjusted to represent different vulnerability profiles across diverse asset types and construction categories. The Beta distribution effectively models the uncertainty in damage estimation at specific hazard intensities, capturing both the expected value and the variance of potential outcomes.

The development process for parametric functions follows a structured methodology that involves categorising assets into similar vulnerability groups based on construction characteristics, occupancy types, or land use categories. For each asset category, the parameters of the chosen distribution undergo estimation procedures using available data or expert assessment. These functions then undergo validation against historical damage data where such information is available, allowing for refinement of the parametric representations.

Parametric functions offer distinct methodological advantages in risk modelling applications. They provide mathematical elegance and computational efficiency, allowing for rapid calculation across large portfolios of assets. Their probabilistic nature enables the representation of uncertainty through complete probability distributions rather than point estimates, reflecting the inherent variability in damage outcomes. These functions integrate seamlessly with probabilistic risk assessment frameworks, supporting advanced risk metrics and uncertainty quantification. This makes parametric functions highly desirable for global banking systems needing to crunch vast numbers of assets at any defined frequency.

However, several limitations must be considered in their application. Standard mathematical distributions may oversimplify complex damage mechanisms that exhibit threshold effects or non-standard response patterns. Parameter estimation presents significant challenges when working with limited empirical data, potentially introducing additional uncertainty. Furthermore, standard distributions may not accurately capture the actual shape of the damage-hazard relationship, especially for specialised assets or under hazard conditions. Within the OS-Climate framework, these parametric functions frequently complement other vulnerability modelling approaches, particularly in contexts where uncertainty quantification is crucial in comprehensive risk assessment.

Uncertainty in Vulnerability

Uncertainty in vulnerability sits at the heart of impact quantification. Even when hazard curves are well specified, our ability to say “this level of flooding produces that level of damage” remains constrained by limited theory, sparse data and methodological choices that materially affect the answer. This uncertainty is not a minor technical detail; it directly shapes the damage distributions that flow into portfolio loss, PD/LGD shifts and, ultimately, capital demand.

A primary source of uncertainty lies in the vulnerability of the elements at risk themselves. Flood damage modelling has historically received less scholarly attention than hazard modelling, leaving relatively weak theoretical and empirical foundations for many damage functions and depth-damage curves. As a result, object-specific predictions for individual buildings or infrastructure assets carry wide error bands, even when the underlying hazard intensity is well characterised. Different aggregation frameworks for combining exposure, sensitivity and coping capacity into composite vulnerability indicators can further diverge, with multiplicative versus additive formulations yielding noticeably different rankings and levels of “vulnerability” for the same set of inputs.

On top of these model-structure issues sits a broader spectrum of uncertainty types. In addition to familiar probabilistic uncertainty around event magnitudes and frequencies, vulnerability assessments face technical and statistical uncertainty (measurement error and limited samples), methodological uncertainty (choice of functional form and aggregation scheme), and fundamental uncertainty about how future events and system responses may differ from the past. Because flood vulnerability draws on hydrology, engineering, economics and social science, each discipline brings its own data limitations and modelling traditions, which have to be reconciled in a single framework. Recognising and, where possible, quantifying these uncertainties is essential: it allows banks to treat vulnerability outputs as distributions rather than point estimates and to carry that uncertainty explicitly through into loss, capital and pricing decisions, rather than assuming a false precision that the science cannot support.

The OS-Climate methodology explicitly accounts for two types of uncertainty in vulnerability:

Aleatory Uncertainty

Aleatory uncertainty represents the inherent randomness or natural variability in how assets respond to hazards, even under seemingly identical conditions. This type of uncertainty is irreducible and cannot be eliminated through additional information or improved models.

In the context of flood vulnerability, aleatory uncertainty manifests in various ways. Two identical buildings subjected to the same flood depth may suffer different levels of damage due to random factors such as water flow dynamics, debris impact, or subtle differences in construction quality. This natural variability persists even with perfect knowledge of building characteristics and flood parameters.

The OS-Climate framework quantifies aleatory uncertainty using probability distributions that capture the range of possible damage outcomes for a given hazard intensity. For example, rather than predicting that a 1-meter flood will cause precisely 30% damage to a specific building type, the model might represent this as a probability distribution with a mean of 30% and a standard deviation that reflects the observed variability in historical flood damage data.

Epistemic Uncertainty

Epistemic uncertainty stems from incomplete knowledge about asset characteristics and their vulnerability to hazards. Unlike aleatory uncertainty, epistemic uncertainty can be reduced through additional data collection or improved modelling.

In the context of physical risk assessment, epistemic uncertainty arises from:

- Limited information about building characteristics (e.g., foundation type, materials, age).
- Incomplete understanding of damage mechanisms.
- Scarce historical damage data for calibration.

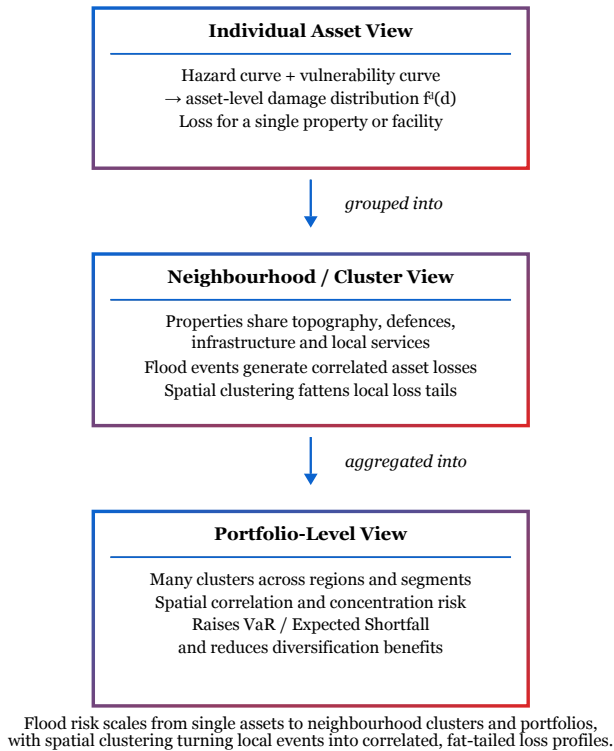
Property Clustering

When we assess flood risk for residential properties, looking at individual properties in isolation tells only part of the story. The spatial clustering of properties creates complex, interconnected risk dynamics that can dramatically amplify the impact of flood events beyond what individual property assessments might suggest.

Properties that cluster in flood-prone areas create a risk multiplier effect. This occurs through several interconnected mechanisms:

- **Correlated Physical Damage:** When properties are closely grouped, they share similar flood risk characteristics. Properties within 1000m of each other (the correlation distance parameter) experience strongly correlated impacts. This means that instead of damages being randomly spread across a portfolio, they tend to occur in concentrated clusters—leading to more significant overall impacts than if properties were considered independent risks. A neighbourhood with 50 homes nearby will often see nearly all properties affected at the same time by a single flood event rather than a random selection. This correlation effect is most pronounced in densely developed floodplains, where property values tend to rise and fall together depending on flood events.
- **Infrastructure Vulnerability Amplification:** Clustered properties typically share critical infrastructure, creating systemic vulnerabilities that amplify flood impacts throughout the neighbourhood. When flood waters overwhelm stormwater systems in a densely populated area, the consequences ripple through all properties connected to that system. The drainage network designed to protect homes becomes a shared point of failure, causing water to back up into multiple properties simultaneously.

Figure 73 – From Individual Assets to Clustered Portfolios



Access becomes another critical issue during flood events in clustered developments. As waters rise, road networks serving entire neighbourhoods can become impassable, cutting off not just individual homes but whole communities from emergency services. This isolation effect extends the impact of flooding well beyond direct water damage, affecting even properties that might remain dry but unreachable.

Perhaps most consequential are the cascading utility disruptions that spread through densely populated areas. A single flooded electrical substation can plunge hundreds of clustered homes into darkness. Water treatment facilities overwhelmed by flood waters can render tap water unsafe across entire subdivisions. These shared service disruptions create prolonged recovery challenges that isolated properties typically don't face.

The failure of shared protective infrastructure represents the most dramatic expression of this vulnerability. When a levee breach or a communal drainage system fails, it doesn't affect just one property but simultaneously unleashes water across countless homes. This synchronised flooding creates resource competition during the emergency response and recovery phases, extending the severity and duration of impacts for all affected properties.

Property clustering creates a robust market value contagion during and after flood events, where banking impacts spread through neighbourhoods much like the flood waters themselves. When a significant portion of homes in a neighbourhood experience flooding, the market reacts not just to the directly damaged properties but to the entire area. Prospective buyers begin to view the whole neighbourhood through the lens of flood risk, creating a collective devaluation effect that touches even homes that remain dry.

Flooding and market stigma

Belanger & Bourdeau-Brien's groundbreaking 2018 study offers significant insights into how flood risk manifests as market stigma in property valuations. By analysing over 600,000 residential properties across England, their research revealed that flood risk creates a complex economic geography where property values respond to actual and perceived hazards.

Properties within designated 100-year floodplains experienced notable price discounts compared to similar properties in safer areas, with the most pronounced effects on waterfront properties. However, proximity to water bodies alone provides an incomplete picture of how markets process flood risk. Factors like elevation differences and localised topographical features create substantial variations in risk perception and resulting price impacts even within the same floodplain.

The study's examination of stigma's temporal nature revealed that post-flood price discounts typically ranged from 10% to 50%, but these effects were not permanent fixtures in the market. Instead, they gradually diminished as the collective memory of specific flood events faded, with most markets showing substantial price recovery within 4-7 years after major floods. This recovery pattern exhibited significant socioeconomic variation, as lower-priced properties demonstrated greater sensitivity and extended recovery periods than higher-priced homes, which frequently benefited from offsetting amenities such as desirable views or waterfront access.

This finding highlights how market responses to flood risk reflect objective hazard assessments and complex interplays among perceived risk, property characteristics, and location-specific amenities that can mitigate negative risk perceptions in specific contexts. Markets typically respond more strongly to recent visible events than abstract probability calculations or hazard maps, disconnecting technical risk assessments and market perceptions.

The physical recovery process influences these market effects, particularly in clustered developments. When entire neighbourhoods require simultaneous repairs, the demand for contractors, materials, and inspections creates bottlenecks that significantly extend reconstruction timelines. Homeowners compete for limited recovery resources, expanding the visible signs of damage throughout the community and prolonging the period of market impact.

Belanger & Bourdeau-Brien also documented a "halo effect" of property devaluation that spreads through clustered developments after flood events. Their analysis reveals that properties located within the same neighbourhood as flooded homes but themselves untouched by waters still experienced statistically significant devaluations—demonstrating that in clustered developments, flood risk impacts extend beyond physical water damage to affect entire market ecosystems.

The study's findings carry significant policy implications for flood risk management. Traditional floodplain designations frequently overlook the nuanced ways in which buyers evaluate and price flood risk. This disconnect highlights the need for more detailed, property-specific risk assessment tools to align market perceptions with hazard profiles better.

Furthermore, the documented pattern of temporary stigma periods reveals opportunities for targeted recovery interventions that could accelerate market normalisation following flood events, potentially reducing economic disruption in affected communities. By emphasising these dynamics, Belanger and Bourdeau-Brien's work provides essential guidance for policymakers aiming to address the financial dimensions of flood vulnerability in an era of extensive new property development.

Quantifying the Clustering Effect

The portfolio impact model captures this clustering effect through the correlation matrix Σ , combined with the shock factor σ :

Belanger and Bourdeau-Brien

$$f_D(d) = \int_{-\infty}^{\infty} f_S(s) f_{D|S}(d, s) ds$$

Where:

$f_D(d)$ represents the probability density function of the damage level d across the portfolio

$f_S(s)$ is the probability density function of the shock intensity s (flood severity)

$f_{D|S}(d, s)$ is the conditional probability density function of damage d given shock s

The integration from $-\infty$ to ∞ accounts for all possible shock scenarios

The portfolio impact model measures how flood risks spread among closely situated properties by analysing their interconnectedness. This approach goes beyond treating each property as an isolated risk. It recognises that when properties are clustered together, the impact of flooding on one affects the others in predictable ways.

The correlation matrix illustrates the strength of the connections between neighbouring properties—when one property suffers from flooding damage, others nearby encounter similar risks due to their proximity and shared environmental characteristics. The shock factor subsequently measures the extent of this ripple effect across the cluster.

This mathematical framework enables us to calculate the overall probability distribution of damage across an entire neighbourhood or development. It combines the individual probabilities of specific shock events with the conditional probabilities of how those shocks translate into actual damage patterns across properties.

The clustering effect can better predict how flooding impacts cascade through developments where homes share similar construction methods, elevation profiles, and proximity to water sources. The model demonstrates that flooding damage doesn't occur randomly across properties but follows patterns that reflect geography and built environment characteristics.

Understanding these clustering dynamics allows insurers, developers, and policymakers to more accurately assess the actual financial vulnerability of residential areas to flood events rather than simply calculating risks for individual properties in isolation:

- **Higher tail risk:** The 95% Value at Risk (VaR) and Expected Shortfall (ES) metrics increase dramatically compared to diverse portfolios of the same size.
- **Greater volatility:** Portfolio impact variance increases with clustering
- **Reduced diversification benefit:** The risk-reducing effect of portfolio diversification diminishes as spatial correlation increases

A portfolio with properties distributed across different watersheds might see a 95% VaR 2-3 times lower than a portfolio of equal value concentrated in a single flood-prone area.

Clustering Patterns in Real Estate Portfolios

Residential property clustering follows systematic patterns that inadvertently maximise flood risk exposure across investment portfolios. The price premium commanded by waterfront views creates high-value property concentrations precisely in the highest-risk areas, generating natural risk accumulations.

Developmental history contributes significantly to risk concentration, as many neighbourhoods were constructed during single development phases, resulting in uniform risk profiles across entire communities sharing identical construction standards and elevations. Infrastructure determinism further reinforces these patterns, as road networks and utility systems naturally follow topographical features like river valleys, creating property density along the same hydrological pathways that channel floodwaters.

These patterns establish natural risk concentration hotspots where property density and flood susceptibility align—precisely the conditions where correlation effects manifest most strongly. Understanding these clustering dynamics informs several key risk management approaches. Geographic diversification across multiple watersheds and flood zones becomes essential for reducing portfolio-level exposure.

Correlation-based pricing mechanisms adjust mortgage rates and insurance premiums based on neighbourhood-level clustering characteristics rather than solely on individual property risk metrics. Targeted mitigation strategies focus infrastructure improvements on protecting entire clusters rather than attempting less cost-effective individual property-level interventions. Advanced stress testing methodologies incorporate portfolio simulations with varying correlation parameters to identify vulnerabilities to specific clustering effects.

As flood patterns shift due to additional property development that converts permeable surfaces to impervious concrete, previously diverse portfolios may become inadvertently clustered regarding risk exposure. Areas once considered to have independent flood risks may begin experiencing correlated flooding as weather patterns and hydrological systems change. This dynamic risk landscape necessitates continuous reassessment of portfolio correlation structures rather than relying on static risk assessments based on historical patterns.

The mathematical analysis of spatial correlation confirms what many risk managers have intuitively understood. When evaluating flood risk, the relative geographic positioning of properties within a portfolio is comparable to their risk profiles. By explicitly modelling these clustering effects, analysts develop a more accurate understanding of portfolio-level flood risk. They can formulate more effective strategies for managing that risk in an increasingly uncertain climate future.

These territorial rating methodologies now feed into insurance premium calculations incorporating several advanced components. Peril-specific territories establish separate geographic segmentation frameworks for flood, wind, fire, and other hazards. Proximity factors enable rating adjustments based on distance to risk features such as coastlines, floodplains, or fault lines.

Topographic variables incorporate elevation, slope, aspect, and other terrain characteristics influencing risk profiles. Hydrological modelling directly integrates outcomes from the flood models discussed in previous chapters. The following chapter will examine These methodological components in greater detail, forming the foundation for comprehensive risk pricing frameworks.

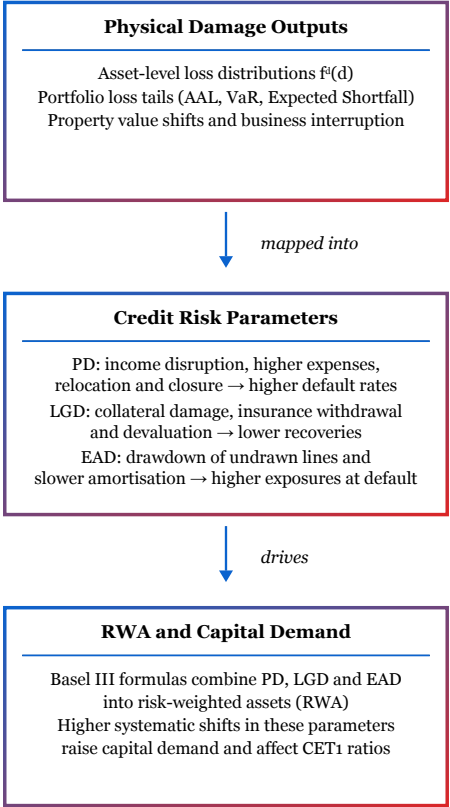
By this point, we have built a complete pathway from flood hazard to portfolio-level physical loss. Hazard curves translate return periods and climate-conditioned scenarios into probabilistic descriptions of flood intensities at specific locations, providing the common language that links OS-Climate/FINOS upstream models with banking-style risk engines. Vulnerability and damage functions then take those intensities and convert them into loss distributions for individual assets, acknowledging that this step rests on imperfect theory, mixed data quality and explicit choices between expert-derived, statistical and parametric approaches.

We have also seen that uncertainty is not a single blur but has structure. Aleatory uncertainty captures the irreducible randomness in how apparently similar assets respond to the same flood, while epistemic uncertainty reflects gaps in our knowledge of asset characteristics, damage mechanisms and model structure. Treating these separately allows us to represent vulnerability as a distribution rather than a point estimate and to carry honest confidence bands through to damage, rather than assuming a precision the science does not support.

Finally, we have moved from individual assets to clustered portfolios, showing how shared topography, infrastructure and market perception turn localised floods into correlated losses across neighbourhoods and loan books. Spatial correlation and clustering raise tail metrics such as Value at Risk and Expected Shortfall, erode diversification benefits and create concentration risk that is invisible if properties are treated as independent points on a map. Taken together,

these elements give us a physically grounded, uncertainty-aware description of how floods damage assets and portfolios; the next chapter builds directly on this foundation to translate those damage distributions into PD, LGD, EAD and capital within the banking book

Figure 74 – Linking Physical Damage to Credit Risk Metrics



Physical flood damage enters the banking book through quantified adjustments to PD, LGD and EAD, which then feed existing Basel III capital and pricing frameworks rather than a separate ESG overlay.

Chapter 8.2 - From Flood Depth to Loss

Damage functions are where hydrological science meets balance-sheet reality. In the previous chapter, we built hazard curves that describe how often different flood intensities occur at a given location. Damage functions complete the next step: they tell us what happens when water of a given depth, duration and velocity actually encounters a building, a piece of infrastructure or an agricultural asset. Without this translation layer, we have scientifically rigorous flood maps but no way to price risk, allocate capital or evaluate mitigation investments.

At their core, damage functions are empirical relationships—usually expressed as curves—that map physical flood intensity to fractional or absolute loss. The most common form is the depth-damage curve: a function that takes inundation depth as input and returns an estimated damage ratio, typically expressed as a percentage of asset value destroyed or impaired. A simple residential depth-damage curve might indicate that fifty centimetres of internal flooding causes damage equivalent to twenty per cent of building value, while one hundred and fifty centimetres might push that figure to sixty per cent. These curves vary by building type, construction standards, contents and use, which is why most damage function libraries stratify by residential, commercial, industrial and infrastructure categories, often with further granularity within each.

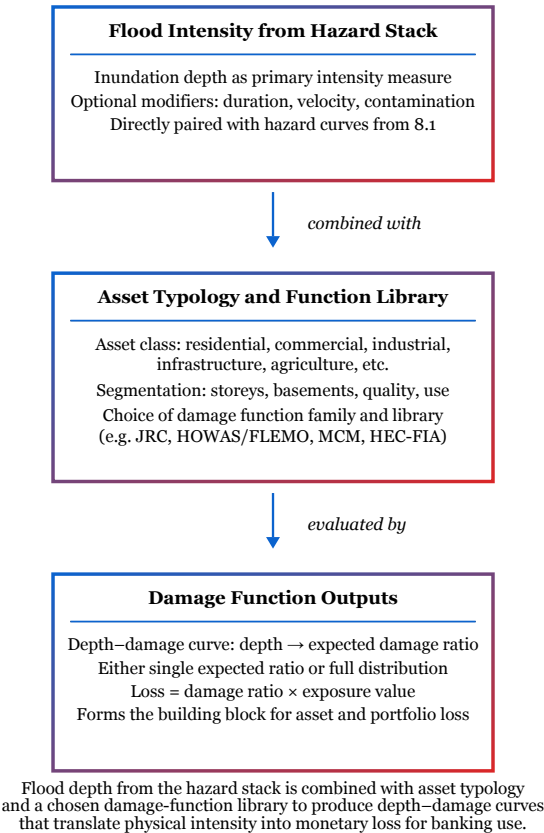
The reason depth dominates as the primary variable is partly physical—deeper water causes more damage—and partly pragmatic: depth is the one flood characteristic we can measure or model with reasonable confidence and for which we have post-event damage data to calibrate against. Duration, flow velocity and contamination all matter, and more sophisticated functions incorporate them, but the core relationship remains depth-driven because that is where the data exists and where hydrological models deliver their most reliable outputs. For banking applications, this focus on depth is convenient: it allows damage functions to be paired directly with the depth-frequency outputs from the hazard stack developed in earlier chapters, creating a computational pipeline from rainfall scenarios through to expected portfolio loss.

Yet damage functions are not physical laws. They are statistical summaries of how buildings have responded to past floods, supplemented by engineering judgment about how similar buildings might respond to future ones. This means they carry uncertainty from multiple sources: sparse calibration data (most locations have experienced few if any severe floods), regional variation in construction practices and materials, differences in occupant behaviour and preparedness, and the challenge of predicting how older functions will perform as flood patterns, building codes and resilience measures evolve. Understanding these limitations is essential for banks that need to justify model choices to auditors, explain risk concentrations to boards, and avoid the false precision that comes from treating an empirically derived curve as if it were a deterministic formula.

In the sections that follow, we examine how damage functions are constructed, which major function libraries exist and are used in practice, and how the OS-Climate and FINOS frameworks have adapted catastrophe-modelling approaches to create open, auditable damage-function pipelines suitable for banking risk management. We also address the practical

question every risk manager faces: when should I use a global synthetic function, when do I need region-specific empirical data, and how do I quantify and communicate the uncertainty that comes with either choice?

Figure 75 – From Flood Depth to Damage Functions



Conceptual Foundation

The conceptual foundation of damage functions rests on a small number of core ideas that are easy to state but subtle in their implications. At a glance, it seems straightforward to say “for this type of building, one metre of water causes about X per cent damage”, yet behind that statement sits a chain of modelling choices about what exactly is being related to what, which physical variables are included or excluded, how assets are classified, and whether we treat damage as a single expected value or as a distribution.

Before turning to the practical question of how damage curves are built in different frameworks, it is therefore useful to be explicit about four elements: what damage functions represent in

probabilistic terms, why depth is usually chosen as the primary intensity measure, how asset typologies and segmentation shape the curves, and what kinds of outputs these functions are intended to produce for use in portfolio loss modelling and banking applications

What Damage Functions Represent

Damage functions provide a compact way to describe how physical flooding translates into monetary loss. They take a small set of hazard variables and asset characteristics and return an expected damage ratio, typically between zero and one, which can then be multiplied by an exposure value to obtain a loss amount. In practice, they are conditional expectations: given a certain flood intensity and asset type, what fraction of value do we expect to lose on average, recognising that individual outcomes will vary around this mean.

Choice of Intensity Measures

For fluvial and pluvial flooding, inundation depth is the dominant intensity measure used in damage functions. Depth is physically meaningful—higher water generally affects more of the building fabric and contents—and it is readily available from hydraulic models and flood maps, which makes it attractive for large-scale modelling. Other factors such as duration, flow velocity and contamination clearly influence damage, especially for structural elements and infrastructure, but are less consistently recorded in post-event datasets; as a result, depth-only functions remain the workhorse in most applied models, with multi-parameter extensions used where data permit.

Asset Typologies and Segmentation

Damage functions are not one-size-fits-all curves; they are defined for specific asset typologies. Residential buildings, commercial properties, industrial facilities, infrastructure and agricultural assets each have distinct depth-damage relationships driven by their construction, layout, contents and critical systems. Even within a given class, further segmentation by building age, quality, number of storeys, presence of basements or ground-floor usage can materially change the shape and steepness of the curve, which is why modern models such as FLEMO or national handbooks provide families of functions rather than a single generic one.

Outputs: From Ratios to Loss Distributions

Conceptually, a damage function can return either a single expected damage ratio or a full distribution of possible ratios for a given intensity level. Traditional implementations often use deterministic curves and treat the output as a point estimate, which is simple but hides aleatory and epistemic uncertainty in vulnerability. More recent approaches, including those in catastrophe and OS-Climate style models, treat the function as defining a probability distribution over damage ratios, allowing Monte Carlo methods to generate loss distributions at asset and portfolio level that explicitly capture variability and model uncertainty.

Construction Approaches for Damage Functions

Damage functions are not discovered; they are constructed. Different research communities and agencies have built them in different ways, depending on the data available, the spatial scale of interest and the decisions they were meant to inform. Three broad approaches dominate practice: empirical functions calibrated from post-event damage data, synthetic functions derived from engineering judgement and component analysis, and hybrid or statistical-learning approaches that combine elements of both. Each approach brings its own strengths, weaknesses and implications for banking use, particularly around transferability, transparency and uncertainty.

Empirical damage functions

Empirical damage functions are built “from the ground up” using observed losses from past floods. After a major event, survey teams or insurers collect data on water depth, building type, use and repair or replacement costs for large numbers of affected properties. These observations are then grouped by asset class and intensity, and statistical methods are used to derive average depth-damage relationships—essentially, the mean damage ratio observed at each depth, often with measures of dispersion.

The HOWAS21 database in Germany is the canonical example: it contains object-specific records of residential, commercial and industrial flood damages, which have been used to derive both simple depth-damage curves and more detailed models such as the FLEMO family. Similar empirical bases underpin national models and manuals in the UK, US and other countries, where agencies have combined engineering inspections with claims data to produce occupancy-specific curves. For banks, the appeal of empirical functions is clear: they are grounded in real events, reflect local construction practices and, when documented properly, provide a transparent audit trail back to observed losses.

However, empirical approaches face three structural limitations. First, they require large, high-quality datasets that are only available in a few regions with repeated flood experience and strong data infrastructure. Second, because they reflect local building types, codes and behaviours, their transferability to other countries or even other regions of the same country is limited; applying German-derived curves directly to, say, Southeast Asia or Eastern Europe risks systematic bias. Third, extreme depths and rare combinations of factors are inherently under-represented in historical data, which means empirical curves are extrapolated precisely where banks care most about tail loss. These limitations do not make empirical functions unusable, but they reinforce the need for careful documentation, local adaptation and explicit uncertainty treatment when they are used in portfolio models.

Synthetic (expert-based) damage functions

Synthetic damage functions take the opposite starting point: instead of asking “what happened in past floods?”, they ask “what would happen to this type of asset under a range of hypothetical water depths?” Engineers and building experts break down a representative property into components—structure, finishes, services, contents—and assess, for each depth band, which

components are likely to be affected and what fraction of their value would be lost. These component-level losses are then aggregated to produce an overall depth-damage curve for the asset type.

This approach underpins several widely used tools. The UK Multi-Coloured Manual combines empirical evidence with engineering judgement to define depth-damage relationships for different residential and non-residential categories, including variants for coastal, fluvial and pluvial flooding and, in some cases, for different salinity or wave conditions. At a larger scale, the JRC global depth-damage functions developed by Huizinga and co-authors use a synthetic methodology informed by international literature and limited empirical data to produce consistent curves for residential, commercial, industrial, infrastructure and agricultural assets worldwide. These global functions are not as precise as country-specific empirical models, but they provide a coherent basis for cross-country analysis when local data are unavailable.

Synthetic functions offer flexibility and coverage where empirical data are sparse or non-existent, and they can be tailored to future building standards or hypothetical asset types that have not yet experienced major floods. Their downside lies in validation and subjectivity: they depend on expert judgement about damage mechanisms, and independent checks against real loss data are often limited to small samples, if they exist at all. For banks, this means synthetic curves should be accompanied by clear documentation of assumptions and, ideally, sensitivity analysis that shows how loss estimates move when those assumptions are varied.

Hybrid and statistical-learning approaches

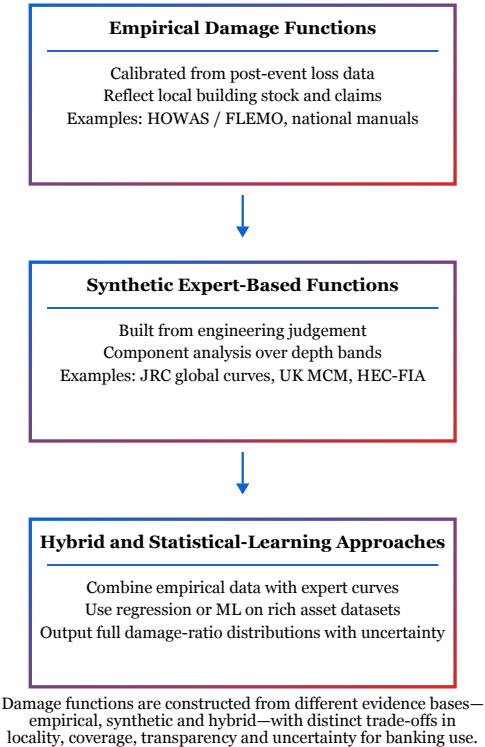
Hybrid approaches attempt to combine the strengths of empirical and synthetic methods while mitigating some of their weaknesses. One strand uses empirical data to calibrate or adjust expert-based curves, for example by shifting or scaling a synthetic function to match observed mean losses in a given region or building category while retaining its overall shape. Another strand uses statistical-learning techniques—such as regression, Random Forests or Bayesian models—to estimate damage as a function of multiple predictors, including depth, building type, floor area, construction quality and precautionary measures, rather than relying on a single depth axis.

Models like FLEMOps+ exemplify this direction: they start from an empirically informed structure, then incorporate additional variables such as building use, contamination, recurrence interval and private precaution to refine loss estimates and capture heterogeneity between ostensibly similar properties. Recent research also explores fully data-driven damage models trained on insurance claims or detailed post-event surveys, which can produce probabilistic damage estimates and quantify epistemic uncertainty through ensemble or Bayesian methods. In the catastrophe-modelling and OS-Climate communities, these ideas are embedded within vulnerability modules that output not just a mean damage ratio but a distribution conditioned on intensity and asset attributes.

For banking applications, hybrid and statistical-learning approaches are attractive because they can explicitly handle multiple drivers of loss and deliver uncertainty information that can be propagated into portfolio risk metrics. Their practical constraints are familiar: they require

sufficiently rich and representative training data, robust validation, and governance frameworks that can explain model behaviour to non-specialists and regulators. Where those conditions are met, they offer a route to more nuanced, asset-specific damage estimates that align naturally with the asset-level hazard and clustering frameworks developed earlier in the book.

Figure 76 – Families of Flood Damage Functions



Key Function Libraries Used in Practice

A small number of function libraries and modelling frameworks underpin most practical flood-damage assessments in use today. For banks, understanding where these come from and how they differ is important, because vendor models, public tools and internal implementations often rely—directly or indirectly—on one or more of them. What follows is not an exhaustive catalogue, but a brief profile of the main families that appear in regulatory studies, catastrophe models and open-source toolchains.

JRC Global Depth-Damage Functions (Huizinga et al.)

The Joint Research Centre (JRC) of the European Commission has developed a suite of global flood depth-damage functions that provide harmonised curves for residential, commercial, industrial, infrastructure and agricultural assets. These functions are constructed using a

predominantly synthetic methodology, informed by international literature and limited empirical data, and expressed as relative damage (fraction of value) as a function of water depth above ground floor.

Their primary strength is coverage: they offer a consistent set of curves that can be applied worldwide, making them a natural choice for global portfolio screening, cross-country scenario analysis and macro-prudential studies where fine local calibration is not feasible. The trade-off is precision at local scale: because they abstract from national building codes, materials and behaviours, they should be treated as first-order approximations and adapted where more specific national or regional curves exist.

HOWAS₂₁ and FLEMO (Germany)

Germany's HOWAS₂₁ database is one of the most detailed empirical resources for flood damage worldwide. It contains object-specific records of residential, commercial and industrial damages from multiple historic flood events, including information on depth, building characteristics and monetary loss. Researchers at GFZ Potsdam have used this database to develop the FLEMO (Flood Loss Estimation MOdel) family, including FLEMOps and FLEMOcs for the private and commercial sectors.

FLEMO models extend simple depth-damage curves by incorporating additional predictors such as building type, construction quality, number of storeys, contamination and private precautionary measures. This makes them powerful tools for detailed regional studies and for exploring the effect of resilience measures on loss, but also ties them closely to German building stock and institutional conditions. When applied elsewhere, they require careful adjustment and, ideally, re-calibration against local data.

HEC-FIA and USACE Curves (United States)

In the United States, the US Army Corps of Engineers' HEC-FIA (Flood Impact Analysis) software provides a standardised set of depth-damage relationships for a range of occupancy classes (residential, commercial, industrial, public) and foundation types. These curves have been assembled from a mixture of empirical claims data, engineering judgement and previous national studies, and are used widely in federal benefit–cost analyses and flood-risk management studies.

For practitioners working with US portfolios, HEC-FIA functions offer regulatory familiarity and a clear provenance, which can be valuable in supervisory dialogue. Their limitation, as with other national libraries, is that they are tailored to US construction and valuation practices; using them in other jurisdictions without adjustment would embed unexamined assumptions about building design, cost structures and insurance coverage.

UK Multi-Coloured Manual (MCM)

The UK “Multi-Coloured Manual” (MCM) and its predecessors provide depth-damage relationships for a wide range of residential and non-residential property categories, developed to support appraisal of flood and coastal defence schemes. The curves reflect a combination of

empirical evidence from UK floods and engineering judgement, with variants that distinguish, for example, between coastal and fluvial flooding and different salinity or wave conditions.

A key feature of the MCM framework is its explicit treatment of different damage components (buildings, contents, vehicles, business interruption) and its ability to incorporate the effect of property-level resilience measures, such as flood doors or raised electrics. This makes it particularly relevant to the resilience discussions in later chapters, but also means that direct application requires UK-specific cost and valuation assumptions to be revisited when used elsewhere.

Catastrophe Models and Oasis-Style Vulnerability Modules

Commercial catastrophe models (e.g. those used by insurers and reinsurers) embed their own proprietary libraries of vulnerability or damage functions, typically structured as intensity–damage relationships for many peril–occupancy combinations. The open-source Oasis Loss Modelling Framework exposes a standardised way to implement such functions, usually as mean damage and coefficient-of-variation by intensity bin, enabling different modelling groups to plug in their own curves while using a common calculation engine.

In the OS-Climate and FINOS context, this catastrophe-modelling heritage is reflected in the physrisk engine, which uses asset-level vulnerability models rather than aggregate GDP damage functions. These models define mean damage curves and, where possible, full damage-ratio distributions as a function of hazard intensity and asset attributes, enabling Monte Carlo simulation of loss distributions under different hazard scenarios. For banks, this architecture creates a bridge between open, auditable damage functions and the portfolio-level risk metrics familiar from insurance and credit modelling.

Pulling these strands together and impact on model governance.

Choosing and governing damage functions is a model-risk problem, not a mere parameter choice. Although the output of a damage function is deceptively simple—a single damage ratio for a given depth and asset type—it embeds a long chain of assumptions about data, engineering judgement, transferability and uncertainty that must be made explicit under frameworks such as SR 11-7. The apparent simplicity of “depth in, percentage loss out” creates a real trap for banks: it invites widespread re-use of a few curves across portfolios, regions and use-cases without sufficient scrutiny of whether the underlying evidence and design are appropriate to each application.

From a governance perspective, the first decision is which family of functions to use for which purpose. Global synthetic libraries such as the JRC functions are well suited to screening and scenario analysis across diverse geographies, but they should not be treated as precise loss predictors for capital calculations in any one country. National or regional models—HOWAS/FLEMO in Germany, HEC-FIA in the US, the Multi-Coloured Manual in the UK—offer richer local calibration and better alignment with building stock and regulatory practice, but their use outside their home context requires explicit justification, adjustment and, where possible, back-testing. Catastrophe-model vulnerability modules and OS-Climate style curves add another

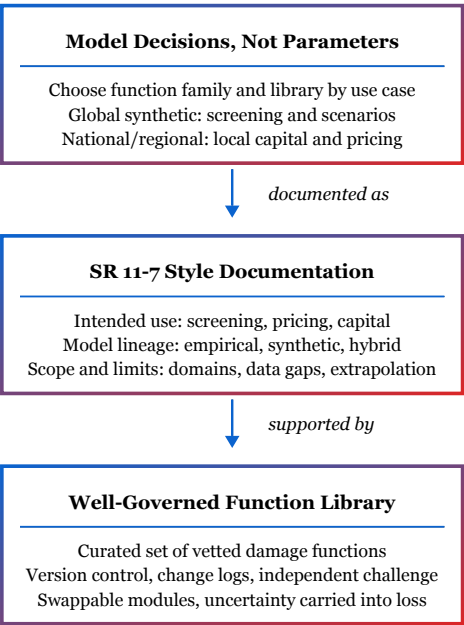
layer: they often provide probabilistic outputs and multi-parameter dependence, but at the cost of greater complexity and a stronger need for transparent documentation.

Under SR 11-7 style expectations, these choices should be framed and documented as full model decisions. That means a clear statement of intended use (screening vs pricing vs capital), model lineage (empirical, synthetic or hybrid; source databases and studies), applicability domain (geographies, asset types, value ranges) and known limitations and uncertainties (data sparsity, extrapolation ranges, missing variables). Validation should go beyond mechanical checks to include reasonableness tests (do loss levels align with known events or insurance experience?), sensitivity analysis (how much do portfolio losses move if alternative curve families are used?) and challenge of transfer assumptions when functions are moved across regions or applied to new asset classes.

Finally, banks need to resist the temptation to treat damage functions as fixed constants baked into black-box vendor models. Open frameworks such as Oasis and OS-Climate demonstrate that vulnerability modules can be swapped, compared, and improved over time, and that uncertainty in damage should be carried forward into portfolio loss distributions rather than collapsed into single-point estimates.

“In practice, this means curating a small, well-governed library of damage functions with clear provenance, maintaining version control and change logs, and ensuring that business users understand both the power and the limits of these deceptively simple curves.”, David Kelly, MKM

Figure 77 – Governing Flood Damage Functions as Models



Banks should treat damage functions as governed models within a curated library, not fixed parameters baked into black-box tools.

Chapter 8.3 - Quantifying Resilience

Up to this point, our impact quantification has treated assets and portfolios largely as passive recipients of flood hazard. Hazard curves, vulnerability functions and clustering dynamics describe what happens when water meets the built environment under current conditions. In practice, however, banks lend to borrowers and asset owners who can change their fate. Resilience is not just a vague aspiration; it can be operationalised, documented and measured through concrete Standard Operating Procedures that systematically reduce damage, shorten downtime and stabilise cash flows.

Standard Operating Procedures sit at the centre of this operational resilience. They codify how organisations prepare for, respond to and recover from flood events: how assets are protected in advance, how people and systems are managed during an event, and how quickly normal operations resume afterwards. From a banking perspective, these procedures are not soft “ESG” features; they are observable, auditable drivers of loss severity and recovery time, and therefore of PD, LGD and EAD. The challenge, and the focus of this chapter, is to move from qualitative checklists to quantitative measures: how to express the presence and quality of SOPs as adjustments to damage distributions, business interruption durations and, ultimately, expected and tail losses.

To do that, we need a framework that links resilience measures to the same machinery developed in earlier chapters. SOP effectiveness must be reflected in vulnerability functions (through lower damage ratios and narrower uncertainty), in business interruption models (through shorter and less variable downtime), and in clustering dynamics (through reduced correlation of failure across similarly exposed assets that are differently prepared). This chapter develops that framework, showing how resilience can be scored, parameterised and incorporated into the hazard–damage–portfolio chain so that banks can recognise, price and, where appropriate, finance genuine improvements in flood resilience.

Quantifying Resilience Using Standard Operating Procedures

Standard Operating Procedures (SOPs) provide a structured approach to quantifying resilience by formalising how systems prepare for, respond to, and recover from disruptive events. When properly implemented, SOPs transform theoretical resilience concepts into measurable outcomes, particularly for built environments facing hazards like flooding.

SOP Fundamental

The fundamental calculation for SOP resilience coverage can be expressed as:

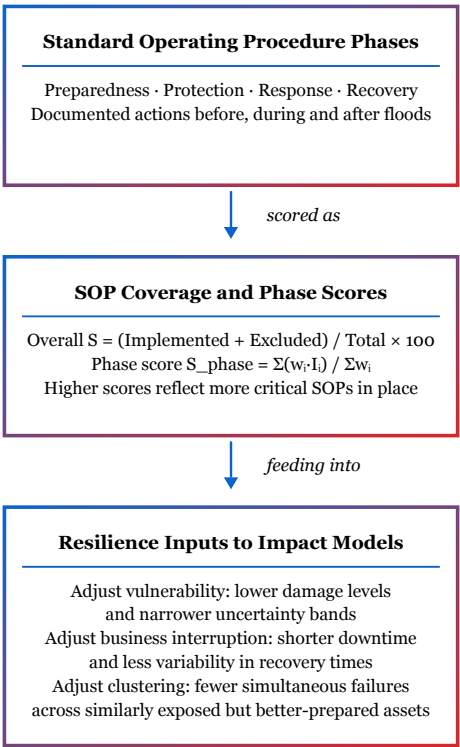
$$S = (\text{Implemented SOPs} + \text{Excluded SOPs}) / (\text{Total Recommended SOPs}) \times 100$$

Where:

- S represents the SOP coverage score (0-100%).
-
- Implemented SOPs are procedures actually in place.
-
- Excluded SOPs are procedures deliberately omitted with justification.
-
- Total Recommended SOPs encompass all procedures advised for the specific context.

This calculation yields a normalised percentage that can be integrated into broader resilience assessments.

Figure 78 – SOP Phases and Resilience Scoring



Resilience Quantification: Assessment

Earlier, we treated vulnerability largely as a function of where a property sits and how water behaves around it. That is only half the story. Two properties exposed to the same hazard and built from similar materials can experience very different loss outcomes depending on how well

they are prepared, how they are operated during an event, and how systematically recovery is managed afterwards. Resilience is the name we give to this active component of vulnerability: the extent to which disciplined preparation and response can bend the damage curve and shorten downtime.

To quantify resilience, we first need a clear baseline. For any given property or portfolio segment, we begin by specifying an unprotected scenario that reflects current or minimal practice. For a property in a flood zone, this might mean accepting that a 1-in-100-year event produces one metre of internal flooding, causes structural and contents damage of order X, and results in complete loss of use for Y days while repairs are undertaken. This baseline is not a prediction set in stone; it is a reference point against which the effect of Standard Operating Procedures can be measured.

Resilience Quantification: SOP Inventory and Analysis

Once the baseline is defined, the task becomes understanding how much we can shift it through better behaviour. Standard Operating Procedures provide the most concrete way to describe and measure that behaviour. Rather than treating resilience as a vague quality, we break it down into documented actions that can be observed, audited and, ultimately, scored. These procedures typically cluster into four phases: preparedness before an event, protection as water approaches, response during the incident itself, and recovery in the days and weeks that follow.

For a given property type, we can therefore construct an SOP inventory that lists the key actions in each phase and assesses whether they are in place and likely to be executed. Preparedness might cover seasonal maintenance, early-warning monitoring and stockpiling of critical supplies. Protection focuses on measures such as temporary or permanent barriers, elevation of critical equipment and pre-planned isolation of utilities. Response encompasses evacuation protocols, communication plans and on-site safety procedures, while recovery deals with structured water removal, drying and decontamination sequences, and the staged restoration of systems and operations. By assigning relative weights to these procedures based on their expected impact on damage ratios and downtime, we can move from a qualitative checklist to a quantitative resilience score that can be fed back into the vulnerability and business-interruption models introduced earlier.

Resilience Implementation Calculation

Phase-specific SOP

For each resilience phase, calculate a phase-specific SOP score:

$$S_{\text{phase}} = (\sum(w_i \times I_i)) / (\sum w_i)$$

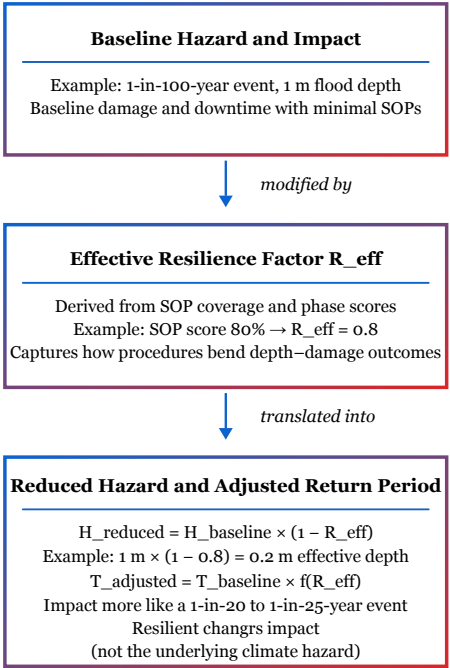
Where:

- S_{phase} is the score for a specific resilience phase.
- w_i is the weight of the i -th SOP.

- I_i is the implementation status (1 for implemented, 0 for not implemented).
- Σ represents the summation across all SOPs in that phase.

This calculation evaluates each SOP in that phase, determines whether it has been implemented (assigning a value of 1) or not (assigning a value of 0), and multiplies this implementation status by the weight assigned to that specific SOP. These weighted values are then summed and divided by the total of all the weights. This results in a score reflecting the implementation level for that particular resilience phase.

Figure 79 – Resilience, Hazard Reduction and Effective Return Period



The resulting S_{phase} score offers a normalised measure (usually between 0 and 1) indicating how effectively weighted SOPs have been implemented within a specific resilience phase. A score closer to 1 suggests that most important SOPs (those with higher weights) are in place, whereas a score nearer to 0 indicates that many significant SOPs remain unimplemented.

This method ensures that more critical procedures (those with higher weights) have a proportionally more significant effect on the overall resilience score for that phase.

Resilience Hazard Reduction

The implementation score translates to hazard reduction through modelling relationships between procedures and outcomes:

Hazard Reduction

$$H_{\text{reduced}} = H_{\text{baseline}} \times (1 - R_{\text{eff}})$$

Where:

- H_{reduced} is the reduced hazard level
- H_{baseline} is the baseline hazard level
- R_{eff} is the effective resilience factor derived from the SOP scores

For our flood example:

- Baseline flood depth = 1m
- SOP implementation score = 80%
- Effective resilience factor = 0.8
- Reduced flood depth = $1\text{m} \times (1 - 0.8) = 0.2\text{m}$ (20cm)

Resilience Quantification: Return Period Translation

The reduced hazard level can then be translated to an adjusted return period:

Return Period Adjustment

$$T_{\text{adjusted}} = T_{\text{baseline}} \times f(R_{\text{eff}})$$

Where:

- T_{adjusted} is the adjusted return period.
- T_{baseline} is the baseline return period.
- $f(R_{\text{eff}})$ is a site-specific function relating resilience to return period adjustment.

In our example, a properly implemented set of SOPs effectively transforms a 1-in-100-year flooding event into a 1-in-20-year event in terms of impact.

Residential Property Flood Resilience

Consider a single-family home in a flood-prone area with the following characteristics:

- It is located in a 1-in-100-year floodplain with a 1m projected flood depth
- Home replacement value: \$350,000

- Contents value: \$75,000
- Displacement cost: \$3,000 per month
- Recovery time without SOPs: 4 months
- The homeowner implements a comprehensive set of SOPs:

Preparatory SOPs

- Installation of flood-resistant materials in lower level
- Regular cleaning of gutters and drainage pathways
- Elevation of electrical systems and HVAC equipment
- Creation of a household emergency plan

Response SOPs

- Deployment of door/window flood barriers
- Moving valuable items to the upper floors
- Secure storage of essential documents in waterproof containers

Recovery SOPs

- The proper sequence for water removal and drying
- Mold prevention procedures

Using the formula and appropriate weights for each SOP category:

$$S_{\text{total}} = ((4 \times w_{\text{prep}}) + (3 \times w_{\text{resp}}) + (2 \times w_{\text{recov}})) / ((5 \times w_{\text{prep}}) + (4 \times w_{\text{resp}}) + (3 \times w_{\text{recov}}))$$

Assuming weights of 0.4 for preparatory, 0.4 for response, and 0.2 for recovery SOPs, the SOP implementation score is approximately 75%.

This translates to:

- Reduced flood depth: 25cm (instead of 1m).
- Shortened recovery time: 1.5 months (instead of 4).

- Damage reduction: \$85,000 to structure and contents.
- Displacement cost reduction: \$7,500.
- Return period adjustment: Impacts equivalent to a 1-in-25-year event.

Integration with Broader Resilience Frameworks

Once SOP effectiveness has been translated into shifts in expected damage and recovery times, it needs to be viewed as one component of a broader resilience picture rather than in isolation. A property or portfolio is not resilient simply because it has good procedures on paper; resilience also depends on the physical robustness of buildings and infrastructure, the financial capacity of owners and tenants to absorb shocks, and the wider social and institutional context that shapes recovery.

SOP-based resilience scores therefore sit alongside structural measures (such as permanent defences and building standards), financial measures (such as insurance coverage, contingent liquidity and access to credit) and community or systems measures (such as redundancy in transport and utilities). Integrating SOP calculations with these other metrics allows banks to see how procedural strength amplifies or is constrained by these wider factors, and to identify cases where relatively modest investments in procedures could unlock disproportionate reductions in expected and tail losses across their portfolios. SOP resilience calculations should be integrated with other resilience metrics for comprehensive assessment:

Total Property Resilience

$$R_{\text{total}} = w_1 \times S + w_2 \times P + w_3 \times A + w_4 \times T$$

Where:

- R_{total} is the total resilience score.
- S is the SOP coverage score.
- P is the physical infrastructure score.
- A is the adaptive capacity score.
- T is the technological readiness score.
- w_1 through w_4 are respective weights.

Quantifying resilience through SOP implementation provides a structured approach to understanding and enhancing a property's ability to withstand disruptive events. Proper procedures transform a 1-in-100-year event into an effective 1-in-25-year event, demonstrating the tangible value of operational resilience for residential properties.

This methodology allows homeowners, insurers, and policymakers to make informed decisions about resilience investments based on measurable outcomes rather than theoretical frameworks alone. This could result in reduced insurance premiums, decreased recovery costs, and enhanced

property values. The next step is to ensure that these fields are included in a property's CDM design.

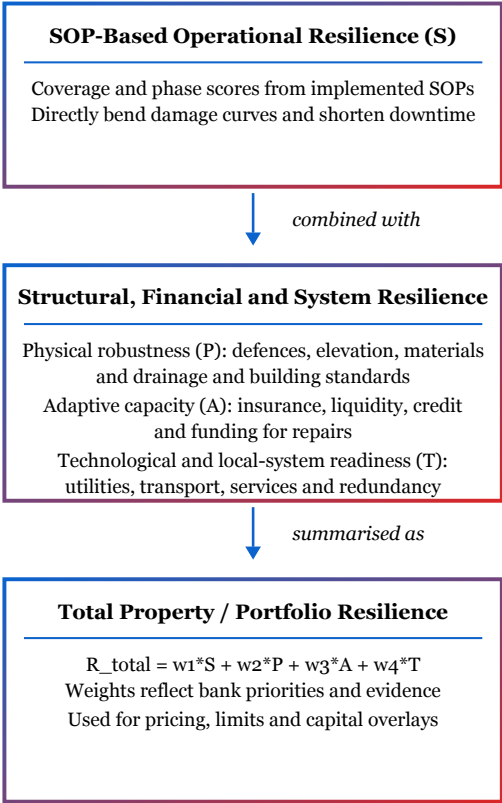
Beyond SOP

There are several other resilience dimensions materially influence how a flooded area fares and how quickly it recovers. The first is structural resilience: the permanent, physical measures built into properties and local infrastructure. Elevated floor levels, flood-resistant materials, sacrificial ground-floor spaces, sustainable drainage systems and properly maintained embankments or walls all change the relationship between flood depth and damage, sometimes dramatically. In modelling terms, structural measures shift and flatten the vulnerability curve before any procedural response is triggered.

A second dimension is financial resilience. Even well-designed buildings and well-written procedures can fail to deliver if owners, tenants or operators lack the financial capacity to act on them. Adequate and durable insurance coverage, access to emergency credit lines, liquidity buffers and pre-agreed funding for repairs all affect how quickly damage can be addressed and how long business interruption lasts. These factors do not change the hazard itself, but they strongly influence whether a temporary loss becomes a permanent impairment and whether borrowers can continue to service debt through and after the event.

The third dimension is local system resilience, which encompasses the wider urban fabric in which individual properties sit. Municipal and district authorities, local utilities, transport networks, schools, shops and hospitals all play a role in either cushioning or amplifying the impact of flooding. Well-designed drainage at street level can prevent property-level defences from being overwhelmed; robust power and water systems can keep large areas habitable; accessible schools, healthcare and retail services can stabilise communities and labour markets during recovery. Conversely, poorly planned road layouts, critical facilities in flood-prone zones, or fragile utility networks can turn a contained event into a protracted crisis. When banks assess resilience at portfolio level, these structural, financial and local-system metrics complement SOP-based scores, together providing a richer view of how flood risk will actually manifest in losses and how much can be improved through targeted investment and better planning.

Figure 80 - Components of Total Property Resilience



Chapter 9 - Insurance Risk Assessment

Insurance calculation represents a critical junction in the physical risk assessment process, where the scientific understanding of hazards developed in previous chapters converges with banking risk transfer mechanisms. The preceding chapters have established a comprehensive framework for understanding flood risk—from the fundamental principles of weather prediction and hydrological modelling, hydrological routing and hydraulic modelling to the detailed assessment of flood hazards and property-specific vulnerabilities. Insurance modelling builds directly upon this foundation, transforming these physical insights into economic terms through structured mathematical frameworks.

"Insurance serves as the essential bridge between physical risk science and banking markets—it translates the complexities of hydrodynamic modelling and property vulnerability into an annual premium reflecting a precise risk quantification. While mortgages extend risk over decades, insurance distils it into a single year, creating both a temporal mismatch and a powerful market signal about the true cost of hazards." — Swiss Re Institute.

This connection is particularly evident in territorial rating methodologies, which practically implement the hydrological and hydraulic modelling discussed in earlier chapters. The flood modelling approaches—including riverbank breach modelling, digital elevation analysis, and flow path simulation—directly inform how insurers delineate rating territories and assign relative risk factors. These territories essentially translate complex hydrological realities into financially actionable zones for risk pricing.

The insurance industry is uniquely positioned in the physical risk landscape due to its one-year time horizon. Unlike mortgages, which commit capital over decade, insurance contracts typically last just twelve months, allowing for rapid adaptation to emerging risk information.

This fundamental difference creates challenges and opportunities—insurers can quickly incorporate new physical risk data into their pricing models. Still, this same flexibility allows them to withdraw coverage entirely from areas they deem too risky, potentially leaving longer-term mortgage exposures unprotected.

This chapter explores how insurers bridge the gap between scientific risk assessment and banking quantification. We begin with premium calculation approaches that translate territorial risk assessments derived from hydrological and hydraulic models into specific pricing structures. We then examine how recent advancements in territorial rating methodologies have enabled more granular, property-specific risk assessments that more accurately reflect the underlying physical realities modelled in previous chapters.

Premium Calculation

Calculating insurance premiums represents the practical application of risk modelling principles within the constraints of market realities, regulatory requirements, and business objectives. For property insurance, particularly in areas prone to flooding and other natural hazards, advanced premium calculation methods are essential for ensuring insurer solvency and market accessibility.

The fundamental insurance premium equation comprises several distinct components:

$$\text{Premium} = \text{Pure Premium} + \text{Expense Loading} + \text{Risk Loading} + \text{Profit Margin}$$

Where:

Pure Premium: The expected loss cost, as determined by comprehensive risk modelling

Expense Loading: Operational costs, including acquisition, administration, and claims handling

Risk Loading: Additional charge for uncertainty and volatility in the loss distribution

Profit Margin: Required return on capital to support the business

The pure premium component derives directly from the risk modelling process through the application of credibility theory, which balances the predictive value of specific risk data against broader class experience:

Formula 18: Pure Premium

$$Z = n / (n + K)$$

Where:

Z represents the credibility factor.

n is the number of exposure units.

K is a constant reflecting the expected variance of the risk class.

The final pure premium calculation then becomes:

$$PP = Z \times \text{Individual Experience} + (1 - Z) \times \text{Class Experience}$$

This approach ensures that properties with sufficient historical data receive individualised rates, while those with limited experience benefit from the stability of class ratings.

Territorial Rating Refinement

Territorial rating has long been a cornerstone variable for property insurance. It is traditionally defined by administrative building blocks such as postal codes, census blocks, or municipal boundaries.

Building on the flood modelling techniques discussed in previous chapters, modern approaches employ significantly more granular spatial segmentation.

The territorial rating process involves two critical phases:

Determining the boundaries of each territory

Establishing the rate relativities for these territories

Traditional territory-based methods faced an inherent challenge: balancing the need for territories large enough to provide statistical credibility while small enough to encompass regions with relatively uniform risk exposure. Key considerations for effective territorial boundaries include:

Credibility thresholds: Ensuring sufficient exposure and claim volume

Risk homogeneity: Minimizing variance within territories

Intuitive boundaries: Creating defensible geographic delineations

Regulatory compliance: Adhering to jurisdictional constraints

Modern flood insurers abandon the conventional zone-based rating approach in favor of coordinate-based systems that represent flood risk as a continuous gradient across landscapes.

This approach uses property-specific latitude and longitude coordinates integrated with high-resolution digital elevation models to assess flood exposure with unprecedented precision.

As Mitchell-Wallace et al. note in their practitioner's guide, this creates "continuous risk surfaces that better reflect localised risk gradients," enabling insurers to differentiate between properties even within the same postal code. As we have covered in the previous chapters, modern approaches to hydrology and hydraulic modelling enable advanced tools that allow the following:

Micro grid-based pricing: using uniform spatial grids down to 5m that transcend administrative boundaries and allow different Insurance for two properties in the same Zipcode (Postcode)

Risk-based clustering: Employing machine learning to identify regions of similar risk profiles that have more measurable historic events

Dynamic boundaries: Adjusting territory definitions based on emerging loss patterns and changing hazard conditions

Multi-level territories: Implementing hierarchical structures that balance granularity with credibility

The evolution toward these micro-territories has particular relevance for flood modelling, where risk can vary dramatically within short distances based on elevation, drainage infrastructure, and proximity to water bodies. These approaches enable the incorporation of spatial correlation in premium calculations, recognising that geographic proximity often entails similar flood risk exposures.

Others have begun modelling territorial signals directly at the individual risk level, creating a continuous risk surface rather than discrete territories. This approach involves smoothing and aggregating nearby risks until sufficient credibility is reached, creating dynamic micro-territories that adapt to emerging patterns. These methodologies typically employ holdout validation techniques to ensure model accuracy, with insurers reserving portions of their data to verify that territorial refinements genuinely have indicative power.

The advantages of these advanced geospatial approaches extend beyond technical sophistication. Insurers implementing these methods report significantly increased accuracy in risk assessment, with greater differentiation and more significant properties even within traditionally defined territories. Perhaps more importantly, these approaches enable insurers to respond to localised changes in risk levels without requiring broad rate changes across entire regions.

This granularity promotes fairness and precision in pricing, ensuring policyholders pay rates more closely aligned with their actual risk exposure. From an operational perspective, these systems typically enable incredible speed and efficiency in risk assessments and policy quotes, as territory assignment becomes an automated, algorithmic process rather than a manual lookup procedure.

Multivariate Class Rating

Modern premium calculation employs multivariate rating plans that simultaneously consider numerous risk factors through generalised linear models (GLM):

Formula 19: Multivariate Premium Calculation

$$\log(\text{Premium}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where:

$\log(\text{Premium})$ is the natural logarithm of the premium amount.

β_0 is the intercept or base rate coefficient.

$\beta_1, \beta_2, \dots, \beta_n$ are the regression coefficients that represent the effect of each risk factor.

$X_1, X_2, \dots, X_n$ are the risk factors or predictor variables (both continuous and categorical).

These models incorporate both continuous variables (e.g., distance to coast) and categorical factors (e.g., construction type) through:

Main effects: Direct impact of individual factors.

Interaction terms: Combined effects of factor pairs.

Offset variables: External constraints on the rating structure.

Smoothing functions: Techniques like cubic splines for handling nonlinear relationships.

Advanced implementations: extend GLMs through techniques such as:

Generalised additive models (GAMs) for capturing complex nonlinear relationships.

Mixed-effects models for addressing hierarchical data structures.

Elastic net regularisation for variable selection and coefficient stability.

Cost of Capital and Risk Loading

Property insurance—particularly for catastrophe-exposed regions—requires substantial capital reserves to cover potential losses. The risk-loading component of premiums must reflect this capital requirement:

$\text{Risk Loading} = \text{CoC} \times \text{Required Capital}.$

Where CoC represents the cost of capital (typically exceeding risk-free rates by 6-12 percentage points), and Required Capital is determined through:

$\text{Required Capital} = \text{TVaR}_{99.5} - \text{Expected Loss}$

Where $\text{TVaR}_{99.5}$ (Tail Value at Risk) represents the average loss in the worst 0.5% of scenarios. This approach ensures that the premium adequately compensates for the extreme tail risk characteristic of property catastrophe exposures.

Solvency II regulatory framework

The European Union formally implemented Solvency II on January 1, 2016, following nearly a decade of development and several postponements from its originally planned 2012 launch. All 27 EU member states were required to adopt this regulatory framework for insurance and reinsurance companies. While initially implementing Solvency II as an EU member, the UK has maintained these regulations post-Brexit through the UK Solvency II regime. However, some modifications have been introduced over time.

Under Solvency II, insurers must account for catastrophe risk in their Solvency Capital Requirement (SCR) using either the standard formula or internal models. The standard formula utilises predefined scenarios and correlation matrices, with catastrophe risk integrated within underwriting modules (non-life, life, and health). For non-life catastrophe risk, insurers apply 1-in-200-year loss scenarios across various perils and regions, reporting capital requirements both before and after implementing risk mitigation strategies like reinsurance. Life catastrophe risk involves more straightforward calculations, such as applying 0.15% to the total capital at risk across policies.

Alternatively, larger insurers may develop internal models that reflect their specific risk profiles more accurately. These regulator-approved models use Monte Carlo simulations to model loss distributions with company-specific exposure data and reinsurance structures. While the

standard formula offers simplicity with fixed scenarios and correlations, internal models provide greater precision but require more complex data and validation processes.

Both approaches aim to ensure insurers maintain sufficient capital reserves to withstand extreme events at the 99.5% confidence level (1-in-200-year Value at Risk), balancing policyholder protection with operational feasibility.

This regulatory charge directly influences premium pricing, as insurers must generate sufficient returns to compensate for this committed capital. Thus, the regulatory view of extreme physical risk effectively becomes a direct component of consumer premium costs.

While these advancements mentioned earlier offer improved risk segmentation—with some insurers reporting 15-25% reductions in cross-territory rate error compared to conventional methods—they also face regulatory scrutiny. Insurance commissioners increasingly demand transparency in these models, requiring detailed documentation of how geographic risk factors are isolated and validated before approving rate filings based on these new methodologies.

The ongoing evolution of flood premium modelling represents a material leap forward in risk-based pricing accuracy, potentially reducing cross-subsidization while better-aligning incentives for property-level flood mitigation investments.

Experience and Schedule Rating

Beyond model-driven base rates, premium calculation often incorporates more nuanced methodologies to account for individual risk characteristics and historical performance. Two key approaches that enable this refinement are experience rating and schedule rating.

Experience rating adjusts premiums based on a policyholder's actual loss history relative to expected losses for similar risks. This approach recognises that historical performance is a reliable predictor of future claims activity.

Calculation Methodology

The experience modification factor is typically calculated as follows:

$$\text{Experience Modification} = (\text{Actual Losses} + \text{ESF}) / (\text{Expected Losses} + \text{ESF})$$

Where:

Actual Losses: The policyholder's historical claims over a defined period (typically 3-5 years)

Expected Losses: Industry average losses for similar risk profiles

ESF (Experience Stabilization Factor): A parameter that dampens the impact of random fluctuations, crucial for smaller portfolios where a single large claim could create disproportionate premium volatility

The resulting factor directly modifies the base premium:

Factor > 1.0: Premium surcharge for worse-than-expected loss experience

Factor = 1.0: No adjustment (experience aligns with expectations)

Factor < 1.0: Premium credit for better-than-expected loss experience

Credibility Weighting

The ESF functions as a credibility mechanism that balances the statistical significance of individual experience against broader risk class data. As exposure volume increases, the formula gives greater weight to personal experience:

$$Z = n / (n + \text{ESF})$$

Z represents the credibility assigned to individual experience, and n reflects exposure units. Exposure units represent the volume of risk-bearing activity (e.g., payroll, property value, or operational scale) used to quantify an insured entity's exposure to potential losses. When calculating premiums, this parameter determines how much weight an insurer assigns to the entity's loss experience versus industry-wide data.

Example: Workers' Compensation

A construction firm with a \$5M annual payroll (n = 500 units at \$10,000/unit) and an ESF of 1,000 would have:

$$Z = 500 / (500 + 1,000) = 0.33$$

Here, 33% of the premium adjustment would derive from the firm's claims, while 67% would use industry data.

If payroll grows to \$15M (n = 1,500):

$$Z = 1,500 / (1,500 + 1,000) = 0.6$$

Now, 60% of the weight is on the firm's experience, reflecting greater statistical confidence.

Practical Application

For property insurance, particularly in flood-prone areas, experience rating might consider frequency and severity of previous flood claims, loss mitigation responses at earlier events, time elapsed since the last significant claim, and loss ratio compared to premium history.

Example: A commercial property

A commercial property with \$10M in coverage has experienced two minor flood claims totalling \$125,000 over the past decade, compared to an expected loss of \$200,000 for similar properties. With an ESF of \$75,000, their experience modification would be:

$$(\$125,000 + \$75,000) / (\$200,000 + \$75,000) = \$200,000 / \$275,000 = 0.73$$

This 0.73 modification factor would reduce their base premium by 27%, reflecting their better-than-expected loss history.

Schedule Rating

While experience rating looks backward at actual loss history, schedule rating evaluates current risk characteristics that statistical models or territorial classifications may not fully capture.

Implementation Approach

A schedule rating applies structured credits or debits to the base premium based on specific risk factors. Insurance underwriters evaluate properties against defined criteria, with typical adjustment ranges of $\pm 25\%$ from the base rate.

Key Rating Factors

Schedule rating for flood insurance typically evaluates several critical categories. Loss control measures include evaluating property-specific defences such as elevated critical systems, backflow prevention devices, and temporary flood barrier deployment capabilities. Construction features assessment focuses on the building's inherent resilience, examining foundation integrity, water-resistant materials usage, and structural reinforcements against hydrostatic pressure.

Management practices evaluation encompasses organisational readiness through emergency response planning, staff training protocols, and ground-floor exposure minimisation strategies. Maintenance programs review examines the property owner's commitment to regularly inspecting water intrusion points, maintaining drainage systems, and adhering to manufacturer schedules for flood defence systems.

Practical Application

Example: A manufacturing facility located in a moderate flood zone might receive the following schedule rating net computed as the sum of adjustments:

Table 5: Insurance Premium Adjustment

This 15% net credit would apply to the base premium (after any experience rating adjustments), reflecting specific risk mitigation efforts despite some management deficiencies.

Premium Optimization

Contemporary insurance pricing extends beyond pure risk assessment to incorporate strategic business considerations through premium optimisation techniques. This approach balances actuarial indications with market realities to achieve broader organisational objectives.

Market-Responsive Pricing

Premium optimisation leverages demand elasticity modelling to understand how price changes affect purchasing behaviour across different market segments. By analysing price sensitivity, insurers can identify segments where modest premium increases may have minimal impact on retention while applying more competitive pricing in highly elastic segments to drive growth.

The optimisation function typically takes the form:

$$\text{Maximize: } \sum_i P_i(r_i) \times (r_i - LR_i)$$

Where:

P_i represents the purchase probability at rate r_i , and LR_i the expected loss ratio.

Competitive Positioning

Advanced optimisation models incorporate competitive intelligence to position premiums strategically within the market landscape. This requires data collection on competitor pricing and regular market basket analyses to understand relative value propositions.

Insurers typically establish target competitive positions based on product features, service levels, and brand strength, then optimise premiums to maintain these relative positions while achieving profitability targets.

Customer Lifetime Value

Rather than optimising short-term profitability, contemporary approaches increasingly focus on maximising customer lifetime value (CLV).

Customer Lifetime Value

$$CLV = \sum_t (P_t \times (r_t - LR_t - E_t)) / (1 + d)^t$$

Where:

P_t (Probability of renewal): The likelihood that a customer continues their policy in period t . This reflects customer retention rates.

r_t (Premium): The amount the customer pays for insurance coverage in period t .

LR_t (Expected loss ratio): The anticipated claims costs as a proportion of premium. This represents the core insurance cost.

E_t (Expense loading): All operational costs associated with servicing the customer (administration, customer service, etc.).

$(r_t - LR_t - E_t)$: This calculates the profit margin per customer in period t .

d (Discount rate): The rate used to convert future profits into present value, accounting for the time value of money.

$(1 + d)^t$: The discount factor that decreases the value of future profits based on how far in the future they occur.

$\sum_t$: Summation over all periods of the customer relationship.

Customer Lifetime Value (CLV) represents the total financial contribution a customer is expected to make to a company throughout their entire relationship. This concept has become increasingly crucial in insurance companies shifting from short-term profitability models to longer-term value maximisation approaches.

In the insurance context, CLV considers several key elements:

- **Renewal Probability:** The likelihood that a customer will continue their policy over time, reflecting customer loyalty and satisfaction.
- **Premium Payments:** The revenue generated from the customer's insurance payments, which may change over time.
- **Expected Claims Costs:** The anticipated payouts for customer claims, which directly impact profitability.
- **Operational Expenses:** The costs associated with servicing the customer, including administration, customer service, and overhead.
- **Time Value of Money:** Future profits are discounted to present value, acknowledging that money received in the future is worth less than money received today.

This approach allows insurers to make more strategic decisions. For instance, they might offer lower initial premiums to customer segments that data suggests will develop into profitable, long-term relationships. This is especially valuable when cross-selling opportunities exist across multiple insurance products.

Insurers can build more sustainable business models that prioritise customer retention and relationship development by focusing on lifetime value rather than immediate profitability. This might mean accepting lower margins or even initial losses on new customers with the understanding that the long-term relationship will ultimately be profitable.

Constrained Optimization

Premium optimisation operates within multiple constraint systems:

Regulatory constraints on rate changes and rating factor usage.

Business constraints on growth and retention targets.

Operational constraints on implementation capabilities.

Risk constraints on portfolio diversification requirements.

optimisation models employ techniques like linear and nonlinear programming, multi-objective optimisation, and stochastic dynamic programming to navigate these complex constraint environments.

Experience and Schedule Rating

Modern premium optimisation systems integrate with experience and schedule rating methodologies to create a cohesive pricing framework:

Calibrated Elasticity Models

Advanced systems calibrate elasticity models based on historical experience rating data, recognising that price sensitivity often varies based on loss history. Policyholders with favourable experience modifications typically demonstrate higher retention rates even when facing modest premium increases, while those with poor experience may be more price-sensitive.

Optimised Schedule Rating Factors

Rather than applying uniform schedule rating approaches, optimisation techniques identify the most influential schedule rating factors for different market segments. This allows underwriters to focus on the most impactful risk characteristics during property inspections and provides more precise guidance on credit allocation priorities.

Simulation-Based Optimization

Leading insurers employ Monte Carlo simulations that model the combined impact of experience rating, schedule rating, and market factors on portfolio performance. These simulations allow for robust sensitivity analysis and scenario testing before implementing rating changes.

Regulatory Considerations

The premium calculation must operate within regulatory frameworks that vary substantially by jurisdiction. Key regulatory constraints include:

Rate adequacy requirements: Ensuring premiums are sufficient to cover expected costs.

Rate equity standards: Prohibiting unfair discrimination between similar risks.

Rate stability provisions: Limiting the magnitude of premium changes.

Filing requirements: Prior approval vs. file-and-use vs. use-and-file systems.

In many jurisdictions, insurers must demonstrate the actuarial soundness of their premium calculation methodologies through:

Statistical justification of rating factors.

Analysis of indicated vs. selected rate changes.

Compliance with specific limitations on rating variables.

Documentation of modelling methodologies.

These regulatory considerations create a complex constraint environment for premium calculation, necessitating optimisation approaches that balance actuarial indications with regulatory requirements.

Technology-Enabled Pricing Innovations

The insurance industry is undergoing a significant transformation in calculating premiums for flood risk. Technological advancements that allow for more nuanced risk assessment are driving this transformation. These innovations reshape traditional pricing, creating opportunities and challenges for insurers, policyholders, and regulators.

Real-Time Data Integration: The Dynamic Premium

Historically, flood insurance pricing relied on static risk assessments updated infrequently, sometimes only every few years. Insurers are increasingly incorporating real-time data streams that provide continuously updated risk information. Weather forecasts and alerts feed directly into pricing algorithms, allowing premiums to reflect imminent weather patterns rather than just historical averages.

Property modification data—such as newly installed flood barriers or drainage improvements—can be captured and reflected in premiums more quickly. Even occupancy and utilisation patterns are being monitored to adjust risk profiles based on whether properties are continuously occupied or seasonally vacant.

This dynamic approach offers several advantages: premiums more accurately reflect current risk conditions, policyholders can see more immediate banking benefits from risk mitigation investments, and insurers can manage their exposure more effectively during peak risk seasons. However, this approach also introduces volatility into premiums that may frustrate consumers accustomed to stable pricing. Additionally, the constant influx of data creates significant challenges for actuarial teams to validate and incorporate responsibly, potentially leading to unintended biases or errors if not properly managed.

Parametric Pricing: Objectifying Risk

Parametric components are increasingly supplementing traditional indemnity-based insurance models. These components trigger rate adjustments based on objective, measurable parameters rather than subjective assessments. Distance-based flood premium components, for example, might automatically adjust rates based on precise measurements from flood plains or water bodies rather than relying on broad zone classifications. Wind-speed triggered adjustments can modulate premiums based on recorded wind velocities in a region. Most innovatively, satellite-derived vegetation indices are being used to assess ground cover and permeability, factors that significantly impact flood risk but were previously difficult to quantify at scale.

Introducing parametric elements brings greater transparency to pricing—a clear benefit for consumer understanding and regulatory oversight. These objective measures also reduce disputes over claims and pricing decisions. However, parametric approaches may sometimes oversimplify complex risk factors and create discontinuities in pricing that seem arbitrary to consumers. For instance, properties on opposite sides of a distance threshold might see dramatically different premiums despite minimal practical difference in risk.

“Parametric insurance is the foundation for designing a Physical Risk Swap.”-David Kelly, MKM.

Machine Learning: The Algorithmic Revolution

The most profound change in premium calculation comes from machine learning applications that enhance predictive accuracy beyond what traditional actuarial methods could achieve. Gradient boosting algorithms define more granular rate classes by identifying subtle patterns in risk data that human analysts might miss. Neural networks improve claims cost prediction by recognising complex relationships between seemingly unrelated factors. Reinforcement learning algorithms optimise portfolios, balancing risk exposure across different property types and geographical areas.

These advanced techniques deliver remarkable improvements in pricing accuracy, allowing insurers to match premiums to actual risk better and expand coverage to previously uninsurable properties. They also enable more personalised pricing that rewards risk-reducing behaviours. The downside, however, is the "black box" nature of many machine learning models, which creates significant challenges in explaining rate decisions to consumers and regulators. There are also legitimate concerns about algorithmic bias potentially perpetuating or even amplifying existing inequities in insurance access and pricing

Evolving Risk Cost Assessment

The integration of evolving risk cost patterns into household insurance models reflects a growing recognition of the compounding financial pressures from natural hazards, even as hazard frequency remains relatively stable. In 2023, insured losses from natural catastrophes reached \$108 billion, as the Swiss Re Institute reported. This marked the fourth consecutive year that such losses exceeded \$100 billion globally.

A record 142 natural catastrophes were reported during the year, with severe convective storms (SCS) representing the largest share of losses at \$64 billion. These storms, which include events such as hailstorms and tornadoes, had a particularly significant impact in the United States, contributing to 85% of SCS-related losses, while Europe experienced the fastest growth in such losses.

The most significant single event was the earthquake in Turkey and Syria, which caused \$6.2 billion in insured losses. Nevertheless, most events in 2023 resulted in medium-severity losses ranging from \$1 billion to \$5 billion. The increasing frequency and severity of these occurrences align with a long-term trend of rising insured losses driven by urbanisation, economic growth, and climate change. Swiss Re estimates that annual insured losses could double within a decade due to intensifying weather hazards and greater exposure in vulnerable regions.

To address these challenges, Swiss Re emphasises the importance of mitigation and adaptation measures, such as enforcing stricter building codes, constructing flood defences, and fostering collaboration between insurers, governments, and communities to reduce risk exposure.

Exposing property developments into high-risk zones is a significant driver of increased costs. Urbanisation and housing demand have pushed construction into floodplains, wildfire-prone regions, and coastal areas, where over 700 U.S. Superfund sites alone face flood risks. Commercial real estate professionals increasingly prioritise climate resilience (46%), yet many properties in disaster-prone areas still encounter elevated insurance premiums and limited coverage options.

In the multifamily housing sector, insurers specifically cite "weather risk" and aging infrastructure as reasons for premium increases ranging from 14% to 45% annually. These significant price hikes have compelled 93% of housing providers to raise deductibles or reduce coverage merely to manage costs.

Construction Quality Concerns

Cost-cutting measures in construction significantly amplify damage severity when natural hazards occur. Issues include:

- Increased property values and development in high-risk areas.
- Higher concentration of properties.
- Rising construction and repair costs.
- Greater concentration of wealth in vulnerable regions.
- More complex building systems and materials.

For example, hailstorms in Italy (2023) caused \$5.5 billion in insured losses—a European record—partly attributed to inadequate hail-resistant building standards. Similarly, construction projects in catastrophe-exposed regions face 20–30% higher insurance premiums if they lack proper risk mitigation protocols.

Banking Sector Implications

The rising costs of natural hazard insurance directly affect mortgage affordability and credit risk. Financial institutions now confront:

- Collateral devaluation: Properties in flood zones or wildfire corridors experience value declines of 3.6–10%.

- Increased default risks: Following catastrophic events, mortgage delinquencies increase by 15–30% in households in disaster-stricken areas.
- Capital constraints: As insurers retreat from high-risk markets (Florida and Texas spring to mind), banks must either absorb more risk or reduce lending in these areas.

The Cost-Frequency Paradox

Swiss Re projects that insured losses could double within a decade, emphasising that socioeconomic factors—not just hazard frequency—drive cost escalation. This underscores the need for models that prioritise exposure growth and construction quality alongside climatic shifts.

The disproportionate rise in costs relative to event frequency demonstrates that vulnerability and exposure have become the dominant factors in determining loss potential. This requires a fundamental reassessment of how risk is priced in household insurance and mortgage underwriting.

To address these changing cost patterns, insurance models now incorporate:

- Dynamic replacement cost estimation accounting for building material inflation.
- Density-adjusted exposure models reflecting wealth concentration.
- Time-dependent vulnerability functions addressing changing building practices.
- Granular rating territories more accurately reflect local risk variations.

Cost Frequency Bayesian

These adjustments typically employ Bayesian hierarchical models that integrate historical data with socioeconomic factors:

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

Where

$p(\theta|y)$ is the posterior probability distribution - the updated probability of model parameters θ given the observed data y .

$\propto$ means "proportional to" - indicating the relationship without including the normalising constant.

$p(y|\theta)$ represents the likelihood function - the probability of observing the data y given parameters θ .

$p(\theta)$ is the prior probability distribution - the initial belief about the parameters before observing new data.

θ (theta) represents the model parameters (e.g., vulnerability factors, exposure growth rates).

y represents the observed data (e.g., historical losses, claims data)

The Bayesian approach is very generic and something anyone looking at market data timeseries would recognise. The approach to evolving insurance premiums begins with prior knowledge, where insurers establish initial risk estimates based on historical patterns and expert judgment. When new evidence emerges captured in the likelihood function from claims data, property changes, or environmental shifts), the model updates beliefs by proportionally weighting this evidence against prior assumptions to produce a posterior probability distribution.

The crucial aspect is that this weighting isn't fixed—more reliable or robust evidence receives more significant influence in reshaping premium calculations, while uncertain data has less impact. This creates a continuously refining system where each update incorporates all previously learned information, allowing premiums to evolve gradually rather than overreacting to short-term fluctuations while still remaining responsive to genuine emerging trends in risk factors.

Broad Territorial Rating

Modern insurance risk assessment has evolved significantly from historical broad territorial rating approaches, now leveraging property-specific geospatial analysis powered by advanced machine learning techniques. This transformation combines high-resolution spatial data with computational methods to create granular risk profiles.

Traditional rating systems grouped properties into large geographic zones with uniform risk assumptions. This "one-size-fits-all" approach often resulted in similar properties facing different premiums based solely on arbitrary boundaries like zip codes or counties.

Contemporary models now analyse individual properties using:

Sub-metre resolution DEMs for precise floodwater flow modelling: High-definition topographic maps capturing elevation changes with 10-30cm accuracy, enabling detailed water movement simulation and identifying micro-topography features that affect flood vulnerability.

LiDAR-derived building characteristics: Creates 3D digital representations that automatically extract critical structural features influencing wind, hail, and snow damage potential without requiring physical inspection.

Satellite-based vegetation analysis: - Monitors vegetation density, type, and moisture content; calculates defensible space metrics and fuel load profiles; and tracks seasonal changes in wildfire exposure through regular imaging.

Hydrological flow models Simulates water accumulation and movement during rainfall events, accounting for drainage capacity, soil saturation, and infrastructure while providing time-based flood progression scenarios instead of static zone designations.

The predictive engine combines ensemble machine learning methods that handle complex spatial relationships. The model combines outputs from multiple decision trees (typically 100-1000) into a single prediction.

Each decision tree processes specific spatial features like elevation gradients and building shapes. The final prediction is calculated by adding together all these individual tree outputs, with each tree's contribution weighted based on its predictive accuracy. This approach allows the system to capture complex interactions between different geographical and structural factors, creating a comprehensive risk assessment that's far more nuanced than traditional rating methods.

Formula 22: Predictive Tree

$$\hat{f}(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

Where:

$h_m(x)$ = Individual decision tree processing spatial features (e.g., elevation gradients, building shapes).

γ_m = Weighting factor optimising tree contributions.

M = Total number of trees in the ensemble.

This architecture is an advancement at modelling non-linear interactions between hundreds of geospatial variables:

- Slope derivatives from DEMs: Calculates elevation change rates across distances, identifies flow acceleration zones where water velocity increases, and detects natural drainage channels that may not appear on standard maps.
- Hydraulic connectivity indices: Measure water travel efficiency between points, identify properties vulnerable to distant flooding impacts, and account for underground drainage systems and culverts that affect water flow patterns.

- Urban heat island effects: Quantifies temperature amplification in densely built environments, correlates building density and surface materials with temperature patterns and identifies areas with increased risk of heat-related infrastructure stress based on spatial arrangement.

Behavioral Risk Components

Beyond physical hazard characteristics, effective household insurance modelling must account for human behavioural factors that influence risk outcomes:

Risk perception variables derived from demographic data.

Mitigation implementation rates by region and property type.

Claim filing propensity based on past insurance history.

These factors significantly impact loss experience beyond what physical models alone would predict, necessitating the integration of socioeconomic data into comprehensive risk models.

Insurance modelling thus represents a multidisciplinary integration of actuarial science, hazard modelling, geospatial analysis, and behavioural economics. The resulting framework provides the foundation for premium calculation, coverage design, and capital allocation across insurance markets.

Common Data Models as Drivers of Resilience and Risk Precision

Developing more granular and frequent data points plus more flood and insurance models begins the critical journey toward linking lower premiums or more effective insurance with tangible incentives to improve resilience.

While representing significant progress, the true potential lies in standardising data structures through Common Data Models (CDMs) so that all stakeholders operate from a shared definition of reality. Open source initiatives such as FINOS for banking and OASIS for insurance are ideally positioned to facilitate agreement among members on standardised property definitions, creating unified fields to drive calculations for resilience measurements.

CDMs establish the causal foundations for next-generation insurance modelling by creating directional relationships between physical attributes, hazard exposures, and loss outcomes. Unlike traditional models that rely heavily on correlative patterns in historical data, CDM-driven approaches enable a causal understanding of risk factors. This shift from correlation to causation significantly enhances premium calculations' predictive power and explainability, especially in expanding property urbanisation, where historical patterns may become less reliable indicators of future risk.

The transformative power of CDMs lies in their ability to connect disparate data sources through standardised ontologies. For flood risk specifically, CDMs can integrate building characteristics, terrain data, infrastructure performance metrics, and meteorological measurements into a unified framework that preserves the semantic relationships between these elements. This structured approach enables more causal modelling, where the impact of specific interventions (such as improved drainage systems or flood barriers) can be quantified with greater precision than previously possible.

Machine learning algorithms applied to CDM-structured data can identify subtle, causal relationships that traditional actuarial methods might miss. Unlike conventional AI applications' "black box" concerns. This combination of advanced analytics with standardised data models represents a fundamental advancement in risk assessment methodology, allowing insurers to develop more targeted and effective pricing strategies that precisely reflect individual risk profiles.

“CDM-based machine learning maintains explainability by operating on semantically consistent data structures with clearly defined relationships.”- Johnny Mattimore, MKM.

The revolutionary potential emerges when envisioning local governments maintaining detailed information on taxpayer properties, infrastructure, and terrain characteristics within CDM frameworks, enabling optimised resilience spending that directly translates to lower insurance premiums. This creates a transformative economic model where reduced insurance costs could offset increases in local taxation, leaving property owners financially advantaged and building community resilience

“CDMs facilitate real-time data integration in ways that ad hoc approaches cannot match. By establishing standardised interfaces for continuous data streams, insurers can incorporate emerging risk information without the required extensive data transformation.”- David Kelly, MKM.

This standardisation dramatically reduces the technical complexity and potential errors associated with dynamic premium adjustments, making real-time risk-based pricing more feasible and reliable.

“Perhaps most importantly, CDMs enable transparent risk pricing by clarifying the causal factors influencing premium calculations.”- David Kelly, MKM.

When all stakeholders—insurers, policyholders, regulators, and municipal governments—share the same structured understanding of risk drivers, incentives naturally align toward meaningful risk reduction. Property owners gain clear visibility into how specific resilience improvements translate to premium reductions, while insurers can more confidently offer discounts for measures with demonstrable causal impact on expected losses.

Standardised data frameworks built on CDMs allow insurers to precisely calibrate premium discounts against specific resilience improvements, creating transparent economic incentives aligned with individual and collective risk reduction. This represents the foundation for a virtuous cycle where data standardisation drives improved risk assessment, enabling more targeted resilience investments and ultimately reducing losses and insurance costs across entire communities.

As insurance models evolve from traditional actuarial approaches toward machine learning applications, the quality and structure of underlying data become increasingly critical. CDMs provide the essential foundation for this evolution, ensuring that advances in analytical techniques translate to genuine improvements in risk assessment rather than merely amplifying existing data biases or correlative patterns.

By establishing causal clarity with standardised data structures, CDMs are revolutionising property insurance by shifting their role from passive risk compensation to active risk prevention. By standardising data structures among stakeholders – banks, insurers, regulators, and risk modellers – CDMs create causal clarity that reveals root risk drivers while enabling predictive analytics at scale.

Chapter 10.1 - Flood Model Landscape

In earlier phases of modern finance, a “model” was typically a single, self-contained black box guarded by a small group of quants or traders. It often sat on a local machine or shared drive, surrounded by a narrow circle of specialists. Documentation, if it existed, was uneven. The intellectual property was perceived to reside inside the black box itself: the code, the calibration tricks, and the undocumented choices that differentiated one desk or vendor from another.

Over time, that perception became increasingly difficult to sustain. The mathematical foundations of many widely used models were normalised into industry standards and public academic material. Option pricing, term-structure modelling, credit portfolio models and, later, catastrophe and climate models were described in journals, textbooks, conference proceedings, and open teaching resources. What had once been seen as proprietary insight gradually migrated into the public domain in the form of shared methods, benchmark formulae, and reference implementations.

Flood risk provides a clear illustration of this shift. Many leading flood model vendors are effectively spin-offs from academia, built on decades of university research in hydrology, hydraulics, and geospatial analysis. Their teams include professors and researchers whose work is published and scrutinised in the open literature. Even when commercial models are implemented in proprietary code, the underlying model landscape—catchment definitions, river networks, elevation models, rainfall–runoff methods, and inundation techniques—is anchored in methods that are extensively documented and widely taught. In the case of groups such as Fathom from the University of Bristol and similar ventures elsewhere, much of the scientific scaffolding is visible in the public domain.

A similar pattern can be observed across other hazards and asset classes. Earthquake engines draw on openly available ground-motion prediction equations and fault catalogues; climate-conditioned hazard models rely on global climate model outputs and scenario frameworks published by international bodies; derivatives models employ stochastic processes and numerical schemes that have been dissected in the literature for decades. As a result, the “secret sauce” has moved away from the raw mathematics and towards data curation, computational engineering, integration, and how these components are deployed in real portfolios and contracts.

At this stage, it becomes essential to understand not only individual vendors but also the public bodies that maintain and curate key elements of the model landscape. National meteorological and hydrological services, geological surveys, standards bodies, and international organisations provide reference data, hazard catalogues, climate scenarios, and methodological guidance that underpin many commercial and open models. The modern model landscape is therefore a layered construct: commercial and in-house engines and implementations built on top of scientific methods and data that are increasingly specified, standardised, or overseen by public and quasi-public institutions.

The remainder of this chapter describes that landscape in more detail, moving from the core computational engines through the science engines they host, and then to the deployed model instances used in practice. This sets the scene for the next chapter, which looks at one specific ecosystem through which these components are assembled and distributed.

The modern model landscape can be understood as three distinct but interlocking layers: core engines, science engines, and deployed model instances. Each layer solves a different problem, and confusing them is one of the main reasons why model inventories become unmanageable.

Core engines

Core engines are the computational workhorses. They are designed to take in large volumes of data, run through many scenarios or events, apply financial logic, and return results fast enough for practical use. In their purest form, core engines are agnostic about which particular hazard, market, or risk they are representing. They provide the machinery for simulation, aggregation, discounting, and reporting, while remaining largely indifferent to whether they are processing interest-rate paths, earthquake events, or flood depths.

Technically, a core engine implements the generic parts of the model pipeline: random number generation and scenario handling, time stepping or event looping, application of financial terms, and portfolio-level aggregation. It is usually optimised for throughput and stability rather than mathematical novelty. Crucially, the same engine can host many different science engines. A single platform might run a stochastic interest-rate model in the morning and a climate-conditioned flood loss simulation in the afternoon, simply because both ultimately require drawing scenarios, applying them to exposures, and rolling up resulting cashflows or losses.

From a landscape perspective, there are relatively few core engines in any serious institution. They are expensive to build, hard to maintain, and deeply integrated into surrounding systems. As methods proliferate and model counts rise, these engines become the natural anchor points: they are where computational control and architectural discipline must live, even though they say nothing on their own about how a given peril behaves or how a particular product is structured.

Science engines

Science engines sit one layer above. They answer the question: how does the world behave, at least as far as the model is concerned. In catastrophe and climate risk, science engines describe the hazard: the frequency and severity of storms, the behaviour of rivers and floodplains, the shaking produced by earthquakes, the effect of wind on buildings. In derivatives and banking, they describe the evolution of interest rates, credit spreads, equity prices, and default processes. In all cases, they translate physical, economic, or financial theory into a form that can be executed by a core engine.

These engines are where most of the intellectual content resides. They encode the choice of stochastic processes, partial differential equations, hydrological or meteorological models,

damage functions, and calibration procedures. They make decisions about which academic results to adopt, how to adapt them to noisy data, and how to approximate phenomena that cannot be observed directly. Even here, however, the trend has been towards normalisation. Many flood models adopt similar rainfall–runoff structures and inundation solvers. Many climate-conditioned cat models draw on the same families of global climate models and scenario frameworks. Many interest-rate models rely on a small set of standard processes and numerical schemes. The differentiation lies in exactly how these ingredients are chosen, combined, and tuned.

A science engine does not, by itself, know where it is being applied or to whom. A global hurricane model, a European flood model, or a SABR volatility model can in principle be deployed in many contexts. It is only when a science engine is tied to specific geography, exposure, and financial structure that it becomes the model that end-users recognise in daily business. The science engine therefore plays a dual role: it is both a reusable library of methods and a reference object that sits behind many superficially different “models” in the inventory.

Deployed model instances

The final layer is the one that business users and regulators usually see: the deployed model instance. This is the concrete object that appears in a model inventory, has a name, and is used for pricing, risk measurement, capital, or disclosure. In practice, a deployed instance is not a new mathematical construction. It is a particular pairing of a core engine and a science engine, configured for a specific deployment context.

That deployment context has three main elements. First, there is geography: the part of the world to which the model is applied. A global flood engine becomes a European river model, a German regional model, or a Thames catchment model only once the relevant spatial domain and data sets are selected. Second, there is the asset and vulnerability specification: the portfolios, lines of business, or instruments to which the hazard is applied, together with the vulnerability or loss functions that translate hazard intensity into damage or valuation impact. Third, there is the financial vehicle: the way the resulting losses or cashflows are structured in legal and economic terms, whether as insurance treaties, catastrophe bonds, weather derivatives, capital models, or internal pricing tools.

For clarity, this book refers to such an object as a CSD model instance, highlighting its three components: Core engine, Science engine, and Deployment. Two instances that share the same core and science engines but differ in geography, asset mix, or financial structure are counted as separate CSDs, because they will behave differently in practice and may matter for different business purposes. Conversely, many CSDs in a model inventory may ultimately trace back to the same small set of engines and scientific methods, despite having different names and owners in the organisation.

Seen through this lens, the modern model landscape is no longer a flat list of unrelated black boxes. It is a layered architecture in which a handful of core engines host a library of science

engines, which in turn give rise to hundreds of CSD model instances once they are deployed into specific geographies, asset classes, and financial vehicles.

The layered view of engines and instances leads naturally to a small set of dimensions that define any deployed model. Each dimension is simple enough on its own, but the combinations grow rapidly, which explains why inventories expand from a handful of engines to hundreds of named models.

The first dimension is the core engine. This is the computational platform that runs the model: the system that generates scenarios or events, applies them to portfolios, and aggregates results. A single bank might use one core engine for derivatives pricing, another for enterprise market risk, and a third for physical risk or catastrophe losses. Each of these engines can, in principle, host many different scientific methods. For example, the same risk platform may run both a historical simulation VaR and a Monte Carlo interest-rate model simply by swapping the science layer while keeping the core engine unchanged.

The second dimension is the science engine. This describes the behaviour of the underlying risk factors. In capital markets, this might be a SABR volatility model, a Hull-White interest-rate model, or a credit portfolio model. In physical risk, it could be a global hurricane model, a Europe flood model, or an earthquake ground-motion model. These engines encapsulate the theoretical and empirical work from statistics, physics, hydrology, seismology, or financial mathematics that has been progressively normalised into industry and academic standards. A single science engine can support many applications; for instance, the same flood engine can be used for national insurance portfolios, infrastructure finance, and sovereign risk analysis.

The third dimension is geography. A model that uses the same engine and science can behave differently depending on where it is applied. A storm model over the North Atlantic does not face the same boundary conditions or exposure patterns as a storm model over the Western Pacific. Similarly, a flood engine applied to a large continental river system in Europe will see different hydrological regimes from a coastal catchment in Asia. In practice, geography may be specified at the level of region, country, river basin, or even individual city, each selection effectively creating a new deployed instance while reusing the same underlying methods.

The fourth dimension is asset type and vulnerability. Hazard alone does not create loss; it must interact with exposed assets and their susceptibility to damage or valuation impact. In physical risk, this means distinguishing between residential buildings, commercial property, critical infrastructure, energy assets, and agricultural land, each with its own vulnerability curves or damage functions. In financial risk, it may mean portfolios of mortgages, corporate loans, sovereign bonds, or derivatives positions with different sensitivities to the same shock. Changing the portfolio composition or the vulnerability set, even with identical hazard, produces a new

model instance in the sense that outputs and their relevance to capital, pricing, or disclosure decisions will be different.

The fifth dimension is the financial vehicle. The same hazard and exposure can be structured in multiple ways: as primary insurance policies, reinsurance treaties, catastrophe bonds, weather derivatives, or internal capital models. Each structure defines how losses are shared, when triggers occur, and how cashflows are shaped over time. For example, a hurricane model deployed for a CAT bond with a parametric trigger will yield different risk measures from the same model deployed for an indemnity reinsurance layer, even though the underlying hazard and exposure are identical. In the derivatives domain, a single yield-curve engine may price swaps, swaptions, caps, and structured notes, each with distinct payoff profiles and hedging implications.

A sixth, and often implicit, dimension is purpose. The same CSD configuration—core engine, science engine, geography, assets, and financial vehicle—may be used for pricing, internal risk management, regulatory capital, or public disclosure. While the mathematical machinery does not change, the standards of evidence, documentation, and acceptable approximation do. A model instance used solely for exploratory analytics does not sit in the same risk category as one used to determine regulatory capital or to support external climate risk disclosures, even if all other dimensions coincide.

Taken together, these dimensions explain why model counts grow so quickly. If an institution has five core engines and twenty science engines, that already gives one hundred engine combinations before any deployment choices are made. Adding ten meaningful geographical groupings, five distinct asset or line-of-business clusters, and four common financial structures creates thousands of potential configurations. Not all combinations are realised, but the number of feasible CSD model instances grows combinatorially as each new geography, asset class, or structure is added. What appears as “700 models” in an inventory is often the visible surface of this combinatorial landscape: a relatively small set of engines and scientific methods, recombined across many deployment contexts to meet diverse business needs.

CSD Model Instances – Core, Science, Deployment

The layered structure and dimensions described above can be condensed into a single, practical construct: the CSD model instance. In this book, a deployed model will be treated as a CSD, defined by three components: a Core engine, a Science engine, and a Deployment context. This is the level at which most institutions count “how many models they have”, even if the underlying engines and methods are shared.

The Core component specifies which engine is performing the computation. It identifies the platform that runs scenarios or events, applies financial logic, and aggregates results. Two instances that share the same core engine rely on the same simulation and calculation machinery, even if everything else differs. The Science component identifies which hazard or analytics engine is being used on top of that platform. This determines how physical, economic, or financial processes are represented: which storm model, flood model, interest-rate model, credit model, or climate scenario engine is actually driving the behaviour of the risk factors.

The Deployment component captures how that engine–science pair is applied in practice. It combines geography, asset type and vulnerability, and financial vehicle into a single notion: where in the world the model operates, which portfolios or lines of business it affects, and how the resulting losses or cashflows are structured. A global flood engine running on a generic risk platform becomes a specific CSD instance only when tied, for example, to Central European river basins, a residential mortgage portfolio, and an indemnity reinsurance programme. Change any of those elements and a new CSD exists, even though the core and science layers remain the same.

In symbolic terms, a CSD instance can be thought of as:

where Deployment is itself a tuple of geography, assets/vulnerability, and financial vehicle. This abstraction applies equally to derivatives, catastrophe risk, climate analytics, and weather: a swaption pricing configuration, a hurricane CAT bond model, and a flood capital model are all CSDs assembled from the same basic ingredients.

Seen in this way, the proliferation of models in inventories is no longer mysterious. A firm with a handful of core engines and a limited library of science engines can still end up with hundreds of CSD instances once those engines are deployed across many geographies, asset classes, and financial structures. The CSD concept therefore provides a precise language for describing what each “model” actually is in a complex landscape, while making clear how much of that landscape rests on a relatively small set of shared components.

The Public Foundations of the Model Landscape

A useful way to introduce this section is to note that the modern model landscape is not simply a private marketplace of proprietary black boxes. A great deal of the scientific content sits on top of material that is already public, standardised, or at least widely shared. Flood modelling, for example, depends on hydrology, hydraulics, terrain data, rainfall–runoff relationships, river networks, and vulnerability functions that are heavily informed by academic literature and public sector datasets. Climate modelling relies on the scenario frameworks and documentation

produced by international scientific and policy bodies, while earthquake and wind models draw on similar combinations of public science, reference data, and engineering knowledge.

That matters because it changes how the landscape should be understood. The competitive question is less often “who invented the mathematics?” and more often “who has assembled the best combination of data, calibration, implementation quality, and usability?” In practice, the science has been progressively normalised, while differentiation has shifted toward curation, packaging, software architecture, and deployment. The result is a model landscape that is increasingly legible, increasingly modular, and increasingly supported by public infrastructure rather than opaque invention.

This is the point at which Oasis enters the picture naturally. Oasis is an open-source catastrophe modelling framework built around open data standards, a shared engine, and a community of model developers, users, and institutions. Its purpose is not merely to provide software; it is to make the model landscape more interoperable by separating exposure data, results data, and model logic into standard components. That makes it a concrete example of the broader shift from isolated model black boxes to shared public infrastructure supporting many deployed model instances.

A further point to make here is that the public foundation is not only academic, but institutional. National meteorological services, geological agencies, standards bodies, and sector initiatives all contribute to the landscape by maintaining reference data, hazard catalogues, scenario sets, and model formats. Oasis fits into that ecosystem because it provides the connective tissue between these public and semi-public ingredients and the operational needs of insurers, banks, governments, and model developers.

The practical implication is that a reader should now be ready to see Oasis not as an isolated platform, but as one visible expression of a wider public model landscape. It sits at the point where public science, open standards, and commercial deployment begin to overlap, and it provides a useful transition into the dedicated Oasis chapter that follows.

Over time, the same forces that turned black-box models into structured model landscapes also created the conditions for Oasis to emerge. What began as a practical attempt to make catastrophe modelling more transparent and more usable across a wider market evolved into an open-source platform, a shared simulation engine, and a set of open data standards that could separate exposure, hazard, and results into reusable components. In that sense, Oasis did not appear as an isolated software project; it grew out of a broader recognition that the industry needed a more open and interoperable way to assemble catastrophe models from common scientific ingredients, public and private data sources, and standardised outputs. Its evolution from a not-for-profit initiative in 2012 to a mature open platform with model providers,

community governance, and standardised exposure and results formats makes it the clearest practical expression of the shift from proprietary black boxes to a shared model landscape.

THE EVOLUTION OF THE MODEL LANDSCAPE

In earlier phases of modern finance, a “model” was typically a single, self-contained black box guarded by a small group of quants or traders. It often sat on a local machine or shared drive, surrounded by a narrow circle of specialists. Documentation, if it existed, was uneven. The intellectual property was perceived to reside inside the black box itself: the code, the calibration tricks, and the undocumented choices that differentiated one desk or vendor from another.

Over time, that perception became increasingly difficult to sustain. The mathematical foundations of many widely used models were normalised into industry standards and public academic material. Option pricing, term structure modelling, credit portfolio models and, later, catastrophe and climate models were described in journals, textbooks, conference proceedings, and open teaching resources. What had once been seen as proprietary insight gradually migrated into the public domain in the form of shared methods, benchmark formulae, and reference implementations.

Flood risk provides a clear illustration of this shift. Many leading flood model vendors are effectively spin-offs from academia, built on decades of university research in hydrology, hydraulics, and geospatial analysis. Their teams include professors and researchers whose work is published and scrutinised in the open literature. Even when commercial models are implemented in proprietary code, the underlying model landscape—catchment definitions, river networks, elevation models, rainfall–runoff methods, and inundation techniques—is anchored in methods that are extensively documented and widely taught.

A similar pattern can be observed across other hazards and asset classes. Earthquake engines draw on openly available ground-motion prediction equations and fault catalogues; climate-conditioned hazard models rely on global climate model outputs and scenario frameworks published by international bodies; derivatives models employ stochastic processes and numerical schemes that have been dissected in the literature for decades. As a result, the “secret sauce” has moved away from the raw mathematics and towards data curation, computational engineering, integration, and how these components are deployed in real portfolios and contracts.

At this stage, it becomes essential to understand not only individual vendors but also the public bodies that maintain and curate key elements of the model landscape. National meteorological and hydrological services, geological surveys, standards bodies, and international organisations provide reference data, hazard catalogues, climate scenarios, and methodological guidance that underpin many commercial and open models. The modern model landscape is therefore a layered construct: commercial and in-house engines and implementations built on top of scientific methods and data that are increasingly specified, standardised, or overseen by public and quasi-public institutions.

The same forces that turned black-box models into structured model landscapes also created the conditions for Oasis to emerge. What began as a practical attempt to make catastrophe modelling more transparent and more usable across a wider market evolved into an open-source platform, a shared simulation engine, and a set of open data standards that could separate exposure, hazard, and results into reusable components. In that sense, Oasis did not appear as an isolated software project; it grew out of a broader recognition that the industry needed a more open and interoperable way to assemble catastrophe models from common scientific ingredients, public and private data sources, and standardised outputs. Its evolution from a not-for-profit initiative in 2012 to a mature open platform with model providers, community governance, and standardised exposure and results formats makes it one of the clearest practical expressions of the shift from proprietary black boxes to a shared model landscape.

Over time, that shared landscape has had consequences well beyond transparency. Open models and open frameworks have changed how the market thinks about competition, access, and price discovery. In derivatives, the availability of common mathematical conventions and open analytical tools helped move pricing away from isolated desk-specific craft and towards repeatable market infrastructure. In catastrophe and cat bond markets, the availability of open modelling frameworks has had a similar effect: it has lowered barriers to entry, increased the number of credible risk views, and made it easier for sponsors, investors, and intermediaries to compare structures and interrogate assumptions.

That shift matters especially in cat bond pricing. A market that once depended on a small number of internally held views can now be supported by multiple model views, alternative hazard assumptions, and increasingly accessible modelling infrastructure. Open models do not remove uncertainty; they make it more visible, more comparable, and more contestable. That is important because cat bond pricing sits at the junction of science, structure, and capital markets. The more transparent the modelling substrate, the easier it becomes for issuers, investors, and arrangers to form their own view of risk rather than simply accept a single house view. In that sense, the growth of open models has been transformative not only for technical modelling practice but for market development itself.

This is why the next part of the book turns to Oasis in more detail. Oasis is not just a platform; it is a practical manifestation of the broader evolution of the model landscape. It shows how shared engines, public standards, and a community of model providers can turn a once fragmented and opaque field into something more modular, more interoperable, and more accessible to the market as a whole.

Chapter 10.2 - OASIS and Parametric Insurance

OASIS LMF Overview and Core Architecture

The OASIS Loss Modelling Framework is an open-source platform for catastrophe risk modelling. Available through GitHub repositories under the BSD 3-Clause licence, OASIS LMF provides modular catastrophe modelling capabilities for risk assessment applications. This chapter provides an overview of what OASIS does and how it can be adopted to calculate flood potentials for parametric insurance at each node.

Understanding the OASIS Data Flow Architecture

The OASIS LMF Data Flow & Requirements diagram (to be included) illustrates the framework's four-stage processing architecture, built around standardised data inputs and typical modular computational components for catastrophe modelling applications.

Catastrophe insurance, by its very nature, covers extreme events and is therefore priced based on a range of potential events drawn from a large database of actual occurrences, which are essentially synthetic scenarios. The best way to think about catastrophe insurance is to imagine a Pacific island like Mauritius, whose economy relies on exporting sugar and derivatives such as rum. An insurance policy against a tornado that could damage the port, connecting roads, and communication infrastructure would demonstrate good stewardship, and such a policy would be regarded by insurance and flood modellers as a one-off project.

The aim is to move down the tail into extreme but plausible flood events, such as across a river like the Thames, while building a framework that calculates a mark-to-market value of each parametric insurance based on a combination of changes in the market's perception of long-term risk and shifts in short-term weather patterns.

At this stage, it is helpful to understand how OASIS functions. Like all models we discuss, it fortunately has three components:-

- **Input Data Components:** Four input data streams, each serving specific functions in the risk assessment process.
 - **Open Exposure Data (OED)** provides property and portfolio information, including location coordinates, asset values, and policy terms for parametric trigger zone definition.

- The Hazard Module contains event footprints and intensity measures that define how flood events manifest across geographic areas.
- Vulnerability Models establish relationships between flood intensity and expected damage ratios
- Financial Module captures policy terms, deductibles, and coverage structures that determine final payout calculations.
- Processing Engine: The central OASIS LMF Core Engine coordinates interactions between input data streams, using algorithms to combine hazard, vulnerability, and financial information.
- Monte Carlo Sampling enables stochastic simulation of thousands of potential event scenarios, providing statistical foundations for parametric trigger probability calculations.
- The Financial Calc component applies policy terms recursively through hierarchical insurance structures, accommodating the layering typical of commercial parametric products.
- Output Generation: The framework produces risk outputs designed to support various applications
 - Ground-up Loss calculations provide baseline property damage estimates before policy terms application.
 - Insured Losses reflect actual payout obligations under policy conditions.
 - Risk Outputs encompass probability distributions and statistical metrics.
 - Average Annual Loss (AAL) and Loss Exceedance Curves (LEC) provide standard industry metrics for portfolio assessment.

Core Data Requirements for Thames Implementation

The OASIS model stack is generally quite straightforward and primarily focuses on generating extreme tail events. The challenge for Thames river parametric insurance lies in its need for careful attention to data quality and format compliance. For advance work that considers asset portfolios rather than gauge readings, modified vulnerability functions that account for basis risk—the potential disconnection between gauge measurements and actual property floodins.

- **Location Data and Basis:** Every property within parametric coverage zones requires precise geocoding to enable correlation with Thames gauge measurements. The OED Location file demands mandatory fields including

LocNumber (unique property identifier), CountryCode (GBR for UK properties), LocPerilsCovered (flood peril specification), and BuildingTIV (total insured value).

- **Property Geocoding Requirements:** Thames parametric insurance requires property coordinates primarily for zone assignment rather than detailed flood risk assessment. Basic geocoding to postcode level may suffice for zone allocation, though more precise coordinates enable better basis risk quantification. Commercial geocoding services provide adequate accuracy for parametric zone assignment without requiring expensive survey-grade positioning data.
- **Elevation and Topographic Data Integration:** Elevation data primarily assesses basis risk between gauge readings and property flood likelihood rather than for detailed hydraulic modelling. Basic Digital Elevation Models from public sources provide sufficient resolution for an initial basis risk assessment. More detailed LiDAR data can improve basis risk quantification, but may not justify additional costs for gauge-centric parametric products.
- **Event Data Architecture:** Thames flood events require representation through OASIS event sets that capture relationships between gauge readings and spatial flood patterns. Each synthetic event must specify event probability, affected areaperil_id values (geographic zones), and intensity distributions representing potential flood depths across the Thames catchment.
- **Vulnerability Relationships:** While parametric insurance has a fixed payoff when a gauge reading hits a trigger, correlation matrices that capture spatial dependencies across the Thames system, where event triggers are likely to happen in clusters.

Hazard Module Implementation Challenges

The Hazard Module is OASIS LMF's most computationally intensive component, responsible for translating meteorological and hydrological inputs into spatially distributed flood risk assessments.

This module must process relationships between upstream gauge readings, downstream impacts, and property-specific flood risks across different topographic and urban environments for Thames parametric applications.

- **Footprint File Complexity:** Thames flood footprints require representation of flood mechanisms, including fluvial flooding from gauge exceedance, tidal influences in lower reaches, and urban flooding from drainage system overwhelm. Each footprint must specify event_id (unique event identifier),

areaperil_id (geographic zone affected), intensity_bin_id (discretised flood depth), and probability (likelihood of intensity). The Thames system's nature often requires millions of footprint records for coverage, creating computational and storage challenges.

- **Intensity Measure Discretisation:** OASIS employs discretised intensity bins rather than continuous distributions, requiring calibration for Thames gauges. Flood depth bins must capture thresholds for parametric triggers whilst maintaining resolution for correlation with property damage. Typical implementations employ 20-50 intensity bins, but Thames applications often require finer resolution to capture differences between minor flooding and trigger activation.
- **Spatial Correlation Matrices:** The Thames system's interconnected nature demands spatial correlation modelling where gauge readings at Reading influence flooding patterns throughout downstream reaches. OASIS handles these correlations through matrix calculations that can overwhelm available computational resources for daily operational pricing. The framework's histogram approach, whilst theoretically sound, proves computationally intensive for Thames-wide correlation analysis.
- **Temporal Correlation Considerations:** Thames flood events exhibit temporal correlation structures that affect parametric trigger timing and duration. Gauge readings may remain elevated for extended periods following peak conditions, creating opportunities for duration-based triggers. ML timeseries models can learn optimal trigger duration settings based on historical correlation between gauge readings and flood impacts.
- **Urban Drainage Interactions:** Thames parametric applications must account for urban drainage system interactions that can decouple property flooding from gauge readings. Combined sewer systems, surface water drainage, and groundwater interactions create correlation patterns that vary with rainfall intensity and infrastructure capacity. ML models can learn these interaction patterns from historical events and adjust parametric pricing accordingly.

Output Specifications and Limitations

OASIS generates output suites designed for traditional insurance applications, but parametric insurance requires adaptation of standard output formats. The framework's detailed loss distributions become a limitation when parametric products require simple trigger/no-trigger decisions based on gauge thresholds.

- **Standard Output Limitations:** Traditional OASIS outputs focus on damage ratios and monetary losses, poorly suited for parametric triggers based on physical measurements. AAL calculations assume traditional indemnity relationships that don't apply to parametric products. LEC outputs provide insight into portfolio risk but require interpretation for parametric trigger frequency analysis.
- **Parametric-Specific Requirements:** Parametric user case requires custom output processing to translate OASIS results into trigger probabilities and expected payout frequencies. This processing must account for multiple trigger conditions, duration thresholds, and geographic correlation effects that standard OASIS outputs don't directly address. The computational burden of generating parametric-specific outputs often exceeds that of the core OASIS calculations.

The conclusion here is that for very early day adoption, OASIS is an appropriate model as it has a credible track record with the insurance industry and will get a capital markets solution to first base. Still, the limitations will become apparent when contract maturity stretches beyond one year and when basis trading between index and portfolio becomes the standard approach.

Data Sourcing and Preparation Challenges

The transition from OASIS LMF's theoretical capabilities to Thames parametric insurance reveals the challenge of data acquisition, validation, and integration. A practical approach begins with basic, publicly available datasets before incorporating more detailed proprietary data sources. This staged implementation allows for system validation and performance assessment before committing to expensive data acquisition programmes.

Staged Data Implementation Strategy

- **Phase 1: Public Data Foundation** Initial Thames parametric implementation should utilise publicly available datasets to establish baseline functionality. Environment Agency gauge data provides the primary trigger mechanism through their open API. Ordnance Survey Open Data supplies basic elevation models and geographic boundaries. Historical flood outlines from government sources establish preliminary flood zones. This foundation enables initial system development and testing without substantial data acquisition costs.
- **Phase 2: Enhanced Commercial Data** Once basic functionality is established, commercial data sources can improve model accuracy. High-resolution LiDAR elevation data provides precise topographic information.

Detailed property databases enhance geocoding accuracy. Commercial weather datasets extend historical records and provide enhanced forecasting capabilities. Insurance industry loss data enables trigger correlation validation.

- **Phase 3: Real-Time Integration** A Final implementation that incorporates real-time data streams for intra-day pricing is a long way off, apart from live gauge monitoring. Incorporating short-term weather pattern changes into the pricing using Nvidia's Earth-2 at its source would be computationally intensive, and the business case would reside on having a huge number of outstanding contracts. We can, however, imagine that will be the direction of travel.

Event Set Development

Creating event sets for Thames parametric insurance demands integrating historical observations, synthetic event generation, and forward-looking projections that capture changing flood patterns. This process proves more involved than standard catastrophe modelling applications due to parametric insurance's reliance on gauge-based triggers rather than generalised damage relationships.

- **Historical Event Catalogue Construction:** Thames historical data spans varying periods across different monitoring stations, creating challenges for consistent event set development. The Environment Agency's gauge network provides records extending back 20-150 years depending on location, but data quality varies across periods and stations. Early records often lack the temporal resolution necessary for parametric trigger analysis, whilst recent data may not capture long-term extremes for pricing models.
- **Synthetic Event Generation:** OASIS parametric applications require synthetic event sets that extend beyond historical observations to capture potential future scenarios. Thames synthetic events must preserve spatial and temporal correlations between upstream and downstream stations whilst generating statistically plausible extreme events that exceed historical experience.

The generation process employs statistical techniques including copula-based dependence modelling, extreme value theory applications, and Monte Carlo simulation methods. However, these techniques often struggle with Thames-specific challenges, including tidal influences, urban drainage interactions, and infrastructure modifications that alter flood patterns over time. The resulting synthetic events may preserve statistical properties whilst failing to capture physical realities that determine actual parametric trigger behaviour.

Environment Agency Data Integration

Real-time integration with Environment Agency flood monitoring systems is a dependency for operational Thames parametric insurance. The Agency's monitoring network provides data streams, but integration challenges can impact system reliability and pricing accuracy.

- **API Integration Complexities:** The Environment Agency provides real-time flood data through multiple API endpoints that require integration procedures. The main flood-monitoring API delivers water level measurements, flood warnings, and station metadata through RESTful interfaces updated every 15 minutes during normal conditions and more frequently during flood events. We hope like the API to continue to operate during a flood event.
- **Data Quality and Validation Procedures:** Environment Agency gauge data requires validation before use in parametric pricing calculations. Raw measurements can contain errors from sensor malfunctions, maintenance activities, and extreme event impacts that damage monitoring equipment.
- **Historical Data Consistency:** Thames parametric models require consistent historical datasets that may span decades of monitoring records. However, gauge station relocations, equipment upgrades, and measurement methodology changes create discontinuities that affect model calibration. The Environment Agency maintains metadata documenting these changes, but integrating this information into consistent datasets requires manual processing that cannot be easily automated.

Parametric Contract Specification under OED

Adapting OASIS Open Exposure Data standards for parametric insurance applications requires significant extensions to standard data formats. Traditional OED structures focus on property values and damage relationships poorly suited for parametric trigger mechanisms.

- **Location File Extensions:** Parametric applications require additional location file fields that capture trigger zone assignments, gauge dependencies, and basis risk factors. Standard OED location files must be extended with ParametricZoneID (linking properties to specific gauge coverage areas), TriggerGaugeID (identifying primary gauge for trigger determination), BasisRiskFactor (quantifying correlation strength between gauge and property), and ElevationAOD (elevation above Ordnance Survey Datum for flood correlation analysis).

- **Account File Modifications:** Parametric policy terms require account file modifications that specify trigger conditions, payout structures, and coverage limitations. New fields must capture TriggerType (gauge level, duration, or combination), TriggerThreshold (specific gauge readings for activation), PayoutStructure (fixed amount or tiered based on exceedance), and AggregateLimit (maximum annual payout regardless of trigger frequency).
- **Financial Module Adaptations:** Standard OASIS financial calculations assume damage-based loss distributions poorly suited for parametric trigger mechanisms. Parametric applications require modified financial modules that evaluate trigger conditions against gauge measurements rather than calculating damage ratios from intensity distributions. This modification requires significant custom development that may not be easily portable across different OASIS implementations.

The complexity of OED extensions for parametric applications often necessitates custom data formats that compromise OASIS LMF's standardisation objectives. These customisations can create maintenance burdens and reduce interoperability with standard OASIS tools and models.

Core Engine Processing and Sensitivity Analysis

The OASIS LMF Core Engine is the computational heart of the framework, coordinating interactions between hazard, vulnerability, and financial components to generate risk assessments. For Thames parametric insurance, this engine must process large datasets whilst maintaining precision for daily pricing operations. However, the engine's sensitivity to input data variations can impact parametric pricing stability and trigger reliability.

Processing Architecture and Computational Demands

The OASIS Core Engine employs a multi-stage processing architecture designed to handle the calculations required for catastrophe risk assessment. Understanding this architecture's computational demands is necessary for operational Thames parametric implementation, as processing times and resource requirements often exceed initial estimates.

- **Data Loading and Preprocessing:** The engine begins by loading and validating all input data streams for gauge-based parametric pricing. For Thames gauge implementation, the datasets are relatively modest compared to traditional catastrophe modelling. Gauge location data includes coordinates, elevation, and catchment area definitions for approximately 50-100 Thames monitoring stations. Historical gauge readings span decades but represent simple timeseries data measured in megabytes rather than gigabytes. Preprocessing includes data consistency checks, missing value identification, and format validation for gauge

measurements. These validation steps often identify data quality issues that necessitate returning to earlier data preparation stages, creating iterative cycles that extend processing time beyond theoretical estimates.

- **Hazard-Vulnerability Convolution:** The convolution process is simplified for gauge-based parametric pricing compared to traditional applications. Rather than evaluating thousands of property locations, the system processes trigger relationships for individual gauges against synthetic event scenarios. Each gauge requires evaluation against multiple flood event scenarios, but the computational burden is limited by the number of gauges (50-100) rather than property counts.
- **Monte Carlo Simulation Processing:** OASIS employs Monte Carlo techniques to generate statistical samples from probability distributions, enabling risk assessment under uncertainty. For Thames parametric applications, this sampling must preserve spatial and temporal correlations between gauge readings whilst generating thousands of potential trigger scenarios for statistical analysis.

The sampling process requires attention to random number generation, correlation preservation, and convergence criteria that ensure stable results across multiple model runs. However, achieving adequate convergence for parametric trigger probability calculations often requires sample sizes that exceed OASIS default settings, creating computational burdens.

Sensitivity Analysis and Model Stability

OASIS LMF output stability depends on input data quality and consistency, but parametric applications often exhibit enhanced sensitivity to data variations that can impact pricing and trigger reliability. Understanding these sensitivities is necessary for operational risk management.

- **Input Data Sensitivity:** Thames parametric pricing exhibits sensitivity to gauge data quality, with minor measurement errors potentially affecting trigger probability calculations. Systematic gauge calibration errors can bias trigger frequencies throughout entire pricing models, while temporary sensor malfunctions can create anomalous results that require manual intervention to correct. Historical data quality variations across different time periods can create temporal biases in parametric pricing that may not become apparent until extreme events test trigger reliability. Missing data periods often require gap-filling techniques that introduce additional uncertainties propagating through parametric calculations in unpredictable ways.
- **Model Parameter Sensitivity:** OASIS parametric models often exhibit high sensitivity to correlation parameters that capture relationships between gauge readings and property flooding. Small changes in correlation assumptions can alter trigger probability calculations, creating pricing instability that complicates operational applications. Vulnerability function parameters relating gauge readings to trigger activation probabilities often require calibration against limited historical data, creating parameter uncertainties propagating through model calculations. Monte Carlo sampling parameters, including sample size, convergence criteria,

and random seed selection, can impact results for parametric applications that focus on tail event probabilities.

- **Processing Algorithm Sensitivity:** The OASIS Core Engine uses various numerical algorithms for probability calculations, integration, and sampling, which can be sensitive to computational precision and implementation details. Parametric applications that focus on extreme tail events may be especially vulnerable to numerical precision issues that impact trigger probability calculations.

The Bottom Line: Good Enough to Get to First Base

OASIS passes what we might call the "Ronseal Test" - it does what it says on the tin. For practitioners looking to launch Thames parametric insurance, the framework provides sufficient credibility and statistical rigour to get capital markets comfortable with flood risk pricing. While not purpose-built for parametric applications, OASIS offers the industry acceptance and regulatory familiarity necessary for market entry.

The framework's track record in catastrophe insurance creates immediate credibility with reinsurers and regulators who understand its methodology. This credibility matters more than computational elegance when launching novel insurance products that capital markets haven't yet embraced. OASIS transforms Thames flood scenarios into the standardised risk metrics that institutional investors expect, using Monte Carlo simulation techniques that have proven reliability across decades of catastrophe modelling applications.

However, catastrophe insurance typically involves one-off annual policy calculations, while capital markets demand daily mark-to-market pricing. This shift exposes computational limitations that become apparent when OASIS's correlation matrix calculations, designed for annual policy assessments, struggle with the processing demands of continuous pricing operations.

Thames parametric insurance requires precise correlation modelling between gauge triggers, yet historical flood data provides limited validation for these statistical relationships. When one gauge hits trigger levels, downstream responses follow hydrological patterns that are well-understood physically but poorly quantified statistically. These correlation assumptions drive portfolio risk calculations, creating pricing sensitivities that compound across multiple policies during major flood events.

Basis risk quantification presents another layer of complexity. The relationship between gauge readings and actual property flooding depends on topographical factors that current elevation models capture incompletely. Water absorption during heavy precipitation and flow patterns down the Thames require terrain data improvements that go beyond standard Digital Elevation Models. OASIS can incorporate these relationships through modified vulnerability functions,

but the underlying data quality limitations affect pricing accuracy in ways that become apparent only during actual flood events.

The scarcity of extreme events presents additional calibration challenges. Unlike earthquakes or hurricanes, which benefit from global datasets that enable robust tail modelling, the Thames flood history remains geographically limited. Synthetic event generation techniques address this gap through copula-based dependence modelling and applications of extreme value theory, but these methods introduce model risks that can accumulate over multi-year contracts. The statistical techniques maintain their mathematical properties while potentially overlooking physical realities that influence actual trigger behaviour during unprecedented flood scenarios.

In its current form, extending OASIS applications beyond one-year contracts exposes limitations that do not affect traditional insurance applications. Multi-year parametric contracts require stable pricing and correlation modelling, which OASIS handles sufficiently for annual assessments but struggles to sustain over longer time horizons. The framework's sensitivity to input parameter variations introduces pricing volatility that becomes problematic for longer-term capital market applications.

Yet these limitations shouldn't overshadow OASIS's fundamental value proposition for early-stage parametric insurance development. The framework successfully translates hydrological scenarios into financial risk metrics using industry-standard methodologies. One-year parametric insurance contracts can be priced consistently with their capital market equivalents within acceptable error bounds. This provides the market validation necessary to establish parametric insurance as a viable risk transfer mechanism.

While substantial, the computational demands remain manageable for initial implementations focusing on Thames gauge coverage. Processing requirements scale with gauge numbers rather than property portfolio sizes, creating more predictable resource demands than traditional catastrophe modelling applications. The framework's modular architecture allows for staged implementation that can begin with basic functionality and enhance sophistication as market volumes justify additional computational investments.

Looking forward, the industry trajectory points toward market-based calibration rather than purely historical simulation. As parametric insurance transaction volumes increase, pricing data from physical risk swaps will provide calibration foundations that historical flood records alone cannot support. This evolution will eventually reduce dependence on synthetic event generation and correlation modelling techniques that currently strain OASIS's computational capabilities.

For Thames parametric insurance launching in the current market environment, OASIS provides the necessary foundation despite its limitations. The framework offers statistical credibility, regulatory acceptance, and operational functionality sufficient to prove market demand and establish pricing relationships. These capabilities create the market entry conditions necessary for parametric insurance adoption, even while acknowledging that purpose-built platforms will likely replace adapted catastrophe modelling frameworks as the market matures.

The assessment ultimately comes down to practical market realities rather than technical perfection. OASIS enables Thames parametric insurance to reach first base by providing credible risk assessment methodologies that capital markets can understand and regulators can approve. This foundation proves sufficient for early adoption while the parametric insurance ecosystem develops the scale and sophistication necessary to justify custom modelling platforms optimised for daily pricing and real-time risk assessment applications.

Chapter 11 - Mortgage Risk Assessment

Mortgage valuation stands at the intersection of real estate appraisal and banking risk assessment. While traditional property valuation focuses on determining the market or lending value of the underlying collateral, modern mortgage valuation extends beyond this to assess the banking instrument itself. This comprehensive approach accounts for the complex interplay between borrower characteristics, property values, and macroeconomic conditions, determining the probability and impact of default events.

"Creating a reliable mark-to-market framework for mortgages is not merely an accounting exercise—it's fundamental to banking stability. A robust valuation methodology provides the transparency that allows capital to flow efficiently even during periods of stress." — William C. Dudley, Former President of the Federal Reserve Bank of New York.

Three primary banking modelling frameworks have emerged as the foundation for mortgage valuation: discounted cash flow analysis, option-adjusted spread methodology, and stochastic house price models. These approaches share common mathematical underpinnings but differ in conceptualising and quantifying risk. The discounted cash flow method projects expected payment streams adjusted for default probabilities and recovery values. Option-adjusted spread methodology treats default as an embedded option exercised by the borrower under certain conditions. Stochastic models directly simulate the evolution of key variables like house prices and interest rates to derive default probabilities under various scenarios.

Capital markets technology has recently introduced an alternative approach that draws from credit default swap (CDS) pricing methodologies. This framework centres on affordability ratios as primary drivers of default probability, establishing a direct relationship between payment burden and credit spread. By implementing piecewise functions that capture the exponential increase in default risk as affordability deteriorates, this approach offers particular advantages when modelling the impacts of physical hazards such as flooding. Incorporating insurance costs, income disruptions, and property value adjustments following such events becomes more straightforward within this framework.

A comprehensive mortgage valuation synthesises these approaches, recognising that default risk stems from two fundamental sources:

Inability to pay: income-related default. Imagine shocks to someone's income-making cash flow, such as redundancy and divorce.

Strategic default: equity-related. Imagine a mortgage holder leaving the keys on the kitchen table and leaving since the property is now uninsurable.

By modelling both the probability of default (PD) and loss-given default (LGD) components dynamically throughout the mortgage's life, these models provide a robust foundation for pricing, risk management, and capital adequacy assessment in residential mortgage lending.

Discounted Cash Flow with PD/LGD Integration

The discounted cash flow approach begins with the fundamental premise that a mortgage's value equals the present value of its expected future cash flows. However, these cash flows must be adjusted for the possibility of default and the resulting losses.

We first calculate the scheduled payment for each period according to the mortgage terms. We then adjust this expected cash flow by subtracting the expected loss from default, which equals the probability of default in that period multiplied by the loss given default and the outstanding balance.

Formula 23: Discount Cash Flow

The formula for the expected cash flow in period t becomes:

$$ECF(t) = \text{Scheduled Payment}(t) \times (1 - PD(t)) - \text{Outstanding Balance}(t) \times PD(t) \times LGD(t)$$

Where:

$ECF(t)$ is the expected cash flow in period t .

Scheduled Payment(t) is the contractual mortgage payment in period t .

$PD(t)$ is the probability of default in period t (expressed as a decimal between 0 and 1).

Outstanding Balance(t) is the remaining principal balance in period t .

$LGD(t)$ is the loss given default in period t (expressed as a decimal between 0 and 1).

These expected cash flows are then discounted using an appropriate risk-adjusted rate to arrive at the present value. The mortgage value is the sum of these discounted expected cash flows across all periods:

$$\text{Mortgage Value} = \sum ECF(t) / (1 + r)^t$$

Where:

Σ represents the summation across all time periods t .

r is the risk-adjusted discount rate (expressed as a decimal).

t is the time period (typically in months or years)

The PD component evolves dynamically as the loan ages. Initially, it reflects the borrower's credit profile and loan characteristics at origination. Over time, it adjusts based on the changing loan-to-value ratio as property values fluctuate and the loan amortises. Macroeconomic factors like unemployment and interest rate environments further modify the default probability.

The LGD component captures the severity of loss when default occurs. It starts with the expected property value at the time of default but accounts for foreclosure costs, property maintenance, and the time value of money during the recovery process. In high-value markets with quick property turnover, LGD might be relatively low. Conversely, in declining markets with extended foreclosure timelines, LGD can be substantial.

Option-Adjusted Spread (OAS) with PD/LGD Factors

The OAS approach recognises that mortgages contain embedded options—the borrower's option to prepay and, implicitly, the option to default. We simulate numerous possible future interest rates and house price scenarios to value these options properly.

We determine whether default occurs in each simulated path based on the evolving conditions. Default typically happens when the borrower crosses certain thresholds - when home equity becomes sufficiently negative (strategic default) or payments become unaffordable relative to income (payment default).

Formula 24: Mortgage PD and LGD

For path i at time t , the probability of default is modeled as a function of the simulated loan-to-value ratio and other triggers:

$$PD(i,t) = f(LTV(i,t), \text{Payment_Burden}(i,t), \text{Unemployment}(i,t))$$

Where:

$PD(i,t)$ is the probability of default for simulation path i at time t .

$f()$ represents a functional relationship (often logistic regression).

$LTV(i,t)$ is the loan-to-value ratio for path i at time t .

$\text{Payment_Burden}(i,t)$ is the ratio of mortgage payment to income for path i at time t .

$\text{Unemployment}(i,t)$ is the simulated unemployment rate for path i at time t .

Similarly, the loss severity depends on the simulated house price and market conditions:

$$LGD(i,t) = g(\text{House_Price}(i,t), \text{Market_Liquidity}(i,t), \text{Foreclosure_Timeline})$$

Where:

$LGD(i,t)$ is the loss given default for simulation path i at time t .

$g()$ represents a functional relationship.

House_Price(i,t) is the simulated house price for path i at time t .

Market_Liquidity(i,t) is a measure of real estate market liquidity for path i at time t .

Foreclosure_Timeline is the expected time to complete foreclosure proceedings.

We calculate the present value of cash flows along each path, adjusting for these path-specific default probabilities and loss severities. The mortgage value equals the average present value across all simulated paths:

$$\text{Mortgage Value} = (1/N) \times \sum_i \sum_t \text{CF}(i,t) / (1 + r(i,t))^t$$

Where:

N is the total number of simulation paths.

$\sum_i$ represents summation across all simulation paths i .

$\sum_t$ represents summation across all time periods t .

$\text{CF}(i,t)$ is the cash flow for simulation path i at time t .

$r(i,t)$ is the discount rate for simulation path i at time t

The "option-adjusted spread" is the additional yield required above risk-free rates to make the present value equal to the market price. This spread captures all risks, including default and prepayment risks and their correlation.

Stochastic House Price and Interest Rate Model

The stochastic model takes a more interest rate swaption approach, directly modelling house prices and interest rates as correlated random processes.

Formula 25: Stochastic House Price

House prices might follow geometric Brownian motion, while interest rates could follow a mean-reverting process:

$$dH/H = \mu_h dt + \sigma_h dW_h \quad dr = \alpha(\theta - r)dt + \sigma_r dW_r$$

Where

Wiener processes W_h and W_r .

correlate ρ .

Default occurs when the property value falls below a threshold relative to the loan balance or when other borrower-specific triggers are activated. This framework allows us to calculate the probability that the borrower will cross the default boundary during any given period.

The mortgage value is determined by solving a partial differential equation that accounts for the stochastic processes and boundary conditions created by default possibilities. The partial differential equation above incorporates the probability of default and the loss severity upon default, which vary with the property value at default time.

Rather than generating explicit cash flows, this approach directly models the evolving mortgage value over time. The present value emerges naturally from the solution to the PDE, which accounts for all future possibilities in a continuous-time framework.

The advantage of this approach is its ability to capture complex interactions between interest rates and house prices, especially for high LTV mortgages where default risk is susceptible to property value changes.

Synthesis of Approaches

“The most valuation models combine elements of all three approaches. They might use the stochastic model’s combined elements of all three approaches. They might use the stochastic model’s mathematical rigour to generate realistic scenarios, apply option-theoretic principles to

model borrower behaviour, and structure the results in an intuitive discounted cash flow framework for banking analysis.”-David Kelly, MKM.

The key insight is that mortgage value depends fundamentally on the timing and magnitude of cash flows, which are significantly shaped by default probabilities and loss severities. These PD and LGD factors are dynamic, responding to evolving loan characteristics, borrower circumstances, property values, and macroeconomic conditions.

Given the timing of cashflows - the interest payment, the pre-payment and the proceeds of sale given default, a Monte Carlo simulator is appropriate to value the mortgage:-

Path Generation: The model simulates hundreds of potential future economic scenarios by generating random interest rate paths, projecting house price movements, and modelling key economic indicators like unemployment rates and market liquidity conditions that affect borrower behaviour.

Default Risk Modelling: For each simulated path and period, the model calculates the probability of default by evaluating the borrower's incentives and constraints, including the current loan-to-value ratio, payment affordability relative to income, local unemployment conditions, and individual credit quality factors.

Loss Severity Assessment: When defaults occur in the simulation, the model estimates the percentage of principal that would be lost by analysing the projected property value, current market liquidity conditions, foreclosure timelines specific to the location, legal recovery processes, and any mortgage insurance protection.

Cash Flow Projection: The model calculates the expected cash flow at each point by combining scheduled payments with adjustments for both prepayment behaviour (driven by refinancing opportunities) and default outcomes (including recovery amounts), creating a probability-weighted cash flow projection for each path.

Discounted Valuation: The final mortgage valuation is derived by discounting each path's adjusted cash flows using path-specific discount rates (incorporating the option-adjusted spread),

then averaging these present values across all simulated paths to arrive at a comprehensive risk-adjusted price.

The Capital Markets Approach Using CDS

This approach borrows from credit default swap (CDS) pricing methodology, where the mortgage is valued based on a credit spread that reflects the borrower's default probability. Unlike traditional mortgage models that rely heavily on LTV ratios and house prices, this model centres on affordability as the primary driver of default risk.

The core of this model is the relationship between the affordability ratio (payment burden relative to income) and credit spread. The code implements a piecewise function where:

The credit spread increases linearly with the ratio at low affordability ratios (below 15%).

There's a steeper linear relationship in the moderate range (15-30%).

At high affordability ratios (above 30%), the credit spread grows exponentially, reflecting rapidly increasing default risk when housing costs consume too much income.

The loan pricing function then uses these credit spreads to derive hazard rates (instantaneous default probabilities), determining survival probabilities over time. The loan value is calculated as the present value of expected cash flows (weighted by survival probabilities) minus the present value of expected losses (outstanding balance minus recovery value, weighted by default probabilities).

Integration with PD/LGD Framework

This model elegantly handles both PD and LGD components:

Probability of Default (PD): The credit spread derived from the affordability ratio, is a proxy for default probability converted to a hazard rate using $h = 1 - \exp(-\text{spread}/2)$ for semi-annual periods. Survival probabilities are calculated recursively, with each period's survival probability depending on previous periods.

Loss Given Default (LGD): The model applies a haircut parameter (h) to the property value to determine recovery. LGD is calculated as $\max(0, \text{outstanding_balance} - h \cdot \text{property_value})$. This dynamically changes as the loan amortises, reflecting the evolving equity position

Advantages of Physical Risk Assessment

The key advantage of this approach, especially when considering physical risks like flooding, is that it can easily incorporate additional costs that affect affordability. When a flood event occurs, it can impact a mortgage in several ways:

Increased insurance costs directly affect the affordability ratio.

Temporary income disruption during recovery.

Property value declines due to realised flood risk.

Dynamically adjusting the insurance parameter (I) after a flood event.

Modifying the affordability ratio to reflect temporary income impacts.

Adjusting the haircut parameter (h) to reflect reduced recovery value in flood-prone areas.

The loan pricing function generates several valuable outputs for risk assessment:

Credit spreads over time.

Hazard rates (instantaneous default probabilities).

Survival probabilities.

Expected loss given default at each point in time.

Present value of the loan accounting for default risk.

The scenario analysis capability allows you to assess how different LTV ratios and income levels affect loan value, which is particularly useful for stress testing under various physical risk scenarios.

This capital markets approach offers a more direct link between affordability shocks (which can result from physical risk events) and credit quality than traditional mortgage models. Focusing on affordability rather than just collateral value better captures the dual impact of physical risks on property values and household finances.

This approach could be a valuable bridge between traditional mortgage valuation techniques and forward-looking physical risk assessment.

Chapter 12 - Credit Default Swaps

The credit default swap market represents one of the most dramatic stories in modern finance—a journey from a creative capital relief mechanism at JPMorgan to a \$62 trillion market that became central to the 2008 financial crisis, subsequently transformed through comprehensive standardisation, central clearing, and operational reforms.

Understanding the evolution of Credit Default Swaps provides essential context for Physical Risk Swaps. The CDS market's thirty-year journey—from innovative capital management tool through systemic crisis to mature, regulated infrastructure—offers a template for how novel risk transfer instruments achieve market adoption. The infrastructure developed for CDS—pricing services, legal standardisation, central clearing, and trade compression—provides the foundation upon which Physical Risk Swaps can build.

This chapter traces the complete evolution of CDS from its introduction in 1994 through the development of critical infrastructure including Markit Partners as pricing agent, ISDA's legal standardisation framework, and the implementation of central clearing and compression to reduce outstanding notional amounts and counterparty risk. The lessons learned inform the development pathway for PRS as a genuinely tradeable, capital-efficient risk transfer mechanism.

Origins and Early Innovation: 1994–2000

The modern credit default swap was invented in 1994 by a team of JPMorgan bankers led by Blythe Masters in response to a specific capital management challenge. Following the 1989 Exxon Valdez oil spill, Exxon faced potential punitive damages of \$5 billion and approached JPMorgan for a \$4.8 billion credit line. The bank faced a dilemma: maintaining the valuable client relationship required extending the credit, but Basel I capital requirements mandated that JPMorgan hold 8% of the loan amount in reserve capital, tying up approximately \$384 million in regulatory capital.

Masters and her team engineered an innovative solution: JPMorgan convinced the European Bank of Reconstruction and Development (EBRD) to assume the credit risk associated with Exxon defaulting on the loan in exchange for a premium payment. This transaction effectively transferred the credit risk off JPMorgan's balance sheet without transferring the loan itself, allowing the bank to reduce its regulatory capital requirements whilst maintaining the client relationship. This structure became known as the world's first modern credit default swap.

Strategic Motivations

The invention of CDS was driven by multiple strategic imperatives beyond the specific Exxon transaction. Banks sought to optimise regulatory capital efficiency under Basel I rules, which did not differentiate between investment-grade and speculative-grade borrowers in capital requirements. Banks wanted to maintain lending relationships with important clients without accumulating excessive concentrations of credit risk. The technology allowed banks to monetise

their credit assessment capabilities by packaging and selling credit risk whilst continuing to earn relationship fees.

The BISTRO Revolution

JPMorgan’s innovation accelerated dramatically in December 1997 with the launch of the Broad Index Secured Trust Offering (BISTRO), a \$700 million synthetic collateralised debt obligation referencing a portfolio of more than 300 corporate and public finance credits across Europe and North America. BISTRO represented a technological leap forward by bundling multiple credit exposures into tranches with different risk profiles, allowing investors to purchase specific slices of credit risk whilst JPMorgan hedged its overall portfolio exposure.

The BISTRO structure demonstrated remarkable capital efficiency: JPMorgan needed to raise only \$700 million to provide protection on a \$9.7 billion portfolio because the underlying loans were deemed sufficiently secure. Moody’s credit rating agency validated this assessment in 1997, marking the beginning of a new era in banking where institutions could systematically shift risk off their balance sheets whilst simultaneously generating income and freeing capital for additional lending. The Federal Reserve permitted JPMorgan to secure regulatory capital relief for BISTRO deals, cementing the viability of this approach.

Market Expansion and Growing Pains: 2000–2007

The CDS market exhibited exponential growth between 2001 and 2007. Outstanding notional principal grew from \$632 billion in 2001 to approximately \$20 trillion by 2006, ultimately reaching a peak of over \$62.2 trillion by the end of 2007. This dramatic expansion reflected increasing demand from multiple market participants. Banks accounted for 81% of all protection sold and 63% of protection bought in 2000, but by 2006 their market shares had fallen to 59% and 44% respectively as hedge funds, insurance companies, and other institutional investors entered the market.

Gross Notional Outstanding

Year	Gross Notional Outstanding	Annual Growth
2001	\$632 billion	—
2004	\$8.4 trillion	138%
2006	\$20 trillion	54%
2007 (Peak)	\$62.2 trillion	211%
2010	\$26.3 trillion	-58%
2019	\$7.8 trillion	-70%

2023	\$9.3 trillion	19%
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The Operational Crisis of 2004–2005

The rapid market growth created severe operational challenges that threatened market stability. By 2003, the average derivatives dealer maintained a confirmation backlog of 21 business days for credit derivatives and nine days for vanilla interest rate swaps. The situation worsened in 2004, when confirmation backlogs at some firms exceeded capital by a factor of two, creating enormous unquantified counterparty risk exposures.

The crisis stemmed from antiquated infrastructure that had not kept pace with market volumes. Multiple trading systems operated in silos, manual processes predominated, and reconciliations depended heavily on spreadsheets. The 2005 downgrades of General Motors and Ford exposed these deficiencies dramatically, as firms struggled to process the surge in CDS activity. Regulatory intervention became necessary when the UK Financial Services Authority sent letters to all supervised institutions active in credit derivatives in February 2005, expressing concern about the level of unconfirmed trades and calling for industry initiatives to address the problem.

Infrastructure Response

The industry responded with significant investments in automation and standardisation. The number of personnel involved in processing trades increased from 0.67 per trader in 2004 to 1.25 in 2006. More importantly, electronic confirmation platforms, particularly DTCC's Deriv/SERV system, achieved widespread adoption. By the end of 2006, outstanding credit derivatives confirmations at large firms had fallen from 25 business days in 2004 to just six days.

The Trade Information Warehouse (TIW), launched by DTCC in November 2006, became the centralised electronic database holding legally confirmed records for virtually all CDS contracts in the global marketplace. This golden record infrastructure would prove essential for managing the 2008 crisis and subsequent market reforms.

The Rise of Markit as Market Infrastructure

Reference Entity Database (RED)

Markit emerged as a critical infrastructure provider in the early 2000s through its Reference Entity Database (RED), which addressed the fundamental challenge of identifying and tracking credit reference entities across the fragmented CDS market. The RED platform provided legally verified reference data across credit, loan, and fixed income asset classes, establishing itself as an integral part of credit workflows.

The RED service assigned unique six-digit alphanumeric codes (RED codes) to reference entities and nine-digit RED Pair codes to entity-obligation pairings. These codes became the

market standard for confirming trades on MarkitSERV and other trade matching and clearing platforms. The DTCC mandated the use of RED index codes for all CDS index trade confirmations, cementing Markit's role as the authoritative source for reference data. Legal verification was conducted through a scrubbing process undertaken by the law firm Allen & Overy in conjunction with Markit, ensuring that the relationships between reference entities and reference obligations were legally sound.

Pricing and Index Administration

Beyond reference data, Markit established itself as the dominant pricing agent for CDS instruments. The firm's pricing data services provided independent valuations of CDS single names, indices, and tranches to support price discovery, risk management, compliance, research, and valuations. Coverage included approximately 2,200 CDS entities calculated with observable data and 1,600 evaluated entities based on bond pricing information, with full-term structure and recovery rate assumptions. Historical pricing data extended back to August 2005, providing essential time-series information for risk management and valuation.

Markit's role expanded significantly in November 2007 when it acquired the CDX and iTraxx families of credit indices. These synthetic indices, which had originated separately as TRAC-X (formed in 2003 from JPMorgan's JECI and HYDI indices and Morgan Stanley's Synthetic TRACERS) and iBoxx, became the most liquid instruments in the CDS market. As administrator and calculation agent for these indices, Markit determined index composition, managed the semi-annual roll process, and calculated index factors following credit events. The indices roll every six months in March and September, with each new series featuring updated constituents, and liquidity typically concentrating in the most recent on-the-run series.

Markit's pricing services became deeply integrated with central counterparties. LCH CDS Clear established its daily settlement price for all cleared CDS instruments using a pricing process developed in partnership with Markit, with clearing members required to contribute prices each day for positions they hold. This pricing process determined daily settlement prices used for mark-to-market valuations across the cleared CDS market.

ISDA's Legal Standardisation Framework

The 2003 Credit Derivatives Definitions

The International Swaps and Derivatives Association (ISDA) played the central role in creating the legal framework governing CDS transactions. The ISDA Master Agreement, first developed in the 1980s for various OTC derivatives, provided the foundation for CDS documentation. This master agreement established standard terms and conditions applicable to all transactions between counterparties, including payment mechanics, events of default, termination provisions, and dispute resolution procedures.

The 2003 ISDA Credit Derivatives Definitions represented a major step forward in standardisation, providing the basic framework for documenting CDS transactions and

establishing common definitions for key concepts including credit events, succession events, deliverable obligations, and settlement procedures. These definitions were designed for use in confirmations of individual CDS transactions governed by ISDA Master Agreements and became the market standard. The standardisation facilitated by the 2003 Definitions enabled the rapid market growth in the mid-2000s by reducing negotiation time and legal uncertainty.

The Big Bang Protocol: 2009

The 2008 financial crisis exposed critical deficiencies in CDS documentation and settlement procedures. The collapse of Lehman Brothers in September 2008 created enormous uncertainty about settlement procedures, with \$400 billion in CDS contracts outstanding on Lehman debt and fears that the opacity of the market could trigger cascading failures. The industry conducted an ad hoc auction on 10 October 2008, which successfully determined a recovery rate of 8.625% and reduced required cash flows from \$72 billion gross notional to \$5.2 billion net settlement, demonstrating both the value and the necessity of standardised auction procedures.

ISDA responded with the Big Bang Protocol, published on 12 March 2009, with an adherence period closing on 7 April 2009. The Protocol represented the final step in hardwiring auction settlement terms into standard CDS documentation. Key changes included:

- **Auction Settlement Hardwiring:** The Protocol incorporated auction settlement as a standard method for calculating payments following a credit event, eliminating the need for separate auction protocols for each default. The auction process created a permanent governance structure to oversee future auctions.
- **Determination Committees:** The Protocol established Credit Derivatives Determination Committees with fifteen members (typically ten dealers and five buy-side institutions) to make binding determinations on whether credit events and succession events had occurred, whether auctions should be held, and which obligations qualified as deliverable. This centralised decision-making replaced the previous bilateral determination process and provided greater certainty and transparency.
- **Fixed Coupons:** The Protocol standardised premium payments to fixed coupons of 100 basis points for investment-grade reference entities and 500 basis points for high-yield reference entities, with upfront payments to adjust for differences between the fixed coupon and market spread. This standardisation made CDS contracts more fungible and facilitated multilateral netting and compression.
- **Standardised Settlement Dates:** The Protocol established standard quarterly coupon dates on the 20th of March, June, September, and December (credit IMM dates), with semi-annual roll conventions implemented for single-name CDS every March and September.
- **Lookback Provisions:** The Protocol implemented 60-day lookback provisions for credit events and 90-day lookback provisions for succession events, meaning events could be effective even if they occurred prior to the trade date of a CDS transaction.

The Big Bang Protocol achieved extraordinarily high adherence rates, with most major market participants adhering by the 7 April deadline. The changes facilitated central clearing by standardising contract terms and settlement procedures, making CDS contracts more suitable for multilateral netting.

The Small Bang and 2014 Definitions

The Small Bang Protocol, which closed for adherence on 27 July 2009, extended similar standardisation to European sovereign and other non-North American corporate CDS contracts. The Protocol addressed specific issues that had emerged in the European sovereign debt markets and provided a second opportunity for market participants to adhere to the Big Bang Protocol.

ISDA published the 2014 Credit Derivatives Definitions in February 2014, with implementation beginning on 22 September 2014. The New Definitions constituted a major reform addressing numerous issues identified during the previous decade, particularly in the context of the Eurozone sovereign debt crisis. Key innovations included the Senior/Subordinated CDS split (insulating senior CDS contracts from subordinated CDS contracts), Asset Package Delivery provisions for sovereigns and financial institutions, Standard Reference Obligations enabling parties to trade without specifying a reference obligation, and the Universal Successor concept for corporate restructurings.

Central Clearing and Counterparty Risk Mitigation

The Case for Central Counterparties

The 2008 financial crisis revealed the enormous counterparty risk exposures embedded in the bilateral CDS market. When Lehman Brothers filed for Chapter 11 bankruptcy on 15 September 2008, market participants feared that sellers of CDS contracts would face large losses based on gross notional values, even though net positions were substantially smaller due to offsetting long and short positions. The opacity of the bilateral market meant that participants could not quickly ascertain their true exposures or the exposures of their counterparties.

In markets served by central counterparties (CCPs), the Lehman default was handled far more efficiently. LCH.Clearnet Limited's SwapClear, a CCP for OTC interest rate swaps, unwound Lehman's \$9 trillion in open positions smoothly using a centralised and tested default-management process that involved hedging exposures and auctioning the hedged portfolio to surviving participants. Adequate margin had already been collected from Lehman, so recourse to the default fund proved unnecessary. This stark contrast between cleared and uncleared markets led regulators and industry participants to prioritise central clearing for CDS.

The Mechanics of Central Clearing

A central clearing counterparty interposes itself between counterparties to contracts traded in financial markets, becoming the buyer to every seller and the seller to every buyer through legal

mechanisms such as novation or open offer. This substitution of the CCP as common counterparty enables several risk reduction mechanisms:

- **Multilateral Netting:** CCPs net offsetting positions across all clearing members, significantly reducing gross exposures. For example, if Dealer A buys \$100 million of protection from Dealer B, Dealer B buys \$100 million of protection from Dealer C, and Dealer C buys \$100 million of protection from Dealer A, all on the same reference entity, a CCP can eliminate this circular exposure entirely through multilateral netting.
- **Margin Requirements:** CCPs require initial margin (collateral posted upfront) and variation margin (daily mark-to-market settlements) from all clearing members. This collateralisation reduces the build-up of large uncollateralised exposures that characterised the bilateral market.
- **Default Fund and Loss Mutualization:** CCPs maintain guarantee funds contributed by clearing members that can be used to cover losses exceeding a defaulting member's margin deposits. This risk mutualization provides additional protection against severe defaults.
- **Standardised Risk Management:** CCPs implement consistent, conservative risk management practices across all participants, eliminating the variability in bilateral collateral agreements and reducing operational complexity.

The Establishment of CDS Clearinghouses

ICE Trust (later renamed ICE Clear Credit) was established in March 2009 as the first CCP specifically designed for CDS clearing, working with twelve prominent CDS dealers. By mid-2010, ICE Trust had cleared \$3.5 trillion notional of index-based CDS contracts. LCH.Clearnet launched LCH CDS Clear as a competing service, focusing initially on European CDS markets.

The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 mandated central clearing for certain classes of standardised swaps, including specific categories of CDS. The Commodity Futures Trading Commission issued final rules in 2012 requiring certain classes of credit default swaps and interest rate swaps to be cleared by Derivatives Clearing Organizations registered with the Commission. The first clearing mandates came into force in 2013. The mandates apply to untranchised CDS indices in both North American (CDX) and European (iTraxx) series, covering investment-grade and high-yield indices in various tenors.

By the first half of 2015, approximately 75% of average daily interest rate derivatives notional volume was cleared, and 78% of average daily CDS index notional volume was cleared. As of 2023, 58.7% of corporate single-name CDS traded notional required to be reported under SEC rules was cleared. ICE Clear Credit has maintained dominance in CDS clearing, particularly for USD-denominated products, holding 99% market share for USD-denominated single-name CDS and 96% share for USD-denominated indices. In September 2024, ICE Clear Credit processed a record of more than \$1.1 trillion USD equivalent notional amount in a single day.

Trade Compression and Notional Reduction

The Compression Imperative

The gross notional outstanding in the CDS market understates actual economic risk because many trades represent offsetting positions that generate no net exposure. When a dealer accumulates both long and short protection positions on the same reference entity as clients enter and exit positions, the bilateral gross notional grows even though the dealer's net risk remains unchanged or decreases. This excess notional creates operational risks, increases capital requirements under banking regulations, and complicates portfolio management despite not representing genuine economic exposure.

Portfolio compression is a post-trade netting technique through which market participants can modify or remove outstanding contracts and create new ones in order to reduce overall gross notional position without modifying net positions. Compression reduces counterparty risks of derivatives portfolios without changing their market risks. The process operates through multilateral cycles in which multiple participants simultaneously submit their portfolios, and optimisation algorithms identify redundant trades that can be eliminated or replaced with trades of smaller notional value.

TriOptima's triReduce Service

TriOptima emerged as the dominant provider of compression services through its triReduce platform, which offers multilateral, risk-constrained compression for both cleared and uncleared OTC derivatives. The service operates across multiple asset classes including credit derivatives (single names, indices, and tranches), interest rate swaps, FX forwards, and commodity derivatives.

The efficiency of compression depends on the network structure of trading relationships and the presence of closed chains of intermediation. In dealer-dominated markets, compression can be highly effective because dealers intermediate large volumes of offsetting trades. Research on European OTC derivatives markets found that on average, approximately 75% of market gross notional relates to excess, with even conservative multilateral compression approaches capable of eliminating up to 98% of this excess.

By 2015, TriOptima had eliminated more than \$524 trillion notional principal across multiple asset classes in cleared and uncleared environments. In January 2021 alone, \$541 billion of gross notional was eliminated through triReduce compression service—more than double the amount achieved the previous year.

Impact on Market Size

Portfolio compression and trade netting explain much of the dramatic decline in CDS gross notional outstanding from the 2007 peak of \$62.2 trillion. By mid-year 2010, outstanding gross notional had fallen to \$26.3 trillion, and by early 2012 it stood at \$25.5 trillion. The Bank for

International Settlements reported in 2018 that the market decline was initially driven by compression of contracts and reduction of exposures following the financial crisis, with continued compression and increased central clearing contributing to further reductions.

Research shows that dealers drove both the sharp pre-crisis rise in notional amounts and the compression that occurred around the crisis. The continued evolution of CDS compression and netting contributed to a reduction in gross notional from the 2008 financial crisis levels to \$7.8 trillion by mid-2019. More recently, growth has picked up again, with notional CDS outstanding increasing to \$9.3 trillion by June 2023, though this remains 85% below the 2007 peak.

The 2008 Financial Crisis: CDS in the Spotlight

AIG's Credit Default Swap Crisis

American International Group's near-collapse in September 2008 exemplified the systemic risks embedded in the CDS market. AIG Financial Products (AIGFP) had sold enormous quantities of credit protection on collateralised debt obligations containing subprime mortgages without posting initial collateral, setting aside capital reserves, or hedging its exposure. This represented a profound failure in corporate governance and risk management practices.

As housing prices declined and subprime mortgages defaulted, AIG faced mark-to-market losses on its CDS book totalling \$11.5 billion by mid-August 2008. Credit rating downgrades by S&P and Moody's on 15 September 2008 triggered ratings-based collateral calls of \$14.5 billion, with Goldman Sachs alone requesting \$5.5 billion. AIG's total collateral demands reached \$32 billion, and the company had already posted \$14.5 billion in collateral, leaving it unable to meet the additional calls.

The Federal Reserve concluded that AIG was systemically important because its failure would leave banks holding credit protection from AIG without that protection, forcing them to recognise losses and potentially triggering a cascade of failures. Additionally, AIG's \$1.6 trillion over-the-counter derivatives portfolio, of which \$400 billion was concentrated in twelve large counterparties, created spillover risks. The government initially provided an \$85 billion emergency credit facility, which ultimately expanded to more than \$182 billion in total commitments to prevent AIG's collapse.

The Lehman Brothers CDS Auction

Lehman Brothers' bankruptcy filing on 15 September 2008 created the largest CDS settlement event to that date, with approximately \$400 billion in gross notional value of CDS contracts outstanding. In the days prior to the 10 October 2008 auction, 358 parties signed up to adhere to the auction protocol, participating either directly as one of fourteen securities dealers or as customers of those dealers.

The auction process, conducted in two stages over the course of one day, successfully determined a final recovery price of 8.625% for Lehman's senior bonds. This meant that sellers of credit

protection paid buyers 91.375% of the notional amount. The \$400 billion in gross notional was reduced through multilateral netting to a net cash flow of just \$5.2 billion from CDS sellers to buyers—a more than twelve-fold reduction that demonstrated the efficiency of multilateral netting and the success of the auction mechanism.

The Lehman auction largely validated the auction process despite some operational challenges. Bond price discovery appeared relatively effective, with the bond price on the day of default (\$18.5) and the day before the auction (\$16) providing fairly accurate predictions of the final CDS auction price (\$8.625), though the bond price moved significantly during the auction period.

Systemic Lessons

The financial crisis exposed critical vulnerabilities in the CDS market that drove subsequent reforms. The lack of transparency in this large, bilateral market created enormous uncertainty about counterparty exposures and potential losses. The absence of standardised settlement procedures created operational risks and the potential for disorderly settlement. Inadequate collateralisation in bilateral relationships allowed large exposures to accumulate. The concentration of CDS exposures in a small number of large, interconnected institutions created systemic risk.

The combined market and regulatory response addressed each of these vulnerabilities through the Big Bang Protocol standardisation, mandatory central clearing, comprehensive reporting to trade repositories, margin requirements for uncleared swaps, and ongoing portfolio compression to reduce gross notional exposures.

Contemporary Market Structure

Current Market Characteristics

As of 2025, the CDS market has evolved into a mature, well-regulated, and transparent infrastructure with gross notional outstanding of approximately \$9.3–14 trillion depending on measurement methodology. Index CDS instruments consistently account for 90% or more of total market activity, with single-name CDS representing a smaller but important market segment for managing specific credit exposures. The most liquid indices include CDX North America Investment Grade, CDX North America High Yield, iTraxx Europe, iTraxx Europe Crossover, and various emerging market and sovereign indices.

Approximately 75–80% of CDS trading is now centrally cleared, with index CDS achieving even higher clearing rates. The DTCC Trade Information Warehouse maintains records for over \$10 trillion in credit derivatives with approximately 70,000 accounts representing counterparties across 95 countries. CDS Kinetics, DTCC's data service, provides weekly stock and volume reports delivering detail on global CDS contract activity.

Regulatory and Operational Maturity

The contemporary CDS market benefits from comprehensive regulatory oversight and robust operational infrastructure. Trade repositories provide regulators with visibility into aggregate exposures and market activity. Mandatory clearing requirements ensure that standardised CDS contracts benefit from multilateral netting and collateralisation at CCPs. Uncleared margin rules require collateral exchange for non-cleared derivatives, reducing counterparty risk in the bilateral market.

Lifecycle event processing through DTCC's Trade Information Warehouse automates corporate actions, succession events, and credit events, with Credit Derivatives Determination Committees providing standardised decisions on whether credit events have occurred and whether auctions should be held. The auction mechanism hardwired into CDS documentation through the Big Bang Protocol ensures orderly settlement when defaults occur. Since January 2024, the Trade Information Warehouse has processed over 36,000 trades for credit, succession, and corporate action events.

Recent Market Resilience

The market has demonstrated resilience through recent stress periods, including the COVID-19 pandemic in 2020 and the 2023 banking sector stress. During the March 2020 market turbulence, CDS index bid-ask spreads widened from approximately 1–2 basis points to around 10 basis points for the most liquid indices, but the market continued to function and liquidity recovered within months. Single-name CDS trading surged in 2023 fuelled by rising interest rates and banking sector concerns, reaching \$3.3 trillion in market activity before declining to more normalised levels.

Lessons for Physical Risk Swaps

The thirty-year history of credit default swaps offers essential lessons for the development of Physical Risk Swaps. The instrument invented by JPMorgan's team in 1994 to optimise regulatory capital evolved into a \$62 trillion market by 2007, growing far faster than the supporting operational and risk management infrastructure could accommodate. The 2008 financial crisis exposed the dangers of this imbalance, with AIG's near-failure and the Lehman Brothers default revealing how opaque, inadequately collateralised bilateral derivatives markets could threaten global financial stability.

The post-crisis transformation of CDS infrastructure demonstrates the capacity of financial markets to adapt and strengthen in response to systemic failures. Markit's reference entity database and pricing services provided the data foundation for standardised credit risk measurement. ISDA's Big Bang and Small Bang protocols, followed by the comprehensive 2014 Credit Derivatives Definitions, created standardised legal frameworks and settlement procedures. Central clearing through ICE Clear Credit and LCH CDS Clear introduced multilateral netting, margining, and default management processes that dramatically reduced counterparty risk. Portfolio compression through TriOptima eliminated over \$500 trillion in

redundant gross notional exposures. The DTCC Trade Information Warehouse became the golden record for CDS contracts, automating lifecycle events and facilitating regulatory reporting.

Physical Risk Swaps can build upon this infrastructure foundation. The following chapter demonstrates how PRS adopts identical pricing mechanics—hazard rates, survival probabilities, premium and protection legs—enabling banks to process flood risk using existing credit risk systems. The legal framework established by ISDA extends naturally to physical risk triggers. The Common Domain Model provides data standardisation. Central clearing infrastructure stands ready to accommodate new instrument types. The path from innovation to institutional adoption has been charted; PRS follows a proven trajectory.

Chapter 13 - Physical Risk Swaps

“Physical Risk Swaps represent a straightforward extension of credit risk infrastructure rather than an exotic commodity derivative. The purpose of integrating PRS into banking is to follow Credit Default Swaps completely”. David K Kelly, MKM

Introduction

In the evolving landscape of climate finance, one critical challenge has remained unaddressed: the fundamental disconnect between long-term lending and short-term insurance coverage. This mismatch creates significant exposure for banking institutions, particularly mortgage lenders, who face escalating physical risks to their collateral assets over periods far beyond standard insurance timeframes.

“Physical Risk Swaps (PRS) represent a pioneering banking innovation designed to bridge this gap. By adapting established capital markets instruments to address climate-related physical risks—particularly flooding—PRS offers a mechanism for banking institutions to effectively transfer and manage these exposures over multi-year horizons that align with their asset portfolios.” — Johnny Mattimore, MKM

The preceding chapter traced the thirty-year evolution of Credit Default Swaps from JPMorgan’s 1994 innovation through the development of comprehensive market infrastructure. That evolution—from creative capital relief mechanism to \$62 trillion market to mature, regulated infrastructure—provides the template for Physical Risk Swaps. PRS adopts identical pricing mechanics, legal frameworks, and operational infrastructure, substituting flood events for credit events whilst maintaining complete compatibility with existing banking systems.

This chapter presents PRS as a natural extension of the CDS framework. The mathematics translate directly: hazard rates become flood probabilities, survival curves become no-flood probabilities, and the premium and protection leg calculations proceed identically. Banks’ existing credit risk infrastructure—XVA desks, pricing models, regulatory capital calculations—can process PRS without modification. The path from innovation to institutional adoption has been charted; PRS follows a proven trajectory.

Inception of Physical Risk Swaps

MKM Research Labs has been working on Sustainable Finance and Physical Risk since partnering with BNP Paribas to lead the OS-Climate “Physical Risk and Resilience” project. The concept of Physical Risk Swaps emerged from this sustained engagement with climate risk challenges facing the banking system.

In November 2023, the phrase “Physical Risk Swap” was coined by the author and Johnny Mattimore in a LinkedIn post entitled “Repricing Mortgages using Flood Insurance Risk

Transfer” as a way of describing a capital markets equivalent of parametric insurance. The natural solution was to create a new swap, building on the history of risk transfer. This eureka moment brought together several foundational principles: parametric insurance as a tradeable security; analogy with CDS product design and market development; analogy with ISDA legal definition of a CDS; and connection back to the origins of tranches, mortgage-backed securities, and collateralised mortgage obligations.

In October 2024, Johnny Mattimore and the author presented “Physical Risk Swap for Flood Risk” as a concept to the FINOS New York gathering. The driving design was to create an acceptable multi-year flood risk insurance protection in capital markets. In February 2025, Johnny Mattimore and the author presented a beta version of the Physical Risk Swap CDM alongside property, mortgage, and household insurance to the FINOS CDM Working Group.

The background to the development of Physical Risk Swaps positioned the instrument in the evolution toward risk transfer mechanisms—occupying the space where sophisticated financial engineering meets climate adaptation. This positioning reflects the conviction that climate physical risk requires not merely disclosure and measurement, but active transfer through liquid market mechanisms.

The Mortgage Lender’s Dilemma

The banking system faces a critical structural challenge in managing physical risks: a fundamental disconnect between long-term lending and short-term insurance coverage. Property owners typically borrow for 10–30 years but can only secure property insurance on annual renewal terms. This temporal mismatch creates significant exposure for banking institutions, particularly mortgage lenders, who face escalating physical risks to their collateral assets over periods far beyond standard insurance timeframes.

This disconnect represents more than just an operational inconvenience—it constitutes a growing systemic risk as continued urbanisation intensifies the financial impact of physical hazards. Banking institutions are required to reconcile the long-duration nature of their asset portfolios with the inherently short-term, annually repriced nature of traditional insurance protection. Without addressing this fundamental mismatch, lenders face increasingly uncertain risk profiles that complicate capital planning, threaten portfolio stability, and potentially restrict credit availability in vulnerable regions.

The business problem that Physical Risk Swaps address encompasses three distinct but interconnected risk layers that emerge from the insurance-lending duration mismatch:

- **Property Owners:** Face significant uncertainty about future insurability and premium costs throughout their mortgage term. Annual insurance renewals create vulnerability to sudden premium increases or, more seriously, outright insurance unavailability following major events or shifts in risk assessment. This uncertainty affects property valuations and introduces potential covenant breaches if properties become uninsurable during the loan term.

- **Mortgage Lenders:** Face direct exposure when borrowers cannot secure ongoing insurance coverage. As properties lose insurance protection, lenders bear increased risk that physical events will impair collateral values without corresponding insurance recovery. This risk compounds at the portfolio level, where geographic concentrations can simultaneously create correlated exposures across multiple properties facing insurance challenges.
- **Banking System:** Faces systemic correlation risk as weather patterns shift against further urbanisation. Entire geographic regions may experience concurrent insurance market disruptions, with insurers either raising premiums to unaffordable levels or withdrawing coverage entirely from higher-risk locations.

“This portfolio-wide concentration risk threatens synchronised collateral devaluation across significant portions of mortgage books.” — David Kelly, MKM

These risks are embedded in existing loan portfolios but remain largely unquantified in traditional risk management systems. Banking institutions must integrate physical risk assessment into their core banking architecture, measuring, monitoring, and managing these exposures with the same rigour applied to market or credit risk. They need mechanisms to translate specialised physical risk data (like flood scores from proprietary vendor models) into quantitative valuations and risk metrics for portfolio management.

For many lenders, flood risk represents the dominant physical hazard threatening collateral impairment. Yet they lack both methodologies to translate flood risk scores into banking terms and established market mechanisms for transferring this risk. This capability gap prevents active management of embedded physical risks and complicates capital allocation to climate-vulnerable regions, potentially accelerating regional economic disparities through credit availability constraints.

Parametric Insurance Foundations

The architecture of Physical Risk Swaps draws substantial inspiration from parametric insurance contracts rather than traditional catastrophe insurance. Parametric insurance triggers payouts based on objective measurements from specific instruments—such as a water level gauge on the wall of a building—without requiring an assessment of actual damages. These contracts establish direct relationships between physical measurements and banking settlements, creating transparent and rapid trigger mechanisms.

A PRS extends this concept by incorporating multiple gauges throughout a catchment area, typically managed and maintained by the relevant environmental authority, ensuring data integrity and operational reliability. This multi-gauge approach creates a comprehensive risk transfer mechanism covering broad geographic areas whilst maintaining objective, verifiable, and legally enforceable contract triggers.

By establishing tiered trigger levels corresponding to different water depths, parametric structures create granular risk segmentation that allows precise pricing and targeted protection. These established insurance constructs provide proven mechanisms for quantifying and

transferring specific segments of the risk distribution without the complications of loss adjustment processes, offering valuable precedent for capital markets approaches that require rapid and unambiguous financial settlement mechanisms. Their performance during flood events demonstrates how objective physical measurements can effectively drive banking settlements in complex environmental scenarios.

The core design of a PRS is a natural evolution of a parametric insurance contract. The only real difference is the legal wrapper, which is governed by two different regulatory frameworks. The cashflows, however, are identical. Unlike other innovations in finance, a PRS is incredibly easy to explain. Its simplicity and intuitive nature make it an elegant solution that will drive uptake.

From CDS to PRS: The Architectural Parallel

Credit Default Swaps established a well-understood pricing framework that Physical Risk Swaps adopt directly. The transition from CDS to PRS pricing requires examining the fundamental components of CDS valuation and demonstrating their direct application to physical risk.

CDS Pricing Mechanics

In a standard CDS contract, pricing involves calculating the present value of two legs: the premium leg (fixed periodic payments made by the protection buyer) and the protection leg (contingent payment made by the protection seller upon a credit event). The value of a CDS is established when the present values of these two legs are equal:

$$\text{PV Premium Leg} = \text{PV Protection Leg}$$

The premium leg payments are weighted by the probability that the reference entity survives until each payment date, whilst the protection leg is weighted by the likelihood of default. This creates a mathematical relationship:

$$\sum S \times \Delta t_i \times DF_i \times PS_i = (1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i$$

Where:

- S is the CDS spread
- Δt_i is the time interval
- DF_i is the discount factor
- PS_i is the probability of survival to time I
- R is the recovery rate.

The par spread represents the equilibrium rate where the present value of protection payments equals the present value of potential default payouts, ensuring the CDS contract has zero net value at inception.

PRS Pricing Mechanics

Physical Risk Swaps adopt this framework precisely, substituting physical events for credit events. The critical adaptation involves replacing default probability curves with physical hazard curves. For a flood PRS, the hazard curve represents the probability of a flood event of a certain magnitude occurring within a specific time frame, where meteorological data expressing “i-in-X-year” events are converted to annual exceedance probabilities.

The hazard rate $h(t)$ represents the conditional probability of the physical event occurring in a small interval Δt , given that it has not occurred before time t :

$$h(t) = \lim_{\Delta t \rightarrow 0} P(t < T \leq t + \Delta t | T > t) / \Delta t$$

For PRS, the survival probability represents the probability that the trigger event (e.g., flood level exceeding the threshold) has not occurred by time t :

$$PS(t) = e^{-\int_0^t h(s) ds}$$

The protection leg is calculated using:

$$PV \text{ Protection Leg} = (1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i$$

Where:

R represents the “recovery value” in the physical risk context—the residual value after the physical event occurs.

The spread for a PRS represents the cost of protection and is calculated identically to a CDS spread:

$$S = [(1 - R) \times \sum (PS_{i-1} - PS_i) \times DF_i] / [\sum \Delta t_i \times DF_i \times PS_i]$$

According to market conventions for standardised contracts, this spread is converted to an upfront payment plus a fixed coupon, mirroring the Big Bang Protocol conventions established for CDS.

Service Lines: CDS to PRS Mapping

The commercial opportunity for PRS mirrors that proven for CDS. The service lines that emerged around credit derivatives—from IHS Markit/S&P—translate directly to physical risk:

Service Category	CDS (IHS Markit/S&P)	PRS (MKM)
Elementary Services	CDS Pricing Data	PRS Pricing Data (Gauge/Bench)
	Reference Entity Data (RED)	Physical Asset Entity including C
	Credit Indices Administration	PR Indices & Migration Matrices
Intermediate Services	Sector & Rating Curves	Regional Catchment (Polygon) I
	Enhanced Sector/Rating Curves	Enhanced Regional Catchment C
	Liquidity & Market Metrics	Liquidity & Market Metrics
	Credit Event/Corporate Action	PR Event/Adaptation Services
	Historical Archives & Surveillan	Historical Archives & Surveillan
Advanced Services	Loan & Securitised Credit Data	Structured Credit PR Enhancem
	Entity Matching & Cross-Refer	Gauge Matching & Cross-Refer
	Portfolio Valuations	Portfolio Valuations
	Trade Processing (MarkitSERV	Trade Processing & Lifecycle

	Research Signals & Factor Libr	Research Signals & Factor Libr
	Custom Curves & Bespoke Proc	Custom Curves & Bespoke Proc
	Regulatory & Compliance Supp	Regulatory & Compliance Supp
Professional Services	Professional Services & Advisory	Professional Services & Advisory
	Platform & Data Integration	Platform & Data Integration

This parallel demonstrates that PRS maps identically to CDS for market instrument standards (trading OTC to documentation via ISDA), market infrastructure (give-up, clearing, compression), and bank risk architecture (models, systems, data, capital calculations and relief). The most significant commercial opportunity lies in the lifecycle of indices, precisely as proven for CDS.

From Return Periods to Hazard Curves

Traditional flood risk assessment employs return periods: a “1-in-100-year flood” indicates an event with 1% annual exceedance probability. Whilst intuitive for communication, return periods require translation into the continuous hazard rate framework that credit risk systems expect.

The Fundamental Relationship

The mathematical relationship between return period and exceedance probability provides the starting point. For a return period τ (in years), the probability of at least one exceedance event in time period t follows:

$$P(X \geq 1) = 1 - e^{-(t/\tau)}$$

For annual assessment ($t = 1$ year), a 100-year return period yields approximately 1% annual exceedance probability. For a trigger level with annual exceedance probability p , the constant hazard rate λ satisfies:

$$1 - e^{(-\lambda)} = p$$

Solving yields $\lambda = -\ln(1-p)$. For small probabilities ($p < 10\%$), the approximation $\lambda \approx p$ holds with minimal error, enabling straightforward translation from return period data to hazard rate parameters.

Gauge-Specific Hazard Curve Construction

Each monitoring gauge possesses unique flood characteristics captured through historical data and climate projections. A gauge might exhibit the following return period profile:

Return Period	Flood Level	Annual Exceedance Probability
10 years	3.5 metres	10.0%
25 years	4.0 metres	4.0%
50 years	4.2 metres	2.0%
100 years	4.8 metres	1.0%
200 years	5.2 metres	0.5%
500 years	5.5 metres	0.2%

These discrete data points fit a continuous curve representing the relationship between flood levels and exceedance probabilities. The trigger level specified in the PRS contract determines which point on this curve applies. A contract triggered at 4.8 metres references the 1% annual exceedance probability; a contract triggered at 5.5 metres references the 0.2% annual exceedance probability.

The probability space for pricing is built by converting traditional flood return periods into annual exceedance probabilities, developing a continuous hazard rate function based on these probabilities, and incorporating weather projections to adjust forward-looking probabilities. Each monitoring gauge has its hazard curve based on historical data and climate projections.

Event Definition and Triggering Mechanism

Credit Default Swaps and Physical Risk Swaps share structural similarities but differ fundamentally in their triggering mechanisms and applications. Both instruments allow for customisable risk transfer, but PRS leverage measurable physical parameters—such as flood depths—to define payout conditions, offering granular flexibility akin to credit risk tranching in CDS markets.

CDS Triggering

CDS contracts are structured to cover credit events across a spectrum of credit grades (AA to B), with payouts triggered by predefined conditions like bankruptcy, failure to pay, or restructuring. Key features include dynamic pricing (CDS spreads adjust intraday based on market perceptions of creditworthiness, enabling trading at intermediate risk levels before a default event) and recovery rates (post-default payouts account for the residual value of debt, typically determined via cash settlement).

PRS Triggering

PRS contracts can define payout triggers for specific flood depths (e.g., 0.5m–1.0m) or catastrophic levels (e.g., >2.0m), analogous to tranching CDS. Crucially, flood events are determined by remote sensors or gauge data, eliminating subjective judgements required in credit events like restructuring. This objectivity represents a significant advantage: whilst CDS requires Determination Committees to adjudicate whether restructuring constitutes a credit event, PRS triggers are binary and verifiable—the gauge either exceeded the threshold or it did not.

A flood PRS typically references specific water level gauges in a catchment area. Many contracts reference multiple gauges, requiring correlation modelling between different gauge locations, settlement conditions based on one or more gauges exceeding thresholds, and weighted payout structures based on the severity and location of flooding. The correlation modelling typically employs copula functions to represent joint probability distributions between gauges, accounting for the fact that flood events often affect multiple locations simultaneously.

Product Tiers: Index, Portfolio, and PRO

The PRS platform offers three product tiers addressing different risk management needs and sophistication levels, mirroring the evolution of CDS from single-name contracts to index products to tranching structures.

Index PRS

The entry-level product provides liquid, standardised protection. Single-gauge triggers (e.g., Thames at Teddington at 5.5m) produce all-or-nothing payouts. At 55bp baseline pricing for a three-year term, Index PRS offers the lowest cost and simplest structure. One-day execution makes this suitable for rapid hedging requirements. The trade-off is basis risk: portfolio-specific characteristics remain unhedged.

Characteristic	Index PRS
Trigger	Single gauge (Thames @ Teddington 5.5m)
Payout	All-or-nothing (£500M or £0)
Price	55bp (baseline)

Execution	1 day
Pros	Liquid market, simple, easy to understand
Cons	High basis risk, binary payout, no customisation
Typical Buyer	Retail investors, quick hedges

Portfolio PRS

The institutional tier provides customised protection matched to specific portfolio characteristics. Multi-gauge weighted triggers (e.g., 60% Thames, 40% Richmond) combined with asset location data eliminate basis risk. At 63bp (8bp above index), the additional cost buys precision. Execution requires two to three weeks for customisation and documentation. Institutional funds and REIT portfolios with defined physical risk exposure benefit most from this tier.

Characteristic	Portfolio PRS
Trigger	Multi-gauge weighted (60/40 Thames/Richmond) + asset locations
Payout	All-or-nothing but portfolio-specific
Price	63bp (+8bp basis vs index)
Execution	2-3 weeks
Pros	Zero basis risk, customised to assets
Cons	More expensive, less liquid, longer execution
Typical Buyer	Institutional funds, REIT portfolios

PRO Tranches (Portfolio Risk Obligation)

The sophisticated tier introduces CDO-style structuring for flood risk. Rather than all-or-nothing payouts, PRO products split exposure into tranches with graduated triggers. This structure enables risk layering: sophisticated hedgers can purchase only the tranches matching their risk appetite, whilst yield-seeking investors can access the equity tranche's attractive returns.

Tranche	Flood Depth	Notional	Yield	Risk Profile
Equity	0-0.2m	£50M	180bp	First loss, high yield
Mezzanine	0.2-1.0m	£150M	85bp	Subordinated
Senior	1.0m+	£300M	35bp	Protected, low yield

The blended cost of 64.5bp across all tranches provides optionality at minimal premium to Portfolio PRS. Each tranche absorbs losses in order: equity absorbs first (0-0.2m flood depth), mezzanine absorbs next (0.2-1.0m), and senior absorbs only catastrophic losses (>1.0m). This mirrors CDO waterfall structures familiar from credit markets.

Portfolio PRS Basis Decomposition

The Portfolio PRS pricing differs from Index PRS due to “basis”—the additional spread reflecting the distance from assets to reference gauge, elevation differences (topography), hydrological correlations (water flow patterns), and model uncertainty and composition effects. Basis is the key innovation that enables precise, customised flood hedging.

Component	Impact	Typical Range	Example (Thames)
Base Index PRS	Baseline	N/A	55bp (3Y)
Model Uncertainty	Spread on model inputs	±2bp	+2bp
Distance to Gauge	Farther = higher risk	+1bp to +6bp	+3bp
Terrain Variation	Flood plain exposure	-1bp to +3bp	+2bp
Elevation Benefit	Higher assets safer	-2bp to 0bp	-1bp
Asset Composition	Diversity effects	-1bp to +2bp	+2bp
Total Basis	Sum of components	±3bp to ±8bp	+8bp
Portfolio PRS	Final price	Index ± Basis	63bp

The basis waterfall provides transparency: every basis point differential receives explicit justification through quantified risk factors. This transparency distinguishes the platform from opaque insurance pricing and supports informed hedging decisions.

The Basis Risk Decision

Consider the Richmond REIT portfolio: £500M across 120 assets. Index PRS at 55bp provides protection triggered by the Thames at Teddington reaching 5.5 metres. However, portfolio assets average 2.3km from the gauge with +15m average elevation. The index hedge covers only 64% of actual portfolio risk (beta 0.64), leaving scenarios where assets flood but the gauge trigger is not reached—basis losses that could reach several million pounds.

Portfolio PRS at 63bp eliminates this basis risk by triggering on actual portfolio flood impact rather than index gauge levels. The additional 8bp annual cost translates to £1.2M over a three-year term for a £500M portfolio—well justified against potential multi-million basis losses. The extra £1.2M premium is well worth eliminating multi-million basis risk exposure. The REIT avoids scenarios where assets flood but the gauge trigger is not hit, leaving them unhedged.

Implementation Framework

The journey from theoretical pricing models to a functioning Physical Risk Swap market requires careful orchestration across regulatory, technological, and banking dimensions. Rather than developing entirely new market infrastructure, the PRS ecosystem leverages existing frameworks whilst introducing targeted innovations to address the unique characteristics of physical risk.

The transformative innovation of PRS lies in extending established risk transfer concepts into banking and capital markets infrastructure. Whilst insurance mechanisms have long addressed physical risks, they remain largely separate from banking capital frameworks and lack the standardisation, liquidity, and regulatory integration necessary for effective banking risk management. PRS bridge this gap by aligning with existing data sets from environmental monitoring and catastrophe modelling whilst structuring the banking instrument according to ISDA standards—the same legal and operational framework that underpins hundreds of trillions in interest rate and credit derivatives.

“Most significantly, PRS development includes regulatory capital methodologies aligned with Basel standards, enabling banks to receive appropriate capital relief for transferred physical risks and creating powerful incentives for market adoption.” — David Kelly, MKM

Leveraging Existing Monitoring Infrastructure

Implementing PRS benefits from substantial existing infrastructure that can be leveraged without requiring extensive new monitoring systems. For example, the Thames River Basin in the UK represents an ideal development environment with its highly populated, economically critical areas and persistent flood risks affecting assets from residential to industrial properties.

The UK already maintains a robust flood monitoring system managed by government agencies. A network of monitoring stations along the Thames River provides real-time water level data at regular intervals. This existing infrastructure offers a solid foundation for flood risk assessment models and creates reliable triggering mechanisms for banking contracts. Rather than building

new monitoring capabilities, PRS implementation can focus on integrating and standardising these data streams for banking applications.

Strategic Partnerships

Strategic partnerships with leading experts in flood modelling allow the adaptation of terrain mapping, flood simulators, and event set development to meet the specific needs of mortgage lending risk assessment. These collaborations enable access to decades of accumulated expertise in hydrological modelling without requiring banking institutions to develop specialised climate science capabilities internally.

The computational demands of PRS valuation require partnerships providing access to high-performance computing capabilities. Whilst gauge-level monitoring data is available in real-time, the complex task of portfolio revaluation requires substantial computing resources, particularly for stress testing across multiple climate scenarios. Cloud-based computing platforms with dedicated physical risk modelling capabilities allow market participants to access analytics without individual infrastructure investments.

For effective implementation, governance frameworks must include cross-disciplinary representation where meteorologists, hydrologists, atmospheric and weather scientists work alongside derivatives structurers and trading strategists to refine risk assessment methodologies. Insurance underwriters with decades of catastrophe experience advise on threshold calibration and recovery value estimation. This cross-disciplinary integration accelerates product development whilst strengthening model validation through diverse perspectives.

ISDA Master Agreement and Market Practices

The cornerstone of derivatives market standardisation for decades—the ISDA Master Agreement—provides the contractual foundation for Physical Risk Swaps. This established legal framework reduces documentation uncertainty and creates immediate familiarity for market participants who have executed trillions in interest rate and credit derivative trades under the same structure. The agreement’s modular design accommodates new definitions and supplements for physical risk events whilst maintaining consistency with existing market conventions for trade confirmation, settlement, and dispute resolution.

FINOS Common Domain Model

This legal standardisation is complemented by the FINOS Common Domain Model (CDM), which offers critical data standardisation across the trade lifecycle. The CDM creates a unified representation of trade events, product definitions, and process models, enabling seamless interoperability between counterparties, clearing houses, and regulatory reporting systems.

By incorporating physical risk parameters and gauge-level monitoring data into this existing framework, PRS transactions benefit from day-one compatibility with established market infrastructure. The CDM implementation ensures that gauge readings, threshold triggers, and

settlement calculations maintain consistency across diverse market participants and technology platforms.

Digital Regulatory Reporting

Most significant for market adoption, Physical Risk Swaps integrate directly with the FINOS Digital Regulatory Reporting (DRR) framework from inception. This pre-established compliance approach eliminates the traditional regulatory uncertainty often accompanying banking innovation. Because PRS instruments are designed to be DRR-compliant from their first transaction, they avoid the regulatory ambiguity and implementation lags that have historically slowed the adoption of new derivative products. Banks can confidently deploy these instruments with clear regulatory treatment, focusing implementation efforts on risk modelling rather than compliance infrastructure.

Developing Market Liquidity

Developing a liquid market for Physical Risk Swaps represents a critical innovation in risk management for physical assets. Following the successful evolution of the Credit Default Swap market, we can establish a roadmap for creating a robust, efficient marketplace for transferring and managing physical risk exposures.

Learning from the CDS Market Evolution

The Credit Default Swap market demonstrated how standardisation, institutional participation, and regulatory frameworks can create a functioning risk transfer mechanism. From its inception in the early 1990s to its peak of \$62 trillion in notional value by 2007, the CDS market grew on the back of infrastructure that PRS can now inherit. Whilst the market subsequently contracted to approximately \$9 trillion, the core infrastructure and liquidity mechanisms remained intact, particularly for index products.

The core design of a PRS is a natural evolution of a parametric insurance contract. The only real difference is the legal wrapper, which is governed by two different regulatory frameworks. The cashflows, however, are identical. Unlike other innovations in finance, a PRS is incredibly easy to explain. Its simplicity and intuitive nature make it an elegant solution that will drive uptake.

Liquidity Development Trajectory

Based on the CDS market experience, we can anticipate the following liquidity development pattern for PRS:

Initial Phase: Concentrated bilateral trading among major dealers using standardised documentation.

Index Development: Creation of diversified indices covering major risk categories and geographies (Thames, Mississippi, Rhine, Han River).

Buy-Side Adoption: Gradual expansion to asset managers, pension funds, and other institutional investors.

Market Bifurcation: Emergence of highly liquid index products alongside more specialised single-name PRS.

Electronic Execution: Migration from voice trading to electronic platforms as volume increases.

Market Maturity: Development of secondary market trading, curve trading, and basis trading strategies.

Central Clearing Advantages

Early central clearing provides significant advantages for market development, as demonstrated by the CDS market's post-crisis evolution. Central clearing delivers counterparty risk reduction through a central counterparty, netting efficiency improving capital efficiency through multilateral netting, standardised margining creating predictable and efficient collateral requirements, and position portability facilitating the transfer of positions between counterparties.

The involvement of major banking institutions is critical for initial liquidity, requiring a coordinated launch and ongoing commitment to make markets by major banks that create an immediate two-way market.

Market Impact: Beyond Risk Transfer

Whilst mortgage lenders are the primary beneficiaries, Physical Risk Swaps have broader implications for property markets, climate resilience, and banking stability.

The introduction of PRS creates significant positive externalities beyond direct risk transfer benefits. By reducing long-term risk exposure, lenders may be more willing to maintain mortgage availability in flood-prone areas, improving property market liquidity in regions that might otherwise face credit contraction. This sustained credit provision supports property values and enables ongoing investment in areas requiring adaptation rather than abandonment.

PRS pricing provides valuable market-based information about perceived flood risks, creating transparent signals that inform urban planning, infrastructure investment, and adaptation priorities. Regional pricing differentials will highlight areas where flood mitigation infrastructure could deliver the most significant economic benefits as the market develops, potentially influencing public investment decisions and encouraging preventative measures rather than post-disaster recovery.

The availability of PRS may incentivise community-wide resilience improvements by creating banking rewards for risk reduction. As lenders better manage their risks, they may offer more favourable terms to borrowers implementing flood resilience measures, creating economic incentives for adaptation investments.

The broader banking system benefits from improved risk transparency and reduced correlation of flood-related losses. By distributing concentrated physical risks more widely through capital markets, PRS minimises the potential for synchronised institutional stress during significant weather events. This distribution mechanism supports banking stability objectives by preventing the concentration of weather risks in institutions with limited capacity to bear them.

Market Size and Commercial Opportunity

The potential scale of the Physical Risk Swap market is substantial, with estimates suggesting it could develop into a multi-trillion dollar notional market as significant capital markets firms integrate these instruments into their offerings. This projected growth follows patterns observed in other successful derivatives markets.

For comparison, CDS currently represents approximately \$10 trillion in gross notional, whilst interest rate derivatives are in the hundreds of trillions. The annual notional value of PRS transactions may realistically grow into the trillions of USD as the market matures.

First-mover advantage creates network effects and establishes market convention around pricing methodologies. This would position early entrants as physical risk/climate risk market infrastructure—analogue to interest rate and credit derivative utilities—capturing transaction fees, index licensing revenue, and calculation agent mandates.

Conclusion

Physical Risk Swaps represent not merely a technical banking innovation but a critical step toward a banking system that properly accounts for and manages physical risks—ultimately supporting both economic stability and climate adaptation in the decades ahead.

The successful implementation of PRS requires collaboration between atmospheric and weather scientists, hazard assessment experts, banking engineers, and market participants to develop robust pricing models that accurately capture the complex dynamics of physical risks whilst maintaining the standardisation and liquidity necessary for market adoption. With the foundation of ISDA documentation and the FINOS CDM, the market can focus on the unique aspects of physical hazard risk modelling rather than reinventing established market infrastructure.

The following chapter presents the operational platform that brings these concepts to life: the PRS Pricing Platform with its real-time gauge monitoring, hazard curve construction, portfolio basis calculation, and historical validation capabilities.

Chapter 14 - Extreme Flood Events

The Gumbel distribution has been mentioned in the MKM Framework Phases and briefly covers extreme events such as floods. Gumbel is a fundamental probability distribution in extreme value theory, widely used to model the maximum (or minimum) values in samples of independent and identically distributed random variables. Before examining a real example of a PRS, it is worthwhile to revisit this distribution in more detail, as it is crucial when transitioning from a one-year insurance horizon to a multi-year one. In practical applications, the Gumbel distribution is excellent for modelling phenomena such as annual flood peaks, maximum wind speeds, precipitation intensities, and other environmental extremes. The distribution is named after Emil Julius Gumbel (1891-1966), who developed foundational work in extreme value statistics. It belongs to the family of extreme value distributions identified by Fisher and Tippet, classified as Type I extreme value distribution.

Parametric Structure

The Gumbel distribution has two parameters:

- **Location parameter:** It controls the central tendency or the modal behaviour of the distribution. In flood modelling, it usually represents the "threshold" level around which extreme values tend to gather.
- **Scale parameter:** It controls the width or dispersion of the distribution. Larger values produce wider, flatter distributions; smaller values produce narrower, more concentrated ones.

For flood applications, these parameters are often estimated from historical annual maximum water level data using methods such as maximum likelihood estimation (MLE) or probability-weighted moments (PWM).

Probability Density Function (PDF)

The probability density function for the maximum-type Gumbel distribution is given by:

$$f(x) = (1/\beta) \times \exp(-(x-\mu)/\beta) \times \exp[-\exp(-(x-\mu)/\beta)], \quad x \in \mathbb{R}$$

Where:

- $(1/\beta) \times \exp(-(x-\mu)/\beta)$ is the "base" exponential decay term
- $\exp[-\exp(-(x-\mu)/\beta)]$ is the "double exponential" component that creates the characteristic heavy-tail behaviour

The PDF is “non-symmetric” and exhibits a longer right tail for maximum-type Gumbel, making it ideal for capturing rare, extreme events that occur more frequently than exponential models would predict

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) gives the probability that the random variable X is less than or equal to a value x :

$$F(x) = \exp[-\exp(-(x-\mu)/\beta)]$$

This compact form is one of the elegant properties of the Gumbel distribution.

Note that:

- When $x = \mu$, we have $F(\mu) = \exp(-1) \approx 0.368$ (not 0.5 as in a symmetric distribution)
- When $x \rightarrow -\infty$, $F(x) \rightarrow 0$
- When $x \rightarrow +\infty$, $F(x) \rightarrow 1$

The CDF is strictly increasing and continuous for all real x .

Survival Function

The survival function $S(x)$ represents the probability that the random variable exceeds a given threshold x :

$$S(x) = 1 - F(x) = 1 - \exp[-\exp(-(x-\mu)/\beta)]$$

In flood risk terminology, $S(x)$ is interpreted as the probability of exceedance—the likelihood that water levels will exceed the design level x in any given year. This is the fundamental quantity used in flood frequency analysis.

Quantile Function (Inverse CDF)

The quantile function $F^{(-1)}(p)$ computes the value x corresponding to a given probability $0 < p < 1$. For the Gumbel distribution:

$$F^{(-1)}(p) = \mu - \beta \times \ln(-\ln p)$$

This function is particularly useful in engineering practice. Given an annual exceedance probability p , the quantile function returns the design level—the water height expected to be exceeded with probability p in any given year.

Practical Application: For a 1-in-100 year flood (annual exceedance probability $p = 0.01$):

$$x_{100} = \mu - \beta \times \ln(-\ln 0.01) = \mu - \beta \times \ln(4.605) = \mu + 1.527\beta$$

This means a 1-in-100 year flood level is approximately 1.527β units above the location parameter μ .

Statistical Properties

Mean (Expected Value)

The mean of a Gumbel distributed random variable is:

$$E[X] = \mu + \beta \times \gamma$$

Where:

- γ is the Euler-Mascheroni constant with numerical value $\gamma \approx 0.57721566\dots$

Note that the mean depends on both parameters: it increases linearly with scale parameter β and shifts directly with location parameter μ . The constant γ is a universal property of the Gumbel distribution.

Variance

The variance of a Gumbel distributed random variable is:

$$\text{Var}(X) = (\pi^2/6) \times \beta^2 \approx 1.6449 \times \beta^2$$

The variance depends only on the scale parameter β (not on μ), and is proportional to β^2 . This means doubling the scale parameter increases variance by a factor of four.

The standard deviation is:

$$\sigma = \beta \times \sqrt{(\pi^2/6)} = \beta \times (\pi/\sqrt{6}) \approx 1.2825 \times \beta$$

Higher Moments and Skewness

The Gumbel distribution has fixed skewness (asymmetry coefficient):

$$\text{Skewness} = (12\sqrt{6/\pi^3}) \times \zeta(3) \approx 1.1396$$

where $\zeta(3)$ is the Riemann zeta function. The skewness is always positive, indicating a right-tail bias—the distribution has a longer tail extending toward higher values. This property is independent of the parameters and is inherent to the Gumbel distribution's shape

This indicates moderately heavy tails compared to a normal distribution (which has excess kurtosis of 0), but lighter tails than power-law distributions.

The Generalized Extreme Value (GEV) Distribution

The Gumbel distribution is one of three asymptotic extreme value distributions identified by Fisher and Tippet. These three limiting distributions are:

Type I (Gumbel) : $F(x) = \exp[-\exp(-(x-\mu)/\beta)]$

- Suitable for unbounded distributions (e.g., normal, exponential, log-normal distributions of the underlying data)
- Used for many flood and wind speed applications

2. Type II (Fréchet) : For heavy-tailed underlying distributions

- Suitable for data with polynomial tail decay (e.g., very heavy rainfall distributions)
- Results in $F(x) = \exp[-(x/\mu)^{-\alpha}]$ form

3. Type III (Weibull) : For bounded underlying distributions

- Suitable for data with upper bounds (e.g., minimum lifetimes, maximum strength)
- Results in $F(x) = \exp[-(((\omega-x)/\mu)^\alpha)]$ form

The Generalized Extreme Value (GEV) distribution unifies all three types with a single parametric form controlled by a shape parameter ξ :

$$F(x) = \exp(-(1 + \xi(x-\mu)/\sigma)^{-1/\xi})$$

Where:

$\xi = 0$, the GEV reduces to the Gumbel (Type I) distribution. When $\xi > 0$, it becomes Fréchet; when $\xi < 0$, it becomes Weibull.

Application to Flood Frequency Analysis

Parameter Estimation from Data

Given a historical record of annual maximum water levels $x_1, x_2, \dots, x_n$ (typically spanning 30-100 years), parameters μ and β are estimated using:

Maximum Likelihood Estimation (MLE):

The log-likelihood function is:

$$\ell(\mu, \beta) = -n \times \ln(\beta) - \sum (x_i - \mu) / \beta - \sum \exp(-(x_i - \mu) / \beta)$$

MLE estimators $\hat{\mu}$ and $\hat{\beta}$ maximize this function. Numerical optimization (e.g., Newton-Raphson) is typically required.

Probability-Weighted Moments (PWM):

An alternative approach less sensitive to outliers:

$$\hat{\beta} = (\hat{a}_1 - 2\hat{a}_2) / \ln(2)$$

$$\hat{\mu} = \hat{a}_1 - \gamma \times \hat{\beta}$$

Where:

$\hat{a}_k$ are sample probability-weighted moments.

Return Period Interpretation

A key concept in flood engineering is the return period T (in years), related to annual exceedance probability by:

$$p = 1/T$$

For example:

- 1-in-100 year flood: $p = 1/100 = 0.01$ (1% annual probability)

- 1-in-50 year flood: $p = 1/50 = 0.02$ (2% annual probability)

The corresponding design level (flood height with return period T) is:

$$x_t = \mu - \beta \times \ln(-\ln(1 - 1/T))$$

For large T (rare events), this approximates:

$$x_t \approx \mu + \beta \times \ln(T)$$

showing the logarithmic scaling of extreme values with return period.

Multi-Year Hazard Accumulation

Extension Beyond Annual Maxima

While the classical Gumbel distribution models annual maxima, insurance and risk applications require extending this to multiple years. The probability of at least one exceedance during t years is derived from the survival function:

Simple Poisson Approach:

Assuming independent annual events with annual exceedance probability p :

$$P(\text{at least one event in } t \text{ years}) = 1 - (1-p)^t$$

This approach underestimates tail risk for heavy-tailed distributions.

Gumbel-adjusted Approach (Heavy-Tail Correction):

Incorporating the heavy-tail property of Gumbel-distributed extremes:

$$P(\text{event in } t \text{ years}) = 1 - (1 - p \times (1 + \alpha \times \ln(t)))^t$$

where $\alpha \approx 0.77$ is a Gumbel-specific constant derived from extreme value theory. The $\ln(t)$ term captures the logarithmic growth in tail probability—reflecting that flood extremes accumulate more quickly than exponential models suggest.

Empirical Validation

The Gumbel-adjusted probability approach is validated through:

- Historical back-testing : Comparing predicted multi-year exceedance probabilities against observed flood events in 50-year Thames records
- Comparison with alternatives : Showing 75-120% higher probabilities than simple exponential approaches

- Regulatory alignment : Matching approaches used by UK Environment Agency and European Central Bank climate risk frameworks

Relationship to Other Distributions

Gumbel vs. Normal Distribution

The normal distribution is symmetric and has light tails, making it unsuitable for extreme values. The Gumbel distribution is asymmetric with heavier right tails, providing superior modeling for rare events.

Gumbel vs. Log-Normal Distribution

Log-normal distributions are often used for positive-valued data (e.g., asset prices). However, they do not capture the extreme-value limiting behavior—a log-normal sample's maximum does not follow log-normal distribution asymptotically.

Gumbel vs. Exponential Distribution

The exponential distribution models waiting times and has memoryless property. The Gumbel distribution, by contrast, specifically models the maximum of samples and has no memoryless property. For iid exponential samples, the distribution of their maximum follows Gumbel.

Limitations and Extensions

Assumption of Stationarity

Classical Gumbel fitting assumes flood frequency is stationary—that historical patterns persist into the future. Climate change introduces non-stationarity, with potentially shifting μ and β parameters.

Modern approaches incorporate:

- Trend analysis : Detecting changes in $\mu(t)$ over time
- Climate-adjusted parameters : Modifying μ and β based on climate projections
- Extended return periods : Accounting for changed flood frequency in long-term risk pricing

Spatial Correlation

Multiple gauges on a single river exhibit correlated exceedances. Simple Gumbel fitting at each gauge ignores this dependence. Multivariate extensions and copula methods address spatial correlation.

Seasonal Effects

Some flood peaks are driven by winter rainfall, others by spring snowmelt. Gumbel fitting assumes all annual maxima have same distribution. Seasonal decomposition may improve accuracy in snow-dominated systems.

Multi-year probability (with Gumbel tail adjustment)

$\alpha = 0.77$

$t = 5 \text{ years}$

$\text{multiyear_prob} = 1 - (1 - p \cdot (1 + \alpha \cdot n \cdot \log(t)))^t$

Summary

The Gumbel distribution provides a mathematically rigorous foundation for modeling extreme values in environmental and financial applications. Its key properties—asymmetry, heavy right tail, and non-linear quantile scaling—make it superior to simpler exponential or normal approximations for rare-event modeling.

In Physical Risk Swaps and climate-related financial risk assessment, the Gumbel distribution enables:

- Precise hazard quantification : Converting gauge readings into annual exceedance probabilities with known theoretical properties
- Multi-year accumulation : Calculating contract-level probabilities that account for heavy-tail behavior
- Regulatory alignment: Using standard hydrological methodology accepted by central banks and insurance regulators
- Transparent pricing : Grounding insurance premiums in objective probability distributions rather than opaque loss-adjustment models

The distribution's limitations (stationarity assumption, spatial correlation, climate-induced parameter shifts) drive ongoing research in adaptive extreme value methods, but the Gumbel

framework remains the industry standard for flood frequency analysis and parametric risk transfer.

Chapter 15 - Theoretical Use Case

This chapter demonstrates how Physical Risk Swap theory translates into operational practice through a specific Thames Valley flood risk application. Rather than repeating the CDS pricing framework and mathematical foundations established in Chapter 12, this chapter focuses on:

- Implementation-specific considerations: gauge selection, trigger level calibration, and settlement mechanics
- A two-step pricing approach: establishing 1-year baseline pricing, then extending to multi-year horizons with sophisticated hazard accumulation
- Advanced methodology for probability calculation that captures the heavy-tail characteristics of flood distributions

Part I: One-Year Baseline Pricing

The Thames Valley PRS begins with establishing 1-year pricing using the CDS framework. This baseline serves two functions:

- Market reference price for liquid trading
- Foundation for multi-year contract valuation

Gauge Selection and Trigger Levels

The Thames flood monitoring network comprises Environment Agency gauges at regular intervals. For PRS implementation, five strategic gauge locations provide comprehensive coverage of the Thames Valley:

Location	Distance	Trigger Level	Flood Zone	Historical Peak (Jan 2024)
Oxford (Osney Lock)	0 km	1.80m AOD	Floodplain	2.07m
Reading	70 km	7.25m AOD	Floodplain	7.54m
Windsor	135 km	2.80m AOD	Defended (MWEFAS)	3.02m
London Bridge	160 km	3.50m AOD	Thames Barrier	3.25m

Tilbury	180 km	4.00m AOD	Tidal Estuary	3.60m
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Thames Valley flood monitoring gauges and trigger levels

Trigger levels are established at the "severe flood warning" threshold defined by the Environment Agency. This selection ensures:

- Objectivity: Transparent, legally defensible trigger points
- Verifiability: Real-time gauge data from government sources
- Calibration: Extensive historical data enables hazard curve fitting
- Regulatory Compliance: Direct alignment with emergency preparedness definitions

1-Year Pricing Results

Using estimates of return periods that would lead to an event where the severe warning is breached, thereby triggering a GBP 1,000,000 fixed and immediate payoff, we present the following table. For this purpose, the estimates would be replaced with actual calculations based on methodologies outlined in previous chapters.

The following illustrative 1-year premiums emerge:

Gauge	Return Period	Annual Probability	Pure Risk Cost	Total Premium
Oxford	40 years	2.5%	£25,000	£50.487
Reading	30 years	3.3%	£33,000	£60.883
Windsor	20 years	5.0%	£50,000	£82.975
London Bridge	100 years	1.0%	£10,000	£30.995
Tilbury	200 years	0.5%	£5,000	£24.497

One-year PRS pricing by Thames Valley gauge

The premium structure includes:

- Pure Risk Cost: $AEP \times \text{Payoff}$ (e.g., $3.3\% \times £1m = £33k$ for Reading)
- Risk Margin: 15% of pure cost (captures model uncertainty, tail risk)

- Operating Costs: 13% of risk-adjusted cost (admin + broker)
- Capital Cost: £18,000 annually (15% of payoff @ 12% cost of capital)

Calibration Note

The 1-year annual probabilities reflect historical flood frequency analysis from the Thames Valley:

- January 2024 (Storm Henk): 1-in-30 to 1-in-50 year event
- February 2014 floods: 1-in-15 to 1-in-20 year event
- Major historical events: 1947, 2003, 2013-14

These empirical return periods form the basis for AEP estimates used in pricing.

Part 2: Multi-Year Extension with Sophisticated Hazard Accumulation

Moving from 1-year to multi-year pricing requires addressing a critical question: How does the probability of a flood event accumulate over time?

The naive approach—calculating cumulative probability as —significantly underestimates multi-year risk for flood events. Floods exhibit heavy-tail distributions, meaning extreme events occur more frequently than exponential models predict. Using scaling (as suggested in preliminary models) provides only a modest improvement but lacks theoretical justification.

Flood depths typically follow Gumbel distribution that provide excellent fit for modeled river systems across European river basins (Thames, Rhine, Danube) based on decades of hydrological literature.

The Gumbel distribution captures a key feature of flood behavior: the return period vs magnitude relationship is non-linear. The jump from a 100-year to 200-year flood represents a larger increase in water depth than the jump from a 10-year to 20-year flood. This nonlinearity reflects the accumulated energy in extreme precipitation events.

Mathematical Framework

For a return period (in years), the probability of at least one exceedance during time period follows:

This is the standard Poisson approach. However, the Gumbel adjustment for heavy-tail behavior introduces:

Where:

- Is the 1-year annual exceedance probability
- Gumbel parameter from extreme value theory
- Captures the logarithmic growth in tail probability over time
- is the number of years

This formulation:

1. Reduces to the simple formula when
2. Produces heavier-tail behaviour for (more realistic for floods)
3. Is bounded by physical constraints (cannot exceed 100%)
4. Is calibrated to observed flood frequency distributions

Comparison: Simple vs Gumbel Approach

The following table demonstrates the material difference between methodologies:

Gauge	Years	Simple	Gumbel + Tail	Difference
Oxford	3 years	7.3%	13.2%	80.7%
Oxford	5 years	11.9%	25.0%	110.5%
Reading	3 years	9.6%	17.2%	79.4%
Reading	5 years	15.5%	31.9%	106.4%
Windsor	3 years	14.3%	25.2%	76.8%
Windsor	5 years	22.6%	44.8%	97.9%

Comparison of probability accumulation methodologies

The Gumbel approach typically produces 75-120% higher multi-year probabilities than simple exponential models.

For Reading (1-in-30 year annual risk):

- Simple 5-year probability: 15.45% → pricing impact (low premium)

- Gumbel 5-year probability: 31.88% → pricing impact (~2x higher premium)

Multi-Year Hazard Accumulation: CDS Settlement Mechanics

The multi-year PRS premium incorporates contract termination risk (analogous to credit events in CDS). Once a flood trigger occurs, the contract terminates—no further premiums are due, no further protection is provided. For each year in the contract period, this represents the probability that NO flood trigger has occurred through year .

Present Value of Premium Leg (3-year example and discount factor set at 3%):

For Reading (£60,883 annual premium):

- Year 1: $£60,883 \times 0.9709 \times 1.000 = £59,058$
- Year 2: $£60,883 \times 0.9426 \times 0.967 = £55,513$
- Year 3: $£60,883 \times 0.9151 \times 0.936 = £52,245$
- Total PV Premium = £166,816

For Reading:

- Year 1: $£1,000,000 \times 1.000 \times 0.033 \times 0.9709 = £32,043$
- Year 2: $£1,000,000 \times 0.967 \times 0.033 \times 0.9426 = £30,149$
- Year 3: $£1,000,000 \times 0.936 \times 0.033 \times 0.8903 = £27,519$
- Total PV Protection = £89,711

This implies that over 3 years, the protection seller requires 53.8% additional compensation relative to the premium payments received, accounting for:

- Time value of money (3% discount rate)
- Early termination probability
- Default-free curve dynamics

Market Pricing and Illiquidity Premium

The theoretical spread of 53.8% provides the par pricing for a 3-year Reading PRS. In practice, market pricing includes an illiquidity premium reflecting:

- 3-year contracts: 2% illiquidity premium

- 5-year contracts: 3% illiquidity premium

This accounts for:

- Wider bid-offer spreads on longer-dated contracts
- Lower secondary market liquidity
- Reduced dealer inventory

5-Year Reading PRS Pricing:

- PV Premium = £261,538
- PV Protection = £141,760
- Implied Spread = 54.2%
- Total Premium = $£261,538 \times 1.542 \times 1.03 = £414,570$ total for 5 years
- Annual equivalent = £82,914

Capital Cost Integration

Under Basel III, the capital requirement for a PRS protection seller includes:

For a 3-year £1m PRS:

- Capital Cost = $£150,000 \times 0.12 \times 3 = £54,000$ total
- Amortized annually = £18,000

This is separate from the premium/spread calculation and reflects regulatory capital allocation for risk-bearing capacity.

Multi-Year Pricing Summary Table

The following table summarises pricing across 1, 3, and 5-year maturities:

Gauge	Maturity	Annual Premium	Total Cost	Annual Equivalent
Reading	1-Year	£60,883	£60,883	£60,883
Reading	3-Year	£166,816	£262,400	£87,467

Reading	5-Year	£261,538	£414,570	£82,914
Oxford	1-Year	£50,487	£50,487	£50,487
Oxford	3-Year	£139,337	£223,143	£74,381
Oxford	5-Year	£220,269	£336,509	£67,302
Windsor	1-Year	£82,975	£82,975	£82,975
Windsor	3-Year	£223,389	£362,639	£120,880
Windsor	5-Year	£344,895	£546,532	£109,306

PRS pricing summary across maturities

Key Observations

1. Annual equivalents DECLINE with maturity (e.g., Reading falls from £60,883 to £82,914). This reflects the decreasing probability of contract survival to later years. A flood event in year 1 terminates the contract, reducing expected premium collection in years 2-3.
2. The Gumbel approach produces materially higher multi-year premiums than simple scaling would suggest. For Reading, 5-year probability is 31.9% (Gumbel) vs 15.45% (simple), resulting in ~2x higher expected loss accumulation.
3. Capital costs increase linearly with maturity (£18k p.a. × years) but are less dominant in longer-dated contracts where risk cost dominates.

Geographic Extension and Basis Considerations

The Thames Valley implementation can be extended to other UK river systems (Severn, Trent, Humber) using the identical methodology. Each new river system requires:

- Gauge selection: Identifying monitoring stations that correlate with property exposure
- Hazard curve fitting: Applying the Gumbel distribution to return period data
- Basis estimation: Quantifying the correlation between gauge trigger and actual property flood
- Pricing adjustment: Adding basis spread to reflect location-specific risk

For example, a portfolio of properties 2-3km from the Thames gauge might have:

- Base index price: 55bp
- Distance basis: +3bp (farther = higher risk)

- Elevation benefit: -1bp (higher properties safer)
- Terrain basis: +2bp (floodplain exposure)
- Model uncertainty: +2bp
- Total portfolio-specific price: 61bp vs 55bp index

Conclusion

The Thames Valley PRS represents a transition from insurance-based flood risk management to capital markets risk transfer. By applying CDS infrastructure to parametric flood triggers, this framework enables:

- Duration matching between multi-year mortgage assets and hedging instruments
- Transparent pricing based on hazard curves and terminal probability
- Rapid settlement without loss adjustment complexity
- Regulatory capital recognition for risk transfer effectiveness
- Liquidity through standardisation and ISDA documentation

The two-step approach (1-year baseline + multi-year extension) provides both market transparency and operational simplicity. The Gumbel extreme value methodology captures the heavy-tail behavior of flood distributions with mathematical rigor while remaining implementable within standard banking risk systems.

Extensions beyond Thames Valley to other river systems, peril types, and geographies follow naturally from this framework. The standardization of hazard curves, trigger definitions, and settlement mechanics across rivers creates the foundation for indices and portfolio products that would establish true market liquidity.

Chapter 16 - Banking Supervisory Capital

Origins and Evolution of the Three-Pillar Framework

The Basel regulatory framework for banking supervision emerged from a recognition, forged through successive financial crises in the late 20th century, that single-point capital rules inadequately captured the full spectrum of banking risks. The resulting three-pillar architecture—comprising minimum capital requirements (Pillar 1), supervisory review (Pillar 2), and market discipline (Pillar 3)—represents an attempt to balance quantitative rigour with qualitative judgement and transparency.

The Role of Capital in Banking Supervision

Bank capital serves as a buffer against unexpected losses, protecting depositors and maintaining confidence in the financial system. The fundamental principle underlying risk-based capital requirements is straightforward: institutions holding riskier assets should maintain proportionally higher capital reserves. This approach aims to ensure banks can withstand adverse scenarios while continuing to meet their obligations.

The Basel framework established a minimum capital requirement of 8% of risk-weighted assets (RWA), with risk weights assigned according to the perceived riskiness of different asset classes. Banks may employ standardised approaches, using regulatory-prescribed risk weights, or internal model-based approaches that promise enhanced risk sensitivity in exchange for greater methodological complexity and supervisory scrutiny.

The Tripartite Structure

- **Pillar 1: Minimum Capital Requirements** establishes quantitative thresholds for credit risk, market risk, counterparty risk, and operational risk. Beyond these risk-weighted requirements, the framework incorporates non-risk-sensitive safeguards, including the leverage ratio—designed to prevent excessive balance sheet expansion—and liquidity requirements through the Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR). The LCR ensures institutions maintain sufficient high-quality liquid assets to survive 30-day stress scenarios, whilst the NSFR addresses longer-term stability through maturity structure requirements.
- **Pillar 2: Supervisory Review** introduces qualitative oversight and institution-specific assessment beyond standardised minima. Banks must conduct Internal Capital Adequacy Assessment Processes (ICAAP) and Internal Liquidity Adequacy Assessment Processes (ILAAP), demonstrating that their internal risk management frameworks adequately identify, measure, and manage all material risks. Supervisory authorities evaluate these assessments through the Supervisory Review and Evaluation Process (SREP), with the power to mandate additional capital buffers or operational interventions where necessary.

- Pillar 3: Market Discipline mandates public disclosure of risk profiles, capital positions, and risk management practices. The underlying logic holds that transparent reporting enables market participants—depositors, creditors, investors—to make informed decisions, thereby harnessing market forces to complement regulatory supervision. Well-informed markets should reward prudent risk management whilst penalising excessive risk-taking through pricing adjustments and capital allocation decisions.

The Aftermath of the 2008 Financial Crisis

The 2008 financial crisis exposed critical weaknesses in the Basel II framework, particularly regarding counterparty credit risk in derivative markets and the interconnectedness of major financial institutions. The collapse of Lehman Brothers demonstrated how counterparty exposures could amplify systemic risk through complex chains of interconnected obligations.

Basel III, developed in response, introduced enhanced counterparty credit risk requirements. The Credit Valuation Adjustment (CVA) charge captures the risk of losses arising from deterioration in counterparty creditworthiness, whilst stricter collateral and margin requirements for over-the-counter derivatives aim to reduce systemic exposures. The framework also established capital buffers for systemically important financial institutions (SIFIs), recognising that the failure of such entities poses disproportionate threats to financial stability.

Importantly, the crisis demonstrated the limitations of relying on Pillar 1 capital charges alone. Models underpinning market and credit risk capital requirements often failed under stress, revealing excessive dependence on Value-at-Risk metrics and optimistic assumptions in correlation and diversification benefits. This drew attention to the procyclicality of Pillar 1 approaches, where risk-weighted assets fell in stable periods only to spike during downturns, amplifying systemic vulnerabilities.

As a result, supervisors placed increased emphasis on Pillar 2, which allows for qualitative judgment, stress testing, and supervisory review. Supervisory authorities began to require more forward-looking evaluations of tail risk, liquidity stress, and systemic contagion—factors not sufficiently captured by standardised capital formulas. Stress testing gained prominence as a key supervisory tool, combining quantitative models with scenario analysis that explicitly considered extreme but plausible outcomes. This transition also established the foundation for the current use of stress testing in identifying emerging risks such as climate change.

The collapse of quantitative risk models during the crisis also prompted the formal development of model risk governance as a separate discipline. Banks were found to have insufficient oversight of the assumptions, validation processes, and limitations of the models used to determine regulatory and internal capital. In response, financial institutions and regulators introduced frameworks for independent model validation, managing model inventories, and establishing governance committees. The Federal Reserve's SR 11-7 in the U.S., and later European guidance, formalised the principle that model risk itself is a distinct type of risk that must be managed, not simply accepted.

Together, these developments marked a regulatory and philosophical turning point. Pillar 1 capital rules were tightened and made more robust, but their limits became clearer; Pillar 2 supervisory judgment grew in prominence; and strong model risk governance emerged as essential to ensuring the credibility of both capital adequacy frameworks and wider risk management. These lessons provide a valuable backdrop to current debates over climate-related financial risks, where uncertainty in modelling, widespread systemic exposures, and the limits of quantitative approaches once again demand a combination of capital safeguards, supervisory oversight, and disciplined governance of the models upon which decisions rely.

Stress Testing Under Pillars 1 and 2

Thorough stress testing forms a fundamental part of both Pillar 1 and Pillar 2 frameworks, acting as a proactive addition to retrospective capital calculations. Supervisory authorities develop stringent yet credible stress scenarios specifically designed to evaluate institutional resilience during adverse situations.

Stress Testing Methodology

The stress testing process operates through several interconnected stages. Supervisors construct scenarios incorporating macroeconomic shocks—GDP contraction, unemployment spikes, interest rate movements—alongside specific sectoral or geographic stresses. These scenarios, whilst severe, must remain plausible; their purpose is to test resilience rather than model apocalyptic outcomes.

Banks then project the impact of these scenarios on their portfolios, calculating comprehensive effects on capital ratios while accounting for provisions, losses, and changes in risk weights resulting from rating downgrades or collateral devaluations. Supervisors establish minimum post-stress capital thresholds—typically around 5-6% Common Equity Tier 1 (CET1) ratio, potentially augmented by institution-specific systemic importance buffers.

When stress test results indicate that a bank's capital would fall below the required post-stress minimum, supervisors can mandate pre-stress capital increases. For instance, if an institution begins with a 14% capital ratio but stress testing projects deterioration to 4% against a required 6% minimum, supervisors might require the pre-stress ratio to increase to 16%, ensuring adequate post-stress buffers.

Pillar 2 Supervisory Interventions

Under Pillar 2, supervisory authorities possess extensive intervention powers when stress tests or other assessments reveal inadequate risk management or capital adequacy. These powers range from requiring enhanced reporting and improved control frameworks to mandating dividend restrictions, additional capital raising, asset reductions, business line exits, or even management changes.

The European Central Bank employs a four-step approach to determine Pillar 2 requirements on a risk-by-risk basis, incorporating business model assessments, internal governance evaluations, and comprehensive risk assessments across credit risk, market risk, operational risk, and interest rate risk in the banking book. This granular approach allows supervisors to target specific institutional vulnerabilities rather than applying uniform add-ons.

Model Risk Management and Vendor Data Challenges

The increasing sophistication of risk measurement approaches, particularly internal models for Pillar 1 and 2 calculations, has elevated model risk management to critical importance. Model risk arises from potential adverse consequences of decisions based on incorrect or misused model outputs, whether due to fundamental errors in model design, implementation flaws, or inappropriate application.

Governance Framework for Model Risk

Effective model risk governance operates within a three-lines-of-defence framework. Model developers and users (first line) manage daily development and implementation; risk management functions (second line) oversee and establish minimum standards; internal audit (third line) provides independent evaluation of the framework's effectiveness.

When models greatly influence business decisions—including risk management, capital planning, and regulatory capital calculations—governance frameworks must be equally rigorous. Regulatory guidance emphasises that the complexity of model governance should correspond to the sophistication of model usage, with stricter requirements for models affecting regulatory capital determinations.

The Vendor Data Challenge

When banks employ vendor-supplied models or analytical data—such as flood scores for physical climate risk assessment—model risk management faces unique challenges. The regulatory principle of "buyer beware" places responsibility for model outcomes firmly with the purchasing institution, regardless of external development.

Vendor models present validation difficulties because banks typically lack full access to underlying methodologies, data sources, and computational processes. External models may not permit examination of computer coding and implementation details, forcing reliance on sensitivity analysis, benchmarking, and outcomes testing. For analytical data like flood scores, this creates a "black box" problem where banks receive numerical outputs without complete transparency into risk assessment methodology.

Banks must nonetheless establish systematic validation procedures to understand vendor products' capabilities, applicability, and limitations. This includes assessing whether vendor data adequately represents the institution's specific exposure profile—geographical concentrations, portfolio characteristics, asset types. For flood risk, vendor scores developed using national or regional data may inadequately capture microclimatic conditions or localised hazard characteristics relevant to specific asset locations.

The challenge increases for climate-related risks due to three types of uncertainty: inherent future uncertainty in climate projections, measurement uncertainty caused by data gaps, and model-based uncertainty that requires strong quantitative evaluation. Each step of converting physical climate phenomena into financial risk measures adds more uncertainty that banks need to assess and manage. However, vendor methodologies may lack the full governance structures that meet banking standards

Pillar 3 Climate Risk Disclosures

Climate risk is the most recent and rapidly evolving element within the three-pillar framework. What began as recommendations from the Task Force on Climate-related Financial Disclosures (TCFD) has become a mandatory regulatory requirement, although the shift from disclosure to capital adequacy has faced significant obstacles and has not yet occurred.

From TCFD Recommendations to Regulatory Reality

The Financial Stability Board's TCFD, established in 2015, outlined voluntary recommendations for climate-related financial disclosures across four pillars: governance, strategy, risk management, and metrics and targets. Whilst initially advisory, these recommendations gained traction as institutional investors and regulators recognised climate risks as material financial considerations.

The European Banking Authority published final Implementing Technical Standards (ITS) in January 2022, transforming voluntary guidance into mandatory requirements. Beginning in December 2022, large institutions with securities traded on regulated markets must now disclose environmental, social, and governance (ESG) risks semi-annually. These requirements encompass both qualitative information on risk management strategies and quantitative templates covering transition risks, physical risks, and climate-aligned exposures.

The disclosure framework comprises ten standardised templates addressing: credit risk quality of climate-sensitive exposures; energy efficiency of real estate collateral; institutions' financed emissions (scope 1, 2, and 3); exposures subject to physical climate risks including floods, droughts, and extreme weather; Green Asset Ratio (GAR) measuring portfolio alignment with EU Sustainable Finance Taxonomy; and Banking Book Taxonomy Alignment Ratio (BTAR) for additional alignment metrics.

Implementation followed a phased timeline: December 2022 introduced "light" disclosure requirements covering selected templates; December 2023 extended requirements, including GAR disclosures; June 2024 marked full implementation, incorporating financed emissions (scope 3), BTAR disclosures, and additional taxonomy-related metrics.

Globally, the Basel Committee on Banking Supervision initially proposed integrating climate disclosures into international Pillar 3 requirements. However, the final framework published in June 2025 became entirely voluntary rather than mandatory, largely due to resistance from US regulators. This jurisdictional divergence leaves implementation to national discretion, creating potential regulatory arbitrage opportunities.

The Stuttering Progression to Pillar 2

Following initial consultation and the adoption of Pillar 3 disclosure requirements, progress towards Pillar 2 capital charges has significantly slowed. The expected shift from transparency to capital adequacy remains stalled, hindered by fundamental methodological shortcomings.

The main obstacle lies in methodology. Current approaches lack the sophistication needed to accurately incorporate climate-related risks, especially physical risks, into capital frameworks designed for measurable, short-term exposures. Converting uncertain, long-term climate phenomena into precise capital charges remains hugely difficult when core questions about probability distributions, correlation structures, and transmission mechanisms are unresolved.

Fewer than half of institutions currently include scenario analyses in their Pillar 2 processes, indicating limited maturity in internal risk quantification capabilities. Making climate risks formal Pillar 2 requirements would require substantial enhancement of institutional skills that many banks presently lack. Existing Pillar 3 disclosures are criticised for being "very granular and complex" with limited reliable data, particularly concerning physical risk assessments. Converting these uncertain metrics into exact capital requirements heightens the existing challenges without a clear solution.

The Net Zero Diversion

Conflicting objectives within climate risk regulation further complicate matters. While banking supervisors' core responsibility involves ensuring systemic financial stability, climate risk frameworks increasingly prioritise funding net-zero transition targets. This blending of objectives creates tension between prudential regulation and industrial policy.

The Green Asset Ratio and taxonomy alignment metrics exemplify this tension. While ostensibly measuring transition risk exposure, these metrics also serve as mechanisms to steer capital allocation towards "sustainable" activities as defined by political taxonomies. Banks face pressure not only to measure and manage climate risks but to actively redirect lending towards favoured sectors regardless of pure risk-return considerations.

This shift from financial stability towards climate policy implementation complicates the development of methodologies that accurately reflect actual financial risks. When regulatory

frameworks pursue dual aims—stability and transition acceleration—the outcome risks satisfying neither effectively. Capital requirements calibrated to promote green lending may inadequately reflect genuine risk exposures, while political definitions of "sustainable" activities may diverge from financial materiality.

The stated goal of pushing financial institutions towards net-zero funding creates perverse incentives. Banks may prioritise regulatory compliance over precise risk assessment, while supervisors face pressure to soften requirements that might constrain green lending. This dynamic undermines the fundamental purpose of prudential regulation: maintaining resilient financial institutions capable of withstanding adverse scenarios.

Physical Risk Transition Paths

Physical climate risks—floods, wildfires, hurricanes, sea level rise, temperature extremes—present immediate challenges distinguishable from longer-term climate transition scenarios. These hazards currently impact property values, insurance availability, and collateral quality, creating present-day financial exposures rather than distant contingent liabilities.

Current Physical Risk Landscape

Insured losses from natural catastrophes have grown steadily, with severe convective storms showing particularly marked increases. The Swiss Re Institute sigma reports indicate long-term upward trends in insured losses, driven by economic growth, urbanisation, and increased storm activity, with losses growing 5-7% annually since 1992.

The complexity of physical risk assessment demands increasingly granular analysis at the asset level. Banking institutions must now evaluate thousands or millions of portfolio assets, considering precise physical locations and projecting expected losses across multiple physical hazards. This granular approach represents significant departure from traditional portfolio-level risk assessment methods.

Data and Modelling Challenges

A detailed physical risk analysis requires extensive data. Banking institutions need reliable information on asset locations, hazard forecasts, and potential damage models. Uncertainties and gaps in weather predictions and impact modelling complicate these data needs. Combining various data sources—from high-resolution terrain models to real-time weather data—requires new methods for data management and standardisation.

Current vendor solutions for physical risk assessment vary in complexity and dependability. Flood scores illustrate the data challenges: vendor scores often use proprietary algorithms that combine multiple data sources, climate forecasts, and modelling techniques without full transparency. Banks must verify whether these scores suit their specific portfolios, despite limited insight into the underlying assumptions and calculations.

Methodological Barriers to Pillar 2 Integration

Moving climate risks from Pillar 3 disclosure requirements to Pillar 2 capital charges faces substantial methodological obstacles, particularly regarding physical risks and stress testing compatibility.

The Stress Testing Incompatibility

Traditional bank stress testing assesses short- to medium-term scenarios (usually one to three years) based on established relationships between macroeconomic variables and portfolio performance. Physical climate risks operate on fundamentally different timescales and transmission mechanisms. Acute physical risks—floods, hurricanes, wildfires—manifest as sudden shocks affecting specific geographic areas. While individual events happen quickly, their overall financial impact depends on complex interactions between event frequency, severity, asset exposure, insurance coverage, and recovery processes. Capturing these dynamics in standardised stress scenarios is immensely challenging. Chronic physical risks—sea level rise, temperature changes, shifting precipitation patterns—unfold over decades, leading to gradual changes in asset values, agricultural productivity, and habitability. Translating these slow-moving phenomena into capital adequacy assessments designed for annual horizons causes methodological tension. How should banks capitalise against risks that may not fully materialise for thirty years, especially when discount rates, economic development, and adaptation measures generate significant uncertainties?

Data Adequacy and Quantification Challenges

Current Pillar 3 disclosures are criticised for being "very granular and complex" with limited reliable data, especially concerning physical risk assessments. Converting these uncertain metrics into exact capital requirements for Pillar 2 would increase existing challenges. Less than half of institutions currently incorporate scenario analyses into their Pillar 2 processes, indicating a lack of maturity in internal risk quantification skills.

Making climate risks formal Pillar 2 requirements would require significant improvements in institutional capabilities that many banks do not yet possess. Physical risk quantification faces three main uncertainties: inherent future uncertainty in climate projections; measurement uncertainty due to data gaps; and model-based uncertainty that requires thorough quantitative assessment. Each layer of uncertainty adds to the difficulty of allocating capital precisely.

Regulatory and Jurisdictional Divergence

The voluntary aspect of the Basel framework emphasises notable regulatory disagreement. US regulators' resistance to voluntary climate disclosures indicates stronger opposition to mandatory capital requirements. This jurisdictional fragmentation complicates the establishment of consistent Pillar 2 standards worldwide, creating potential for regulatory arbitrage and uneven playing fields.

Unlike traditional credit, market, and operational risks, which have well-established quantification methods, climate risks pose unique challenges in terms of time horizons, scenario dependencies, and systemic interactions. Developing technical standards for climate risk capital requirements that are both scientifically robust and practically feasible remains an ongoing challenge without a clear resolution pathway.

Conclusion: Towards Market-Based Solutions

The current sequential approach—starting with disclosure requirements in Pillar 3 before progressing to Pillar 2 capital requirements—has faced fundamental challenges. The mismatch between traditional stress testing frameworks and physical risk characteristics, combined with limited methodological sophistication, suggests this route may be ineffective for physical climate risks.

The complexity of existing modelling techniques can be significantly simplified through market mechanisms. Instead of forcing physical climate risks into unsuitable regulatory frameworks that demand complex scenario analysis and long-term projections, creating liquid markets for flood risk allows willing buyers and sellers to set prices collectively through transparent exchanges.

This market-based solution offers notable benefits. Price discovery via actual transactions provides insight into market participants' overall assessment of physical risks, including diverse perspectives on probability, impact, and correlation, without the need for regulatory instructions. Standardised contracts based on clear physical indicators—such as flood depths, wind speeds, and temperature extremes—support transparent pricing while preventing disputes over causation or damage estimation.

Physical Risk Swaps (PRS), similar to Credit Default Swaps (CDS) for credit risk, could act as the mechanism for this market-based risk transfer. These instruments would enable banks to hedge against physical risk concentrations in their portfolios, fostering liquid markets for the redistribution of physical risks. This approach could deliver short-term, quantifiable risk metrics compatible with existing Pillar 1 frameworks, sidestepping the methodological difficulties associated with multi-decade stress testing, and staying focused on real financial stability rather than policy-driven lending targets.

Nonetheless, implementing such market mechanisms requires an understanding of how the current financial system views physical risks, especially within insurance and reinsurance markets that have extensive experience pricing and transferring such exposures. The following chapter examines how the financial system currently handles physical risks, exploring how insurance markets price

catastrophe exposures and how banking institutions can build on these established practices to develop effective risk transfer instruments.

Basel III - BCBS 530 Physical/Climate Risk Principles.

The Basel Committee on Banking Supervision established its Task Force on Climate-related Financial Risks in 2020 to strengthen the regulation, supervision, and practices of banks worldwide in addressing climate-related financial risks. The Committee's work is comprehensive, spanning regulatory, supervisory, and disclosure dimensions across all three pillars of the Basel Framework.

The framework recognises that climate change may result in physical and transition risks affecting individual banks and the broader banking system. Physical risks arise from acute events (floods, wildfires, hurricanes) and chronic changes (rising sea levels, temperature shifts), while transition risks stem from societal changes toward a low-carbon economy, including policy shifts, technological changes, and market sentiment.

A critical insight underlying the Basel approach is that climate-related financial risks are not a separate risk category but rather risk drivers that materialize through traditional financial risk categories—credit risk, market risk, liquidity risk, and operational risk. This transmission channel approach allows climate risks to be addressed within the existing Basel Framework rather than requiring entirely new regulatory structures.

BCBS 530 comprises 18 high-level principles designed to be applied proportionately based on the size, complexity, and risk profile of banks and banking sectors. Principles 1-12 address banks' management of climate-related financial risks, while Principles 13-18 provide guidance for prudential supervisors.

Corporate Governance (Principles 1-3)

Principle 1 requires banks to develop and implement a sound process for understanding and assessing the potential impacts of climate-related risk drivers on their businesses and operating environments. Banks must consider material climate-related financial risks across various time horizons and incorporate these risks into their overall business strategies and risk management frameworks. This includes understanding how physical and transition risk drivers could impact business model resilience over short, medium, and longer terms, and assessing exposure to structural changes in the economy and financial system.

Banks must recognize that climate risks may materialize beyond traditional two- to three-year capital planning horizons but within the maturities of longer-dated positions, while other risks may emerge over much longer timeframes. This temporal complexity demands that boards and senior management take a long-term view and adopt a prudent, dynamic approach to developing risk management capacities.

Principle 2 mandates that the board and senior management clearly assign climate-related responsibilities to members and/or committees and exercise effective oversight of climate-related financial risks. Responsibilities must be identified throughout the organizational structure. Banks should ensure their boards and senior management have adequate understanding of climate risks and appropriate skills to manage them, building capacity through internal workshops or external collaboration where necessary. Where dedicated climate units are established, their responsibilities and interaction with existing governance structures must be clearly defined.

Principle 3 requires banks to adopt appropriate policies, procedures, and controls implemented across the entire organization to ensure effective management of climate-related financial risks. Management of material climate-related financial risks should be embedded in policies, processes, and controls across all relevant functions and business units, including client onboarding and transaction assessment.

Internal Control Framework (Principle 4)

Principle 4 mandates that banks incorporate climate-related financial risks into their internal control frameworks across the three lines of defense to ensure sound, comprehensive, and effective identification, measurement, and mitigation of material climate-related financial risks.

The first line of defense should conduct climate-related risk assessments during client onboarding, credit application and review processes, ongoing monitoring and engagement with clients, and new product or business approval processes. Frontline staff must have adequate awareness to identify potential climate-related financial risks.

The second line of defense (risk function) should undertake climate-related risk assessment and monitoring independently from the first line, challenging initial assessments, while the compliance function ensures adherence to applicable rules and regulations.

The third line of defense (internal audit) should provide independent review and objective assurance of the quality and effectiveness of the overall internal control framework, the first and second lines of defense, and the risk governance framework, considering changes in methodology, business, risk profile, and data quality.

Capital and Liquidity Adequacy (Principle 5)

Principle 5 establishes critical requirements for integrating climate risks into banks' Internal Capital Adequacy Assessment Process (ICAAP) and Internal Liquidity Adequacy Assessment Process (ILAAP), including stress testing programs where appropriate.

Banks must develop processes to evaluate the solvency impact of climate-related financial risks that may materialize within their capital planning horizons. Climate-related financial risks assessed as material over relevant time horizons that may negatively affect capital position (through their impact on traditional risk categories) must be included in the ICAAP.

For liquidity, banks must assess whether climate-related financial risks could cause net cash outflows or depletion of liquidity buffers under both business-as-usual and stressed conditions (considering severe yet plausible scenarios). Material climate-related financial risks that may impair liquidity position must be included in the ILAAP.

The incorporation of climate risks into ICAAP and ILAAP includes, where appropriate, incorporating physical and transition risks relevant to the bank's business model, exposure profile, and business strategy into stress testing programs to evaluate the bank's financial position under severe but plausible scenarios. The Basel Committee recognizes that this incorporation will occur iteratively and progressively as methodologies and data mature.

Risk Management Process (Principle 6)

Principle 6 requires banks to identify, monitor, and manage all climate-related financial risks that could materially impair their financial condition, including capital resources and liquidity positions. Banks must ensure that their risk appetite and risk management frameworks consider all material climate-related financial risks and establish reliable approaches to identifying, measuring, monitoring, and managing those risks.

The board and senior management must ensure climate-related financial risks, where material, are clearly defined and addressed in the bank's risk appetite framework. Banks should regularly assess climate-related financial risks and set clear definitions and thresholds for materiality, recognizing that the risk management framework should capture all material risks with an integrated firm-wide perspective.

These risks may include those posed by concentrations, particularly those related to industry, economic sectors, and geographic regions. As with other material risks, banks should develop appropriate key risk indicators for effective management that align with regular monitoring and escalation arrangements. Where appropriate, banks should consider risk mitigation measures such as establishing internal limits for various types of material climate-related financial risks across credit, market, liquidity, and operational risk profiles.

Management Monitoring and Reporting (Principle 7)

Principle 7 establishes comprehensive requirements for internal risk reporting and data aggregation capabilities. This principle is central to ensuring that climate risks are transparently reported to decision-makers within banks.

Banks' risk data aggregation capabilities must include climate-related financial risks to facilitate the identification and reporting of risk exposures, concentrations, and emerging risks. Banks must have systems in place to collect and aggregate climate-related financial risk data across the banking group as part of their overall data governance and IT infrastructure. Processes must ensure that aggregated data is accurate and reliable, with banks potentially investing in data infrastructure and enhancing existing systems to identify, collect, cleanse, and centralize necessary data.

Banks should consider actively engaging clients and counterparties to collect additional data and develop better understanding of their transition strategies and risk profiles. Where reliable or comparable climate-related data are not available, banks may use reasonable proxies and assumptions as alternatives in their internal reporting as an intermediate step.

Internal reporting should be timely and updated regularly, with banks considering appropriate intervals for updating risk reports given the evolving nature of climate-related financial risks. Banks should develop qualitative and/or quantitative metrics or indicators to assess, monitor, and report climate-related financial risks, making limitations that prevent full climate risk data assessment explicit to stakeholders where relevant.

Comprehensive Management of Credit Risk (Principle 8)

Principle 8 requires banks to understand the impact of climate-related risk drivers on their credit risk profiles and ensure that credit risk management systems and processes consider material climate-related financial risks.

Banks must have clearly articulated credit policies and processes to address material climate-related credit risks, including prudent policies and processes to identify, measure, evaluate, monitor, report, and control or mitigate the impacts of material climate-related risk drivers on credit risk exposures (including counterparty credit risk) on a timely basis. Banks should incorporate consideration of material climate-related financial risks into the entire credit life cycle, including client due diligence during onboarding and ongoing monitoring of clients' risk profiles.

Banks should also identify, measure, evaluate, monitor, report, and manage concentrations within and between risk types associated with climate-related financial risks. For example, banks could use metrics or heatmaps to assess and monitor concentration of exposure to geographies and sectors with higher climate-related risk.

Risk mitigation options may include adjusting credit underwriting criteria, deploying targeted client engagement, or imposing loan limitations or restrictions such as shorter-tenor lending, lower loan-to-value limits, or discounted asset valuations. Banks could also consider setting limits on or applying appropriate alternative risk mitigation techniques to exposures to companies, economic sectors, geographical regions, or segments of products and services that do not align with their business strategy or risk appetite.

Comprehensive Management of Market, Liquidity, Operational and Other Risks (Principles 9-11)

Principle 9 addresses market risk, requiring banks to understand how climate-related risk drivers could affect the value of financial instruments in their portfolios, evaluate potential losses and increased volatility, and establish effective processes to control or mitigate associated impacts. Given the specific characteristics of market risk, analysis of sudden shock scenarios could serve as a useful tool, featuring variation in liquidity across assets exposed to climate-related risk and variation in the speed at which exposures could be closed out.

Principle 10 focuses on liquidity risk, requiring banks to assess the impacts of climate-related financial risks on net cash outflows (e.g., increased drawdowns of credit lines, accelerated deposit withdrawals) or the value of assets comprising liquidity buffers. Where material and appropriate, banks should incorporate these impacts into their calibration of liquidity buffers and liquidity risk management frameworks.

Principle 11 addresses operational risk and other risks, requiring banks to assess the impact of climate-related risk drivers on their operations and ability to continue providing critical operations. Banks are expected to analyze how physical risk drivers can impact business continuity and take material climate-related risks into account when developing business continuity plans. Banks should also assess the impact of climate-related risk drivers on other risks such as strategic, reputational, regulatory compliance, and liability risk, taking such risks into account as part of risk management and strategy-setting processes where material.

Scenario Analysis (Principle 12)

Principle 12 establishes requirements for climate scenario analysis and stress testing. Where appropriate, banks should use scenario analysis to assess the resilience of their business models and strategies to a range of plausible climate-related pathways and determine the impact of climate-related risk drivers on their overall risk profile. These analyses should consider physical and transition risks as drivers of credit, market, operational, and liquidity risks over a range of relevant time horizons.

The objectives of climate scenario analysis should reflect the bank's overall climate risk management objectives as set by its board and senior management. These could include: (i) exploring the impacts of climate change and the transition to a low-carbon economy on the bank's strategy and business model resiliency; (ii) identifying relevant climate-related risk factors; (iii) measuring vulnerability to climate-related risks and estimating exposures and potential losses; (iv) diagnosing data and methodological limitations in climate risk management; and (v) informing the adequacy of the bank's risk management framework, including risk mitigation options.

Scenario analysis should reflect relevant climate-related financial risks for banks, including physical and/or transition risks relevant to the bank's business model, exposure profile, and business strategy. Scenarios should cover a range of plausible pathways, with banks considering potential benefits and limitations of selected scenarios and assumptions (e.g., balance sheet assumptions).

Banks should build sufficient capacity and expertise to conduct climate scenario analysis proportionate to their size, business model, and complexity, with larger and more complex banks expected to have more advanced analytical capability. Scenario analysis should employ a range of time horizons, from short- to long-term, to address different risk management objectives. Shorter timeframes may analyze the crystallization of risk within typical business planning horizons at lower uncertainty levels, while longer timeframes (carrying higher uncertainty) may evaluate the resiliency of strategies and business models to structural changes in the economy, financial system, or distribution of risks.

The field of climate scenario analysis is highly dynamic, and practices are expected to evolve rapidly as climate science advances. Climate scenario models, frameworks, and results should be subject to challenge and regular review by internal and/or external experts and independent functions.

Prudential Regulatory and Supervisory Requirements (Principles 13-15)

Principle 13 requires supervisors to determine that banks' incorporation of material climate-related financial risks into their business strategies, corporate governance, and internal control frameworks is sound and comprehensive. Supervisors should determine that roles and responsibilities for climate-related financial risks, including for the board and senior management, are clearly assigned, adequate, and properly documented in relevant policies, procedures, and controls.

Supervisors should assess the effectiveness of board and senior management oversight and verify that they receive accurate and appropriate internal reporting on material climate-related financial risks. Supervisors should maintain sufficiently frequent contact with board and senior management to develop understanding and assess the bank's long-term approach to addressing climate-related financial risks in a forward-looking manner, challenging assumptions made in setting strategies and business models where necessary.

Principle 14 requires supervisors to determine that banks can adequately identify, monitor, and manage all material climate-related financial risks as part of their assessments of banks' risk appetite and risk management frameworks. Supervisors should review the extent to which banks regularly assess the materiality of climate-related financial risks, supported by appropriate key risk indicators and risk-mitigating measures.

Supervisors should assess the extent to which material climate-related financial risks are included in banks' risk management frameworks and risk appetite, along with appropriate processes and procedures to identify, monitor, and manage such risks. This includes ensuring banks' risk management frameworks account for all material climate-related financial risks and assessing whether banks' data aggregation capabilities and internal reporting practices can facilitate identification and reporting of climate-related risk exposures, concentrations, and emerging risks, as well as banks' ability to deploy a range of risk management approaches.

Principle 15 requires supervisors to determine the extent to which banks regularly identify and assess the impact of climate-related risk drivers on their risk profile and ensure that material climate-related financial risks are adequately considered in their management of credit, market, liquidity, operational, and other types of risk. Supervisors should determine that, where appropriate, banks apply climate scenario analysis.

Supervisors should ensure that banks consider a range of mitigation options to manage and control material climate-related risks. Supervisors should also determine that banks consider climate-related financial risks assessed as material over relevant horizons within their internal capital and liquidity adequacy assessments (ICAAP and ILAAP). Where appropriate,

supervisors should determine that banks have in place a scenario analysis program proportionate to their size, business model, and complexity to assess the resilience of their business models and strategies to a range of plausible climate-related outcomes, reviewing and challenging model assumptions, methodologies, and results as part of the assessment.

Responsibilities, Powers and Functions of Supervisors (Principles 16-18)

Principle 16 requires supervisors to utilize an appropriate range of techniques and tools in conducting supervisory assessments of banks' management of climate-related financial risks and adopt adequate follow-up measures in case of material misalignment with supervisory expectations. Supervisors should set expectations in a manner proportionate to the nature, scale, and complexity of relevant banks' activities. To foster cross-border collaboration, home and host supervisors of cross-border banking groups should share information related to the climate risk resilience of banks and banking groups, leveraging existing frameworks for sharing information and undertaking collaborative work.

Principle 17 requires supervisors to ensure they have adequate resources and capacity to effectively assess banks' management of climate-related financial risks. Supervisors should take regular stock of existing skills and projected requirements, taking into account evolving market practices and supervisory practices, and take timely measures to build adequate expertise in identified skill sets. Where aspects of climate-related risk assessments are outsourced, supervisors should maintain appropriate knowledge to ensure the results of outsourced analysis are credible and realistic.

Principle 18 requires supervisors to consider using climate-related risk scenario analysis, including stress testing, to identify relevant risk factors, size portfolio exposures, identify data gaps, and inform the adequacy of risk management approaches. Where appropriate, supervisors should consider disclosing the findings of these exercises.

Supervisors should clearly articulate their specific objectives for supervisory climate scenario analysis, including stress testing, which could include:

- Exploring the impact of climate change and the transition to a low-carbon economy on supervised banks' strategies and business model resiliency
- Identifying and assessing relevant climate-related risk drivers affecting individual banks or the banking system
- Facilitating information sharing and identifying common data and methodological gaps and limitations in climate-related risk management
- Informing the adequacy of supervised banks' risk management frameworks, including risk mitigation options.

In designing scenario analysis, supervisors should consider material climate-related financial risks, including physical and transition risks, accounting for the nature, scale, and complexity of banks within their jurisdiction. Scenarios should incorporate a range of plausible climate pathways, with supervisors considering a range of time horizons from short- to long-term depending on the objectives of the exercise.

Regulatory Capital and Accounting Treatment for Physical Risk Swaps

Positioning PRS within the Basel framework

Physical Risk Swaps are not a new risk category in regulatory terms; they are derivatives whose cashflows transmit *physical* risk drivers into the familiar channels of credit, market, counterparty and liquidity risk. This aligns directly with the Basel Committees view that climate risks act as risk drivers of existing categories, rather than a separate silo. PRS, therefore, sit naturally inside the existing three-pillar architecture: they change the structure and distribution of credit and market risk on mortgage and property books, and they introduce counterparty and CVA risk to the hedging bank, but they do not require an entirely new capital regime.

From a supervisory perspective, the key question is not whether PRS are “green” or “climate” instruments, but whether they provide effective risk transfer of clearly identified physical hazards, supported by robust data, modelling and legal documentation. When those conditions are met, the same logic that justified capital relief for CDS portfolios can be applied to PRS hedges on physical risk concentrations in mortgage and CRE portfolios.

Pillar 1: credit, market and counterparty capital

Under Pillar 1, capital for PRS arises in three main places:

- Credit risk on the underlying exposures. A bank that buys PRS protection on a flood-exposed mortgage book does not remove the loans from its balance sheet, but it can, in principle, recognise lower loss given default (LGD) or expected loss (EL) if PRS payouts are contractually and operationally aligned with collateral impairment. In an internal-ratings based (IRB) setting, this would mean explicitly modelling PRS protection in LGD and exposure-at-default (EAD) assumptions, subject to supervisory approval.
- Counterparty credit risk and CVA on the PRS. The PRS itself is an over-the-counter derivative with a protection seller as counterparty, so it attracts counterparty credit risk capital and a CVA charge in exactly the same way as a CDS. The hedging bank reduces physical risk on its mortgage collateral, but simultaneously takes on exposure to the solvency and liquidity of the PRS seller and, where relevant, the central counterparty once PRS becomes clearable.
- Market risk on the PRS position. Changes in hazard curves, trigger probabilities and market PRS spreads generate mark-to-market volatility that feeds into the trading book capital framework. In practice, if PRS are booked in the trading book alongside CDS and interest rate

swaps, they will be subject to the standardised or internal models-based market risk charges, using hazard-rate shocks analogous to spread shocks.

In this configuration, capital relief arises if the reduction in credit risk capital on the hedged loan portfolio (through lower EL/LGD or lower stress losses) exceeds the sum of counterparty, CVA and market risk capital on the PRS positions. This mirrors the economics of early CDS capital trades, but with physical risk replacing corporate default risk as the primary driver.

Pillar 2: supervisory overlays and climate-specific expectations

Pillar 2 has become the main locus for supervisory expectations on climate and physical risk, particularly through BCBS 530 and jurisdiction-specific guidance from the ECB, PRA and others. Supervisors now expect banks to:

- Integrate material physical risks into their ICAAP and capital planning horizons, including the impact of floods and other hazards on collateral values, default patterns and concentration risk.
- Use scenario analysis and climate-related stress testing to explore severe but plausible combinations of hazard events, insurance withdrawal and macro-financial feedbacks, even where precise quantification remains uncertain.
- Demonstrate credible model risk governance over physical risk models and vendor data, recognising multiple layers of uncertainty in the flood model stack and the potential for error amplification.

Within this setting, PRS can serve as a Pillar 2 mitigation tool. Where a bank can show that a PRS hedge materially reduces losses under supervisory climate scenarios—for example, by transferring a defined share of flood loss distributions to a well-capitalised protection seller—supervisors have a basis to reduce Pillar 2 add-ons linked to those specific physical risk concentrations. The capital logic is identical to other recognised risk mitigants: the residual risk after PRS hedging, not the gross unhedged exposure, should determine the additional capital buffer.

However, supervisors will expect several conditions before accepting PRS for meaningful Pillar 2 benefit:

- Clear mapping from gauge-level triggers and hazard curves to portfolio-level loss distributions in the bank's models.
- Evidence that PRS payouts are timely and reliably captured in capital projections and stress tests, not treated as ex-post management actions.
- Robust contractual enforceability under ISDA documentation, including clarity about trigger verification, dispute resolution and basis risk.

Pillar 3: disclosure and market discipline

Pillar 3 climate and physical risk disclosures have, so far, been dominated by qualitative descriptions, scenario narratives and high-level metrics, with limited direct linkage to capital and hedging actions. PRS introduce an opportunity to make disclosures more concrete and decision-relevant.

Banks can disclose:

- The notional amount of physical risk exposure hedged through PRS, broken down by geography, hazard type and portfolio segment (e.g. retail mortgages vs CRE).
- The proportion of flood-exposed collateral subject to PRS hedging, and the corresponding reduction in tail losses under internal or supervisory scenarios.
- Qualitative explanations of PRS design features—triggers, hazard curves, basis risk management—that underpin the bank's view of risk transfer effectiveness.

These disclosures allow investors and supervisors to see physical risk management as a capital-relevant activity, rather than an abstract ESG narrative, and they reinforce the linkage between climate risk, hedging strategy and capital resilience.

Accounting treatment and hedge accounting considerations

From an accounting perspective, PRS are derivatives that will typically be measured at fair value through profit or loss, consistent with other swaps documented under ISDA. The hedged items—mortgages, CRE loans or portfolios of property-backed exposures—may sit at amortised cost, fair value through other comprehensive income, or fair value through profit or loss depending on business model and classification choices.

This creates several practical challenges:

- Timing mismatch. PRS payouts are contingent on physical events and may be infrequent, while changes in hazard expectations and spread levels generate continuous mark-to-market volatility on the PRS leg. Loan impairment models recognise expected credit losses that include physical risk, but the linkage to PRS fair value changes is indirect and may not be perfectly aligned in time.
- Basis risk and effectiveness. For hedge accounting under IFRS 9, a bank must demonstrate that changes in the PRS fair value offset changes in the hedged risk on the loan or portfolio within prescribed effectiveness bounds. Because PRS typically reference gauge-level triggers and hazard curves, while loan losses depend on property-level damage and borrower behaviour, there is inherent basis risk to be documented and justified.
- Designation choices. Banks may choose between fair value hedges of specific risk components (e.g. the flood-related element of expected credit losses on a defined portfolio) and cash flow

hedges of variability in expected loss provisions or insurance costs. In both cases, careful documentation of the hedged risk, the hedging relationship and the method for measuring effectiveness is essential.

Where hedge accounting is not pursued—or cannot be supported with sufficient evidence of effectiveness—PRS positions will introduce P&L volatility as hazard expectations and spreads move. Management will then need to explain to boards and investors why this volatility represents economically rational risk transfer rather than speculative activity, and how it interacts with expected credit loss movements in the loan book.

In all cases, the accounting treatment must remain consistent with the core premise of the book: PRS are banking instruments that translate physical risk into banking value using the same legal, operational and accounting machinery that already exists for credit and interest rate derivatives. The challenge is not to invent new rules, but to apply existing accounting and capital frameworks to a new class of risk driver with clarity, transparency and disciplined model governance.

Chapter 17: Climate Stress Testing

The Proliferation of the Institutional Landscape Problem:

“Excuse me but are you the NGFS?”

“Nah we are the NZBA and we hate the CCCA and the NZAOA and the SBFN and the SBTi and the NZBA”

“What? We are the NZBA!”

“I thought we were the GFANZ! Where are they?”

“Over there in between the PRB and SIF Splitters!”

- Ode to Monte Python Life of Brian

The climate stress testing landscape resembles a crowded theatre where everyone has something important to say, yet nobody quite agrees on the script. For banking institutions attempting to navigate physical and transition risks, this cacophony of guidance presents both opportunity and confusion. Understanding who does what—and why so many bodies exist in the first place—proves essential for implementing practical stress testing frameworks that satisfy regulatory expectations whilst genuinely informing risk management decisions.

By 2025, a bank seeking to conduct climate stress testing faces guidance from the Financial Stability Board, scenario frameworks from the Network for Greening the Financial System, methodological recommendations from the United Nations Environment Programme Finance Initiative, sector-specific standards from the Basel Committee on Banking Supervision, supervisory expectations from national regulators, and academic methodologies from institutions including Oxford and the London School of Economics. This proliferation is not accidental—each body serves distinct functions within a complex ecosystem, attempting to translate climate science into financial risk assessment but on occasion does feel all of these bodies are vying for relevance.

The fundamental challenge stems from climate risk's inherent characteristics: forward-looking nature extending decades beyond traditional stress test horizons, deep uncertainty regarding both physical manifestations and policy responses, cross-border transmission mechanisms that transcend national supervisory frameworks, and interconnections spanning the entire financial system rather than concentrating in specific sectors. No single institution possesses the mandate, expertise, or operational capacity to address all these dimensions simultaneously. The resulting division of labour, whilst creating apparent redundancy, reflects genuine functional specialisation.

The Financial Stability Board coordinates at the apex, translating G20 mandates into actionable frameworks for member jurisdictions. Its 2021 Climate Roadmap established four pillars—firm-level disclosures, data availability, vulnerabilities analysis, and regulatory tools—providing the strategic architecture within which other bodies operate. The FSB's January 2025 framework for

assessing climate-related vulnerabilities represents the culmination of this coordination function, synthesising work from multiple standard setters into coherent monitoring approaches applicable across jurisdictions.

The NGFS serves as the primary scenario developer, convening central banks and supervisors to produce harmonised climate scenarios that financial institutions can employ for stress testing. Without such coordination, each institution would develop bespoke scenarios, rendering results incomparable and supervisory assessment impossible. The NGFS scenarios—Current Policies, Delayed Transition, Net Zero 2050, and variations thereof—have become the de facto standard, despite limitations we shall examine presently.

UNEP FI occupies a distinctive niche as convener and methodology developer, bridging climate science and financial practice through collaborative frameworks developed with industry participation. Its credit risk stress testing methodology, developed through 2018 pilots with sixteen global banks, represents the most widely adopted approach for translating climate scenarios into corporate credit risk assessments. The organisation's role extends beyond mere technical guidance to establishing voluntary principles—Principles for Responsible Banking, Principles for Sustainable Insurance—that shape industry practice even where regulatory mandates remain absent.

Sector-specific standard setters—the Basel Committee, International Association of Insurance Supervisors, and International Organization of Securities Commissions—translate these frameworks into supervisory expectations tailored to banking, insurance, and securities markets. The Basel Committee's work on climate-related financial risks, published iteratively since 2021, provides supervisory principles rather than prescriptive rules, recognising that methodologies remain developmental and jurisdictional circumstances vary considerably.

National supervisors then implement programmes reflecting both international frameworks and domestic priorities. The Bank of England's Climate Biennial Exploratory Scenario (CBES), European Central Bank's climate risk stress test, US Federal Reserve's pilot exercises, and programmes from OSFI, APRA, HKMA, and others each adapt global frameworks to specific banking systems, risk profiles, and policy contexts. This jurisdictional variation frustrates standardisation efforts yet allows methodological innovation through natural experimentation.

Academic research centres—particularly Oxford's Sustainable Finance Group, the LSE's Grantham Research Institute, and various central bank research functions—develop and validate underlying methodologies. The TRISK framework from Oxford, CRISK measures from NYU, and enhanced stress testing approaches from LSE represent cutting-edge developments that gradually filter into supervisory practice. This research function proves critical because climate stress testing remains methodologically immature, with fundamental questions regarding appropriate time horizons, scenario construction, and impact transmission still contested.

The United Nations Architecture

Within this ecosystem, UN-convened initiatives occupy peculiar but influential positions. UNEP FI, established in 1992, pioneered engagement between the finance sector and sustainability

considerations long before climate risk entered mainstream risk management discourse. By 2023, more than 400 financial institutions managing assets exceeding USD 80 trillion had joined as members, creating a constituency for voluntary action that often precedes—and shapes—eventual regulatory requirements.

The organisation's most significant contributions emerge through collaborative framework development. The Principles for Responsible Banking, launched in 2019, provide a comprehensive framework for banks to align with the UN Sustainable Development Goals and Paris Agreement. By 2025, over 345 banks globally had signed these principles, committing to set and publish targets in at least two significant impact areas within four years, with boards playing oversight roles. Whilst voluntary, these commitments create market expectations and competitive dynamics that drive practice forward independent of regulatory mandate.

The Principles for Sustainable Insurance, established earlier, similarly shape insurer behaviour regarding environmental, social, and governance considerations. The principles require embedding ESG issues into decision-making, working with clients on ESG risk management, collaborating with stakeholders to advance action, and demonstrating accountability through disclosure. Multiple major insurers—HSBC Life, Lancashire Group, MAPFRE—publish annual progress reports against these principles, creating transparency mechanisms that influence both investor expectations and supervisory focus.

Perhaps most ambitiously, UNEP FI convened the Net-Zero Banking Alliance in 2021, bringing together banks committed to aligning lending and investment portfolios with net-zero emissions by 2050. At its peak, the alliance included major global institutions collectively managing trillions in assets. The October 2024 progress report demonstrated significant momentum, with 97% of participating banks having set initial sectoral targets and nearly two-thirds having published transition plans.

Following policy shifts in the United States during late 2024 and early 2025, and mounting legal challenges regarding potential antitrust implications, major banks began withdrawing. Goldman Sachs departed in December 2024, followed rapidly by Wells Fargo, Citigroup, Bank of America, Morgan Stanley, five major Canadian banks, and five Japanese institutions. By October 2025, the alliance ceased operations entirely—a stark reminder that voluntary commitments prove vulnerable to political winds and legal uncertainty.

This fallback demonstrates that voluntary frameworks, whilst useful for pioneering approaches and building consensus, ultimately require regulatory underpinning for durable implementation. The knowledge and methodologies developed through the NZBA (Net Zero Banking Alliance)—particularly regarding financed emissions measurement, sectoral decarbonisation pathways, and transition plan frameworks—persist within participating institutions and inform emerging regulatory requirements. The collaborative learning process created lasting value even as the formal alliance dissolved.

UNEP FI's climate risk stress testing methodology for corporate credit, developed through intensive collaboration with participating banks and climate scientists, provides step-by-step guidance for translating climate scenarios into probability of default adjustments. Its approach—

using representative borrowers, expert assessments, risk factor pathways, and credit quality indices—has been adopted by institutions including the Bank of Canada and forms the foundation for many banks' internal climate risk assessment frameworks.

The organisation's 2024 Climate Risk Landscape Report provides analysis of climate risk tool markets, identifying persistent challenges including data quality, sectoral coverage limitations, and accessibility imbalances affecting emerging markets. This monitoring function—tracking both tool development and implementation barriers—informs both vendor evolution and regulatory expectations regarding what institutions can reasonably achieve given current capabilities.

From a banking institution's perspective, navigating this landscape requires understanding not merely what each body recommends but how recommendations interrelate and where genuine requirements exist versus best practice guidance. The distinction proves critical: supervisory stress test participation constitutes a regulatory obligation; TCFD-aligned disclosure increasingly carries legal force in major jurisdictions; NGFS scenarios represent expected but not mandated inputs; UNEP FI methodologies provide useful frameworks without regulatory compulsion.

Effective navigation starts with mapping regulatory requirements in relevant jurisdictions. For EU banks, this includes the ECB's supervisory expectations on climate risk, mandatory TCFD-aligned disclosure under the Corporate Sustainability Reporting Directive, scenario analysis requirements for larger institutions, and forthcoming climate-specific Pillar 2 requirements.

UK banks face Bank of England expectations through CBES participation, Prudential Regulation Authority supervisory statements, and enhanced disclosure requirements under UK law. US banks navigate Federal Reserve supervisory guidance, emerging SEC climate disclosure rules (under legal review), and state-level requirements varying considerably across jurisdictions.

International frameworks then overlay these jurisdictional requirements. Basel Committee principles on climate-related financial risks establish supervisory expectations that national authorities implement through local frameworks. IOSCO recommendations on sustainability disclosures influence securities regulators globally. IAIS guidance shapes insurance supervisory approaches across member jurisdictions. These international standards rarely create direct legal obligations but establish benchmarks against which supervisory authorities assess institutional practice.

Voluntary frameworks and industry initiatives occupy a third tier—influential but not mandatory. UNEP FI principles, CDP disclosure, Science Based Targets initiative commitments, and similar programmes create market expectations and competitive dynamics that drive practice even in the absence of a regulatory mandate. Banks publishing TCFD reports commit to scenario analysis; this commitment, whilst initially voluntary, becomes a reputational feature that shareholders and stakeholders monitor.

Academic methodologies represent the frontier—not yet incorporated into supervisory expectations but likely to influence future developments. Understanding emerging approaches

like TRISK transition risk modelling, enhanced stress testing frameworks capturing non-linearities, and nature-related risk assessment methodologies positions institutions to anticipate rather than merely react to evolving regulatory expectations.

The key insight from this institutional landscape is that climate stress testing requirements emerge through complex interaction between regulatory mandate, supervisory expectation, market practice, and methodological development. No single "correct" approach exists; rather, institutions must synthesise guidance from multiple sources into coherent frameworks that satisfy regulatory obligations whilst genuinely informing risk management and strategic decisions.

From Global Frameworks to Bank Implementation

This institutional proliferation creates particular implementation challenges. Banks cannot simply adopt NGFS scenarios, apply UNEP FI methodology, implement TRISK transition risk models, and report using FSB metrics while simultaneously satisfying Basel Committee principles, national supervisory expectations, and TCFD disclosure requirements. The frameworks overlap but do not align perfectly; scenarios from different sources produce divergent results; methodologies require adaptation to institutional circumstances; and resource constraints force prioritisation decisions.

For practical implementation, regulatory requirements obviously take precedence—supervisory stress test participation, mandatory disclosure, and Pillar 2 capital expectations cannot be ignored. Beyond these mandatory elements, institutions must assess which voluntary frameworks provide genuine decision-usefulness versus mere compliance burden. UNEP FI methodology might inform corporate credit risk assessment; NGFS scenarios provide standardised inputs enabling peer comparison; FSB metrics facilitate monitoring of evolving exposures.

The challenge for banks facing a huge cost for implementation lies in building frameworks that satisfy multiple audiences through common foundations rather than creating parallel processes for each requirement. Core climate scenario analysis capabilities can serve supervisory stress tests, TCFD disclosure, internal risk appetite setting, and strategic planning if properly designed.

Emissions measurement systems support financed emissions reporting, transition risk assessment, sectoral target setting, and client engagement. Risk factor databases enable both bottom-up portfolio analysis and top-down scenario application.

This integration proves particularly important because climate risk capabilities remain resource-intensive to build and maintain. Institutions cannot afford separate teams, systems, and processes for each regulatory or voluntary framework. Efficient design requires identifying common requirements across frameworks, building reusable components, and adapting outputs to specific reporting needs rather than creating bespoke processes for each audience.

Connection to Supervisory Capital

The institutional landscape ultimately matters because it determines how climate risks translate into supervisory capital requirements—the subject of Chapter 8 and the critical question for banking institutions. Climate stress testing frameworks, regardless of their institutional origin, aim to quantify potential losses that might necessitate additional capital buffers under Pillar 2 of the Basel framework.

The chain of reasoning proceeds through several steps: climate scenarios project economic conditions; these conditions affect borrower creditworthiness and asset values; changes in credit quality translate into probability of default and loss given default adjustments; these credit parameters determine expected and unexpected losses; and losses potentially require capital increases to maintain target solvency ratios.

Each step involves methodological choices, parameter calibration, and expert judgement—discretion that the institutional landscape attempts to constrain through common frameworks and supervisory guidance.

Understanding this chain clarifies why so many bodies contribute to climate stress testing development. The FSB coordinates because climate risks pose systemic threats requiring global cooperation. The NGFS develops scenarios because banks need common economic projections. UNEP FI provides credit risk methodologies because translating scenarios into credit parameters requires domain expertise. The Basel Committee establishes principles because climate risks ultimately flow through to capital adequacy. National supervisors run programmes because only they can assess institution-specific vulnerabilities and impose capital requirements.

The following sections examine specific methodologies that have emerged from this institutional ecosystem—beginning with UNEP FI's credit risk approach.

UNEP FI Credit Risk Stress Testing Methodology

The challenge of translating climate scenarios into credit risk assessments has vexed banking institutions since climate risk first entered mainstream risk discourse. Traditional stress testing methodologies—developed for economic recessions, market crashes, and liquidity crises—proved inadequate for climate risks extending decades into the future, involving uncertain policy responses, and requiring sector-specific technological understanding beyond typical banking expertise. The UNEP FI credit risk stress testing methodology, developed through collaboration between sixteen global banks, climate scientists, and risk management consultancies during 2018, represents the first systematic attempt to bridge this gap at scale.

Origins and the TCFD Challenge

The methodology emerged directly from the Financial Stability Board's establishment of the Task Force on Climate-Related Financial Disclosures in 2015. The TCFD's final recommendations in June 2017 created a quandary for banking institutions: scenario analysis requirements under the Strategy pillar demanded that organisations describe the resilience of

their strategy under different climate scenarios, including a 2°C or lower pathway. Yet no established frameworks existed for conducting such analysis on corporate credit portfolios comprising thousands of heterogeneous borrowers across diverse sectors.

The sixteen banks that formed the UNEP FI pilot group—including ANZ, Barclays, BBVA, BNP Paribas, Citigroup, and others—faced immediate practical problems. Their risk management functions possessed established capabilities for assessing credit risk under economic scenarios: GDP growth, unemployment rates, interest rate movements, and property price fluctuations. Climate scenarios provided none of these familiar variables. Instead, they offered carbon price trajectories, technology deployment pathways, physical temperature increases, and precipitation pattern changes—variables with no obvious connection to corporate balance sheets or income statements.

Moreover, climate scenarios operated on fundamentally different timescales. Traditional stress tests examine three to five year horizons—periods over which macroeconomic relationships remain relatively stable and historical patterns provide guidance. Climate scenarios extended to 2050 or 2100, timeframes over which technological change, policy evolution, and physical climate manifestations could render historical relationships meaningless. The very concept of a "scenario" shifted from "adverse but plausible economic conditions" to "possible future states of the world conditional on policy assumptions and technological development."

The pilot group therefore required methodology that could achieve several objectives simultaneously: translate climate scenario variables into impacts on corporate financial performance; maintain consistency with existing credit risk frameworks and Basel capital requirements; provide granularity sufficient for portfolio-level assessment yet remain implementable with available data; and produce outputs comparable across institutions despite varying portfolio compositions and risk measurement approaches.

The resulting methodology, published in stages through 2018 and 2019, achieved these objectives through three key innovations: the representative borrower approach for establishing sector-level relationships; risk factor pathways providing structured transmission mechanisms from climate variables to financial impacts; and the climate credit quality index aggregating multiple risk factors into single metrics compatible with existing credit models.

The Representative Borrower Approach

Traditional bottom-up credit modelling requires detailed financial projections for each borrower—hundreds or thousands of companies in typical corporate portfolios. Extending such analysis decades into the future under multiple climate scenarios presents insurmountable data and resource requirements. The representative borrower approach circumvents this problem through intelligent aggregation whilst preserving meaningful differentiation.

The methodology segments the portfolio into homogeneous groups based on sector, geography, and size. For each segment, a representative borrower is identified—either an actual company typical of the group or a synthetic entity with average characteristics. These representative borrowers need not be portfolio constituents; indeed, using publicly traded companies with

available data often proves more practical than attempting detailed analysis of privately held borrowers.

Expert assessments then evaluate how specific climate scenarios would affect each representative borrower's financial performance. These experts—typically combining climate scientists, sector specialists, and credit analysts—assess impacts through structured workshops rather than pure quantitative modelling.

The approach recognises that forecasting individual company performance thirty years hence involves irreducible uncertainty; expert judgement informed by scenario narratives provides more defensible estimates than spurious quantitative precision.

The Climate Credit Quality Index

We are going to do a slight detour to describe how credit impacts are calculated based on the increase cost of either carbon tax and capital expenditure to transition away from high emissions. The physical risk approach is an extension of this, so we need to cover the Climate Credit Quality Index. These IPCC pathways must ultimately translate into credit risk metrics compatible with existing risk frameworks—specifically, probability of default adjustments that feed into expected loss calculations and capital requirements. The climate credit quality index provides this translation, aggregating the disparate RFPs into single metrics that modify baseline credit assessments.

The mathematical formulation centres on expressing net income—the ultimate determinant of credit quality—as a function of the four risk factors. Conceptually:

Net Income = Baseline Revenue - Direct Carbon Costs - Indirect Cost Increases - Adaptation CapEx + Revenue Growth Opportunities

Each term connects to specific risk factors: baseline revenue and opportunities link to the revenue RFP; direct carbon costs flow from emissions × carbon price; indirect costs relate to input price changes; adaptation CapEx follows scenario-specific investment requirements. The relationship between net income and probability of default follows from standard structural credit models—declining net income weakens debt service capacity and increases default risk.

The climate credit quality index formalises this relationship through a scoring function that weights each risk factor by its sensitivity to scenario variables and the borrower's exposure to each factor. The index takes the form:

Climate Index = $\Sigma(\text{Sensitivity}_i \times \text{Risk Factor}_i)$

where each risk factor represents percentage changes in revenues, costs, or capital requirements under the scenario relative to baseline, and sensitivities reflect how changes in each factor affect overall creditworthiness. These sensitivities are calibrated through the expert assessment process—a 10% revenue decline might increase default probability more severely for a highly leveraged firm than for one with conservative capital structure.

The credit risk adjustment then applies through standard probability of default models. The Basel Internal Ratings-Based approach models PD using a Vasicek-type single factor framework where default occurs when a firm's asset value falls below its debt obligations. The climate credit quality index enters this framework as an additional systematic factor affecting asset values:

$$\text{Scenario-adjusted PD} = \Phi(\Phi^{-1}(\text{Through-cycle PD}) + \beta \times \Phi^{-1}(\rho \times \text{Climate Index}))$$

where Φ represents the standard normal cumulative distribution function, β captures the scaling factor calibrated to representative borrower assessments, and ρ represents the asset correlation parameter already embedded in Basel IRB formulas. This formulation preserves consistency with existing capital frameworks—institutions need not abandon tested credit models but rather incorporate climate scenarios as additional risk factors.

The strength of this approach lies in its parsimony. It requires no new parameters beyond what institutions already maintain for Basel compliance: through-cycle PDs by rating grade, asset correlations by exposure type, and loss-given default estimates. The climate-specific inputs—risk factor sensitivities and scenario impacts—layer onto this existing infrastructure rather than requiring parallel systems.

The UNEP FI methodology initially focused exclusively on policy responses and technological change required to limit warming. Physical risk—the impacts of changing climate conditions on borrowers' operations and revenues—required subsequent methodological extension.

“Physical Climate-related Stress has proved challenging because physical impacts manifest locally rather than globally, vary by specific hazard type, and require granular geographic data often unavailable at the portfolio scale.” - David Kelly, MKM

The physical risk extension to transition risk employs similar conceptual structure: representative borrowers in vulnerable sectors and regions, risk factor pathways capturing physical impacts, and credit quality adjustments flowing through to PD modifications. However, the risk factor pathways differ fundamentally. Rather than carbon prices and energy costs, physical risk pathways capture:

- Asset damage from acute events (floods, storms, wildfires) reduces the value of productive capital and requires reconstruction investment. A manufacturing facility in a flood-prone location might face expected annual losses of 2% of asset value, with one-in-20-year events causing 15-20% damage. These losses directly reduce profitability and potentially impair collateral values.
- Business interruption when physical events disrupt operations, eliminating revenues whilst fixed costs continue. A port closed for three weeks following a hurricane loses not only the direct revenue but potentially customers who shift to alternative logistics routes. Supply chain disruptions extend impacts beyond directly affected facilities to downstream customers and upstream suppliers.

- Chronic stress from gradually changing conditions—water stress affecting agricultural yields or industrial processes; temperature increases requiring additional cooling or changing growing seasons; sea level rise threatening coastal facilities over multi-decade periods. These chronic impacts prove harder to model than acute events because effects accumulate gradually and trigger adaptive responses that partially offset damages.
- Input availability and costs when climate impacts affect key inputs. Water stress increases costs for water-intensive industries; heat stress reduces outdoor worker productivity; changing precipitation patterns affect agricultural input costs throughout food value chains. These indirect effects can exceed direct impacts for many sectors.

Implementing physical risk analysis requires geographic granularity absent from policy risk assessment. Carbon prices affect steel manufacturers globally (with regional variations); flood risk affects specific facilities based on precise location relative to water bodies and topography. Banks therefore need facility-level location data for material exposures—information not typically captured in credit systems designed around legal entity headquarters rather than operational geography.

The methodology suggests tiered approaches: conducting detailed facility-level analysis for the largest or most vulnerable exposures; performing sector- and region-level analysis for medium-sized exposures using representative borrower assumptions about typical geographic distribution; and excluding immaterial exposures from explicit physical risk assessment. This risk-based approach recognises that analysing all exposures remains unfeasible with current data availability.

Physical risk scenarios follow different structure than policy scenarios. Rather than coherent global narratives like NGFS pathways, physical risk requires hazard-specific projections: how do flood return periods change under 2°C versus 4°C warming? How does wildfire risk evolve? What coastal areas become routinely inundated by 2050? These projections come from physical climate models rather than integrated assessment models, and typically require downscaling from global climate projections to regional or local hazard assessments—introducing additional uncertainty.

Integration of policy and physical risk remains methodologically immature. The risks clearly interact—physical damages increase adaptation costs, potentially reducing resources available for other adjustments; aggressive policy mitigation limits eventual physical impacts—but modelling these interactions requires coupling distinct frameworks with different temporal dynamics and geographic scales. Most implementations therefore assess risks separately and acknowledge interaction as qualitative consideration rather than quantifying compound effects.

Limitations and Ongoing Challenges

Despite widespread adoption, the UNEP FI methodology faces recognised limitations that constrain interpretation of results and require careful communication to stakeholders. Understanding these limitations proves as important as understanding the methodology itself.

The representative borrower approach provides efficiency but sacrifices firm-specific precision. Two companies in the same sector may face vastly different challenges based on management quality, technological positioning, and strategic foresight. The methodology captures sector-average impacts but may miss significant winners and losers within sectors—precisely the information most valuable for active portfolio management. Banks using the framework for strategic positioning therefore typically supplement representative borrower analysis with detailed assessments of their largest exposures.

Expert judgement, whilst necessary given fundamental uncertainty, introduces subjectivity and potential bias. Different expert panels assessing identical scenarios can reach different conclusions about impacts on representative borrowers. This variability creates challenges for capital planning because it generates uncertainty about uncertainty—are loss projections uncertain because future scenarios genuinely contain irreducible uncertainty, or because expert assessments lack precision? Distinguishing between these proves difficult.

The methodology's long time horizons—necessarily matching climate scenario horizons of 30+ years—exceed traditional risk management and capital planning timeframes by order of magnitude. Banks planning capital requirements consider three to five year horizons; strategic planning extends perhaps ten years; credit underwriting for corporate loans spans five to seven years.

The 2050 endpoint of standard climate scenarios therefore falls well beyond the horizon relevant for most management decisions. This temporal mismatch raises questions about the practical utility of long-dated scenario results versus their role in satisfying disclosure requirements.

Validation poses perhaps the deepest challenge. Traditional credit models validate against historical default experience; climate models have no historical analogue. We cannot observe actual defaults caused by carbon prices reaching USD 500 per tonne because such prices have never existed. Model validation therefore relies on qualitative assessment (does the logic appear sound?), benchmarking (do results align with other approaches?), and sensitivity analysis (do results respond appropriately to input variations?) rather than traditional backtesting. This fundamentally different validation paradigm requires careful explanation to model validation functions accustomed to quantitative performance metrics.

The methodology produces point estimates for each scenario rather than probability distributions around these estimates. A borrower's PD might be projected as 4.2% in 2040 under a delayed scenario, but this single number masks enormous uncertainty. The true outcome could reasonably range from 1% to 10% depending on how specific technological, policy, and competitive dynamics unfold.

Without probability distributions, users may misinterpret point estimates as precise forecasts rather than central tendencies within broad ranges. Some institutions have developed approaches for quantifying this uncertainty—Monte Carlo simulation around key assumptions, sensitivity analysis across parameter ranges—but such enhancements require substantial additional analytical effort.

Finally, the methodology assumes relatively stable relationships between financial metrics and credit risk. The PD equations calibrated on historical data assume that a 20% profit decline affects default probability similarly regardless of cause—whether recession, competitive pressure, or carbon cost increases. Climate change might violate this assumption if it triggers fundamental restructuring of industries, with high failure rates even among previously strong credits. The framework cannot capture such regime change effects within its mathematical structure.

Evolution and Future Development

UNEP FI continues refining the methodology based on implementation experience and emerging research. The 2024 collaboration with Global Credit Data initiated a global benchmarking survey assessing how banks integrate climate risks into credit risk frameworks, providing insights into diverse implementation approaches and common challenges. Results, shared with participating institutions, highlight data infrastructure investments required for systematic implementation.

Integration with other risk dimensions represents a frontier area. The methodology currently treats climate as standalone risk factor; reality involves climate interacting with other portfolio vulnerabilities. A borrower facing both high cyber risk and climate risk might experience compounded stress exceeding simple addition of individual risks. Methodologies for assessing such interactions remain developmental.

The framework is also extending to additional asset classes beyond corporate credit. Mortgage portfolios require different analytical approaches given long maturities and property-specific collateral. Consumer lending faces indirect climate impacts through household income effects. Sovereign exposures require assessing how climate risks affect government fiscal positions—work now progressing through academic channels and beginning to inform supervisory expectations.

Perhaps most significantly, the focus is gradually shifting from pure risk assessment toward opportunity identification. Early implementations emphasised downside risks—which exposures face greatest losses? Maturing implementations increasingly consider positive exposure opportunities—which sectors benefit from climate change adaptation, and how can banks support beneficial change whilst managing downside risks? This shift from risk avoidance toward risk-informed opportunity pursuit better aligns with banks' fundamental intermediation role.

The UNEP FI methodology, for all its limitations, provided the structured starting point that catalysed widespread bank engagement with climate credit risk. It demonstrated that systematic scenario analysis was feasible despite data gaps and methodological uncertainties. It created common language and concepts enabling peer learning. And it established the foundation upon which subsequent developments—including supervisory stress tests and the loss given default adjustments we examine next—have built.

Loss Given Default in Physical Climate Scenarios

The systematic relationship between probability of default and loss given default represents one of the most pernicious features of credit downturns. During benign economic periods, these parameters appear uncorrelated—some defaults recover 80% whilst others recover 20%, seemingly at random. Economic stress reveals the illusion. As default rates climb during recessions, recovery rates simultaneously deteriorate. Banks discover that elevated default frequencies coincide with reduced recoveries, compounding losses precisely when capital buffers face greatest pressure.

This adverse correlation operates through several channels. Deteriorating conditions that push marginal borrowers into default also reduce collateral values—property prices decline, equipment sells at distressed valuations, inventory liquidates at steep discounts. Simultaneously, bankruptcy proceedings lengthen as courts face capacity constraints, legal fees consume larger proportions of recoverable value, and secondary markets for distressed assets become illiquid. The combination produces systematic LGD increases during periods of elevated default rates.

Physical climate events amplify these dynamics through mechanisms that traditional credit models fail to capture. A major flood does not merely damage individual properties—it simultaneously impairs numerous borrowers' creditworthiness and destroys collateral across entire geographic regions. Insurance gaps transfer losses to borrowers and lenders. Secondary markets for damaged assets evaporate. Recovery timeframes extend as reconstruction capacity becomes constrained. The result: physical climate shocks generate precisely the conditions under which LGD and PD move together most severely.

The Physical Risk Problem

Traditional credit risk models treat LGD either as a constant parameter or as a stochastic variable independent of PD. The Basel Internal Ratings-Based approach requires banks to estimate "downturn LGD"—loss severity expected during periods of high defaults—but provides limited guidance on calibration. Many institutions simply add arbitrary buffers to long-run average LGD: if typical recovery rates average 50%, downturn LGD might be set at 60% or 65% based on historical worst-case observations.

This approach proves doubly inadequate for physical climate risk assessment. First, the historical worst-case observations occurred during financial crises or recessions—periods of economic stress but not physical asset destruction at the scale climate scenarios envision. A 2008-style recession reduces property values 30% through demand collapse and credit tightening; a major flood physically destroys properties, potentially eliminating 100% of collateral value for worst-affected borrowers whilst impairing regional property markets for years.

Second, arbitrary LGD buffers fail to scale appropriately with scenario severity. Adding 15 percentage points to baseline LGD might prove adequate for mild recession scenarios but grossly underestimate losses in severe physical shock scenarios. The relationship between default

rate increases and LGD deterioration follows non-linear dynamics that simple add-ons cannot capture.

Physical climate events create conditions potentially more severe than historical financial stress episodes. Hurricane Katrina in 2005 demonstrated this dynamic: mortgage defaults in affected areas spiked as borrowers faced simultaneous income disruption and property damage.

Collateral values dropped dramatically not only due to market price shifts but also because of physical destruction. Insurance payouts proved inadequate because of coverage gaps and disputes. Recovery rates on defaulted mortgages in the worst-affected areas fell below 30%—worse than during the peak of the financial crisis despite being during generally stable macroeconomic conditions.

The 2011 Thailand floods provided similar evidence from commercial lending. Manufacturing facilities across the central plains suffered complete inundation, destroying equipment and inventory. Borrowers defaulted as revenue ceased whilst debt service obligations continued. Recovery rates on secured lending fell dramatically because collateral—specialised manufacturing equipment—had been submerged for weeks and could not be salvaged. Even where physical equipment survived, secondary markets for flood-damaged machinery proved nonexistent. Lenders recovered 20-30 cents on the pound compared to typical 60-70% recoveries for commercial defaults absent physical destruction.

Australian banks' experience with repeated flooding events in Queensland and New South Wales through 2010-2022 similarly demonstrates the dynamic—individual flood events produced localised spikes in mortgage defaults and depressed recovery rates. Properties in flood-affected areas showed recovery rates 15-20 percentage points below comparable properties elsewhere. The persistence of elevated LGD extended beyond immediate aftermath—properties with recent flood history continued showing impaired recovery rates for 3-5 years as buyers demanded substantial discounts and insurance costs remained elevated.

The European Central Bank's 2022 climate stress test highlighted these dynamics at portfolio scale. Banks reported that incorporating physical risk scenarios without adjusting LGD assumptions produced implausibly optimistic loss projections—default rates increased substantially but total losses remained moderate because constant LGD assumptions implied unchanged recovery rates. When banks applied stressed LGD estimates reflecting collateral impairment and market disruption, projected losses increased 40-60% beyond PD-only adjustments.

Frye-Jacobs LGD Model Overview

Jon Frye of the Federal Reserve Bank of Chicago and Michael Jacobs Jr. of the Office of the Comptroller of the Currency developed a methodology addressing precisely this challenge: deriving scenario-adjusted LGD estimates from through-the-cycle parameters using only information already embedded in Basel IRB frameworks. Published in 2012 and refined through subsequent research, the Frye-Jacobs function provides a parsimonious approach to linking conditional LGD to conditional default rate through a single mathematical relationship.

The model rests on four explicit assumptions that balance theoretical rigour with practical applicability. First, it assumes that greater rates of credit loss accompany greater rates of default—the asymptotic distributions of default and loss are comonotonic, meaning they represent non-decreasing functions of the same random variable. This assumption proves much less restrictive than directly assuming LGD and PD move together; it focuses on the observable quantity (total credit loss) rather than making strong assumptions about the unobservable relationship between default probability and loss severity.

Second, both credit loss and default are assumed to follow two-parameter distributions in the asymptotic portfolio. Frye and Jacobs tested this assumption extensively against more complex multi-parameter specifications and found that available credit loss data cannot support additional parameters—the two-parameter specification proves sufficient for explaining observed patterns.

Third, among possible two-parameter distributions (Beta, Lognormal, Vasicek), the Vasicek distribution is selected based on practical considerations: it has explicit formulas for both cumulative distribution function and inverse, constrains all rates below 100%, and its correlation parameter is already estimated within current credit loss models and Basel IRB frameworks. Importantly, empirical testing showed that different distributional choices produce approximately the same LGD-default relationship, differing principally when default rates are very low—precisely where empirical distinction proves most difficult.

Fourth, the asset correlation parameter (ρ) estimated from default data is assumed to apply equally to the loss distribution. This assumption survived explicit testing—allowing correlations to differ produced no statistically significant improvement—and therefore represents an empirically validated simplification rather than mere convenience.

The resulting mathematical formulation expresses conditional LGD (cLGD) as a function of conditional default rate (cDR):

$$\text{cLGD} = \Phi((\Phi^{-1}(\text{EL}/\text{PD}) - \sqrt{\rho} \times \Phi^{-1}(\text{cDR})) / \sqrt{(1-\rho)})$$

where:

- Φ represents the standard normal cumulative distribution function
- Φ^{-1} its inverse
- EL represents expected loss ($\text{PD} \times \text{ELGD}$ on a through-the-cycle basis)
- ρ represents the asset correlation parameter from Basel IRB formulas.

This formulation has elegant properties. When the conditional default rate equals the through-the-cycle PD (unstressed conditions), the function returns the baseline ELGD—LGD remains at its long-run average. As the conditional default rate exceeds the baseline PD (stressed conditions), the function yields higher LGD estimates, with the magnitude depending on the relationship between baseline parameters. When the conditional default rate drops below the baseline (benign conditions), LGD decreases below the baseline—accurately reflecting the full systematic relationship in both directions.

The model introduces a composite parameter called the LGD Risk Index (k) that encapsulates how baseline PD, ELGD, and correlation jointly determine LGD sensitivity to default rate changes:

$$k = (\Phi^{-1}(\text{EL}) - \sqrt{p} \times \Phi^{-1}(\text{PD})) / \sqrt{(1-p)}$$

This index determines the degree to which LGD responds to changes in default rate. When expected loss equals PD (implying ELGD of 100%), k equals zero and LGD remains constant at 100% regardless of default rate—sensibly, because losses cannot exceed 100%. For typical banking exposures with ELGD between 25% and 50% and PD between 1% and 5%, k typically ranges between 0 and 2, producing moderate positive sensitivity of LGD to default rate.

Research demonstrates that expected loss (EL) has greatest impact on k and therefore on conditional LGD, probability of default represents the next most important factor, and asset correlation has least impact within typical ranges (12-24% for most corporate exposures). Exposures with low ELGD and high PD combinations exhibit higher sensitivity—the model extracts greater "penalty" from such exposures during stress scenarios.

Application to Physical Climate Stress Testing

The Frye-Jacobs methodology integrates naturally into physical climate stress testing frameworks through a straightforward sequence. Physical climate scenarios first translate into credit risk impacts through hazard modelling and borrower vulnerability assessment (as described in earlier chapters), producing scenario-specific probability of default estimates at various future time points.

These conditional PDs—elevated above through-the-cycle baselines due to physical climate impacts—become inputs to the Frye-Jacobs function, which generates corresponding conditional LGD estimates that reflect the systematic relationship between default rates and loss severity.

Consider a mortgage portfolio concentrated in a coastal region with baseline characteristics: through-the-cycle PD of 1.5%, expected LGD of 25%, and asset correlation of 15% (typical for residential mortgages under Basel formulas). Under benign conditions, expected loss equals $1.5\% \times 25\% = 0.375\%$. The LGD Risk Index calculates as:

$$k = (\Phi^{-1}(0.00375) - \sqrt{0.15} \times \Phi^{-1}(0.015)) / \sqrt{(1-0.15)} = 0.85$$

Now consider a physical climate scenario where major flooding affects 30% of properties in the portfolio.

Physical risk assessment using estimates of flood depths, damage functions, and business interruption impacts. This analysis estimates that affected borrowers face elevated default probability due to income disruption, property damage, and insurance shortfalls. Conditional PD increases to 8% for the affected segment. Applying Frye-Jacobs:

$$cLGD = \Phi((\Phi^{-1}(0.00375/0.015) - \sqrt{0.15} \times \Phi^{-1}(0.08)) / \sqrt{(1-0.15)}) = 43\%$$

Loss given default increases from 25% baseline to 43%—a 72% relative increase reflecting the systematic deterioration in recovery rates when default rates spike. The economic logic appears sound: flooding damages properties (reducing collateral value), disrupts local property markets (reducing recoverable value even for undamaged collateral), creates insurance disputes that delay settlements, and overwhelms local construction capacity needed for repairs—all factors reducing eventual recoveries.

For commercial real estate exposures facing similar physical shocks, typical baseline parameters might be PD of 2.5%, ELGD of 35%, and correlation of 20%. Consider a scenario where wildfires destroy or severely damage 20% of properties in a portfolio concentrated in high-risk areas. The physical damage assessment indicates total losses for the worst-affected properties, business interruption for others, and regional market disruption. Under this severe physical scenario where conditional PD reaches 12%, the Frye-Jacobs function produces a conditional LGD of 58%—increasing from 35% baseline by 66%. The larger absolute increase reflects commercial real estate's greater vulnerability to localised property market disruptions and longer workout periods.

These LGD adjustments prove materially consequential for loss calculations and capital requirements. Expected credit loss under a scenario equals:

$$ECL = \text{Conditional PD} \times \text{Conditional LGD} \times \text{EAD}$$

For the mortgage example, ignoring LGD adjustment produces ECL of $8\% \times 25\% = 2.0\%$ of exposure. Incorporating the Frye-Jacobs adjustment increases ECL to $8\% \times 43\% = 3.4\%$ —a 70% increase in projected losses from the same default rate assumption. This difference directly affects scenario-based capital planning and buffer sizing.

Practical Implementation Considerations

Implementing Frye-Jacobs adjustments for physical climate risk requires addressing several technical and operational challenges beyond the mathematical formulation. The first concerns parameter availability and quality. The function requires three baseline parameters: through-the-cycle PD, expected LGD, and asset correlation. Whilst Basel IRB banks maintain these parameters by construction, their granularity may not align perfectly with physical climate stress testing requirements.

Basel IRB parameters typically exist at rating grade level (perhaps 10-15 grades from highest to lowest quality) and broad exposure type level (corporate, SME, residential mortgage, etc.). Physical climate stress testing often requires greater granularity—distinguishing between coastal and inland mortgages, between flood-prone and protected commercial properties, between drought-vulnerable and irrigated agricultural land. Banks therefore face choices: aggregate climate scenarios to match available IRB granularity, or develop sub-segment parameters through further analysis and expert judgement.

Most institutions adopt hybrid approaches. For material exposure concentrations warranting detailed analysis—coastal mortgage portfolios, commercial property in wildfire zones, agricultural lending in drought-prone regions—they develop segment-specific parameters through bottom-up analysis of borrower characteristics and collateral attributes. For smaller or less climate-sensitive exposures, they apply standard IRB parameters recognising this introduces conservatism (likely overstating diversity and understating concentration effects) but remains defensible given materiality thresholds.

Asset correlation parameters present particular challenges because Basel formulas specify correlations as functions of PD for certain exposure types. The Basel corporate formula, for instance, sets correlation as:

$$\rho = 0.12 \times (1 + \exp(-50 \times \text{PD})) / (1 - \exp(-50 \times \text{PD})) + 0.24 \times (1 - (1 + \exp(-50 \times \text{PD})) / (1 - \exp(-50 \times \text{PD})))$$

This produces correlations ranging from 24% for very low PD exposures to 12% for very high PD exposures—declining correlation as credit quality deteriorates. For through-the-cycle parameters, this formula applies straightforwardly. For scenario-conditional calculations, institutions must decide whether correlation remains constant at the through-the-cycle level or adjusts based on conditional PD.

The Frye-Jacobs theoretical development assumes correlation remains constant—it represents structural portfolio characteristics rather than scenario-dependent variables. Empirical testing supported this assumption; allowing correlations to vary produced no statistical improvement. Practical implementations therefore typically hold correlation at baseline levels, applying the Basel formula using through-the-cycle PD rather than conditional PD. This approach preserves theoretical consistency and simplifies implementation.

Expected LGD estimation requires particular care for exposures where physical climate risks might directly affect collateral values. Mortgages secured by coastal properties face both default risk (borrower loses income or faces repair costs following flooding) and collateral risk (property value declines due to increased flood risk perception or physical damage). Traditional ELGD estimates based on historical recovery rates may not reflect climate-influenced collateral dynamics.

Some institutions address this by developing climate-adjusted baseline ELGD estimates for vulnerable exposure segments. Coastal mortgages might receive 30% baseline ELGD rather than 25% standard mortgage ELGD, reflecting expectation of reduced collateral value in default scenarios. This adjusted baseline then feeds into Frye-Jacobs calculations, producing conditional LGD that reflects both the baseline collateral risk and the systematic deterioration as default rates increase. This approach requires justification through analysis of collateral value trajectories under different physical hazard assumptions.

Geographic data quality represents the most significant practical challenge. Applying Frye-Jacobs to physical climate risks requires knowing precisely which exposures face which hazards.

This necessitates facility-level location data—latitude/longitude coordinates or detailed addresses—for material exposures. Many bank credit systems capture legal entity headquarters rather than operational locations, creating substantial data gaps.

Building this geographic database requires significant investment: data collection from borrowers, address standardisation, geocoding, and ongoing maintenance as properties are bought, sold, or redeveloped. For large portfolios, this represents hundreds of thousands of individual property records requiring geocoding and linkage to hazard databases. The investment proves worthwhile for material physical climate risk assessment but requires senior management commitment and multi-year implementation timelines.

Limitations and Appropriate Use

Despite its theoretical grounding and empirical validation, the Frye-Jacobs methodology carries limitations that constrain appropriate interpretation and application for physical climate risks. Understanding these limitations proves as important as understanding the mechanics, particularly when communicating results to senior management and boards less familiar with technical credit modelling.

The model was primarily validated using market-based LGD data—post-default bond prices observed shortly after default. For exposures with long workout processes, particularly mortgages and commercial real estate, the relationship between scenario conditions at default and ultimate recovery years later may prove more tenuous. Research has struggled to find significant systematic effects of macroeconomic variables on realised LGDs for workout exposures, potentially because the workout period spans sufficient time that initial stress conditions no longer prevail by resolution.

This validation gap matters particularly for physical climate stress testing given the nature of physical risks. A flood occurs, causes immediate property damage and income disruption, elevates default rates in subsequent months, but resulting defaults may not resolve for 2-4 years as borrowers attempt workouts, modifications, or eventual foreclosure. The Frye-Jacobs relationship assumes that recovery rates at ultimate resolution reflect conditions prevailing when default rates were elevated. For isolated events where recovery occurs during otherwise normal conditions, this assumption may overstate LGD deterioration.

However, for scenarios involving sustained or repeated physical stresses—coastal regions facing annual flood events, drought-affected agricultural areas with multi-year water shortages, wildfire zones with repeated fire seasons—the assumption appears more reasonable. Recovery markets remain persistently impaired, collateral values stay depressed, and workout capacity stays constrained. The systematic relationship between elevated default rates and poor recoveries persists throughout the workout period rather than representing point-in-time correlation.

The model treats exposures within a segment as homogeneous, sharing common correlation structure with the systematic risk factor. In reality, even narrowly defined segments contain heterogeneity—some coastal properties sit behind recently upgraded flood defences whilst others remain vulnerable; some commercial properties carry full insurance whilst others face coverage

gaps; some agricultural borrowers have drought-resistant crops whilst others remain dependent on irrigation. This heterogeneity means that observed default rates and loss rates within segments will show imperfect correlation even under stress.

The Frye-Jacobs framework cannot capture contagion effects where one borrower's default and recovery directly affects another's LGD through mechanisms beyond the systematic factor. Fire sales of similar collateral might depress recovery values for related defaults; closure of major regional employer due to physical climate damage might simultaneously affect employee default rates and local property values. These spillover effects operate atop the systematic relationship rather than through it, potentially generating additional LGD deterioration beyond model predictions.

Extreme scenarios present particular validation challenges. The Frye-Jacobs function was developed and tested using data spanning default rates from near-zero to perhaps 15-20% in worst historical episodes. Physical climate scenarios could envision conditional default rates reaching 30-40% in worst-affected regions under severe scenarios—entire coastal communities facing repeated flooding, agricultural regions experiencing multi-year drought, wildfire zones with annual burn cycles. Model behaviour at these extreme default rates relies on extrapolation beyond empirical experience. The Vasicek distributional assumption ensures LGD remains below 100% but provides limited confidence about the precise functional form in extreme tails.

Users must also recognise that the model provides expected conditional LGD rather than distributions. Two defaults occurring when conditional PD reaches 10% will not both experience exactly the conditional LGD the model produces—one might recover 80% whilst another recovers 20%. The model provides average expected outcome, useful for portfolio-level projections but not for individual exposure forecasting. Capital planning using Frye-Jacobs results should therefore focus on portfolio aggregates with sufficient exposure counts that law of large numbers provides reasonable confidence in expected value realisations.

Finally, the Frye-Jacobs

framework assumes that physical climate risks affect credit parameters but not the fundamental structural relationship between default and loss. This assumption may not hold under genuinely catastrophic physical scenarios. A Category 5 hurricane completely destroying a coastal community creates dynamics potentially beyond the model's conceptual framework—not merely elevated defaults and reduced recoveries, but fundamental breakdown of property markets, mass population displacement, and potential government intervention through special resolution regimes or disaster relief that alters traditional creditor-borrower relationships.

For such extreme scenarios, users should supplement Frye-Jacobs calculations with qualitative assessment of how catastrophic physical events might fundamentally alter credit resolution processes. Historical precedents—Hurricane Katrina, the 2004 Indian Ocean tsunami, the 2011 Tōhoku earthquake—provide some guidance on how credit systems function following truly catastrophic physical destruction, though each event possessed unique characteristics limiting generalisability.

Integration with Capital Planning

The ultimate purpose of Frye-Jacobs LGD adjustments in physical climate stress testing involves informing capital planning decisions and regulatory capital requirements. Banks must translate scenario-based loss projections into assessments of capital adequacy under stress, considering both Pillar 1 minimum requirements and Pillar 2 buffers for risks not fully captured in standardised formulas.

The Basel IRB approach calculates regulatory capital requirements using downturn LGD—loss severity expected when portfolio default rates reach elevated levels. The Frye-Jacobs methodology provides systematic approach for calibrating these downturn LGD estimates. Rather than applying arbitrary buffers to through-the-cycle ELGD, institutions can apply Frye-Jacobs function using conditional default rates corresponding to downturn conditions (typically 99.5th percentile of default rate distribution over economic cycle).

For physical climate risk specifically, Pillar 2 application appears most appropriate. Physical climate scenarios extend beyond traditional business cycle stress and involve risk drivers (flooding, drought, wildfire) not contemplated in standard IRB formulas. Supervisors increasingly expect institutions to demonstrate that capital buffers prove adequate not merely for economic downturns but also for plausible physical climate scenarios. Frye-Jacobs-adjusted loss projections directly inform the sizing of these climate-specific buffers.

The temporal dimension of physical climate scenarios creates additional capital planning complexity. A scenario might show peak losses occurring in 2040–16 years hence—as sea level rise and storm surge combine to overwhelm coastal defences. Should the bank hold additional capital today to cover this potential future loss? Or should it plan to build capital gradually over the intervening period as physical risks materialise? Regulatory expectations remain unclear, with some supervisors suggesting forward-looking capital planning horizons extending 5-10 years whilst others focus on nearer-term (3 year) assessments.

Institutions typically adopt multi-horizon approaches: immediate capital planning addresses scenarios showing material impacts within 3-5 years (severe flood events, near-term wildfire risks, ongoing drought); medium-term strategic planning (5-10 years) incorporates scenarios with later peak losses (coastal flooding as defences age, agricultural impacts from cumulative precipitation changes); and long-term strategic positioning considers scenarios beyond planning horizons primarily for portfolio allocation and risk appetite decisions rather than immediate capital buffer sizing.

Recovery rates also factor into capital planning beyond expected loss calculations. Basel unexpected loss formulas—determining actual capital requirements—incorporate both PD and LGD through correlation structures. Higher LGD implies greater loss volatility conditional on default, increasing required capital even holding expected loss constant. Frye-Jacobs adjustments therefore affect both expected loss provisions (direct impact) and unexpected loss capital (indirect impact through increased loss volatility).

Institutions may also consider whether physical climate risks warrant adjusting through-the-cycle parameters themselves rather than treating elevated conditions as temporary stress. If flood return periods permanently shorten due to climate change—what was 1-in-100 year event becomes 1-in-50 year event—this represents baseline shift rather than temporary stress. Such baseline shifts might warrant adjusting through-the-cycle PD for affected exposures upward, with corresponding impacts on baseline ELGD and capital requirements under both unstressed and stressed conditions.

This distinction proves important for risk appetite and pricing decisions. Temporary stresses warrant holding additional capital buffers and perhaps reducing new originations in vulnerable segments during stress periods. Permanent baseline shifts warrant repricing affected exposures to reflect new risk profile, potentially withdrawing from certain geographies or product types entirely if risk-adjusted returns become inadequate, and adjusting through-the-cycle capital allocations accordingly.

Validation Approaches for Physical Climate Applications

Validating Frye-Jacobs LGD models for physical climate applications presents distinct challenges from traditional credit model validation. Standard approaches—backtesting model predictions against realised outcomes, calibrating parameters to historical experience, assessing predictive accuracy—rely on observing the phenomena being modelled. Physical climate stress scenarios, by definition, describe conditions not yet observed and potentially unprecedented in historical data.

Model validation for physical climate applications therefore emphasises alternative validation dimensions: theoretical soundness, parameter reasonableness, benchmarking, and sensitivity analysis. The theoretical validation examines whether the Frye-Jacobs framework rests on plausible assumptions and logically sound derivations. The four core assumptions—comonotonicity of loss and default, two-parameter distributions, Vasicek specification, and equal correlations—all passed empirical testing in the original research using corporate default data. This pedigree provides foundation for application in physical climate contexts despite lack of climate-specific validation data.

Parameter reasonableness assessment reviews whether baseline inputs (PD, ELGD, correlation) reflect appropriate through-the-cycle calibrations and whether scenario-conditional PDs represent plausible outcomes given physical climate scenario assumptions. This qualitative review cannot validate numerical accuracy but can identify implausible parameter combinations or internal inconsistencies. For instance, if conditional PD projections for a moderate flood scenario exceed those observed following Hurricane Katrina by factors of two or three, reviewers should probe whether scenario assumptions warrant such extreme outcomes or whether model sensitivity to physical hazard inputs appears excessive.

Benchmarking against historical physical events provides the closest available approximation to validation. Hurricane Katrina, Queensland floods, Thailand floods, California wildfires, and other major physical events produced observable default and recovery data that can be compared to Frye-Jacobs predictions. Whilst no historical event perfectly matches future climate scenarios

in severity, geographic extent, or socioeconomic context, comparing model predictions against historical experience builds confidence if approaches converge or flags concerns if results diverge substantially.

For example, if Frye-Jacobs predicts conditional LGD of 45% for mortgages following a flood event of specified severity, but historical flood events of similar severity produced LGD of 25-35%, this discrepancy warrants investigation. Perhaps the model overestimates sensitivity to default rate increases. Perhaps historical events occurred in areas with better insurance coverage than current scenarios assume. Perhaps collateral markets functioned better historically than scenarios envision. Resolving these discrepancies through careful analysis improves both model calibration and scenario specification.

Sensitivity analysis examines how Frye-Jacobs outputs respond to variations in input parameters. Do conditional LGD estimates increase monotonically with conditional PD (they should)? How much difference does correlation assumption make (should be modest within reasonable ranges)? If baseline ELGD increases 5 percentage points due to climate-adjusted collateral valuations, does conditional LGD increase proportionally? These mechanical validation checks ensure the function operates as designed and produces results with sensible properties.

Sensitivity analysis proves particularly important for physical climate applications because scenario uncertainty exceeds typical stress testing uncertainty. A flood scenario might reasonably assume conditional PD anywhere from 6% to 12% depending on assumptions about insurance coverage, business interruption duration, and borrower resilience. Testing Frye-Jacobs results across this range demonstrates whether conclusions prove robust to parameter uncertainty or whether small assumption changes dramatically alter outcomes.

Institutions implementing Frye-Jacobs for physical climate stress testing should document these validation dimensions explicitly, recognising that traditional backtesting remains infeasible for future-looking scenarios. Model risk management functions comfortable with quantitative validation metrics may initially resist qualitative validation approaches. Clear communication about why standard validation proves impossible for physical climate applications, and what alternative validation dimensions provide assurance, facilitates acceptance.

Importantly, validation should assess not merely the Frye-Jacobs function itself but its integration within broader stress testing frameworks. The function represents one component in multi-step process: physical climate scenarios drive hazard projections, which feed into property damage and business interruption assessments, producing conditional PD estimates, which enter Frye-Jacobs to generate conditional LGD, feeding final loss calculations. Validation must ensure consistency across all components and appropriate uncertainty propagation through the chain.

Particular attention should focus on the PD estimation step—how physical hazards translate into borrower default probabilities. This represents the most uncertain component of the framework, requiring substantial expert judgement about borrower resilience, insurance effectiveness, and recovery timelines. Frye-Jacobs then mechanically transforms conditional PD into conditional LGD through well-defined mathematical relationship. If the PD inputs prove unreliable, the

LGD outputs will be similarly unreliable regardless of Frye-Jacobs function validity. Validation must therefore trace back through the entire calculation chain.

Future Developments and Enhancements

Research continues on extending and refining the Frye-Jacobs framework for improved physical climate risk application. Several directions appear particularly promising for enhancing model performance and practical utility.

Incorporation of collateral-specific adjustments represents one frontier. The current Frye-Jacobs formulation treats baseline ELGD as fixed parameter. For exposures where physical climate risks directly impair collateral value—coastal properties facing sea level rise, properties in wildfire-prone areas, agricultural land with permanent water stress—enhancing the model to reflect collateral value deterioration distinct from systematic recovery rate effects could improve accuracy. This might involve pre-adjusting baseline ELGD based on collateral exposure to physical hazards before applying Frye-Jacobs, or developing conditional ELGD functions that vary with physical scenario variables directly rather than merely through default rate correlation.

One proposed approach treats physical collateral damage as deterministic adjustment to baseline ELGD, then applies Frye-Jacobs to capture systematic recovery rate deterioration beyond direct physical damage. For instance, if flooding physically damages 20% of property value on average for affected properties, baseline ELGD might increase from 25% to 40% (assuming 75% collateral coverage becomes 60% coverage after damage). This adjusted baseline then feeds into Frye-Jacobs, producing conditional LGD that reflects both direct physical damage and systematic market deterioration as default rates rise.

This approach requires careful application to avoid double-counting. If conditional PD already reflects borrowers whose property damage makes default more likely, the physical damage has already been incorporated through the PD channel. Additional LGD adjustment should capture only the direct collateral impairment for defaults that occur regardless of physical damage, plus the systematic market effects. Getting this accounting correct requires clear conceptual framework for what each parameter represents.

Geographic granularity enhancements would better capture the localised nature of physical risks. Standard Frye-Jacobs implementation produces single conditional LGD for each exposure segment based on segment average conditional PD. For physical risks affecting specific regions whilst leaving others unaffected—flooding in one river catchment but not adjacent areas, wildfires in particular topographic zones—this produces overly homogeneous results. Developing approaches that maintain geographic heterogeneity whilst preserving computational tractability—perhaps through sub-segment LGD calculations with subsequent aggregation—would improve practical utility for portfolio management.

Integration with insurance market dynamics represents deeper methodological challenge. The Frye-Jacobs framework assumes systematic relationship between default and recovery rates but does not explicitly model insurance as intermediating mechanism. Physical climate events affect credit losses partly through direct property damage but substantially through insurance coverage

gaps, claim disputes, and insurer solvency concerns. As discussed in the FSB framework (from your uploaded document), insurance protection gaps have widened considerably, with only 44% of natural catastrophe losses insured globally in 2024.

Explicitly modelling insurance market dynamics within or alongside Frye-Jacobs calculations could improve physical climate risk assessment. This might involve scenario-specific insurance coverage assumptions (coverage rates declining as physical risks increase and insurers withdraw from high-risk areas), claim payment haircuts (assuming insurers dispute claims more aggressively following major events), or insurer solvency adjustments (reducing assumed recoveries if physical events threaten insurer ability to pay claims). These enhancements require substantial additional complexity but address critical transmission mechanisms for physical climate risks.

Development of climate-specific validation datasets would substantially strengthen empirical foundation. As physical climate events occur and generate default and recovery experience, institutions should systematically capture and share data (appropriately anonymised) to build collective knowledge. Over time—admittedly decades—sufficient experience may accumulate to validate or refine model parameters specifically for physical climate applications rather than relying on validation using financial crisis data.

This data collection requires infrastructure for linking defaults to specific physical events. A default occurring three months after a flood might result from flood impacts or from unrelated factors. Establishing causality requires detailed investigation of default circumstances—whether borrower suffered property damage, experienced business interruption, faced insurance shortfalls. Building systematic datasets requires standardised data collection protocols, industry collaboration, and sustained commitment across multiple institutions and jurisdictions.

Finally, integration with adaptation and mitigation responses represents the deepest methodological challenge. The Frye-Jacobs framework assumes static relationships between default rates and recovery rates. Physical climate scenarios spanning decades involve substantial adaptation—flood defences improve, building codes strengthen, populations relocate, insurance markets adjust. These adaptive responses affect both conditional PD (reducing default probability for adapted exposures) and potentially the PD-LGD relationship itself (improved defences might reduce correlation between defaults and recovery rates if they prevent widespread simultaneous damage).

Capturing these dynamics requires extending the framework beyond its static formulation. One approach involves time-varying parameters: baseline ELGD might decline over time as adaptation investments improve collateral resilience; asset correlations might decrease as geographic diversification increases following climate-driven population movements. Another approach involves scenario-specific parameter adjustments: orderly climate scenarios with sustained adaptation investment might warrant lower correlations than disorderly scenarios with delayed or inadequate adaptation.

These enhancements remain largely conceptual at present, requiring substantial research and validation before practical implementation. They represent the frontier of physical climate risk

modelling rather than current best practice. Institutions should monitor developments in these areas whilst applying current Frye-Jacobs methodology with appropriate recognition of its limitations and the need for expert judgement in interpreting results.

Concluding Observations

The Frye-Jacobs methodology provides the most theoretically grounded and practically implementable approach currently available for incorporating systematic LGD risk into physical climate stress testing. Its parsimony—requiring only parameters already maintained for Basel IRB compliance—enables straightforward implementation whilst its empirical validation using historical credit data provides confidence in the underlying mathematical relationships.

For physical climate risks specifically, the methodology captures essential dynamics: physical events that elevate default rates also impair collateral values and disrupt recovery markets, generating systematic deterioration in loss given default precisely when default probability spikes. These compounding effects substantially increase credit losses beyond what PD-only models would suggest, with LGD adjustments typically contributing 30-50% of incremental scenario losses.

The worked examples—coastal flooding, agricultural drought, wildfire risk—demonstrate material impacts across exposure types and geographies. Banks with concentrated exposures to physical climate risks face potentially substantial capital implications from incorporating Frye-Jacobs adjustments into stress testing frameworks. These findings directly inform strategic decisions about portfolio composition, risk appetite, and capital buffer sizing.

Limitations remain, particularly around validation for unprecedented physical scenarios, treatment of catastrophic events potentially beyond the model's conceptual scope, and integration of adaptation dynamics over multi-decade horizons. Users must apply the methodology with appropriate expert judgement, supplement quantitative results with qualitative assessment, and communicate uncertainty honestly to stakeholders.

The following sections examine how supervisory authorities have incorporated these methodological considerations into mandatory stress testing programmes, and how institutions can integrate Frye-Jacobs-adjusted loss projections into capital planning frameworks that satisfy both regulatory requirements and internal risk management needs.

Chapter 18 - Forward or Backwards?

This chapter argues that catastrophe insurance modelling should prioritise OASIS-type approaches combining historical weather event patterns, synthetic scenario generation, and near-term weather forecasting over climate model integration. Whilst climate models serve important purposes for long-term policy analysis, their fundamental weakness in representing cloud processes and precipitation dynamics makes them poorly suited for operational insurance pricing. The temporal disconnect between weather prediction and climate projection, combined with irreducible uncertainties in cloud feedback mechanisms, suggests that historically-grounded methodologies provide more defensible foundations for parametric insurance applications than climate-conditioned scenarios.

The Temporal Disconnect

Weather forecasting and climate modelling represent fundamentally different scientific endeavours that should not be conflated despite superficial similarities in computational methods and atmospheric physics. Weather prediction constitutes an initial value problem: given current atmospheric conditions measured through observations, numerical models integrate forward in time to project how those conditions will evolve over hours to weeks. The forecast horizon is limited by error growth through chaotic atmospheric dynamics, with predictability typically extending 7–14 days for mid-latitude systems.

Climate modelling addresses a boundary value problem: given altered external forcing conditions (such as increased atmospheric CO₂ concentrations), models project statistical properties of the climate system decades to centuries ahead. Individual weather events become irrelevant noise; the focus shifts to long-term means, variances, and probability distributions. Climate models do not attempt to predict whether 15 July 2050 will be rainy in London—such a prediction proves physically impossible. Instead, they project whether July 2050 will be wetter on average than July 2000, or whether extreme rainfall events will occur more frequently.

This distinction extends beyond timescales to encompass validation methodology, error sources, and predictability limits. Weather forecasts achieve validation through comparison with subsequent observations, enabling systematic model improvement through identification and correction of biases. Forecast skill metrics—root mean square error, anomaly correlation, probability of detection—quantify performance objectively across thousands of verification cases.

Climate projections lack comparable validation opportunities for their primary application. A model projecting 2080 climate conditions cannot be verified until 2080 arrives, by which point the projection's utility for decision-making has expired. Indirect validation approaches—evaluating model skill for historical periods, assessing physical process representation, examining consistency across model ensembles—provide only partial confidence. The climate system may respond to unprecedented forcing (such as rapid CO₂ increase) in ways that historical analogues do not fully constrain.

The error growth mechanisms differ fundamentally. Weather forecast errors accumulate primarily through chaotic dynamics: small uncertainties in initial conditions amplify exponentially through non-linear atmospheric processes. Climate projection errors stem largely from structural uncertainties in representing physical processes—particularly cloud formation, precipitation, and ocean-atmosphere coupling. These structural uncertainties do not decrease through improved initial conditions or increased model resolution; they require better physical understanding of processes inadequately captured by current model parameterisations.

Recent advances in AI-based weather forecasting, exemplified by NVIDIA Earth-2, demonstrate impressive skill improvements for short-term prediction but do not fundamentally alter the temporal divide. These models learn statistical patterns from vast training datasets, enabling rapid predictions that match or exceed traditional numerical weather prediction. However, the learned patterns reflect current climate statistics. Whether these patterns remain valid under substantially altered boundary conditions—the domain of climate projection—cannot be verified until those conditions materialise.

The subseasonal-to-seasonal (S2S) timescale, spanning roughly two weeks to several months, represents contested territory where neither weather prediction nor climate projection approaches achieve reliable skill. This "predictability desert" reflects fundamental atmospheric dynamics: initial condition memory fades within weeks, whilst boundary condition influences (ocean temperatures, soil moisture, snow cover) evolve too slowly to provide strong predictability. Claims that computational advances will bridge this gap must confront the physical limits to predictability imposed by atmospheric chaos.

Cloud Processes: The Irreducible Uncertainty

Cloud feedback represents the largest source of uncertainty in climate sensitivity projections, and this uncertainty has persisted despite decades of intensive research. The Intergovernmental Panel on Climate Change consistently identifies cloud processes as the dominant contributor to inter-model spread in temperature projections for given emissions scenarios. This is not a transient problem awaiting computational solutions; it reflects fundamental difficulties in representing multi-scale physical processes within discrete numerical models.

Clouds influence climate through multiple mechanisms operating simultaneously. Low-level stratiform clouds predominantly reflect incoming solar radiation, exerting cooling influence on surface temperatures. High-level cirrus clouds allow solar radiation to pass whilst trapping outgoing longwave radiation, creating warming effects. Deep convective clouds generate both effects alongside latent heat release that drives atmospheric circulation. The net impact of clouds on climate depends on the balance between these competing mechanisms—a balance that varies spatially, seasonally, and diurnally.

Climate models must parameterise cloud processes because explicit simulation proves computationally prohibitive. Cloud formation involves microphysical processes (aerosol activation, droplet growth, ice crystal formation) operating at micrometre scales, turbulent mixing at metre scales, and convective organisation at kilometre scales. Global climate models operate at 50-100 kilometre resolution, requiring statistical relationships to represent sub-grid

cloud processes. These parameterisations contain adjustable parameters that substantially influence model behaviour.

Boundary-layer clouds exemplify the challenge. Marine stratocumulus clouds cover vast ocean areas and exert substantial cooling through solar reflection. Their representation in climate models remains inadequate despite recognised importance: models disagree on whether warming will increase or decrease stratocumulus coverage. Some models simulate stratocumulus thinning as temperatures rise, reducing cooling influence and amplifying warming (positive feedback). Others project increased coverage, creating negative feedback. This divergence stems from competing processes—enhanced evaporation increasing cloudiness versus stronger inversions suppressing cloud formation—that models balance differently.

Deep convective clouds present similar difficulties. Convection transports heat, moisture, and momentum vertically through the atmosphere, determining precipitation patterns and atmospheric circulation. Models employ convection parameterisations that activate when atmospheric instability exceeds thresholds, but these parameterisations contain substantial uncertainty. The relationship between large-scale environmental conditions and convective precipitation remains poorly constrained, creating divergent model responses to warming.

High-resolution models that explicitly simulate convection (rather than parameterising it) do not resolve cloud uncertainty. Whilst explicit convection improves some aspects of precipitation simulation, fundamental uncertainties remain regarding cloud microphysics, aerosol-cloud interactions, and the organisation of convection into larger-scale systems. Computational resources limit explicit convection models to regional domains or short time periods, precluding their use for century-scale global projections required for climate assessment.

The observational record constrains cloud feedback uncertainty inadequately. Satellite observations of clouds span roughly four decades—the period since comprehensive global monitoring became available. This observational period proves too short for reliably distinguishing long-term trends from natural variability, particularly given substantial multi-decadal climate oscillations. Earlier observations from surface stations and research aircraft provide limited spatial coverage and temporal consistency, precluding robust long-term trend assessment.

Cloud property measurements themselves contain uncertainties. Satellite retrievals of cloud fraction, optical depth, liquid water path, and particle size involve complex algorithms with assumptions about cloud vertical structure and radiative properties. Different satellite instruments and retrieval algorithms produce inconsistent results, complicating trend detection. Ground-based measurements provide higher accuracy at point locations but cannot constrain global cloud behaviour adequately.

The Measurement Challenge: Cloud Top Temperature

Cloud top temperature measurements illustrate observational limitations constraining long-term cloud trend analysis. Modern satellite systems employ infrared radiometry to measure cloud top temperatures globally: sensors detect electromagnetic radiation emitted from cloud tops, with intensity related to temperature according to Planck's law. The Hong Kong Observatory describes the technique: since clouds are generally opaque to specific infrared frequencies, measured radiation temperatures closely represent actual cloud top conditions. Advanced systems like GOES-R use multiple infrared bands simultaneously to account for variations in cloud emissivity and atmospheric interference, retrieving cloud top temperature, height, and pressure for individual pixels.

However, systematic global cloud top temperature observations span only the satellite era from the 1970s onwards. Earlier observations from weather stations and research aircraft provide inadequate spatial coverage and temporal consistency for trend analysis. Four decades of consistent observations prove insufficient for distinguishing anthropogenic trends from natural variability in systems exhibiting multi-decadal oscillations.

The relationship between cloud top temperature and precipitation operates through atmospheric dynamics rather than simple causation. Colder cloud tops indicate higher altitude clouds, often associated with vigorous convection producing intense precipitation. Yet this represents diagnostic correlation for weather forecasting rather than causal relationship suitable for climate projection. Cloud formation involves atmospheric instability, moisture availability, wind patterns, aerosol interactions, and topography—complexities precluding simple extrapolation from short-term observations to long-term trends.

The measurement history limitation extends to other cloud properties essential for climate assessment. Cloud optical depth, liquid water path, ice water content, and particle size distributions all require satellite observations providing at most four decades of consistent data. This temporal coverage proves insufficient for robustly attributing observed changes to anthropogenic forcing versus natural variability.

The CO₂-Cloud-Precipitation Causal Chain

The causal pathway linking atmospheric CO₂ concentrations to regional precipitation patterns passes through cloud processes inadequately constrained by observations or theory. Rising CO₂ increases atmospheric temperatures through well-established radiative physics: carbon dioxide absorbs outgoing longwave radiation, reducing energy loss to space and warming the surface-atmosphere system. This forcing-response relationship operates on firm theoretical grounds validated by laboratory measurements and atmospheric observations.

Subsequent impacts on clouds and precipitation involve complex feedbacks represented uncertainly in models. As temperatures rise, atmospheric moisture capacity increases following

Clausius-Clapeyron relationship: saturation vapour pressure rises approximately 7% per degree Celsius warming. This thermodynamic response suggests precipitation intensity should increase, as warmer atmospheres can support higher moisture contents available for conversion to precipitation.

However, this thermodynamic effect interacts with dynamic changes in atmospheric circulation that models represent with large uncertainty. Increased moisture and latent heat release may intensify existing circulation patterns, or may alter circulation configurations in ways that reduce precipitation in some regions whilst increasing it elsewhere. The moisture-convection-latent-heat feedback cycle creates potential instabilities that models handle through parameterisations containing substantial uncertainty.

Different models predict divergent responses of deep convective clouds to warming. Some simulate increased convective intensity with reduced frequency: individual storms strengthen but occur less often, possibly due to increased atmospheric stability between events. Others project different patterns based on alternative representations of convection-environment interactions. These differences stem from fundamental uncertainties in sub-grid-scale processes that neither observations nor theory adequately constrain.

Regional precipitation projections carry enormous uncertainty ranges reflecting this fundamental indeterminacy. Models project 95th percentile winter precipitation increases approaching 100% in some regions whilst others show decreases of 75%, depending on model formulation and region. This spread reflects genuine uncertainty about circulation responses to warming rather than merely computational limitations amenable to resolution improvements.

Crucially, the inability to validate cloud-precipitation relationships over climate-relevant timescales means that improvements in weather forecasting skill do not automatically translate to improved climate projection capability. Weather models achieve validation through comparison with subsequent observations across thousands of cases, enabling systematic refinement. Climate projections lack comparable validation for their primary application—projecting unprecedented future conditions—forcing reliance on indirect validation approaches that provide only partial confidence.

Frequency Versus Intensity: Model Divergence

Climate model analysis reveals that extreme precipitation events exhibit frequency increases exceeding intensity increases. Studies of 99th percentile precipitation show approximately 20% frequency increase compared to 5% intensity increase from mid-20th century to early 21st century periods. This pattern raises questions about model physics rather than resolving uncertainties about future precipitation behaviour.

The frequency-intensity divergence suggests models capture event probability changes more readily than event magnitude changes. This could reflect genuine physical processes—perhaps atmospheric moisture availability increases event probability whilst circulation changes limit intensity increases. Alternatively, it may indicate model limitations in representing extreme precipitation mechanisms, particularly for rare events exceeding normal model validation ranges.

Substantial inter-model diversity for extreme precipitation further suggests underlying uncertainties in fundamental processes. Models employing different convection parameterisations, cloud microphysics schemes, and boundary layer representations produce divergent extreme precipitation responses despite agreeing on basic thermodynamic relationships. This diversity does not decrease with increased model resolution or complexity; indeed, some analyses suggest uncertainty ranges expand as models incorporate more detailed process representations.

Weather prediction systems focus primarily on intensity forecasting—predicting rainfall amounts for specific events occurring within forecast horizons. Climate models attempt projecting both frequency and intensity changes for statistical distributions decades ahead. The skill demonstrated in weather forecasting does not automatically extend to climate projection, as physical processes controlling event-to-event variability differ from those controlling long-term distributional changes.

The distinction proves crucial for insurance applications. Weather forecasts inform short-term risk management decisions: whether to deploy temporary flood defences, issue evacuation warnings, or adjust operational procedures. Climate projections ostensibly inform long-term capital allocation, infrastructure design, and risk pricing. However, the large uncertainties in precipitation projections—particularly at regional scales relevant for insurance—limit the actionable information climate models provide for these applications.

Predictability Degradation Under Warming

Research demonstrates that warming intensifies weather unpredictability in mid-latitudes, with forecast skill degrading approximately one day per 3°C temperature increase. This finding carries profound implications for extending weather prediction capabilities toward climate-relevant timescales. If deterministic weather prediction limits contract as climate changes, the S2S forecasting gap may widen despite computational advances.

The mechanism driving predictability loss relates to increased atmospheric chaos under warmer conditions. Stronger water vapour feedbacks create enhanced latent heat release during condensation, amplifying vertical motion and accelerating error growth. More vigorous convection generates smaller-scale features that numerical models represent less accurately, introducing additional error sources. The result manifests as faster decorrelation of forecast states from actual atmospheric evolution, reducing useful prediction horizons.

This predictability degradation reinforces the conceptual divide between weather and climate applications. Even hypothetically perfect weather models capturing all relevant atmospheric physics would face shrinking temporal limits on skilful prediction as climate changes. Climate projections must rely on statistical relationships and ensemble averaging rather than deterministic prediction extension, maintaining the methodological gap between initial value and boundary value problem approaches.

For insurance applications, this finding suggests that extending weather forecast capabilities cannot substitute for climate projection needs—if such needs genuinely exist. The two

approaches serve fundamentally different purposes operating on incompatible timescales. Attempts to bridge the gap through S2S forecasting confront physical limits that computational sophistication cannot overcome.

IPCC Scenario Construction: Socio-Economic Narratives Disconnected from Climate Physics

The IPCC scenario development process reveals a fundamental methodological weakness: scenarios derive primarily from socio-economic narratives rather than physical climate model constraints, creating a disconnect between scenario construction and the climate projections ostensibly based upon them.

The Shared Socioeconomic Pathways (SSPs) that replaced earlier SRES scenarios represent storylines about future demographic trends, economic development, technological change, and policy choices. These narratives describe alternative worlds—SSP1 depicts sustainable development with low challenges to mitigation and adaptation, whilst SSP3 envisions regional rivalry with high challenges to both. The pathways then translate these qualitative stories into quantitative projections of emissions, land use, and other climate forcing factors.

However, this translation from narrative to numbers involves assumptions and models from economics, demography, and energy systems analysis—disciplines operating independently from climate physics. The economic models projecting future emissions under different SSPs employ assumptions about technological progress, energy transitions, and policy effectiveness that lack the physical constraints governing atmospheric processes. An economist's model of solar panel adoption rates or carbon capture deployment faces no validation requirements comparable to climate model evaluation against observations.

The scenario construction timeline exposes this disconnect. Socio-economic pathways develop through stakeholder workshops, expert elicitation, and integrated assessment model runs before climate models employ the resulting emissions trajectories. Climate modellers receive emissions scenarios as external inputs rather than co-developing them through iterative processes ensuring physical consistency. This sequential approach creates scenarios that may prove internally inconsistent—economic assumptions might imply energy system transformations that physical resource constraints or technological limitations render implausible.

The "open process" for scenario development that IPCC documentation emphasises involves extensive stakeholder consultation, but stakeholders represent policy, business, and civil society communities rather than physical scientists who might identify implausible assumptions. The resulting scenarios reflect political and economic feasibility judgements more than physical constraints on how rapidly atmospheric composition can change or how climate systems respond to altered forcing.

The RCP (Representative Concentration Pathway) framework that preceded SSPs made this disconnect explicit: pathways specified radiative forcing trajectories without prescribing how societies would achieve them. This agnosticism about socio-economic mechanisms enabled

multiple storylines to reach identical forcing levels through different pathways. However, the framework assumed that specified forcing trajectories could be achieved—an assumption that physical resource constraints, technological limits, or political realities might violate.

For insurance applications, this methodological structure proves particularly problematic. Parametric pricing requires understanding how specific trigger events—flood levels, precipitation amounts, wind speeds—change in frequency and intensity. Yet the scenario construction process prioritises global mean temperature targets and aggregate emissions pathways over regional precipitation patterns or extreme event statistics. The chain from SSP narrative through emissions projections to global temperature change to regional precipitation involves multiple modelling stages, each introducing uncertainties that compound rather than cancel.

The scenario framework also embeds implicit value judgements about plausible futures that may not align with insurance risk assessment requirements. Scenarios emphasising rapid decarbonisation reflect policy aspirations rather than physical projections; scenarios depicting continued high emissions may understate technological disruption potential. Insurance pricing needs unbiased probability distributions across possible futures, but scenario construction reflects "decision-making under uncertainty" framing where scenarios explore policy choices rather than likelihood-weighted outcomes.

The fundamental issue manifests clearly when examining how scenarios connect to regional precipitation projections—the variable most relevant for flood insurance. The chain of reasoning proceeds: SSP narrative → emissions trajectory → global temperature change → regional circulation response → local precipitation changes. Each arrow represents substantial modelling uncertainty, with the circulation response and precipitation changes involving precisely the cloud processes identified earlier as fundamentally uncertain. Yet the scenario construction treats these connections as sufficiently understood to warrant detailed storylines about 2100 emissions levels.

Historical scenario performance further undermines confidence. Past IPCC scenarios have consistently misrepresented actual development trajectories—underestimating renewable energy deployment, misjudging coal consumption trends, and failing to anticipate major geopolitical shifts affecting emissions. These failures don't merely reflect forecasting difficulty; they suggest the narrative-driven scenario construction methodology proves poorly suited even for the socio-economic projections it prioritises, let alone the physical climate projections that follow.

For catastrophe insurance modelling, this methodological structure suggests scenarios provide limited value compared to historical-synthetic approaches. The scenarios ostensibly inform long-term climate projections, but those projections depend on cloud-precipitation physics that models represent inadequately regardless of emissions pathway. The socio-economic narratives driving scenario construction prove largely irrelevant for understanding whether September 2060 will bring Thames flooding—the narrative about global cooperation versus fragmentation matters far less than whether models can project regional autumn precipitation changes, which fundamental cloud uncertainties prevent.

The appropriate conclusion acknowledges scenarios serve legitimate purposes for policy analysis and mitigation planning whilst recognising their severe limitations for insurance pricing. Using SSP-conditioned climate projections to adjust parametric flood triggers introduces dependencies on socio-economic narratives constructed through stakeholder processes disconnected from physical climate constraints, filtered through climate models that inadequately represent cloud-precipitation physics. Historical event catalogues avoid these compounded uncertainties by anchoring directly in observed atmospheric behaviour under varying conditions documented through actual measurements rather than scenario storylines.

The Case for Historical-Synthetic Approaches

Recognition that weather prediction and climate projection operate through fundamentally different mechanisms, combined with irreducible uncertainties in cloud-precipitation relationships, supports prioritising historically-grounded approaches for catastrophe insurance modelling over climate model integration.

Historical event catalogues provide the most defensible foundation for insurance pricing. Weather observations spanning decades document actual atmospheric conditions producing loss-generating events. Precipitation measurements, stream gauge records, wind observations, and temperature data capture environmental conditions during historical catastrophes. These records reflect verified atmospheric states that produced documented impacts, avoiding the model uncertainty inherent in projection-based approaches.

Synthetic event generation extends historical catalogues through statistical techniques preserving observed correlation structures and spatial dependencies. Monte Carlo sampling generates thousands of events statistically consistent with historical observations, providing sample sizes adequate for tail risk assessment. Copula-based dependence models capture spatial correlations between affected regions, ensuring synthetic events exhibit realistic geographical patterns. Extreme value theory enables rigorous extrapolation to return periods exceeding observational records whilst quantifying associated uncertainties through established statistical frameworks.

These synthetic extensions introduce uncertainty, but uncertainty quantified through transparent statistical methods validated against observations. Climate model projections introduce structural uncertainty from process parameterisations that cannot be validated over relevant timescales. The difference proves crucial: statistical uncertainties decrease with additional observations, whilst structural model uncertainties may persist indefinitely absent fundamental physics breakthroughs.

Pattern evolution analysis provides defensible approaches for adjusting historical probabilities based on emerging trends without requiring climate model validation. If systematic changes appear in observed event characteristics—altered seasonality, modified intensity distributions, changed duration patterns—these observed trends inform catalogue adjustments more reliably

than model projections. Statistical techniques including change-point detection, trend analysis with proper attention to autocorrelation, and persistence testing distinguish genuine regime shifts from natural variability.

The validation accessibility of historical-synthetic approaches proves decisive. Catalogue performance can be assessed through backtesting against withheld historical events, evaluating whether catalogue statistics match observed loss frequencies and severities. Spatial correlation accuracy can be verified by comparing synthetic event patterns to actual multi-region events. Extreme value extrapolations can be validated through comparison with independent palaeoclimate records or historical documentary evidence.

Climate-conditioned scenarios lack comparable validation. Projections of 2080 precipitation cannot be verified until 2080, by which point pricing decisions based on those projections have already succeeded or failed. Indirect validation—evaluating models against historical periods, assessing process representation, examining ensemble consistency—provides insufficient confidence for applications requiring quantitative precision.

Computational Tractability and Transparency

Historical-synthetic approaches prove computationally tractable for operational insurance applications, whilst climate model integration faces substantial resource requirements. Event catalogues containing 10,000–100,000 synthetic events provide adequate statistical samples for portfolio analysis whilst remaining manageable with standard computing resources. Modern GPU acceleration enables rapid catastrophe model processing, with complete portfolio analyses completing in minutes rather than hours.

Climate model integration demands ensemble approaches to characterise projection uncertainty, requiring dozens of model runs for individual emissions scenarios. Downscaling procedures translating coarse-resolution climate outputs to insurance-relevant spatial scales introduce additional computational burdens. Bias correction procedures adjusting model outputs to match historical observations require extensive computation whilst introducing methodological uncertainties about whether historical biases persist under altered climate states.

The computational efficiency of historical-synthetic approaches enables operational capabilities that climate-conditioned scenarios cannot match. Rapid pricing updates responding to changing portfolio compositions, immediate sensitivity analyses exploring parameter uncertainties, and portfolio optimisation procedures examining diverse risk transfer strategies all require computational speed available only through historically-grounded methods.

Methodological transparency provides additional advantages for regulatory compliance and customer communication. Historical-synthetic catalogue construction follows clear steps from observational data through statistical processing to final event sets. Model assumptions, parameter selections, and validation results admit comprehensive documentation accessible to technical reviewers without requiring climate science expertise.

Climate-conditioned approaches introduce dependencies on complex model chains complicating regulatory review. Emissions scenario selections, climate model choices, downscaling methodologies, bias correction procedures, and impact assessment techniques each require justification. The resulting documentation spans multiple scientific disciplines, creating review challenges for insurance regulators lacking climate modelling expertise.

Customer communication benefits similarly. Policyholders understand pricing based on documented historical events and transparent statistical extensions without climate science knowledge. Explaining why parametric triggers activate during specific historical analogues proves straightforward. Climate-conditioned pricing requires explaining complex model chains, uncertain projections, and scenario dependencies that may undermine confidence rather than building understanding.

Near-Term Weather Prediction: The Appropriate Integration

Weather forecasting capabilities merit integration into insurance operations, but through mechanisms distinct from pricing foundations. Short-term forecasts inform dynamic risk management: deploying temporary protection measures, issuing early warnings to policyholders, adjusting claims reserves based on developing events, and managing reinsurance exposures during active catastrophe situations.

NVIDIA Earth-2 and similar AI-based forecasting systems provide valuable capabilities for these operational applications. Rapid updates enabling multiple daily forecast cycles, high spatial resolution capturing local variations in hazard intensity, and ensemble predictions quantifying forecast uncertainty all support real-time risk management decisions.

However, these capabilities should not extend to fundamental pricing adjustments based on multi-year forecast skill claims. The temporal horizon where weather forecasts achieve validated skill—typically 7–14 days—proves far too short for insurance pricing applications requiring annual or multi-year perspectives. Subseasonal forecasts beyond two weeks lack sufficient validation for confident operational use, while seasonal forecasts achieve only modest skill for specific variables in limited regions.

The integration should preserve a clear separation: historical-synthetic catalogues supply pricing foundations validated through backtesting, while weather forecasts guide short-term tactical decisions during active event periods. This separation maintains validation accessibility, fostering confidence in the pricing methodology whilst leveraging genuine forecast skill where it exists.

Pattern recognition within recent observations provides another defensible application for analysis techniques. Machine learning algorithms can identify emerging systematic changes in event characteristics warranting catalogue adjustments: shifts in seasonal timing, changes in spatial correlation patterns, altered relationships between meteorological conditions and loss outcomes. These applications operate within observational domains where validation proves feasible, avoiding extrapolation to unprecedented future states.

Regulatory and Communication Advantages

Insurance regulators increasingly scrutinise catastrophe model methodologies, requiring transparent documentation, validation evidence, and sensitivity analyses. Historical-synthetic approaches provide clear audit trails from source observations through statistical processing to final event catalogues. Every processing step admits documentation accessible to reviewers without specialised climate science training.

Validation evidence proves straightforward: backtesting results comparing catalogue predictions to actual loss experience, spatial correlation verification against multi-region historical events, extreme value extrapolation assessment through comparison with independent evidence. These validation approaches operate within established actuarial and statistical frameworks familiar to insurance regulators.

Climate-conditioned scenarios create documentation and validation challenges. Justifying emissions scenario selections requires explaining socio-economic assumptions spanning centuries. Climate model choices involve technical details about atmospheric physics, numerical methods, and parameterisation approaches requiring expert review. Downscaling and bias correction methodologies introduce additional assumptions requiring validation that observations cannot provide for future climate states.

The inability to validate long-term climate projections creates fundamental regulatory challenges. How should regulators assess model appropriateness when validation must rely on indirect evidence? What standards should apply to scenarios projecting conditions unprecedented in observational records? These questions lack clear answers, creating regulatory uncertainty that historical-synthetic approaches avoid.

Customer communication similarly benefits from historical-synthetic transparency. Policyholders grasp pricing based on documented events within living memory or historical record. Understanding why 1953 North Sea flood or 2000 Autumn floods appear in event catalogues requires no climate science expertise. Explaining parametric trigger activation through historical analogues proves straightforward.

Climate-conditioned pricing requires explaining model projections, scenario uncertainties, and complex causal chains from emissions through atmospheric physics to regional impacts. This complexity may undermine rather than build confidence, particularly given public scepticism about climate model reliability fuelled by past projection failures and communication missteps.

Conclusion

The temporal disconnect between weather prediction and climate projection, combined with fundamental uncertainties in cloud processes determining precipitation responses to warming, supports prioritising OASIS-type historical-synthetic approaches for catastrophe insurance modelling over climate model integration. Weather forecasting and climate projection serve different purposes operating on incompatible timescales with distinct validation requirements.

Cloud feedback uncertainty—persisting despite decades of research—creates irreducible indeterminacy in precipitation projections that computational advances cannot resolve.

Historical event catalogues extended through rigorous statistical methods provide defensible insurance pricing foundations validated through backtesting against actual loss experience. Pattern recognition within observations enables identification of systematic changes warranting catalogue adjustments without dependence on unvalidated climate projections. Near-term weather forecasts inform operational risk management within validated skill horizons without influencing fundamental pricing methodology.

This historically-grounded approach positions catastrophe insurance as practical risk transfer based on documented event experience rather than climate projection vehicles dependent on uncertain model outputs. The resulting pricing proves more defensible to regulators, more comprehensible to customers, and more amenable to validation than alternatives requiring integration of climate model projections through precipitation mechanisms that models represent with fundamental, irreducible uncertainty.

Chapter 19 - Model Risk Governance

As banking institutions increasingly incorporate flood risk into their operations, the governance of flood risk models has emerged as a critical challenge. This chapter explores how traditional model risk governance frameworks must evolve to address the unique complexities of flood risk modelling. We examine the "model stack problem" in flood risk assessment, propose governance solutions, and outline collaborative industry approaches to standardisation.

"Integrating current fragmented vendor offering of flood risk information and analytics into banking operations represents a fundamental challenge for model risk governance."
David Kelly, MKM.

As banks face increasing pressure to incorporate natural hazard considerations into their risk assessments and capital allocations, they must ensure their flood modelling approaches meet rigorous governance standards. Key challenges include:

- **Regulatory evolution:** Regulators are expanding existing prudential frameworks to address climate-related risks, making model governance urgently needed when flood models inform capital requirements or lending criteria.
- **Complex decision-making:** Banks face a web of requirements when implementing flood risk models, particularly for scenarios like assessing flood risk in mortgage portfolios, which can lead to adjusting lending criteria.
- **Societal implications:** Poor model governance creates significant societal risks. For example, inappropriate vendor flood scores used to pre-screen UK mortgage applications profoundly impact property markets and individual homeowners.

Overview of Model Risk Governance

"Model Risk can lead to financial loss, poor business and strategic decision making, or damage to a bank's reputation. Model Risk should be managed like other types of risk. Firms should identify the sources of risk and assess the magnitude."

Model risk governance in banking grew out of painful experience, and SR 11-7 is the moment when that experience crystallised into a common language and set of expectations. For a flood risk practitioner, this history explains why banks sound so doctrinaire about models: they have already seen what happens when clever models run ahead of governance.

Before SR 11-7: models without a framework

In the years leading up to the 2008 crisis, banks relied heavily on quantitative models for pricing, risk measurement, and regulatory capital, but model governance was largely informal and fragmented. Individual desks or business lines often owned their own models, validated them

internally, and documented them to highly variable standards, with no consistent inventory or enterprise view.

This worked passably in benign conditions, but several episodes exposed how fragile the setup was. Long-Term Capital Management in 1998 showed that models calibrated to normal markets could not cope with extreme conditions and liquidity dry-ups, even when designed by Nobel laureates. The subprime boom and bust then revealed systemic weaknesses: mortgage and securitisation models consistently underestimated default and correlation risks, while rating-agency models conferred AAA labels on structures that later proved highly fragile.

During the 2008 crisis itself, model failures were not mainly about coding errors. They were failures of risk management around models: inappropriate use, blind faith in outputs, poor documentation, and a lack of independent challenge. JP Morgan's "London Whale" losses, for example, involved a complex hedging strategy where internal risk models understated true exposure and where flaws in model choice and calibration went unchecked until losses exceeded 6 billion.

Why SR 11-7 was issued and what it changed

In response, US regulators issued SR 11-7, "Guidance on Model Risk Management", in April 2011, jointly from the Federal Reserve and the OCC. It is short and readable by regulatory standards, which is part of why it became the de facto global benchmark for model risk governance.

SR 11-7 did three important things that matter for a flood-to-PRS audience:

- **Model:** It broadened the definition of model risk beyond technical mistakes to include inappropriate use and unavoidable limitations.
- **Validation:** It required independent validation and "effective challenge" across the full model lifecycle, not just at go-live.
- **Responsibility:** It made clear that outsourcing does not outsource responsibility so that banks remain on the hook for model risk even when they buy or rent vendor models.

The definition of a model in SR 11-7 has become the standard reference for banks and vendors alike: a quantitative method that uses statistical, economic, financial or mathematical theories and assumptions to turn input data into quantitative estimates for decision-making, with explicit input, processing, and reporting components.

The guidance also pushed banks to formalise three structural elements:

- A model inventory covering all in-scope models.
- A clear model lifecycle with documented development, validation, implementation, monitoring, and retirement.

- Explicit roles and accountability, including model owners, independent validators, and a governance committee (often called a Model Risk Committee).

Over the next decade, these SR 11-7 principles spread beyond the US, influencing European and UK expectations even where the document itself is not formally referenced. Concepts such as effective challenge, model tiering by materiality and complexity, and documented limitations of use are now standard vocabulary for bank validators talking to any vendor, including climate and flood specialists.

What is a model and why flood tools fall into scope

SR 11-7's model definition cuts through technology labels. A model is any quantitative method that:

- Inputs: Ingests data and asserts assumptions (e.g. rainfall, river flow, elevation grids, property attributes).
- Processes: Applies a processing component that combines theories and algorithms (e.g. hydrological routing, hydraulic simulation, statistical frequency analysis, depth-damage curves).
- Outputs: Produces quantitative outputs used for decisions (e.g. flood probabilities, depth maps, loss estimates, risk scores).

By this standard, many flood tools used in banks are clearly models:

- Hazard models that translate rainfall into inundation depths and frequencies.
- Vulnerability models that convert water levels into building damage.
- Portfolio aggregation models that map site-level impacts into portfolio loss distributions or capital inputs.

In contrast, pure infrastructure systems—databases, GIS viewers, simple transactional systems—are usually not models if they do not apply theoretical frameworks to generate new estimates. The moment a flood system starts inferring missing values, extrapolating risk, or combining hazard and exposure into a numeric score that affects underwriting or PRS pricing, SR 11-7 thinking applies.

This classification matters because once something is called a model, it is pulled into the governance framework: inventory, validation, monitoring, documentation, and committee oversight. For a flood modeller entering banking, the first surprise is often that what felt like “just analysis” is now treated as a governed model artefact.

The model lifecycle in SR 11-7 terms

SR 11-7 encourages banks to view models as going through a lifecycle, with risk management at each stage rather than a one-off approval. A typical lifecycle, applied to flood and climate tools, has six stages:

1. Define the business case

- Clarify what decision the model supports: screening properties for lending, pricing flood-linked swaps, setting risk appetite, or informing climate disclosures.
- Document intended use, scope, and key limitations from the outset, to reduce misuse later.

2. Develop the model

- Choose methodology, build the analytical core, treat data, and test behaviour in development environments.
- Produce development documentation that explains theoretical foundations, assumptions (for example, stationary vs non-stationary hydrology), and known weaknesses.

3. Validate the model

- An independent party (a model validation team or at least a peer not involved in development) reviews methodology, tests implementation, and challenges design choices.
- Validation results in findings, conditions of use, and an overall risk rating or “tier” reflecting materiality and complexity.

4. Implement in production

- Embed the model into production systems: connect data feeds, schedule runs, and expose outputs into workflows.
- Perform regression and integration testing to ensure the production instance behaves as the validated design, not a subtly altered variant.

5. Monitor performance

- Track stability, accuracy, and operational reliability over time.
- For flood models, this may include back-testing against events, tracking drift in input data quality, or monitoring override patterns and run-time performance.

6. Manage model risk and decommission

- Use governance forums to review monitoring results, decide on remediation, restrict use, or retire models that no longer meet standards.

- Maintain a model inventory so that the institution always knows which models exist, who owns them, and what their risk status is.

In climate and flood contexts, this lifecycle is crucial because both the physical system and the regulatory environment evolve. A model that was adequate under older climate data or capital rules may become materially misleading under updated hazard projections or new regulatory expectations, even if the code has not changed.

Why validation is not the same as testing

SR 11-7 draws a clear line between testing and validation, which many non-bank specialists initially miss. This distinction is central enough to repeat explicitly.

Testing asks: does the system work as specified?

- Does the code run without crashing, produce outputs in the right format, and handle typical edge cases?
- Are performance requirements met (for example, can the flood engine process the portfolio overnight)?

Validation asks: is this the right model, used in the right way?

- Is the methodology appropriate for the intended decision and data environment?
- Are assumptions reasonable, transparent, and consistent with alternative methods and domain knowledge?
- Does the model behave sensibly under stress and at the edges of the data space?
- Do users understand limitations and conditions of use, or are they likely to over-trust the outputs?

For flood/climate examples:

- A fluvial model may be technically well-implemented (passes tests) but misused if applied to coastal risks without acknowledging limitations; validation is where that mismatch is challenged.
- A PRS pricing engine might run efficiently but rely on hazard inputs calibrated to historical climate, while being deployed for long-dated mortgages; validation should ask whether that time-horizon mismatch is acceptable.
- Crucially, SR 11-7 insists that validation is independent from development. Development teams are naturally invested in their models and may unconsciously downplay weaknesses, whereas

validators are expected to own the risk associated with approving the model and therefore to surface limitations and usage constraints explicitly.

The nine central validation questions

SR II-7 itself is principle-based, but practitioners have distilled its spirit into nine practical questions that structure a validation and that vendors now see repeatedly. They apply as much to flood and climate models as to pricing libraries:

1. Does the model do what it says on the tin?
 - Alignment between documentation, theoretical description, and actual implementation.
 - For flood, does “1-in-100 year depth” really reflect the statistical definition claimed, or is it an approximate label?
2. Does the model provide timely, accurate, and stable results?
 - Outputs must be numerically correct under the chosen methodology and operationally reliable.
 - For flood portfolios, the engine must complete within required time windows without unstable, jumpy outputs for similar inputs.
3. Does the model consume excessive computational resources?
 - Excessive run-time or resource use becomes a risk in its own right, encouraging unsafe shortcuts.
 - For instance, halving the number of Monte Carlo scenarios to meet a batch window changes uncertainty properties that must be understood and justified.
4. Are there controls over manual overrides?
 - Overrides (for example, adjusting scores for local levees) are not automatically bad but must be rare, justified, logged, and reviewed.
 - Systematic, undocumented overrides suggest deeper issues either with the model or with its deployment.
5. How is the output consumed for risk decision-making?
 - Do users treat a flood score as a binary accept/reject flag, or understand it as an uncertain indicator among other factors?

- Is there guidance about how outputs feed underwriting, pricing, capital allocation, or disclosures?

6. Is the model's output used as input to another model?

- This is the "string of pearls" problem: hazard → damage → capital → pricing.
- Validators must consider how uncertainties and assumptions propagate across the chain, not just model-by-model correctness.

7. What are the model's weaknesses and limitations of use?

- Every model has contexts where it performs poorly—small urban catchments, groundwater-dominated flooding, non-stationary climates, limited gauge coverage.
- SR II-7 culture values explicit, documented limitations more than sales narratives claiming perfection.

8. What type of monitoring is required?

- Once in production, how will performance be checked over time?
- For flood models this might include comparing modelled vs observed events, tracking drift in exposure data, and monitoring the frequency and pattern of overrides.

9. Is the overall model risk acceptable?

- After considering methodology, implementation, usage, limitations, and monitoring, is it sensible to use this model for the proposed purpose?
- A simpler, transparent model whose weaknesses are well-understood may be preferable to an opaque one that is hard to challenge.

These questions are not about box-ticking but about recreating, in a structured way, the critical thinking that was missing in pre-crisis failures. For a flood specialist, they roughly map to: do we know what this model does, where it is strong or weak, how we will keep an eye on it, and how much risk we are willing to take on its shoulders?

Governance: how SR II-7 structures responsibility

SR II-7 does not prescribe a single organisation chart but it strongly implies a governance architecture that most banks and mature vendors now mirror. Key components, applied to flood and climate models, are:

Model inventory and tiering

- A central list of models, including flood hazard engines, vulnerability modules, aggregation tools, and PRS pricing models.
- Each model assigned a tier based on materiality and complexity, so that high-impact models (for example those feeding capital) get more intensive validation and monitoring.

Defined roles (often supported by a RACI)

- Model owner: accountable for performance, documentation, and responding to issues.
- Independent validator: provides effective challenge and documents findings and recommendations.
- Operations / IT: responsible for running models, handling production incidents, and collecting monitoring metrics.
- Model Risk Committee (MRC): a decision-making forum that reviews artefacts, signs off on risk tiers, and escalates material issues to senior leadership.

Templates and artefacts

- Standardised development and validation templates ensure that key aspects—assumptions, limitations, tests, monitoring plans—are captured consistently.
- Minutes and records from the MRC provide evidence of real challenge, not just rubber-stamping, which is vital when regulators or bank clients review governance.

Culture of “warts and all” transparency

- SR 11-7-style governance rewards candour about model weaknesses; a model that comes with a clear “handle with care” label is more likely to pass validation than one presented as flawless.
- For vendors, embracing this culture turns model governance from a compliance tax into a commercial differentiator: banks prefer partners who can speak SR 11-7 fluently.

For a flood risk reader, this governance structure is the institutional analogue of good scientific practice and peer review: rigorous, documented challenge, clear ownership, and repeatable processes, but applied to the entire lifecycle of models that now move capital and credit as well as water and debris.

Model Governance for Climate Risks

While this book is focused on the present hazard of flood risk, it is worth noting how the banking industry takes model governance in new areas of risk management of which it is worth adopting as the industry introduces PRS.

The Commodity Futures Trading Commission (CFTC) issued a groundbreaking report in September 2020 titled *Managing Climate Risk in the U.S. Financial System*, authored by the Climate-Related Market Risk Subcommittee, chaired by Bob Litterman. This report marked the first comprehensive effort by a U.S. financial regulator to address climate-related risks in financial markets.

Litterman, a former Goldman Sachs executive known for his work in risk management, led the diverse committee of 34 experts from financial institutions, academia, and non-governmental organisations.

The report emphasises that effective model governance is essential for managing climate-related financial risks. It acknowledges the unique challenges of climate risk modelling, including data limitations, long time horizons, and deep uncertainty about future scenarios.

The committee recommends that financial institutions establish robust governance frameworks for climate risk models, with clear board-level oversight and integration into existing enterprise risk management systems. According to the report, models must balance standardisation with analytical flexibility, while avoiding false precision in their outputs.

These governance principles aim to ensure that climate risk assessment becomes a mainstream component of financial risk management across the U.S. financial system.

Risk Management and Oversight

- **Board Oversight:** Establish clearly defined oversight responsibilities for climate risk at the board of directors level.
- **Integration with Existing Frameworks:** Address climate-related risks through existing risk management frameworks and appropriate corporate governance.
- **Enterprise Risk Management:** Integrate climate risks into Enterprise Risk Management (ERM) and Own Risk Solvency Assessments (ORSA) processes

Scenario Analysis and Stress Testing

- **Consistent Scenarios:** For assessment, use a consistent and common set of broad climate risk scenarios, guidelines, and assumptions.
- **Analytical Discretion:** Allow firms to decide how they perform scenario analysis, balancing standardisation with flexibility.
- **Continuous Improvement:** Establish mechanisms for ongoing refinement as science, data, tools, and conditions evolve.
- **In-house Capabilities:** Develop internal capabilities to analyse climate scenarios, understand underlying assumptions, and recognise limitations.

- **Avoid False Precision:** Recognise that quantitative results should be treated as illustrative rather than precise in climate scenario analysis.
- **Proportional Analysis:** Ensure the scope, depth, and complexity of analyses are proportionate to the materiality of the impact measured.

Flood Risk Model Stack Problem

At the heart of flood risk modelling lies the "model stack problem." As established in earlier chapter, modern flood risk assessment involves the output of one model becoming the input of another, creating a chain of dependencies that amplify uncertainties throughout the modelling process. For flood risk specifically, this stack includes:

- **Weather pattern models:** Generating precipitation forecasts and historical distributions.
- **Hydrological models:** Translating precipitation into river flows and surface water accumulation.
- **Hydraulic models:** Determining how water moves across landscapes and through built environments.
- **Impact models:** Assessing damage to properties based on flood characteristics.
- **Banking impact models:** Translating physical damage into economic and banking consequences

From a model governance perspective, the practice of hydrologic and hydraulic modelling embodies a certain level of paradox: We use deterministic equations to describe inherently uncertain processes. We apply simplifications to represent indescribably complex systems. We seek precision in a domain where perfect prediction is impossible. Apart from that it should work perfectly fine so long as we understand and measure uncertainty” - David Kelly, MKM.

Key Characteristics of the Flood Model Stack

Cascading dependencies: Physical process models feed into statistical models, which feed into banking impact models, creating multiple layers of uncertainty. For example, the previous example of the Thames flood model requires output from weather pattern timeseries to generate precipitation measurements, which feed hydrological models producing water flow predictions, ultimately leading to real-time flood level monitoring.

- **Simplification of complex processes:** Fundamental physical processes like cloud formation and precipitation patterns directly influencing flooding are often simplified due to computational limitations. As noted earlier, these simplifications significantly impact flood risk assessment.

- **Uncertainty amplification:** Each model in the stack contains its uncertainties, which compound as they move through the chain. Our analysis of flood models showed how errors in flow estimation could be magnified for in-depth calculations, creating significant variations in estimated property damage.
- **Cross-disciplinary integration:** Models originating from different scientific traditions (climate science, hydrology, economics, finance) with varying validation standards are linked, creating governance challenges across domains.
- **Temporal mismatch:** Models operating at different time scales (from hours for weather to decades for climate) must be integrated coherently, as highlighted in our examination of the mortgage-insurance duration mismatch as described in the previous chapter on PRS.

Model Risk and Complexity

The interconnected nature of model stacks in banking creates unique governance challenges that traditional frameworks struggle to address. Banks operate in increasingly complex regulatory and technical environments where models serve multiple purposes—from credit decisions to capital planning—and inform high-stakes business decisions. These models often form intricate webs of dependencies similar to the flood model stack but with additional banking and regulatory complexities.

Model Dependency Management

Banking institutions typically maintain hundreds or thousands of models across departments, creating a tangled web of dependencies. These models form hierarchical structures, sometimes called a “string-of-pearls”, where there is high dependency. Consider the following:-

- Market data models generate timeseries of market simulations that collectively create market data.
- Mid-level models link the market data to risk factors used for instrument pricing and hedging.
- Risk models aggregate the risk factors of the originated risk and the hedges, leaving a basis required for P&L attribution and VaR.
- Stress-based models that define tail events are used in the aggregate for capital allocation.

Each layer inherits assumptions and uncertainties from previous layers, often without transparent documentation of these inherited limitations. For example, a mortgage pricing model might depend on interest rate projections, which rely on economic growth forecasts—each introducing its uncertainty profile that compounds through the chain.

Model Boundary Ambiguity

In complex banking environments, the boundaries between models become increasingly blurred. This creates several governance challenges:

- **Unclear ownership:** When multiple teams contribute to different components of an interconnected model system, responsibility for overall performance becomes diffuse.
- **Validation gaps:** Components at the boundaries between models often receive less scrutiny during validation exercises.
- **Inconsistent assumptions:** Adjacent models frequently operate under contradictory assumptions about the same underlying process

A particularly problematic boundary exists where statistical models interface with expert judgment. In flood risk assessment, as in many banking contexts, human experts often modify model outputs based on experience, creating a "grey area" where governance controls are difficult to apply consistently.

Uncertainty propagation

As highlighted in the flood model stack, uncertainty propagation becomes increasingly important—and challenging—in complex model environments. Banks face particular difficulties with:

- Identifying the sources of model uncertainty.
- Quantifying how uncertainty grows through model chains.
- Attributing performance issues to specific components within the stack.

While techniques like Monte Carlo simulation can help quantify overall uncertainty, attributing this uncertainty to specific model components remains extremely difficult. This creates significant challenges for model improvement efforts as teams struggle to identify which components would benefit most from refinement. Perhaps most concerning from a governance perspective are the emergent properties that arise from complex model interactions.

“Banking history is replete with examples where seemingly well-governed individual models created systemic risks.”- David Kelly, MKM.

The following red flags worth calling out in any complex model instances tend to be a combination:

- Models that individually capture risk factors accurately may collectively miss correlation effects during stress periods.
- Feedback loops between market and risk models can amplify volatility during crises.

- Optimisation of individual models may lead to collectively suboptimal outcomes.

These emergent risks often elude traditional governance frameworks focused on individual model performance rather than systemic interactions.

Complex model environment

Banks face significant operational challenges in governing complex model environments such as:

- Specialised expertise: Effective governance requires rare combinations of domain expertise (e.g., climate science, banking, risk management).
- Resource constraints: Validation resources are typically limited, requiring difficult prioritisation decisions.
- Technical infrastructure: Managing model dependencies requires technical infrastructure that many banks lack

Toward Integrated Model Governance

Addressing these governance challenges requires approaches that explicitly recognise model interdependencies and embrace the complexity inherent in modern banking environments. Key principles include:

- System-level validation: Complementing component-level validation with holistic assessments of model systems.
- Consistent metadata: Developing standardised ways to document model assumptions, limitations, and dependencies.
- Transparent lineage: Creating clear documentation of how data and assumptions flow through model chains.
- Adaptive governance: Implementing frameworks that can evolve as models and methodologies change.
- Cross-functional oversight: Establishing governance bodies with expertise spanning relevant domains.

Case Study: Flood Risk Modelling

The following case study on flood risk modelling illustrates how these principles can be applied in practice, highlighting successes and ongoing challenges in governing complex model environments with significant dependencies.

Our UK flood risk modelling analysis provides an example of a physical risk model stack. The process typically involves the following checklist across the model risk governance process:

- **Measuring typical flow rates:** Officers measure typical flood events in each area, noting how the land responds to rainfall. They manually define scores for pre-defined factors like how quickly water flows, how much is absorbed, and how much runs off.
- **Calibrating flow distributions:** Using data measured in the first step, statisticians create a distribution of potential flows, including the variability of maximum values and their frequency.
- **Estimating flood flow frequency:** The next step combines the catchment profiles with the statistical models to assess the relative increase in water flow for less frequent events than typical flow rates.
- **Flood heights by location:** Modellers then take the flow estimates under different probabilities and adapt fluid dynamic simulations to measure how a rainwater burst leads to a rise in water in each location.
- **Inter-model dependency:** The compatibility of underlying assumptions across the different model types (measurement, statistical, fluid dynamics) is often inconsistent. Statistical models are sometimes shoehorned into data, creating human inconsistency.
- **Error amplification:** Final simulations depend on multiple upstream models with distinct error profiles. As shown in our Thames flood analysis, the simulation utilises output from the initial three steps, which consist of perfectly smooth and error-free distributions of future flow severity and frequency. Subsequently, it incorporates its assumptions about landscape response, thereby amplifying errors.
- **Granularity challenges:** Minor differences between two locations can lead to significantly different results, regardless of model sophistication. Chapter 5 revealed how all those affected by floods point out how some properties avoid being affected while those nearby are inundated.
- **Underlying assumptions:** Clear documentation of assumptions and construction of climate variable distributions (e.g., tails of the distribution of precipitation at a catchment area).
- **Data source lineage:** Transparent information about data sources (public and vendor), measurement techniques, curation processes, quality checks, and update frequency. This is particularly important for flood models that rely on high-resolution LiDAR data and real-time gauge measurements.
- **Input data derivation:** Clear explanation of how vendor models derive their inputs, especially for crucial parameters like catchment response rates and infiltration capacities.
- **Portfolio aggregation:** Documentation of how portfolios of assets are aggregated by location, which is essential for understanding concentration risk in flood-prone areas.

- **Model limitations:** Description of known model weaknesses, output uncertainties, and limitations of appropriate use, particularly regarding the granularity issues identified in our flood model analysis.
- **Model use:** Define and document where and how flood model outputs should be used, distinguishing between screening tools and decision-making instruments.
- **Output interpretation guidelines:** Provide clear guidance on how consumers should interpret outputs within their risk-based processes, particularly for flood scores that may appear deceptively precise.
- **Educate end users:** Ensure that risk managers, loan officers, and senior decision-makers fully understand the technical limitations of flood models and the practical constraints on their utility, especially regarding property-specific characteristics that models may not capture.
- **Validation frameworks:** Develop robust validation approaches designed explicitly for flood risk models, incorporating historical flood data and forward-looking climate projections.
- **Uncertainty protocols:** Establish methods for effectively communicating model uncertainty to decision-makers, recognising the compounding uncertainties throughout the model stack.

Vendor due diligence for physical risk models

Physical risk vendors now sit directly on the critical path between weather, hydrology and banking capital. Their data and models flow into loan underwriting, collateral valuation, risk transfer design and, ultimately, regulatory capital and disclosures. From a supervisor's perspective, this makes banks fully accountable for vendor model failures, irrespective of who owns the IP.

The practical implication is simple: there is no delegation of model risk. Banks must treat key vendors as extensions of their own model landscape, applying proportionate governance that recognises vendor outputs as inputs to material internal models. Vendor due diligence is therefore not a procurement nicety; it is a core model-risk control.

The starting point is to classify vendor solutions by materiality and complexity in the bank's context, not the vendor's marketing. On materiality, the key questions are whether vendor outputs feed into credit decisions, pricing, loan covenants, PRS structuring, capital models, ICAAP or regulatory stress tests, and whether material errors in hazard or loss estimates would affect solvency metrics, risk appetite limits or supervisory expectations on physical risk management. On complexity, the focus is on whether the vendor relies on stacked models with multiple layers of assumptions and potential error amplification, and whether AI or machine-learning components reduce interpretability or complicate back-testing and sensitivity analysis.

High-materiality, high-complexity vendor models, such as flood-loss engines driving PRS pricing or capital stress tests, should be treated as Tier 1 from a bank perspective, triggering maximum

scrutiny. Lower-impact components, such as simple postcode-level hazard flags used for marketing or advisory, may warrant lighter treatment with proportionate checks.

Once tiered, banks can structure vendor due diligence along four dimensions: governance, methodology, data and operations.

On governance and culture, the core question is whether the vendor has internalised model risk management or merely responds ad hoc to questionnaires. Banks should look for the existence of a model risk committee or equivalent with clear terms of reference, a model inventory and owner assignments; written policies covering model development, validation, monitoring, change management and decommissioning; evidence of effective challenge in the form of documented findings and remediation plans; and a willingness to document known weaknesses and limitations of use rather than presenting models as flawless. A vendor that cannot articulate its own model-risk framework should be assumed to transfer the full governance burden onto the bank.

On methodology and the model stack, physical risk vendors typically operate a chain of models spanning weather, hydrology, hydraulics, vulnerability and economic impact, with cascading assumptions. Due diligence needs to unpack this stack and identify where uncertainty enters and accumulates. Banks should expect clear conceptual descriptions of each layer and how outputs from one feed into the next, justification of major modelling choices such as distributional assumptions and treatment of defences, sensitivity analysis across the stack, and a transparent approach to non-stationarity over mortgage and PRS horizons.

On data, coverage and representativeness, vendor claims often pivot on proprietary data and global coverage, but due diligence must test whether the dataset is fit for the bank's purpose rather than simply large. Core questions include the primary sources for hazard and exposure data and how gaps are handled, the effective resolution in the bank's key geographies and how it compares with collateral and PRS trigger definitions, how historical events are used to calibrate and validate models, and how data lineage, versioning and traceability are supported when outputs enter capital calculations or disclosures. Banks should insist on data dictionaries and schema descriptions that support integration with their chosen common data models.

On operations, monitoring and change management, the vendor must demonstrate that its models behave as predictable, supportable production assets rather than research experiments. Areas to probe include monitoring indicators such as performance versus realised events, stability of scores and operational incidents; change control processes for testing, documenting and communicating model updates; and incident-handling playbooks for model failures or major mis-predictions, including client notification, root-cause analysis and remediation.

Effective due diligence is not a one-off gate, but the start of a governed relationship. For key physical-risk vendors, banks should seek contractual recognition that vendor tools are in-scope models for banking use, with explicit obligations around documentation, transparency, incident reporting and prior notification of material changes. Regular joint governance touchpoints between the bank's validation team and the vendor's model-risk leadership help maintain a shared view of limitations, monitoring and planned enhancements. A tiered engagement model allows

more intensive review and documentation for vendor components feeding PRS pricing, ICAAP or capital models, and lighter-touch oversight for ancillary analytics.

Banks that adopt this structured approach will find that vendors with mature model-risk governance become strategic partners in physical-risk integration, while those without it remain trapped in low-value, reporting-only niches. For PRS and broader banking use of physical risk, this distinction will determine which vendors can cross the line from climate storytelling into solvency-relevant risk infrastructure.

Chapter 20 - CDM and Data Framework

The banking industry's approach to physical risk assessment stands at a critical juncture where the need for standardised data models intersects with increasingly complex environmental challenges. The Common Domain Model (CDM) emerges as a crucial framework for addressing this complexity, particularly in the context of flood risk assessment and management, one of the most significant physical risks facing banking institutions today.

“The Common Domain Model represents more than a technical standard - it bridges weather timeseries creation and banking markets, physical reality and banking value, data and decision. Without this bridge, we cannot build the market infrastructure needed for managing physical risk in the decades to come.” - David Kelly, MKM.

The Common Domain Model (CDM) originated as a collaborative effort by three significant banking trade associations: the International Swaps and Derivatives Association (ISDA), the International Capital Market Association (ICMA), and the International Securities Lending Association (ISLA). These organisations developed the CDM to standardise the representation of banking products, lifecycle events, and processes, aiming to streamline operational efficiency, enhance data consistency, and reduce costs and fragmentation within the banking markets.

In September 2022, ISDA, ICMA, and ISLA selected the Fintech Open Source Foundation (FINOS) to host and manage the CDM as an open-source project, marking a critical shift towards fostering community-driven innovation and widespread industry adoption. The transition to FINOS included creating a centralised repository under the Community Specification Licence to allow banking institutions and other stakeholders to collaborate transparently and inclusively.

FINOS officially launched the CDM project in February 2023, integrating it fully into its ecosystem. This collaboration aligns with FINOS's broader mission to promote open-source standards for interoperability, regulatory alignment, and innovation in banking services. The CDM has since evolved into a key tool for automating trade processing, managing banking transactions, and supporting digital frameworks like smart contracts and regulatory reporting.

Core Components of the CDM

The CDM defines and organises the banking transaction lifecycle into the following hierarchical models to ensure efficiency, transparency, and standardisation:

- **Product Model:** Describes banking instruments like contracts used to transfer banking risk.
- **Event Model:** Represents lifecycle events such as trade executions and modifications.
- **Process Model:** Lays the groundwork for automating and standardising industry processes.
- **Reference Data Model:** Captures foundational details such as parties, legal entities, and rate indexes essential for modelling trades and agreements.

From CDS to PRS: Deal Lifecycle Overview

The development of Physical Risk Swaps is best understood by starting from a familiar reference point: the operational lifecycle of a standard Credit Default Swap. The CDS market provides a mature template for how a new risk transfer instrument moves from bilateral innovation to scalable, standardised infrastructure. In this section, we briefly outline the key stages of the CDS lifecycle and set up the analogy that the remainder of the chapter will translate into a Common Data Model for PRS.

At a high level, a CDS trade passes through four main phases in practice: pre-trade structuring and pricing, trade execution and confirmation, ongoing lifecycle management, and termination or portfolio compression. Each phase depends on clear reference data, unambiguous contract terms, and agreed mechanics for events and settlements—exactly the ingredients that a CDM must encode if PRS is to achieve similar robustness.

Pre-trade: reference data, pricing, and term-sheet

Before a CDS is transacted, dealers and clients align on a small set of critical choices: the reference entity (for example, a specific corporate or sovereign), the standard contract terms defined by ISDA, and the market spread reflecting the reference entity's default risk at a given maturity. In modern markets, this pre-trade process relies heavily on centralised reference data (such as Markit RED codes) and standard coupon levels, which enable the construction, quotation, and risk management of CDS consistently across institutions. The output of this phase is effectively a well-specified term sheet that can be booked and priced in a standard way.

Trade execution and confirmation

Once terms are agreed, execution occurs bilaterally or via electronic platforms, followed by confirmation through standardised documentation that references the ISDA Definitions. Over time, the industry has moved from manual confirms and spreadsheet reconciliations to electronic confirmation services and trade warehouses that hold the golden record for each contract. This combination of legal standardisation and electronic infrastructure dramatically reduced operational risk and backlogs when volumes scaled, and it is precisely this pattern that PRS must emulate to become a viable market instrument.

Ongoing lifecycle: premiums, events, collateral, and clearing

After execution, a CDS enters a long phase of routine and event-driven lifecycle management. Protection buyers pay periodic premiums, protection sellers stand ready to compensate for defined credit events, and both sides manage collateral under CSAs or central clearing arrangements. When a credit event occurs, auction mechanisms and determination committees standardise how recovery rates are established and how cashflows are settled, ensuring that thousands of contracts referencing the same entity can be resolved in a coordinated way. This lifecycle framework—regular cashflows punctuated by standardised event handling—is the

operational backbone that a PRS must replicate for physical risk events instead of corporate defaults.

Termination, novation, and compression

Finally, CDS portfolios are actively managed through trade unwinds, novations to new counterparties, and multilateral compression services that eliminate redundant trades while preserving net risk. These techniques have been crucial in shrinking gross notional exposure and maintaining a manageable operational footprint for the market. For PRS, an equivalent set of lifecycle tools will be needed to keep portfolios tractable as banks originate, hedge, and recycle physical risk exposures across many counterparties over time.

The purpose of this chapter is to take this familiar CDS lifecycle and re-express it in the language of a Common Data Model for Physical Risk Swaps. Rather than starting from pricing formulas or hydrology, the focus here is on the data structures and contractual representations that allow a PRS to behave like a CDS operationally, while referencing gauges, flood stages, and hazard portfolios instead of reference entities and credit events.

Why a Common Data Model Is Essential for PRS

The previous chapters have shown that banks face fragmented, inconsistent physical risk information: different vendors use incompatible hazard metrics, model assumptions are opaque, and outputs are rarely aligned with banking time horizons or portfolio structures. This fragmentation makes it difficult to aggregate risk across portfolios, compare results across suppliers, or design scalable risk transfer instruments that multiple institutions can use with confidence. Physical Risk Swaps can only become a true market if the underlying hazards, exposures, and payouts are described in a common language that every participant can interpret in the same way.

ISDA and FINOS already provide much of the institutional blueprint for such a language. ISDA's legal frameworks and definitions illustrate how standardised documentation can turn a bespoke innovation like CDS into a globally fungible product, while FINOS projects demonstrate how open, shared data models and reference implementations can be maintained as industry utilities rather than proprietary one-offs. Together, they provide a legal and data foundation on which a Common Domain Model for physical risk can sit, enabling PRS contracts to be defined, exchanged, and processed consistently across banks, insurers, vendors, and regulators.

For Physical Risk Swaps, this implies several specific requirements that a Common Data Model must satisfy. First, it must explicitly and separately represent hazard, exposure, and payout: gauges and their thresholds, portfolios of at-risk properties, and the contractual mapping from observed events to cashflows all need clear, machine-readable definitions. Second, it must be interoperable across institutions and data providers so that a gauge or flood stage means the same thing whether it comes from a public agency, a commercial vendor, or an internal bank model. Third, it should reuse familiar derivatives constructs—such as legs, schedules, notional

amounts, and party identifiers—so that PRS can plug into existing pricing, booking, and risk systems with minimal friction. The remainder of this chapter takes these requirements and translates them into concrete CDM structures for gauges and Physical Risk Swaps.

Core CDM Concepts for Physical Risk Swaps

Designing a Common Data Model for Physical Risk Swaps begins with a clear separation of three core concepts: how physical hazards are observed, how exposures are organised, and how contracts transform observed events into cashflows. In the PRS context, this translates into distinct CDM objects for flood gauges, gauge sets and portfolios, and the swap contract itself, each with its own responsibilities but all linked by stable identifiers.

- **Hazard observation layer:** At the base of the stack lies the hazard observation layer, represented by a Flood Gauge CDM that standardises how individual sensors, satellite feeds, or weather stations are described. Each gauge has a unique identifier, historical statistics such as maximum recorded levels and exceedance frequencies, detailed metadata about ownership, certification, and maintenance, and formally defined flood stages like Alert, Warning, and Severe thresholds. This object anchors the connection between the physical world and banking, ensuring that when a PRS refers to a particular gauge and trigger level, all parties share the same understanding of what that means in measurement terms.
- **Exposure layer:** The exposure layer is represented by the Property CDM and Mortgage CDM. Property CDM provides a comprehensive schema for asset-level characteristics, including identifiers (UPRN, PropertyID), location and geocoding, construction details, occupancy, valuation and detailed flood-related attributes such as Environment Agency flood zones, overall flood risk classification, distances to water bodies and ground level elevation. Mortgage CDM then links directly to PropertyID and UPRN, adding complete loan, borrower and performance information, including original terms, current LTV, arrears and revaluation slots. This pairing allows banks to trace any PRS-protected loss back from a swap payout to specific properties and mortgages, maintaining consistency between physical risk, collateral valuation and loan performance within a single data framework
- **Contract layer:** On top of this sits the contract and portfolio layer, where Physical Risk Swap CDM links hazard and exposure into tradable instruments. The swap schema reuses familiar derivatives concepts—Header with TradeType and parties, LegData with fixed or floating legs, ScheduleData with standard start, end and tenor fields—while introducing a GaugeSet that enumerates the basket of gauges, their indices and the PayoutSevereFlood amounts for each. In a multi-hazard setting, the same pattern can reference cyclone events or property-level triggers: a GaugeSet can be interpreted generically as the portfolio of physical triggers (gauges, storm footprints, property loss thresholds) that drive payouts, while the swap object encodes how those triggers are aggregated, scaled and settled between counterparties. In combination, these CDM concepts allow a PRS to mirror the operational clarity of CDS while remaining native to physical risk data, property collateral and mortgage exposures

Flood Gauge CDM: Standardising Hazard Observation

Flood gauges provide the most direct measurement of river and water levels, yet historically their metadata, quality controls and trigger definitions have varied widely between agencies and vendors. The Flood Gauge Common Data Model is designed to remove this fragmentation by defining a single, canonical structure for all gauges used in Physical Risk Swaps. It does this by combining a minimal but rigorous header, statistical attributes that support calibration, rich sensor metadata for governance, and standardised flood stage thresholds that can be referenced directly in contracts.

At its core, the Flood Gauge CDM centres on a Header section with a unique GaugeID that unambiguously identifies each sensor across the ecosystem. This identifier becomes the join key that links raw observations, historical statistics, swap gauge baskets and regulatory reports. Around this anchor, the SensorStats block records hydrologically relevant summary historical high water level, the date of that high, the last time the highest alert level was exceeded, and the frequency of such exceedances over a defined lookback window. These fields provide the empirical backbone for estimating return periods and calibrating the probability of severe events, allowing risk models to connect directly to the physical measurement record.

The SensorDetails section records the governance and operational context needed by model risk and supervisory teams. It details the data source type (such as sensor gauge, satellite, or weather station), gauge owner, specific gauge type (like radar or pressure transducer), manufacturer, installation and inspection dates, maintenance schedule, operational status, and certification status. It also includes comprehensive geospatial information, covering latitude, longitude, and elevation, ensuring flood depths and levels can be accurately interpreted in hydrodynamic models.

Lastly, the FloodStage block defines national or regional thresholds such as UK Flood Alert, Flood Warning, and Severe Flood Warning levels as clear numeric triggers. These stage definitions act as a link between continuous water level measurements and the discrete trigger conditions used in PRS contracts, ensuring that when a swap indicates a payout at “Severe Flood Warning” for a given GaugeID, all parties and systems share a common, officially encoded understanding.

Physical Risk Swap CDM: Contract and GaugeSet Schema

The Physical Risk Swap Common Data Model defines how a PRS contract is represented so that trading, risk and operations systems can all interpret the same trade in an identical way. It does this by reusing familiar derivatives patterns for parties, legs and schedules, and adding a GaugeSet section that links the contract to the underlying hazard observations defined in the gauge and event models.

- **Header:** The Header block contains the high-level contract attributes: TradeType, CounterParty, PartyId, ValuationDate, GaugeSetID, ProtectionStart and flags such as SettlesAccrual and PaysAtDefaultTime. These fields align closely with standard OTC derivatives documentation, making it straightforward to integrate PRS alongside CDS and interest rate swaps in booking and valuation engines.
- **Leg:** The Leg Data then specifies whether the leg is Fixed or Float, together with payer direction, currency, notional, day count convention, payment convention and any fixed rate, allowing the premium side of a PRS to behave like a conventional swap leg.
- **Schedule:** The Schedule Data completes the temporal definition with StartDate, EndDate, Tenor, calendar and business-day rules, ensuring that accruals, payment dates and roll conventions follow established market practice.
- **Gauge:** The distinctive feature of the PRS CDM is the GaugeSet section, which binds the contract to physical hazard measurements. GaugeSet includes an identifier for the netting set, a GaugeBasketSize, and a series of Gauge1...GaugeN structures, each containing a GaugeIndex, GaugeID and PayoutSevereFlood value.

GaugeID must correspond to a valid Flood Gauge CDM instance, so that the swap's trigger points can be traced back to a fully specified sensor with known location, history and stage thresholds. PayoutSevereFlood encodes the notional payout associated with that gauge when the severe stage trigger is met, allowing a PRS to define basket payouts as a sum or function of multiple triggered gauges. Taken together, this schema allows a PRS contract to look and behave like a standardised swap from a banking systems perspective, while being natively tied to gauge-level physical risk data through a well-defined GaugeSet.

PRS Deal Lifecycle as Data Transformations

Once the core CDM objects are defined, the Physical Risk Swap lifecycle can be viewed as a sequence of structured data transformations rather than ad hoc file exchanges or bespoke interfaces. Each phase of the lifecycle—origination, validation, execution, event handling and portfolio maintenance—operates by creating, enriching and linking FloodGauge, Property, Mortgage and PhysicalSwap records in a consistent way.

- **Origination:** In the origination phase, raw hazard, property and mortgage data are first normalised into their respective CDM forms. Gauge measurements and metadata from agencies or vendors are mapped into FloodGauge CDM instances keyed by GaugeID, while property portfolios are standardised into Property CDM records with UPRN and PropertyID, and associated loans are encoded as Mortgage CDM records linking back to those same identifiers. A prospective PRS is then constructed by assembling a GaugeSet that selects the relevant gauges or events and defines PayoutSevereFlood amounts, and by creating a PhysicalSwap object that references this GaugeSet alongside standardised Header, LegData and ScheduleData fields.

- **Validation:** Before such a trade is booked, a pre-trade validation step uses schema-aware checks to ensure consistency and completeness. Functions such as `validate_gauge` and `validate_swap` confirm that required fields are present, menu values are valid, basket sizes are plausible and each `GaugeID` in the `GaugeSet` corresponds to a valid `FloodGauge` record. Only CDM-conformant trades proceed to execution, where the `PhysicalSwap` instance becomes the golden record for downstream systems, carrying all contractual terms and links to hazard and exposure data.
- **Daily Valuation:** During the daily valuation and collateral phase, market data and updated hazard probabilities are applied to the `PhysicalSwap` to produce a mark-to-market consistent with the `GaugeSet`, `Property` and `Mortgage` data. The CDM structure allows valuation engines to combine current gauge statistics, storm scenarios and portfolio exposures into expected-loss and spread estimates, and to write back valuation and margin requirements in a standard way that can be consumed by risk, collateral and accounting systems. Fixed-leg cashflows from the premium payer are generated according to `LegData` and `ScheduleData`, and their present value is balanced against the expected protection leg, just as in a CDS, with the difference driving variation margin where CSAs or clearing arrangements are in place.
- **Trigger:** The trigger and settlement mechanics operate by updating hazard and exposure objects and then applying contract logic defined via the `GaugeSet`. When an observed event causes a gauge level or storm attribute to cross a defined threshold—such as a Severe Flood Warning level for a specific `GaugeID` or a cyclone intensity and surge combination—the `FloodGauge` or storm event CDM records are updated with the new measurements and timestamps. The swap engine then evaluates the `GaugeSet`: for each gauge or event that has met its trigger condition, the corresponding `PayoutSevereFlood` or other parametric payout amount is activated and rolled up into the protection leg cashflow, which is recorded and settled between counterparties according to the currency and payment conventions defined in the contract.
- **Closeout:** The closure and portfolio maintenance phase mirrors CDS practice but remains data-driven. As mortgages amortise, properties are revalued or assets are sold, the `Mortgage` and `Property` CDM records are updated, allowing banks to measure how much PRS protection remains relevant to the evolving collateral pool. Trades can be unwound, novated or compressed by operating directly on `PhysicalSwap` records—terminating positions, reassigning `PartyId` fields, or replacing multiple swaps with a smaller set that preserves net exposure. At maturity or early termination, the swap's remaining premium schedule is truncated, any residual trigger-related settlements are processed, and the `PhysicalSwap` is marked as completed, leaving a full CDM audit trail from origination through daily valuation, trigger events, cash movements and final close-out.

The CDM structure also facilitates explicit counterparty risk management by making collateral and exposure data machine-readable and consistent across trades. Since each `PhysicalSwap` clearly specifies parties, notional, mark-to-market value, and protection leg sensitivity to trigger events, banks can compile counterparty exposures across portfolios and directly link them to CSAs, margin models, and risk limits. Daily valuation outputs can be integrated into standard

collateral workflows, ensuring that variation margin reflects both changes in hazard probabilities and shifts in portfolio composition, rather than relying on opaque, trade-by-trade spreadsheets.

Looking forward, a natural progression is the integration of PRS into central clearing infrastructures, echoing the path that CDS has already taken. A CDM-defined PhysicalSwap, with standardised terms and clearly specified GaugeSets, is inherently suitable for novation to a central counterparty, which could then handle multilateral netting, margining, and default procedures across numerous PRS trades. While central clearing for physical risk derivatives remains a future goal, designing PRS to be CDM-compliant from the beginning ensures that communication with potential clearing houses—whether via file interfaces or real-time messaging layers introduced in the next chapter—will be straightforward once market depth and regulatory frameworks support cleared PRS products.

Governance and Standardisation Under FINOS CDM

For Physical Risk Swaps to scale beyond bilateral experiments, the underlying data structures must be governed as shared industry standards rather than private schemas. ISDA's experience with CDS—codifying definitions, credit events, auction procedures and standard coupons—shows how legal standardisation can transform an opaque, bespoke product into a fungible, centrally clearable market. FINOS extends this principle into the data domain by providing open governance processes for Common Domain Models, allowing banks, vendors and supervisors to converge on a single representation for instruments, events and reference data that evolves in a controlled, transparent way.

Positioning the FloodGauge, Property, Mortgage and PhysicalSwap schemas as extensions to a FINOS-hosted CDM ensures that any enhancements—such as adding new hazard types, supporting compound events or introducing more granular trigger logic—follow a documented proposal and review process rather than ad hoc versioning. Industry participants can propose changes, assess impacts on backward compatibility and align implementations, while regulators gain a clear line of sight into how physical risk data are structured and used. This open governance model reduces vendor lock-in, improves comparability of risk metrics across institutions and gives central clearers a stable target format for future PRS products, mirroring the path taken by CDS infrastructure but with data standardisation embedded from the outset.

Preview: From CDM Objects to Bank Integration and FDC₃

The Common Data Model structures outlined in this chapter are intended not only for static documentation but also to serve as payloads that flow through bank systems in real time. FloodGauge, Property, Mortgage and PhysicalSwap records offer a consistent, machine-readable view of hazards, collateral, and contracts that trading, risk, finance, and regulatory platforms can all process without the need for custom translations. Since these objects reuse familiar derivatives concepts for parties, legs, and schedules, they can be integrated alongside

existing CDS and IRS workflows, while still providing the additional detail needed for physical risk triggers and portfolio-level loss attribution.

In the wider integration architecture, FDC₃ and similar messaging standards become the coordination layer that moves CDM objects and references between applications and services. In a mature implementation, hazard updates, daily PRS valuations, collateral calls and portfolio revaluations would be published as CDM-conformant messages on shared channels, allowing deal capture, risk analytics, collateral management and regulatory reporting tools to subscribe and react in a consistent way. The next chapter builds on this foundation by showing how banks can embed these CDM objects into their integration fabric—using FDC₃ and related standards—to achieve real-time hazard awareness, automated PRS lifecycle processing and seamless linkage between physical risk, mortgage portfolios and capital markets hedging.

Chapter 21 - Banking Interoperability

Before physical risk data can flow into banking systems—before flood probabilities can influence collateral valuations or weather patterns can trigger capital adjustments—a fundamental technical prerequisite must be satisfied: the applications that comprise the modern financial desktop must be capable of communicating with one another. This chapter examines the evolution of desktop interoperability standards, focusing on the Financial Desktop Connectivity and Collaboration Consortium (FDC3) and its role within the Fintech Open Source Foundation (FINOS), establishing why these standards represent essential infrastructure for any serious integration of physical risk into banking operations.

The problem of application silos on financial desktops is not new. For decades, traders and financial professionals have been forced to manually copy and paste information between applications, creating operational risk and hampering productivity. A credit analyst might view weather data in one system, property valuations in another, and loan exposure in a third—yet these systems cannot communicate directly, requiring manual transcription that introduces errors and delays.

The integration of physical risk data into banking systems amplifies this challenge considerably. Physical risk assessment requires the orchestration of multiple data sources and analytical tools: weather forecast systems, hydrological models, property databases, mortgage portfolios, and credit risk engines. Without standardised interoperability, each integration becomes a bespoke engineering project, creating maintenance burdens that scale multiplicatively with system complexity.

The Problem of Application Silos

The modern financial desktop has evolved through decades of organic growth, with each new application addressing specific business needs whilst remaining isolated from its neighbours. A typical trading floor might deploy dozens of applications: execution management systems, order management systems, market data terminals, risk analytics platforms, compliance monitoring tools, and client relationship management systems. Each operates within its own data silo, unable to share context with applications running alongside it.

Consider the workflow of a portfolio manager assessing physical risk exposure. She might begin by reviewing weather forecasts in a meteorological application, then switch to a mapping tool to visualise property locations, followed by consultation of a risk model output, and finally reference the loan management system to understand portfolio exposure. Each transition requires manual data transfer—copying property identifiers, re-entering coordinates, and transcribing risk scores. The cognitive overhead alone represents a significant productivity drain; the potential for transcription errors creates genuine operational risk.

This problem compounds at the institutional level. Large banks may operate thousands of internal applications, each requiring integration with dozens of others. Without standards, the

integration burden follows a quadratic growth pattern: N applications require $N \times (N-1)/2$ bilateral integrations. For a bank with 500 applications, this implies nearly 125,000 potential integration points. Even with aggressive prioritisation, the engineering resources required become prohibitive.

The vendor ecosystem exacerbates these challenges. Financial institutions deploy applications from hundreds of external providers, each with proprietary data formats and integration requirements. A market data vendor, an analytics provider, and an execution platform may each define “instrument identifier” differently, requiring translation layers wherever these systems must interact. The resulting integration architecture resembles a tangled web of point-to-point connections, fragile to change and expensive to maintain.

Origins of Desktop Interoperability

The financial services industry has a precedent for solving integration challenges through open standards. The Financial Information eXchange (FIX) protocol, developed in the early 1990s, standardised electronic trading communications and transformed how market participants interact. Before FIX, each counterparty relationship required bespoke integration; after FIX, a single implementation enabled connectivity with any conformant partner. The productivity gains were transformative, enabling the explosion of electronic trading that followed.

FDC₃ emerged from the recognition that a similar transformation was needed for desktop applications. In October 2017, over twenty banks and capital markets solution providers convened to create the FDC₃ working group, led by OpenFin under the vision of Nicholas Kolba, who is recognised as the technical creator and co-founder of FDC₃. Kolba, who had extensive experience as a technology director at Thomson Reuters before joining OpenFin, identified the fundamental inefficiency: despite sophisticated individual applications, the desktop as a whole remained unintegrated.

The founding vision was ambitious yet straightforward: create vendor-neutral standards similar to FIX, but focused on desktop connectivity and collaboration. The goal was to enable plug-and-play interoperability where applications could communicate using a common language without requiring bilateral agreements between vendors. At OpenFin, desktop interoperability had been part of their software since 2012 through their fin.desktop.ClientBus API, but Kolba recognised that true industry transformation required an open standard managed by a neutral governance body.

By May 2018, when FDC₃ was formally contributed to the Fintech Open Source Foundation (FINOS), the initiative had attracted forty firms participating in collaborative development. This rapid adoption signalled clear market demand for standardised desktop interoperability. The contribution to FINOS ensured that an independent, neutral third party with expertise in open source would steward the long-term growth and adoption of FDC₃ standards.

FINOS and Open Source Governance

The Fintech Open Source Foundation (FINOS) serves as the neutral home for open collaboration in financial services technology. Operating under the Linux Foundation, FINOS provides governance frameworks, community infrastructure, and legal clarity that enable competitors to collaborate on shared challenges whilst preserving commercial differentiation in implementation.

FINOS governance follows established open source principles adapted for financial services requirements. Contributors include major global banks, technology vendors, and independent developers, all operating under transparent processes with clear intellectual property frameworks. The Community Specification Licence provides the legal foundation for collaborative standard development, ensuring that participants can implement and extend the standard without proprietary encumbrances.

Beyond FDC₃, FINOS hosts several initiatives relevant to physical risk integration. The Common Domain Model (CDM) standardises the representation of financial products, lifecycle events, and processes across derivatives, bonds, and securities lending. TraderX provides a reference implementation demonstrating how modern financial applications can leverage open standards within cloud-native architectures. Together, these initiatives create a comprehensive ecosystem for financial technology standardisation.

The governing board includes representatives from major financial institutions including Morgan Stanley, Goldman Sachs, and Bank of America, alongside technology companies and independent experts. This balanced governance ensures that the standard evolves to meet genuine industry needs rather than serving narrow commercial interests. The conformance certification programme provides neutral assessment of implementation quality, giving deployers confidence in vendor claims whilst giving vendors clear targets for development.

FDC₃ Technical Architecture

The FDC₃ standard consists of five complementary specifications that work together to enable comprehensive desktop interoperability. Understanding each component is essential to grasping how FDC₃ functions as an integrated ecosystem for financial application communication.

Desktop Agent API

The Desktop Agent serves as the central nervous system of any FDC₃ implementation. It is a software component that implements the FDC₃ API and acts as a router for messages between applications. Without a desktop agent, FDC₃-enabled applications remain isolated—they are merely a cluster of applications with potential for communication but no mechanism to realise it.

The Desktop Agent API defines standardised methods for applications to discover other applications, raise intents, share context data, and manage channels. From FDC₃ 2.0 onward, the API introduced transactional capabilities that enable workflows requiring confirmation,

approval, or multi-step processes. The hub-and-spoke architecture, where the agent serves as a centralised router, provides several advantages: the agent maintains state about running applications, their channel memberships, supported intents, and understood context types.

Intents

Intents represent standardised actions that applications can request from the desktop ecosystem. Where context data represents the nouns of FDC₃ communication, intents serve as the verbs—the actions that applications can perform. The standard defines intents such as ViewChart, ViewQuote, StartChat, ViewInstrument, and ViewAnalysis, each representing a common workflow action in financial services.

Intent-based automation enables seamless transitions between workflow steps. When a user identifies an instrument of interest in a research application, raising a ViewQuote intent allows any conformant pricing application to display the relevant data without manual instrument entry. The originating application need not know which pricing application will respond; it simply expresses intent, and the desktop agent routes to an appropriate handler.

Context Data

Context Data defines standardised schemas for information exchanged between applications. Rather than requiring each application pair to agree on data formats, FDC₃ provides common definitions for financial entities: instruments (identified by ticker, ISIN, FIGI, or other schemes), contacts, organisations, portfolios, positions, and countries. When an application broadcasts context, all listening applications receive structured data they can interpret without translation.

The context schema is extensible, allowing implementations to define custom context types for domain-specific needs. For physical risk applications, this extensibility proves essential: flood risk scores, property coordinates, catchment identifiers, and weather event parameters can all be represented as FDC₃ context, enabling seamless integration with the broader financial application ecosystem.

Channels

Channels provide coordination mechanisms that enable applications to share context selectively. Rather than broadcasting to all applications indiscriminately, applications can join specific channels, receiving only context relevant to their current workflow. The standard defines system channels (colour-coded shared channels available to all applications) and application channels (created by applications for specific purposes).

Channel architecture supports complex multi-instrument workflows. A trader might work with multiple positions simultaneously, assigning each to a different channel. Applications can then display information relevant to each position without confusion, synchronising their views as the trader switches focus between channels.

App Directory

The App Directory is a REST-based standard that enables discovery and distribution of FDC3-enabled applications. It serves as a catalogue describing available applications, their supported intents, handled context types, and launch requirements. FDC3 2.0 significantly improved the App Directory by standardising application records to work across multiple desktop agents, making applications discoverable without proprietary vendor app stores.

The specification supports multiple app directories—including in-house directories, vendor-provided directories, and third-party aggregated directories. FINOS plans to host its own directory of FDC3-enabled applications, providing a neutral registry for the ecosystem. This distributed architecture ensures that no single party controls application distribution whilst maintaining discoverability across implementations.

Industry Adoption

Adoption of FDC3 among financial institutions has accelerated significantly, particularly on the sell side, with major banks recognising the strategic value of standardised interoperability. Citi, UBS, State Street, NatWest Markets, and Bank of America are all officially using technology built on FDC3. The buy side has also engaged deeply, with Wellington Management serving on the FINOS Governing Board and participating in proof-of-concept work demonstrating high-trust transactional use cases.

BlackRock's Aladdin platform achieved FDC3 2.0 Conformance in September 2024, a landmark milestone given Aladdin's position as one of the world's most widely-used institutional investment platforms. This certification ensures that third-party vendor applications adhering to FDC3 can seamlessly interact with Aladdin applications regardless of provider. Morgan Stanley has demonstrated exceptional commitment by achieving FDC3 2.0 Conformance with their open-source ComposeUI platform, contributing .NET bindings to the standard, and taking on co-maintainer responsibilities.

Bank of Montreal (BMO) provides a compelling case study of FDC3 benefits. Having implemented FDC3 across 73 applications spanning all lines of business, BMO reports that development cycles are fifty percent faster minimum and resources have been freed to work on revenue-generating initiatives rather than integration projects. This transformation from integration burden to implementation efficiency exemplifies the standard's value proposition.

The desktop interoperability platform provider landscape has consolidated around FDC3 support. The three major vendors—OpenFin, Glue42, and Finsemble—all provide FDC3-compliant implementations. In 2023, Glue42 and Finsemble merged to form interop.io, creating a unified platform treating the Glue42 API, Finsemble API, and FDC3 API as first-class citizens. This consolidation signals market maturation around the open standard.

FDC3 2.2 and Browser Support

The release of FDC3 2.2 in late 2024 marks a watershed moment in the standard's evolution by introducing experimental support for browser-resident desktop agents. Historically, FDC3 was designed for Electron-based container platforms that provide greater control over web security

and processes by injecting FDC₃ API implementations. These Preload Desktop Agents inject scripts into web windows to provide access to the FDC₃ API.

FDC₃ 2.2 introduces two critical new specifications to enable browser-based implementations. The Web Connection Protocol (WCP) governs how browser-based applications discover and connect to desktop agents without relying on script injection. The Desktop Agent Communication Protocol (DACP) defines standardised message exchanges for all FDC₃ API operations, enabling applications to communicate with desktop agents through various transport mechanisms.

The implications are profound: FDC₃ can now extend beyond the traditional trader desktop to support a broader variety of financial workflows across web applications, mobile interfaces, and potentially AI-powered tools that can be trained on the open-source standard without intellectual property concerns. For physical risk integration, browser support enables deployment scenarios where risk analytics run in cloud environments, delivering results through standard FDC₃ mechanisms to any conformant desktop application.

The Common Domain Model

Whilst FDC₃ provides the communication infrastructure and workflow orchestration layer—the how applications talk to each other—a complementary standard addresses what they exchange. The Common Domain Model (CDM), also hosted at FINOS, standardises the representation of financial products, lifecycle events, and processes.

The CDM originated as a collaborative effort by three significant banking trade associations: the International Swaps and Derivatives Association (ISDA), the International Capital Market Association (ICMA), and the International Securities Lending Association (ISLA). These organisations developed the CDM to standardise the representation of banking products, lifecycle events, and processes, aiming to streamline operational efficiency, enhance data consistency, and reduce costs and fragmentation within banking markets.

In September 2022, ISDA, ICMA, and ISLA selected FINOS to host and manage the CDM as an open-source project, marking a critical shift towards fostering community-driven innovation and widespread industry adoption. The CDM uses a custom domain-specific language called Rune to model financial products and lifecycle events. Running the model produces executable libraries in multiple programming languages, including Java, Python, DAML, TypeScript, and C++, outputting standard JSON objects representing trades or products that can be exchanged between parties running different languages without requiring mapping or conversion.

The synergy between FDC₃ and CDM is complementary and powerful. When combined, these standards enable true end-to-end interoperability where applications can not only invoke each other's functionality through FDC₃ intents but also exchange rich, semantically consistent financial data through CDM-based context objects. For physical risk applications, this combination proves particularly valuable: physical risk data can be expressed in CDM formats, enabling integration with existing banking infrastructure whilst maintaining semantic precision.

Why Interoperability Matters for Physical Risk

The integration of physical risk into banking systems represents precisely the kind of multi-application workflow that FDC3 was designed to enable. Consider the data flow required for a comprehensive physical risk assessment: weather forecast data feeds hydrological models, which inform property-level flood probabilities, which adjust collateral valuations, which impact capital calculations. Each step involves different applications, often from different vendors, traditionally requiring bespoke integration.

With FDC3 infrastructure in place, this workflow transforms fundamentally. A weather event alert can raise a standard intent that triggers flood model recalculation. The resulting risk scores, expressed as FDC3 context, propagate automatically to property valuation systems. Updated valuations flow through to loan management applications, where capital impact calculations can proceed without manual intervention. The entire cascade, from weather observation to capital adjustment, operates through standardised protocols rather than custom integration.

This architectural approach delivers several concrete benefits for physical risk integration. Development efficiency improves dramatically: rather than building custom integrations between each pair of systems, developers implement against the FDC3 standard once and achieve interoperability with the entire ecosystem. BMO's experience of fifty percent faster development cycles indicates the magnitude of these efficiency gains.

Vendor flexibility increases substantially. Banks can select best-of-breed components for each part of the physical risk workflow—weather data from one provider, flood modelling from another, property analytics from a third—confident that standardised interoperability will enable integration without bespoke engineering. The App Directory mechanism ensures that new applications can be discovered and incorporated without reconfiguring the entire system.

Operational resilience improves through standardisation. When integrations follow common patterns, monitoring and debugging become more tractable. The Desktop Agent's central role in message routing provides a natural point for logging, auditing, and troubleshooting. When problems occur, they manifest in standardised ways that support systematic diagnosis rather than requiring investigation of bespoke integration code.

Security and Trust

Traditional FDC3 implementations focused on low-trust use cases such as viewing news or setting context to a particular instrument because the standard initially lacked robust security features. Deeper transactional integrations, particularly between buy-side and sell-side firms, require stronger guarantees around identity verification, authorisation, and data confidentiality.

Development work on signed contexts and encrypted channels enables high-trust use cases. In this model, each party maintains asymmetric key pairs with public keys made available for verification. When a buy-side firm sends a query for product information, their application signs the context with their private key. The sell-side firm can verify the signature against the public

key, ensuring authenticity and preventing trading activity exposure. Encrypted channels protect confidential information and sensitive transactions from leakage.

For physical risk applications, these security features prove essential. Property-level risk assessments may constitute sensitive business intelligence. Portfolio exposure data certainly requires protection. The ability to exchange this information through authenticated, encrypted channels enables the kind of inter-firm collaboration that effective physical risk management may require—reinsurance coordination, syndicated loan assessment, or counterparty risk evaluation—whilst maintaining appropriate confidentiality.

Desktop Agent Bridging

Desktop Agent Bridging, introduced in FDC₃ 2.1, extends FDC₃ beyond single-agent boundaries by connecting multiple desktop agents through a standalone bridge component. The bridge acts as a server receiving and forwarding requests, enabling workflows that span applications hosted by multiple desktop agents—whether running on the same PC or across multiple machines on a network.

NatWest Markets contributed the BackPlane project, providing the first open-source implementation of a desktop agent bridge. BackPlane enables interoperability between in-house desktop agents, vendor-supplied agents, and even monolithic legacy .NET applications that do not run in desktop agents. This bridging capability proves particularly valuable for physical risk integration, where data may originate from specialised scientific computing environments that cannot practically run within standard desktop containers.

The bridging architecture supports progressive adoption. Organisations need not convert all applications to FDC₃ simultaneously; instead, bridges can expose FDC₃ interfaces for legacy systems whilst native FDC₃ applications communicate directly. This hybrid approach reduces implementation risk and enables gradual modernisation rather than requiring wholesale system replacement.

TraderX Reference Implementation

TraderX is a FINOS reference application designed to teach and demonstrate desktop interoperability patterns. Rather than providing production-ready software, TraderX offers a sample trading application where developers can learn about financial application architecture, understand workflow patterns, and experiment with FDC₃ integration in a controlled environment.

The application demonstrates modern patterns including event-driven architecture, microservices communication, and containerisation. TraderX has been integrated with FDC₃ through multiple community initiatives, including the 2024 FINOS Tech Sprint which focused on streamlined research-to-execution workflows.

At the 2024-2025 Open Source in Finance Forum hackathon, the winning team successfully integrated TraderX with the Common Domain Model, creating standardised and enriched trade

data workflows. Trades created in the TraderX application were enriched with CDM metadata fields, creating an auditable, standardised representation of trade lifecycle events. This integration showcased how CDM adoption within a trading application built on FDC3-compatible architecture streamlines trade processing and enhances data quality through consistent semantics.

For organisations considering physical risk integration, TraderX provides a valuable learning resource. The patterns demonstrated—context sharing, intent-based automation, channel coordination—apply directly to physical risk workflows. Developers can experiment with these patterns in the TraderX sandbox before implementing them in production systems.

Challenges and Limitations

Despite significant momentum, FDC3 adoption faces organisational and technical barriers. Decision-making in large financial institutions proceeds slowly, buying cycles are lengthy, and budgets are limited. Uncertainty about which vendors to trust or which version of the standard to implement can cause firms to defer FDC3 investments in favour of competing priorities.

User experience standardisation remains an ongoing challenge. Whilst FDC3 standardises the technical protocols for interoperability, it does not mandate user interface patterns or interaction designs. Without common UX patterns for FDC3 actions, users may encounter inconsistent, unpredictable interfaces that hamper adoption despite technical interoperability. The FDC3 community has recognised this gap and is encouraging contributions to form more mature UX patterns that make FDC3 actions discoverable, predictable, and universally recognised.

The browser support introduced in FDC3 2.2 opens exciting possibilities but also introduces complexity. Browser security models differ significantly from container environments, constraining what desktop agents can control. The experimental designation for web-related features acknowledges that the community is still learning how best to implement these capabilities.

Mobile support represents another frontier. Whilst FDC3 was designed initially for desktop environments, the principles of context sharing and intent-based automation apply equally to mobile workflows. Extending FDC3 to mobile devices would enable seamless transitions between desktop and mobile contexts, supporting the hybrid work patterns increasingly common in financial services.

Implications for Banking Integration

The availability of mature interoperability standards fundamentally changes the economics of physical risk integration. Where previously each integration required bespoke engineering, standardisation transforms the integration problem into a spoke-and-hub architecture where each application integrates once with the standard and achieves interoperability with the entire ecosystem.

For the physical risk integration framework presented in this book, FDC₃ provides essential infrastructure. The precipitation forecasts generated through Bayesian Monte Carlo methods can be exposed through standard FDC₃ context types. Flood risk assessments can trigger standard intents that activate downstream applications. Capital impact calculations can receive context automatically as risk assessments update. The entire Phase 6 integration layer, which transforms meteorological analysis into banking risk metrics, can leverage these standards rather than requiring custom integration engineering.

Looking forward, the combination of FDC₃ for desktop interoperability, CDM for data standardisation, and TraderX as a reference implementation creates a comprehensive open-source stack for financial services innovation. Developers can build applications that communicate through FDC₃, exchange data using CDM, and reference TraderX architectures for cloud-native microservices patterns—all governed under FINOS with enterprise-grade licensing and community support.

This standardisation infrastructure represents a prerequisite for the market-wide adoption of physical risk management capabilities. Individual banks can implement physical risk systems using proprietary integration; industry-wide transformation requires common standards that enable ecosystem-scale interoperability. FDC₃ and its companion standards provide that foundation.

Conclusion

Desktop interoperability standards represent essential infrastructure for integrating physical risk into banking systems. The application silos that characterise modern financial desktops create barriers to the multi-application workflows that physical risk assessment requires. FDC₃, governed under FINOS with broad industry participation, provides the technical foundation to overcome these barriers.

The standard's architecture—centred on the Desktop Agent as a communication router, with standardised intents as verbs, context data as nouns, channels as coordination mechanisms, and app directories as discovery infrastructure—provides a comprehensive foundation for interoperability. The value proposition spans reduced integration costs, improved user productivity, vendor neutrality, and future-proof scalability.

The synergy between FDC₃ and the Common Domain Model creates particularly powerful capabilities for physical risk integration. FDC₃ enables applications to communicate; CDM ensures they speak the same semantic language. Together, these standards enable true end-to-end interoperability where physical risk data flows seamlessly from meteorological sources through banking systems to capital calculations.

With this interoperability infrastructure understood, the following chapter examines Phase 6 of the MKM Bayesian Monte Carlo framework: the specific mechanisms through which precipitation forecasts transform into banking risk metrics, leveraging these interoperability standards to achieve operational deployment within existing banking infrastructure.

Chapter 22 - Integration to Banking

Chapter 23 - Operationalisation of PRS lifecycle

Chapter 24: Final Thoughts

The journey through the cutting-edge developments in flood risk modelling reveals a landscape transformed by artificial intelligence and interdisciplinary collaboration. What emerges is not merely a technological evolution but a fundamental reconceptualisation of how we understand and manage flood risk across the physical-financial continuum.

The integration of AI-enhanced weather prediction, hydrological modelling, and banking-compatible timeseries frameworks represents perhaps the most significant advancement in flood risk management since the introduction of computational fluid dynamics. This coherent model stack approach dissolves the artificial boundaries between scientific domains and banking analysis, allowing us to see risk as nature does: as a continuous flow from cloud movements through flood timeseries to parametric attribution.

While the technical achievements are remarkable, the path forward depends equally on standardisation and governance. The FINOS Common Domain Model (CDM) provides the essential scaffolding for data standardisation, creating a shared language that enables physical risk information to flow seamlessly through banking systems. This standardisation is not merely a technical convenience but a prerequisite for market development, regulatory compliance, and ultimately, societal resilience.

In parallel, legal standardisation through ISDA documentation brings the clarity and certainty that market participants require. By adapting established frameworks for innovative instruments like Physical Risk Swaps, we build on decades of market evolution rather than starting from scratch. This approach accelerates adoption and ensures compatibility with existing risk management frameworks.

Perhaps most critically, robust model governance must support every aspect of this integrated architecture. As uncertainty propagates through multiple model layers, transparency and validation become not only regulatory requirements but also essential components of a trustworthy system. From neural networks predicting precipitation patterns to the complex valuation models for physical risk instruments, each component must undergo rigorous governance that recognises both its individual characteristics and its role in the broader system.

The convergence of cutting-edge AI techniques with established banking frameworks represents a measured optimism rather than unbridled techno-utopianism. We recognise the challenges: computational demands remain substantial, data quality varies significantly across regions, and model interoperability requires continued attention. Yet these challenges appear increasingly surmountable through collaborative effort and technological innovation.

As we conclude this exploration of future developments, we can observe with quiet confidence that the tools required for comprehensive flood risk management are rapidly materialising. The integration of physical science and banking analysis through standardised timeseries provides a pathway to more resilient communities, more efficient capital allocation, and ultimately, a more sustainable relationship with our changing environment climate.

I will now end this book with another quote!

“This is not the end of the journey but rather the beginning of a new paradigm—one that promises to transform how we anticipate, mitigate, and respond to one of humanity's oldest and most persistent challenges.”- David Kelly, MKM.

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Glossary of Terms

A

Chronic Physical Events: Sudden and severe weather occurrences, such as floods, storms, or hurricanes, can immediately damage properties and infrastructure.

Adaptation Planning: Developing strategies to adjust to actual or anticipated climate change effects is especially important for property-level flood resilience measures.

Affordability Ratio: The percentage of income allocated to housing costs, including mortgage payments and insurance premiums, utilised in mortgage valuation models to evaluate default risk.

Aleatory Uncertainty is the inherent randomness in natural processes that cannot be mitigated through additional information or improved models; it contrasts with epistemic uncertainty.

Annual Exceedance Probability: The likelihood that a specified flood level will be surpassed in any given year, often expressed as a percentage (e.g., 1% annual exceedance probability for a "100-year flood").

Asset-Level Analysis: A thorough risk assessment carried out at the level of individual properties or assets, contrasting with portfolio-level analysis.

Atmospheric State Variables: Essential parameters that describe atmospheric conditions, including temperature, pressure, humidity, and wind vectors, which are used in weather prediction models.

B

Basel III/IV: International regulatory frameworks for bank capital requirements that increasingly include climate and physical risk considerations. These regulations establish standards for how banks must measure, disclose, and hold capital against various risks, with evolving requirements for incorporating physical and transition climate risks into banking operations.

Bayesian Approaches: Statistical methods that update probability estimates as new data becomes available; widely used in weather prediction and flood risk assessment to integrate prior knowledge and new observations.

Bayesian Inference: Statistical method used to update the probability estimate for a hypothesis as more evidence or data becomes available. It is based on Bayes' Theorem, which provides a way to calculate the probability of a hypothesis (or model) given observed data. As the industry gathers more data, we update our belief (posterior probability).

Beta Distribution: A family of continuous probability distributions defined on the interval $[0,1]$, commonly used to model damage ratios in vulnerability functions.

Black Box: A system or model whose internal workings are not transparent or easily understood; a concern with specific AI models used in risk assessment.

Boundary Conditions are the values of variables at the edges of a modelled domain, which are essential for solving hydrological and hydraulic models.

Boundary layer in weather refers to the thin layer of air at the Earth's surface that is directly influenced by the surface itself (land, water, vegetation, etc.). This layer is typically around 1 to 2 kilometres (km) thick, although it can vary depending on factors like time of day, weather conditions, and geography. It plays a crucial role in weather dynamics because it's where most weather phenomena, like temperature, moisture, and wind, interact with the Earth's surface.

Building Arrangement Effects: The impact of building configurations in urban environments on flood flow paths and property-specific flood exposure.

Business Interruption: Banking losses arising from the inability to operate a business during and after a flood event; a critical component of comprehensive flood impact assessment.

C

Capital markets Approach: A methodology for valuing physical risk that adapts techniques from banking derivatives markets, particularly credit default swaps, to quantify and price flood risk.

Catastrophe Modelling: Analytical techniques that estimate the physical, social, and economic impacts of natural disasters by combining hazard, exposure, and vulnerability components.

CDM (Common Domain Model): A standardised data representation framework developed through FINOS that establishes consistent definitions for weather events, flood events, properties, mortgages, and other elements of physical risk assessment.

Climate Biennial Exploratory Scenario (CBES): A stress-testing exercise conducted by the Bank of England to assess how climate change might impact the UK banking system. It evaluates financial institutions' resilience to transition risks and physical risks across multiple scenarios and time horizons, enabling regulators to understand potential systemic vulnerabilities.

Coastal Flooding is the inundation of land areas along coastlines, typically caused by storm surges, high tides, or tsunamis. It has distinct characteristics that set it apart from riverine flooding.

Compound Flood Events: Situations in which multiple flood mechanisms occur simultaneously or in close succession, resulting in more complex and severe impacts than single-mechanism events. Examples include combined coastal and riverine flooding during storms or pluvial flooding coinciding with high groundwater levels, creating amplified impacts through their interaction.

Convective storm is a type of storm that forms due to the upward movement (convection) of warm, moist air. This process creates instability in the atmosphere, which leads to the development of storms such as thunderstorms, tornadoes, and hailstorms. Convection occurs when warmer air near the Earth's surface rises because it is less dense than the cooler air above it.

Convolution Integral: A mathematical operation used in hydrograph analysis that combines excess rainfall with the unit hydrograph to calculate direct runoff. It enables hydrologists to predict streamflow response to complex rainfall patterns by treating them as a sequence of simple inputs, each generating its own scaled hydrograph response.

Convolutional Neural Networks: A class of deep neural networks primarily used for analysing visual data, such as images and videos. They are particularly powerful for tasks like image classification, object detection, and segmentation but have also been adapted to other domains like speech and text processing. CNNs are designed to automatically and adaptively learn spatial hierarchies of features from input data.

Copula Functions: Mathematical tools that describe the dependence between random variables, beneficial for modeling the correlation between flood-causing factors. Copulas separate the marginal distributions of individual variables from their joint dependency structure, enabling more flexible and accurate representation of complex relationships like those between precipitation, soil moisture, and river levels.

Correlation Distance Parameter: The spatial distance (typically around 1000m) within which properties experience intensely correlated flood impacts is crucial for understanding portfolio-level risk.

Coupled partial differential equations (PDEs) are a system of two or more partial differential equations that are interrelated or "coupled" together. These equations involve multiple unknown functions that depend on the same set of independent variables (such as time and space) and are linked through their terms. The solutions to these equations are dependent on one another.

Credit Default Swap (CDS): A banking derivative that protects against the risk of default by a particular entity, serving as a model for the development of Physical Risk Swaps.

Credibility Theory: An actuarial approach that balances individual risk experience with broader class experience when setting insurance premiums.

D

Damage Functions: Mathematical relationships that translate flood characteristics (depth, duration, velocity) into expected property damage, typically expressed as a percentage of property value.

Default Probability (PD): The likelihood that a borrower will default on required mortgage payments, potentially influenced by flood events through income disruption or property devaluation.

Designed neural network architectures refer to the various specific configurations of neural networks that are created and optimised for particular tasks or types of data. These architectures determine how the neurons (or nodes) are connected, how data flows through the network, and how the network learns from the data. Different architectures are suited to various types of problems, and they are typically chosen based on the nature of the input data, the computational resources available, and the desired output.

Digital Elevation Models (DEMs): Three-dimensional representations of terrain surfaces used in flood risk assessment to determine water flow paths and inundation zones.

Digital Regulatory Reporting (DRR): A framework within FINOS that standardizes the regulatory reporting process, integrated with Physical Risk Swaps to ensure compliance. DRR transforms regulatory rules into machine-executable code, streamlining reporting while ensuring consistency across institutions.

Discounted Cash Flow Analysis: A valuation method that estimates the value of an investment based on its expected future cash flows, adjusted for the time value of money and risk.

Disturbance Absorption Thresholds: The magnitude of flood events that natural systems can accommodate while maintaining their essential functions; exceeding this threshold may sharply reduce natural resilience.

Distribution Path Modelling: An approach that characterizes the entire space of potential weather pattern evolutions, especially those that could lead to extreme precipitation events. Rather than producing single forecasts, this method maps the full probability distribution of possible weather trajectories, providing a more complete picture of potential flood-generating conditions.

Downscaling: Producing high-resolution local forecasts from coarse global predictions is essential for capturing localised precipitation patterns that drive flood events.

Duration-Depth-Velocity Matrices: Three-dimensional classification schemes that acknowledge that flood impacts depend on water depth, flow velocity, and event duration.

Dynamic Pressure Considerations: Analysing the hydrodynamic forces exerted by moving floodwaters on structures is critical in high-velocity flood scenarios.

Dynamic Wave Approach: A hydraulic modelling technique that employs the full Saint-Venant equations, including all acceleration and pressure terms, to provide the most complete representation of flow dynamics. While computationally intensive, this approach captures complex hydrodynamic effects crucial for accurate flood modeling in rapidly changing conditions.

E

Earth-2 Initiative: NVIDIA's extensive project aims to create a digital twin of Earth's systems by combining global weather models with hydrological and impact assessments for integrated climate risk modelling.

Enterprise Risk Management (ERM) Integration: The process of incorporating climate-related risks into an organization's comprehensive framework for assessing and addressing all risks that affect its business objectives. This ensures climate risks are evaluated alongside traditional risks within established governance structures, rather than being managed in isolation.

Environmental Resilience Factors: These natural systems buffer, absorb, or amplify flood hazards before they impact vulnerable assets, including wetlands, forests, and soil systems.

Epistemic Uncertainty: This type of uncertainty arises from incomplete knowledge about asset characteristics and their vulnerability to hazards; unlike aleatory uncertainty, it can be mitigated through additional data or improved modelling.

Event_CDM: A component of the Common Domain Model that standardises the representation of extreme weather event pathways, utilising advanced AI platform components.

Exceedance Probability: This refers to the likelihood that a specific hazard intensity (such as flood depth) will be met or surpassed within a given time frame, typically one year.

Expected Cash Flow (ECF): In mortgage valuation, the scheduled payment is adjusted for the probability of default and loss given by default.

Expected Shortfall (ES), also known as **Conditional Value at Risk (CVaR),** is the expected loss in the worst $\alpha\%$ of cases. Unlike **Value at Risk (VaR),** which only measures the minimum loss at a given confidence level, ES calculates the average of all losses that exceed this VaR threshold, providing a more comprehensive picture of tail risk and addressing VaR's failure to capture the severity of extreme losses beyond the threshold.

Exposure Analysis: This process identifies assets (buildings, infrastructure, etc.) that could be impacted by flooding, including their location, value, and characteristics.

Exposure Assessment Integration: This involves a comprehensive analysis of all potential paths by which floodwaters could reach a property, incorporating topographic, locational, and built environment factors.

Extreme Event Prediction: These are specialised approaches for forecasting rare but severe weather events, often necessitating methodologies different from those used for typical conditions.

Extreme Value Theory (EVT) is a statistical framework for modelling and analysing the stochastic behaviour of extreme events—rare occurrences that lie in the tails of probability distributions. These events, though infrequent, often have significant societal, financial, or environmental impacts, making their quantification critical for risk management across diverse fields

F

Finite Difference Approximation: A numerical method used in hydrological routing to solve differential equations by replacing continuous derivatives with discrete differences. In flood modelling, this approach enables practical computation of water movement through complex river systems and across floodplains.

FINOS (Fintech Open Source Foundation): An industry consortium that promotes open collaboration in banking services technology, including developing the Common Domain Model for physical risk.

Fixed-Floating Exchange: The principle behind Physical Risk Swaps is that one party pays a fixed rate over the contract period while the other makes contingent payments based on realised flood levels.

Flash Flooding: Rapid-onset flooding characterised by high water velocities, typically occurring within six hours of the initiating event (heavy rainfall or infrastructure failure).

Flood Classification Systems: Methodologies for categorising flood events based on source mechanism, temporal characteristics, spatial extent, and other relevant factors.

Flood Depth: The height of floodwater above the ground or floor level is a primary factor in determining damage to structures.

Flood Gauges CDM: A standardised representation of flood monitoring devices within the Common Domain Model, creating definitive trigger points for banking contracts.

Flood Insurance Rate Maps (FIRMs): Official maps produced by the Federal Emergency Management Agency delineate flood hazard areas for insurance and regulatory purposes.

Flood Re: A not-for-profit reinsurance scheme established in the UK to ensure affordable flood insurance remains available to households in high-risk areas. Designed with a planned 25-year lifespan, Flood Re creates a transitional mechanism to move from subsidised to risk-reflective pricing by 2039.

Flood Risk Swaps: Banking instruments designed to transfer flood risk from one party to another, similar to credit default swaps but triggered by physical events rather than credit events.

Fluvial (Riverine) Flooding: Inundation is caused when water exceeds the capacity of river channels, resulting in overbank flow across the floodplain.

FourCastNet: NVIDIA's neural network approach to global weather forecasting can generate predictions up to 500 times faster than traditional numerical methods.

G

Gauge-Based Triggering System: This is a mechanism used in Physical Risk Swaps, where payouts are determined by water level measurements at specific river or flood gauges.

Generalised linear models (GLMs) are statistical models utilised in premium calculation that simultaneously consider multiple risk factors.

Generative Adversarial Networks (GANs) are a class of machine learning models designed for unsupervised learning, enabling synthetic data generation that mimics real-world distributions. GANs employ two neural networks—a “generator” and a “discriminator”—trained adversarially to improve each other iteratively.

Geographic Diversification: A risk management strategy that spreads property exposures across various watersheds and flood zones to reduce portfolio-level correlation effects.

Geomorphological Unit Hydrograph (GUH): An approach that connects hydrograph shape to catchment physical characteristics by incorporating stream order statistics and network width functions.

Gradient Boosting Algorithms: Machine learning techniques that create predictive models by combining multiple weak models, particularly effective for defining detailed insurance rate classes.

Graph Neural Networks: Specialised deep learning architectures that depict river systems as connected graphs, with nodes representing gauge locations and edges representing river reaches. This approach captures the topological relationships between different parts of a river network, improving the modelling of how flood waves propagate through the system.

GRIB2 Format: a standardised file format for storing and distributing gridded meteorological data, commonly used for weather model outputs and historical climate data. GRIB2 utilises compression techniques to efficiently store large volumes of data but requires specialised software for access and interpretation.

Groundwater Flooding is inundation caused by a rise in the water table to the ground surface, typically following extended periods of high precipitation.

Growth-at-Risk (GaR): A framework that associates banking conditions with the distribution of future economic growth, increasingly integrating physical risk factors.

H

Hazard Curve: A graphical representation of the relationship between hazard intensity (such as flood depth) and its probability of occurrence, which forms the foundation of physical risk assessment.

Hazard Identification: The first step in risk assessment involves recognising and characterising potential sources and mechanisms of flooding.

Hazard Rate: In the context of Physical Risk Swaps, the conditional probability of a physical event occurring within a small time interval, given that it has not happened previously.

HEC-RAS (Hydrologic Engineering Center's River Analysis System): Industry-standard software developed by the U.S. Army Corps of Engineers for hydraulic modelling, capable of simulating one-dimensional and two-dimensional water flow.

Hidden Markov Models (HMMs): Probabilistic models that treat precipitation as a system transitioning among hidden weather states, each associated with distinct precipitation characteristics.

High-Resolution Rapid Refresh (HRRR): This is an operational weather prediction model with a 3km spatial resolution that provides frequently updated forecasts, valuable for near-real-time risk assessment.

Hydraulic Modelling: The simulation of water movement based on fluid dynamics principles, typically solving the Saint-Venant equations to determine water surface elevations and flow velocities.

Hydrological Modelling: Depicting how precipitation transforms into runoff and river flow, incorporating factors such as infiltration, evaporation, and groundwater movement.

Hydrostatic Pressure Distribution: An assumption in the Saint-Venant equations that pressure varies linearly with depth at any point in a water column. This simplification enables practical hydraulic modelling while maintaining sufficient accuracy for most flood simulation applications.

I

Impact Quantification: Converting physical hazard characteristics into measurable effects on property, infrastructure, and economic systems.

Importance Sampling: A variance reduction technique used in Monte Carlo simulations that focuses computational resources on regions of the state space most relevant to risk assessment. This approach improves efficiency by concentrating sampling in areas of the probability distribution that have the greatest impact on flood risk, particularly in the tails of distributions that represent extreme events.

Infrastructure Failure Flood Types: Classifications for floods resulting from the failure of human-made structures such as dams, levees, or urban drainage systems.

Insurance Premium Equation: The mathematical formula that combines pure premium, expense loading, risk loading, and profit margin to determine the cost of insurance coverage.

Intensity-Duration-Frequency (IDF) Curves: Hydrological modelling uses graphic representations of the relationship between rainfall intensity, duration, and frequency.

Intensity-Duration-Frequency (IDF) Relationships: Mathematical functions that relate rainfall intensity to storm duration and recurrence frequency. These relationships, typically presented as curves or equations, provide essential inputs for hydrological modelling and are fundamental to flood hazard assessment.

ISDA Master Agreement: The standardised contract used for over-the-counter derivatives transactions, providing the contractual foundation for Physical Risk Swaps.

J

Joint Probability Analysis: Statistical techniques considering the simultaneous occurrence of multiple flood-causing factors, such as high river levels coinciding with heavy rainfall.

K

Kinematic Wave Approximation: A simplified form of the Saint-Venant equations that neglects acceleration and pressure terms, applicable primarily to steep slopes where gravity and friction forces dominate flow behaviour.

L

LISFLOOD: an open-source distributed hydrological model developed by the European Commission's Joint Research Centre to simulate flood events.

Loan-to-Value (LTV) Ratio: This ratio of mortgage debt to property value is a key factor in mortgage risk assessment and can be significantly affected by flood events.

Local Adaptation refers to tailoring global or regional models to capture location-specific weather or flood behaviour in modelling contexts.

Local Loss Model (LLM): A focused approach to risk assessment that applies detailed vulnerability functions to specific assets or small geographic areas. This methodology provides high-resolution loss estimates by considering individual property characteristics and their particular exposure to flood hazards.

Loss Given Default (LGD): This is the amount a lender loses when a borrower defaults on a loan, typically expressed as a percentage of the outstanding debt.

Lumped Models: Hydrological models that treat watersheds as single units with uniform characteristics, sacrificing spatial detail for computational efficiency.

M

Manning's n Coefficients: Parameters used in hydraulic modelling to represent the roughness or resistance to the flow of a channel or floodplain surface.

Market Value Correlation: The tendency for property values in a neighbourhood to rise or fall together following flood events, even affecting properties that weren't directly damaged.

Material Vulnerability Differentiation: Analysis of how different building materials respond to flood exposure, from structural elements to interior finishes.

Meteorological Models: Computational frameworks that simulate atmospheric conditions to predict weather patterns, particularly precipitation events that may lead to flooding.

Microtopography: Small-scale terrain features (curbs, berms, garden walls) significantly influence urban flood pathways despite their modest size.

Model Risk Governance is the framework of policies, controls, and procedures for managing risks that rely on models. It is crucial for flood risk assessment, where multiple models are linked.

Monte Carlo Simulation is a computational technique that uses repeated random sampling to obtain numerical results. It is handy for exploring the range of possible flood scenarios.

Mortgage Component: A standardised structure within the Common Domain Model that defines how physical risk affects loan-to-value calculations, default probabilities, and valuation impacts.

Multi-criteria Evaluation Frameworks: Assessment methodologies that incorporate multiple factors related to property resilience, weighted according to their relative importance.

Multi-gauge Prediction: The process of generating consistent water level forecasts across an entire network of monitoring points, capturing how flood waves propagate through river systems.

Multivariate Class Rating: Insurance premium calculation approaches that simultaneously consider numerous risk factors through statistical models like generalised linear models.

Muskingum Method: A hydrological flow routing model based on weighted storage, used to calculate the outflow of a river reach given an inflow hydrograph. The method conceptualises a river reach as having both prism storage (regular channel volume) and wedge storage (additional volume from flood waves), making it particularly useful for flood wave propagation modeling.

N

National Flood Insurance Program (NFIP): A U.S. government program established in 1968 to provide flood insurance to property owners in participating communities. It combines insurance availability with floodplain management requirements and employs Flood Insurance Rate Maps (FIRMs) to determine premium rates and regulate development.

Nash Cascade: A conceptual rainfall-runoff model that represents a catchment as a series of linear reservoirs, each with the same storage coefficient. This approach produces an instantaneous unit hydrograph that captures the essential features of catchment response while maintaining mathematical tractability.

Navier-Stokes Equations: The fundamental equations of fluid dynamics that describe the motion of viscous fluid substances, forming the theoretical basis for hydraulic modelling.

Neural Networks: Computational models inspired by the human brain that can learn patterns from data are increasingly applied to weather prediction and flood risk assessment.

Non-linear Gauge Response Modelling: The use of advanced statistical techniques to capture the complex, non-linear relationship between precipitation and water levels at monitoring points.

Non-stationarity: The concept that statistical properties of environmental systems change over time is particularly important in climate contexts where historical patterns may not represent future conditions.

O

One-dimensional (1D) Hydraulic Models: Simulation frameworks that represent water flow along a single spatial dimension, typically the channel centerline, using cross-sections to capture geometry.

Option-Adjusted Spread (OAS): A methodology that values embedded options in banking instruments by simulating numerous future scenarios for interest rates and housing prices.

Orographic Effects: The influence of mountains and other terrain features on precipitation patterns, often creating localised heavy rainfall that traditional weather models struggle to account for. **OS-Climate Framework:** An open-source collaborative initiative for climate risk assessment that standardises approaches to hazard curve development and vulnerability modelling.

Orographic precipitation is a type of precipitation that occurs when moist air is forced to rise over a mountain range or other elevated terrain. As the air rises, it cools and condenses, leading to the formation of clouds and, eventually, precipitation.

OS-SFT (Open-Source Sustainable Finance Taxonomy): A project that aims to harmonize regulatory interpretation across jurisdictions by creating standardized classifications for sustainable finance activities. This taxonomy proposed by Johnny Mattimore supports consistent assessment and reporting of physical risk impacts on financial assets across different regulatory regimes.

Overbank Flow: When water in a river channel exceeds capacity and spills onto the adjacent floodplain.

Own Risk Solvency Assessments (ORSA) Integration: The inclusion of hazard-related factors within an insurer's internal process for evaluating its risk management framework and current and future solvency positions. This integration ensures that potential climate impacts on capital adequacy and financial stability are systematically assessed and reported to regulators.

P

Parametric Insurance is insurance contracts that pay out based on predefined, objective measurements (such as water level at a gauge) rather than assessed damages, providing the foundation for Physical Risk Swaps.

Parametric Vulnerability Functions are mathematical distributions (like Beta, Gamma, or Lognormal) used to model the relationship between hazard intensity and damage, providing expected values and uncertainty ranges.

Physical Models: Hydrological approaches based on fundamental water movement and conservation equations demand significant data and computational resources but provide robust theoretical foundations.

Physical Risk Swap (PRS): A banking derivative instrument that transfers flood risk from mortgage lenders to counterparties better positioned to bear it. It is structured similarly to credit default swaps but triggered by physical events.

Physics-informed Neural Networks (PINNs): Machine learning models that combine neural network architectures with explicit constraints derived from the laws of physics, preserving physical realism while improving computational efficiency.

Pluvial (Surface Water) Flooding: Inundation is caused when rainfall overwhelms drainage systems or cannot infiltrate the ground, causing water to pool or flow over the land surface.

Portfolio Aggregation Techniques are methodologies for combining individual property-level risk assessments into comprehensive portfolio-wide measures. These techniques account for spatial correlation, exposure concentration, and diversification effects to determine overall risk profiles for mortgage or insurance portfolios.

Probability of Default (PD): In mortgage assessment, flood events may exacerbate a borrower's likelihood of defaulting on their loan due to property damage or disruption to income.

Product, Valuation, and Market Data (PVM): A classification system banking institutions use to process banking instruments, offering a framework for integrating physical risk into existing systems.

Property Clustering: The geographic concentration of properties in flood-prone areas, creating risk multiplier effects through correlated physical damage and market value impacts.

Property Component: A standardised structure within the Common Domain Model that captures essential physical characteristics, including geolocation, elevation, building specifications, and flood defence relationships.

Property-Level Flood Resilience (PFR): Specific measures implemented at individual buildings to reduce flood damage, including structural modifications, deployable barriers, and water-resistant materials. PFR approaches can significantly reduce loss potential through either resistance (keeping water out) or resilience (minimising damage when water enters).

Pure Premium: An insurance premium's expected loss cost component is determined through comprehensive risk modelling.

PVM (Product, Valuation, and Market Data): A classification system used by banking institutions to process banking instruments, offering a framework for integrating physical risk into existing systems. PVM provides the structural organization for how physical risk factors are incorporated into product definitions, valuation methodologies, and market data flows.

Q

Quasi-Monte Carlo (QMC) Methods: Computational techniques that systematically explore state spaces through carefully constructed low-discrepancy sequences, providing more uniform coverage than conventional random sampling.

Queuing Loss Model: A mathematical framework for calculating the probability of system failure when demand exceeds capacity, applied to urban drainage systems during flood events.

R

Rainfall-Runoff Models: Hydrological frameworks that transform precipitation inputs into water flow outputs across landscapes, incorporating factors like infiltration, evaporation, and groundwater interaction.

Rating Territories: Geographic zones used by insurers to classify and price flood risk, traditionally defined by administrative boundaries but increasingly based on hydrological and topographical characteristics. Modern approaches use high-resolution data to create micro-territories that reflect actual risk gradients more accurately.

Recurrence Interval: The average time between flood events of a specified magnitude, often expressed in years (e.g., "100-year flood").

Recurrent Neural Networks (RNNs) are machine learning architectures designed explicitly for sequential data like timeseries. They can learn temporal dependencies in water level predictions.

Regularisation: Machine learning techniques prevent overfitting by incorporating a penalty term into the loss function, which is crucial for developing robust flood prediction models.

Regulatory Capital Methodologies are frameworks that determine how much capital financial institutions must hold against specific risks. These methodologies are evolving for physical risks like flooding to incorporate hazard frequency, severity, and correlation effects within established frameworks like Basel standards.

Resilience Certification Programs: Standardized verification of implemented flood protection measures and their expected performance, increasingly recognised by insurance markets.

Return Period: The inverse of the annual exceedance probability; a 100-year return period corresponds to a 1% annual exceedance probability.

Risk-Reflective Pricing: Insurance premium calculation that accurately reflects a property's actual flood risk based on detailed hazard, exposure, and vulnerability assessment. This approach contrasts with subsidised or community-rated pricing by charging higher premiums for higher-risk properties, creating economic incentives for risk reduction.

Riverbank Breach Modelling: Specialized simulation of failures in natural riverbanks or engineered levees, often producing rapid inundation with distinct hazard characteristics.

Risk Loading: A component of an insurance premium that reflects uncertainty and volatility in the loss distribution, crucial for properties exposed to catastrophes.

Runoff Generation Process: The transformation of precipitation into water movement across landscapes, encompassing surface runoff, subsurface flow, and groundwater contribution.

S

Saint-Venant Equations: A system of partial differential equations describing one-dimensional unsteady flow in open channels, forming the mathematical foundation for hydraulic modelling.

Schedule Rating: An insurance pricing approach that applies credits or debits for specific risk characteristics not captured in standard rating factors, such as loss control measures or construction features.

SCS Dimensionless Unit Hydrograph: A standardized unit hydrograph developed by the Soil Conservation Service (now Natural Resources Conservation Service) that represents the temporal distribution of runoff from a unit of excess rainfall. It provides a template that can be scaled based on watershed characteristics, making it widely applicable across diverse landscapes.

Self-Organisation Capacity: The ability of natural systems to dynamically adjust to changing conditions, distinguishing resilient natural flood defences from engineered structures.

Semi-Distributed Models: Hydrological approaches that segment watersheds into sub-basins with unique characteristics, balancing computational efficiency and the representation of spatial variability.

Sensitivity Analysis: A systematic investigation into how changes in model inputs affect outputs is essential for understanding uncertainty in flood risk assessment.

Simplifying Complex Processes: The essential reduction of intricate physical processes (such as cloud formation) in models because of computational limitations, leading to downstream effects on flood risk assessment.

Solvency Capital Requirement (SCR): Under the Solvency II regulatory framework, insurers must hold a specific amount of capital for catastrophe risk, directly influencing premium pricing.

Spatially Distributed Hydrological Model: A modelling approach that divides watersheds into a grid of cells with unique characteristics to simulate water movement across landscapes with high

spatial resolution. These models can capture how terrain, soil, and land use variations influence runoff patterns and flood development.

Special Flood Hazard Areas (SFHAs): Designated zones on Flood Insurance Rate Maps where the National Flood Insurance Program's floodplain management regulations must be enforced and the mandatory purchase of flood insurance applies. These areas face at least a 1% annual chance of flooding.

Snyder's Method: An empirical approach to synthetic unit hydrograph development that relates key hydrograph parameters to watershed physical characteristics. Developed in the 1930s, this method uses regional coefficients to estimate lag time and peak discharge based on watershed length, centroid distance, and other measurable features.

Spatial coherence refers to the consistency or correlation of a physical quantity (such as light, sound, temperature, or other fields) over space. It describes how well a property or signal is correlated across different locations in space, and it's often used to analyse the structure of waves, signals, or phenomena as they propagate or vary over space.

Spatial Correlation: The tendency for nearby properties to experience similar flood impacts due to shared exposure characteristics, creating risk concentration in mortgage portfolios.

Stochastic Differential Equations (SDEs): Mathematical equations model random processes that evolve and are applied to weather pattern evolution in flood risk assessment.

Stochastic House Price Model: A mathematical approach that treats property values as random processes to simulate numerous possible future scenarios for mortgage valuation.

Strategic Default: A deliberate decision by a mortgage holder to stop making payments despite having the financial ability to continue, typically when a property's value falls significantly below the outstanding loan balance. In flood-prone areas, this can occur when properties become uninsurable or suffer substantial devaluation following flood events.

Structural Vulnerability Analysis: Assessment of how building materials, critical system placements, foundation types, and other physical characteristics influence flood resilience.

Survival Probability: In Physical Risk Swaps, the probability that the trigger event (e.g., flood level exceeding threshold) has not occurred by a specific time.

Synthetic Unit Hydrograph: Techniques for estimating flood response without observed data, including the SCS Dimensionless Unit Hydrograph and Snyder's Method.

T

Tail Value at Risk (TVaR): A risk measure reflecting the average loss in the worst scenarios (typically the worst 0.5%) utilised to determine the necessary capital for catastrophe exposures.

Temporal Dynamics: Evaluating how flood exposure shifts over time due to natural processes and human interventions, encompassing coastal erosion, riverbed migration, and infrastructure modifications.

Territorial Rating: An insurance pricing methodology that employs geographic location as a primary rating factor, increasingly enhanced through high-resolution flood risk data.

Thames Estuary 2100 Project: A comprehensive initiative linking climate projections to banking consequences, establishing a flexible adaptation pathway for infrastructure investment.

Timeseries Analysis involves analysing data points gathered or recorded at successive intervals, which is crucial for comprehending precipitation patterns and flood frequency.

Transformer Architectures: Advanced machine learning models initially designed for natural language processing but increasingly employed in sequential flood prediction problems.

Two-dimensional (2D) Hydraulic Models: Simulation frameworks that partition floodplains into grids or meshes, solving shallow water equations to ascertain water depth and velocity vectors throughout the domain.

U

Uncertainty Propagation: The process by which errors and uncertainty in input data and model structure are transmitted and potentially amplified through the modelling chain.

Unit Hydrograph: A mathematical method for understanding how catchments convert excess rainfall into runoff, illustrating the runoff response to a unit of rainfall input.

Urban Density: The concentration of buildings and impervious surfaces in developed areas, fundamentally altering hydrological processes by increasing runoff volume and peak flows.

Urban-Specific Flood Typologies: Specialized classification systems for urban flooding that incorporate drainage capacity exceedance, surface water flow paths, and infrastructure failure modes.

V

Validation Frameworks: Systematic methods for assessing model performance against observed data, particularly crucial for flood risk models that include forward-looking climate projections.

Value at Risk (VaR): A statistical measure of investment risk, indicating the maximum expected loss over a specified period at a given confidence level.

Variational Inference: A technique in machine learning that enables networks to learn and update probability distributions over weather patterns rather than producing single-point predictions.

Vine Copula Structure: A flexible method for modelling high-dimensional dependencies by breaking them down into bivariate relationships, helpful in capturing complex interactions between atmospheric variables.

Vulnerability Assessment: The process of evaluating how susceptible properties, infrastructure, and communities are to flood damage based on their physical, social, and economic characteristics.

Vulnerability Curves are mathematical functions that link hazard intensity (such as flood depth) to expected damage, typically represented as a percentage of asset value.

Vulnerability Function Testing: Empirical validation of damage prediction models compared with actual loss data from historical flood events.

W

Weather Pattern Distributions: Statistical representations of possible atmospheric conditions, focusing on those that could lead to extreme precipitation events.

Weather Pattern Timeseries: Chronological sequences of meteorological data that act as inputs to hydrological models and provide the foundation for integrated flood risk assessment.

Weighted Payout Structures: Settlement mechanisms in Physical Risk Swaps based on the severity and location of flooding, measured by multiple gauges within a catchment area.

Wetland Buffer Capacity: Natural wetland systems' ability to temporarily store floodwaters and release them gradually, thus reducing downstream peak flows.

X

X-Year Flood: A flood with a 1-in-X probability of occurring in any given year (e.g., a 100-year flood has a 1% annual probability of occurrence).

Y

Year-over-year growth: In mortgage risk assessment, the comparative analysis of property values or default rates across consecutive annual periods is utilised to identify emerging trends in flood-affected areas.

Yield Curve: In banking contexts, a curve illustrates the relationship between interest rates and various maturity lengths, similar to hazard curves that depict the relationship between flood magnitude and return period.

Z

Z-Score: A statistical measure that expresses a value's relationship to the mean of a group, used in flood risk analysis to identify statistical anomalies in gauge readings or damage patterns.

Zarr Format: A cloud-optimized data format for storing and accessing large arrays of meteorological data, enabling efficient processing of specific geographic subsets for regional flood analysis.

Zone-Based Pricing: Traditional flood insurance rating methods assign premiums based on broad risk zones, contrasted with property-specific assessment approaches that capture individual risk characteristics.

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